

Asia Utilities and Clean Energy Strategy

1H13E Outlook: Beneficiary of Gradual Economic Recovery

- Industry Overview
- Company Update
- Equities

- **Summary** — With economic growth set to recover in China and Asia, albeit moderately, and inflation and interest rates likely to stay low in 2013, the utilities sector's profits and volumes should increase while costs stay subdued. Top picks: *CRP, Huadian Power, CKI, ENN, Guangdong Investment and KEPCO*.
- **China IPPs preferred in pro-stability push** — Aside from the margin expansion induced by lower coal costs, IPPs Huaneng and CRP have positive free cash flows – a first since 03. CRP's under-performance vs. Huaneng is unjustified, in our view, as both have large capacity in coastal regions, where coal costs are lower. Huadian is a top buy as 4Q12E profits should rise with past overdues now cleared.
- **HK utilities a proxy for bond investment** — The SOC mid-term review in 2013 will likely maintain status quo. Positive catalysts would be M&As. HK utilities have become more appealing: HK\$ bond yield has halved to 0.7% in 10M12 and is hostage to the low US\$ rate. CKI does not have plans to issue new shares. At 10.7x FY13E PE, it is 22% below Power Asset despite sharing almost similar profit profile.
- **China gas sector: Opening the floodgates** — China's NG consumption should rise 16% yoy in 2013E, driven by additional supply from new infrastructure and substitution of costlier oil-related fuels. We like ENN for its resilient profits, with 50% new residential connections from existing buildings, and positive FCF.
- **Lower profits for China's power equipment sector** — China's coal fired equipment demand is low. Its 2020 nuclear target is 58GW, down from 70-80GW.
- **Renewables** — 2013E is a consolidation year; CDM revenue likely to fall.
- **Regional** — We like KEPCO on fuel cost cut, resumption of nuclear units, tariff hike.
- **Top Sells** — Dongfang Electric (lower 2013E profit), HKCG (risk from upstream projects), Goldwind (low profitability) and Indian power names.

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Figure 1. Asian Utilities — Top Buys

	Mkt Cap (US\$m)	Price 28-Nov-12	Target Price	Upside (%)	FY13E			
					PE	PB	ROE	Yld (%)
CR Power	10,538	HK\$17.16	HK\$21.00	22	10.1	1.4	14.4%	3.2
Huadian Power	2,273	HK\$2.39	HK\$2.80	17	7.3	0.8	10.7%	4.2
CKI	14,911	HK\$46.30	HK\$54.50	18	10.7	1.7	16.3%	3.9
ENN	4,649	HK\$33.75	HK\$39.00	16	16.0	2.9	19.9%	1.6
Guangdong Inv	4,987	HK\$6.20	HK\$7.00	13	11.7	1.5	13.7%	3.2
KEPCO	15,790	₩26,700	₩35,000	31	9.5	0.3	3.4%	3.3

Source: Citi Research estimates

See Appendix A-1 for Analyst Certification, Important Disclosures and non-US research analyst disclosures.

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1H13E Asian Utility & Clean Energy Outlook

Figure 2. Asian Utility — 1H13E Outlook by Country (✓ = positive; 0 = neutral; X = negative)

	Hong Kong	China				Korea	Malaysia	Philippines	Indonesia	India
	Power	Power	Power Equip	Gas	Wind	Power	Power	Power	Gas	Power/ Gas
Overall Outlook	Overweight ✓	Overweight ✓	Underweight (from neutral) X	Overweight ✓	Neutral (from Underweight) 0	Overweight ✓	Overweight ✓	Neutral 0	Underweight X	Underweight/ Neutral X/0
Major event	SOC mid-term review in 2013E to maintain status quo	Pro-stability economic push under new govt leadership	Sales volume cut; PRC 2020 nuclear capacity also cut to 58GW, from 70-80GW	Most gas companies guiding flat yoy connection fee in 2013E	Grid curtailment down 2ppts yoy to 16% in 2013E; but CDM revenue to fall in 2013E	Presidential election in Dec 2012E	General election before April 2013E	Election in May 2013; privatization is expected afterwards	Regulatory uncertainty on volume	Inadequate local fuel supply
Our view	0	0	X	✓	0	0	0	✓	X	X
Power/gas tariff	CLP plans 40% tariff hike with fuel cost to rise 2.5x, not for PAH	No tariff change has been planned by NDRC so far	Lower ASP for new orders amid competition	Retail price hike to lag upstream increase by 2-3 m/s for C&I users & 6-12 m/s for residential users	Renewable energy quota system might be announced in 1H13E, but moderate	Assumed next tariff hike of 5% effective on 1 July 2013	No tariff hike prior to general election	Upward pressure in spot price in medium term amid tighter supply	35% price hike at Western Java in Sep 2012; next tariff change in Mar 2013	Power: Needs 45-60% retail tariff hike to cover cost, too high / Gas: End users highly price sensitive (power & fertilizers)
Our view	✓	0	X	✓	0	✓	0	✓	✓	X
Fuel/raw material cost	No earnings impact from fuel cost change with full pass-through mechanism	Unit fuel cost to drop 2-5% yoy in 2013E, with QHD spot price -15% yoy in 1H13E but +10% yoy in 2H13E	Higher fixed cost per unit under reduced sales volume	Upstream NG price hike in 1H13E to pass through more gas from imports at high costs	Steady wind turbine sales price with ample supply in a buyer market	Unit fuel cost to drop 12% yoy in 2013E on lower unit coal, LNG and oil costs	Higher subsidized NG unit cost, if any, to be offset by tariff hike	Fuel cost pass through under PPAs	Spread uncertainty reduced after price hike in Sep 2012	Power: Imported coal unit cost 2.5x local's Gas: Higher cost from LNG imports
Our view	0	✓	X	X	0	✓	0	0	0	X
Margin trend	Stable	Up	Down	Down	Down	Up	Stable	Flat	Down	Down
Our view	0	X	X	X	X	✓	0	0	X	X
FY13E profit Growth (yoy)	+14.7%	+25.4%	-11.2%	+6.3% (HKCG -3%; others + over 20%)	+19.0%	Turnaround	+15.2%	+2.9%	+13.5%	Power: +9.1% Gas: +2.0%
Our view	✓	✓	X	✓	✓	✓	✓	X	✓	0 / X
Capacity growth	PAH to add new gas-fired unit in 2014-18E to cut emission; no new unit for CLP	National capacity to add 6-7% yoy in 2013E, no shortage is expected	New orders flow down yoy in 2H12E, both from China & exports (esp. India)	PRC NG usage up 16-17 pa in 2012-13E, sustainable growth	PRC new wind capacity flat yoy at 14GW in 2013E	Demand +4% in FY13E; reserve margin only 4%; MKE to take action if reserve <4GW	Reserve margin high at 34% for 2013E, no new plant in 2013-15E	New capacity needed in Luzon in 2014E. Shortage at Mindanao of 200MW in 2013E	Distribution volume growth of 6% yoy in 2013E	Power: 9.4GW (4%) added in Apr-Oct 12, below target; Gas: Local KG output cut in 2013E
Our view	✓	✓	X	✓	0	✓	✓		X	0
Interest rates in 1H13E	+10bps	unchanged	unchanged	unchanged	unchanged	-25bps	unchanged	unchanged	unchanged	-50bps
Our view	X	0	0	0	0	✓	0	0	0	✓
FY13E P/E	12.6x	9.0x	10.1x	20.6x	11.6x	9.7x	9.9x	14.7x	17.5x	Power: 13.6x/ Gas: 11.4x
Our view	0	✓	0	X	0	✓	✓	X	X	X / 0

Source: Company data, Citi Research estimates

Figure 3. Asia Utilities — Top Buys (Price as of 28 Nov 2012)

	Target Price	Upside (%)	FY13E				Positive catalysts
			PER (x)	PB (x)	ROE (%)	Yield (%)	
CR Power	HK\$21.00	22%	10.1	1.4	14.4%	3.2%	■ Prime beneficiary of more price cuts on seaborne coal, with 60% capacity in Guangdong and Jiangsu ■ Concerns overdone regarding lower profits from coal mines, which represent only 14% of 2012E net profit ■ Valuation premium narrows against peers despite better ROE
Huadian Power	HK\$2.80	17%	7.3	0.8	10.7%	4.2%	■ Profitability to improve from 4Q12E after clearing historical cost burden in 9M12 ■ Unit fuel cost cut of -5% pa in 2012-13E, the most among peers, with largest spot exposure & only 5% from key contracts ■ 2013E PER and PB look expensive, and yield attractive
CKI	HK\$54.50	18%	10.7	1.7	16.3%	3.9%	■ Too much concern on share placement risk as free float has reached management target of 25% after placement in Jul 2012 ■ Acquisitions to raise 2013E EPS by up to 15%, assuming 14% ROE ■ 22% discount to Power Asset on 2013E P/E looks unjustified as both have 90% of profits coming from the same sources
ENN	HK\$39.00	16%	16.0	2.9	19.9%	1.6%	■ High and sustainable NG sales volume growth of over 20% pa in 2012-14E ■ Resilient new household connections at 1.05m in 2013E, flat yoy, including 50% from old buildings and 50% from new ones ■ Valuation of 0.8x 2013E PEG looks undemanding, with positive free cashflow hence no share placement risk
Guangdong Inv.	HK\$7.00	13%	11.7	1.5	13.7%	3.2%	■ High quality earnings with 68% from water distribution and 17% from property investment in 1H12 ■ Financially sound for expansion at an 8% net debt to equity ratio by end-1H12 ■ Discounted valuation against other water names and HK utilities
KEPCO	W35,000	31%	9.5	0.3	3.4%	3.3%	■ Two one-off losses from nuclear decommissioning and suspension of 3 out of 23 nuclear units in 4Q12E non-recurring ■ To turn profitable in 2013E on unit fuel cost cut yoy and tariff hike in mid-2013E, after presidential election in Dec 2012 ■ Cheapest PB among Asian utilities

Source: Citi Research estimates

Figure 4. Asian Utilities — Top Sells (Price as of 28 Nov 2012)

	Target Price	Upside/Downside (%)	FY13E			Yield (%)	Negative catalysts
			PER (x)	PB (x)	ROE (%)		
HKCG	HK\$17.00	-19%	25.6	3.7	15.0%	2.0%	- Lower recurrent profit growth (below 10% pa) than pure PRC gas names but similar PER - Profit of new energy business to rise to 14% in 2013E, to lead to higher earnings volatility - A bonus share for every 10 shares paid in 2011 not value accretive on a fundamental basis
Dongfang Electric	HK\$9.00	-31%	10.9	1.2	11.4%	0.9%	- Net profit to fall 20% yoy in 2013E on revenue and margin cuts - Weak new orders flow yoy in 2012E and no recovery in 2013E - Losing market share in PRC nuclear equipment for focusing on 2G technology while new government approval to use 3G ones
Goldwind	HK\$2.30	-24%	21.7	0.5	2.3%	1.0%	- L-shaped industry recovery in next 2-3 years - Increased sales in 4Q12E for seasonal reasons but unsustainable in 1H13E and on thin GP margin
Adani Power	P34.00	-32%	109.7	1.8	-1.8%	0.0%	- Most profits from wind farm disposal, which is unpredictable - Extremely high fuel cost - A high net debt to equity ratio of 554% by end FY13E - Looks expensive on FY13E PER and PB

Source: Citi Research estimates

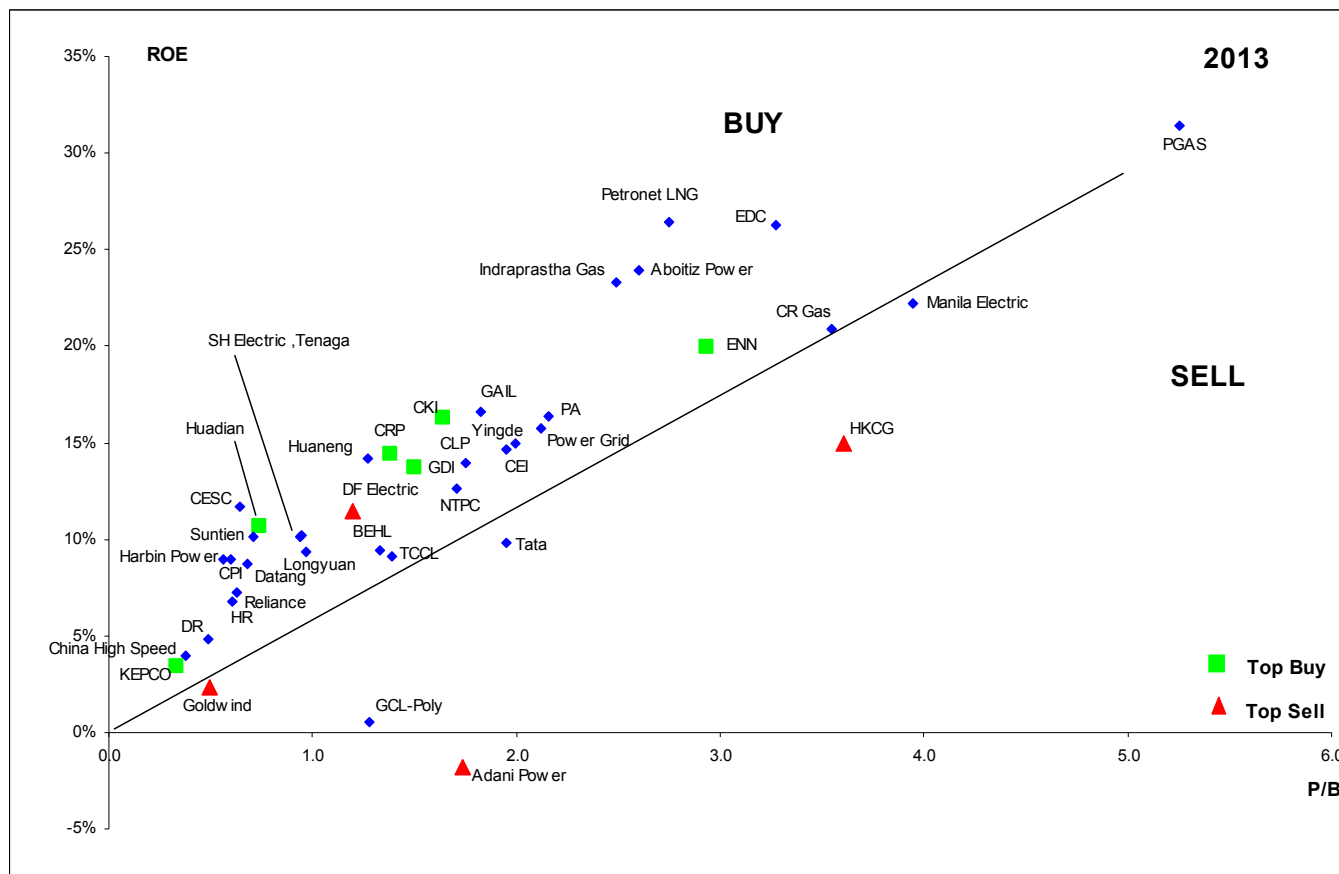
Citi Asian Utility Model Portfolio

Figure 5. Citi Model Portfolio of Asian Utilities against MSCI Utility Sector Weightings

Stock	Country	MSCI weighting	Recommended weighting	+/- against MSCI	Comment
China Resources Power	China	3.5%	7.0%	3.5%	Higher profit from coal price cut
CKI	Hong Kong	2.5%	5.5%	3.0%	Potential earnings increase from acquisitions
KEPCO	Korea	5.4%	8.0%	2.6%	Increase in profit from fuel cost cuts
ENN Energy	China	2.8%	5.0%	2.2%	Resilient EPS growth of 20% pa in 2012-14E
Power Assets Holdings	Hong Kong	10.1%	12.0%	1.9%	Potential earnings increase from acquisitions
Huaneng	China	2.2%	4.0%	1.8%	Higher profit from coal price cuts
Guangdong Investment	China	1.7%	3.0%	1.3%	Gradual completion of four property projects in 2014-17E
China Resources Gas	China	1.4%	2.5%	1.1%	High (20-25%) net profit growth from more NG sales
Petronas Gas	Malaysia	3.2%	4.0%	0.8%	Receiving 80% of India's LNG imports with rising volume
Energy Development	Philippines	1.0%	1.5%	0.5%	Capacity addition in Dec 2012 & M&As in 2H13E
Reliance Infrastructure	India	0.8%	1.2%	0.4%	Two roads to come online in FY13E
Tenaga Nasional	Malaysia	5.3%	5.5%	0.2%	Higher profit from coal price cut
Datang	China	0.9%	1.0%	0.1%	Potential losses from coal-to-gas & chemical projects
YTL Corp	Malaysia	2.3%	2.3%	0.0%	No coverage
KOGAS	Korea	1.5%	1.5%	0.0%	No coverage
Restricted name	China	1.5%	1.5%	0.0%	On restricted list
Glow Energy	Thailand	1.0%	1.0%	0.0%	No coverage
YTL Power International	Malaysia	0.9%	0.9%	0.0%	No coverage
Power Grid Corp. of India	India	2.0%	2.0%	0.0%	Most defensive play in Indian power sector
NTPC	India	1.5%	1.3%	-0.2%	Less fuel cost risk than its privately owned peers
Manila Electric Co.	Philippines	0.8%	0.6%	-0.2%	High valuations
GAIL	India	2.1%	1.8%	-0.3%	Lacks growth & catalyst
Aboitiz Power Corp.	Philippines	1.3%	1.0%	-0.3%	Gain from higher spot prices amid shortage
Reliance Power	India	0.8%	0.5%	-0.3%	No coverage
Contact Energy	NZ	1.3%	0.5%	-0.8%	Low growth in a developed market
Tata Power	India	1.5%	0.5%	-1.0%	Suffering from local fuel shortage
SP Ausnet	Australia	1.5%	0.5%	-1.0%	Low growth in a developed market
China Longyuan	China	1.2%	0.0%	-1.2%	2013E EPS cut from less CDM revenue
Perusahaan Gas Negara	Indonesia	4.4%	2.5%	-1.9%	High valuation versus 6% yoy volume growth in 2013E
APA Group	Australia	2.9%	1.0%	-1.9%	Low Australian NG consumption growth
CLP Holdings	Hong Kong	13.0%	10.2%	-2.8%	Low profitability in Australia and India
AGL Energy	Australia	6.6%	3.0%	-3.6%	Competitive retail market with high churn rate (16%)
Hong Kong & China Gas	Hong Kong	11.2%	7.2%	-4.0%	Increase earnings risk from more new energy businesses
		100.0%	100.0%	0.0%	

Source: MSCI and Citi Research estimates Note: We assign neutral weightings for non-rated stocks

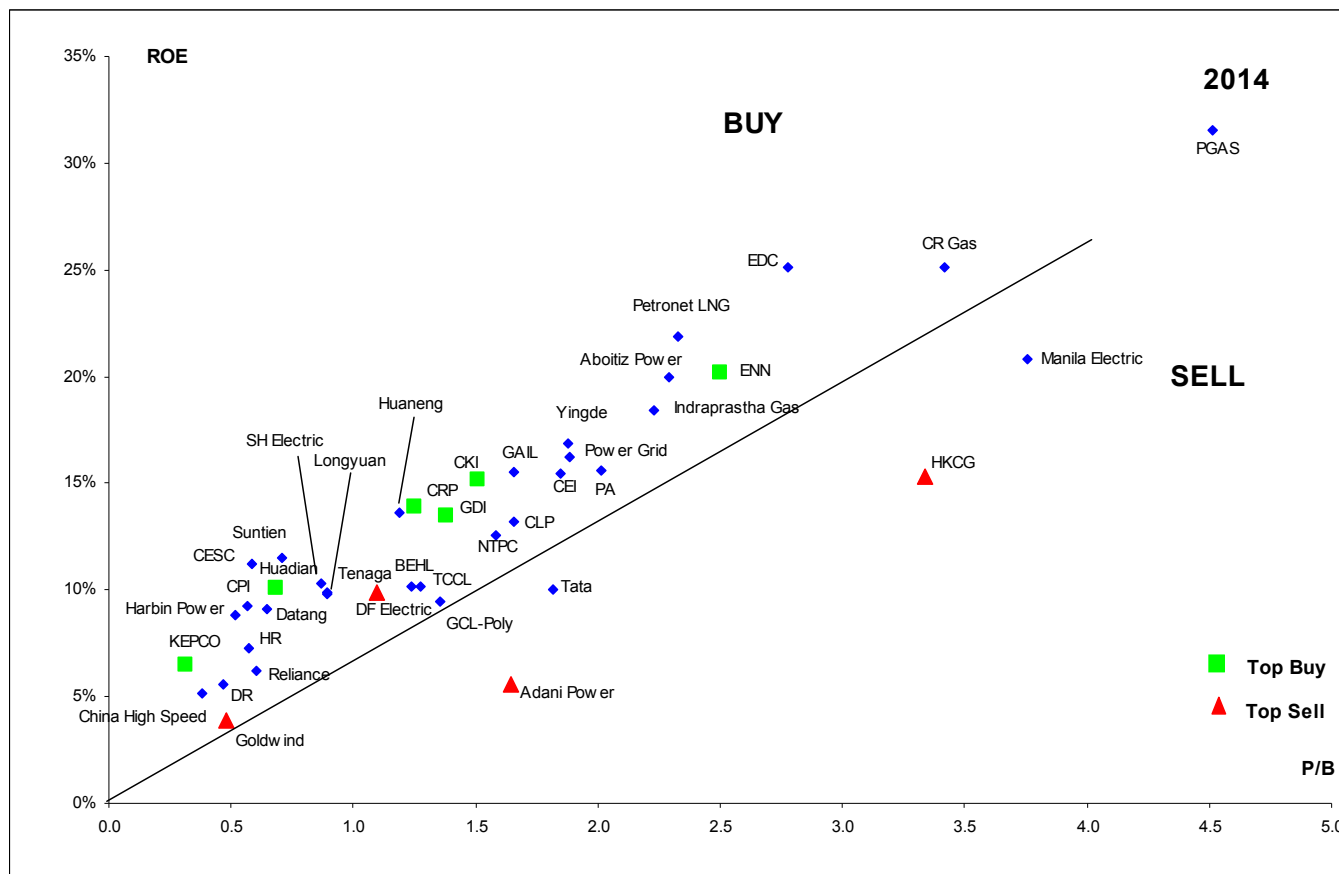
Figure 6. 2013E PB vs. ROE



Source: Citi Research

While our top buys in the Asian utilities sector are primarily based on valuations and potential catalysts in 1H13E, for reference we also examine profitability using an ROE vs. P/B approach. The chart above, based on 2013 estimates, supports our top buy calls on CRP, Huadian, CKI, ENN, Guangdong Investment and KEPCO, as well as top sell calls on HKCG and Goldwind.

Figure 7. Asian Utilities — 2014E PB vs. ROE



Source: Citi Research

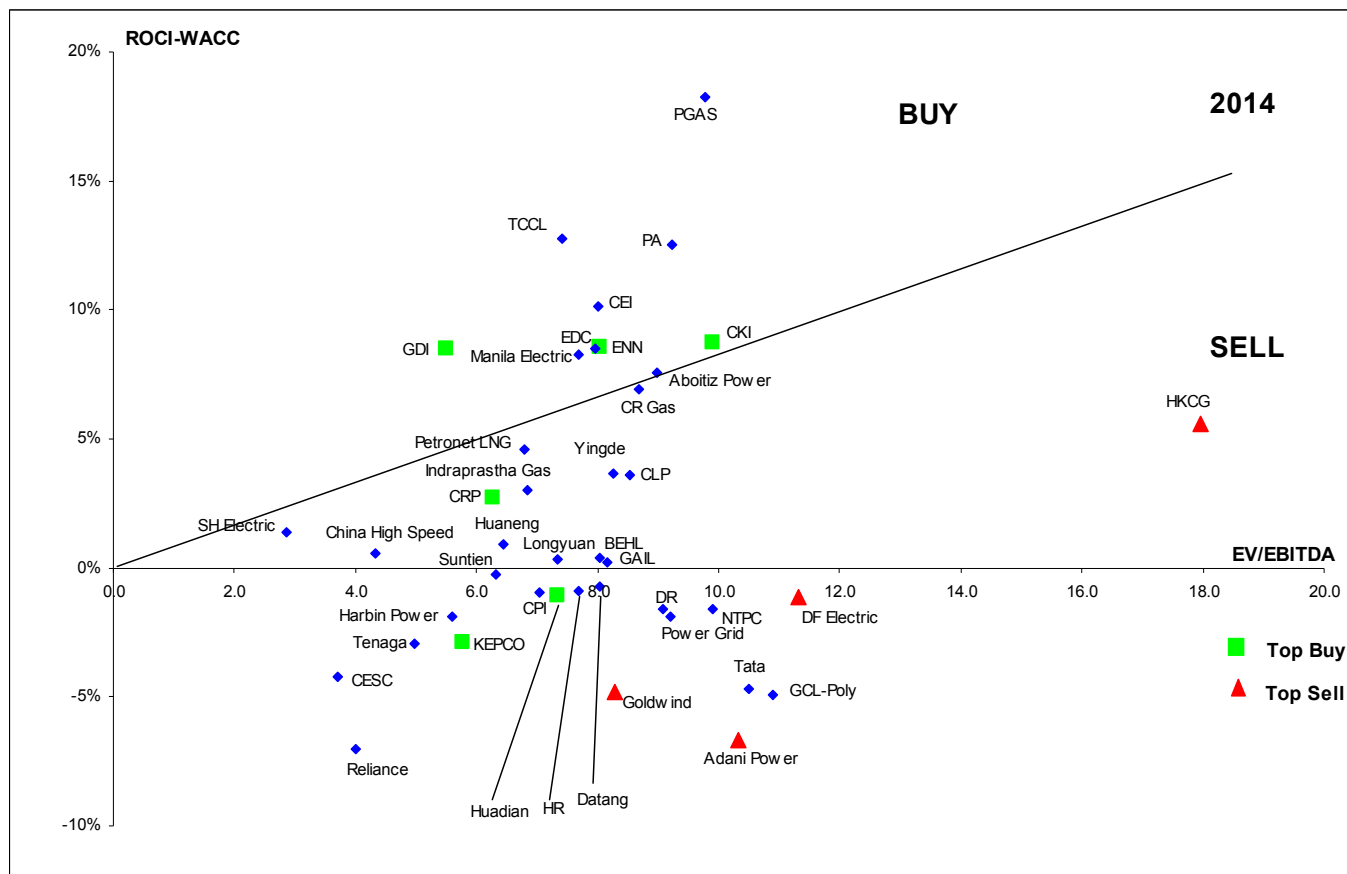
Using estimates for FY14E, the above chart supports all our six top buy calls – Huadian, CRP, CKI, ENN, Guangdong Investment and KEPCO, as well as our two top sells – HKG and Adani Power.

2013



We also check our Top Buys and Sells against their value creation ability (ROIC – WACC) vs. valuations (EV/EBITDA). The chart above, based on FY13E, supports our top buy calls on CKI, ENN and Guangdong Investment, but not on CRP, Huadian Power and KEPCO. The chart supports our all four top sell calls on HKCG, Dongfang Electric, Goldwind and Adani Power.

Figure 9. Asian Utilities — 2014E ROCI-WACC vs. EV/EBITDA



Source: Citi Research

Using FY14E data, the chart above endorses our top-buy ideas ENN, CKI and Guangdong Investment. The chart supports our all four top-sell calls on HKCG, Dongfang Electric, Goldwind and Adani Power.

Figure 10. Summary of Rating and Target Price Changes

Company	Code	New Rating	Old Rating	Chg	New TP	Old TP	Chg	Reason
ENN	2688.HK	BUY	BUY	Nil	39.00	31.50	24%	Rollover after lifting of restriction
PA	0006.HK	BUY	BUY	Nil	77.00	68.50	12%	Benchmarking to lower bond yields
Guangdong Inv.	0270.HK	BUY	BUY	Nil	7.00	6.50	8%	Rollover into 2013E
CRP	0836.HK	BUY	BUY	Nil	21.00	20.00	5%	Factoring lower unit fuel cost
Goldwind	2208.HK	SELL	SELL	Nil	2.30	2.20	5%	More profits in 4Q12E from wind farm sales & seasonal factor
CLP	0002.HK	NEUTRAL	NEUTRAL	Nil	70.00	68.00	3%	Rollover partly offset by profit cuts from Australia & India
CKI	1038.HK	BUY	BUY	Nil	54.50	53.20	2%	More profit from UK power networks for labor cut & larger asset base
CPI	2380.HK	BUY	BUY	Nil	2.50	2.50	0%	Nil
Datang	0991.HK	BUY	BUY	Nil	3.30	3.30	0%	Nil
Datang Renew.	1798.HK	BUY	BUY	Nil	1.08	1.08	0%	Nil
DF Electric	1072.HK	SELL	SELL	Nil	9.00	9.00	0%	Nil
HKCG	0003.HK	SELL	SELL	Nil	17.00	17.00	0%	Nil
Huadian	1071.HK	BUY	BUY	Nil	2.80	2.80	0%	Nil
Huaneng	0902.HK	BUY	BUY	Nil	7.50	7.50	0%	Nil
Huaneng Renew.	0958.HK	BUY	BUY	Nil	1.25	1.25	0%	Nil
KEPCO	015760.KS	BUY	BUY	Nil	35,000	35,000	0%	Nil
Longyuan	0916.HK	BUY	BUY	Nil	5.6	5.6	0%	Upgrade to Buy for replacing share issuance by perpetual bonds
SH Electric	2727.HK	BUY	BUY	Nil	3.5	3.5	0%	Nil
Harbin Power	1133.HK	BUY	BUY	Nil	7.50	8.30	-10%	Postponed downcycle with earnings decline in 2013E

Source: Citi Research estimates

Figure 11. Asia Pacific Utilities — Valuation Sheet

	FY	CIR	Price	Target	Total	Mkt-Cap	Avg t/o	EV	Installed	EV/MW	Share Price Movement					EPS Growth			ROE	ROA	Net D/E		
											RIC	End	Rating	28 Nov 2012	Price	Return	US\$m	US\$ m				US\$ m	capacity
Hong Kong																							
CLP	0002.HK	Dec	Neutral	HK\$67.10	HK\$70.00	8%	20,832	20.2	26,486	13,946	1.90	1%	2%	3%	5%	-3%	7%	24%	1%	13.9%	5.5%	77%	
Power Assets	0006.HK	Dec	Buy	HK\$67.85	HK\$77.00	17%	18,685	22.9	21,249	4,071	5.22	0%	3%	11%	24%	14%	8%	8%	2%	16.4%	10.0%	26%	
CKI	1038.HK	Dec	Buy	HK\$46.30	HK\$54.50	21%	14,911	13.6	15,313	n/a	n/a	1%	1%	-1%	6%	6%	18%	10%	2%	16.3%	12.3%	-1%	
China (Power)																							
Huaneng	0902.HK	Dec	Buy	HK\$6.28	HK\$7.50	24%	11,811	13.6	33,205	52,478	0.63	-3%	3%	17%	28%	53%	431%	26%	3%	14.2%	3.0%	226%	
Datang	0991.HK	Dec	Buy	HK\$2.73	HK\$3.30	26%	4,689	3.7	28,034	28,377	0.99	0%	0%	3%	1%	4%	62%	15%	10%	8.7%	1.3%	302%	
Huadian	1071.HK	Dec	Buy	HK\$2.39	HK\$2.80	19%	2,273	2.4	16,616	23,752	0.70	4%	13%	-1%	42%	66%	1092%	101%	2%	10.7%	1.2%	421%	
CRP	0836.HK	Dec	Buy	HK\$17.16	HK\$21.00	25%	10,538	12.5	20,341	21,612	0.94	1%	3%	3%	25%	14%	57%	16%	7%	14.4%	4.3%	146%	
CPI	2380.HK	Dec	Buy	HK\$2.03	HK\$2.50	30%	1,454	1.7	7,487	12,461	0.60	-4%	-4%	-4%	32%	26%	132%	7%	9%	8.9%	1.9%	220%	
China (Power equipment)																							
Dongfang Electric	1072.HK	Dec	Sell	HK\$13.02	HK\$9.00	-30%	3,938	9.7	7,983	n/a	n/a	-1%	4%	14%	-23%	-44%	-20%	-21%	-5%	11.4%	2.6%	132%	
Harbin Power	1133.HK	Dec	Buy	HK\$6.35	HK\$7.50	21%	554	3.2	2,163	n/a	n/a	1%	-2%	2%	-11%	-10%	-2%	-11%	6%	8.9%	2.1%	51%	
Shanghai Electric	2727.HK	Dec	Buy	HK\$3.10	HK\$3.50	16%	5,129	9.1	3,760	n/a	n/a	2%	-3%	-3%	-3%	-4%	6%	-4%	9%	10.1%	2.9%	-38%	
China (Gas)																							
Beijing Enterprises	0392.HK	Dec	Buy	HK\$50.10	HK\$57.30	16%	7,354	6.3	7,960	n/a	n/a	0%	2%	3%	15%	20%	17%	22%	18%	9.6%	4.6%	27%	
CR Gas	1193.HK	Dec	Neutral	HK\$16.40	HK\$15.80	-3%	4,706	11.3	4,850	n/a	n/a	-3%	1%	15%	17%	41%	18%	15%	25%	20.9%	5.5%	53%	
ENN Energy	2688.HK	Dec	Buy	HK\$33.75	HK\$39.00	17%	4,649	6.9	5,396	n/a	n/a	-1%	5%	14%	25%	25%	19%	21%	19%	19.9%	6.0%	57%	
HKCG	0003.HK	Dec	Sell	HK\$20.90	HK\$17.00	-17%	23,436	13.9	25,263	n/a	n/a	3%	2%	13%	25%	27%	19%	-7%	10%	15.0%	8.1%	24%	
Towngas China	1083.HK	Dec	Buy	HK\$6.49	HK\$7.20	12%	2,060	1.1	2,429	n/a	n/a	3%	7%	12%	25%	49%	14%	24%	19%	9.4%	5.0%	27%	
Yingde Gas	2168.HK	Dec	Neutral	HK\$7.58	HK\$7.00	-6%	1,767	3.2	1,842	n/a	n/a	5%	3%	13%	13%	-8%	-21%	21%	18%	15.0%	6.4%	102%	
China (Clean Energy)																							
China High Speed	0658.HK	Dec	3H	HK\$2.53	HK\$2.00	-21%	445	3.2	496	n/a	n/a	2%	-3%	0%	-20%	-32%	-51%	7%	29%	3.9%	1.4%	78%	
Goldwind	2208.HK	Dec	Sell	HK\$3.01	HK\$2.30	-23%	1,047	0.7	83	n/a	n/a	0%	2%	17%	-10%	-32%	-59%	21%	75%	2.3%	0.8%	50%	
China Longyuan	0916.HK	Dec	Buy	HK\$5.19	HK\$5.60	10%	4,999	4.7	10,610	5,937	1.79	8%	4%	1%	16%	-9%	9%	-6%	13%	9.3%	2.5%	170%	
Datang Renewables	1798.HK	Dec	Buy	HK\$0.84	HK\$1.08	29%	788	0.5	4,239	4,028	1.05	5%	2%	9%	-19%	-41%	-72%	121%	20%	4.9%	0.7%	327%	
Huaneng Renewables	0958.HK	Dec	Buy	HK\$1.17	HK\$1.25	8%	1,275	0.7	4,241	2,523	1.68	4%	16%	26%	-15%	-33%	-54%	52%	14%	6.8%	1.5%	216%	
China Suntien	0956.HK	Dec	Buy	HK\$1.51	HK\$1.95	33%	631	0.7	1,112	2,524	0.44	-3%	1%	5%	9%	1%	-5%	19%	16%	9.7%	3.0%	143%	
Guangdong Investment	0270.HK	Dec	Buy	HK\$6.20	HK\$7.00	16%	4,987	5.1	5,360	2,525	2.12	2%	1%	12%	18%	32%	18%	-6%	8%	13.7%	9.0%	-8%	
China Everbright Int'l	0257.HK	Dec	Buy	HK\$3.70	HK\$4.60	24%	1,928	9.0	2,303	2,525	0.91	-2%	-7%	-5%	1%	30%	29%	0%	11%	14.6%	6.6%	36%	
GCL-Poly	3800.HK	Dec	3H	HK\$1.27	HK\$1.00	-21%	2,536	35.8	3,395	n/a	n/a	-2%	-4%	7%	-26%	-37%	-134%	106%	1500%	0.5%	0.1%	153%	
India (Power)																							
Adani Power	ADAN.BO	Mar	Sell	Rs49.85	Rs34.00	-32%	1,951	2.5	4,072	6,600	0.62	6%	5%	29%	-2%	-32%	-118%	207%	244%	-1.8%	-0.2%	554%	
CESC	CESC.BO	Mar	Sell	Rs293.35	Rs300.00	4%	658	4.0	1,035	975	1.06	6%	4%	-5%	8%	12%	13%	8%	7%	11.7%	5.0%	42%	
NTPC	NTPC.BO	Mar	Neutral	Rs158.60	Rs173.00	12%	23,474	8.4	28,680	31,000	0.93	-2%	-6%	-8%	10%	-1%	1%	9%	7%	12.6%	6.6%	47%	
Reliance	RLIN.BO	Mar	Buy	Rs470.70	Rs620.00	34%	2,222	21.7	3,073	885	3.47	3%	-4%	1%	-1%	15%	64%	-14%	-10%	7.2%	3.1%	37%	
Tata Power	TPPW.BO	Mar	Sell	Rs103.35	Rs85.00	-16%	4,402	7.6	5,427	2,368	2.29	3%	-1%	5%	12%	10%	21%	9%	9%	9.8%	4.5%	74%	
Power Grid	PGRD.BO	Mar	Buy	Rs118.40	Rs140.00	20%	9,840	7.7	16,786	n/a	n/a	-2%	4%	-2%	11%	22%	29%	20%	15%	15.7%	3.9%	232%	
India (Gas)																							
GAIL	GAIL.BO	Mar	Neutral	Rs339.40	Rs389.00	17%	7,728	6.9	7,128	n/a	n/a	-2%	-7%	-7%	4%	-13%	3%	4%	4%	16.6%	8.9%	33%	
Indraprastha Gas	IGAS.BO	Mar	1H	Rs258.35	Rs300.00	18%	649	3.4	635	n/a	n/a	1%	3%	2%	25%	-36%	18%	2%	-10%	23.3%	12.0%	59%	
Petronet LNG	PLNG.BO	Mar	Buy	Rs160.15	Rs180.00	14%	2,156	4.0	2,638	n/a	n/a	2%	-4%	5%	22%	-2%	71%	-3%	-1%	26.4%	10.7%	64%	
Rest of Asia																							
PGAS - ID	PGAS.JK	Dec	Sell	Rp4,650.0	Rp4,375.0	-2%	11,742	11.4	12,068	n/a	n/a	1%	3%	27%	25%	58%	18%	10%	11%	37.7%	22.2%	-36%	
KEPCO - KR	015760.KS	Dec	Buy	W26,700	W35,000	31%	15,790	39.7	49,904	63,529	0.79	-1%	-4%	6%	19%	3%	46%	199%	96%	3.4%	1.2%	105%	
Tenaga - MY	TENA.KL	Aug	Buy	RM6.93	RM8.75	29%	12,536	13.5	16,856	10,836	1.56	0%	0%	2%	6%	30%	335%	18%	10%	10.2%	4.3%	34%	
Aboitiz - PH	AP.PS	Dec	Buy	P35.05	P36.00	6%	6,306	1.9	6,324	2,051	3.08	1%	6%	4%	3%	20%	9%	-5%	-4%	23.9%	12.4%	-7%	
EDC - PH	EDC.PS	Dec	Buy	P7.09	P7.65	10%	3,250	2.8	4,093	n/a	n/a	4%	13%	20%	22%	18%	66%	25%	16%	26.3%	9.2%	96%	
MER - PH	MER.PS	Dec	Sell	P264.20	P235.00	-9%	7,286	3.6	7,311	n/a	n/a	3%	-5%	7%	18%	8%	6%	0%	-1%	22.2%	7.2%	-36%	
Wgt average (by mkt cap)												1.04	1%	1%	6%	14%	14%	58%	24%	29%	14.6%	6.7%	74%

Source: Company Reports and Citi Research estimates

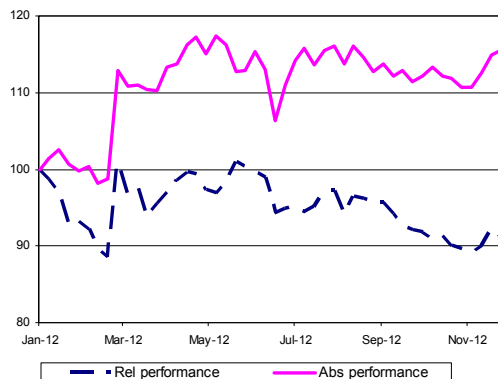
Figure 12. Asia Pacific Utilities — Valuation Sheet (cont'd)

	RIC	Business Type	P/E (x)				P/B (x)				EV/EBITDA (x)				Yield (%)			
			FY11	FY12E	FY13E	FY14E	FY11	FY12E	FY13E	FY14E	FY11	FY12E	FY13E	FY14E	FY11	FY12E	FY13E	FY14E
Hong Kong																		
CLP	0002.HK	Generation	17.4	16.3	13.1	13.0	2.0	1.9	1.8	1.7	9.8	9.8	8.7	8.6	3.8%	3.8%	4.0%	4.3%
Power Assets	0006.HK	Generation	16.0	14.7	13.7	13.3	2.5	2.3	2.2	2.0	18.9	18.8	16.9	15.4	3.4%	3.7%	4.0%	4.1%
CKI	1038.HK	Infrastructure	14.0	11.8	10.7	10.5	2.1	1.8	1.7	1.5	76.5	67.0	59.3	57.3	3.3%	3.7%	3.9%	4.1%
China (Power)																		
Huaneng	0902.HK	Generation	60.1	11.3	8.9	8.7	1.4	1.3	1.2	1.1	9.6	7.1	6.5	6.4	1.0%	4.4%	5.6%	5.8%
Datang	0991.HK	Generation	14.8	9.2	8.0	7.3	0.7	0.7	0.7	0.6	10.1	8.7	8.0	8.0	5.0%	4.9%	5.6%	6.2%
Huadian	1071.HK	Generation	n/a	14.8	7.3	7.2	0.8	0.8	0.8	0.7	11.2	8.4	7.3	7.4	0.0%	2.1%	4.2%	4.3%
CRP	0836.HK	Generation	18.3	11.6	10.1	9.4	1.7	1.5	1.4	1.2	10.3	7.8	6.8	6.2	1.7%	2.7%	3.2%	3.4%
CPI	2380.HK	Generation	16.4	7.1	6.6	6.1	0.6	0.6	0.6	0.5	11.0	8.0	7.5	7.0	2.8%	6.4%	6.9%	7.5%
China (Power equipment)																		
Dongfang Electric	1072.HK	Power equipment	6.9	8.6	10.9	11.4	1.5	1.3	1.2	1.1	8.7	10.3	11.2	11.3	1.5%	1.2%	0.9%	0.9%
Harbin Power	1133.HK	Power equipment	5.7	5.8	6.5	6.2	0.7	0.6	0.6	0.5	6.5	5.8	6.1	5.6	2.7%	2.7%	2.7%	2.7%
Shanghai Electric	2727.HK	Power equipment	9.8	9.2	9.6	8.8	1.1	1.0	0.9	0.9	3.3	3.3	3.4	2.8	3.1%	3.2%	3.1%	3.4%
China (Gas)																		
Beijing Enterprises	0392.HK	Gas	21.4	18.4	15.1	12.8	1.5	1.4	1.3	1.2	10.9	10.1	8.7	7.6	1.4%	1.7%	2.1%	2.5%
CR Gas	1193.HK	Gas	25.0	21.2	18.3	14.7	4.0	4.1	3.6	3.3	16.9	11.2	9.8	8.6	0.7%	1.0%	1.2%	1.5%
ENN Energy	2688.HK	Gas	23.0	19.2	16.0	13.4	4.0	3.4	2.9	2.5	12.1	10.2	9.0	7.9	1.1%	1.3%	1.6%	1.9%
HKCG	0003.HK	Gas	28.2	23.7	25.6	23.2	4.4	4.0	3.7	3.4	25.4	21.4	19.8	18.2	2.3%	1.7%	2.0%	2.2%
Towngas China	1083.HK	Gas	22.5	19.7	15.9	13.4	1.7	1.6	1.4	1.3	14.2	11.6	9.5	7.7	0.8%	0.9%	1.1%	1.3%
Yingde Gas	2168.HK	Gas	13.3	16.9	13.9	11.8	2.2	2.1	2.0	1.9	8.2	10.6	9.1	8.4	2.1%	2.6%	3.2%	4.0%
China (Clean Energy)																		
China High Speed	0658.HK	Wind equipment	5.0	10.2	9.5	7.4	0.4	0.4	0.4	0.4	2.5	4.3	4.4	4.3	13.5%	0.0%	3.6%	4.1%
Goldwind	2208.HK	Wind equipment	10.7	26.2	21.7	12.4	0.5	0.5	0.5	0.5	2.8	8.4	9.5	8.2	2.1%	0.8%	1.0%	1.8%
China Longyuan	0916.HK	Wind generation	11.8	10.9	11.5	10.2	1.2	1.1	1.0	1.0	8.7	7.7	7.9	7.5	1.7%	1.8%	1.7%	1.9%
Datang Renewables	1798.HK	Wind generation	6.7	24.3	11.0	9.2	0.5	0.5	0.5	0.5	8.7	10.3	9.5	9.1	5.8%	0.8%	1.8%	2.2%
Huaneng Renewables	0958.HK	Wind generation	6.7	14.5	9.6	8.5	0.7	0.7	0.6	0.6	8.1	8.3	8.0	7.7	0.0%	1.4%	2.1%	2.4%
China Suntien	0956.HK	Wind generation	8.8	9.2	7.7	6.7	0.8	0.8	0.7	0.7	6.1	7.8	7.7	7.0	4.8%	5.0%	5.3%	5.5%
Guangdong Investment	0270.HK	Water	12.9	11.0	11.7	10.8	1.8	1.7	1.5	1.4	7.2	6.3	6.1	5.6	2.9%	3.1%	3.2%	3.4%
China Everbright Int'l	0257.HK	Environmental	17.1	13.2	13.2	11.8	2.2	2.0	1.9	1.8	10.9	10.3	8.5	7.6	1.2%	1.7%	1.7%	1.9%
GCL-Poly	3800.HK	Solar	4.6	n/a	216.5	13.5	1.0	1.1	1.2	1.3	3.5	37.7	15.4	10.8	4.4%	0.0%	0.1%	1.5%
India (Power)																		
Adani Power	ADAN.BO	Generation	20.8	n/a	109.7	31.9	1.7	1.8	1.8	1.7	22.1	30.6	16.1	10.5	0.0%	0.0%	0.0%	0.0%
CESC	CESC.BO	Generation	7.5	6.6	6.1	5.8	0.9	0.8	0.7	0.6	4.8	4.5	4.2	3.8	1.4%	1.7%	1.7%	1.7%
NTPC	NTPC.BO	Generation	15.0	14.9	13.6	12.7	1.9	1.8	1.7	1.5	11.0	11.5	10.5	9.7	2.4%	2.5%	2.6%	2.8%
Reliance	RLIN.BO	Generation	13.0	7.9	9.2	10.2	0.7	0.7	0.6	0.6	5.8	3.2	4.5	4.2	1.7%	1.9%	2.1%	2.4%
Tata Power	TTPW.BO	Generation	28.1	23.2	21.2	19.5	2.3	2.2	2.0	1.9	16.2	14.0	11.8	10.8	1.2%	1.2%	1.3%	1.3%
Power Grid	PGRD.BO	Transmission	21.5	16.8	14.0	12.1	2.6	2.3	2.1	1.9	12.5	11.1	9.9	9.2	1.5%	1.8%	1.9%	2.0%
India (Gas)																		
GAIL	GAIL.BO	Gas	12.1	11.8	11.3	11.0	2.2	2.0	1.8	1.6	8.0	8.1	7.9	8.0	2.2%	2.6%	2.7%	2.9%
Indraprastha Gas	IGAS.BO	Gas	13.9	11.8	11.6	12.9	3.6	2.9	2.5	2.2	7.7	6.7	6.6	6.9	1.9%	1.9%	2.3%	2.7%
Petronet LNG	PLNG.BO	Gas	19.4	11.4	11.7	11.7	4.5	3.4	2.8	2.4	11.4	7.7	7.5	6.9	1.2%	1.6%	1.9%	1.9%
Rest of Asia																		
PGAS - ID	PGAS.JK	Gas	17.4	14.7	13.4	12.1	6.8	5.6	4.6	3.9	12.0	10.1	8.8	7.6	3.5%	4.1%	4.5%	5.0%
KEPCO - KR	015760.KS	Generation	n/a	n/a	9.5	4.8	0.3	0.3	0.3	0.3	12.6	12.9	6.9	5.7	0.0%	0.0%	3.3%	6.4%
Tenaga - MY	TENA.KL	Generation	52.1	12.0	10.2	9.3	1.2	1.1	1.0	0.9	10.0	5.9	5.4	5.0	0.5%	2.9%	2.9%	3.2%
Aboitiz - PH	AP.PS	Generation	12.2	11.2	11.8	12.3	3.8	3.0	2.6	2.3	9.5	8.1	8.7	9.1	3.8%	2.9%	3.2%	3.0%
EDC - PH	EDC.PS	Geothermal	29.9	17.9	14.4	12.4	4.9	4.1	3.4	2.9	13.2	10.2	9.0	8.2	2.3%	2.0%	2.2%	2.8%
MER - PH	MER.PS	Transmission	19.7	18.6	18.5	18.7	4.7	4.3	4.1	3.9	9.0	8.4	8.2	8.1	3.0%	3.3%	4.0%	4.0%
Wgt avg (by mkt cap)			20.0	13.9	16.2	12.5	2.4	2.2	2.0	1.8	16.0	14.7	12.8	11.9	2.3%	2.6%	3.1%	3.5%

Source: Company Reports and Citi Research estimates

Aboitiz Power (AP.PS)

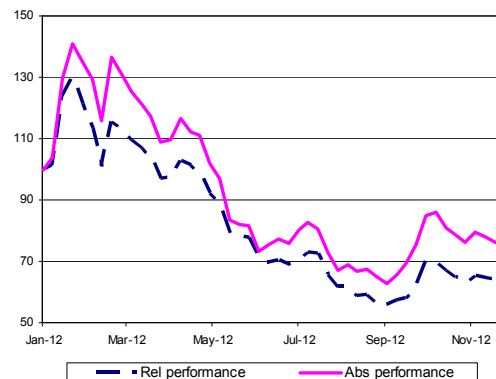
Price performance relative to country index



Price on 28 Nov	P35.05
12-month target	P36.00
Rating	Buy
Market Cap (US\$m)	6,306
Avg daily Turnover (US\$m)	1.9

- Potential upside from an upward trend in spot market prices and from power-shortage situations
- To add at least 300MW of new generation capacity by 2014, 200MW by 2015 and 300MW by 2017
- Key risk: Earnings weakness in 2013E on higher steam cost and slowing sales of ancillary services

Adani Power (ADAN.BO)

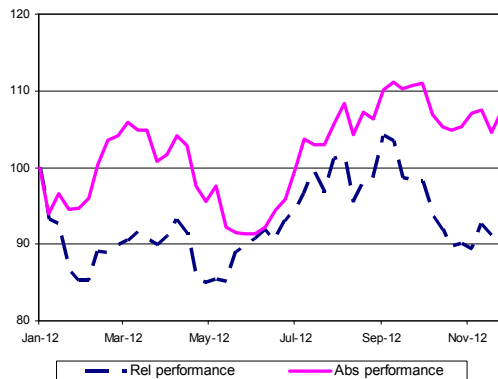


Price on 28 Nov	Rs49.85
12-month target	Rs34.00
Rating	Sell
Market Cap (US\$m)	1,951
Avg daily Turnover (US\$m)	2.5

- Extremely high fuel cost
- High net debt to equity ratio of 554% by end FY13E
- Looks expensive at FY14E PER and PB

Beijing Enterprises (0392.HK)

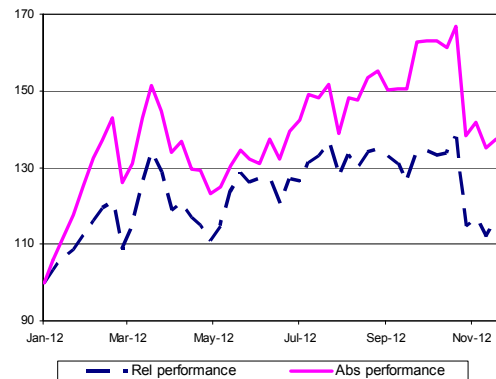
Price performance relative to country index



Price on 28 Nov	HK\$50.10
12-month target	HK\$57.30
Rating	Buy
Market Cap (US\$m)	7,354
Avg daily Turnover (US\$m)	6.3

- Mid-teens NG sales volume growth in 2013E, after 17% growth in 2012E, driven by ramp of cogen gas power plant
- Shaanxi-Beijing pipelines to drive higher gas transmissions
- Positioning for waste-to-energy opportunities via Beijing Development HK (154 HK, not rated)

CESC (CESC.BO)

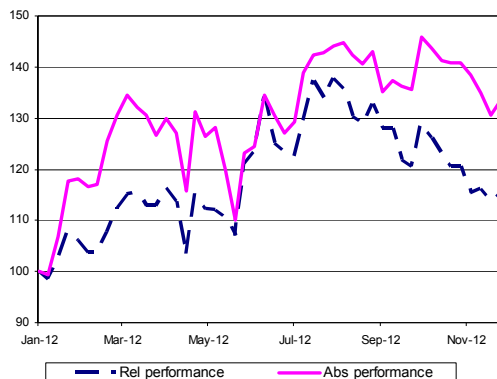


Price on 28 Nov	Rs293.35
12-month target	Rs300.00
Rating	Sell
Market Cap (US\$m)	658
Avg daily Turnover (US\$m)	4.0

- Continues to engage in unrelated diversification - has acquired FSL (a BPO company)
- Spencer's retail continues to lose money
- Kolkata distribution business continues to generate healthy cash, but cash is being dissipated in non-related businesses

China Everbright Int'l (0257.HK)

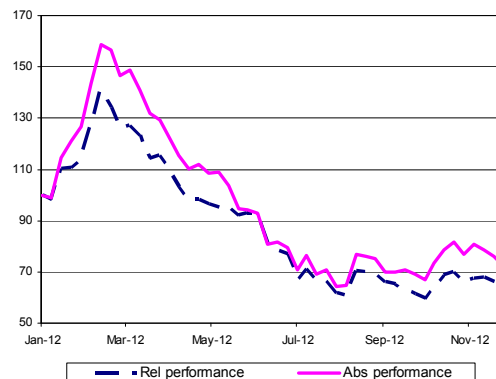
Price performance relative to country index



Price on 28 Nov HK\$3.70
12-month target HK\$4.60
Rating Buy
Market Cap (US\$m) 1,928
Avg daily Turnover (US\$m) 9.0

- Ramping capacity to handle PRC environmental issues including wastewater and solid waste projects
- Rising operating profits from existing plants
- Completed equity fundraising in mid-2012; removed share price overhang

China High Speed Transmission (0658.HK)



Price on 28 Nov HK\$2.53
12-month target HK\$2.00
Rating Sell/ High Risk
Market Cap (US\$m) 445
Avg daily Turnover (US\$m) 3.2

- No improvement in domestic order demand, as sector undergoes inventory reduction
- Focusing on export orders ahead of CDM reduction
- Investment in coal-mining and agriculture will take additional time to deliver results

CKI (1038.HK)

Price performance relative to country index



Price on 28 Nov HK\$46.30
12-month target HK\$54.50
Rating Buy
Market Cap (US\$m) 14,911
Avg daily Turnover (US\$m) 13.6

- Too much concern on share placement risk as free float has reached management target of 25% after placement in Jul 2012
- Acquisitions to raise 2013E EPS by up to 15%, assuming 14% ROE
- 2013E PER looks unjustified at 20% discount to Power Asset's while both have 90% profits coming from same sources

CLP (0002.HK)

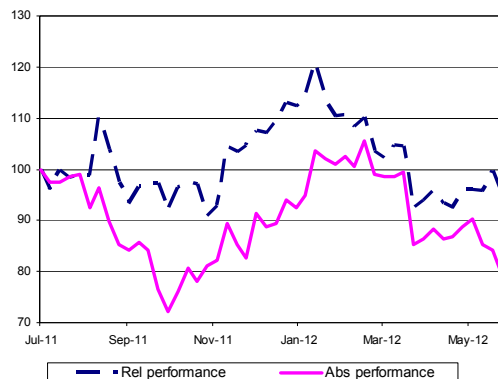


Price on 28 Nov HK\$67.10
12-month target HK\$70.00
Rating Neutral
Market Cap (US\$m) 20,832
Avg daily Turnover (US\$m) 20.2

- SOC interim review in 2013E likely statue quo hence limited risk
- Lower margin of Australian retail power business in NSW from 1 July 2013 under new retail price calculation formula
- Indian coal shortage persists, but some improvement in 4Q12-1Q13E

CPI (2380.HK)

Price performance relative to country index



Price on 28 Nov

HK\$2.03

12-month target

HK\$2.50

Rating

Buy

Market Cap (US\$m)

1,454

Avg daily Turnover (US\$m)

1.7

- Hydropower profit +2x yoy in 9M12E
- Unit fuel cost -1% yoy in 2012E and -2% yoy in 2013
- Inexpensive valuations at 7x 2013E PER and 0.6x PB

CR Gas (1193.HK)



HK\$16.40

HK\$15.80

Neutral

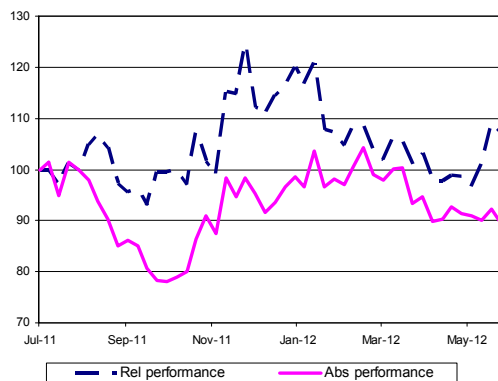
4,706

11.3

- Low EPS growth after recent share placement and completion of asset injection from parent
- Tianjin JV should see earnings contribution after 2-3 years
- More expensive valuations than peers

CRP (0836.HK)

Price performance relative to country index



Price on 28 Nov

HK\$17.16

12-month target

HK\$21.00

Rating

Buy

Market Cap (US\$m)

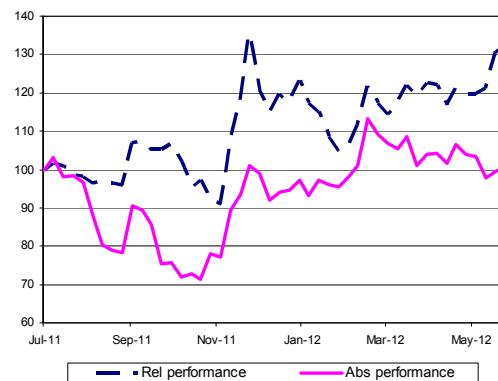
10,538

Avg daily Turnover (US\$m)

12.5

- Prime beneficiary of more price cut of seaborne coal
- Concern overdone regarding lower profit from coal mines, which represent only 14% 2012E net profit
- Narrower valuation premium against peers despite better ROE

Datang (0991.HK)



HK\$2.73

HK\$3.30

Buy

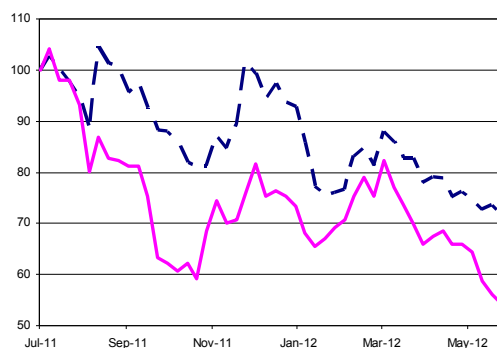
4,689

3.7

- Declining unit fuel cost driving profit growth in 2012-13E
- Potential loss of Keqi coal-to-gas as NG sales price to PetroChina is being squeezed
- Possible loss for coal-to-chemical project due to poor execution, unable to keep utilization above breakeven of 70%

Datang Renewables (1798.HK)

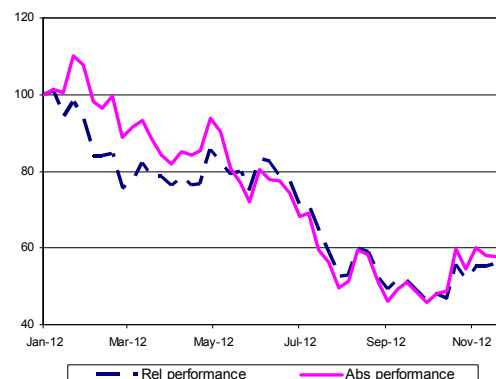
Price performance relative to country index



Price on 28 Nov HK\$0.84
12-month target HK\$1.08
Rating Buy
Market Cap (US\$m) 788
Avg daily Turnover (US\$m) 0.5

- Gross generation up over 25% yoy in 9M12, despite utilization cut 8% yoy
- Grid curtailment trough in 2012E, slight improvement in 2013E
- The most inexpensive PRC nationwide wind farm operator

Dongfang Electric (1072.HK)

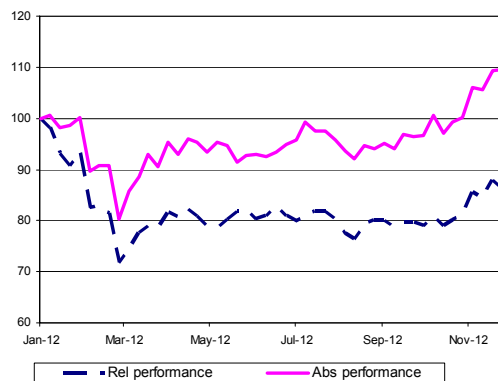


HK\$13.02
HK\$9.00
Sell
3,938
9.7

- Lower GP margin in 2013E, dragged by contractions at wind equipment and engineering services
- Losing PRC market share in nuclear equipment on changing technology from 2G to 3G
- Net profit cuts of 20% pa in 2012-13E on revenue and margin cuts

Energy Development (EDC.PS)

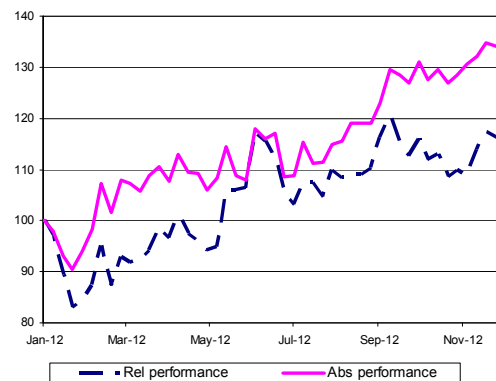
Price performance relative to country index



Price on 28 Nov P7.09
12-month target P7.65
Rating Buy
Market Cap (US\$m) 3,250
Avg daily Turnover (US\$m) 2.

- Catalysts: 1) delivering on earnings and 2) commissioning of BacMan plant by yearend (will drive 25% earnings growth in 2013)
- Upside in the medium to long-term could come from its venture into wind power and overseas greenfield projects.
- Renewable energy play: low-cost producer and is not exposed to fuel price risk

ENN Energy (2688.HK)

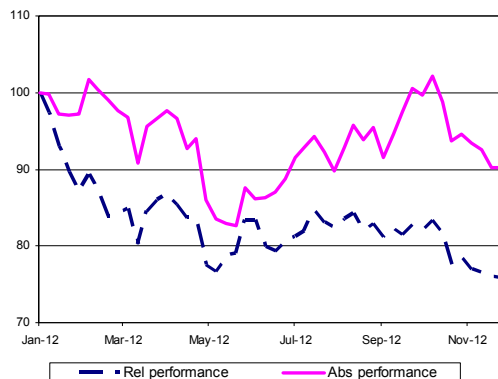


HK\$33.75
HK\$39.00
Buy
4,649
6.9

- High and sustainable NG sales volume growth of over 20% pa in 2012-14E
- Resilient new household connections at 1.05m in 2013E, flat yoy, including 50% from old buildings and 50% from new ones
- Looks undemanding at 0.8x 2013E PEG with positive free cashflow hence no share issuance risk

GAIL (GAIL.BO)

Price performance relative to country index



Price on 28 Nov	Rs339.40
12-month target	Rs389.00
Rating	Neutral
Market Cap (US\$m)	7,728
Avg daily Turnover (US\$m)	6.9

- Lacks near-term catalysts; vols could only pick up post 4Q once Dabhol and Kochi LNG terminals are commissioned
- Stands to benefit from the gov't decision to cap LPG cylinder subsidy, but rollback is a risk
- Valn for gas business at ~11x P/E does not look compelling enough given constrained growth outlook

GCL-Poly (3800.HK)

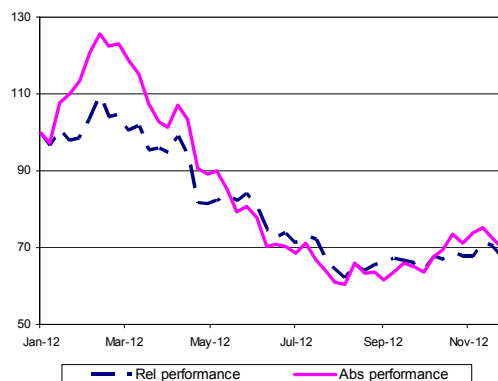


Price on 28 Nov	HK\$1.27
12-month target	HK\$1.00
Rating	Sell/ High Risk
Market Cap (US\$m)	2,536
Avg daily Turnover (US\$m)	35.8

- Sharp drop in poly and wafer shipments as company faces challenging demand and poly/wafer pricing
- Despite selling of US solar projects, rising inventory may put more pressure on company's cash flows
- Opportunity in developing solar farms in China in 2013

Goldwind (2208.HK)

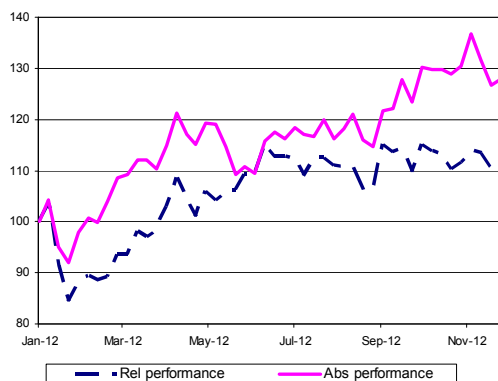
Price performance relative to country index



Price on 28 Nov	HK\$3.01
12-month target	HK\$2.30
Rating	Sell
Market Cap (US\$m)	1,047
Avg daily Turnover (US\$m)	0.7

- L-shaped industry recovery in next 2-3 years
- More sales in 4Q12E for seasonal reason but unsustainable in 1H13E and on thin GP margin
- Most profit from wind farm disposal, which is unpredictable

Guangdong Investment (0270.HK)

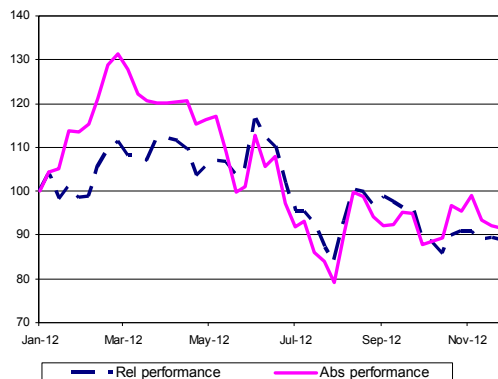


Price on 28 Nov	HK\$6.20
12-month target	HK\$7.00
Rating	Buy
Market Cap (US\$m)	4,987
Avg daily Turnover (US\$m)	5.1

- High quality earnings with 68% from water distribution and 17% from property investment
- Financially sound for expansion at 8% net debt to equity ratio by end 1H12
- Discounted valuation against other water names and HK utilities

Harbin Electric (1133.HK)

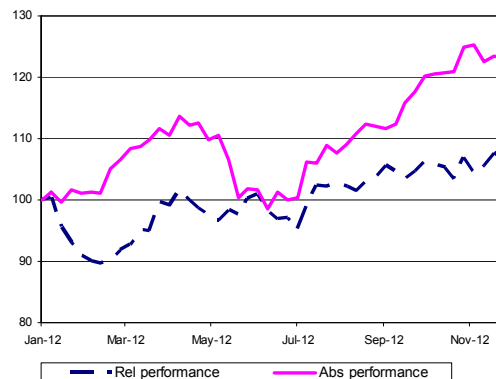
Price performance relative to country index



Price on 28 Nov	HK\$6.35
12-month target	HK\$7.50
Rating	Buy
Market Cap (US\$m)	554
Avg daily Turnover (US\$m)	3.2

- Declining revenue in 2012-13E for weak coal-fired equipment demand and small scale of new energy equipment business
- Margin contraction in 2H12-FY13E owing to ASP cuts
- Weak new orders flow of only Rmb7bn in 3Q12, equal to 29% that of 1H12

HKCG (0003.HK)

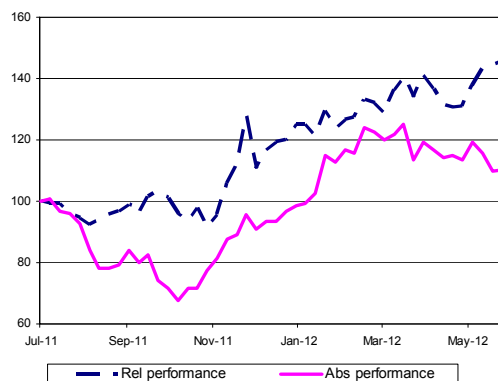


Price on 28 Nov	HK\$20.90
12-month target	HK\$17.00
Rating	Sell
Market Cap (US\$m)	23,436
Avg daily Turnover (US\$m)	13.9

- Lower recurrent profit growth (below 10% pa) than pure PRC gas names but similar PER
- Profit of new energy business to rise to 14% in 2013E, to lead to higher earnings volatility
- A bonus share for every 10 shares paid in 2011 not value accretive on a fundamental basis

Huadian (1071.HK)

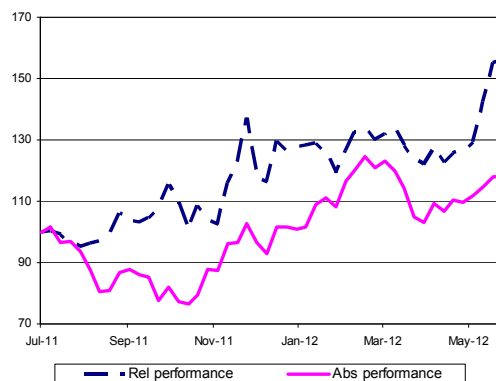
Price performance relative to country index



Price on 28 Nov	HK\$2.39
12-month target	HK\$2.80
Rating	Buy
Market Cap (US\$m)	2,273
Avg daily Turnover (US\$m)	2.4

- Profitability to improve from 4Q12E after paying historical cost burden in 9M12
- Declining unit fuel cost of -5% pa in 2012-13E
- Looks inexpensive on 2013E PER and PB

Huaneng (0902.HK)

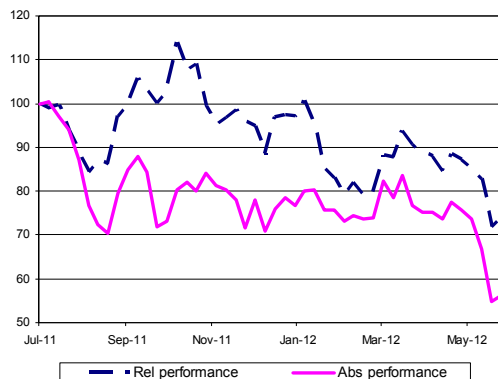


Price on 28 Nov	HK\$6.28
12-month target	HK\$7.50
Rating	Buy
Market Cap (US\$m)	11,811
Avg daily Turnover (US\$m)	13.6

- Generation started showing positive growth in Oct 2012
- Unit fuel cost to drop at least 5% yoy in 2012E and was down mom in Oct
- Positive free cashflow under curtailed capex, the first time since 2003

Huaneng Renewables (0958.HK)

Price performance relative to country index



Price on 28 Nov

HK\$1.17

12-month target

HK\$1.25

Rating

Buy

Market Cap (US\$m)

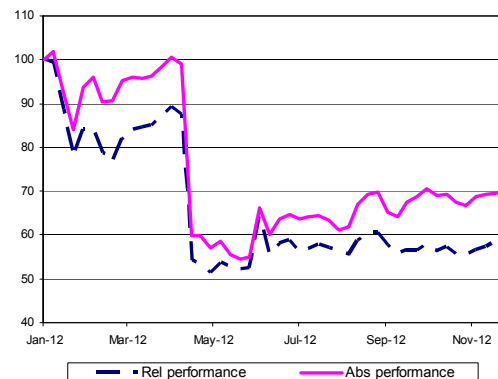
1,275

Avg daily Turnover (US\$m)

0.7

- Gross generation +15% yoy in 9M12 driven by capacity addition
- Provision possible in 2H12E on CDM revenue booked in 1H12 due to default of fixed-price contracts
- No share issuance dilution risk for trading below 1.0x PB

Indraprastha Gas (IGAS.BO)



Rs258.35

Rs300.00

Buy/ High Risk

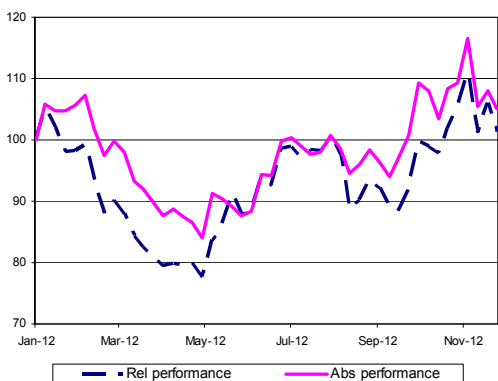
649

3.4

- Fundamental strength continues on the back of strong EBITDA/scm led by regular price hikes
- PNGRB order, however, likely to remain an overhang as the matter is still sub-judice
- Maintain Buy, but a 'high risk' one

KEPCO (015760.KS)

Price performance relative to country index



Price on 28 Nov

₩26,700

12-month target

₩35,000

Rating

Buy

Market Cap (US\$m)

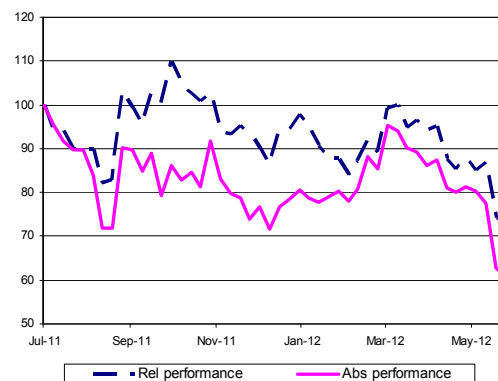
15,790

Avg daily Turnover (US\$m)

39.7

- Two one-off losses from nuclear decommissioning and suspension of 3 out of 23 nuclear units in 4Q12E non-recurring
- Profitable in 2013E on unit fuel cost cut yoy
- Tariff hike in mid 2013E, after presidential election in Dec 2012

Longyuan (0916.HK)



HK\$5.19

HK\$5.60

Buy

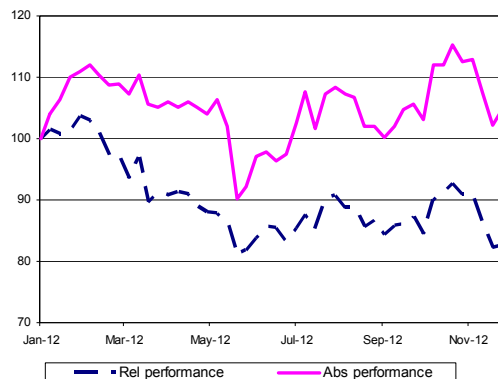
4,999

4.7

- Share issuance overhang removed by issuing perpetual bonds instead
- Expecting perpetual bond issuance to have good market response amid ample liquidity and heightened bond prices
- Looks inexpensive in terms of PB

Manila Electric (MER.PS)

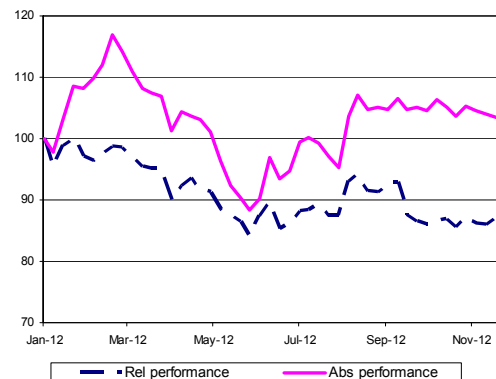
Price performance relative to country index



Price on 28 Nov	P264.20
12-month target	P235.00
Rating	Sell
Market Cap (US\$m)	7,286
Avg daily Turnover (US\$m)	3.6

- Lackluster earnings growth outlook in 2013-14E on the back of flattish distribution tariff
- Valuations appear steep: 2013E PE of 18x vs Philippine utility average of 12.5x
- Earnings impact from power generation could come in 4-5 years' time

NTPC (NTPC.BO)

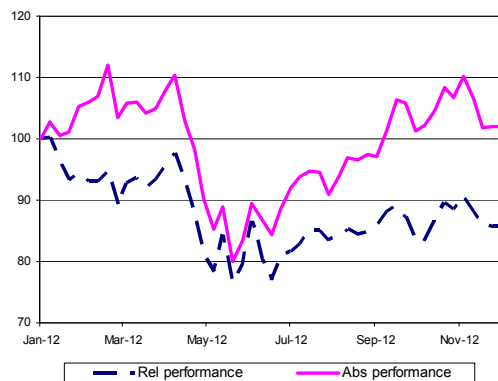


Price on 28 Nov	Rs158.60
12-month target	Rs173.00
Rating	Neutral
Market Cap (US\$m)	23,474
Avg daily Turnover (US\$m)	8.4

- Business remains steady - working on adding 14GW capacity in 12th plan
- Improvement at margin - CIL's coal production is up 9% YTD which has helped fuel supply
- However PLF / PAF still continue to be impacted by low coal supplies, adversely impacting RoEs

Petronet LNG (PLNG.BO)

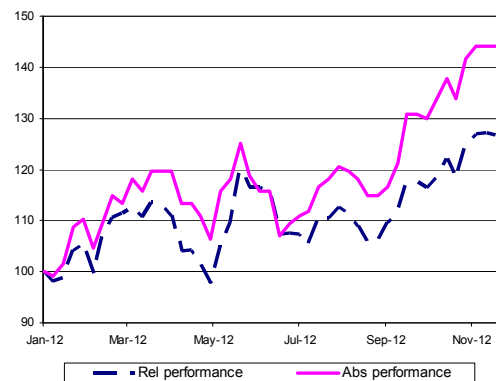
Price performance relative to country index



Price on 28 Nov	Rs160.15
12-month target	Rs180.00
Rating	Buy
Market Cap (US\$m)	2,156
Avg daily Turnover (US\$m)	4.0

- At the forefront of LNG regas capacity build-out; will still control ~50% of the market by FY17E
- Near-term earnings under pressure as new capacities kick in, but will gain from operating leverage and first mover advantage
- Remain positive on the long-term LNG outlook for India and believe regulatory risk is limited

PGAS (PGAS.JK)

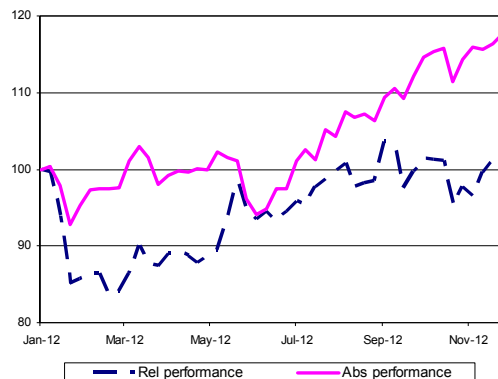


Price on 28 Nov	Rp4,600
12-month target	Rp4,375
Rating	Sell
Market Cap (US\$m)	11,742
Avg daily Turnover (US\$m)	11.4

- Regulatory uncertainty remains a major concern, as it will impact near term volumes
- Valuations are at +1std above mean, limiting upside from current levels
- Earnings and share prices sensitive to volume/Spread changes

Power Assets (0006.HK)

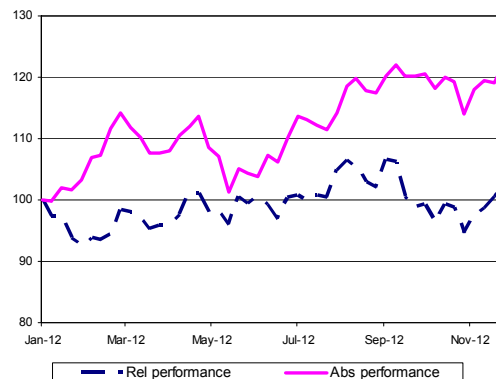
Price performance relative to country index



Price on 28 Nov HK\$67.85
12-month target HK\$77.00
Rating Buy
Market Cap (US\$m) 18,685
Avg daily Turnover (US\$m) 22.9

- Potential acquisition of regulated utility assets in the UK to raise FY13E net profit up to 11%
- Net debt to equity ratio at 26% by end FY12E, able to spend HK\$30bn for new deals while keeping its A+ credit rating
- Attractive yield against other low risk asset classes such as high quality HK\$ bonds and deposits

Power Grid Corp of India (PGRD.BO)

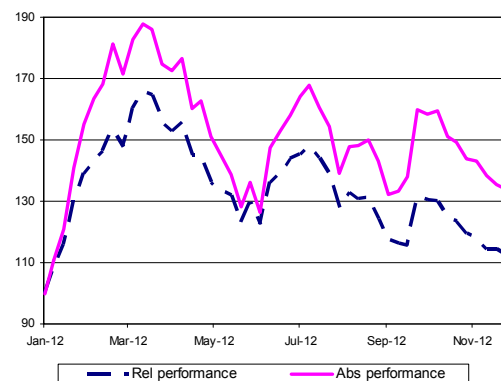


Rs118.40
Rs140.00
Buy
9,840
7.7

- Safe business model, in our view. Set to deliver 17% EPS growth over FY12-15E with average RoEs of 16%
- 12th plan target capex is ~2x of 11th plan target (which has been achieved successfully)
- Chances of equity raising are low

Reliance Infrastructure (RLIN.BO)

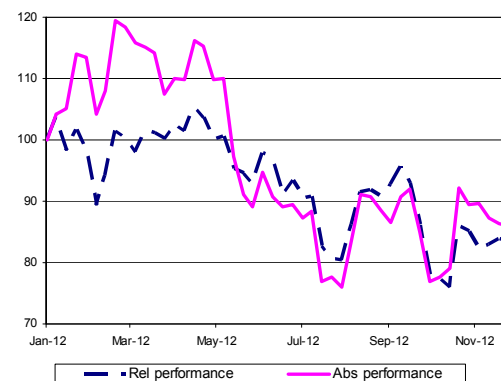
Price performance relative to country index



Price on 28 Nov Rs470.70
12-month target Rs620.00
Rating Buy
Market Cap (US\$m) 2,222
Avg daily Turnover (US\$m) 21.7

- Adjusted cons BV of Rs658 provides valuation support
- Mumbai and Delhi tariff hikes have come through
- Execution on diverse infrastructure assets remains more or less on track

Shanghai Electric (2727.HK)

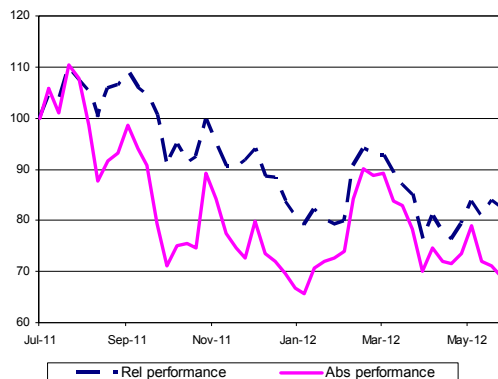


HK\$3.10
HK\$3.50
Buy
5,129
9.1

- 3Q12 profit missed expectations mainly due to lower revenue
- Revenue to rebound qoq in 4Q12E
- New orders to rise qoq in 4Q12E

Suntien (0956.HK)

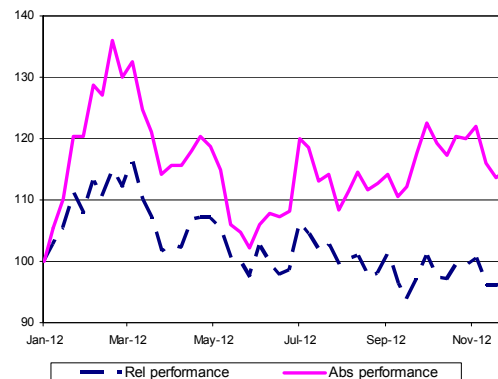
Price performance relative to country index



Price on 28 Nov HK\$1.51
12-month target HK\$1.95
Rating Buy
Market Cap (US\$m) 631
Avg daily Turnover (US\$m) 0.7

- Earnings growth to resume in 2013E, after a slowdown in gas volume and wind installs in 2012E
- Facing limited wind grid-curtailment issues; achievable wind farm installation target, based on central approvals
- Gas volume to improve on city-gas demand and potential new gas supplies in the coming 1-2 years

Tata Power (TTPW.BO)

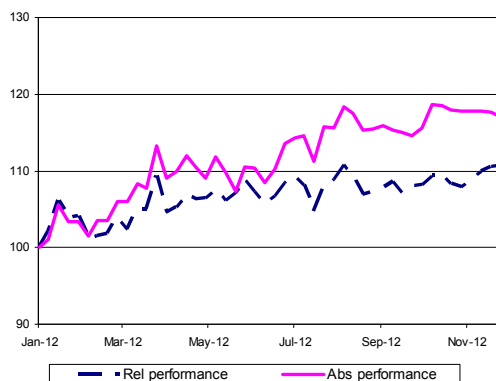


Rs103.35
Rs85.00
Sell
4,402
7.6

- Trading at P/E of 17x-20x over FY13-FY15E and P/BV of 1.7x with average RoE of 9.5%
- Mundra UMPP tariff revision chances are low; stock is already factoring in a high chance of revision of tariff
- Mundra UMPP and Maithon projects continue to make losses

Tenaga (TENA.KL)

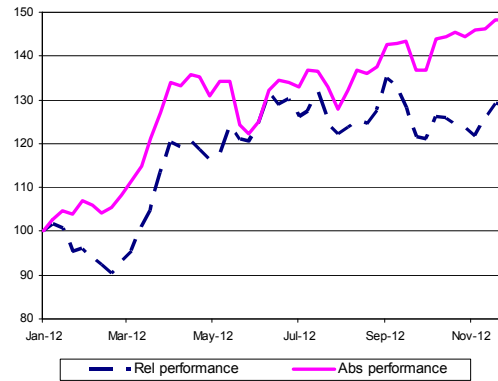
Price performance relative to country index



Price on 28 Nov RM6.93
12-month target RM8.75
Rating Buy
Market Cap (US\$m) 12,536
Avg daily Turnover (US\$m) 13.5

- Positive on tariff adjustment but only after general election
- Melaka Plant delayed to 1Q13, coal price to be below US\$100/tonne
- Limited details on PPA extensions but it will likely be neutral for Tenaga as the savings will be passed back

Towngas China (1083.HK)

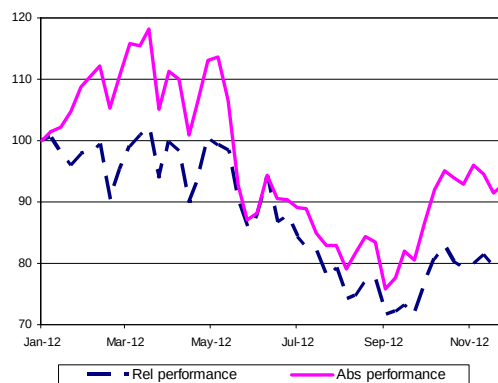


HK\$6.49
HK\$7.20
Buy
2,060
1.1

- Improving ROE on better operating metrics and rising gas volumes driving further re-rating
- Projects in the Northeast to benefit from higher gas supplies
- Maintain stable new connection fees, driven by higher number of affordable housing developments

Yingde Gas (2168.HK)

Price performance relative to country index



Price on 28 Nov	HK\$7.58
12-month target	HK\$7.00
Rating	Neutral
Market Cap (US\$m)	1,767
Avg daily Turnover (US\$m)	3.2

- On-site customers face uncertainty in the steel market; merchant market remains challenged after sharp drop in solar demand
- Strong demand in certain projects; resumption of production in Inner Mongolia
- Developing self-supported merchant market, which may reduce volatility for its merchant gas sales

Sector

China Power Sector

Favorable Operating Environment under Pro-stability

The key message from the 18th Party Congress held in Nov was that policy will be more pro-stability than pro-reform. Reform ahead therefore would not be very significant but incremental. Without much economic stimulus, we forecast PRC power demand growth to remain moderate. We have lowered our annual PRC power demand growth assumption in 2013-14E from 9% to 7%, similar to the forecast of China Electricity Council at 7.0-7.5%. The more moderate growth is positive to Chinese power companies, whose net profits have more sensitivity to coal price than demand growth. We think the PRC coal price will be largely stable in 1H13E and any rebound in winter will be short lived. Reduced capex has resulted in positive free cashflow for some of the Chinese power companies, which has not been seen since 2003.

Asian Utilities

China Power Sector

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Summary

Figure 1. PRC IPPs: FY13E Net Profit Sensitivity

(Rmbm)	Huaneng	Datang	Huadian	CRP	CPI	Average
Base case net profit *	7,932	3,647	1,927	8,078	1,365	n/a
Case 1. net profit change with extra 1% cut in average coal price % change	454 5.7%	157 4.3%	280 14.5%	291 3.6%	71 5.2%	n/a 6.7%
Case 2. net profit change with extra 1% hike in power tariff % change	746 9.4%	299 8.2%	626 32.5%	460 5.7%	125 9.2%	n/a 13.0%
Case 3. net profit change with extra 1% cut in utilization hours % change	-272 -3.4%	-110 -3.0%	-121 -6.3%	-174 -2.2%	-73 -5.3%	n/a -4.0%
Case 4. net profit change with extra 25bps interest rate cut % change	242 3.0%	213 5.8%	287 14.9%	205 2.5%	43 3.2%	n/a 5.9%

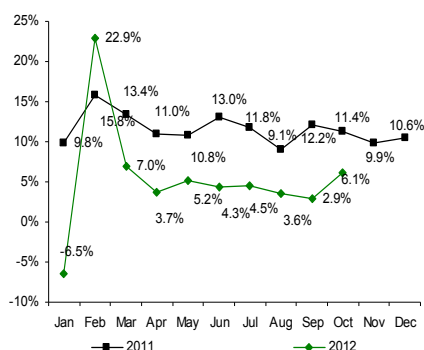
Source: Company Reports and Citi Research Estimates

* In Rmbm except CRP which is in HK\$m

** Key assumptions - (i) unit fuel cost ranges from -2% yoy (Huaneng, Datang, CRP & CPI) to -5% yoy (for Huadian) in 2013E; (ii) average 6.5% tariff hike on 1 Dec 2011, but nil in 2012-13E

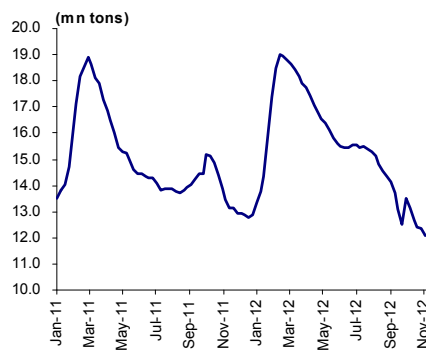
Lowering PRC Power Demand Growth Assumptions

Figure 2. Decelerated PRC monthly power demand growth yoy



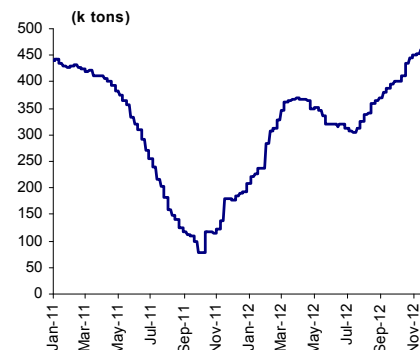
Source: China Electricity Council

Figure 3. Destocking at PRC steel industry



Source: CEIC

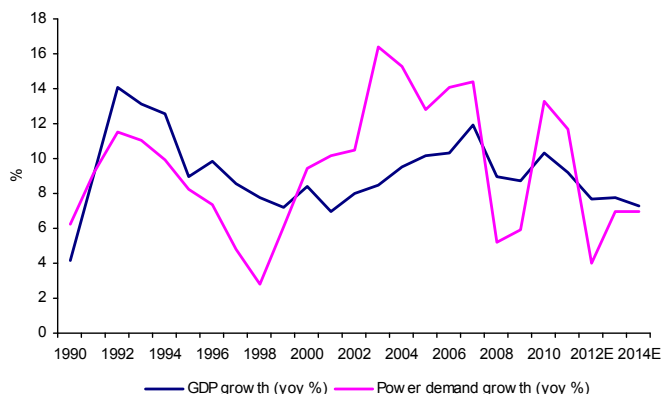
Figure 4. Restocking at aluminum industry



Source: CEIC

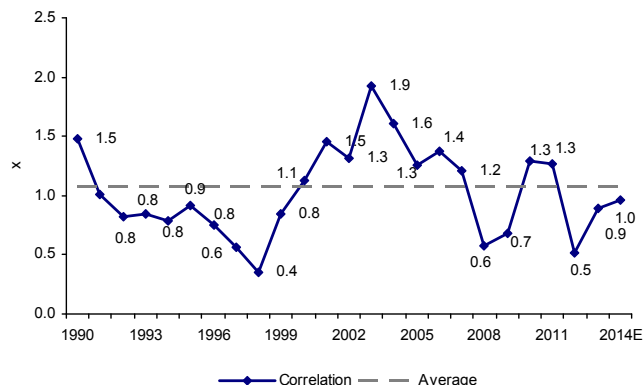
Chinese power demand growth has decelerated since March 2012. Without much economic stimulus ahead, we expect the curtailed growth rate to persist, though the low base might mildly accelerate the growth rate from 4.9% in 10M12 to 7% in 2013E.

Figure 5. PRC annual power demand and GDP growth



Source: NDRC, China Electricity Council and CIRA Estimates

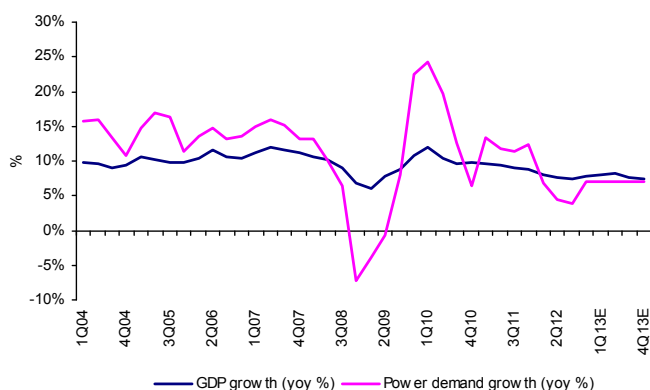
Figure 6. Multiple of PRC annual power demand vs. GDP growth



Source: NDRC, China Electricity Council and CIRA Estimates

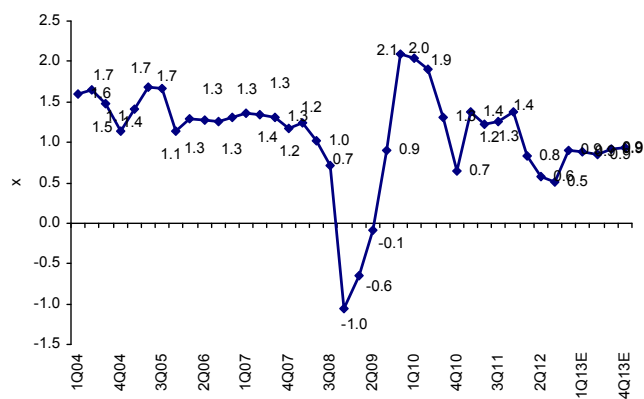
The correlation between PRC annual power demand and GDP growth was high at 80% in 2005-11. Regarding PRC GDP outlook, Citi is expecting it to slow to 7.7-7.8% in 2012-13E, from 9.2% in 2011. We expect power demand growth to decelerate to 4% in 2012E and 7% in 2013E, from 11.7% in 2011.

Figure 7. PRC quarterly power demand and GDP growth



Source: NDRC, China Electricity Council and CIRA Estimates

Figure 8. Multiple of PRC quarterly power demand vs. GDP growth

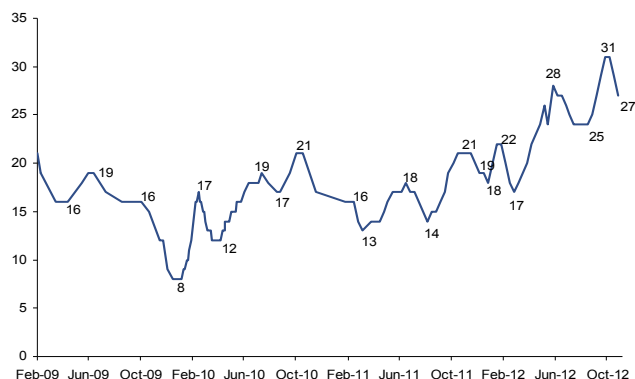


Source: NDRC, China Electricity Council and CIRA Estimates

NDRC has no concern of power shortage in 2013E based on our recent update, expecting lower power demand growth ahead,

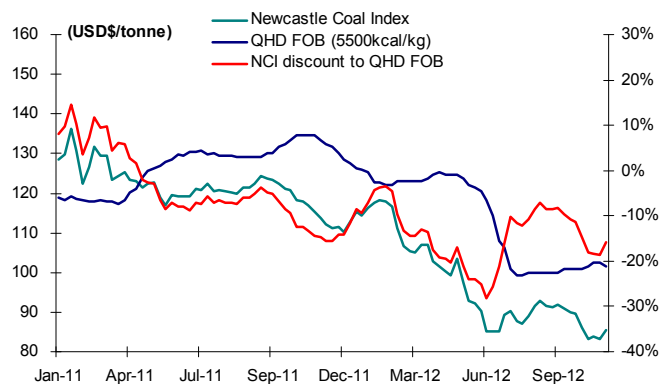
No Fundamentals of Coal Price Rise beyond Seasonal Fluctuation

Figure 9. High coal inventory at PRC coal-fired power plants at 27 days in late Nov, 80% above optimal level of 15 days



Source: SXCoal

Figure 10. Pricing discount of Newcastle coal vs. Chinese Qinhuangdao widened to 16% by Nov, from 2% a year ago



Source: SXCoal

According to our recent update with China Coal Trading & Logistic Centre, they expect:

- (i) **Spot coal price lacks momentum to rise in next 6-9 months.** The prices might rise slightly during winter for seasonal reasons, but the upward momentum would be unsustainable after CNY holiday in Feb 2013;
- (ii) **Coal supply is ample without railway transport bottleneck.** Coal transport by trucks on highway has been substantially curtailed. Destocking has been done hence inventories at mines and ports are normal. But the inventory at power plants is high due to weak electricity demand and inventory pile up for winter demand; and
- (iii) Key contracted coal price setting by market mechanism from Jan 2013, instead of by NDRC before, would not have much impact on coal price near term as the prevailing contracted and spot prices are close.

Figure 11. PRC IPPs coal supply mix in 2012E

	Huaneng	Datang	Huadian	CRP **	CPI	Average
Raw coal consumption (m tonnes)	140	95	80	78	15	n/a
Key contracts	27%	35%	5%	25%	50%	28%
Import	21%	10%	2%	6%	0%	8%
Own mines	6%	20%	0%	22%	0%	10%
Spot market & normal contracts *	46%	35%	93%	47%	50%	54%
Total	100%	100%	100%	100%	100%	100%

* Normal contracts with prices with reference to spot prices

** CRP with 50% coal supply from key and normal contracts with no breakdown; we assume half from each

Source: Company Reports and Citi Research Estimates

We generally assume PRC IPPs unit fuel cost to drop 2% yoy in 2013E (=-3%*2/3+0%*1/3) based on the following assumptions:

- (i) Coal supply – nearly two-thirds from spot market, normal contracts and imports where prices are correlated to spot prices; and around one-third from key contracts where prices are fixed by NDRC, at Rmb595/tonne for 5,500kcal/kg at Qinhuangdao port in 2012;
- (ii) Key contracted prices - we assume the price of Rmb595/tonne to largely remain in 2013E despite key contracted price intervention to be cancelled in Jan 2013. This is because the key contracted price is close to the spot one (currently Rmb637/tonne);
- (iii) Spot price - It was Rmb760/tonne for thermal coal of 5,500kcal/kg at Qinhuangdao port in 1H12. We assume it at Rmb635/tonne in 2H12E, Rmb650/tonne in 1H13E and Rmb700/tonne in 2H13E. The year average would be Rmb695/tonne in 2012E and Rmb675/tonne in 2013E, -3% yoy. If PRC power demand growth is shy of our expectation of 7% yoy next year, the spot coal price in 2H13E could be lower than our assumption.

Positive Free Cash Flow amid Trimmed Capex

Figure 12. PRC IPP capex cut in 2012E and to be flat in 2013E

(Rmbm)	2011	2012E	YoY	2013E	YoY
Huaneng	28,893	15,708	-46%	16,263	4%
Datang	26,010	23,495	-10%	23,607	0%
Huadian	17,592	15,565	-12%	15,374	-1%
CRP	15,029	12,667	-16%	15,008	18%
CPI	4,028	6,697	66%	6,788	1%
Total	91,552	74,132	-19%	77,040	4%

Source: Citi Research

Figure 13. PRC IPPs free cash flow

(Rmbm)	2011	2012E	YoY	2013E	YoY
Huaneng	-15,489	3,128	-120%	5,474	75%
Datang	-18,453	-10,389	-44%	-6,952	-33%
Huadian	-13,232	-9,656	-27%	-6,339	-34%
CRP	-5,953	2,565	-143%	260	-90%
CPI	-2,678	-2,291	-14%	-2,720	19%

Source: Citi Research

IPPs have cut their capex for:

- (i) **High gearing** average 294% net debt to equity ratio by end 1H12. Since 2011, the five big Chinese power generating groups have been requested by SASAC in their performance appraisal to keep their debt to asset ratio below 85%. Exceeding the gearing cap has led to more stringent spending control by the regulators; and

Figure 14. Debt to asset ratio of big five Chinese power generating groups by end 2011

(Rmbbn)	Total assets end 2011	Profit before tax (2011)	Ratio
China Datang Group	616.1	1.6	87.8%
China Huadian Group	522.8	2.2	86.8%
China Power Int'l Group	503.4	2.6	85.9%
China Huaneng Group	751.3	6.1	85.0%
China Guodian Group	646.0	6.1	83.6%
Average			85.8%

Source: State Electricity Regulatory Commission

- (ii) **High earnings volatility** of PRC coal-fired power plants inversely correlated to coal prices and tariff setting is beyond their control; and
- (iii) **No power shortage** in China hence NDRC is selective in new coal-fired power project approval.

Some PRC IPPs such as Huaneng and CR Power have positive free cash flow in 2012E, the first time since 2003. Some of them also offer attractive yield with stable dividend payable ratio.

PB Average 20% below Historical Mean

Figure 15. Huaneng historical PB chart



Source: Company Reports and Citi Research Estimates

Figure 16. Datang historical PB chart



Source: Company Reports and Citi Research Estimates

Figure 17. Huadian historical PB chart



Source: Company Reports and Citi Research Estimates

Figure 18. CRP historical PB chart



Source: Company Reports and Citi Research Estimates

Figure 19. CPI historical PB chart

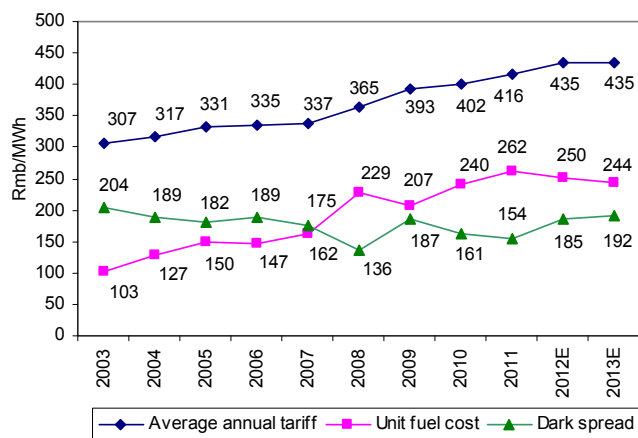


Source: Company Reports and Citi Research Estimates

Source: Company Reports and Citi Research Estimates

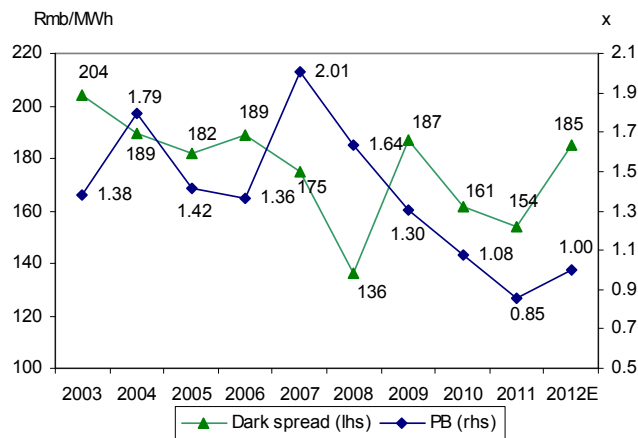
We believe that the margin recovery would revalue PRC IPPs to trade toward their historical average valuations.

Figure 20. Dark spread (=average tariff – unit fuel cost) rebound in 2012E



Source: Citi Research and various companies

Figure 21. 5 HK listed PRC IPPs – Higher dark spread, higher PB



Source: Citi Research and various companies

Stock Pick – Huadian & CRP

We like PRC IPPs for margin improvements stemming from spot coal price declines. Huadian Power therefore is our top pick for its high net profit sensitivity to coal prices. Its profitability would also improve from 4Q12E after paying historical cost burden in 9M12.

For quality-oriented investors looking for better ROE and more stable earnings, CRP would be more suitable among PRC IPPs, in our view. Similar to Huaneng, CRP has over 60% power plants in such coastal provinces as Guangdong and Jiangsu where the coal price has dropped more than inland ones. We think the market is concerned that too much of its profit will be cut from coal mines, which would contribute only 14% of CRP's profit in 2012E.

China Power Equipment Sector

Earnings Cuts in 2013E

Summary

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We remain cautious on China's power equipment sector, expecting weak new order flow to persist in 1H13E for two reasons: 1) In the domestic market, the NDRC sees ample power generating capacity, with no worry of shortages amid moderate power demand growth (7% pa in 2013-14E); and 2) In the largest export market, India, new project development is being delayed in the face of coal shortages. Resumption of new nuclear project approvals with a reduced target of 58GW operational capacity by end 2020, from 70-80GW before, and construction of more gas fired power plant for emission cut in China might render some minor bright spot but not significant enough to reverse the earnings cut trend in 2013. In the sector, Shanghai Electric would have the least earnings cut, mitigated by its most diversified earning base with one third profit from the non-power equipment sector.

Figure 1. Power supply-demand equilibrium in China

National Power Generating Capacity					
	FY10	FY11	FY12E	FY13E	FY14E
Year beginning	874,070	966,413	1,055,759	1,130,759	1,205,759
Add: new capacity	103,343	94,355	80,000	80,000	85,000
Less: closure	-11,000	-5,000	-5,000	-5,000	-5,000
Year end	966,413	1,055,759	1,130,759	1,205,759	1,285,759
yoy	10.6%	9.2%	7.1%	6.6%	6.6%

National Power Generation Breakdown by Fuel Type					
	FY10	FY11	FY12E	FY13E	FY14E
Coal	3,252	3,691	3,673	3,986	4,199
Hydro	687	663	814	765	809
Nature Gas	78	105	128	154	183
Nuclear Power	75	87	102	122	167
Wind Power	49	73	93	119	149
Solar	0	3	9	16	23
Others	92	92	92	92	92
Total	4,228	4,722	4,911	5,507	5,622
yoy	15.4%	11.7%	4.0%	7.0%	7.0%

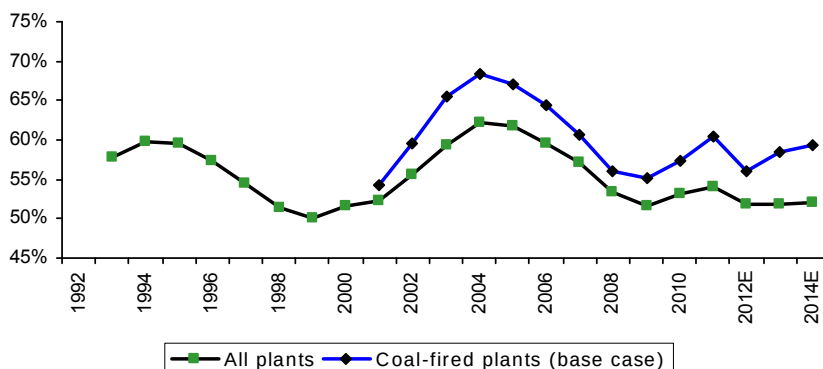
Annual Capacity Addition Breakdown by Fuel Type (plus End FY11 breakdown)					
	End FY11	FY11	FY12E	FY13E	FY14E
Coal	732,802	54,555	37,667	33,500	31,772
Hydro	230,514	14,456	14,486	15,000	15,000
Natural gas	32,653	6,229	7,347	8,000	9,000
Nuclear	12,572	1,752	2,000	4,000	9,428
Wind	45,051	15,476	15,000	14,000	14,000
Solar	2,143	1,883	4,000	4,500	4,500
Others	24	4	1,000	1,000	1,300
Total	1,055,759	94,355	80,000	80,000	85,000

Annual Utilization Hours of Chinese Power Plants					
	FY10	FY11	FY12E	FY13E	FY14E
Coal	5,031	5,294	4,911	5,116	5,205
Hydro	3,404	3,028	3,500	3,100	3,100
Nuclear Power	7,924	7,772	7,672	7,672	7,672
Wind Power	2,047	1,903	1,870	1,870	1,920
Solar Power	1,400	1,500	1,500	1,500	1,500
Others	3,750	3,750	3,750	3,750	3,750
All plants	4,650	4,731	4,544	4,546	4,562
yoy(all units)	2.7%	1.7%	-4.0%	0.1%	0.3%
yoy(coal-fired)	4.0%	5.2%	-7.4%	4.3%	1.7%

Source: China Electricity Council and Citi Research Estimates

Weak New Orders Flow from Domestic Market

Figure 2. Average utilization hours of Chinese power plants



Source: Company Reports and Citi Research Estimates

NDRC does not worry power shortages in China in 2012-13E. From 2000, the annual average utilizations of PRC coal-fired power plants were between 54% and 68%. We forecast the utilization of 56-59% in 2012-14, hence shortages in the period would be of limited scale in term of geography and period – to recap, China had a power shortage during the summer in 7-8 provinces in 2011 and 2-3 provinces in 2012; we assume the situation in 2013E to be similar to that in 2011.

Figure 3. Investment in PRC power industry breakdown (Rmb bn)

Investments	2007	2008	2009	2010	2011	YoY	10M12	YoY
Power Grid	245	288	385	341	311	-9%	278	+8%
Power Plants	304	288	371	364	309	-15%	268	-3%
Breakdown:								
- thermal	n/a	156	149	131	92	-30%	72	-12%
- hydropower	n/a	71	87	79	78	-1%	91	+29%
- nuclear	n/a	28	58	63	66	4%	60	-1%
- wind	n/a	32	76	89	64	-28%	42	-25%
- others	n/a	1	2	2	10	416%	4	-58%

Source: China Electricity Council and Citi Research Estimates

Investment in PRC power plants was -3% yoy to Rmb268bn in 10M12, including Rmb72bn (-12% yoy) for coal-fired plants, Rmb91bn (+29% yoy) for hydropower plants, Rmb60bn (-1% yoy) for nuclear units and Rmb42bn (-25% yoy) for wind farms. According to the NDRC, China would continue to focus on changing the power-generation fuel mix to lower the proportion of electricity output from coal in order to cut emissions. We forecast PRC power generation from coal to drop from 78% in 2011 to 71% in 2020.

Figure 4. CO₂ emission of various primary energy per unit of MJ

Fuel type	CO ₂ emission (kg)
Standard coal	0.100
Petroleum	0.068
Diesel	0.069
Heavy crude oil	0.079
LPG	0.065
Natural gas	0.056

Source: National Energy Bureau

Figure 5. Energy content conversion table for primary energy

Energy type	Unit	Unit energy content (MJ)	Energy content comparison (with NG=1)
Natural gas	Nm3	35.53	1.00
LPG	Nm3	108.68	3.06
Diesel	L	39.59	1.11
Petroleum	L	31.617	0.89
Heavy crude oil	L	39.768	1.12
Standard coal	kg	29.26	0.82

Source: National Energy Bureau

Revenue from Nuclear Equipment Plateau Ahead

Figure 6. PRC nuclear power plants in operation

Project	Province	Capacity (MW)	Type	Operator	Commercial operation
Daya Bay 1&2	Guangdong	1,968	PWR	CGNPC	1994
Qinshan Phase I	Zhejiang	310	PWR(CNP-300)	CNNC	Apr.1994
Qinshan Phase II, 1-2	Zhejiang	1,300	PWR(CNP-600)	CNNC	2002; 2004; 2010
Qinshan Phase II, 3-4	Zhejiang	1,320	CNP-600	CNNC	2010; 2012
Qinshan Phase III, 1&2	Zhejiang	1,400	PWR(Candu 6)	CNNC	2002; 2003
Ling Ao Phase I, 1&2	Guangdong	1,981	PWR	CGNPC	2002; 2003
Tian Wan, 1&2	Jiangsu	2,120	PWR(VVER-1000)	CNNC	2007
Ling Ao Phase II, 1	Guangdong	1,086	PWR(CPR-1000)	CGNPC	Sep.2010
Ling Ao Phase II, 2	Guangdong	1,086	CRP-1000	CGNPC	Aug.2011
Total		12,570			

Source: China Electricity Council

Figure 7. PRC nuclear power plants under construction

	Province	Capacity (MW)	Technology	Operator	Construction since	Planned operation
3rd Gen or above technology						
Sanmen, 1&2	Zhejiang	2,500	AP-1000	CNNC	Mar 2009, Dec 2009	Nov.2013; Sep.2014
Haiyang, 1&2	Shandong	2,500	AP-1000	CPI	Sep 2009, Jun 2010	May.2014; Mar.2015
Sub-total		5,000				
2nd Gen technology						
Hongyanhe, 1-4	Liaoning	4,480	CPR-1000	CGNPC	Aug.2007 - Aug.2009	Oct 2012; 2013; 2014
Ningde, 1-4	Fujian	4,360	CPR-1000	CGNPC and Datang	Feb.2008 - Sep.2010	Dec 2012; 2013; 2014; 2015
Fuqing, 1&2	Fujian	2,160	CPR-1000	CNNC	Nov 2008 - Jun 2010	Oct 2013; Aug 2014
Yangjiang	Guangdong	4,320	CPR-1000	CGNPC	Dec 2008 - Mar 2011	Aug 2013; 2014; 2015; 2016
Fangjiashan 1&2	Zhejiang	2,160	CPR-1000	CNNC	Dec 2008; Jul 2009	Dec 2013; Oct 2014
Hongyanhe, 5&6	Liaoning	2,160	CPR-1000	CGNPC	2011	2015
Fangchenggang, 1&2	Guangxi	2,160	CPR-1000	CGNPC	Jul.2010; 2011	2015; 2016
Fuqing, 3&4	Fujian	2,160	CPR-1000	CNNC	Jul.2010; 2012	Jul 2015; May 2016
Hongshiding (Rushan), 1&2	Shandong	2,160	CPR-1000	CNEC/CNNC	N/A	2015
Sub-total		26,120				
Total		31,120				

Source: Company Reports and Citi Research Estimates

On the summons of Premier Wen Jiabao, the State Council approved “Energy development 12th 5-year plan”, “Nuclear safety plan (2011-20)” and “Nuclear mid-long term development plan (2011-20)” on 24 Oct. The State Council agreed nuclear power development was essential in China, but safety is the first priority. The three key points about nuclear are:

- (1) **To progressively resume normal nuclear plant construction in China**, but no PRC nuclear capacity target for 2020 has been mentioned in the plan – DF expects new nuclear order flow to resume, after new nuclear project approvals were suspended since the Fukushima nuclear accident in March 2011. But the amount of new nuclear orders might be lower than those in 2008-10, in our view, as the future capacity addition would be progressive. but not aggressive, for safety reasons;
- (2) **To select sites scientifically**, with new nuclear plants to be built in a few selected sites in coastal provinces, not in inland ones, in the 12th 5-year period (2011-15) – Of orders on hand, DF has Rmb3-4bn (or 3%) belonging to three planned inland nuclear projects; namely the nuclear island of Xianning nuclear project in Hubei under CGNPC, the nuclear island of Taohuajiang nuclear project in Hunan under CNNC and the conventional island of Pengze nuclear project in Jiangxi. DF has started production of the equipment of these orders and expects them to be transferred to other nuclear projects of these state-owned nuclear power plant developers, rather than cancelled. DF has got 10% customer deposit of these orders, which would help to mitigate financial impact, if any; and
- (3) **To approve the construction of new nuclear plants with 3G standards** in order to raise the entrance barrier and demand the highest global safety standards – This direction is more negative to DF than its peers as the majority of the DF's orders are 2G while those of Shanghai Electric are mostly 3G.

Figure 8. PRC nuclear power plants in planning

	Province	Capacity (MW)	Technology	Operator	Construction since	Planned operation	Impact of new policy
3rd Gen in coastal areas							
Xudabao, 1&2	Liaoning	2,500	AP-1000	CNNC and Datang	N/A	N/A	Possible to be built in future
Sanmen, 3&4	Zhejiang	2,500	AP-1000	CNNC	N/A	N/A	Possible to be built in future
Haiyang, 3&4	Shandong	2,500	AP-1000	CPI	N/A	N/A	Possible to be built in future
Longyou(Zhexi), 1&2	Zhejiang	2,500	AP-1000	CNNC	N/A	N/A	Possible to be built in future
Zhangzhou, 1&2	Fujian	2,500	AP-1000	CNNC and Guodian	N/A	N/A	Possible to be built in future
Shaoquan, 1-4	Guangdong	5,000	AP-1000	CGNPC	N/A	N/A	Possible to be built in future
Taishan, 1&3	Guangdong	3,540	ERP	CGNPC	N/A	N/A	Possible to be built in future
Sanming, 1&2	Fujian	1,760	BN-800	CNNC	N/A	N/A	Possible to be built in future
Shandong Shidaowan	Shandong	210	HTR-PM	Huaneng	N/A	N/A	Possible to be built in future
Sub-total		23,010					
3rd Gen in inland areas							
Xianning(Dafan), 1&2	Hubei	2,500	AP-1000	CGNPC	N/A	N/A	Not build for in coastal province
Taohuajiang, 1-4	Hunan	5,000	AP-1000	CNNC	N/A	N/A	Not build for in coastal province
Pengze, 1&2	Jiangxi	2,500	AP-1000	CPI	N/A	N/A	Not build for in coastal province
Xiaomoshan, 1&2	Hunan	2,500	AP-1000	CPI	N/A	N/A	Not build for in coastal province
Yanjiaoshan	Jiangxi	2,500	AP-1000	CNNC	N/A	N/A	Not build for in coastal province
Wuhu, 1&2	Anhui	2,500	AP-1000	CGNPC	N/A	N/A	Not build for in coastal province
Sub-total		17,500					
2nd Gen technology							
Ningde, 5&6	Fujian	2,160	CPR-1000	CGNPC	N/A	N/A	Not build for no 2G approval
Fuqing, 5&6	Fujian	2,160	CPR-1000/CNP1000	CNNC	N/A	N/A	Not build for no 2G approval
Lianyungang, 1&2	Jiangsu	2,160	CRP-1000	CGNPC	N/A	N/A	Not build for no 2G approval
Shanwei(Lufeng), 1&2	Guangdong	2,160	CRP-1001	CGNPC	N/A	N/A	Not build for no 2G approval
Tianwan, 3&4	Jiangsu	2,120	VVER-1000 (AES-91)	CNNC	N/A	N/A	Not build for no 2G approval
Tianwan, 5&6	Jiangsu	2,400	VVER-1200	CNNC	N/A	N/A	Not build for no 2G approval
Sub-total		13,160					
Total		53,670					

Source: Company Reports and Citi Research Estimates

PRC Government has cut its 2020's nuclear target to 58GW operational capacity, from 80GW, and 30GW under construction, which is the same as before. This implies new nuclear orders of 5GW pa in 2013-15E, assuming (i) China to have 43GW operational nuclear capacity by completing projects under construction; and (ii) nuclear capacity which is to be operational by end 2020 need to have their equipment placed in or before 2015. New orders of 5-6GW pa would mean nuclear revenue of PRC power equipment makers are unlikely to have much upside from those in 2012-13E, even though an increase in the domestic proportion would offset part of the reduction in total new orders. The 2020's nuclear capacity target might be reviewed after commissioning of China's first two nuclear plants using 3G technology; namely, Sanmen and Haiyang nuclear plans in Zhejiang and Shandong, to be completed in 2014E.

Figure 9. Capex of Chinese nuclear power plants

Technology	2G	3G (AP1000)
Total capex (Rmbm/MW)	12	17
Total capex breakdown:		
- equipment cost	50%	n/a
- construction cost	15%	n/a
- installation cost	10%	n/a
- other costs	25%	n/a
Equipment cost (Rmbm/MW)	6	10
Equipment cost breakdown:		
- nuclear island	50%	n/a
- conventional island	30%	n/a
- other equipment	20%	n/a

Source: National Energy Bureau

Weak Overseas New Orders Flow

Figure 10. Overseas orders of Chinese power equipment companies and their exposure to India

	Dongfang Electric	Shanghai Electric	Harbin Electric
Orders on hand by end 1H12 (Rmb bn)	147	238	127
% of order on hands for export	18%	40%	39%
Order on hands by end 1H12 for export (Rmbbn)	27	97	49
Order on hands by end 1H12 for export by countries:			
- India	45%	59%	65%
- other countries	55%	41%	35%
Major customer in India	Several (including Abhijeet)	Reliance (for 78% Indian orders)	N/A
Orders on hand by end 1H12 for export by product types:			
- coal-fired power equipment	80%	Nearly 100%	Nearly 100%
- hydropower, gas & other equipment	20%	Minimal	Minimal
Customer deposit of overseas orders	10-15%	10-15%	10%
New orders recognition method	After contract signed & customer deposit received	Once after contract signed	After contract signed & customer deposit received

Source: Company Reports and Citi Research Estimates

Most of the planned power projects in India are set to be delayed due to inadequate local fuel supply at economical pricing, in particular those from the private sector. Fuel issues are adversely affecting about 20GW operational capacity. For instance, Lanco not only has local coal supply problems but also gas supply concerns. Over half of Reliance's projects are being delayed. Coal imports cannot help much due to higher pricing and technical limitations of boilers which prevent the consumption of large quantities of low-ash imported coal – domestic coal in India has high ash content and most Indian boilers are designed to consume such coal. Additionally, the implementation of a 19% import duty plus US\$/India Rupee appreciation have lowered the marketability of Chinese power equipment in India.

Figure 11. Comparison of PRC power equipment makers

	Dongfang Electric	Shanghai Electric	Harbin Electric*
Production volume in 9M12 (GW)	24.6	n/a	12.6
Thermal	19.1	n/a	9.2
Hydro	5.5	n/a	3.4
Power equipment order backlog by end Sep 12 (Rmb bn)	146.6	238.0	117.0
Coal	67.7	83.5	n/a
Nuclear	15.4	17.0	n/a
Hydro	14.3	0.0	n/a
Wind	4.4	5.2	n/a
Export (mostly coal)	27.0	95.0	49.0
New orders in 9M12 (Rmb bn)	34.6	21.1	24.5
Coal	12.4	10.5	10.9
Wind	2.7	2.9	0.9
Gas	9.7	4.2	1.6
Hydro	3.5	0.0	1.8
Nuclear	1.0	0.3	2.9
Export (mostly coal)	5.1	0.0	8.2
Financial data in 1H12 (Rmb bn)			
Total revenue	20	36	11
Revenue mix:			
- Thermal	47%	37%	59%
- Hydro	8%	0%	16%
- Wind	17%	3%	0%
- Nuclear	10%	6%	0%
- Others	18%	54%	26%
% of revenue from export	21%	22%	15%
GP margin	20%	19%	22%
- Thermal	22%	21%	22%
- Hydro	21%	0%	36%
- Wind	16%	-1%	0%
- Nuclear	19%	21%	0%
OP margin	7%	8%	8%
Net Margin	6%	5%	6%
Account receivable days	160	137	248
Account payable days	207	149	333

*New orders and orders backlog of Harbin Electric were 1H12 data

Source: Company Reports and Citi Research Estimates

China Gas Sector

Opening the Floodgates

Summary

Asian Utilities

China Gas Sector

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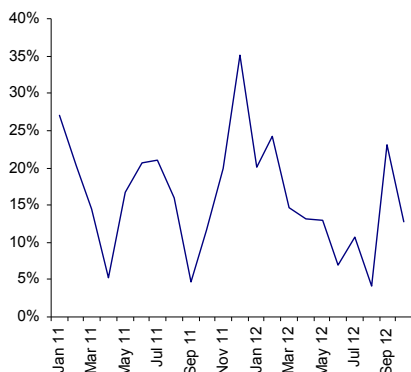
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We are positive on Chinese gas companies for their high supply-push volume growth story (PRC NG consumption +14% yoy in 10M12) at inexpensive prices against other energy sources to persist in 2013E. The incremental supply would come from new long-distance pipelines and LNG terminals under construction to augment 16% yoy in the year. Residential connection fee risk seems priced-in with most downstream gas distributors guiding no growth yoy in 2013E amid the property slowdown. Uncertainty could come from the long-awaited NG city-gate price hikes and a time delay might exist to pass through the incremental cost of the next upstream price hike to end users. The new NG price setting mechanism being tested in Guangdong and Guangxi has not been very meaningful so far due to the tiny quantity of only 5bcm, which was lower than expected subsequent to the pipeline construction delay, probably due to the loss-making issue for upstream supplier PetroChina. As the sector has been re-rated rewarding its high earnings growth amid the PRC's decelerated economic growth in 2012E, we think it more suitable for long-term investors ahead to capture the ~20% p.a. growth in 2013-14E.

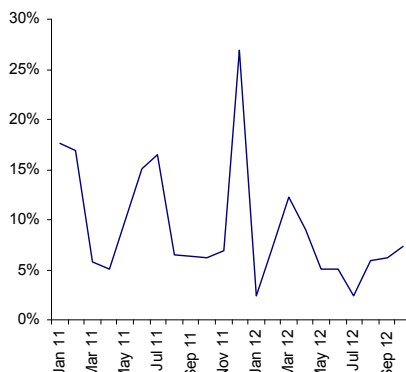
Resilient Growth in 2012E and Sustainable in 2013E

Figure 1. PRC monthly NG consumption yoy



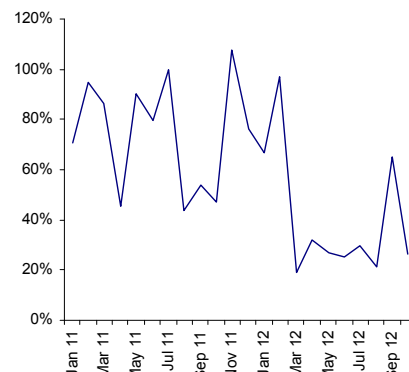
Source: NDRC, PRC Customs, Argus Asia Gas, Citi

Figure 2. PRC monthly NG domestic supply yoy



Source: NDRC, PRC Customs, Argus Asia Gas, Citi

Figure 3. PRC monthly NG import yoy



Source: NDRC, PRC Customs, Argus Asia Gas, Citi

In Oct, PRC NG consumption was +12.7% yoy to 12.4bcm, with the supply comprising 8.8bcm (+7.3% yoy) from domestic sources and 3.9bcm (+26.4% yoy) imports. In 10M12, PRC NG consumption was +14.1% yoy to 118.7bcm, with the supply including 87.8bcm (+6.4% yoy) from domestic sources and 34.4bcm (+37.6% yoy) imports. These high growth rates have been sustainable since the use of NG is more economical than such other energy sources as diesel and gasoline, facilitating downstream gas distributors recruiting new customers using NG, substituting the reduced demand, if any, from the old customers.

Figure 4. Chinese NG Supply vs. Demand - Majority of Supply from Domestic Gas Fields, while Supply from Imports is Increasing

(unit: bcm)	2010	2011	2012E	2013E	2015E	2011-12E CAGR	2011-15E CAGR
Demand	111	130	152	180	240	17%	16%
Supply:							
Domestic sources: (1)	95	104	112	120	150	8%	10%
- gas field	n/a	n/a	n/a	n/a	105	n/a	n/a
- coal-bed methane	n/a	n/a	n/a	n/a	20	n/a	n/a
- coal gasification **	n/a	n/a	n/a	n/a	25	n/a	n/a
Import: (2)	16	29	43	60	90	48%	33%
- Central Asia	4	12	23	35	45	92%	39%
- Russia	0	0	0	0	0	n/a	n/a
- Burma Road	0	0	0	0	12	n/a	n/a
- LNG terminals*	12	17	20	25	33	18%	18%
Total supply (1)+(2)	111	133	155	180	240	17%	16%
YoY	27%	20%	17%	16%	15%		
Supply breakdown:							
- domestic	86%	78%	72%	67%	63%		
- import	14%	22%	28%	33%	38%		
NG in % of PRC primary energy	4%	5%	6%	6%	8%		
% of PRC energy from non-fossil fuel	8%	8%	9%	9%	11%		

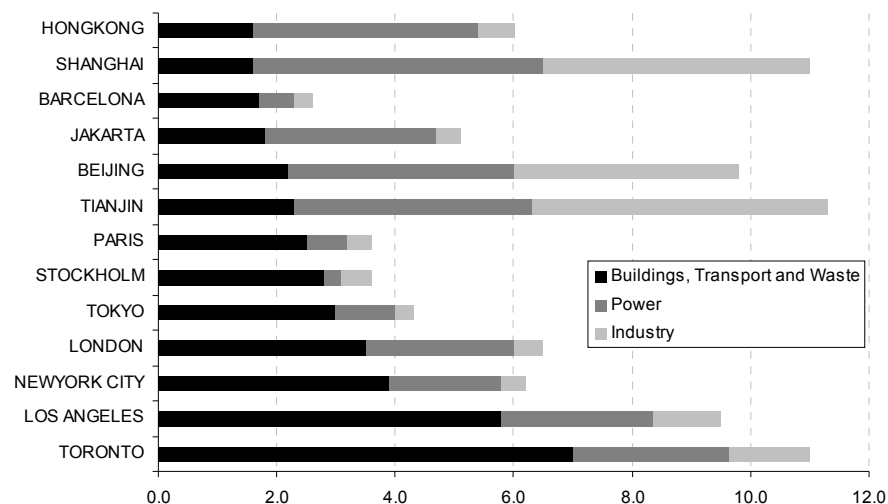
* Many small LNG plants were installed in 2012, with CNPC saying that Chinese gas liquefaction capacity totaled 21.5m m³/day in July 2012, up from 12.3m m³/day in 2011

** 6.5bcm from shale gas by 2015E

Source: NDRC, World Oil & Gas, NDRC, Guangdong Oil & Gas Association and CIRA Estimates

We maintain our long-term positive view on the PRC NG sector for sustainable high growth in the run-up to 2015E. The PRC environmental protection target is to lower the amount of carbon dioxide produced per unit of GDP by 17% in the same period. The use of NG could be more conducive to meet the target than fuels with the same heat value; the CO₂ emissions of NG are lower than coal by 42%, fuel oil by 31%, diesel by 29% and crude oil by 27%. China is set to cut energy consumption per unit of GDP output by 16% in 2011-15E, lower greenfield gas emission by 17% in the same period, and raise percentage of energy from fossil fuel to 11.4% in 2015E. While imported pipeline gas from central Asia and internationally traded LNG have been key components of the recent rise in gas supplies, the government is pushing for an increase in other forms of domestic gas production too. For instance, it said in early November that it planned to offer financial incentives of Rmb0.4/m³ of shale gas production in the run-up to 2015.

Figure 5. Comparison of greenhouse gas emission in global major cities in 2011 - High in China



X-axis unit: ton of CO2 equivalent per capita

Source: World Bank

NDRC forecasts China will consume 240bcm in 2015E, resulting in a 17% five-year CAGR in 2010-15E. New NG supply sources would comprise:

- **17 national long-distance pipelines** such as West-to-East Pipeline No.IV, China-Myanmar Pipeline and Shaanxi-Beijing Pipeline No.IV would be completed and launched in the run-up to 2015. The capacity of the West-to-East pipeline No.3 would rise to 25bcm by end 2015E, which would boost the total capacity of the three West-to-East pipelines to 55bcm. Negotiation to import NG from Russia has also commenced, though is unlikely to commence operation before end 2015 as no agreement on the NG sales price has been reached - Russia NG Company is offering to sell NG at US\$0.35/m³ to China, which accepts to buy at no more than US\$0.25/m³; and

Figure 6. Major domestic gas pipelines

Name	Operator	Route	Annual capacity (bcm)	Completed
Shaanxi-Beijing No.1	CNPC	Ordos Basin-Beijing	3	1997
Se-Ning-Lan No.1	CNPC	Qinghai-Gansu	3	2001
West-East No.1	CNPC	Xinjiang-Shanghai	12	2004
Zhong-Wu	CNPC	Chongqing-Hubei	3	2004
Shaanxi-Beijing No.2	CNPC	Ordos Basin-Beijing	17	2005
Qing-Ha	CNPC	Daqing-Harbin	5	2007
West-East No.2	CNPC	Xinjiang-Guangzhou	30	2009
Yong-Tang-Qin	CNPC	Yongqing (Hebei)-Qinhuangdao	9	2009
Se-Ning-Lan No.2	CNPC	Qinghai-Gansu	5	2010
Sichuan East	Sinopec	Sichuan-Shanghai	12	2010
Yu-Ji	Sinopec	Inner Mongolia-Jinan	3	2010
Shaanxi-Beijing No.3	CNPC	Ordos Basin-Beijing	15	2011
Shen-Qin	CNPC	Liaoning-Heilongjiang	9	2011

Source: NDRC, CNPC

- **LNG terminal** in Jiangsu province with capacity of 3.5m tons pa to handle LNG from Qatar and that in Liaoning province with capacity of 3m tons pa to deal with LNG from Australia have commenced operation in 2012. Construction has commenced for LNG terminals at Zhuhai, Guangdong and Qingdao, Shandong.

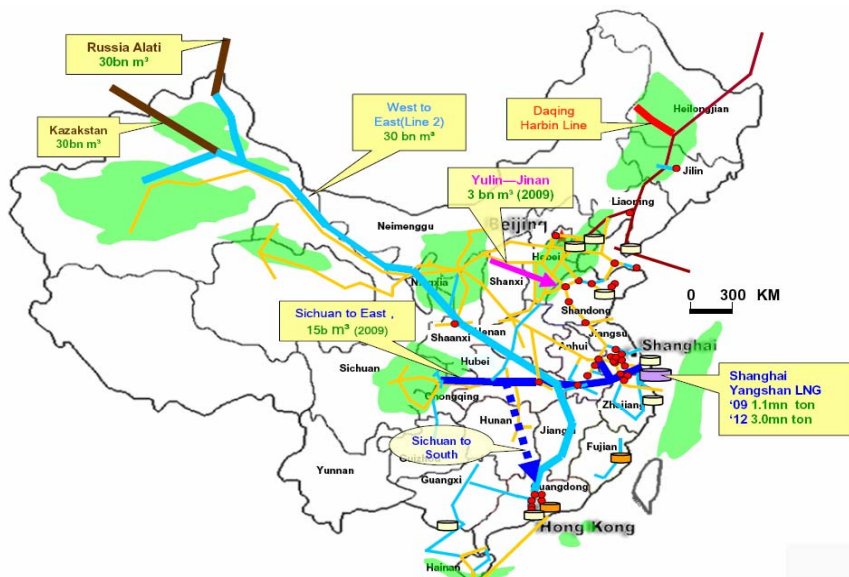
Figure 7. Key information of PRC operational LNG terminals

Terminal	Owner	Annual capacity (m tons)	Primary suppliers
Rudong, Jiangsu	CNPC/Pacific Oil & Gas	3.5	Qatar
Dalian, Liaoning	CNPC/Dalian Port	3.0	Australia, Qatar
Putian, Fujian	CNOOC/Fujian Investment & Development	2.6	Indonesia, Malaysia
Yangshan, Shanghai	CNOOC/Shenergy	3.0	Malaysia
Dapeng, Guangdong	CNOOC/BP	3.7	Australia

Source: Company Reports and Citi Research Estimates

To distribute the NG supply from these new sources, China aims to expand its gas pipelines network from 40,000km by end-2010 to 70,000km by end-2015E.

Figure 8. Chinese NG supply network



Source: Hong Kong and China Gas, Citi Research

In 2020E, NG consumption would be 55% for commercial & industrial production, 15% for electricity generation, 20% for residential, 6% for vehicle fuel and 4% for other raw materials; this implies rising proportion of NG usage for electricity generation and vehicle fuel in coming years against the prevailing sales mix of the listed companies.

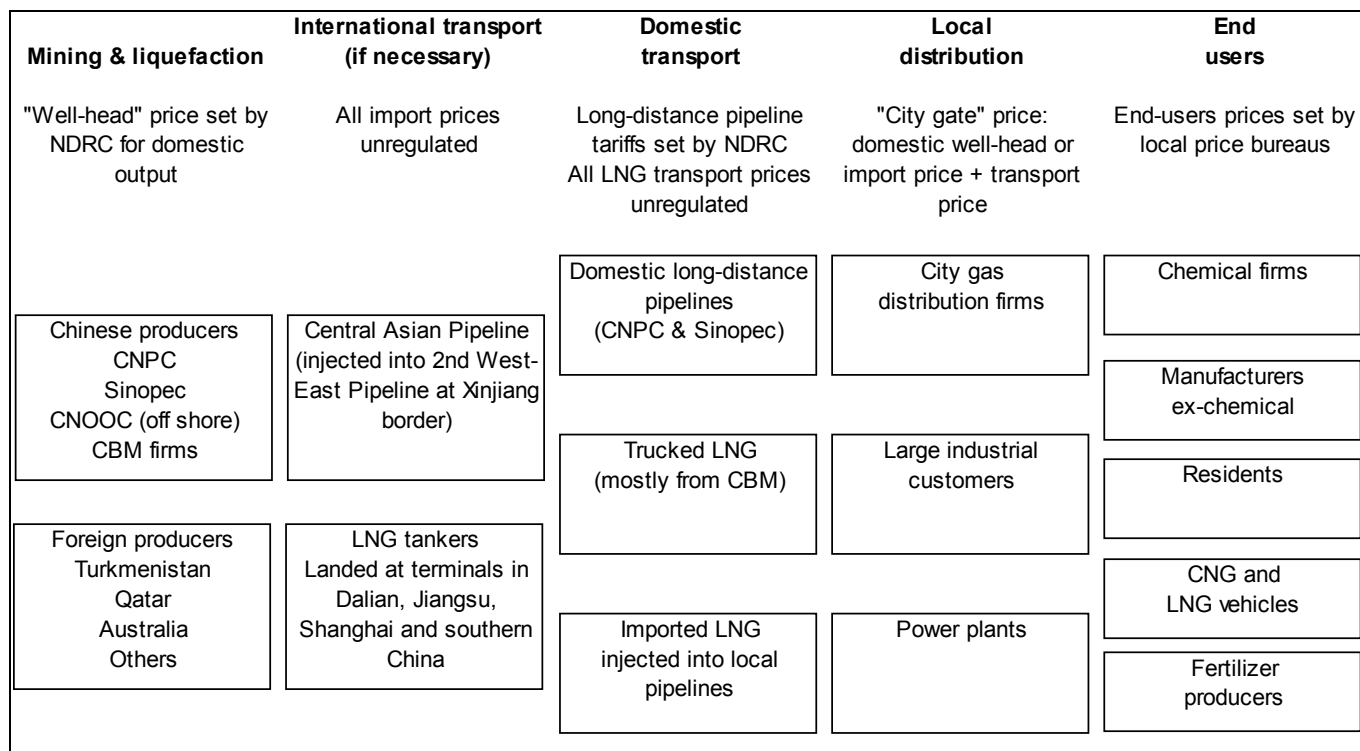
Figure 9. Gas sales usage mix in China in 2012E

	Residential	Commercial	Industrial	Vehicles	Heating	Total
CR Gas	30%	N/A	59%	11%	0%	100%
Towngas China	26%	15%	54%	5%	0%	100%
HKCG	24%	21%	50%	5%	0%	100%
Beijing Enterprises	19%	N/A	27%	7%	47%	100%
ENN	15%	36%	36%	13%	0%	100%
Peer A	13%	66%	12%	9%	0%	100%
Average	21%	35%	40%	8%	8%	100%

Source: Company Reports and Citi Research Estimates

Long Awaited Price Hike to Happen in 1H13E

Figure 10. China's natural gas value chain



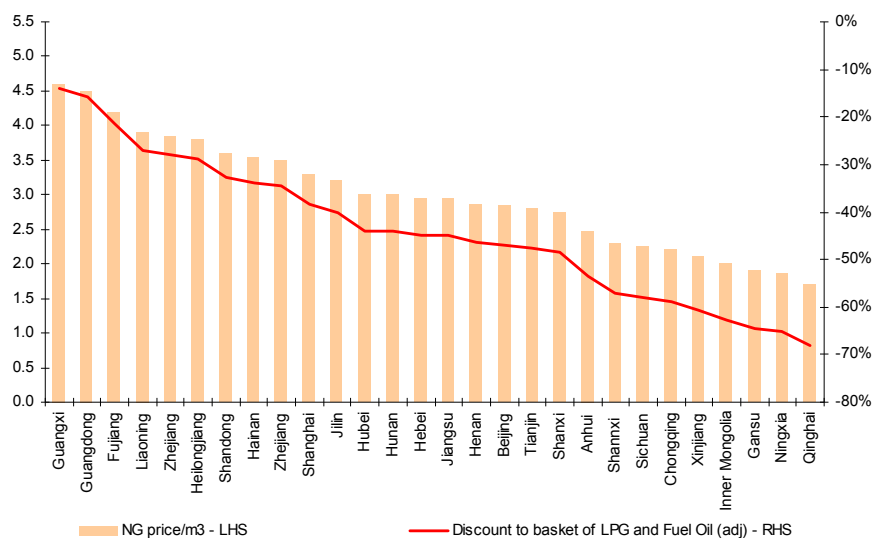
Source: Citi Research

The NDRC has launched trial runs of the new NG pricing reform in two provinces in Southern China, Guangdong and Guangxi, effective 26 Dec 2011. The purpose of the reform is to set PRC NG city gate ceiling prices with reference to the market prices of its substitutes (namely fuel oil and LPG), replacing the prevailing cost-plus basis to reflect supply-demand equilibrium. PetroChina Group incurred a cumulative loss of Rmb21.4bn by importing NG and LNG at end-2011, as it buys Turkmen gas from Central Asia at US\$0.3/m³ but has to resell it at a fixed price of US\$0.122-0.165/m³, regulated by the NDRC. A more market-oriented NG price setting mechanism therefore is essential to pass through the incremental cost to end-users, in particular Chinese NG supply through imports looks set to rise from 22% in 2011 to 38% in 2015E. PRC NG is 50-70% cheaper than product substitutes, compared with 30% in other countries, according to PetroChina Group.

New NG city gate ceiling prices are based on the weight average of fuel oil and LPG prices in Shanghai, with weightings of 60% and 40% respectively, under the new mechanism. The prices at Shanghai were chosen because Shanghai is regarded as the center of the PRC NG market, with NG supply from both W-to-E long distance pipelines and imports from a local LNG terminal. The actual city gate prices could be set below the ceilings, depending on negotiations between buyers and sellers. The price ceiling would be adjusted annually at the beginning, but adjustments could be more frequent on a semiannually or quarterly basis in future. We expect city gate prices to rise steadily with China having more NG from imports. Globally, there is high correlation between crude oil and NG prices.

But testing of the new pricing mechanism has not been very meaningful so far due to the tiny quantity. According to Guangdong Oil & Gas Association, only 5bcm was supplied from the W-to-E pipeline No.2 under the new pricing regime at Rmb2.74/m³. The volume was below expectation probably due to the loss-making issue for upstream supplier PetroChina at this price. The association added that there is no review of the testing up to Nov 2012.

Figure 11. PRC provincial NG retail prices versus LPG & fuel oil prices (per unit of energy)



Source: CEIC

The new pricing regime would be more difficult in other provinces than in Guangdong and Guangxi. The chart above shows that the two provinces have the least retail NG price discounts against LPG and fuel oil. Retail NG price hike pressure that arose from the use of the new NG price setting mechanism in these two provinces is less than elsewhere. The new mechanism therefore would take around 2-3 years for nationwide implementation, according to a *China Business Daily* report on May 5. Accordingly, we think the next nationwide tariff hike, if it happens in 1H13E, might follow the old method with increments decided by NDRC.

Connection Fee No Growth in FY13E

Figure 12. Revenue and gross profit from residential connection in FY12E

(Rmbm*)	Beijing Ent.	HKCG	Peer A	CR Gas	ENN	TCCL
2012E total revenue	35,356	24,979	21,913	16,827	17,644	4,936
2012E revenue from residential connection	0	1,925	2,534	3,442	3,754	1,112
2012E GP margin of residential connection	N/A	60%	65%	69%	62%	50%
2012E gross profit from residential connection	N/A	1,155	1,647	2,361	2,327	556
2012E total gross profit	7,791	6,493	4,293	5,084	4,820	1,157
% of 2012E gross profit from residential connection fee	Nil	18%	38%	46%	48%	48%

* HK\$m for HKCG, CR Gas, Beijing Enterprises, Peer B, TCCL; Rmbm for ENN

Source: Company Reports and Citi Research Estimates

PRC downstream gas distributors have average 33% gross profit from residential connection fees in 2012E. Residential connection fees came from new properties (50-80%) and old properties (20-50%). The weakening property market in China could reduce new connections from new properties. In view of the property market slowdown, most downstream gas distributors are guiding no connection growth yoy in 2013E. We think this is in the price. Among the gas companies, Beijing Enterprises is immune from residential connection fee cut risk as it no longer charges a connection fee in Beijing; HKCG also has lower risk for it has half of profit from Hong Kong without a connection fee.

Figure 13. Chinese new property development

		2011	YoY	10M12	YoY	2012E	YoY
Real Estate Investment Completed	Rmbbn	6,174	27.9%	5,763	15.4%	6,949	12.5%
GFA Construction completed	M sqm	892	13.3%	583	17.3%	982	10.0%
GFA under construction	M sqm	5,080	25.3%	5,381	13.3%	5,715	12.5%
GFA commencement for construction	M sqm	1,901	16.2%	1,468	-8.5%	1,735	-8.7%
Transaction amount	Rmbbn	5,912	12.1%	4,630	5.6%	5,978	1.1%
GFA Sold	M sqm	1,099	4.9%	787	-1.1%	1,059	-3.7%
YoY Growth in Residential Property price	%	5.3%		2.5%		-5%	

Source: CEIC and Citi Research Estimates

Valuations Look Undemanding from PEG Perspective

Figure 14. PRC Downstream Gas Companies – Valuation Comparison (Share price as at 28 Nov 2012)

		Price	Mkt-Cap	ROE		P/E (x)		P/B (x)		Yield (%)		PEG
	RIC		US\$m	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E	
BJ Enterprises (Buy)	0392.HK	HK\$50.10	7,354	8.4%	9.6%	18.4	15.1	1.4	1.3	1.7%	2.1%	1.0
ENN (Buy)	2688.HK	HK\$33.75	4,649	19.6%	19.9%	19.2	16.0	3.4	2.9	1.3%	1.6%	0.9
CR Gas (Neutral)	1193.HK	HK\$16.40	4,706	18.8%	20.9%	21.2	18.3	4.1	3.6	1.0%	1.2%	0.7
Kunlun Energy (Buy)	0135.HK	HK\$15.66	16,263	19.0%	17.2%	16.8	14.7	2.7	2.4	1.4%	1.7%	0.7
HKCG (Sell)	0003.HK	HK\$20.90	23,436	16.8%	15.0%	23.7	25.6	4.0	3.7	1.7%	2.0%	3.2
Towngas China (Buy)	1083.HK	HK\$6.49	2,060	8.1%	9.4%	19.7	15.9	1.6	1.4	0.9%	1.1%	1.0
Average				15.1%	15.3%	19.8	17.6	2.9	2.6	1.3%	1.6%	1.3

Source: Citi Research estimates and Factset for non rated (NR) companies

The average FY13E PER of Chinese and HK gas distribution companies is rather demanding at 17.6x, though on PEG they look more appealing on an average of 1.3.

Figure 15. Comparison of average project size of PRC downstream gas distribution companies

Company	Beijing Ent.	HKCG	CR Gas	Towngas China	ENN	Peer A
Number of projects	1	34	73	66	104	115
Connectable population in 2012 (in m)	n/a	n/a	n/a	n/a	53.14	62.49
Average connectable population per project in 2012 (in m)	n/a	n/a	n/a	n/a	0.51	0.54
Total connected population in 2012 (in m)	4.32	13.20	10.50	6.67	20.45	6.61
Average connected population per project in 2012 (in m)	4.32	0.39	0.14	0.10	0.20	0.06
Average household penetration rate (in %)	n/a	n/a	n/a	n/a	38.5%	34.8%
Total NG sales volume in 2012 (in bcm)	6.47	10.30	7.22	4.67	5.01	4.78
Average NG sales volume per project in 2012 (in bcm)	6.47	0.30	0.10	0.07	0.05	0.04
Ranking (in term of average per project NG sales volume)	1	2	3	4	5	6

Source: Company Reports and CIRA Estimates

Figure 16. Natural gas approximate conversion factors

From	To					
	billion cubic meters NG	billion cubic feet NG	million tonnes oil equivalent	million tonnes LNG	trillion British thermal units	million barrels oil equivalent
			Multiple by			
1 billion cubic meters NG	1	35.3	0.9	0.73	36	6.29
1 billion cubic feet NG	0.028	1	0.026	0.021	1.03	0.18
1 million tonnes oil equivalent	1.111	39.2	1	0.805	40.4	7.33
1 million tonnes LNG	1.38	48.7	1.23	1	52	8.68
1 trillion British thermal units	0.028	0.98	0.025	0.02	1	0.17
1 million barrels oil equivalent	0.16	5.61	0.14	0.12	5.8	1

Source: BP Global Statistics (** 28 m3 = 1 mmbtu)

China Wind Sector

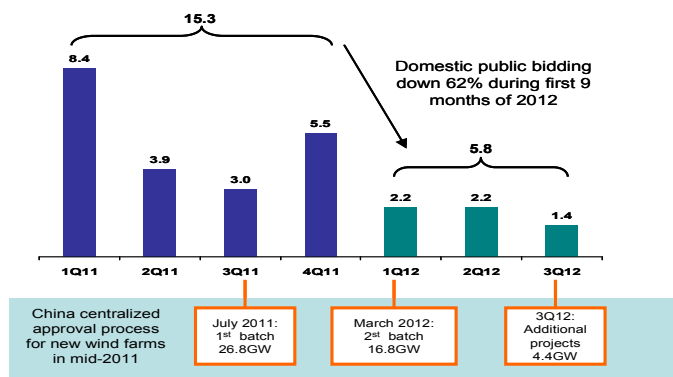
A Consolidation Phase in 2013E

We think 2013E will be a consolidation year for the China wind sector with (i) annual wind capacity addition flat yoy at 14GW, regulated by NDRC to avoid excessive installation to mitigate grid curtailment, while operators have trimmed their capex under high gearing; (ii) utilization to be sustainable at current level after declining in previous years due to exacerbating grid curtailments since capacity addition has become more orderly as stipulated above; and (iii) the expectation on CDM revenue has been substantially curtailed after spot price fall in 2012E, hence becoming more realistic. Renewables Quotas System might be implemented in 1H13E but would not raise the wind farm utilization much as the quotas would be moderate to facilitate execution considering grid curtailment issue. The sector therefore would not have much positive news in 1H13E, and valuation is the key to stock pick. Wind equipment makers are undergoing a tough L-sharp recovery to eliminate excessive production capacity; Goldwind should have some profit rebound in 4Q12E due to seasonal reasons, which would be unsustainable in 1H13E.

Summary

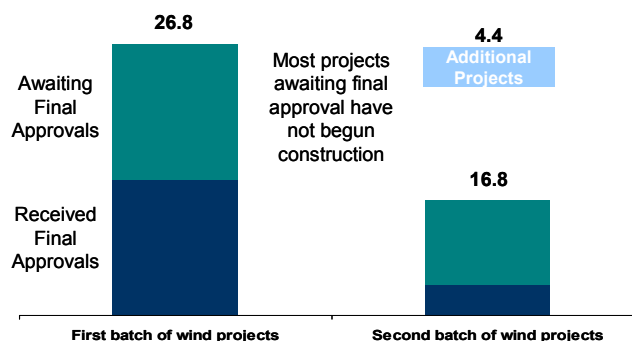
Lower PRC Wind Capacity Growth

Figure 1. Quarterly Domestic Public Tenders (GW)



Source: Goldwind and Citi Research

Figure 2. Domestic Wind Farms Projects Proposed by NEA (GW)



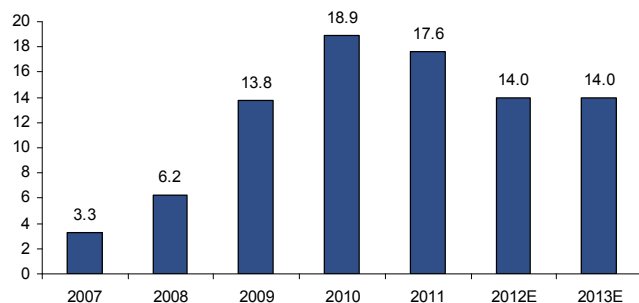
Source: Goldwind and Citi Research

NDRC's approval of new wind farm projects has been slowing. According to China Wind Energy Association, this was in view of:

- (i) Tightened credit for PRC wind-farm operators;
- (ii) Elimination of unqualified and small PRC WTG makers; and
- (iii) Slowdown in new installation to ease grid curtailment – according to Goldwind, 80% of wind turbine generator orders tendered in China in 9M12 are for installation in low wind speed regions.

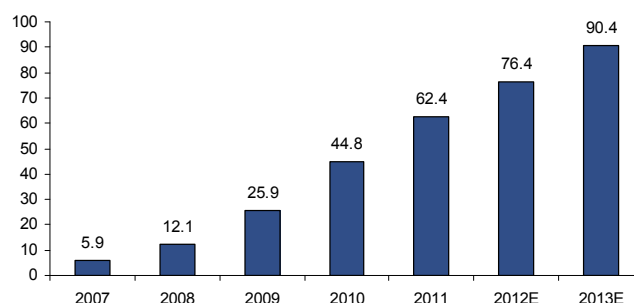
The association expects more regulatory approval in 2014-15E when the small and inefficient WTG manufacturers have exited and grid curtailment might be mitigated after the completion of new ultra-high voltage transmission lines.

Figure 3. PRC annual new wind farm installation (GW)



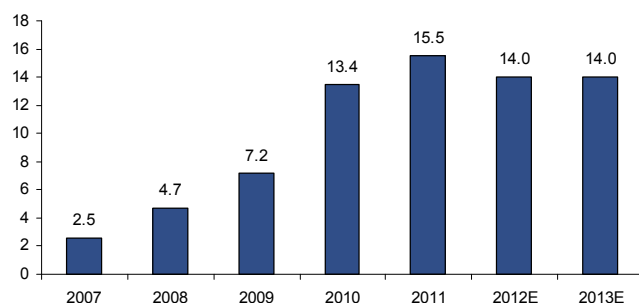
Source: BTM and CIRA Estimates

Figure 4. PRC cumulative wind farm installation by year-end (GW)



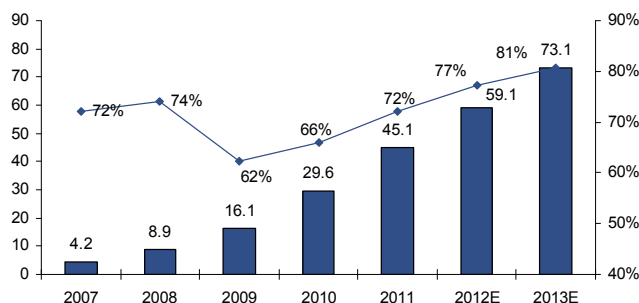
Source: BTM and CIRA Estimates

Figure 5. PRC annual new wind capacity with grid connection (GW)



Source: BTM and CIRA Estimates

Figure 6. PRC cumulative wind capacity with grid connection (GW)



Source: BTM and CIRA Estimates

Grid Curtailment to Mildly Improve in 2013E, but Still High

Figure 7. Grid curtailment ratio of PRC wind farm operators

	9M11	2011	1H12	9M12	2012E	2013E
Nationwide players:						
Longyuan	n/a	12%	16%	n/a	15%	14%
Huaneng Renewables	n/a	12%	24%	19%	21%	18%
Datang Renewables	13%	17%	26%	21%	23%	20%
Average	n/a	14%	22%	n/a	20%	17%
Suntien (in Hebei province)	n/a	4%	5%	5%	5%	4%

Source: Company Reports and Citi Research Estimates

Grid curtailment has been exacerbated since 4Q11 when much new wind capacity was installed. While the problem showed some improvement in 3Q12 compared to that in 1H12 owing to relatively weak wind power during the summer, it has re-emerged in 4Q12E, although it is slightly better than a year ago. The situation is the most severe in Inner Mongolia (East) and Northeast China where there were many wind farms of 6.6GW and 17.0GW respectively by end 1H12, representing 31.2% and 16.5% total power generating capacity for these regions.

PRC wind farm operators expect mild grid curtailment to improve in 2013E as capacity addition has slowed. But the improvement would not be material dragged by decelerated PRC power demand growth. Installation of cross-provincial ultra-high voltage transmission lines to dispatch electricity from wind farms in Northern China to end users in coastal provinces would also takes years with progress slower than expected during to technical issues with transmission efficiency lower than previously planned.

No High Hopes for a Renewables Quotas System (RPS)

To incentivize the use of clean energy, the Chinese government has a plan to launch a renewable quota system to make it mandatory for grid companies to buy a certain proportion of their electricity from renewable energy power plants. This would help to mitigate the grid curtailment problem of PRC wind farms. The Chinese government has issued a draft of this new policy to various wind farm operators and grid companies for consultation.

The renewable energy quotas system might be launched in 2013E. The system will divided the 31 provinces in China into four tiers, subject to various percentages of electricity from renewable energies. But the system would be relatively mild and moderate in order to facilitate execution. We therefore do not expect RPS to raise utilization of PRC wind farms much in 2013E.

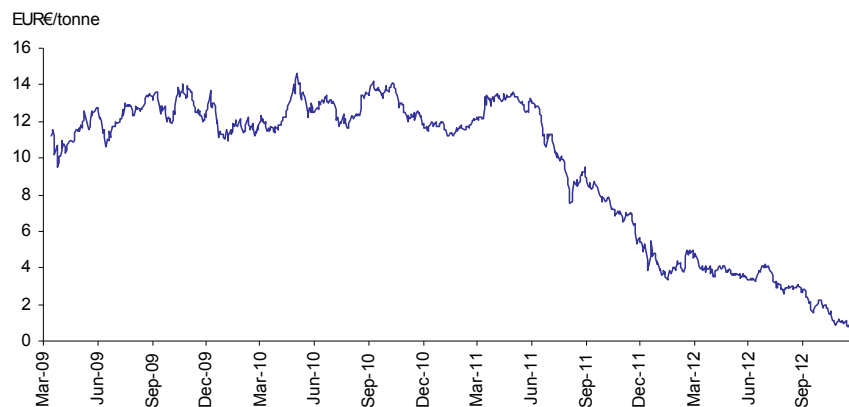
Figure 8. Percentage of electricity from renewable energies in 31 provinces in China

% of electricity from renewable energies	Number of provinces	Tier
15%	2	1
10%	7	1
6%	7	2
4%	3	3
3%	7	3
1%	5	4

Source: NDRC

CDM Revenue Post 2012E at High Risk

Figure 9. ICE Daily CER Index –CDM spot price



Source: Bloomberg (Ticker: ICEDCER index)

The CDM price has dropped over 83% year to date from €4.34/tonne to €0.75/tonne on 28 Nov amid a weak European economy, leading to expectation of lowered demand for carbon emission quotas. These three HK-listed PRC wind farm operators had an average 18% 1H12 pre-tax profit from CDM income, which is paid by Europe. We think that this revenue source faces high downside risk in 2013E and is assumed to be -75% yoy in our model.

Figure 10. % of 1H12 profit from CDM revenue of PRC wind farm operators

Company	Stock code	CDM revenue (Rmbm)	Profit before tax (Rmbm)	%
Datang Renewables	1798 HK (Buy)	71	106	67%
Longyuan	916 HK (Buy)	447	1,988	22%
Huaneng Renewables	958 HK (Buy)	65	303	21%
Jingneng	579 HK (not rated)	83	538	15%
Suntien	956 HK (Buy)	68	526	13%
Average				18%

Source: Companies

NDRC plans to set up carbon trading markets with trial runs in seven cities and provinces (namely, Beijing, Tianjin, Shanghai, Chongqing, Guangdong, Hubei and Shenzhen) by end 2013E and the nationwide market by 2015E. But the fundamentals look premature as China currently has only a limited amount of voluntary carbon reduction trading in the country. Problems from three aspects, including, (i) how to calculate greenhouse gas emission volume; (ii) how to allocate greenhouse gas emission quotas; and (iii) how to set up trading mechanism, would also have to be resolved.

China Solar Sector

Investment Outlook – Subsidizing Rural Solar Installs

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China's Ministry of Finance announced plans in Nov 2012 to approve additional solar power projects based on the existing framework under the Golden Sun and Solar BIPV policies. The subsidy will be tied to a project completion date of 30 Jun 2013, while a new subsidy rate for rural solar farm installs was introduced. This is constructive for Chinese solar project contractors such as China Singyes Solar Technologies (0750.HK), while manufacturing supply chain companies, like GCL-Poly (3800.HK) will need to see better global end-demand before we see any ASP or profit recovery.

While a China demand ramp will help domestic orders, China will likely account for 10-20%, or 3-6GW, of global demand in 2012. Even though 2013E growth looks to be strong and could reach upwards of 10GW (including the inventory re-stock), new approvals in China are unlikely to help offset the weak global end-demand outlook and improve profitability of the manufacturers. Both manufacturers and installers have been struggling due to uncertainty in new subsidies, potential anti-dumping duties, and heavy competition due to overcapacity. We believe the solar equipment exporters will still face an uphill battle to reduce losses.

Figure 1. China's 12th Five Year Plan (12-5) on Solar Power Capacity

Category	Capacity (GW)	Target (GW)	Target (GW)	Key focus in 2011-2015
	2010	2015	2020	
Solar Power Station	0.45	11	23	In Qinghai, Gansu, Xinjiang, Inner Mongolia, Tibet, Ningxia, Shaanxi, Yunnan, as well as regions suitable for building photovoltaic plants in North and Northeast part of China. The construction will be tied to large-scale hydro power and wind power developments
Photovoltaic (PV) Power Station	0.45	10	20	
Solar Thermal Power Station	0	1	3	Build pilot solar-thermal power generation plants with plenty of solar, land and water resources
Distributed Power Production System	0.41	10	27	
				Build distributed on-grid photovoltaic system in industry park, economic development zone, and roof of large public buildings in the East China and Central China. Build independent PV systems in remote regions facing the electricity shortage in Tibet, Qinghai, Gansu, Shaanxi, Xinjiang, Yunnan and Sichuan as well as on outlying islands. Expanding electricity supply from PV system to city lighting and traffic signals systems
Total Power Generation	0.86	21	50	

Source: China National Energy Administration, Citi Research

On October 23, 2012, Mr. Sicheng Wang, from the Energy Research Institute of NDRC, highlighted challenges faced by the PV manufacturing sector. While polysilicon prices fell to below US\$20/kg level, module prices also fell 45% over 12 months, to €0.5/W as manufacturers face significant over-capacity. This has resulted in further losses by the manufacturers, which are facing additional pressure from inventory write-downs.

Prices for crystalline-silicon based solar materials and equipment fell sharply in 2012 due to overcapacity and lingering uncertainty in end-demand. We remain concerned that the rapid ramp-up in production capacity in the past two years may lead to weak margins.

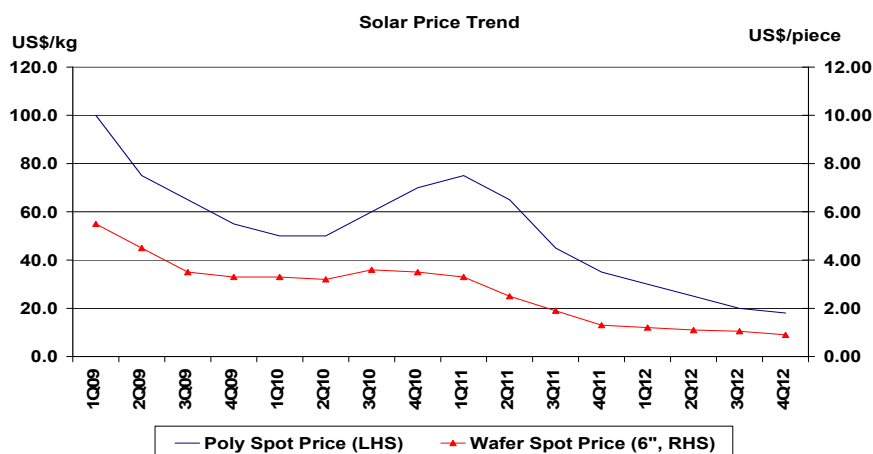
Figure 2. Solar Supply Chain – Price Trend

	1Q10	2Q10	3Q10	4Q10	1Q11	2Q11	3Q11	4Q11	1Q12	2Q12	3Q12	4Q12
Poly Spot Price (per kg)	US\$50	US\$50	US\$60	US\$70	US\$60-90	US\$55-75	US\$40-50	US\$30-40	US\$25-32	US\$22-28	US\$20-24	US\$15-20
Poly Contract Price (per kg)			US\$60	US\$60	US\$50	US\$50	US\$45	US\$35-40	US\$25-30	US\$25	US\$20	US\$18
Wafer Spot Price (6")		US\$3.2	US\$3.6	US\$3.5	US\$3.3	US\$2.5	US\$1.9	US\$1.3	US\$1.1-1.2	US\$1.0-1.2	US\$1.0-1.1	US\$0.85-1.0
Wafer Spot Price (Per Watt)		US\$0.8	US\$0.9	US\$0.9	US\$0.8	US\$0.6	US\$0.48	US\$0.38	US\$0.30	US\$0.26	US\$0.25	US\$0.22
Cell Spot Price (Per Watt)	US\$1.3	US\$1.35	US\$1.4	US\$1.35	US\$1.2	US\$0.9	US\$0.68	US\$0.58	US\$0.50	US\$0.47	US\$0.42	US\$0.35
Module Spot Price (Per Watt)		US\$1.8	US\$1.85	US\$1.75	US\$1.65	US\$1.3	US\$1.10	US\$1.00	US\$0.85	US\$0.82	US\$0.75	US\$0.68

Source: Citi Research

In essence, many solar polysilicon makers are facing an endurance test, where scale (and cost) leaders face operating losses, while their customers face imminent production cuts. Demand ramp-up from China/Japan was unable to offset weak European demand. With the sector facing demand and trade war uncertainty, we remain cautious on GCL-Poly and OCI, but expect both to be survivors of the downturn.

Figure 3. Polysilicon Price Trend (1Q09-4Q12)

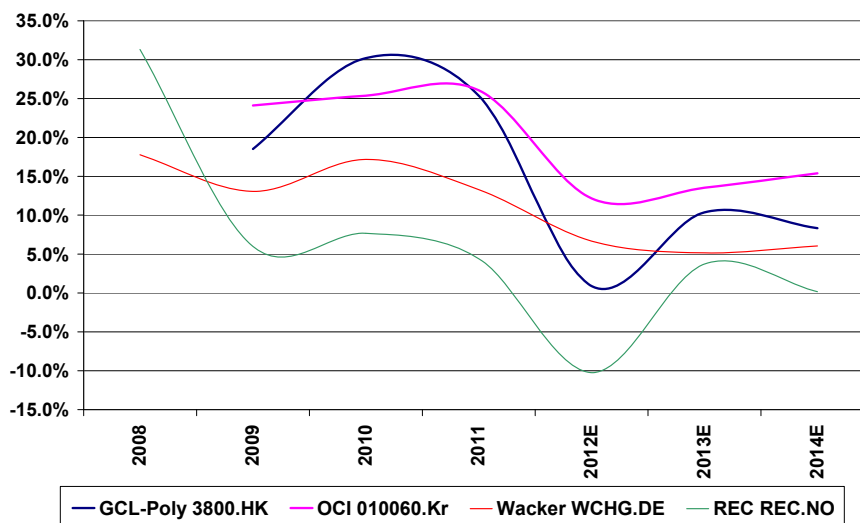


Source: Citi Research

The sharp ramp in capacity in the past two years, in essence, creates a situation that is similar to the DRAM and the TFT-LCD sectors – the more they produce, the lower the price – while coordinated production cuts remain unlikely in view of pressure to reduce their operating losses.

As a result, margins for the solar sector have collapsed and we believe operating margins for the top players may remain structurally lower than 2009-2010.

Figure 4. EBIT Margin Trend for Polysilicon Producers (2008-2014E)



Source: Company Data; Citi Research Estimates

Potential Catalysts

While the sector continues to suffer from oversupply, we highlight few potential catalysts which may help the supply-chain manufacturers to see a sustained margin improvements:

1. End-demand improvement in new markets

While the market remains focused on the subsidies from Europe and trade war concerns, we see a potential for demand improvement as prices reach investment threshold for energy-intensive industries, with or without government intervention, and as effective supply reduces further on weak prices. Falling prices are also driving higher IRR for some markets with high solar utilization hours. While incremental improvements take time, sharper pick-up and better visibility in end-demand in China, Japan, and US are pre-requisites to pricing and demand up-ticks, as they become the major end-demand countries in 2013E.

2. Resolutions from trade investigations

We expect local manufacturers will continue to pressure governments to raise costs of imports, via trade investigations and potential anti-dumping tariffs, which will impact profitability of export-driven downstream producers. The US has already announced an anti-dumping tariff against Chinese modules, but the trade discussions are far from over, as Europeans are also appealing for trade restrictions against Chinese products.

The China Ministry of Commerce had alleged the US was breaking World Trade Organization (WTO) rules with regards to the clean energy projects in the US. The ruling stated that certain US projects received subsidies that violated WTO rules, specifically in Massachusetts, Ohio, Washington, New Jersey and California. The Ministry of Commerce called to "give China renewable energy products fair treatment", according to an article on its website.

Besides the complaint with the WTO, China has also been investigating US and Korea polysilicon producers, as announced on 20 July by the Ministry of Commerce. On both the US and Korea manufacturers, concerns relate to the cost structure for polysilicon production, and whether or not they are selling their products below their production costs and whether this could result in anti-dumping charges. The investigation period is between 1 July 2011 and 30 June 2012, and will focus on solar-grade polysilicon. For the US market, there is also another investigation into one federal project and 16 provincial government subsidies, which may have broken the law regarding anti-dumping charges. For the Korean manufacturers, besides concerns over product dumping, there have also been wide discussions in the local press of an investigation on whether they may have breached any rules regarding the transfer pricing mechanism. This could be a response to concerns over below market pricing when companies sell into China, which may have an impact on their tax liability. The investigations can last until 20 July 2013 and may be extended to 20 January 2014.

3. Solar Farm Developments as Earnings Driver

In the next 12 months, we see sector profits as increasingly coming from construction and operation of solar power plants, which have historically been captured by smaller domestic players. Both GCL-Poly and OCI have increased investment into project construction.

Figure 5. Recent Projects Announced by GCL-Poly

Name of Project/Contract	Size	Announced
Datong, Shanxi ground-mount project	310MW	27-Aug-12
Datong, Shanxi roof-top project	30MW	27-Aug-12
China Merchants New Energy Letter of Intent (LOI)	973MW	2-Aug-12
SPS Alpaugh 50, SPS Alpaugh North Disposals	92MW	27-Jul-12
ISOFOTON MOU	1,000MW	18-May-12

Source: Company Information, Citi Research

Unfortunately, as all the manufacturers appear to be shifting to the more profitable project development areas, it is not yet possible to identify those players with significant cost advantages in solar farm construction.

GCL-Poly has taken a step further, and is working with NRG Energy to set up a joint venture (Sunora Energy) in Phoenix, Arizona, to develop more efficient pre-assembled systems, which include both panels and mounting systems, which may reduce the cost by 15-40%, according to the CEO of Sunora. If successful, it will allow Sunora to achieve significant cost advantages relative to its peers and could help the joint venture secure more US-based projects.

Hong Kong Power Sector

Proxy of Bond Investments; Overseas Add Catalysts

Summary

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Share prices of HK utility companies have gone up an average 18% year to date (up to 22 Nov 2012) in response to declining US\$ and HK\$ bond yields by 12% to 1.75% and by 49% to 0.71% in the same period. With bond rates at historical lows, further downside seems limited; similarly, as no US\$ rate hike is expected in the run-up to 2015E, a massive share price retreat is also not expected in 1H13E. We regard the SOC mid-term review as a storm in a tea cup with little investment risk, as the permitted rate of return of Hong Kong power companies has been fixed and could not be altered without their consent. Overseas acquisition therefore remains the key positive catalyst to divert share price performance. Our top pick is CKI for this theme. We think the market's concern about CKI's new share issuance risk is overdone after the significant underperformance of CKI's share price against PAH's while profit sources for both companies overlap by 90%.

SOC Mid-Term Review in HK – Storm in a Tea Cup

The electricity market is regulated by the Hong Kong government so that the two power companies could make a permitted return of 9.99% ROA in the run-up to end 2018. The two power companies in Hong Kong are vertically integrated running power generation, transmission, distribution and retailing. Each has monopoly status in their service areas with no competition. Capex of the two power companies in Hong Kong is regulated by the Hong Kong government under a 5-year capex plan with the existing plan to be terminated by end 2013.

The Hong Kong government will proceed with a mid term review in 2013E with the two Hong Kong power companies to decide the SOC capex in the next five-year period. Additionally, the development plan of the future local electricity market will also be discussed. While there is uncertainty regarding the outcome subject to negotiation, we believe that:

1. **The 9.99% ROA permitted rate of return would be intact** in the run-up to 2018. Any amendment of this high return must be mutually agreed by both the government and the power companies. The power companies are reluctant to make any change on this per our understanding from them. But **the Hong Kong government might propose a lower return for new capex**, which would be subject to negotiation with the two power companies;
2. **Any possible opening of the HK electricity market would be remote after 2018**. Under the prevailing bilateral contracts signed between the Hong Kong government and the two power companies, any change of the power market after 2018 must be decided before 2015. But we do not expect the power market to be opened because (i) power grids in Hong Kong are owned by the two power companies, and the Hong Kong government has no intention to buy back the grids. This is shown by the government not buying back the privately run local tunnels despite the smaller expenditures involved; and (ii) no merger of the Hong Kong power market with the mainland market is considered taking reliability into account;
3. **Power generation mix would be reviewed for emission cut**; currently, power generation mix in Hong Kong is 56% coal, 27% NG and 17% nuclear.

Figure 1. Fuel mix for power consumption in Hong Kong

Year	2012 (CLP + Power Assets)	2012 (CLP)	2012 (Power Assets)
Coal	56%	50%	70%
Natural gas	27%	25%	30%
Nuclear	17%	25%	0%

Source: CLP, Power Assets and HK Government

HK government would like to raise the electricity mix from nuclear and the planned 2,160MW Lufeng nuclear plant in Guangdong province could be a suitable target to supply electricity to Hong Kong while the plant itself is rather far away from Hong Kong, hence should not pose a concern to Hong Kong residents. Unfortunately, construction of this plant has been suspended following the Fukushima nuclear accident. Though the PRC government has announced it will resume new nuclear power project approval ahead, this project would not be processed unless its technology is upgraded from 2G to 3G. Completion of this project therefore would only be around 2020, if not later.

CLP and PAH will try to include offshore wind farms in the mid-term SOC review in 2013E. To lower tariff hike impact, their offshore wind farms would be scaled down, from 200MW to 100-150MW for CLP and half of it for PAH. The capex per MW of the offshore wind farms would be high around HK\$25m/MW. Surprisingly, based on the wind data collected in 2012E so far (at over 7m per second in summer and 8m per second in winter), utilization of the offshore wind farms in Hong Kong could be relatively high at over 2,500 hours pa. The Hong Kong government plans to have 4% energy from renewable by 2020E, with the majority to come from waste to energy projects; and

- SOC capex in 2014-18E will have more upside for PAH than CLP.** We assume HK\$14.5bn for PAH in the period, 18% above that in 2009-13E, due to the construction of a new gas-fired power generating unit called L10 assumed at HK\$2.0-2.5bn. For CLP, we assume its SOC capex in 2014-18E to be similar to that of 2009-13E, with the benefit of higher raw material offset by the lack of new fixture development (note: part of CLP's SOC capex in 2009-13E was for FGD installation, which was completed, hence would not be recurring in the next five-year period).

Judging from the above, we think the regulatory regime in the Hong Kong power industry would be largely status-quo in 2013E.

Overseas Expansion to Drive Earnings Uplift

Figure 2. FY12E profit mix of Hong Kong utility companies by country

Country	CKI	Power Assets	CLP	HKCG
UK	59%	38%	0%	0%
HK	18%	47%	68%	50%
Australia	14%	8%	21%	0%
China	7%	5%	8%	50%
Canada	1%	2%	0%	0%
New Zealand	1%	1%	0%	0%
Southeast Asia & Taiwan	0%	0%	2%	0%
India	0%	0%	1%	0%
Total	100%	100%	100%	100%

Source: Company Reports and Citi Research Estimates

HK based utility companies are expanding their operations overseas as businesses in Hong Kong are matured. Financially they are also well capitalized to do so, endorsed by strong recurrent cash inflow from Hong Kong. In FY12E, profit from overseas accounted for 86% of total profit for CKI, 62% for Power Assets, 50% for HKCG, and 32% for CLP. Geographically, their overseas expansions are located primarily in the Asia-Pacific region for CLP, English-speaking developed countries (e.g., Australia, New Zealand, the UK and Canada) for CKI and Power Assets, as well as mainland China for HKCG.

Among the Hong Kong based utility companies, we view CKI and Power Assets as prime beneficiaries of overseas expansion as they are financially robust for expansion and interested in brownfield acquisitions, which would likely generate profit shortly after acquisitions. Their deal selection criteria are as follows:

1. **Country** – Power Assets and CKI prefer to invest in regulated utility businesses in English-speaking developed countries with well-established regulatory regimes, in particular Australia, New Zealand, the UK, and Canada;
2. **Business** – Power Assets is especially keen on regulated low-risk assets such as power, gas and water distribution. So far, the company has invested in power and gas distribution businesses in Australia and the UK respectively;
3. **Deal Size** – The deal size for CKI and Power Assets could be as high as GBP4bn each with reference to their joint bid for the two power networks in the UK sold by E.On in March, although they did not win the bid. They prefer to gain major stakes in the acquired assets. If minor stakes of larger assets are acquired, then veto power on the director board would be sought; and
4. **Returns** – Power Assets prefers operational projects so it can start creating profit in the 1st or 2nd term after acquisition. The investment hurdle risk is double-digit ROI.

Figure 3. List of utility assets held by infrastructure funds and companies that meet CKI's deal selection criteria

Fund/ Company Name	Assets Held	Country	Financial Statistics	Description of Asset Held	Other Stakeholders	Stake
Berkshire Hathaway - listed	CE Electric UK	UK	NA	Electricity distribution network operator in North East, Yorkshire	N/A	89.9%
Colonial First State & JP Morgan infrastructure fund	Electricity North West	UK	Net asset by end-Mar 11: GBP545m	Electricity distribution network operator in North West serving 5.1m customers	N/A	100%
Iberdrola - listed	Scottish Power Energy Network	UK & Scotland	Net asset by end-Mar 11: GBP2,366m	Operates transmission and distribution network in 4 regions with 3.5m customers	N/A	100%
Scottish and Southern Energy - listed	Scottish Hydro Electric & Southern Electric Power	UK & Scotland	NA	Operates transmission and distribution network in 2 regions	N/A	100%
Scottish and Southern Energy - listed	Scotia Gas Network	UK & Scotland	Net asset by end-Mar 2011: GBP173m	With 74,000km gas pipelines in Scotland, south and southeast England	Canadian pension fund (50%)	50%
N/A	Synergen Power	Ireland	N/A	Operate a 400MW combined cycle gas-powered generation plant, supplying 8% of power demand	N/A	N/A
Bord Gais	Bord Gais Energy	Ireland	N/A	Retail arm of Bord Gais, selling gas and power to over 600k gas customers and 300k electricity customers	N/A	N/A
Queensland Government	Powerlink	Australia	Net asset by end-Jun 2011: A\$2,120m	Power transmission service provider with network capacity of 50,000GWh per year	N/A	100%
Queensland Government	Energex	Australia	Net asset by end-Jun 2011: A\$2,922m	Owns 53,000km of power lines and cables, serving 1.3m customers	N/A	100%
Queensland Government	Ergon Energy	Australia	Net asset by end-Jun 2011: A\$3,315m	Electricity distributor, retailer and generator with 150km power lines serving 680k customers	N/A	100%
Capstone Infrastructure Corporation	Cardinal Power	Canada	EBITDA for 2011: CAD30m	A combined cycle gas cogeneration plant with installed capacity of 156MW	N/A	100%
Capstone Infrastructure Corporation	Erie Shores Wind Farm	Canada	EBITDA for 2011: CAD20m	Wind farm with a capacity of 99MW and generated 228GWh in 2010	N/A	100%
Capstone Infrastructure Corporation	Hydro power facilities	Canada	EBITDA for 2010: CAD9m	Owns 4 hydro power plants which generated 151GWh in 2010	N/A	100%
ATCO Group	ATCO Gas	Canada	N/A	Alberta based, province-wide natural gas distribution company	N/A	100%
Enbridge Inc	Gazifere	Canada	N/A	One of the two distributors of natural gas in Quebec serving 32k customers	N/A	N/A
Trencap S.E.S.	Gaz Metro Limited Partnership	Canada	Total asset by end-Jun 2010: CAD3.6bn	One of Canada's largest natural gas distributor serving 180k customers in Quebec	Enbridge (38.89%)	61%
Spectra Energy	Union Gas	Canada	Total asset: CAD5.6bn	A natural gas storage, transmission and distribution company based in Ontario	N/A	100%
ATCO Group	ATCO Pipelines	Canada	N/A	Provides gas transportation and distribution across Western Canadian Sedimentary Basin and Alberta	N/A	100%
Hasting Funds Mgt Ltd & JPMorgan Asset Mgt	Southwest Generation	U.S.A.	EBITDA for 2011: USD74.1m	Owns 7 gas-fired plants in the US with an aggregate capacity of 974MW	NA	100%
Alinda Capital	NorTes Gas Storage	U.S.A.	N/A	2 storage facilities at Texas providing gas storage services	N/A	100%
Alinda Capital	SourceGas LLC	U.S.A.	N/A	Owns regulated local gas distribution co. in Arkansas, Colorado, Nebraska & Wyoming serving >1.1m users	GE Energy Financial Services	50%
MidAmerican Energy Holdings Company	Northern Natural Gas	U.S.A.	N/A	Operates an interstate NG pipelines (15,000 miles) with design capacity of 5.5bcf/day, based in Nebraska	N/A	100%
MidAmerican Energy Holdings Company	Kern River Gas Transmission	U.S.A.	N/A	Operates an interstate NG pipeline (1,708 miles) with design capacity of 2.1bcf/day, based in Salt Lake City	N/A	100%
Utilities of Australia Pty Ltd	Perth airport	Australia	N/A	Australia's 4th largest airport in terms of passenger traffic	Hastings Funds AREF AIF (29.74%)	55.6%
Australian Infrastructure Fund	Queensland airport	Australia	N/A	Owns and operates Gold Coast, Townsville and Mount Isa airports	N/A	49.1%
Australian Infrastructure Fund	Port of Portland	Australia	N/A	Owns several marine terminals and 4 airports	N/A	50.0%
Australian Infrastructure Fund	Port of Geelong	Australia	N/A	Largest regional port in Victoria, Australia	N/A	35.0%
Macquarie European Infrastructure Fund	Bristol International Airport	UK	N/A	With 77 scheduled routes and 59 charter destinations	Bristol Airport (Bermuda) Limited	N/A
AMP Core Capital Infrastructure Fund	Melbourne & Launceston Airport	Australia	N/A	Both airports are operated under a 50-year long-term lease	Hastings (20%), RREEF (17.5%), Other funds (37.5%)	25%

Source: Company Information, Citi Research

Expanded Yield against High Quality Bonds

Figure 4. HK utility share prices negatively correlated to interest rates in 1998-2012

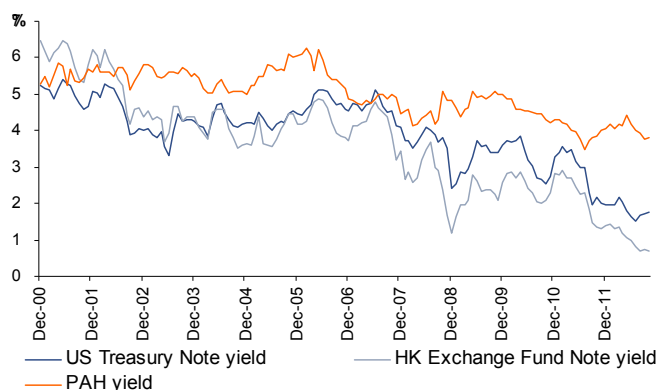
	US\$ bond yield	HK\$ bond yield	HK\$ prime rate	3 months HIBOR
CLP	-79%	-84%	-50%	-55%
HKE	-88%	-88%	-59%	-63%
CKI	-81%	-79%	-44%	-48%
HKCG	-71%	-84%	-53%	-57%
HSI Index	-41%	-58%	-21%	-31%

* Up to 23 Nov 2012

Source: Citi Research Estimates

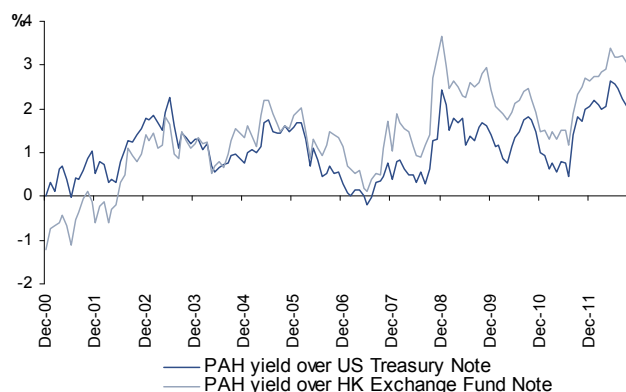
Hong Kong utility stocks are yield plays in view of their high payout ratio (55% for PAH, 62% for CLP and 43% for CKI in FY12E) and strong recurrent cash inflow from local operation. It is not surprising that the share price performance of these utility companies are highly negatively correlated (71-88%) to US\$ bond yields, versus 41% for the Hang Seng Index in the last 10 years.

Figure 5. Comparison of Power Assets' yield, US\$ treasury note yield & HK\$ exchange fund notes yield



Source: Citi Research, Federal Reserve, Hong Kong Monetary Authority

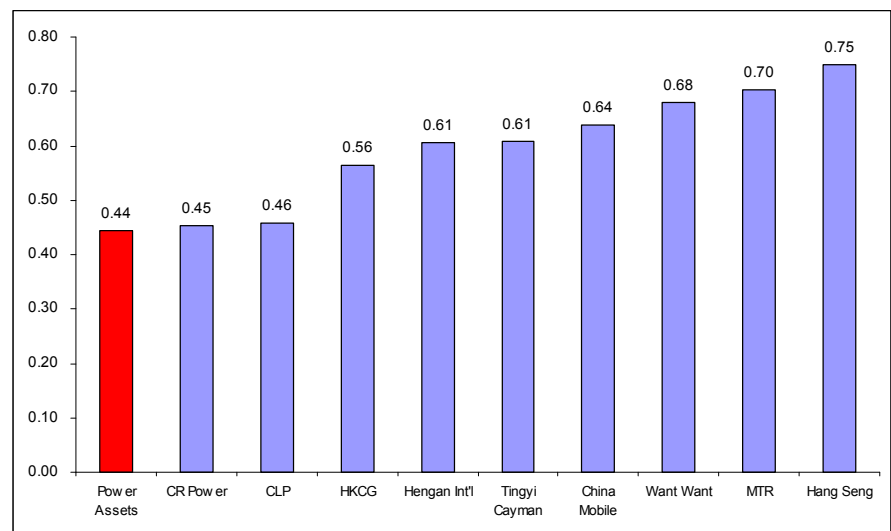
Figure 6. Widened spread of Power Asset's yield against US\$ treasury note yield and HK\$ exchange fund notes yield



Source: Citi Research

The premium of PAH's dividend yield has been widened against 10-year US\$ treasury note yield and HK\$ exchange bond fund note to 204bps and 308bps respectively, versus average 137bps and 220bps in last five years as well as 119bps and 170 bps in last 10 years. The Federal Reserve is likely to keep US interest rates at the current near-zero rate until 2015, minutes from its September rate-setting meeting have shown.

Figure 7. The 10 HSI Constituent Stocks with lowest beta



Source: Bloomberg

India Electric Utilities - Overview

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Across-the-board tariff hikes have helped at the margin

- Since Apr 2012, 22 SEBs (accounting for ~85% of power sales in the country) have raised tariffs by an average of 2-37%. This follows a 1-35% average tariff increase in FY12. SEBs have also approved a formula for passing through any variation in coal and power-purchase costs.

SEB debt restructuring has finally been put in place

- SEB debt restructuring involves 1) mandatory conversion of state government loans into equity, 2) restructuring of short-term loans into long-term loans; 3) issuance of state-backed bonds, and 4) interest subvention by Government of India (GoI), subject to adherence to the stringent conditions of reduction in AT&C losses.
- The debt restructuring package is not mandatory and states have the option not to accept it. If a state doesn't avail this, it has an option to enter negotiations with banks and should provide subsidy to DISCOMs. The four states of TN, Rajasthan, Haryana and UP account for ~70% of total accumulated losses (Rs2.4trn).

Fuel-supply issue remains unresolved

- Under the new Fuel Supply Agreement (FSAs), Coal India Limited (CIL) will meet 80% of requirements; however, a part of this would be imported. Initially, imported coal would be 15% and will progressively reduce to 5% by FY17.
- Coal imported would be on cost-plus basis and power companies would have to pay a 2% service charge to CIL. Power companies will have to make advance payments for imported coal and will not be allowed to reject the coal once an order is placed.
- In case CIL supplies less than Annual Contracted Quantity (ACQ), it will pay a penalty, which will be 1.5% of the price of the short-supply volume. As the supplies fall below 65% the penalty would increase to 40% of the short volumes, in a phased manner from 5%.
- FSA will be signed for plants which achieve a commercial operation date (CoD) after Dec 09 till Mar15 for 20 years.

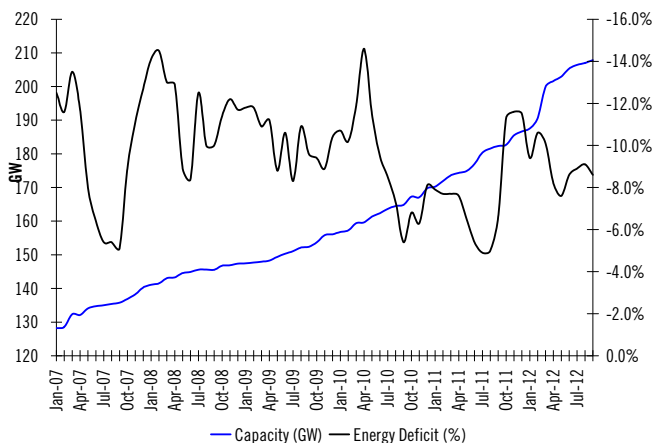
Coal price pooling is facing resistance from states and DISCOMs

- The Ministry of Power (MoP) and Central Electricity Authority (CEA) are proposing pooling of coal prices (uniformly loading of cost of imported coal over all users). However, the proposal is facing stiff resistance from states such as West Bengal, Chhattisgarh and Orissa.
- So far, supplies to projects which have achieved CoD after Apr 09 are on an MoU basis, without a guarantee of a specific supply limit. As a result, ~6GW of capacity commissioned by the private sector since April-2011 (based on supplies from Coal India) is operating at PLFs of 21-75%.

Despite capacity addition, deficit has stayed high

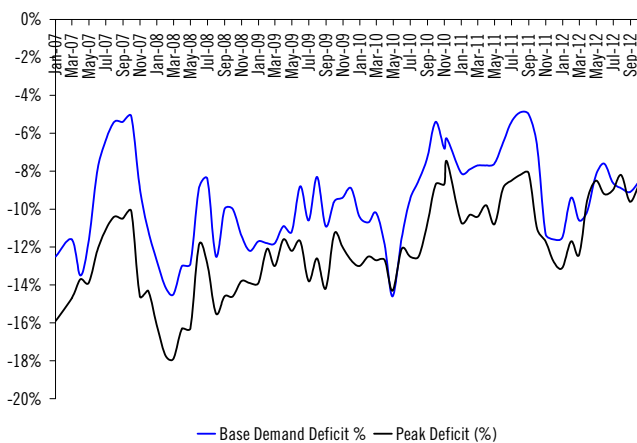
- India has ~209.3GW of installed capacity as of October 2012. ~9.4GW of capacity has been added since March 2012. However, despite a healthy pace of capacity addition, power deficits have remained elevated ~9% levels. This is due to growing demand and fuel shortage (which has caused new capacity to run at low PLFs).

Figure 1. India – Installed Capacity and Deficit



Source: CEA, Citi Research

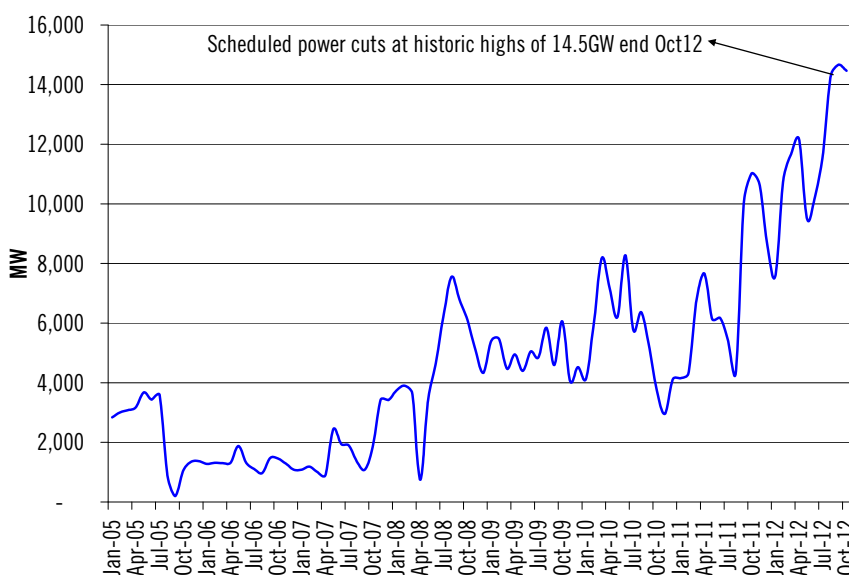
Figure 2. India – Base and Peak Deficits



Source: CEA, Citi Research

- **Scheduled notified power cuts to industries at historic highs** — Scheduled notified power cuts to HT/ LT industrial consumers are at a historic high of 14GW+ in October 2012. This figure has gone off the charts starting September 2011. We wonder how bad the situation is with regard to the unscheduled non-notified power cuts in India.

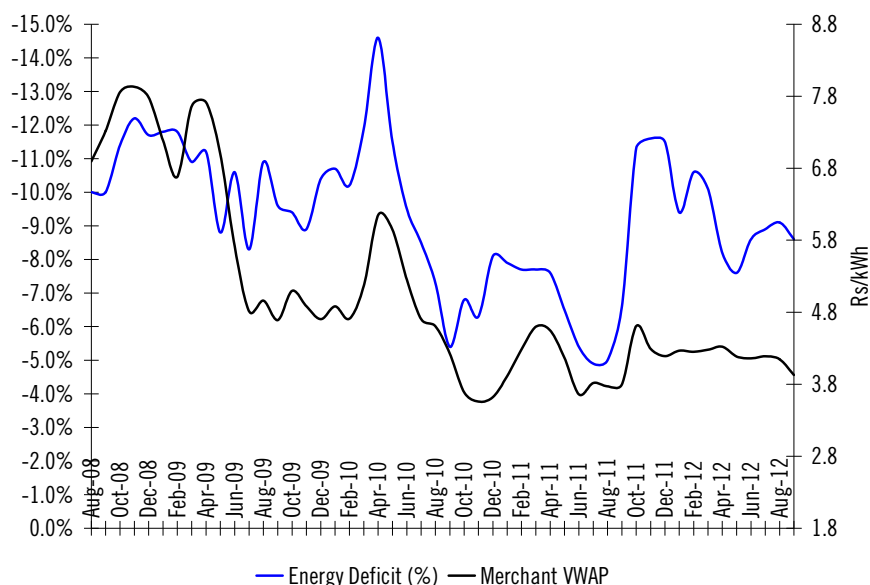
Figure 3. India – Scheduled Notified Power Cuts



Source: CEA,, Citi Research

- **Merchant prices are holding up** — Merchant power prices have declined from their peak 3-4 years ago due to deteriorating financial health of DISCOMs. However growing fuel shortages, persistently high deficits, and general cost inflation have supported merchant prices at ~Rs4/kwh.

Figure 4. Merchant Prices Seem to Have Found a Bottom



Source: Citi Research

Regulatory regime is in a flux and faces risks of becoming impractical and convoluted

- In response to fuel shortages and poor financial health of IPPs (inability to pass-on higher fuel costs), the authorities have been trying to evolve new power procurement guidelines for DISCOMs. These guidelines could fundamentally alter the India Electric Utility landscape.
- **Existing regime allows fuel cost pass-throughs and works just fine** — Both Case-1 & Case-2 bids in current regime have escalable (at CERC rates) & non-escalable energy charges. It is up to the generator to suitably apportion the same between the two. IPPs are suffering as: (1) in an intensely competitive environment, IPPs apportioned a very small portion of fuel costs to the escalable component (nil in some cases) to achieve a lower levelized tariff to win the bid and (2) idiosyncratic jumps in fuel costs cannot be passed on. All that needs to be done is to clearly define what constitutes force majeure or change the law so there are no controversies.
- **Complete pass-through could create new issues** — Draft Case-2 PPA envisages complete pass-through of fuel costs and competitive bidding on fixed costs only. This solves the problem of non-linear idiosyncratic jumps in fuel costs. However, it will (1) inhibit competition on the basis of fuel costs and will be detrimental to consumer interests, (2) reduce incentives to take risks to procure fuel efficiently, (3) lead to gold plating (a problem in all pass-through regimes).
- **Case-2 provisions could discourage private investment** — (1) If a company has captive mines abroad, why would it sell coal at a 30% discount to Indian power projects when it can sell coal at the benchmark prices outside India? (2) Complete cost pass-throughs will discourage mine acquisitions abroad and tying up of long-term contracts at remunerative costs.

- **.....and provide too much protection to debt investors** — In case of disruption in fuel supply, 70% of fixed capacity charges will be paid and, in the case of certain force majeure events, 80% of debt will be protected under Case-2. This will protect debt investors but antagonize DISCOMs and end consumers who will pay for power that is not being supplied to them.
- **Problems in Case-1 fuel-wise segregation** — This will be difficult to implement in practice given it will be cumbersome for DISCOMs and could also lead to collusion among the limited number of generators who have captive coal.

Overall, regulatory changes, fuel shortages, poor financial health of DISCOMs continue to impact the sector. We believe these problems will still take some more time to resolve. In this context PGCIL (PGRD.BO; Rs118.40; 1) and JPVL (JAPR.BO; Rs37.80; 1) are our top picks in the sector. Tata Power (TTPW.BO; Rs103.35; 3) and Adani Power (ADAN.BO; Rs49.85; 3) are our top sells.

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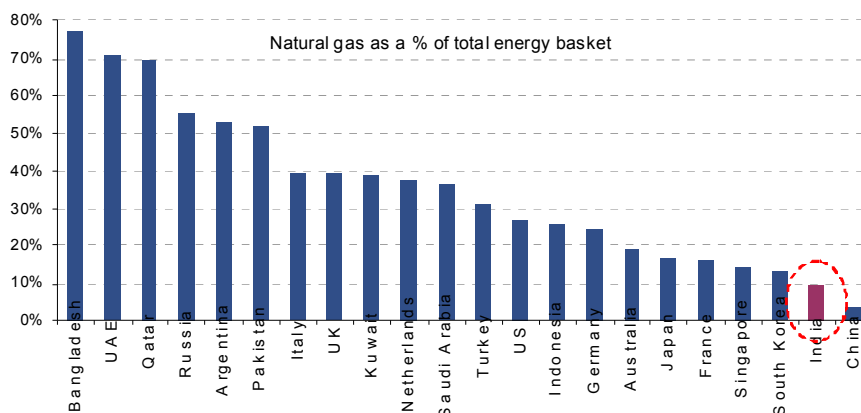
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India Gas Sector

Gas demand in India remains strong and largely supply-led. With KG gas output continuing to fall and an uninspiring outlook for other domestic sources, LNG will continue to be the way forward for the sector in India. While regulatory challenges have abated to some extent in recent months, some unknowns still remain.

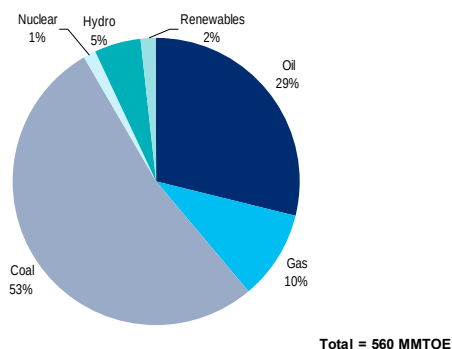
- **India's energy consumption set to grow** – India's primary energy consumption is set to grow ~2x from ~0.5 BTOE in 2011 (*source: BP Statistical Review, Jun'12*) to ~1 BTOE in 2030. Gas currently forms a measly ~11% of the total energy pie in India, as against the global average of ~25% (see Figure 1). Therefore, India's gas consumption is likely to display robust growth – as it gains increasing share in an expanding pie. While >80% of energy needs are currently supplied by crude and coal, both these sources are likely to face crunches in supplies on the domestic side, with ramp-up in production lagging demand growth by a large margin.

Figure 1. India's % Gas Consumption vs. Global Consumption



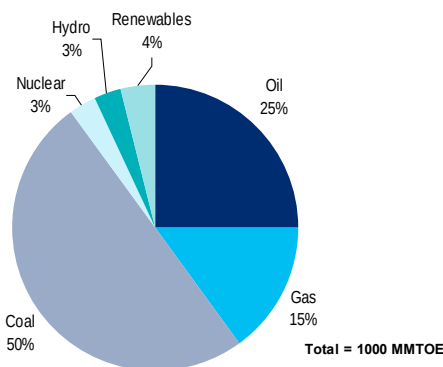
Source: Citi Research, BP Statistical Review

Figure 2. India's Energy Mix is skewed towards coal and oil (2011)



Source: Citi Research, BP Statistical Review

Figure 3. India's Energy Consumption Forecast 2030

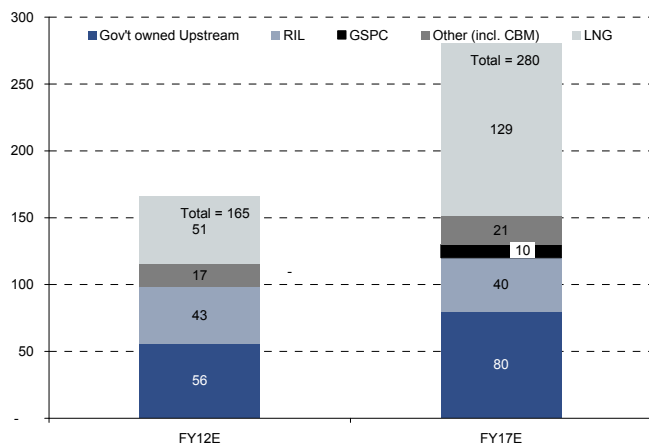


Source: Citi Research, BP Energy Outlook

- **The LNG story is one of secular growth** – India's current gas needs are under-supplied, a scenario that is unlikely to change given the forecast growth in

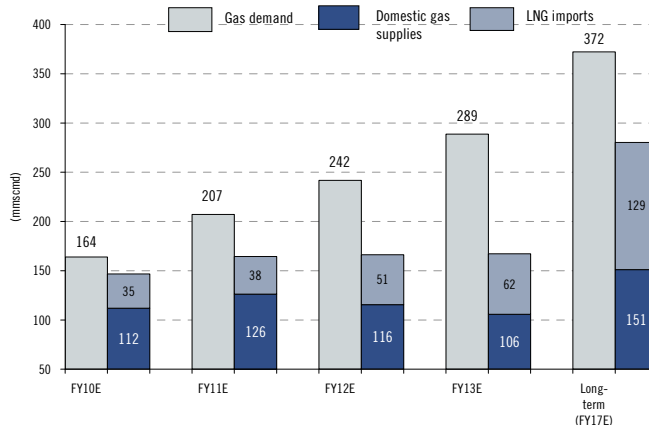
domestic gas which has witnessed some significant downward revision in recent times as production from RIL's KG-D6 block, the single largest domestic supply source, has plummeted sharply (see Figure 6).

Figure 4. India gas supply break-up



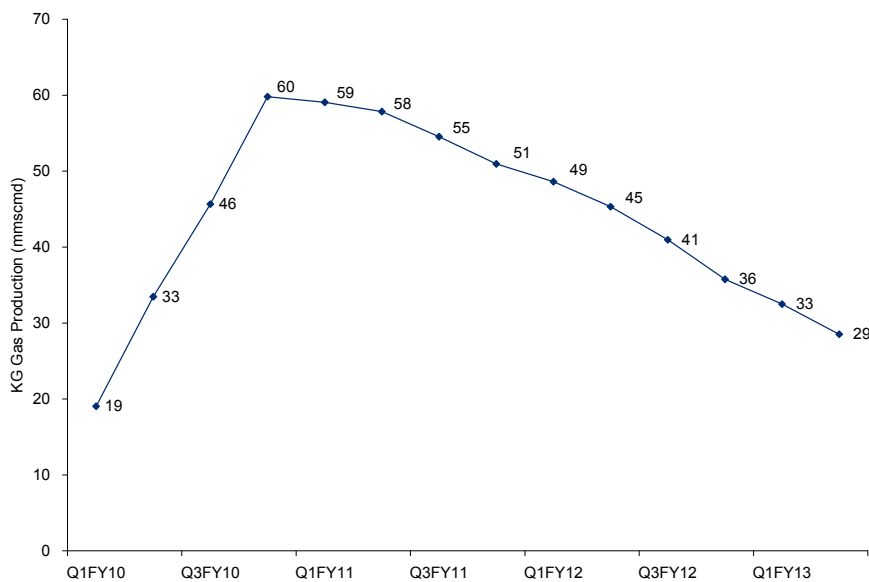
Source: Citi Research

Figure 5. Demand for gas in India will continue to outpace supply



Source: Citi Research

Figure 6. KG gas production continues to fall



Source: Citi Research

With growing energy consumption and increasing shift towards gas, this deficit in gas supply will only get exacerbated. The only alternative for meeting unmet demand, therefore, seems to be imports of LNG since India is not currently fed by any trans-national pipelines. LNG's share in the gas mix is likely to rise from ~30% now to ~45% by FY17E, in our view. Given the constraints on infrastructure, it is believed that gas supply in India will create its own demand.

Figure 7. India – Gas Demand Projections

Gas demand (mmscmd)	FY09E	FY10E	FY11E	FY12E	FY13E	LT (FY17E+)	Comments
Power	35	54	78	97	130	177	Expected demand from planned greenfield/brownfield expansions
Fertilizer	35	44	59	68	71	74	Expected demand from planned greenfield/brownfield expansions as well as switch-overs from naphtha/fuel oil to gas
City Gas	10	12	15	16	20	30	Assuming 50 new cities to be auctioned by PNGRB in the next 5-6 years; 0.3 mmscmd gas required per new city
Industrial	12	18	18	20	21	42	Current demand trends and expected users along new pipelines
Petrochem/Refineries/Internal Consumption/Steel	23	36	38	41	47	50	Expected demand from capacity expansions of refineries and petchem plants having pipeline connectivities
Total demand	115	164	207	242	289	372	

Source: Citi Research

Figure 8. India – Gas Supply Projections

Gas supply (mmscmd)	FY09	FY10	FY11E	FY12E	FY13E	FY14E	LT (FY17E+)	Comments
ONGC	49	50	49	50	52	55	70	Addition from new fields under development and KG-DWN-98/2
RIL (KG/NEC)	-	39	55	43	30	25	40	Incremental gas from satellite fields
Other CBM	0	-	-	1	2	3	5	CBM gas from GEECL, Essar Oil, and others
GSPC	-	-	-	-	-	3	10	As per current FDP for Deen-Dayal + Production from Cambay
PMT	16	14	11	11	11	11	11	Assumed to remain largely flat
Others	9	9	10	12	11	11	15	Supplies from Ravva, other JV fields to largely remain constant
Total domestic supply - (1)	74	112	126	116	106	108	151	Domestic supplies only (excl. LNG imports)
LNG - firm contracts - (2)	19	23	30	33	34	34	45	Taking into account only firm LNG contracts by Petronet
LNG - spot (3)	12	12	8	17	28	35	84	Estimated spot LNG imports at Dahej and Kochi (Petronet), Hazira (Shell), Dabhol (GAIL), Mundra, Pipavav
Total domestic supply + LNG - (1+2+3)	104	147	164	166	167	177	280	

Source: Citi Research

- **Large investments in LNG capacity are planned** – To take advantage of the forecast surge in LNG demand, a number of firms have planned LNG import terminals in India and existing players are looking to expand existing facilities, potentially increasing total LNG regassification in the country ~4x over the next 4-5 years. A number of players are also likely to set up FSRUs, which are cheaper offshore alternatives to on-land LNG terminals and take less time to construct. Increasing availability of LNG combined with improved infrastructure to transport it will bring hitherto unconnected areas in India into the gas-fold, driving increasing acceptance of LNG as a fuel.

Figure 9. LNG terminal capacity expansions in India

LNG capacity (MMTPA)	FY12	FY13	FY14	FY15	FY16	FY17	Comments
PLNG - Dahej	10.0	10.0	12.5	12.5	17.5	17.5	First stage from 10 to 12.5 MMTPA with second jetty and then to 15 MMTPA with regas facilities (actual volumes could, however, go up to ~17-18 MMTPA in the long-term)
Shell - Hazira	3.6	3.6	5.0	5.0	5.0	5.0	Nameplate capacity to be extended to 5 MMTPA by 2013E
GAIL - Dabhol		1.5	1.5	5.0	5.0	5.0	Commissioning delayed to Dec-12, fully operational only after completion of breakwater in FY14
PLNG - Kochi		5.0	5.0	5.0	5.0	5.0	Expected commissioning by Mar-13
GAIL - FSRU in AP				3.5	3.5	3.5	Targeting commissioning by early 2014E
Swan Energy FSRU				3.0	3.0	3.0	Expected commissioning by 2014E, expected financial closure by end-CY12E, Rs25bn project, D/E 3:1
GSPC/Adani - Mundra					5.0	5.0	Expected commissioning in FY16E, EPC contracts to be awarded in 6-12 months
PLNG - Gangavaram						5.0	Expected to be operation by FY17E, likely to reach financial closure in 6-8 months, PLNG considering FSRU in interim
IOC - Ennore						5.0	Expected commissioning in FY17E, environmental and FEED studies undertaken
Total Capacity (MMTPA)	13.6	20.1	24.0	34.0	44.0	54.0	

Source: Citi Research, Company Reports

- **Improving distribution infrastructure** – Demand for gas will increase with the synergetic development of pipeline and consumer-side infrastructure (CGD systems etc.) along with LNG import terminals. This is something that gas aggregators/marketers such as GAIL and GSPC realize and given the paucity of domestic gas, they have been building pipelines connecting LNG terminals in order to ensure good utilisation for their pipelines. India's total pipeline network length is expected to increase from ~11,000 kms currently to >25,000 kms in the next five years.

Risks

- **Regulation:** While gas pipeline transmission and distribution tariffs are already regulated (12% and 14% post-tax project IRRs, respectively), some proposals for further regulation in the gas space have been suggested which are currently under discussion. Following a directive by the gov't, the PNGRB (downstream gas regulator) had asked gas marketing firms to provide details on the marketing margins that they charge. The regulation of marketing margins may be more justifiable for players like GAIL and RIL (who market domestic gas which does not involve any pricing risk and where the allocation of gas is already done by the gov't). However, we believe it is unlikely to impact marketers of LNG which is procured and sold at market-determined prices. Further, the Delhi High Court's decision in the IGL vs. PNGRB case (currently sub-judice in the Supreme Court) substantiated that the PNGRB does not have jurisdiction over final selling prices of gas – fixing marketing margins would be tantamount to fixing the final selling price of gas. In any case, for the PNGRB to be able to regulate gas, it first needs to be a notified product under the PNGRB Act, a process that would require an amendment to the Act itself.
- **LNG pricing:** One of the largest risks for LNG as a fuel for the Indian market is the highly price sensitive nature of end-use industries, esp. power and fertilizer. At current long-term LNG prices of ~US\$14/mmbtu, cost of generation for power can be as high as ~Rs10/kWh, which would be too high for several of the distribution entities (the cash starved state electricity boards) to purchase unless there is meaningful progress in power pricing reforms. LNG can therefore only be viable for peaking power requirements rather than base load power generation. Similarly, given urea fertilizers in India are subsidised by the gov't, production costs would be uneconomically high if based on expensive LNG, directly impacting the gov't's subsidy burden. These concerns are partly allayed by the gov't's domestic gas utilization and allocation policy which accords fertilizer and power sectors the highest priority for (cheaper) domestic gas.
- **LNG terminal regulation** :Though LNG terminals have thus far not come under regulatory oversight (justifiably so, since the gov't decides neither the pricing nor the allocation of LNG, leaving LNG importers to compete in a relatively free market), there have recently been talks about the PNGRB moving to regulate some aspects of the LNG regassification business.
 - LNG terminal authorisation: The gov't could put in place a set of eligibility criteria for a player (minimum capacity, physical specs, network, etc.) to set up an LNG terminal, which would then have to be authorized and registered by the PNGRB. While incumbents such as PLNG may benefit here at the expense of new terminals, this could have negative implications for longer term regulation of regas margins for these entities.

- Open access requirements: LNG terminals could be asked to set aside a sizeable (~1/3rd) of their capacity for use by third parties at 'fair and competitive terms'. The gov't could also determine the charges to be paid by third-parties for this access or leave this process to market forces.

Top pick

Petronet LNG (PLNG.BO; Rs160.25; 1)– The demand for gas, esp. in LNG form, is expected to remain robust in India over the next decade. While LNG import and regassification is a high fixed-cost business like most infrastructure business models, once the investment phase has been completed and capacity utilisations increase, margins and cash flows will improve markedly. PLNG is a unique play on the LNG opportunity in the country (capacity expected to increase from 10 MMTPA to 27 MMTPA over the next five years), and while it is currently in the investment phase, it has a distinct first-mover advantage, is already reaping the benefits of past investments, and is likely to continue reporting gains from operating leverage in the years to come. We remain long-term buyers of the stock, though the stock's recent run (up 30% in six months) admittedly caps near-term upside, with near-term earnings growth being muted till the upcoming Kochi expansion ramps up. The long-term thesis on PLNG, however, remains intact, buoyed by our outlook for LNG and coupled with our view of limited regulatory risks for the company.

Indonesia Utility

From One Regulatory Uncertainty to Another

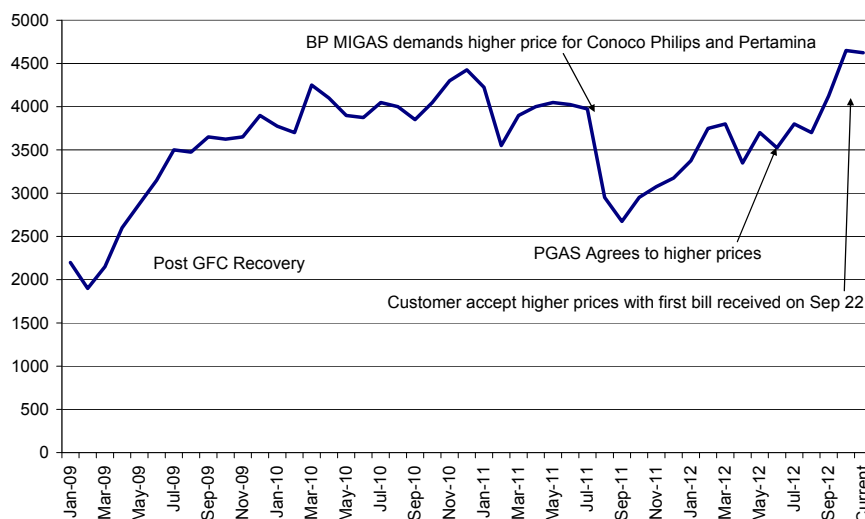
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The Indonesia Energy sector has been facing a stream of uncertainties as policy makers try to balance domestic energy demand with exports needed to balance Current Account. PGAS (PGAS.JK; Rp4,625; 3), the sole utility stock, seems to have overcome these uncertainties with a share price rise of 46% (YTD) and 31% (since June). We maintain a Sell rating on the stock based on 1) Volume uncertainty due to disbanding of the Upstream Regulator and 2) Valuation at +1std above mean (both forward PER and EV/EBTIDA).

Figure 1. Share Price Trend (Rp) and Events



Source: Bloomberg and Citi Research

BP MIGAS Abolition

The Constitutional Court of Indonesia abolished Upstream Regulator, BP MIGAS, on 13 November 2012. The decision was taken because the Court believes that BP MIGAS, instead of running natural resources for the people's welfare, was handing them over to state-owned or private companies, through production-sharing contracts.

These developments are in our opinion likely to raise concerns among foreign Exploration and Production Companies. While there are both negatives and positives for PGAS, we believe the uncertainty is negative for the risk/return profile of the sector, particularly with valuations already at +1 std.

Government ministers have reassured that the decision will not impact the current agreements. Both the staff and functions of BP MIGAS have been transferred to the Ministry of Energy.

However, both ex officials of BP MIGAS and Indonesia Petroleum Association have raised concerns over the near- and long-term impact. They even mention that this may also hinder liquefied natural gas (LNG) ship shipments from Indonesia. Quoting Jakarta Post, there are over 29 concessions slated to expire between 2013 and 2021 such as Mahakam Block in East Kalimantan.

From a PGAS perspective,

- Negatives include 1) New E&P activity will slow down, impacting additional future gas. International LNG exporters may be more cautious when dealing with Indonesian importers.
- Potential positive include: Existing E&P companies may want to maximize their production in near future, due to uncertainty of further changes. Some may opt to sell their Indonesian assets at lower valuations, with PGAS as a potential acquirer.

Tariff Issue almost resolved

The tariff issue seems to have been resolved. The decision by BP MIGAS to raise selling prices of main suppliers by almost 200% had raised concerns about Distribution Spread. Initial attempts to raise customer tariff had been stiffly resisted by customers. However, the involvement of Minister of Energy seems to have convinced the customers in West Java, including PLN, to accept a 50% increase in two phases. The first phase of 35% has been implemented since Sep 2012. The remainder will be from April 2013. These increases will keep Distribution Spreads at USD4.2/btu. The table below gives the timetable of increases by key suppliers.

Figure 2. Tariff Revision as per agreement with BP Migas

Producer	Buyer	Volume btu	Current Tariff USD	Revised Tariff USD
Pertamina	PGN	250	2.23	5.5 (2012), 6 (2013), +4% pa till 2017
ConocoPhillips (Grissik)	PGN	412	1.85	5.61 (till March 2013), 6 (till March 2014), 6.5, +3% pa till 2023
ConocoPhillips Indonesia	Petronas	263	3.10	6, +2.5%pa
PHE WMO	PLN	123	3.00	6.05, +3% pa
Medco E&P Indonesia	PLN Indralaya	19-24	2.68	4.3-5.3
Pertamina EP	Pusri	166	3.64	5.89, +3% pa
JOB Pertamina Talisman	Pembangunan	10	5.40	5.75, +3% pa

Source: Bisnis Indonesia

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Malaysia Power Sector

1H13 Outlook: Tariff Hike Unlikely, Gas Supply to Improve

- **Buy Tenaga** — Key events to look out for in 1H13 are stability in the resumption of gas supply, especially with the commissioning of the Melaka re-gasification plant (scheduled for 1Q2013) and details on the Prai Power Plant and PPA Extensions. We do not expect a tariff hike in the next twelve months. We have a Buy rating on Tenaga as we believe Tenaga's outlook will continue to improve with the imminent increase in gas supply and assurance of continued government support.
- **Demand Outlook** — Tenaga's management expects demand growth of +4–5% YoY in FY13E, vs 4.3% FY12A and Citi Research est: +5.2%. We expect consumption to continue to grow in 1H13, mainly on the back of ETP-related projects. We expect GDP growth to average 5.2% both in 2012E and 2013E.
- **Supply Outlook** — No major new additions to power generation supply are due in 2012–13. The last addition to the national grid was Phase 2 of the 700-MW coal-fired Jimah Power Plant in July 2009. Meanwhile, no brownout is expected, with reserve margin estimated at 34% for 2013 vs. 38% in 2012.
- **Generation Cost Outlook** — We believe generation cost will trend down in 1H13 from 2H12 on the back of improved gas supply and the entry of gas supply from the Melaka re-gasification plant (which will bring in 500 mmscfd of gas supply, of which 200 mmscfd will go to the power sector). Tenaga's management expects coal price to be less than at US\$100/mt for FY13E, vs our assumption of US\$105/mt for FY13E (FY12A: US\$103.6/mt).
- **Tariff Hike Unlikely** — Tenaga's management has expressed its view that the likelihood of a tariff hike in the next twelve months is low, particularly with an upcoming general election. However, management is also optimistic that new supplies from Petronas' LNG re-gasification plant would have neutral impact on Tenaga with the government either giving the nod to tariff adjustments or paying compensation should Petronas charge market price for the gas.

Figure 1. Malaysia Power Sector

Company	RIC Rating	Price (RM) 28 Nov12	TP (RM)	Mkt Cap (US\$m)	ADTV (US\$m)	PE (x) 2013	PB (x) 2013	ROE (%) 2013	Yield (%) 2013
Tenaga Nasional	TENA.KL Buy	6.93	8.75	12,567	13.5	9.8	0.9	10.1	3.0
YTL Power	YTLP.KL NR	1.52	NA	3,660	1.2	9.3	1.1	0.1	3.3
Weighted avg						9.7	1.0	7.8	3.1

Source: Citi Research estimates for Rated companies and IBES estimates for Non-rated Companies

1H13 Outlook

Key events to look out for in 1H13 are stability in the resumption of gas supply, especially with the commissioning of the Melaka re-gasification plant (scheduled for 1Q2013), and details on the Prai Power Plant and PPA Extensions. We do not expect a tariff hike in the next twelve months. We have a Buy recommendation on Tenaga as we believe Tenaga's outlook will continue to improve with the imminent increase in gas supply and assurance of continued government support.

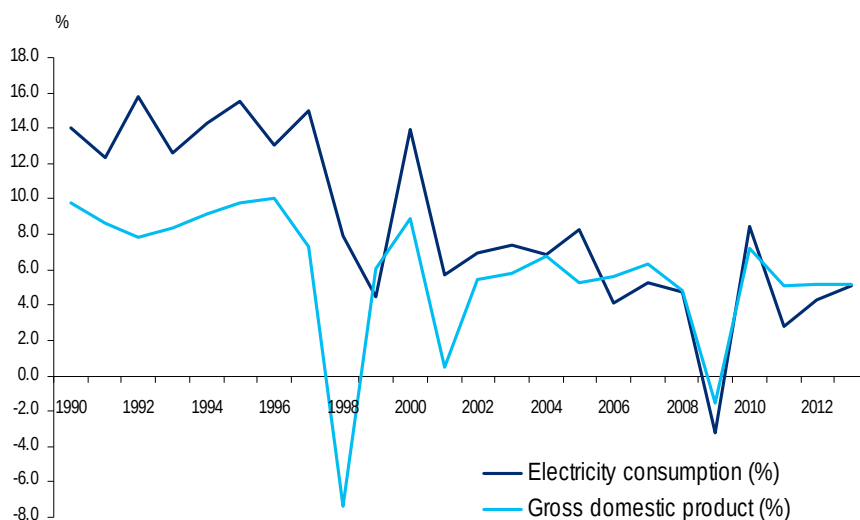
Demand Outlook

Tenaga's management expects demand growth of +4–5% YoY in FY13E, vs 4.3% FY12A and Citi Research est: +5.2%, in line with Citi Economics GDP growth estimates. We expect consumption to continue to grow in 1H13, mainly on the back of growing demand from the cement and steel industries as it sees growing demand for its products from projects related to the government's Economic Transformation Program (ETP).

The key ETP project is the RM25bn Klang Valley MRT SBK Line project which has seen accelerated progress, with an estimated 90% of total contract value already awarded during the period January–October 2012. Total construction contracts awarded YTD to listed companies has grown from RM8.9bn in 10M2011 to RM32.4bn in 10M2012.

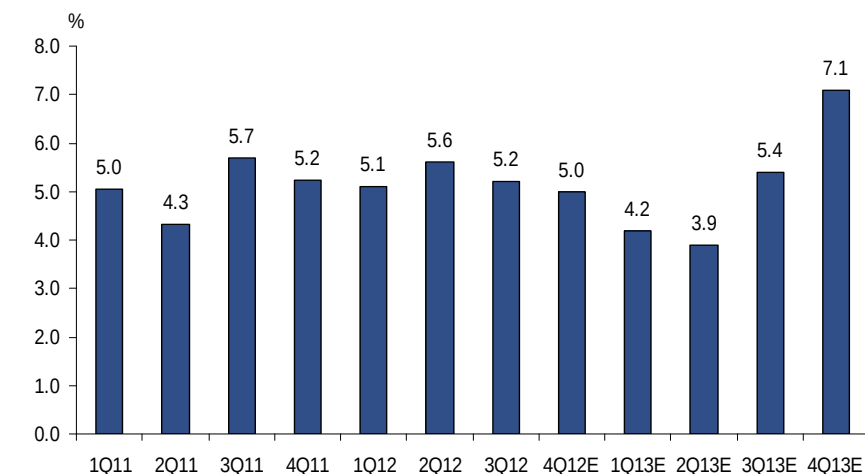
We expect GDP growth to average 5.2% both in 2012E and 2013E. Malaysia recorded a 3Q12A GDP growth of 5.2% YoY, led by the domestic demand growth with a 11.4% YoY growth. The construction sector grew 18.3% in 3Q12A (2Q:22.2%), in tandem with the growth we have seen in construction contract awards.

Figure 2. Electricity Consumption and Gross Domestic Product Growth



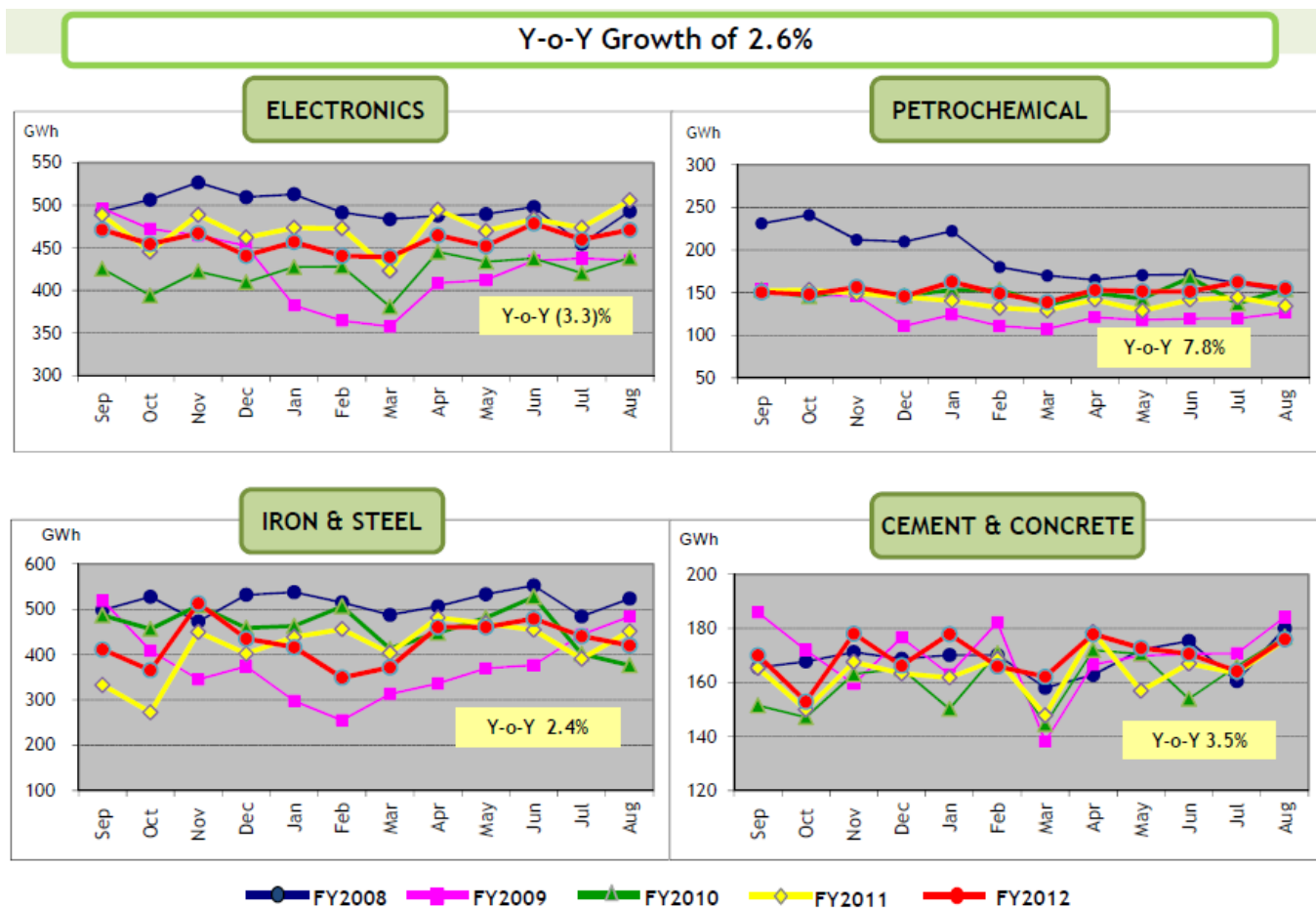
Source: Bank Negara Malaysia (BNM), Company Reports, and Citi Research Estimates

Figure 3. Malaysia GDP Growth (Quarterly Forecast)



Source: BNM, Citi Research estimates

Figure 4. Analysis of Demand from the Industrial Sector



* Source : Top 1,000 PRIME customers database (PRIME customer YTD unit sales equivalent to 35.1% of total YTD unit sales)
* PRIME customers for 4 sectors above attributes 42.8% from the whole PRIME customers YTD unit sales

Source: Company, Citi Research

Supply Outlook

No major new additions to power generation supply are due in 2012–13. The last addition to the national grid was Phase 2 of the 700-MW coal-fired Jimah Power Plant in July 2009. New supply is expected only from 2015 onwards with two hydro power plants and two coal-fired units. The hydro power plants are Hulu Terengganu (250MW) and Ulu Jelai (372MW) hydroelectric dams, both to be built by Tenaga Nasional. Meanwhile, no brownout is expected, with reserve margin estimated at 34% for 2013 vs. 38% in 2012.

Tenaga is also building a 1,000MW coal-fired plant as an extension to its 2,100MW Janamanjung power plant and was recently selected by the Energy Commission (EC) as the preferred builder for the 1,071 MW Prai plant. Another 1,000MW coal-fired extension plant is also being built at Tanjung Bin (owned by listed MMC). The construction contracts for these two plants were awarded in June 2011 and February 2012 respectively.

Prai Plant Award

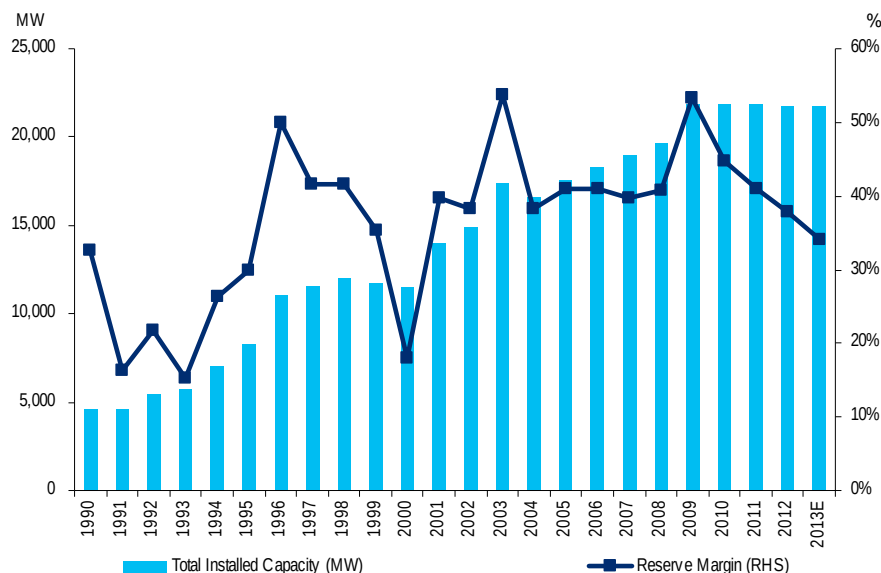
On 9 October, Tenaga received a letter from the EC confirming that it has been selected as the preferred bidder to build, own and operate a 1,071 MW combined cycle gas turbine power plant in Prai, Pulau Pinang with the levelised tariff at 34.7 sen/kWh (at the gas price of RM42.24/GJ). The plant is to be commissioned by 1 March 2016 (see Fig 8). The Letter of Award will be issued upon successful completion of negotiations, which commenced in the week of 15 October and to be completed within 21 days.

Figure 5. Details of the Project (Prai Power)

Capacity	: 2 x 535 MW
Main Fuel	: Natural Gas
Commercial Operation Date	: 1st January 2016
Project Duration	: 32 months
Technology	: Siemens' latest H-Class technology gas turbine with a supercritical, once-through Benson-type Heat Recovery Steam Generator (HRSG), with a combined-cycle efficiency of greater than 60%
Emission	: In compliance with DOE's new Clean Air Regulation

Source: Company and Citi Research

Figure 6. Peninsular Malaysia National Grid Reserve Margin



Source: Company Reports and Citi Research Estimates

Generation Cost Outlook: Trending Down

We believe generation cost will trend down in 1H13 from 2H12 on the back of improved gas supply and the entry of gas supply from the Melaka re-gasification plant (which will bring in 500 mmscfd of gas supply, of which 200 mmscfd will go to the power sector). The commissioning of gas supply from the Melaka LNG Re-gasification Plant has been pushed to 1Q2013, vs the previous guidance of September/October 2012. The daily supply of gas to the power sector averaged about 1,000mmscfd in 4QFY12, vs. the minimum pre-agreed of 1,250mmscfd at subsidized rate.

Management is optimistic that new supplies from Petronas' LNG re-gasification plant would have neutral impact on Tenaga with the govt either giving the nod to tariff adjustments or paying compensation should Petronas charge market price for the gas.

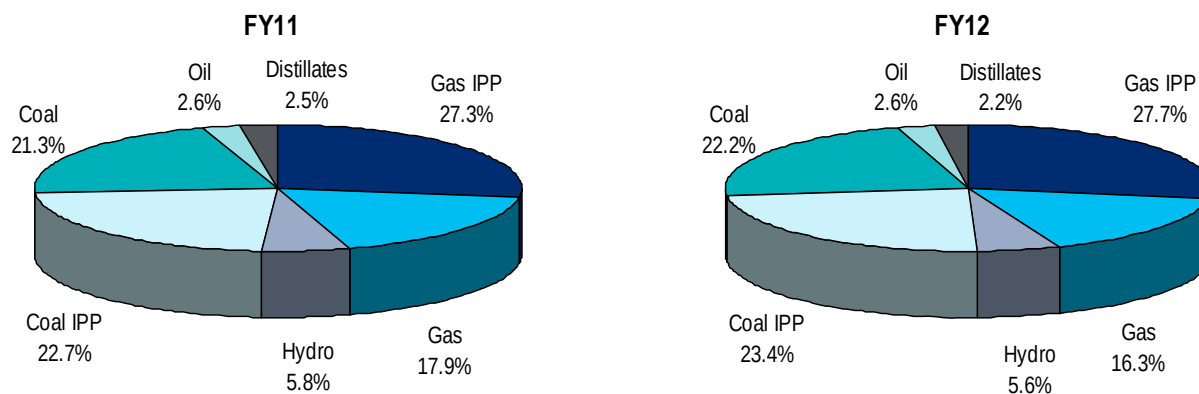
To recap, Tenaga received RM2.023bn in compensation from Petronas and the government under the fuel cost-sharing mechanism announced in early Dec 2011. The gas curtailment resulted in an estimated RM3.034bn in additional cost incurred by Tenaga due to the use of more expensive alternative fuel oil and distillates from 1 Jan 2010 to 31 Oct 2011. Generally, it cost approximately RM0.11 to generate a KWh using gas against RM0.14 and RM0.58 using coal and fuel oil, respectively.

Tenaga expects to receive a total of RM1.47bn in compensation from the government and Petronas for the RM2.21bn spent in FY12. RM0.72bn has been paid for the seven months to March 2012 while a remaining RM0.75bn for the five months to August has not yet been paid.

We do not expect gas prices to be raised ahead of the general elections, as this is typically coupled with an electricity tariff hike, which we believe is unlikely in the next twelve months.

Tenaga's management expects coal price to be less than at US\$100/mt for FY13E, vs our assumption of US\$105/mt for FY13E (FY12A: US\$103.6/mt). Coal consumption has peaked at 20.8m tonnes in FY12A given the low supply of gas. 63% of Tenaga's FY13E coal procurement of 21.3m tonnes is index-linked, 1% at fixed price and a remaining 36% is being negotiated or sourced.

Figure 7. Peninsular Malaysia Generation Mix – FY12 vs. FY11: Gas Curtailment Since 2011 Forced Industry Players To Generate Power From Fuel Oil and Distillates



Source: Company Reports and Citi Research

Figure 8. Newcastle Coal Index



Source: Global Coal, Citi Research

Regulatory Risk: Moderating

Tariff Hike: Unlikely In Next 12 Months

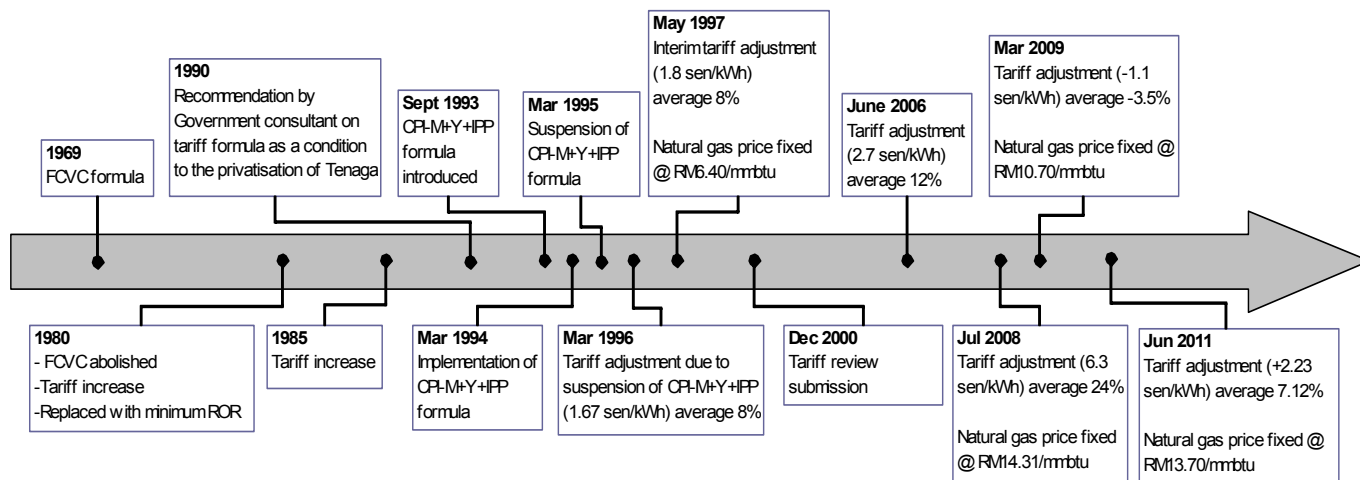
Tenaga's management has been engaging the government and Petronas to work towards a holistic fuel cost pass-through formula that would include prices of coal (now fixed at US\$85/mt) and natural gas. Based on management estimates, a 12% tariff hike would be needed if [1] regulated gas price is raised by RM3/mmbtu to RM16.70; [2] new gas supply is purchased at RM44; and coal cost is raised from US\$85/mt to market price. The last time Tenaga raised electricity tariff was in Jun 2011 by an average of 7.12%. Although management is optimistic the government would proceed with the tariff revision, management believes the rate hike will not happen before the general elections (GE).

We believe that this means that Tenaga's gas supply from the Melaka re-gasification plant will likely be at a subsidized rate from the government. Gas from the Melaka plant will be supplied at current market prices, which are ~3x higher than the current subsidized gas price of RM13.70/mmbtu.

Although the government planned to review the price of gas and tariff every six months (i.e. mid and early/end-year) since 2008, execution has not been consistent. In fact, the tariff has been adjusted three times (i.e. July 2008, Mar 2009 & June 2011) since then.

Generally, tariff adjustments are designed to be neutral (i.e. to compensate for higher gas and coal generation costs), though in the previous review, Tenaga did benefit with a small net positive impact. However, if the reviews are conducted in a more consistent and clear manner, uncertainties surrounding Tenaga's margin, especially in relation to the cost of coal, would be addressed. In the absence of a fixed tariff-setting formula, this is the next best thing Tenaga could hope for.

Figure 9. An Overview of the Tariff Setting Mechanism and History



Note: FCVC - Fuel cost variable cost; CPI - Consumer Price Index; M - Efficiency; Y - Fuel cost; IPP - IPP costs

Source: Company Reports and Citi Research

PPA Renegotiations Concluded, Awaiting Details

The EC has offered Genting Sanyen Power and Segari Energy Ventures 10-year extensions to operate their existing plants while TNB Pasir Gudang has been given a 5-year extension (see Fig 11). These successful bidders will be required to sign a new Power Purchase Agreement (PPA) with terms that have been standardized by the EC.

Tenaga's management believes that any savings from lower capacity payments as a result of the PPA extensions will likely be neutral for Tenaga as the savings will be passed back. However, details remain scant as the government has yet to reveal its plans.

Figure 10. Genting Sanyen Power, Segari Energy Ventures and TNB Pasir Gudang Extension Period and levelised tariff

Power Plants	Capacity, levelised tariff and period
Genting Sanyen Power	: 675 MW; levelised tariff 35.3 sen/kWh for 10 years
Segari Energy Ventures	: 1,303 MW; levelised tariff 36.3 sen/kWh for 10 years
TNB Pasir Gudang	: 275 MW; levelised tariff 37.4 sen/kWh for 5 years

Note: The levelised tariff is derived using projected 2016 market gas price of RM42.24/GJ

Source: Suruhanjaya Tenaga and Citi Research

Figure 11. First-Generation IPPs & Expiring PPAs

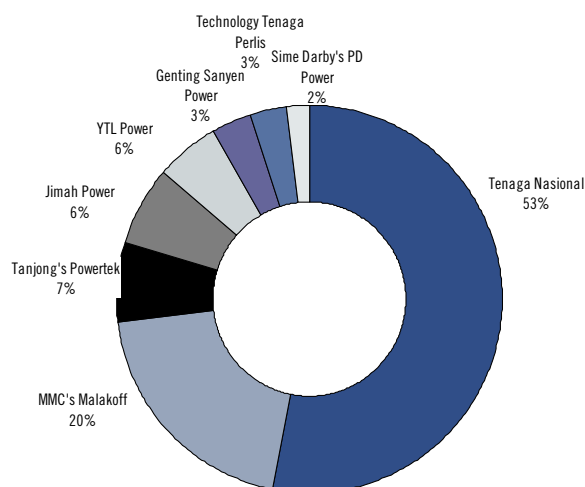
IPPs	Year Initial	Capacity (MW)	Fuel	Type	PPA	PPA Expiry
YTL Power (YTLPG)	1996	1,212	Gas	CCGT	Take or Pay	2015
Segari Energy Ventures (SEV)	1997	1,303	Gas	CCGT	Cap & Energy	2017*
Genting Sanyen Power (GSP)	1996	720	Gas	CCGT	Cap & Energy	2015*
Powertek	1996	440	Gas	OCGT	Cap & Energy	2016
Port Dickson Power (PDP)	1996	440	Gas	OCGT	Cap & Energy	2016

* SEV and GSP's PPAs have been extended by 10 years, subject to new terms by the EC

Source: Company and Citi Research

Figure 12. Key Players In Peninsular Malaysia Power Sector

Company	2012 % of total	2012 % of IPP	2012 MW
Tenaga Nasional	53	n/a	11,586
MMC's Malakoff	20	43	4,393
Tanjong's Powertek	7	14	1,490
Jimah Power	6	14	1,400
YTL Power	6	12	1,212
Genting Sanyen Power	3	7	704
Technology Tenaga Perlis	3	6	650
Sime Darby's PD Power	2	4	440
Total	100	100	21,875



Source: Company Reports and Citi Research Estimates

M&A: Nothing Major Expected

No major M&A deals were struck in 2012. Generally, power players are open to opportunities, but we believe a substantial deal is hard to come by and competition is intense. Tenaga has expressed an interest in looking into power assets overseas (i.e. the Middle-East and Egypt were mentioned). We believe its interests in overseas projects will likely to be limited to investments and joint-ventures (i.e. non-controlling stakes).

Tenaga Nasional

(TENA.KL; RM6.94; 1)

Valuation

We have a RM8.75 target price for Tenaga, valuing the stock at a 10% discount to our DCF valuation of RM9.72 calculated using a weighted average cost of capital of about 9.6% from a 12% equity discount rate (4.5% risk-free, 7.5% equity risk premium and beta of 1.0), a 5.7% cost of debt and 3% terminal growth rate. We consider DCF appropriate to value utilities such as Tenaga, as they have high fixed-cost bases, and given that the relationship between variable costs (particularly fuel) and revenue is reasonably stable in the long run. The DCF value would, of course, vary with the discount rate used. This valuation is also sensitive to differing long-term demand growth rate and tariff assumptions.

Risks

Our earnings forecasts and DCF valuation rely on electricity demand growth (which is influenced by GDP), electricity tariff and fuel costs assumptions. Hence, downside risks to our target price include: 1) lower-than-expected circular electricity consumption growth; 2) the government not paying compensation for gas curtailment; 3) higher-than-expected fuel costs, especially coal costs; and 4) weaker ringgit against US dollar and Japanese yen that would raise cost of imports including fuel and plant and equipment.

Philippine Utilities

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We expect overall sector earnings to be flattish YoY in 2013E. While we expect EDC to grow 25% YoY in 2013E driven by the incremental earnings contribution of a new plant, we expect overall sector growth to suffer a drag from weakness in Aboitiz Power on higher steam supply cost and slowing sales of ancillary services.

We remain positive on the medium- to long-term prospects of the industry since in our view the changing dynamics – tightening power supply, high cost of traditional fossil fuel, increasing preference for renewable energy – present attractive opportunities for growth and value accretion.

With the Philippines facing the risk of an impending power supply shortage, there is a need to develop new capacity in the next 2-3 years:

- This presents Philippine utility companies with an opportunity to expand their asset portfolios and create new sources of growth.
- Supply constraints could also lead to higher selling prices of power, providing a boost to revenues and margins of generation companies.

Impending power supply shortage

Figure 1. Capacity requirements for 2010-2030 per region

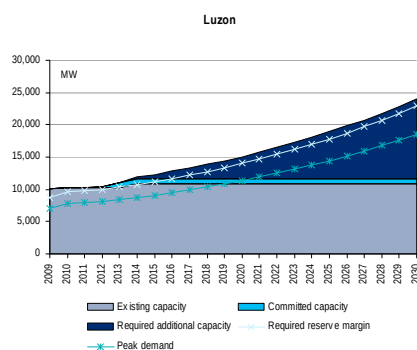
MW	Luzon	Visayas	Mindanao
New capacity needed	11,900	2,150	2,500
Committed	600	654	100
Indicative	3,448	186	581

Source: Department of Energy

Need for new capacity

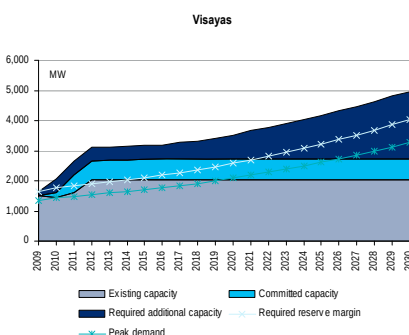
Current indications from the government point to adequate supply of power in the Luzon (northern region, 65% of country's GDP) and Visayas (central region, 17% of country's GDP) with reserve margins currently at 27% and 33%, respectively. However, it expects supply to hit critical levels in 2014 for Luzon and 2017 for Visayas assuming an average peak demand growth of 4.4% p.a. The Mindanao (southern region), on the other hand, is already facing tight supply with current reserve margin of 10% (experiencing 2-4 hours of rotating blackouts).

Figure 2. Supply and demand in Luzon



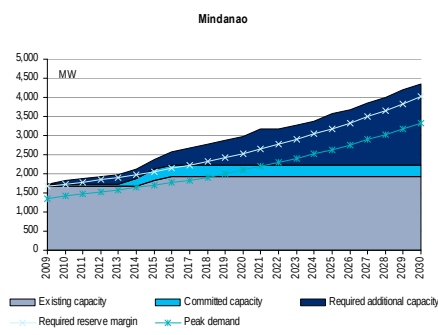
Source: Department of Energy

Figure 3. Supply and demand in Visayas



Source: Department of Energy

Figure 4. Supply and demand in Mindanao



Source: Department of Energy

In anticipation of the expected shortfall in power supply in the next 3-5 years, an increasing number of companies have indicated plans of building new power plants. However, there remain several stumbling blocks that could lengthen the process the development of these projects:

- Obtaining necessary permits and environmental clearances from both the local and national governments,

- Negotiating with the National Grid Corporation of the Philippines (holds the franchise to the transmission grid) to connect their power plants to the transmission grid (which also requires regulatory approval),
- Signing fuel supply contracts and purchase power agreements, and
- Securing financing

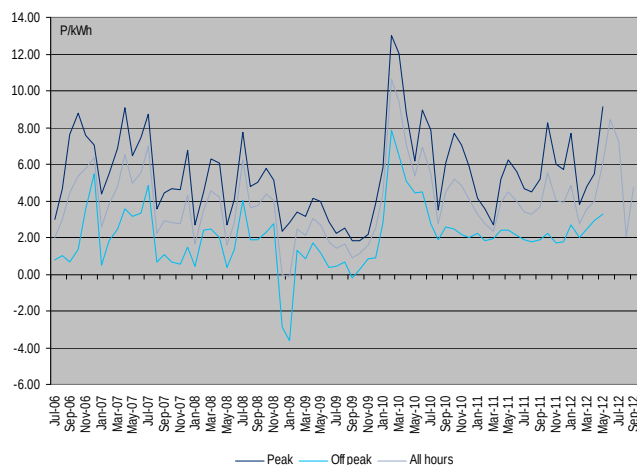
Potential upward pressure on spot prices

Generation plants (located in the Luzon and Visayas grids) are able to sell non-contracted capacity in the Wholesale electricity Spot Market (WESM). Spot market transactions, on average, account for about 15% of total energy transactions.

Spot prices went up 40% YoY in 9M12 on the back of robust electricity demand (high single-digit vs 4% historical average) from residential, commercial and industrial customers. While positive economic indicators point healthy electricity demand in 2013, a sharp increase in spot market prices may be mitigated by an increase in power supply with the commissioning of a 600 MW coal-fired plant (by GN Power) in 1Q13.

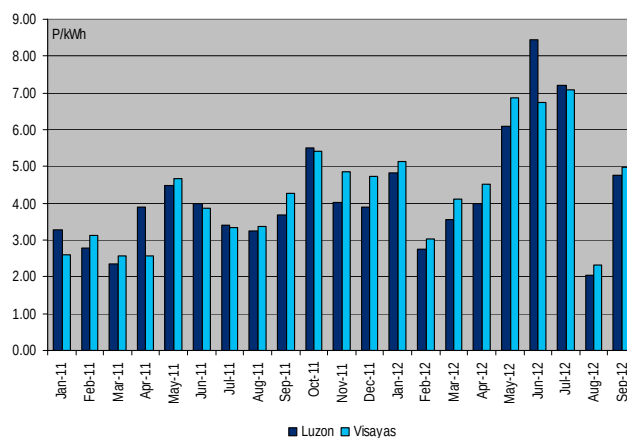
In the medium term, there could be upward pressure in spot prices as supply becomes tighter, especially if economic growth and infrastructure developments accelerate.

Figure 5. Spot market prices (Luzon grid)



Source: Wholesale Electricity Spot Market

Figure 6. All-hour average spot prices – Luzon and Visayas grids



Source: Wholesale Electricity Spot Market

Some progress on government initiatives

1) Approval of the FIT

On 27 July 2012, the Energy Regulatory Commission (ERC) approved the initial Feed-in Tariffs (FIT). The FIT rates are guaranteed payments over a 20-year period given to developers to cover production of energy from renewable sources, particularly run-of-river hydro, biomass, wind and solar. FIT mechanism, however, still yet to be released by the ERC.

- Approved FIT rates are lower than original proposal – The FIT rates are:
P5.90/kWh for run-of-river hydro, P6.63/kWh for biomass, P8.53/kWh for wind,

and P9.68/kWh for solar (FIT for ocean resources was deferred pending further study). To compare, generation tariffs of recently signed power supply agreements for coal-fired plants are P4.50-5.00/kWh. The approved rates are about 18% lower than the rates originally proposed by NREB as the ERC updated construction costs to reflect downtrend in costs and adopted higher capacity factors.

- Guaranteed return of 16.4% – The FIT rates are based on an equity IRR of 16.44% (17% for biomass to account for fuel risk), which we think is decent and largely in line with the target IRRs for ongoing generation projects. Electric distribution utilities (e.g. Meralco) have a regulated return of around 15%.
- Accelerating RE developments – The FIT is a critical component to jumpstart RE projects, especially amid an increasing need to build new capacity and demand for clean energy. The FIT will assure investors of future cash flows and adequate returns for their projects. This will also enable RE developers to gain access to funding from financial institutions, given predictable and guaranteed cash flows.

2) Privatization of remaining government-owned assets and contracts

While much of the future growth of power generation companies will likely come from greenfield and expansion projects, several assets remain and IPP administrator contracts are up for privatization, which the leading industry players are keen to acquire. Privatization of these, however, has been delayed.

The government has indicated a resumption of privatization efforts, starting with the bidding out of the Unified Leyte IPPA contract (which will be broken down into several contracts) later this year. However, with elections coming up next year (i.e. May 2013), a break in the roll-out of the privatization program could be likely.

Figure 7. Government assets and contracted capacities for privatization

Project	Type	Capacity (MW)	Location	Interest parties	Additional info
IPP Administrator (IPPA) contract					
Naga	diesel	149.0	Cebu, Visayas	San Miguel Corp, SPC Power Corp	plant is owned by SPC Power Corp
Unified Leyte	geothermal	588.0	Leyte, Visayas	San Miguel Corp, EDC/First Gen, Aboitiz Power, Ayala Corp	plant is owned by Energy Development Corp
Caliraya-Botocan-Kalayaan I&II (CBK)	hydro	728.0	Laguna, Luzon	San Miguel Corp, Aboitiz Power	plant is owned by J Power; pumped storage facility
Casacnan	hydro	140.0	Nueva Ecija, Luzon	San Miguel Corp, EDC/First Gen	plant is owned by the National Irrigation Administration
Western Mindanao Power Corp	oil	100.0	Zamboanga del Sur, Mindanao		plant is owned by Aboitiz Power
Southern Philippines Power Corp	oil	55.0	General Santos, Mindanao		plant is owned by Aboitiz Power
Mindanao Coal	coal	200.0	Misamis Oriental, Mindanao		plant is owned by STEAG State Power (34%-owned by Aboitiz Power)
Mt. Apo 1 & 2	geothermal	92.5	Kidapawan, Mindanao		plant is owned by Marubeni Philippines
Asset					
Agus-Pulangui	hydro	982.1	Lanao del Sur-Bukidnon, Mindanao	Aboitiz Power, EDC/First Gen	provides more than half of the electricity requirements in Mindanao
Sucat	oil	850.0	Parañaque, Luzon		decommissioned
Power barges 101-104	oil	128.0	Visayas		
Malaya	oil	650.0	Rizal, Luzon	San Miguel Corp	

Source: PSALM, Department of Energy, Aboitiz Power, San Miguel, Philippine Daily Inquirer

Among the assets/contracts generating the most interest from private investors are the IPP administrator contracts for Unified Leyte, CBK and Casecnan, and the Agus-Pulangui hydro complex.

Successful acquisition of these assets and contracts could serve as a catalyst for Philippine utility names, as this could provide a new source of significant growth in a relatively shorter period of time (vs greenfield projects) and could potentially enhance value. Timing, however, remains uncertain at this point.

3) Implementation of Open Access

On 24 September 2012, the ERC has resolved to set the initial implementation of Retail Competition and Open Access (RCOA) on 26 December 2012. The first six months will be considered a transition period with full implementation (with IT systems in place) targeted for December 2013. The RCOA is a scheme that will allow large power users (with average monthly peak demand of 1MW) the right to choose their own electricity suppliers.

Stock picks

- **Top pick - Energy Development Corp (EDC.PS, TP of P7.65, Buy).** EDC is the largest geothermal producer in the Philippines and the third-largest in the world. It produces low-cost energy, which is readily dispatched. We think key catalysts are materializing (i.e. delivering on earnings, commissioning of BacMan by year-end) and that the stock is attractively valued – 2013E PE of 14x on 2012-14E EPS CAGR of 20% (PEG of 0.7x).
- **Aboitiz Power (AP.PS, TP of P36.00, Buy).** AP has an attractive and diversified portfolio of assets and is well-positioned to benefit from asset reflation and growth opportunities arising from an impending power supply shortage in the country in the medium to long term. It continues to expand its asset portfolio via greenfield projects and is actively looking for acquisition opportunities. Key risk, however, is potential earnings weakness in the near term on the back of higher steam supply cost (under the current steam supply contract for the 467 MW Tiwi-MakBan plant, steam cost will be pegged to coal starting May 2013; AP is in discussions with steam supplier Chevron to revise contract terms) and slowing sale of ancillary services amidst increasing competition (note that ancillary revenues have been a key growth driver in 2012).
- **Meralco (MER.PS, TP of P235.00, Sell).** We have a Sell rating on Meralco largely due to its rich valuations and limited earnings growth prospects in 2012-2014 as distribution tariffs, a major driver of earnings, are likely to remain flat. The stock is currently trading at a 2013 PE of 18.0x vs Philippine utility average of 12.5x.

Companies

Adani Power (ADAN.BO)

Sell: Business Model Not Panning Out In Reality

Sell	3
Price (27 Nov 12)	Rs49.85
Target price	Rs34.00
Expected share price return	-31.8%
Expected dividend yield	0.0%
Expected total return	-31.8%
Market Cap	Rs108,675M
	US\$1,951M

Price Performance
(RIC: ADAN.BO, BB: ADANI IN)



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- **Maintain Sell** — The business model on paper has not panned out in reality. Key spoilers have been (1) slower ramp-up, (2) higher auxiliary consumption, (3) lower merchant sales, (4) higher fuel costs, (5) higher O&M costs, (6) duty on Mundra sales, (7) significant cost over-runs. The company is staring at an overleveraged balance sheet, PPAs with limited maneuverability, coal shortages and high fuel costs.
- **Takeaways from FY12 Balance Sheet** — (1) Lower profitability and cost over-runs have led to the net debt/ equity ballooning up to 5.8x end FY12 from 3.7x end FY11, (2) End FY12, 67% of the loans on the B/S were foreign currency loans which have not been hedged, (3) 4.6GW Mundra project has had cost overruns of 31%, (4) Debtors and inventory days of sales remain high at 84 and 74 days respectively, (5) Capex of Rs86.7bn, 136.8bn and Rs140.6bn in FY10, FY11 and FY12 respectively.
- **Extremely high fuel costs similar to JSW Energy** — At ~Rs2.3/kWh per unit generated, fuel costs are extremely high. At least JSW Energy is not stuck with too many PPAs with limited maneuverability and has a majority of sales in the lucrative merchant market.
- **Tariff renegotiation going on** — Adani Power's petition with Central Electricity Regulatory Commission (CERC) is being heard and tariff renegotiation is still pending.
- **Upside risks to our Sell call** — (1) Pass through of costs in low-priced PPAs, (2) significant fall in international coal prices, and (3) allocation of coal mine for Tiroda.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Mar	(RsM)	(Rs)	(%)	(x)	(x)	(%)	(%)
2011A	5,214	2.39	206.5	20.8	1.7	8.6	0.0
2012A	-924	-0.42	-117.7	nm	1.8	-1.5	0.0
2013E	991	0.45	207.2	109.7	1.8	1.7	0.0
2014E	3,406	1.56	243.9	31.9	1.7	5.6	0.0
2015E	7,951	3.65	133.4	13.7	1.5	11.9	0.0

Source: Powered by dataCentral

Beijing Enterprises Holdings (0392.HK)

Rising Gas Volumes Drive Sustainable Earnings Growth

Buy	1
Price (28 Nov 12)	HK\$50.10
Target price	HK\$57.30
from HK\$56.50	
Expected share price return	14.4%
Expected dividend yield	2.1%
Expected total return	16.4%
Market Cap	HK\$56,992M
	US\$7,354M

Price Performance (RIC: 0392.HK, BB: 392 HK)



■ **Maintain Buy** — We are positive on Beijing Enterprises (BEHL), which should benefit from gas volume growth driven by the Beijing gas-fired power plants, and could reach 17% Y/Y gas volume growth in 2012E, after accounting for the additional days for the heating season this year. Full year gas volume in 2012 may reach 7.5 billion m³. BEHL is also unaffected by gas connection fees. As for its gas transmission business, the company remains on track to see 20% Y/Y growth in 2012E. Its recent moves on water asset and toll road also show a more streamlined operation. Maintain Buy with new target price of HK\$57.30, and we also revise up our 2013/14E earnings outlook by 1%/4%.

■ **Water asset transfer** — BEHL announced in Sep it will transfer its water assets (Beijing Water Plant, BE Water (Weifang), BE Water (Hainan), BE Water (BVI)) to BE Water by new share issue, for HK\$1.26bn. BE Water issued share capital will increase by 11.2%, and BEHL's stake in BE Water will rise to 49.76%, from 44.11%.

■ **Transfer of toll road** — The company is transferring the toll road asset to BE Group, or controlling shareholder, for RMB1.114billion. BEHL will record a disposal gain of RMB150million (HK\$185million), or 5.7% of our 2012E earnings. The parent will make six payments over 2.5 years, on a half-yearly basis. We see this transfer as helping streamline its operations and removing the negative earnings impact.

■ **Growth catalysts** — We continue to see its growth driven by the natural gas volumes, which will likely benefit in 2H12 on earlier heating use, and in the next 2-3 years should be driven by higher use of gas-fired power plants in Beijing. We expect the Shannxi-Beijing pipelines, where BEHL has 40% ownership, to offer sustainable earnings contribution on ramp of the No.3 pipeline.

■ **Earnings revision** — We revise our 2013/14E EPS forecasts by 1%/4% to HK\$3.33/HK\$3.92 respectively, to factor in stronger gas volume growth.

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Statistical Abstract

Year to 31 Dec	Net Profit (HK\$M)	Diluted EPS (HK\$)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	2,639	2.27	12.6	22.1	1.7	8.0	1.4
2011A	2,776	2.34	2.9	21.4	1.5	7.7	1.4
2012E	3,236	2.73	16.6	18.4	1.4	8.4	1.7
2013E	3,950	3.33	22.1	15.1	1.3	9.6	2.1
2014E	4,659	3.92	17.9	12.8	1.2	10.5	2.5

Source: Powered by dataCentral

0392.HK: Fiscal year end 31-Dec						Price: HK\$50.10; TP: HK\$57.30; Market Cap: HK\$56,992m; Recomm: Buy					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	27,613	30,472	35,356	41,828	49,799	PE (x)	22.1	21.4	18.4	15.1	12.8
Cost of sales	-21,214	-23,738	-27,565	-32,892	-39,276	PB (x)	1.7	1.5	1.4	1.3	1.2
Gross profit	6,399	6,734	7,791	8,936	10,523	EV/EBITDA (x)	13.4	10.9	10.1	8.7	7.6
Gross Margin (%)	23.2	22.1	22.0	21.4	21.1	FCF yield (%)	8.0	-5.5	-0.4	-0.9	0.7
EBITDA (Adj)	4,122	4,949	5,714	6,725	7,706	Dividend yield (%)	1.4	1.4	1.7	2.1	2.5
EBITDA Margin (Adj) (%)	14.9	16.2	16.2	16.1	15.5	Payout ratio (%)	31	30	31	31	31
Depreciation	-1,383	-1,621	-1,827	-2,041	-2,186	ROE (%)	8.0	7.7	8.4	9.6	10.5
Amortisation	-105	-106	-112	-117	-123	Cashflow (HK\$m)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	2,633	3,222	3,776	4,567	5,397	EBITDA	4,122	4,949	5,714	6,725	7,706
EBIT Margin (Adj) (%)	9.5	10.6	10.7	10.9	10.8	Working capital	1,429	-3,725	-191	-993	-1,218
Net interest	-374	-647	-890	-1,009	-1,090	Other	1,391	-680	-808	-986	-1,166
Associates	1,365	1,674	1,961	2,446	2,703	Operating cashflow	6,942	543	4,716	4,746	5,321
Non-op/Except	171	0	0	0	0	Capex	-2,250	-3,797	-4,973	-5,279	-4,909
Pre-tax profit	3,795	4,249	4,846	6,004	7,010	Net acq/disposals	-2,199	88	-18	-19	-19
Tax	-685	-583	-706	-879	-1,054	Other	-2,839	1,632	-10	-10	-11
Extraord./Min.Int./Pref.div.	-470	-889	-904	-1,175	-1,298	Investing cashflow	-7,288	-2,077	-5,001	-5,308	-4,939
Reported net profit	2,639	2,776	3,236	3,950	4,659	Dividends paid	-1,263	-1,678	-1,853	-2,344	-2,691
Net Margin (%)	9.6	9.1	9.2	9.4	9.4	Financing cashflow	6,217	147	-498	-1,510	-2,301
Core NPAT	2,639	2,776	3,236	3,950	4,659	Net change in cash	6,226	-1,388	-783	-2,072	-1,919
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	4,692	-3,255	-257	-532	412
Reported EPS (\$)	2.27	2.34	2.73	3.33	3.92						
Core EPS (\$)	2.27	2.34	2.73	3.33	3.92						
DPS (\$)	0.70	0.70	0.84	1.04	1.23						
CFPS (\$)	5.89	0.46	3.97	4.00	4.48						
FCFPS (\$)	4.00	-2.74	-0.22	-0.45	0.35						
BVPS (\$)	30.13	33.06	35.07	37.50	40.36						
Wtd avg ord shares (m)	1,137	1,137	1,137	1,137	1,137						
Wtd avg diluted shares (m)	1,188	1,188	1,188	1,188	1,188						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	14.1	10.4	16.0	18.3	19.1						
EBIT (Adj) (%)	-5.3	22.4	17.2	20.9	18.2						
Core NPAT (%)	12.2	5.2	16.6	22.1	17.9						
Core EPS (%)	12.6	2.9	16.6	22.1	17.9						
Balance Sheet (HK\$m)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	14,447	13,655	12,566	10,864	10,256						
Accounts receivables	1,347	1,586	1,770	2,094	2,493						
Inventory	3,727	5,286	6,138	7,653	9,138						
Net fixed & other tangibles	26,300	29,697	33,217	36,177	37,702						
Goodwill & intangibles	8,516	8,702	8,702	8,702	8,702						
Financial & other assets	12,693	18,430	20,409	22,874	25,596						
Total assets	67,029	77,355	82,802	88,363	93,887						
Accounts payable	4,554	1,905	2,740	3,598	4,296						
Short-term debt	2,320	5,705	7,950	9,792	11,272						
Long-term debt	10,170	14,543	14,543	14,543	14,543						
Provisions & other liab	9,048	10,006	10,085	10,165	10,245						
Total liabilities	26,093	32,159	35,318	38,098	40,356						
Shareholders' equity	34,268	37,610	39,890	42,661	45,915						
Minority interests	6,668	7,587	7,595	7,604	7,615						
Total equity	40,936	45,197	47,484	50,265	53,531						
Net debt	-1,956	6,593	9,927	13,472	15,559						
Net debt to equity (%)	-4.8	14.6	20.9	26.8	29.1						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Policy Driven Growth in 2013

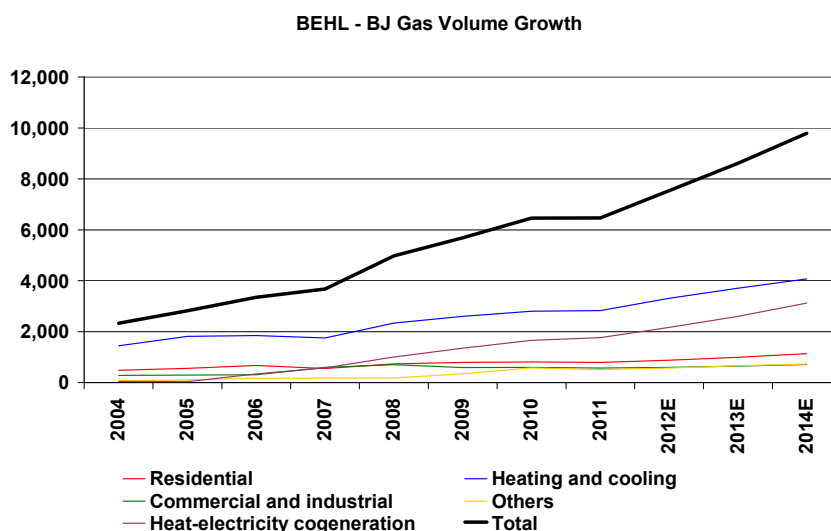
Rising Beijing Gas Volumes

We believe gas volume growth for co-gen power plant is key to drive gas volume and sales growth for BEHL, as co-gen gas usage may rise from 1.8bn m³ in 2011 to 2.2bn m³, after the ramp-up of the Huaneng co-generation plant, and will continue to be a key earnings driver for BEHL.

While the growth in brewery in 2012 has been somewhat disappointing, we do expect better earnings growth in 2013E, as the brewery should still have room to increase its pricing to pass through raw material cost increases and as the company expands its scale and product offerings.

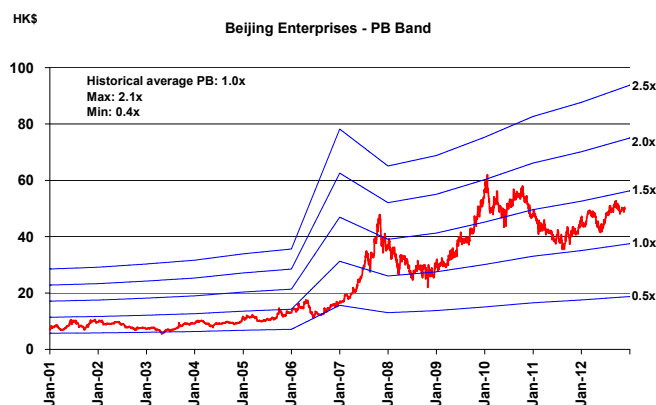
As its water and waste management business grow, we do expect increasing contribution from these two segments to complement its strong growth in the natural gas business in the coming 12-24 months.

Figure 1. Beijing Enterprises – Sustained Gas Volume Growth in 2012-14E



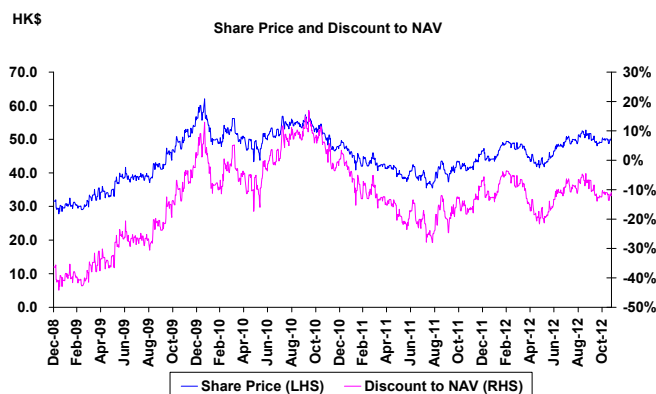
Source: Citi Research

Figure 2. Beijing Enterprises – Forward PB Band (2001-2012E)



Source: Citi Research

Figure 3. Beijing Enterprises – Discount to NAV



Source: Citi Research

Figure 4. Beijing Enterprises – Key Assumptions

Key assumptions	2009	2010	2011	2012E	2013E	2014E
Gas consumption:						
Residential	794	815	790	885	1,000	1,140
% chg	7.2%	2.7%	-3.1%	12.0%	13.0%	14.0%
Heating and cooling	2,603	2,803	2,830	3,311	3,708	4,079
% chg	11.3%	7.7%	1.0%	17.0%	12.0%	10.0%
Commercial	593	601	575	604	646	711
% chg	-15.7%	1.3%	-4.4%	5.0%	7.0%	10.0%
Industrial	347	580	510	571	657	736
% chg	90.9%	67.0%	-12.1%	12.0%	15.0%	12.0%
Heat-electricity cogeneration	1,352	1,660	1,765	2,171	2,605	3,126
% chg	33.9%	22.7%	6.3%	23.0%	20.0%	20.0%
Total	5,690	6,459	6,470	7,542	8,616	9,792
% chg	14.4%	13.5%	0.2%	16.6%	14.2%	13.6%
Gas revenue mix:						
Residential	1,627	1,671	1,620	1,865	2,170	2,548
% chg	7.2%	2.7%	-3.1%	15.1%	16.4%	17.4%
Heating and cooling	5,086	5,743	6,452	7,587	9,007	10,503
% chg	14.5%	12.9%	12.3%	17.6%	18.7%	16.6%
Commercial	1,513	1,585	1,715	1,809	2,052	2,393
% chg	-15.7%	4.8%	8.2%	5.5%	13.4%	16.6%
Industrial	816	1,448	1,463	1,647	2,007	2,383
% chg	90.9%	77.4%	1.0%	12.6%	21.9%	18.7%
Heat-electricity cogeneration	2,502	3,285	4,015	4,963	6,313	8,030
% chg	27.0%	31.3%	22.2%	23.6%	27.2%	27.2%
Total	11,544	13,733	15,264	17,871	21,550	25,856
% chg	13.7%	19.0%	11.2%	17.1%	20.6%	20.0%
Unit prices:						
ave price of households	2.05	2.05	2.05	2.11	2.17	2.24
% of chg	0.0%	0.0%	0.0%	2.8%	3.0%	3.0%
ave price of heating and cooling	1.95	2.05	2.28	2.29	2.43	2.57
% of chg	2.8%	4.9%	11.3%	0.5%	6.0%	6.0%
ave price of commercial	2.55	2.64	2.98	3.00	3.18	3.37
% of chg	0.0%	3.4%	13.1%	0.5%	6.0%	6.0%
ave price of industrial	2.35	2.50	2.87	2.88	3.06	3.24
% of chg	0.0%	6.3%	14.9%	0.5%	6.0%	6.0%
ave price of heat-elect cogen	1.85	1.98	2.27	2.29	2.42	2.57
% of chg	-5.1%	7.0%	14.9%	0.5%	6.0%	6.0%
Gas cost	1.56	1.66	1.66	1.74	1.85	1.92
% of chg	0.2%	6.4%	0.0%	5.0%	6.0%	4.0%
Brewery operation						
Average selling price for beer	1.93	1.94	2.19	2.30	2.41	2.53
% chg in ASP	5.1%	0.6%	13.0%	4.8%	4.8%	4.8%
Volume sold (billion litres)	4.67	5.03	5.51	6.12	6.86	7.68
% chg in volume sold	10.7%	7.7%	9.5%	11.0%	12.0%	12.0%
Interest rate	4.7%	4.2%	4.8%	4.8%	5.3%	5.5%
Exchange rate (HK\$/RMB)	1.12	1.16	1.21	1.24	1.24	1.24

Source: Citi Research estimates, Company

Figure 5. Beijing Enterprises – DCF Model

(HKD m)	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
FCF		-591	-192	1,693	3,142	4,111	4,943	5,861	7,843	8,289
Discount factor		1.00	1.08	1.16	1.25	1.35	1.45	1.57	1.69	1.82
Net debt	-6,593	-591	-178	1,457	2,509	3,046	3,397	3,737	4,640	4,549
Terminal Value:										
NAV in 2021E	4,726									
Expected terminal growth rate	3.9%									
Expected Capital Employed on new investments	15.00%									
WACC	7.79%									
Terminal Value	89,658									
Discount factor	1.82									
NPV of terminal value	49,209									
NPV (HK\$)	65,182									
No. of share (m)	1,138									
Price/Share (HK\$)	57.30									

Source: Citi Research

Figure 6. Beijing Enterprises – Income Statement (2009-2014E, HK\$ Millions)

Profit and loss	2009	2010	2011	2012E	2013E	2014E
Revenue	24,208	27,613	30,472	35,356	41,828	49,799
Natural Gas	12,617	14,119	16,460	19,283	23,352	28,130
Water	1,970	2,405	196	190	1	1
Yanjing Beer	9,744	11,012	13,394	15,584	18,295	21,477
Toll Roads	400	372	198	129	129	136
COGS	(18,390)	(21,214)	(23,738)	(27,565)	(32,892)	(39,276)
Gross profit	5,819	6,399	6,734	7,791	8,936	10,523
Selling and distribution costs	(1,360)	(1,653)	(2,200)	(2,494)	(2,794)	(3,161)
Administrative expenses	(1,794)	(2,117)	(2,442)	(2,338)	(2,612)	(3,054)
Other income and gains, net	441	208	1,112	981	1,200	1,278
Other operating expenses, net	(325)	(203)	18	(164)	(163)	(190)
Operating profit	2,339	2,425	2,110	3,776	4,567	5,397
Finance costs	(364)	(374)	(647)	(890)	(1,009)	(1,090)
Gain on deemed disposal of interest in subsidiaries	105	171	0	0	0	0
Dilution losses on share reforms of subsidiaries	0	0	0	0	0	0
Share of profits and losses of jointly-controlled entities	1,092	1,169	1,300	81	83	85
Share of profits and losses of associates	(8)	196	373	1,879	2,363	2,619
Profit before tax	3,605	3,795	4,249	4,846	6,004	7,010
Tax	(559)	(685)	(583)	(706)	(879)	(1,054)
Profit for continuing operations	3,046	3,110	3,665	4,140	5,125	5,956
Profit for discontinuing operations	0	0	0	0	0	0
Profit after tax	3,046	3,110	3,665	4,140	5,125	5,956
Minority interest	(648)	(470)	(889)	(904)	(1,175)	(1,298)
Net income	2,399	2,639	2,776	3,236	3,950	4,659

Source: Citi Research estimates, Company

Figure 7. Beijing Enterprises – Balance Sheet (2009-2014E, HK\$ Millions)

Balance sheet	2009	2010	2011E	2012E	2013E	2014E
Property, plant and equipment	19,045	22,244	25,242	28,694	31,562	32,974
Investment properties	204	216	681	681	681	681
Goodwill	8,649	7,246	7,454	7,454	7,454	7,454
Other intangible assets	1,724	1,270	1,249	1,249	1,249	1,249
Interests in jointly-controlled entities	5,397	6,102	211	292	375	460
Interest in associates	900	3,110	12,574	14,453	16,816	19,435
Pledged deposits	0	0	0	0	0	0
Available-for-sale investments	290	1,005	917	936	954	974
Other non-current assets	6,718	4,056	4,455	4,524	4,615	4,728
Total non-current assets	42,928	45,249	52,782	58,282	63,706	67,953
Inventories	2,995	3,727	5,286	6,138	7,653	9,138
Trade and bills receivables	1,098	1,347	1,586	1,770	2,094	2,493
Prepayments, deposits and other receivables	1,654	900	2,313	2,313	2,313	2,313
Taxes recoverable	85	304	589	589	589	589
Pledge deposits	118	126	37	37	37	37
Other current assets	741	930	1,108	1,108	1,108	1,108
Cash and cash equivalents	9,486	14,492	13,655	12,566	10,864	10,256
Total current assets	16,177	21,826	24,573	24,520	24,657	25,934
Bank and other borrowings	5,264	7,227	11,769	11,769	11,769	11,769
Bonds	2,721	2,943	2,774	2,774	2,774	2,774
Other long term liabilities	805	768	958	958	958	958
Deferred tax liabilities	413	364	371	371	371	371
Total non-current liabilities	9,203	11,302	15,872	15,872	15,872	15,872
Trade and bills payables	1,408	4,554	1,905	2,740	3,598	4,296
Other payables and accruals	5,437	6,862	7,878	7,956	8,036	8,116
Taxes payable	954	995	676	676	676	676
Bank and other borrowings	3,038	2,320	5,705	7,950	9,792	11,272
Other current liabilities	48	59	124	124	124	124
Total current liabilities	10,885	14,791	16,287	19,446	22,226	24,484
Minority interest	7,712	6,668	7,587	7,595	7,604	7,615
Issued capital	114	114	114	114	114	114
Reserves	30,680	33,642	36,984	39,264	42,035	45,290
Proposed final dividend	512	512	512	512	512	512
Total equity	31,305	34,268	37,610	39,890	42,661	45,915

Source: Citi Research estimates, Company

We revise our 2013/14E EPS by 1%/4% respectively as we have set more optimistic assumptions on the growth of its gas volumes, and as the company continues to streamline its business model. We believe the company will continue to utilize its subsidiaries including BE Water and Beijing Development (HK) to drive growth in the wastewater and waste-to-energy business segments.

Figure 8. Beijing Enterprises – Earnings Revision

	Net Profit (HK\$ mils)		Diluted EPS (HK\$)		
	Old	New	Old	New	% change
2012E	3,241	3,236	2.73	2.73	-0.2%
2013E	3,898	3,950	3.28	3.33	1.3%
2014E	4,475	4,659	3.77	3.92	4.1%

Source: Citi Research estimates

Figure 9. BEHL – Citi Estimates vs. Consensus

	Citigroup	Consensus	% Difference
2012E Sales (HK\$mn)	35,139	35,329	-1%
2012E EBIT (HK\$mn)	3,710	3,618	3%
2012E Net Income (HK\$mn)	3,241	3,232	0%
2012E EPS (HK\$)	\$2.73	\$2.81	-3%
2013E Sales (HK\$mn)	41,700	41,605	0%
2013E EBIT (HK\$mn)	4,426	4,283	3%
2013E Net Income (HK\$mn)	3,898	3,964	-2%
2013E EPS (HK\$)	\$3.28	\$3.43	-4%
2014E Sales (HK\$mn)	49,522	49,258	1%
2014E EBIT (HK\$mn)	4,981	5,011	-1%
2014E Net Income (HK\$mn)	4,475	4,800	-7%
2014E EPS (HK\$)	\$3.77	\$4.16	-9%

Source: Citi Research, Bloomberg consensus

Beijing Enterprises Holdings

Company description

Beijing Enterprises Holdings (BEHL) has businesses in city gas distribution and upstream, water and environmental-related services, beer brewing, toll roads and technology businesses. Its exposure to city gas and gas upstream is mainly under Beijing Gas, which also has 40% ownership of PetroChina Beijing Pipeline Co. with access to mid-stream gas distribution.

Investment strategy

We assign a Buy (1) rating to Beijing Enterprises Holdings (BEHL) shares, with a target price of HK\$57.30. We expect the company's ramp of its pipeline projects in the coming two years will provide additional capacity for Beijing city-expansion projects, and also prepare the company for longer-term growth in NG distribution. We also believe that Beijing's plan to convert its coal-fired power plants, while mostly anticipated, is likely to materialize in the near-term while the increase capacity of co-gen plants will be a short-term catalyst. By managing the growth of its gas projects and improving profitability in its other business segments, we see BEHL delivering 15-20% earnings CAGR in the coming 3 years.

Valuation

Our HK\$57.30 target price for BEHL is based on a DCF analysis of its main businesses. We use the DCF model for our valuation for its gas business, given its longer-term cash flow and potential to benefit from increasing NG supply in the coming 2-3 years. The gas portion of the business is the key earnings growth driver, and accounts for over 70% of the company's value. We expect this portion of the business to grow as the new transmission business grows, with the new Shaanxi-Beijing No. 3 and No. 4 line providing additional upside. Water treatment is the second largest part of BEHL's business profile, which will be consolidated after accounting changes in 2013. We expect the company's gas distribution and its stake in BE Water will be key and account for a growing portion of BEHL's earnings profile.

Risks

The risks to BEHL's share-price include delays to its gas pipeline projects, which may affect the availability of NG to its customers. Other risks factors include delays in implementation when price rises at the city-gate level, weather-related impacts on gas usage, changes to government policies, and a sharp slowdown in China's economy that might lead to a reduction in C&I demand and reduce projections of gas growth. Although China has been using its policy tools to curb rising property prices, at this point we still expect the use of cleaner fuels such as natural gas to be a key priority in the near term. Any of these risks could prevent the shares from reaching our target price.

Cheung Kong Infrastructure Holdings (1038.HK)

Buy	1
Price (28 Nov 12)	HK\$46.30
Target price	HK\$54.50
from HK\$53.20	
Expected share price return	17.7%
Expected dividend yield	3.7%
Expected total return	21.4%
Market Cap	HK\$115,558M
	US\$14,911M

Price Performance (RIC: 1038.HK, BB: 1038 HK)



Too Much Concern on Share Placement Risk

■ **Summary** – CKI's share price has underperformed its closest peer Power Assets (6 HK, Buy) since late July as investors remain concerned about share dilution risk. This looks unjustified given the company has reached its target of nearly 25% free cash flow after the share placement. We believe CKI's fundamentals are sound and expect potential M&As to raise its FY13E EPS by up to 15.2%. Its 10.7x FY13E PER and 3.9% yield look inexpensive. We raise FY13-14E earnings and TP to HK\$54.50 (from HK\$53.20) on higher profit assumptions from UK Power Networks.

■ **Net profit to +4% hoh to HK\$4.9bn in 2H12E** – We forecast CKI's net profit would increase 4% hoh to HK\$4.9bn in 2H12E on (i) more profit from UK Power Network cutting its headcount by 100 in 2H12; since acquisition in Oct 2010, the number of its employees has fallen by 13% to 4,800; & (ii) profit from its cement and concrete business in HK & China to increase by HK\$50m hoh on higher sales prices. These should offset the profit cut from CHEDHA in Australia from the smart meter installation in 2H12E, with the incremental cost to be paid by users later.

■ **Net profit to +10% yoy in 2013E** – Growth should come from full year profit of Wales & West Utilities (WWU) bought in Sep 2012, more profit from UK Power Network on its asset base expansion and labor cost cuts, as well as reversal of the time difference at CHEDHA; these should offset the earnings cut from Northern Gas Network after return reset becomes effective on 1 Apr 2013. Deferred tax gain from UK will likely halve yoy in 2013E as corp tax rate reduces by only 1% vs. 2% in 2012.

■ **Targeting acquisitions in 1H13E** – CKI's FY13E EPS could rise by up to 15% from expansions, assuming 14% ROE and with gearing not exceeding its historical peak. By end 2012, we forecast CKI's see-through gearing at 85%, well below its peak of 140% in 2005, and forecast its consolidated net debt/equity ratio at 7%, including perpetual securities as equity. CKI has no equity issuance plan after raising its free cash flow to nearly 25% in late July.

Asian Utilities

Hong Kong Utilities

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Statistical Abstract

FYE	Net Profit	EPS	P/E	D. Yld
31 Dec	(HK\$M)	(HK\$)	(x)	(%)
2009A	5,588	2.48	18.7	2.6
2010A	5,028	2.23	20.8	2.9
2011A	7,745	3.31	14.0	3.3
2012E	9,572	3.92	11.8	3.7
2013E	10,542	4.32	10.7	3.9
2014E	10,780	4.42	10.5	4.1

Source: Powered by dataCentral

Earnings Revision

FYE	Net Profit	EPS	%	DPS
31 Dec	(HK\$M)	(HK\$)	chg	(HK\$)
2012E	9,572	3.92	-1.0	1.70
Prev	9,667	3.96		1.70
2013E	10,542	4.32	1.8	1.80
Prev	10,357	4.25		1.80
2014E	10,780	4.42	3.8	1.90
Prev	10,388	4.26		1.90

Source: Powered by dataCentral

1038.HK: Fiscal year end 31-Dec						Price: HK\$46.30; TP: HK\$54.50; Market Cap: HK\$115,558m; Recomm: Buy					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	2,814	3,493	3,813	4,234	4,245	PE (x)	20.8	14.0	11.8	10.7	10.5
Cost of sales	-657	-759	-855	-1,068	-1,069	PB (x)	2.3	2.1	1.8	1.7	1.5
Gross profit	2,157	2,734	2,958	3,166	3,176	EV/EBITDA (x)	95.6	76.5	67.0	59.3	57.3
Gross Margin (%)	76.7	78.3	77.6	74.8	74.8	FCF yield (%)	0.4	-0.2	2.2	3.5	2.8
EBITDA (Adj)	1,210	1,623	1,859	2,040	2,069	Dividend yield (%)	2.9	3.3	3.7	3.9	4.1
EBITDA Margin (Adj) (%)	43.0	46.5	48.8	48.2	48.7	Payout ratio (%)	60	46	43	42	43
Depreciation	-79	-69	-51	-69	-86	ROE (%)	11.4	15.9	16.9	16.3	15.2
Amortisation	-9	0	0	0	0	Cashflow (HK\$m)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	1,122	1,554	1,808	1,971	1,984	EBITDA	1,210	1,623	1,859	2,040	2,069
EBIT Margin (Adj) (%)	39.9	44.5	47.4	46.6	46.7	Working capital	458	256	253	557	54
Net interest	-450	-575	-540	-356	-356	Other	-1,099	-1,713	779	1,741	1,389
Associates	4,367	7,397	8,959	9,610	9,835	Operating cashflow	569	166	2,891	4,338	3,512
Non-op/Except	136	-110	0	0	0	Capex	-141	-353	-353	-353	-353
Pre-tax profit	5,175	8,266	10,227	11,225	11,462	Net acq/disposals	-14,042	-9,811	-2,448	0	0
Tax	-8	6	7	8	8	Other	2,674	3,658	5,407	5,454	5,469
Extraord./Min.Int./Pref.div.	-139	-527	-663	-691	-691	Investing cashflow	-11,509	-6,506	2,606	5,101	5,116
Reported net profit	5,028	7,745	9,572	10,542	10,780	Dividends paid	-2,727	-3,108	-4,147	-4,391	-4,635
Net Margin (%)	178.7	nm	nm	nm	nm	Financing cashflow	5,812	5,909	-5,110	-4,748	-4,992
Core NPAT	5,028	7,745	9,572	10,542	10,780	Net change in cash	-5,128	-431	387	4,692	3,637
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	428	-187	2,538	3,985	3,159
Reported EPS (\$)	2.23	3.31	3.92	4.32	4.42						
Core EPS (\$)	2.23	3.31	3.92	4.32	4.42						
DPS (\$)	1.33	1.53	1.70	1.80	1.90						
CFPS (\$)	0.25	0.07	1.19	1.78	1.44						
FCFPS (\$)	0.19	-0.08	1.04	1.63	1.29						
BVPS (\$)	20.26	22.12	25.33	27.85	30.37						
Wtd avg ord shares (m)	2,254	2,339	2,440	2,440	2,440						
Wtd avg diluted shares (m)	2,254	2,339	2,440	2,440	2,440						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	29.0	24.1	9.2	11.0	0.3						
EBIT (Adj) (%)	129.6	38.5	16.4	9.0	0.6						
Core NPAT (%)	-10.0	54.0	23.6	10.1	2.3						
Core EPS (%)	-10.0	48.5	18.5	10.1	2.3						
Balance Sheet (HK\$m)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	5,438	5,947	5,286	8,072	10,185						
Accounts receivables	529	524	515	514	464						
Inventory	143	223	243	270	271						
Net fixed & other tangibles	52,780	64,512	70,342	74,273	78,368						
Goodwill & intangibles	151	0	0	0	0						
Financial & other assets	5,219	5,617	5,617	5,617	5,617						
Total assets	64,260	76,823	82,003	88,747	94,905						
Accounts payable	1,670	2,086	2,350	2,934	2,938						
Short-term debt	1,228	11,342	3,999	3,999	3,999						
Long-term debt	7,259	3,126	3,126	3,126	3,126						
Provisions & other liab	416	497	403	402	402						
Total liabilities	10,573	17,051	9,878	10,460	10,464						
Shareholders' equity	53,606	59,677	72,020	78,171	84,316						
Minority interests	81	95	105	115	125						
Total equity	53,687	59,772	72,125	78,286	84,441						
Net debt	3,049	8,522	1,839	-947	-3,060						
Net debt to equity (%)	5.7	14.3	2.5	-1.2	-3.6						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

DCF Target Price & Valuation Tables

Mild earnings and TP revisions – We lower FY12 net profit forecast for CKI by -1% on less profit assumptions from CHEHDA, but raise FY13-14E net profit by 2-4% on higher profit assumptions from UK Power Networks. We raise our TP to HK\$54.50 (from HK\$53.20) factoring in the FY13-14E profit uplifts.

Figure 1. CKI – DCF and Profit Contribution Breakdown

CKI breakdown by project	DCF breakdown				Profit (HK\$m)			
	Value (HK\$m)	Per share (HK\$)	By project	By country	2011	2012E	2013E	2014E
Hong Kong:								
Power Assets (at HK\$67/share)	55,582	22.8	42%		3,503	3,793	4,088	4,188
Alliance Construction Materials	1,857	0.8	1%		88	119	132	139
Anderson Asia	1,756	0.7	1%		87	105	127	111
Green Island Cement	2,184	0.9	2%		102	136	164	140
GITIC Green Island Cement	47	0.0	0%		3	5	5	5
Siquijor Limestone Quarry	28	0.0	0%	46%	2	2	2	2
Australia:								
ETSA Utilities	5,865	2.4	4%		860	796	831	868
Citipower	746	0.3	1%		154	149	169	190
Powercor	1,698	0.7	1%		138	131	149	168
Spark & Envestra	1,193	0.5	1%		152	158	165	171
Water distribution	15	0.0	0%	7%	1	1	1	1
The UK:								
UK Power Grid	35,771	14.7	27%		2,840	3,270	3,351	3,442
Seabank	1,821	0.7	1%		87	172	171	171
Northern Gas Network	12,792	5.2	10%		531	559	554	552
Wales & West Utilities	2,962	1.2	2%		0	67	255	255
Cambridge Water	0	0.0	0%		136	0	0	0
Southern Water	919	0.4	1%		12	12	12	12
Northumbrian Water Group	9,897	4.1	7%	48%	130	1,148	1,135	1,065
China toll road								
Changsha Bridges, Hunan	602	0.2	0%		37	41	44	47
Jiangmen Chaolian Bridge	273	0.1	0%		16	18	20	21
Jiangmen Jiangsha Highway	521	0.2	0%		29	33	37	40
National Highway 107	812	0.3	1%		49	56	60	64
Panyu Beidou Bridge	251	0.1	0%		15	17	19	20
Shantou Bay Bridge	775	0.3	1%		47	53	57	61
Shen-Shan Highway & others	1,818	0.7	1%	4%	185	194	192	190
New Zealand - power distribution	1,489	0.6	1%	1%	73	73	73	73
Canada - power generation	1,363	0.6	1%	1%	156	156	156	156
Sub-total	143,037	58.6	108%	108%	9,434	11,265	11,968	12,151
Hutchison bonds	1,900	0.8	1%	1%				
Others: Interest income + FX + Other income					-1,171	-1,041	-745	-691
Net cash	-1,839	-0.8	-1%	-1%				
Less: Perpetual bond	-10,140	-4.2	-8%	-8%	-517	-653	-681	-681
Total (=target price / Net Profit)	132,959	54.5	100%	100%	7,745	9,572	10,542	10,780

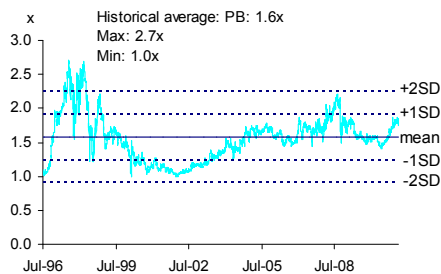
Source: Company Reports and Citi Research Estimates

Figure 2. CKI's historical PER chart



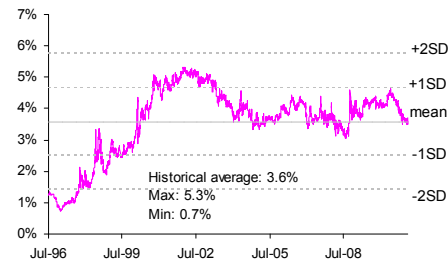
Source: Citi Research

Figure 3. CKI's historical PB chart



Source: Citi Research

Figure 4. CKI's historical dividend yield chart



Source: Citi Research

Figure 5. Sensitivity Analysis - CKI's potential up to 15.2% FY13E net profit enhancement effect from potential acquisitions

Capex of potential acquisition in FY11E	FY13E net profit	Profit uplift from base case	Net debt/equity ratio by end FY13E (excluding off B/S debt)	Net debt/equity ratio by end FY13E (including off B/S debt)
Nil (base case)	10,542	N/A	Net cash	73.5%
HK\$10bn	10,942	3.8%	4.5%	88.2%
HK\$20bn	11,342	7.6%	10.4%	102.9%
HK\$30bn	11,742	11.4%	16.7%	117.6%
HK\$30bn	12,142	15.2%	22.1%	132.3%

* Assuming capex to be financed 60% by debt and 40% by equity; ** Assuming 14% ROE for potential acquisitions and 4% pa equity funding cost

*** CKI had HK\$51bn off balance sheet borrowing by end 2012E

Source: Company Reports and Citi Research Estimates

Figure 6. CKI's proportional off balance debts under its project companies

	Country	Loans in local currencies (m)	Exchange rate vs.HK\$	Stake	Loans in HK\$m	S&P rating
Northern Gas Network	The UK	1,100	12.19	47.1%	6,313	BBB+
UK Power Network	The UK	3,200	12.19	40%	17,060	BBB+
Northumbrian Water Group	The UK	2,600	12.19	40%	12,673	BBB+
WWU	The UK	1,312	12.19	30%	4,798	A-/BBB
Australian power grids	Australia	5,500	7.94	23.1%	10,088	A-
Total					50,932	

Source: Company Reports and Citi Research Estimates

Figure 7. HK Utilities – Valuation Comparison (Prices as at 27 Nov 2012)

	RIC	Rating	Price		PE		P/B		Net D/E		Yield	
			Current	Target	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E
HKCG	0003.HK	Sell	HK\$21.00	HK\$17.00	23.8	25.7	4.0	3.7	30.7%	26.8%	1.7%	1.9%
CLP	0002.HK	Neutral	HK\$67.20	HK\$70.00	16.3	13.1	1.9	1.8	81.2%	76.9%	3.8%	4.0%
Power Asset	0006.HK	Buy	HK\$68.10	HK\$77.00	14.8	13.7	2.3	2.2	30.2%	26.4%	3.7%	4.0%
CKI	1038.HK	Buy	HK\$46.25	HK\$54.50	11.8	10.7	1.8	1.7	14.3%	-1.2%	3.7%	3.9%
Average					16.7	15.8	2.5	2.3	39.1%	32.2%	3.2%	3.5%

Source: Bloomberg and Citi Research Estimates

Figure 8. CKI's sources and uses of proceeds in past two years

Inflow			Outflow		
Date	Source	HK\$m	Date	Asset acquired	HK\$m
Sep-10	6.625% p.a. Perpetual Capital Securities	7,676	Jun-10	Seabank Power Network	-2,500
Jul-11	84.5m CKI share placement at HK\$40.41/share	3,400	Oct-10	UK Power Network	-12,040
Feb-12	7% p.a. Fixed Rate Callable Perpetual Securities	2,291	Apr-11	Meridian Cogeneration Plant (Canada)	-359
Mar-12	50.9m CKI share placement at HK\$45.75/sh	2,300	Oct-11	Northumbrian Water Group	-10,620
Jul-12	50m CKI share placement at HK\$46/sh	2,300	Sep-12	WWU	-7,285
		17,967			-32,804

Source: Company Reports and Citi Research Estimates

Figure 9. CKI's net profit forecasts – Citi vs. consensus

Net profit (HK\$m)	Citi	Consensus	% diff
FY12E	9,572	9,274	3.2%
FY13E	10,542	9,936	6.1%
FY14E	10,780	10,186	5.8%

Source: Bloomberg and Citi Research Estimates

Financial Statements & Key Assumptions

Figure 10. CKI's income statement (HK\$m)

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Core revenue	2,814	3,493	3,813	4,234	4,245
Infrastructure material sale	1,508	1,743	2,182	2,594	2,598
Water supply	260	184	0	0	0
Interest income from associates	842	1,360	1,420	1,422	1,424
Investments in securities	204	206	211	218	224
Other revenue	531	701	442	461	527
Interest Income	271	234	216	235	301
Disposal gain	0	241	0	0	0
Other income	260	226	226	226	226
Total operating expenses	-2,223	-2,640	-2,446	-2,723	-2,789
Staff costs	-317	-334	-344	-354	-365
Depreciation	-79	-69	-51	-69	-86
Amortization	-9	0	0	0	0
Raw material used	-657	-759	-855	-1,068	-1,069
Fair value change of securities	8	-25	0	0	0
Other operating expenses	-1,169	-1,453	-1,197	-1,233	-1,270
Operating profit	1,122	1,554	1,808	1,971	1,984
Finance costs	-450	-575	-540	-356	-356
Exchange loss	136	-110	0	0	0
Share of results of associates	4,034	6,974	8,486	9,102	9,297
Share of results of JCE	333	423	472	508	538
PBT	5,175	8,266	10,227	11,225	11,462
Tax	-8	6	7	8	8
MI	-6	-10	-10	-10	-10
Net profit	5,161	8,262	10,225	11,223	11,461
Coupon to US\$1bn MTN	-133	-517	-653	-681	-681
Net Profit of Equity Shareholders	5,028	7,745	9,572	10,542	10,780
% of change	-10.0%	54.0%	23.6%	10.1%	2.3%

Source: Company Reports and Citi Research Estimates

Figure 11. Key ratios

Valuations	FY10	FY11	FY12E	FY13E	FY14E
PER(x)	24.4	16.5	13.9	12.6	12.3
PEG(x)	0.7	1.5	0.7	0.4	0.0
EV/EBITDA(x)	22.4	13.7	12.4	11.6	11.6
Price/Book (x)	2.7	2.5	2.2	2.0	1.8
Payout ratio	60%	46%	43%	42%	43%
Yield	2.4%	2.8%	3.1%	3.3%	3.5%
Profitability					
EBITDA margin	193%	248%	278%	270%	273%
EBIT margin	190%	246%	277%	268%	271%
Net profit margin	183%	237%	268%	265%	270%
Interest coverage (x)	21.6	23.2	36.1	58.8	59.9
Dividend coverage (x)	1.7	2.2	2.3	2.4	2.3
Return					
ROE	11.4%	15.9%	16.9%	16.3%	15.2%
ROA	8.7%	11.0%	12.1%	12.3%	11.7%
Gearing					
Debt/Equity	15.8%	24.2%	11.5%	10.5%	9.6%
Net Debt/Equity (ex-MI)	5.7%	14.3%	3.0%	net cash	net cash
Debt to Asset	13.2%	18.8%	8.7%	8.0%	7.5%
Net Debt/Asset (ex-MI)	4.7%	11.1%	2.2%	net cash	net cash

Source: Company Reports and Citi Research Estimates

Figure 12. CKI's balance sheet (HK\$m)

As at 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
PP&E	1,276	845	1,147	1,432	1,699
Investment properties	186	206	206	206	206
Interest in associates	50,573	62,504	68,031	71,679	75,506
Interests in JCEs	707	942	942	942	942
Investment in securities	4,824	5,197	5,197	5,197	5,197
Derivative financial instruments	209	158	158	158	158
Goodwill	151	0	0	0	0
Other non-current assets	38	15	15	15	15
Total non-current assets	57,964	69,867	75,697	79,628	83,723
Inventories	143	223	243	270	271
Derivative financial instruments	186	262	262	262	262
Debtors and prepayments	529	524	515	514	464
Bank balances and deposits	5,438	5,947	5,286	8,072	10,185
Total current assets	6,296	6,956	6,307	9,118	11,182
Bank loans	1,228	11,342	3,999	3,999	3,999
Derivative financial instruments	53	12	12	12	12
Creditors and accruals	1,670	2,086	2,350	2,934	2,938
Taxation	107	87	(7)	(8)	(8)
Total current liabilities	3,058	13,527	6,354	6,936	6,940
Bank loans	7,259	3,126	3,126	3,126	3,126
Derivative financial instruments	2	201	201	201	201
Deferred tax liabilities	254	187	187	187	187
Other non-current liabilities	0	10	10	10	10
Total non-current liabilities	7,515	3,524	3,524	3,524	3,524
Net Assets	53,687	59,772	72,125	78,286	84,441
Share capital	2,254	2,339	2,439	2,439	2,439
Reserves	43,419	49,405	59,357	65,508	71,653
Shareholders equity	45,673	51,744	61,796	67,947	74,092
Perpetual capital securities	7,933	7,933	10,224	10,224	10,224
Minority interests	81	95	105	115	125
Total equity	53,687	59,772	72,125	78,286	84,441

Source: Company Reports and Citi Research Estimates

Figure 13. CKI's key assumptions

As at 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Power Assets profit (HK\$m)	7,194	9,075	9,827	10,590	10,850
yoy	7.4%	26.1%	8.3%	7.8%	2.5%
Power Assets SOC's return	10.0%	10.0%	10.0%	10.0%	10.0%
Power Assets SOC's Capex (HK\$m)	2,431	2,887	2,310	1,922	2,191
CPI	3%	5%	3%	3%	3%
AUD/HK\$	7.06	7.80	7.90	7.90	7.90
GBP/HK\$	11.97	12.60	12.30	12.30	12.30

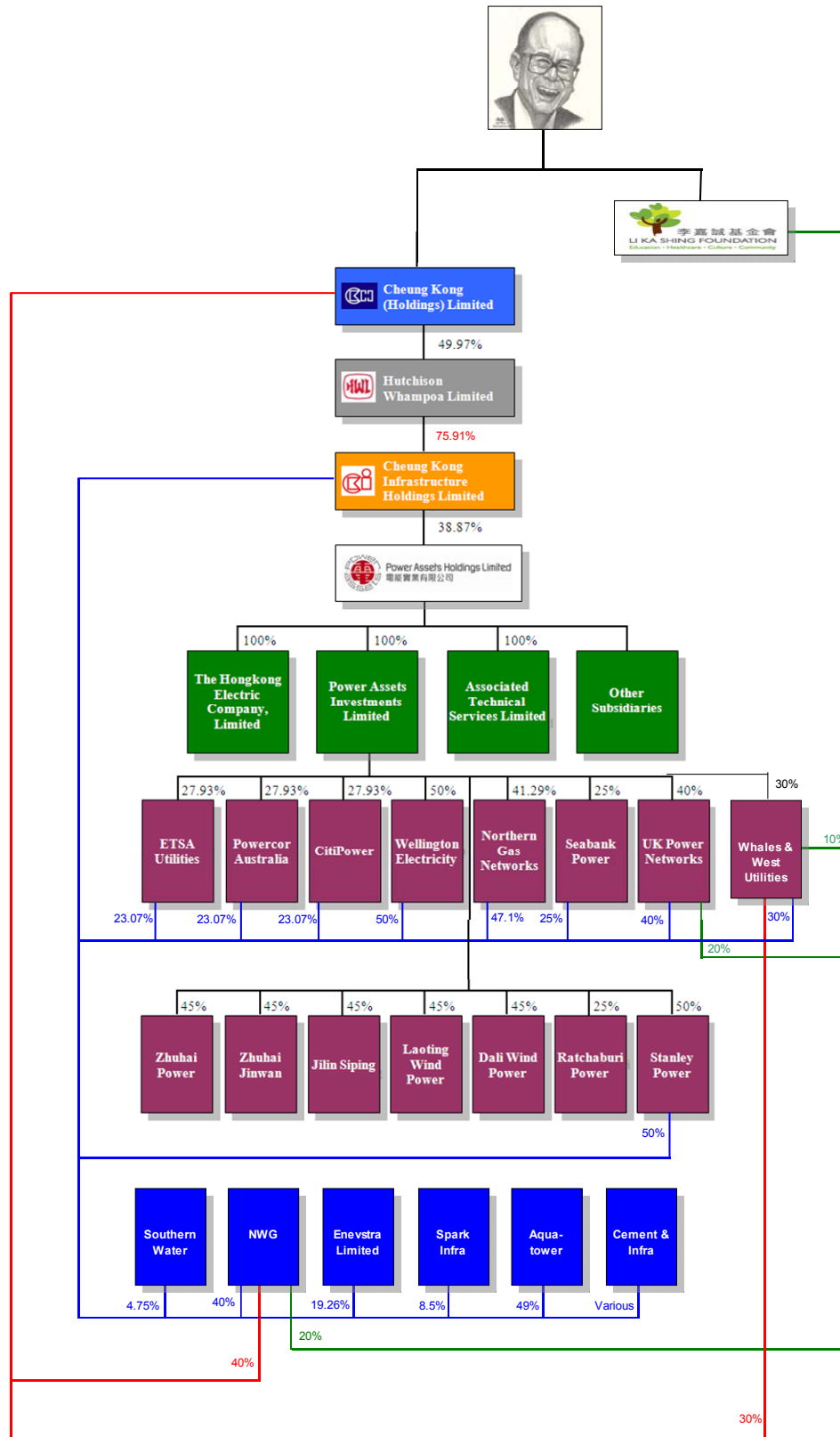
Source: Company Reports and Citi Research Estimates

Figure 14. CKI's cashflow statement (HK\$m)

As at 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Operating activities					
Profit before tax	5,175	8,266	10,227	11,225	11,462
Profit of associates	-4,034	-6,974	-8,486	-9,102	-9,297
Profit of JCEs	-333	-423	-472	-508	-538
Interest income from associates	-842	-1,360	0	0	0
Other interest income	-382	-326	-216	-235	-301
Finance costs	450	575	540	356	356
Depreciation	70	69	51	69	86
Amortization of leasehold land	9	0	0	0	0
Unrealized exchange (gain) / loss	284	210	0	0	0
Return from JCEs	218	191	472	508	538
Interest received from associates	644	1,304	216	235	301
Others	93	-174	0	0	0
Cash generated from operations	1,352	1,358	2,331	2,548	2,608
Chg in working capital	458	256	253	557	54
Chg in inventories	27	-82	-20	-27	-1
Chg in debtors and prepayments	33	-176	9	0	50
Chg in creditors and accruals	388	464	264	583	4
Exchange translation differences	10	50	0	0	0
Operating cash flow	1,810	1,614	2,585	3,105	2,661
Income taxes paid	19	8	-87	7	8
Net operating cash flow	1,829	1,622	2,498	3,112	2,669
Investing activities					
Capex	-141	-353	-353	-353	-353
Dividends from associates	2,385	3,058	5,407	5,454	5,469
Disposal of PP&E	1,214	1,166	0	0	0
Acquisitions of associates	-14,973	-10,863	-2,448	0	0
Advances to associates	-5	-5	0	0	0
Advance to JCEs	-3	-5	0	0	0
Repayments from JCEs	16	16	0	0	0
Purchases of listed securities	-283	-114	0	0	0
Interest received	279	230	0	0	0
Others	2	364	0	0	0
Investing cash flow	-11,509	-6,506	2,606	5,101	5,116
Bank loans	3,026	7,441	0	0	0
Repayment of bank loans	-1,722	-1,299	-7,343	0	0
Finance costs paid	-441	-535	-540	-356	-356
Dividend paid	-2,727	-3,108	-4,147	-4,391	-4,635
Equity issuance	7,676	3,410	4,629	0	0
Financing cash flow	-1,864	2,499	-12,030	-4,748	-4,992
Chg in cash & equivalents	-11,544	-2,385	-6,927	3,466	2,794
Cash b/f	9,306	5,438	5,947	5,286	8,072
Cash c/f	5,438	5,947	5,286	8,072	10,185
Free Cash Flow	1,688	1,269	2,145	2,759	2,316

Source: Company Reports and Citi Research Estimates

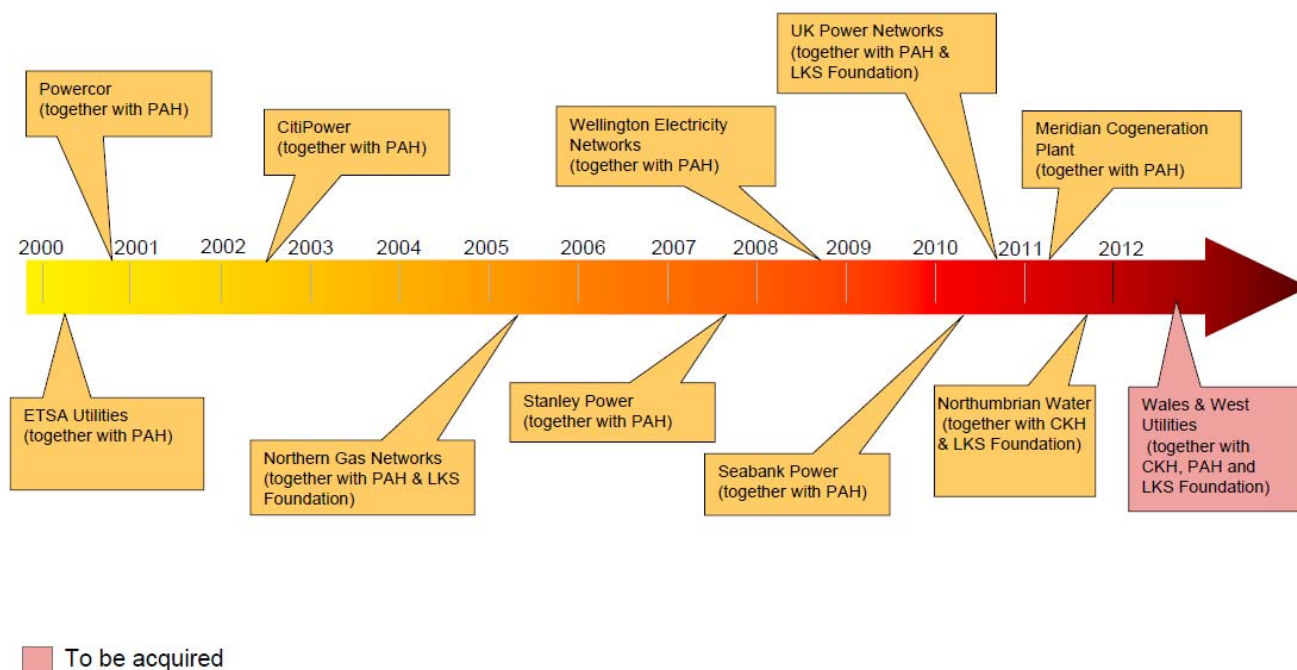
Figure 15. CKI's group structure and its joint investments with related companies



Source: Citi Research and PAH

Appendix II – Acquisition Track Record

Figure 16. CKI's Acquisition Track Record since 2000



Source: CKI

Cheung Kong Infrastructure Holdings

Company description

CKI is the largest publicly listed infrastructure company in Hong Kong, with investments in energy, transportation, water utilities and infrastructure-related businesses in Hong Kong, China, Australia, New Zealand, Canada and the UK. It holds a 38.87% stake in Power Assets (formerly known as Hongkong Electric), making it the largest shareholder, which accounted for 40% of its net profit in FY12E. After selling its power business in China to HKE in April FY09, CKI is looking to invest in Australia, Europe and North America, which could generate high single-digit un-leveraged IRRs and mid-to-high single-digit free cash flow yield.

Investment strategy

We rate CKI a Buy. Potential positive catalysts are mainly large acquisitions, which are expected to have high single-digit un-leveraged IRRs and mid-to-high single-digit free cash flow yield. Also, CKI's valuations appear inexpensive against historical averages and peer valuations, with the benefits of potential expansion likely not yet priced in.

China Everbright Int'l (0257.HK)

Tackling China's Environmental Issues: Waste-to-Energy Plants to See Construction Ramp in 2013

Buy	1
Price (28 Nov 12)	HK\$3.70
Target price	HK\$4.60
from HK\$4.65	
Expected share price return	24.3%
Expected dividend yield	1.6%
Expected total return	25.9%
Market Cap	HK\$14,940M
	US\$1,928M

Price Performance

(RIC: 0257.HK, BB: 257 HK)



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■ **Maintain Buy** — We see China Everbright Int'l (CEI) as a key beneficiary of China's increasing focus on environmental protection during the 12th 5-year plan (2011-2015). CEI's strong track record in waste-to-energy (WTE) developments should put it at an advantage to peers, as the central government tightens environmental approval and emission standards for WTE plants. CEI's existing projects should also see improving operating cash flows. We re-iterate our Buy rating, with revised target price at HK\$4.60 (from HK\$4.65).

■ **Analyst day takeaways** — CEI sees three projects (Xinyu surface water, Suzhou Phase III, and Dezhou) to complete by year-end. Its nine new projects obtained in 2012 and two existing ones (Huidong, Yixing Phase II) will more than double its waste-to-energy (WtE) capacity to 20,000 tonnes/day from 8,150 t/d currently.

■ **Drivers in 2013** — CEI will likely see a sharp jump in construction revenues, as 7 of 9 pending projects are in the process of site formation and should see sharper construction ramp starting in 1H13, which would drive 34% recurring earnings growth. We anticipate CEI to focus on coastal projects, with higher tipping fees.

■ **Wastewater and other segments** — CEI's Xinyi Surface Water BT and Dezhou Nanyunhe Wastewater projects should complete construction by year-end, which should add to its construction revenue in 2H12. While 2012E earnings include the one-off gains from the bridge sale, we believe CEI's business should see sustainable 10-20% growth driven by higher construction growth in 2013-14E and increasing profits from its operations, especially from its newer WTE projects.

■ **Earnings revision** — We lower our 2012/13/14E EPS estimates by 2%/6%/6% to HK\$0.28/HK\$0.27/HK\$0.29 respectively, to factor in slightly better operating margins and dilution impact from its earlier share placement.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(HK\$M)	(HK\$)	(%)	(x)	(x)	(%)	(%)
2010A	616	0.17	49.0	22.2	2.5	12.4	0.7
2011A	801	0.22	30.0	17.1	2.2	13.9	1.2
2012E	1,045	0.28	27.2	13.4	2.1	15.7	1.6
2013E	1,068	0.27	-3.8	13.9	2.0	14.6	1.6
2014E	1,186	0.29	11.1	12.6	1.9	15.5	1.7

Source: Powered by dataCentral

0257.HK: Fiscal year end 31-Dec						Price: HK\$3.70; TP: HK\$4.60; Market Cap: HK\$14,940m; Recomm: Buy					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	2,929	3,664	3,384	4,383	4,660	PE (x)	22.2	17.1	13.4	13.9	12.6
Cost of sales	-1,784	-2,088	-1,747	-2,478	-2,561	PB (x)	2.5	2.2	2.1	2.0	1.9
Gross profit	1,146	1,575	1,637	1,905	2,099	EV/EBITDA (x)	14.0	10.9	10.0	8.1	7.2
Gross Margin (%)	39.1	43.0	48.4	43.5	45.0	FCF yield (%)	-6.0	0.1	4.1	7.3	7.0
EBITDA (Adj)	1,064	1,440	1,568	1,827	2,008	Dividend yield (%)	0.7	1.2	1.6	1.6	1.7
EBITDA Margin (Adj) (%)	36.3	39.3	46.3	41.7	43.1	Payout ratio (%)	15	21	22	22	21
Depreciation	-17	-27	-33	-49	-49	ROE (%)	12.4	13.9	15.7	14.6	15.5
Amortisation	-24	-31	-34	-37	-41	Cashflow (HK\$m)					
EBIT (Adj)	1,023	1,382	1,501	1,741	1,917	EBITDA	1,064	1,440	1,568	1,827	2,008
EBIT Margin (Adj) (%)	34.9	37.7	44.4	39.7	41.1	Working capital	-1,643	-832	-435	-8	-220
Net interest	-178	-247	-305	-314	-333	Other	3	-80	-62	-265	-299
Associates	-1	0	0	50	50	Operating cashflow					
Non-op/Except	0	0	250	0	0	Capex	-251	-518	-498	-473	-450
Pre-tax profit	845	1,135	1,447	1,477	1,634	Net acq/disposals	2	0	0	0	0
Tax	-192	-290	-355	-362	-401	Other	-263	-356	-427	-513	-410
Extraord./Min.Int./Pref.div.	-37	-44	-46	-47	-47	Investing cashflow					
Reported net profit	616	801	1,045	1,068	1,186	Dividends paid	-82	-130	-165	-225	-237
Net Margin (%)	21.0	21.9	30.9	24.4	25.5	Financing cashflow					
Core NPAT	616	801	1,045	1,068	1,186	Net change in cash	-602	343	1,470	547	387
Per share data						Free cashflow to s/holders					
Reported EPS (\$)	0.17	0.22	0.28	0.27	0.29	-826	9	573	1,081	1,039	
Core EPS (\$)	0.17	0.22	0.28	0.27	0.29						
DPS (\$)	0.03	0.05	0.06	0.06	0.06						
CFPS (\$)	-0.16	0.14	0.28	0.39	0.37						
FCFPS (\$)	-0.22	0.00	0.15	0.27	0.26						
BVPS (\$)	1.46	1.68	1.77	1.88	1.94						
Wtd avg ord shares (m)	3,643	3,667	3,791	4,026	4,026						
Wtd avg diluted shares (m)	3,696	3,697	3,792	4,026	4,026						
Growth rates											
Sales revenue (%)	65.9	25.1	-7.6	29.5	6.3						
EBIT (Adj) (%)	51.1	35.1	8.7	16.0	10.1						
Core NPAT (%)	65.8	30.0	30.4	2.1	11.1						
Core EPS (%)	49.0	30.0	27.2	-3.8	11.1						
Balance Sheet (HK\$m)											
Cash & cash equiv.	1,380	1,900	3,370	3,918	4,305						
Accounts receivables	674	1,016	753	988	1,050						
Inventory	21	43	27	38	39						
Net fixed & other tangibles	5,043	6,851	6,243	5,934	5,919						
Goodwill & intangibles	685	1,116	1,332	1,654	1,737						
Financial & other assets	3,067	2,954	3,747	4,183	4,501						
Total assets	10,870	13,880	15,473	16,715	17,551						
Accounts payable	580	868	750	1,064	1,100						
Short-term debt	732	1,064	1,170	1,182	1,300						
Long-term debt	3,155	4,151	4,551	5,051	5,297						
Provisions & other liab	655	1,135	1,359	1,301	1,446						
Total liabilities	5,122	7,218	7,831	8,598	9,143						
Shareholders' equity	5,338	6,190	7,124	7,552	7,796						
Minority interests	411	472	518	565	612						
Total equity	5,749	6,662	7,642	8,116	8,408						
Net debt	2,506	3,315	2,351	2,316	2,293						
Net debt to equity (%)	43.6	49.8	30.8	28.5	27.3						

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For definitions of the items in this table, please click [here](#).

Construction Ramp in 2013

We re-iterate our Buy on China Everbright (CEI), and we expect CEI to see a strong capacity ramp, driven by its strong pipeline for China's waste-to-energy projects. Although construction revenue fell 54% Y/Y in 1H12, a 60% rise in operation services resulted in 14% Y/Y earnings growth for its continued operations. We expect WtE construction revenue to see an 87% Y/Y ramp-up in 2013, as the company looks to double its existing WTE capacity from 8,150 tonnes/day to 20,000 tonnes/day.

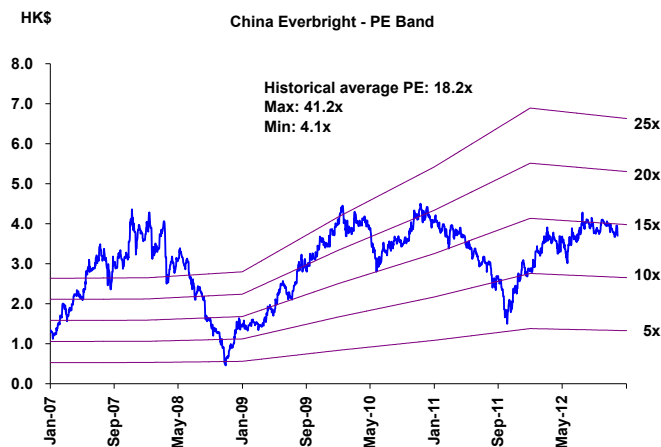
Development in In-House Waste Furnace

In addition to the traditional construction methods, CEI has also been developing technology on grate furnaces, and has started manufacturing its own grate furnaces to meet the demand from its waste-to-energy plants. Aside from cost considerations, this investment will also help develop technologies to meet the type of waste generated in China. Compared with the traditional overseas suppliers, CEI has specifically targeted the following areas:

- **Treatment of municipal waste** – China's municipal waste has not yet adopted source separation in many cases, leading to waste that typically carries high water content and lower heat value. This will require furnace design to specifically allocate space for handling leachate (remaining liquid from waste decomposition), and for maintaining the temperature of the furnace.
- **Modular design** – CEI is using a modular design to meet the challenges in transporting equipment to the WTE location, which will help reduce the installation period.
- **Automatic burning control** – By using the latest technologies in control systems, CEI is able to develop automated burn control to handle various kinds of waste materials and cycles.
- **Durability of grate stoker** – By using durable grate stoker, it will reduce replacement needs to below 5% per year.
- **Turning grate stoker** – By allowing grate stoker to rotate, it will help improve the energy produced from the same amount of municipal waste (fuel) and improve energy capture.
- **Standardized products** – Standardized production of a range of sizes (from 150t/d to 500t/d), which will help meet the needs of the different sizes of cities in China.
- **Room for further development** – Wide range of heat value (4500kJ/kg-9200kJ/kg) will allow the plant to adopt to changing needs of the cities.
- **Reliability** – Support reliable, continuous operation. CEI grate furnace has a target annual operation time of over 8,000 hours.

By mid-2012, CEI has already put 6 sets of equipment in operation, while it is in the process of producing more equipment to meet the company's increasing mandates for new WTE projects.

Figure 1. CEI – Forward PE Band



Source: Citi Research

Figure 2. CEI – Forward PB Band



Source: Citi Research

Compared to consensus, our 2012E earnings estimate is in-line, after factoring in higher margin assumption for its operations, and we remain cautious on the construction revenue growth of its waste-to-energy projects in 2012E, leading to lower top-line revenue.

Figure 3. China Everbright – Earnings Estimates of Citi and Consensus

	Citigroup	Consensus	% Difference
2012E Sales (HKD mn)	3,384	3,899	-13%
2012E EBITDA (HKD mn)	1,568	1,586	-1%
2012E Net income (HKD mn)	1,045	1,049	0%
2012E EPS (HKD)	0.28	0.27	2%
2013E Sales (HKD mn)	4,383	5,658	-23%
2013E EBITDA (HKD mn)	1,827	1,938	-6%
2013E Net income (HKD mn)	1,068	1,164	-8%
2013E EPS (HKD)	0.27	0.30	-12%
2014E Sales (HKD mn)	4,660	6,036	-23%
2014E EBITDA (HKD mn)	2,008	2,309	-13%
2014E Net income (HKD mn)	1,186	1,337	-11%
2014E EPS (HKD)	0.29	0.34	-14%

Source: Citi Research; IBES for consensus estimates

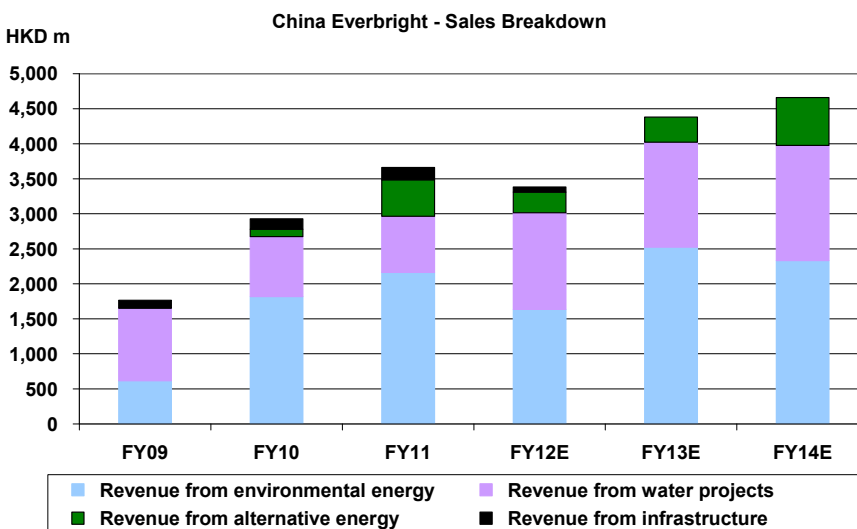
In our model, we assume 2012 earnings will include the disposal gain from its bridge business, which we classify as investment/disposal gain. We now revise our margin assumption to factor in much lower construction revenue in 2012 and higher profit contribution from operation services. We also forecast the tax rate to remain at 25%. We expect CEI's net profit to rise by 30% in 2012 and 2% in 2013.

Figure 4. China Everbright - Income Statement (2009-2014E, HK\$ Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Turnover	1,766	2,929	3,664	3,384	4,383	4,660
Revenue from environmental energy	615	1,816	2,161	1,636	2,524	2,333
Revenue from water projects	1,036	857	805	1,380	1,498	1,643
Revenue from alternative energy	0	107	520	293	361	684
Revenue from infrastructure	114	148	177	74	0	0
Cost of sales	(991)	(1,784)	(2,088)	(1,747)	(2,478)	(2,561)
Gross profit	775	1,146	1,575	1,637	1,905	2,099
Other income	62	62	60	95	104	115
Distribution costs	0	0	0	0	0	0
Administration expenses	(165)	(187)	(252)	(230)	(267)	(295)
EBIT	677	1,023	1,382	1,501	1,741	1,917
Gain / (Loss) on disposal	0	0	0	250	0	0
Finance costs	(170)	(178)	(247)	(305)	(314)	(333)
Profit before taxation	508	845	1,135	1,447	1,477	1,634
Taxation	(99)	(192)	(290)	(355)	(362)	(401)
Minority interest	(37)	(37)	(44)	(46)	(47)	(47)
Net profits	372	616	801	1,045	1,068	1,186
Turnover YoY (%)	-5.2%	65.9%	25.1%	-7.6%	29.5%	6.3%
Net Profit YoY (%)	9.7%	65.8%	30.0%	30.4%	2.1%	11.1%

Source: Company reports, Citi Research estimates

Figure 5. China Everbright – Sales Breakdown (2009-2014E)



Source: Citi Research

Figure 6. China Everbright – Key Ratios (%)

Y/e Dec	2009	2010	2011	2012E	2013E	2014E
Gross margin	43.9	39.1	43.0	48.4	43.5	45.0
Operating margin	38.3	34.9	37.7	44.4	39.7	41.1
Pre-tax margin	28.7	28.8	31.0	42.7	33.7	35.1
Net profit margin	21.1	21.0	21.9	30.9	24.4	25.5
Effective tax rate	19.5	22.7	25.5	24.5	24.5	24.5
ROE (%)	8.1	11.5	12.9	14.7	14.1	15.2
Net debt-to-equity (%)	22.0	45.5	55.1	34.3	31.9	30.6

Source: Company reports, Citi Research estimates

Figure 7. China Everbright - Balance Sheet (2009-2014E, HK\$ Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Bank balances and cash	1,944	1,341	1,684	3,155	3,702	4,089
Pledged bank deposits	29	39	216	216	216	216
Account receivables	604	674	1,016	753	988	1,050
Inventories	13	21	43	27	38	39
Others	354	408	554	996	1,046	1,067
Current assets	2,944	2,484	3,513	5,147	5,989	6,462
PP&E	168	414	888	1,353	1,778	2,178
Investment & Others	1,781	2,659	2,400	2,548	2,930	3,223
Intangible asset	600	685	1,116	48	50	52
Non-current assets	5,723	8,387	10,366	10,326	10,725	11,090
Total assets	8,667	10,870	13,880	15,473	16,715	17,551
Account payables	375	580	868	750	1,064	1,100
Borrowings - amt due within one year	696	732	1,064	1,170	1,182	1,300
Taxation payable	10	29	52	57	63	69
Others	107	274	555	722	649	779
Current liabilities	1,188	1,615	2,539	2,700	2,959	3,249
Borrowings - amt due after one year	2,405	3,208	4,206	4,612	5,115	5,365
Non-current liabilities	2,405	3,208	4,206	4,612	5,115	5,365
Total liabilities	3,593	4,823	6,746	7,312	8,074	8,614
Shareholders equity	4,573	5,338	6,190	7,124	7,552	7,796
Minority interest	357	411	472	518	565	612
Total Equity	4,930	5,749	6,662	7,642	8,116	8,408

Source: Company reports, Citi Research estimates

We forecast CEI to see improving free cash flow in 2012, driven by higher profit before tax, disposal gain, and lower demand in working capital.

Figure 8. China Everbright - Cash Flow Statement (2009-2014E, HK\$ Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Operating activities						
Profit before tax	508	845	1,135	1,447	1,477	1,634
Adjustments	154	164	227	305	314	333
EBIT	662	1,008	1,362	1,751	1,791	1,967
Exceptional items	(0)	46	13	28	31	34
Depreciation & amortisation	39	41	58	33	49	49
Change in working capital	(1,055)	(1,643)	(832)	(435)	(8)	(220)
Operating cash flow	(355)	(547)	600	1,411	1,900	1,872
Net operating cash flow	(374)	(575)	528	1,071	1,554	1,489
Capex	(13)	(251)	(518)	(498)	(473)	(450)
Free cash flow	(388)	(826)	9	573	1,081	1,039
Change in equity	1,456	7	8	1,237	0	0
Change in debt	490	707	1,011	512	515	368
Other investing activities	58	(262)	(356)	(427)	(513)	(410)
Investing cash flow	45	(513)	(874)	(926)	(986)	(860)
Dividends paid	(65)	(82)	(130)	(165)	(225)	(237)
Financing cash flow	1,711	455	642	1,325	(21)	(242)
Beginning cash	562	1,944	1,341	1,684	3,155	3,702
Change in cash	1,382	(633)	296	1,470	547	387
Ending cash	1,944	1,341	1,684	3,155	3,702	4,089

Source: Company reports, Citi Research estimates

China Everbright Int'l

Company description

China Everbright (CEI) has operations in environmental energy (waste-to-energy), environmental protection industrial parks, wastewater treatment, alternative energy production, infrastructure construction, and property investment. This portfolio of environmental protection businesses should allow CEI to benefit from China's efforts to improve standards of living in urban areas. Over the past two years, the company has started to increase its investments in the environmental energy and alternative energy sectors, which should be the drivers for growth in coming years as growth in water projects slows.

Investment strategy

We have a Buy rating on China Everbright (CEI) and a target price of HK\$4.60. CEI more than doubled its turnover in 2011 relative to 2009, driven by its aggressive ramp-up of water and waste-to-energy projects. Although we forecast its waste-to-energy construction revenue to fall in 2012E, CEI should see faster ramp-up of wastewater projects in 2012, and better contribution from its operation services. Overall, we expect net profit will rise by 30% Y/Y in 2012, driven by higher margins and better control of expenses. CEI's investments in waste-to-energy and wastewater projects will allow the company to capture incremental growth in China's environmental business in the coming few years.

Valuation

Our target price of HK\$4.60 is based on 17x 2013E earnings, comparable to its historic average of 17x forward earnings and at a premium to global peers at 13x 2013E. We believe a premium to peers is warranted given CEI's above-peer ROE metric and its sustained new project pipelines. We also note that as the company expands its project pipeline, construction revenue growth may slow and the company may rely more on operating revenues that are recurring and hence more predictable.

Risks

We see the biggest downside risks that could prevent the China Everbright stock from reaching our target price as being: 1) delayed execution or a slowdown of its construction projects, given that construction revenue will continue to account for a significant part of the company's revenue in the coming years; 2) changes in direction of China's government policy, which has been a major driver for new project developments; 3) higher processing costs for its waste-to-energy and wastewater plant, driven by higher fuel costs and/or more restrictive regulations.

China Longyuan Power Group (0916.HK)

Share-Issuance Overhang Removed

Buy	1
Price (28 Nov 12)	HK\$5.00
Target price	HK\$5.60
Expected share price return	12.0%
Expected dividend yield	1.7%
Expected total return	13.7%
Market Cap	HK\$37,321M
	US\$4,816M

Price Performance (RIC: 0916.HK, BB: 916 HK)



■ **Summary** – Longyuan plans to issue US\$300m perpetual bonds instead of new shares. This would remove the overhang of its recent share price underperformance of +1% in last three months, versus its peers (+26% for Huaneng Renewables and +9% for Datang Renewables). Its 1.0x FY13E PB looks undemanding against 9.3% ROE. We expect Longyuan's share price to catch up without the overhang.

■ **Perpetual bond issuance** – Longyuan plans to issue US\$300m perpetual bonds in 4Q12, replacing its original plan to issue up to 30% (or 813m) new H-shares. The interest rate would be decided by book building and with a 5% escalating clause – for instance, if the interest rate is 6% pa in Year 1-3, it will be 6.3% pa [=6%*(1+5%)] in Year 4-6 and 6.615% pa [=6.3%*(1+5%)] in Year 7-9. The will be a redemption option. We think that the bond issuance will meet a good response amid ample liquidity and heightened bond prices in Hong Kong.

■ **Share issuance overhang removed** – Longyuan's share price was +1% over the last 3 months, underperforming peers (Huaneng Renewables +26% and Datang Renewables +9%), owing to the share-issue concern. To recap, it planned to issue up to 813m (30%) new H-shares at up to a 20% discount to its share price or 1.0x PB (=HK\$4.30/share by end 1H12), whichever the higher. We expect Longyuan's share price to catch up with its peers after cancelling the share issuance.

■ **Mild earnings impact from bond issuance** – We expect the interest rate of the US\$ perpetual bonds will be similar to Longyuan's debt-cost average of 5.5% pa (mostly Rmb) in 3Q12, hence not much earnings impact on equity shareholders.

■ **Carbon revenue-cut risk in our model** – Longyuan's CDM revenue was +12% yoy to Rmb613m in 9M12 and it targets Rmb800-900m in FY12E. We assume a 75% fall yoy in 2013 on spot CDM price cuts (-81% to €0.84/tonne year-to-date) and the end of the Kyoto Treaty by end 2012, when all fixed-price contracts change to floating ones. If it had no CDM revenue, our FY13E EPS would drop 7%.

Asian Utilities

China Power Sector

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Statistical Abstract

Year to 31 Dec	Net Profit (RmbM)	Diluted EPS (Rmb)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	2,008	0.269	54.6	14.9	1.3	8.9	1.3
2011A	2,638	0.353	31.4	11.4	1.2	10.8	1.7
2012E	2,869	0.384	8.8	10.4	1.1	10.8	1.9
2013E	2,696	0.361	-6.0	11.1	1.0	9.3	1.8
2014E	3,058	0.410	13.4	9.8	0.9	9.8	2.0

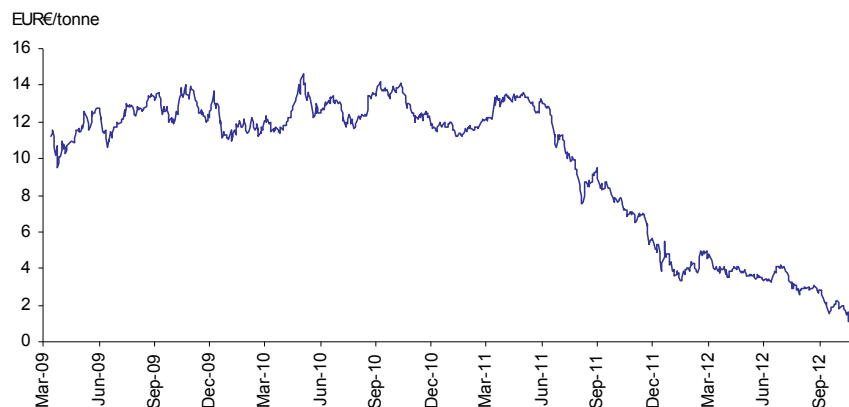
Source: Powered by dataCentral

0916.HK: Fiscal year end 31-Dec						Price: HK\$5.00; TP: HK\$5.60; Market Cap: HK\$37,321m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	15,204	17,431	20,686	22,354	24,548	PE (x)	14.9	11.4	10.4	11.1	9.8
Cost of sales	-2,737	-2,877	-2,994	-3,054	-3,054	PB (x)	1.3	1.2	1.1	1.0	0.9
Gross profit	12,467	14,554	17,692	19,300	21,494	EV/EBITDA (x)	8.6	8.5	7.6	7.8	7.4
Gross Margin (%)	82.0	83.5	85.5	86.3	87.6	FCF yield (%)	-46.1	-36.2	-23.2	-10.0	-5.1
EBITDA (Adj)	6,309	7,882	10,142	11,021	12,473	Dividend yield (%)	1.3	1.7	1.9	1.8	2.0
EBITDA Margin (Adj) (%)	41.5	45.2	49.0	49.3	50.8	Payout ratio (%)	20	20	20	20	20
Depreciation	-2,237	-2,845	-3,841	-4,472	-5,102	ROE (%)	8.9	10.8	10.8	9.3	9.8
Amortisation	0	0	0	0	0						
EBIT (Adj)	4,073	5,036	6,301	6,549	7,371	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT Margin (Adj) (%)	26.8	28.9	30.5	29.3	30.0	EBITDA	6,309	7,882	10,142	11,021	12,473
Net interest	-1,100	-1,487	-2,533	-3,000	-3,329	Working capital	-1,732	-1,012	-901	-726	-658
Associates	228	60	101	117	119	Other	-556	-806	-364	-373	-424
Non-op/Except	0	0	0	0	0	Operating cashflow	4,021	6,064	8,877	9,922	11,390
Pre-tax profit	3,200	3,609	3,869	3,666	4,160	Capex	-17,842	-16,913	-15,822	-12,904	-12,904
Tax	-441	-305	-364	-373	-424	Net acq/disposals	0	0	0	0	0
Extraord./Min.Int./Pref.div.	-751	-667	-636	-598	-678	Other	145	400	87	115	116
Reported net profit	2,008	2,638	2,869	2,696	3,058	Investing cashflow	-17,697	-16,513	-15,735	-12,789	-12,788
Net Margin (%)	13.2	15.1	13.9	12.1	12.5	Dividends paid	0	-403	-515	-560	-526
Core NPAT	2,008	2,638	2,869	2,696	3,058	Financing cashflow	1,313	10,002	7,865	3,325	1,029
						Net change in cash	-12,420	-448	1,007	458	-369
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-13,821	-10,850	-6,945	-2,982	-1,514
Reported EPS (Rmb)	0.269	0.353	0.384	0.361	0.410						
Core EPS (Rmb)	0.269	0.353	0.384	0.361	0.410						
DPS (Rmb)	0.054	0.069	0.075	0.071	0.080						
CFPS (Rmb)	0.539	0.812	1.189	1.329	1.526						
FCFPS (Rmb)	-1.852	-1.454	-0.930	-0.399	-0.203						
BVPS (Rmb)	3.119	3.415	3.730	4.016	4.356						
Wtd avg ord shares (m)	7,464	7,464	7,464	7,464	7,464						
Wtd avg diluted shares (m)	7,464	7,464	7,464	7,464	7,464						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	47.4	14.6	18.7	8.1	9.8						
EBIT (Adj) (%)	42.5	23.7	25.1	3.9	12.5						
Core NPAT (%)	124.6	31.4	8.8	-6.0	13.4						
Core EPS (%)	54.6	31.4	8.8	-6.0	13.4						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	4,338	3,677	4,684	5,142	4,773						
Accounts receivables	3,496	5,157	6,112	6,857	7,541						
Inventory	636	890	926	945	945						
Net fixed & other tangibles	51,735	62,678	74,659	83,091	90,893						
Goodwill & intangibles	7,673	8,174	8,174	8,174	8,174						
Financial & other assets	6,796	9,531	9,632	9,749	9,868						
Total assets	74,675	90,107	104,186	113,958	122,193						
Accounts payable	1,515	1,597	1,662	1,695	1,695						
Short-term debt	17,200	16,369	16,369	16,369	16,369						
Long-term debt	19,975	31,308	42,308	49,308	54,308						
Provisions & other liab	8,560	10,967	10,991	10,996	11,022						
Total liabilities	47,250	60,241	71,331	78,368	83,394						
Shareholders' equity	23,281	25,490	27,844	29,980	32,511						
Minority interests	4,144	4,375	5,012	5,610	6,288						
Total equity	27,425	29,866	32,856	35,589	38,799						
Net debt	32,837	44,001	53,993	60,535	65,904						
Net debt to equity (%)	119.7	147.3	164.3	170.1	169.9						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

CDM Revenue to Shrink in 2013E

Figure 1. ICE Daily CER Index –CDM spot price



Source: Bloomberg (ticker - ICEDCER Index)

Longyuan forecast its 2013E CDM revenue to be 60-70% that of 2012E, assuming (i) spot carbon emission prices to rebound from current €1/tonne to €4-5/tonne in 2013E and (ii) taking into account its increasing registered CDM capacity, from 6,052MW by end 2011 to 8,357MW on 19 Oct 2012, +38% year-to-date. Citi assumes Longyuan's CDM revenue to fall 75% yoy in 2013E expecting demand to remain weak in Europe and supply to increase from developing countries. Our estimate of Longyuan's 2013E net profit is 17% below consensus.

Additionally, the CDM market in China had its first batch of government-directed transactions with four cement companies for testing in Guangzhou recently. But Longyuan expects no material CDM revenue from China until after 2014. China has selected seven provinces and municipality cities for CDM trading tests in 2013.

Figure 2. Longyuan's DCF calculation (Rmbm)

DCF- in the run up to 2020	2012E	2013E	2014E	2015E	2016E	2017E	2018E	2019E	2020E
Net operating cashflow		10,037	11,506	12,852	14,191	15,536	16,890	18,242	19,606
Less: capex		-12,904	-12,904	-12,904	-12,904	-12,904	-12,904	-12,904	-12,904
FCF		-2,867	-1,398	-52	1,287	2,632	3,986	5,338	6,702
Discount factor		1.00	1.07	1.14	1.22	1.31	1.40	1.49	1.60
Net debt by end 2012E	-56,581	-2,867	-1,307	-45	1,054	2,015	2,853	3,574	4,197
DCF - Terminal Value									
NAV in 2021E		6,970							
Expected terminal growth rate		4.0%							
Capital Employed on new capex		9.5%							
WACC		6.9%							
Terminal Value		138,508							
Discount factor		1.71							
NPV of terminal value		81,136							
Total DCF									
DCF - in the run up to 2020E	-47,107								
DCF- from 2021E		81,136							
Total DCF		34,030							
Exchange Rate		1.23							
Total DCF (in HK\$m)		41,800							
No. of share (m)		7,464							
DCF / share (HK\$)		5.60							

Source: Company Reports and Citi Research Estimates

Utilization +5% YoY in Oct, -3% YoY in 10M12

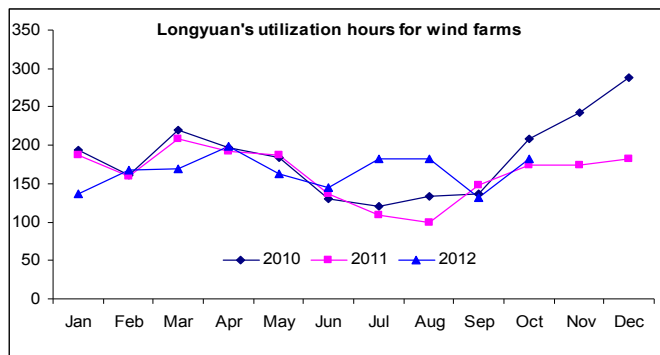
Figure 3. Longyuan's monthly utilization hours +5% yoy in Oct 2012 but -3% yoy in 10M12

Province	Power Network	Oct 11	Oct 12	yoy	10M11	10M12	yoy
Yunnan	Southern	142	160	13%	1,956	2,461	26%
Shanxi	Northern	0	333	n/a	0	2,410	na
Xinjiang	Northwest	177	227	28%	1,892	2,319	23%
Fujian	Eastern	383	298	-22%	2,515	2,154	-14%
Jiangsu	Eastern	151	132	-13%	1,674	1,776	6%
Tianjin	Northern	157	183	16%	0	1,620	na
Hebei	Northern	163	190	17%	1,932	1,615	-16%
Inner Mongolia	Northern	171	202	18%	1,408	1,589	13%
Zhejiang	Eastern	163	139	-15%	1,273	1,518	19%
Shandong	Northern	140	197	40%	716	1,394	95%
Heilongjiang	Northeast	236	188	-20%	1,529	1,394	-9%
Anhui	Eastern	155	106	-31%	983	1,347	37%
Ningxia	Northwest	0	149	n/a	0	1,294	na
Liaoning	Northeast	172	184	7%	1,600	1,281	-20%
Gansu	Northwest	87	132	53%	1,531	1,272	-17%
Hainan	Southern	208	187	-10%	1,276	1,203	-6%
Jilin	Northeast	137	137	0%	1,288	1,049	-19%
Guizhou	Southern	0	177	n/a	0	638	na
Total		174	183	5%	1,601	1,551	-3%

Source: Company Reports and Citi Research Estimates

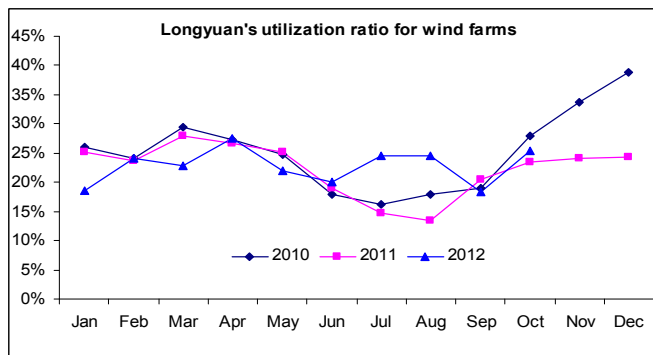
Longyuan's consolidated generation was +9.8% yoy to 2.38M MWh in Oct, including +30% yoy to 1.59M MWh from wind but -20% yoy to 0.74M MWh from coal. Average utilization of its wind farms was +5% yoy to 183 hours in Oct, including increments in eight provinces but decline in another six ones.

Figure 4. Longyuan's monthly average wind farm utilization hour(s)



Source: Company Reports and Citi Research Estimates

Figure 5. Longyuan's monthly average wind farm utilization rate (%)



Source: Company Reports and Citi Research Estimates

Figure 6. Renewable Energy Companies — Valuation Comparison (Pricing date 28 Nov)

Company	Stock code	Citi Rating	Share Price	Target Price	Upside/downside	Mkt cap (US\$m)	PER		PB		ROE 2012E
							2012E	2013E	2012E	2013E	
China Longyuan	916 HK	Buy	HK\$5.00	HK\$5.60	12%	4,999	10.9	11.5	1.1	1.0	10.8%
EDP Renovaveis	EDPR PL	Buy	EUR3.70	EUR4.60	24%	4,180	23.2	17.0	0.6	0.6	2.6%
Acciona	ANA SM	Neutral	EUR48.00	EUR50.00	4%	3,560	20.3	13.6	0.5	0.5	2.7%
Huaneng Renewables	958 HK	Buy	HK\$1.17	HK\$1.25	7%	1,275	14.5	9.6	0.7	0.6	4.7%
Datang Renewable Power	1798 HK	Buy	HK\$0.84	HK\$1.08	29%	788	24.3	11.0	0.5	0.5	2.2%
China Suntien	956 HK	Buy	HK\$1.51	HK\$1.95	29%	631	9.2	7.7	0.8	0.7	8.3%
Simple average							17.1	11.7	0.7	0.6	5.0%
Weighted average							16.2	12.3	0.7	0.7	5.3%

Source: Company Reports and Citi Research Estimates

Financial Statements and Key Assumptions

Figure 7. Consolidated Income Statement

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12F	FY13F	FY14F
Sales of electricity	8,544	10,270	12,751	14,504	16,032
Sales of steam	311	319	335	352	370
Service concession construction revenue	1,450	793	833	874	918
Sales of electricity equipment	262	289	318	350	385
Sales of coal	3,276	4,109	4,520	4,972	5,469
CERs and VERs income	392	727	950	238	238
Other revenue	968	922	979	1,064	1,136
Total revenue	15,204	17,431	20,686	22,354	24,548
Coal consumption	-2,737	-2,877	-2,994	-3,054	-3,054
Service concession construction costs	-1,450	-793	-833	-874	-918
Repairs and maintenance	-184	-254	-318	-347	-388
Depreciation	-2,237	-2,845	-3,841	-4,472	-5,102
Labor cost	-666	-769	-1,046	-1,204	-1,337
S&A expenses	-222	-294	-369	-407	-436
Other operating expenses	-3,635	-4,562	-4,984	-5,447	-5,942
Operating profit	4,073	5,036	6,301	6,549	7,371
Interest income	79	400	87	115	116
Interest expense	-1,179	-1,887	-2,620	-3,115	-3,445
Profit from associates	228	60	101	117	119
Profit before taxation	3,200	3,609	3,869	3,666	4,160
Taxation	-441	-305	-364	-373	-424
Minority interest	-751	-667	-636	-598	-678
Net profit	2,008	2,638	2,869	2,696	3,058

Source: Company Reports and Citi Research Estimates

Figure 8. Longyuan's net profit estimates – Citi vs. Consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	2,869	2,905	-1%
FY13E	2,696	3,223	-16%
FY14E	3,058	3,890	-21%

Source: Citi Research and Bloomberg

Our FY12E net profit is in line with consensus. And our FY13-14E net profits are 16-21% below consensus as we assume Longyuan's CDM revenue to -75% yoy in 2013E for lowered CDM unit sale prices.

Figure 9. Key Ratios

Year-end 31 Dec (%)	FY10	FY11	FY12F	FY13F	FY14F
Gross margin (%)	56.5	61.2	61.4	60.9	61.5
Operating margin (%)	26.8	28.9	30.5	29.3	30.0
Operating margin (%) - wind plants	69.0	60.6	54.0	53.2	52.3
Operating margin (%) - coal fired	10.3	7.3	6.9	7.0	7.0
Net margin (%)	12.9	14.7	13.7	11.9	12.3
Net debt to total capital (x)	0.3	0.5	0.5	0.5	0.5
Net debt to equity (%)	128	156	172	177	177
Interest coverage ratio (x)	3.7	2.9	2.5	2.2	2.2
ROA (%)	2.8	3.2	3.0	2.5	2.6
ROE (%)	8.9	10.8	10.8	9.3	9.8
Three-year EPS CAGR (%)	30.2	10.3	5.0	6.4	13.4

Source: Company Reports and Citi Research Estimates

Figure 10. Key Assumptions

Capacity (MW)	FY10	FY11	FY12F	FY13F	FY14F
Total Installed capacity - year end	8,812	10,970	12,831	14,431	16,031
yoy change (%)	30.6	24.5	17.0	12.5	11.1
Wind power installed capacity	6,895	8,995	10,636	12,236	13,836
yoy change (%)	42.4	30.5	18.2	15.0	13.1
Coal fired installed capacity	1,875	1,875	1,875	1,875	1,875
yoy change (%)	0.0	0.0	0.0	0.0	0.0
Solar & biomass installed capacity	42	100	320	320	320
yoy change (%)	45.0	138.4	220.2	0.0	0.0
Consolidated capacity - year end	8,473	10,573	12,492	14,092	15,692
yoy change (%)	32.2	24.8	18.1	12.8	11.4
Wind power consolidated capacity	6,556	8,598	10,297	11,897	13,497
yoy change (%)	45.6	31.2	19.8	15.5	13.4
Coal fired consolidated capacity	1,875	1,875	1,875	1,875	1,875
yoy change (%)	0.0	0.0	0.0	0.0	0.0
Solar & biomass consolidated capacity	42	100	320	320	320
yoy change (%)	45.0	138.4	220.2	0.0	0.0
Attributable capacity - year end	6,349	8,178	9,833	11,209	12,586
yoy change (%)	46.8	28.8	20.2	14.0	12.3
Wind power attributable capacity	5,768	7,597	9,159	10,535	11,912
yoy change (%)	53.5	31.7	20.6	15.0	13.1
Coal fired attributable capacity	539	539	539	539	539
yoy change (%)	0.0	0.0	0.0	0.0	0.0
Others attributable capacity	42	42	135	135	135
yoy change (%)	50.5	0.0	224.5	0.0	0.0
Generation volume (M MWh)					
Consolidated gross generation	21.6	25.3	29.9	33.7	37.0
Wind power	10.1	13.4	17.2	20.6	23.9
Coal fired	11.4	11.7	11.9	12.1	12.1
Others	0.1	0.2	0.9	1.0	1.0
Consolidated net generation	20.2	23.7	28.1	31.6	34.7
yoy change (%)	26.7	17.2	18.6	12.5	9.8
Wind power	9.4	12.5	16.2	19.5	22.6
Coal fired	10.7	11.0	11.1	11.4	11.4
Others	0.1	0.1	0.7	0.7	0.7
Utilization					
Utilization hours - consolidated wind plants	2,217	2,026	1,924	1,946	1,963
yoy change (%)	-3.3	-8.6	-5.0	1.2	0.9
Utilization rate (%)	25.3	23.1	22.0	22.2	22.4
Utilization hours - consolidated coal fired plants	6,055	6,266	6,321	6,448	6,448
yoy change (%)	4.1	3.5	0.9	2.0	0.0
Utilization rate (%)	69.1	71.5	72.2	73.6	73.6
Tariff (Rmb/MWh)					
Average consolidated tariff - wind power (before 17% VAT)	572	578	578	581	580
yoy change (%)	0.9	1.1	0.1	0.5	-0.1
Average consolidated tariff - coal fired (before 17% VAT)	423	425	448	448	448
yoy change (%)	0.6	0.5	5.4	0.0	0.0
Coal price					
Unit fuel cost (Rmb/MWh) - consolidated plants	256	260	269	269	269
yoy change (%)	14.3	1.6	3.2	0.0	0.0
Standard coal cost (Rmb/ton)	755	793	817	817	817
yoy change (%)	14.5	5.0	3.0	0.0	0.0
Net coal consumption rate (g/kWh)	295	293	292	290	289
yoy change (%)	-0.9	-0.5	-0.5	-0.5	-0.5

Source: Company Reports and Citi Research Estimates

China Longyuan Power Group

Company description

Longyuan is the largest wind power generator in China in terms of installed wind power capacity by end 2011. Its wind farms are mainly located in the three Northeast Provinces, Inner Mongolia, the Southeast coastal provinces, Xinjiang, Gansu and Hebei. It also has two coal-fired power plants in Jiangsu. In FY11, 83% of its operating profit came from wind farms. Its parent company, Guodian Power Group, is one of the Big 5 Chinese IPP groups.

Investment strategy

We rate Longyuan a Buy assuming the company to have wind capacity growth of at least 1.6GW pa in 2012-15E to drive profit growth. Being a first mover, Longyuan has the lowest grid curtailment ratio among its peers. Its earning quality would also improve with profits from wind farms to rise from 78% in 2009 to over 90% in 2012E, implying lower fuel cost risk.

Valuation

Our HK\$5.60 target price for Longyuan is based on a DCF model, which we believe is suitable for power companies such as Longyuan that operate in a relatively stable industry with predictable cash flows. Our model factors in earnings forecasts up to 2020 and a terminal growth rate of 5% thereafter. Our WACC for Longyuan is 6.9%, which assumes a risk-free rate of 5.2%, a market risk premium of 6.8%, an equity beta of 0.85x (according to Bloomberg), cost of debt of 6.5%, a target debt-to-capital ratio of 67%, and a 25% corporate-tax rate. At our DCF-based target price, Longyuan would trade at a 2013E P/E of 13.5x and P/B of 1.2x.

Risks

Key downside risks that could prevent the shares from reaching our target price include: 1) much lower utilization hours of the wind farms due to unpredictable weather conditions; 2) less than expected capacity addition; and 3) Rmb interest rate hike.

China Power International (2380.HK)

Diversified Profit from Both Coal-Fired & Hydropower Generation

Buy	1
Price (28 Nov 12)	HK\$2.03
Target price	HK\$2.50
Expected share price return	23.2%
Expected dividend yield	6.4%
Expected total return	29.6%
Market Cap	HK\$11,267M
	US\$1,454M

Price Performance (RIC: 2380.HK, BB: 2380 HK)



■ **Summary** – We raise our China Power International (CPI) net profit estimates by 1%/7%/10% for FY12E/FY13E/FY14E on lower unit fuel cost. Attributable net profit of its hydropower plants was +207% yoy to Rmb309m at 9M12 amid strong rainfall. We estimate its FY12E net profit breakdown as 71% from coal-fired and 29% from hydropower. Our DCF TP is HK\$2.50. CPI is inexpensive at 6.6x FY13E PER and 0.6x PB, which are below their historical means by 48% and 40%, respectively.

■ **Hydropower profit +2x yoy in 9M12** – CPI expects its hydropower net generation to be +40% yoy to 12M MWh in 2012E, including 10.6M MWh in 10M12, amid more rainfall volume. Net profit of its 63%-held hydropower arm, Wuling Hydropower Power Ltd, was +207% yoy to Rmb491m at 9M12, of which 63% belonged to CPI.

■ **Declining unit fuel cost of coal-fired plants** – CPI would consume 15m tonnes of raw coal in 2012E, including 50% from key contracts, 20% from normal contracts and 30% from spot market. Its average unit fuel cost of coal-fired plants dropped from Rmb266/MWh in 1H12 to Rmb259/MWh in 10M12, including Rmb239/MWh in Oct; we assume it to be -1% yoy to Rmb254/MWh in FY12E and -2% yoy in FY13E.

■ **Utilization cut of coal-fired plants** – CPI's consolidated net generation of its coal-fired plants would be +3% yoy to 40M MWh in 2012E, including 30.5M MWh in 10M12. The new target is 7-8% below its target set at the year beginning amid a weakened economy and substitution by high hydropower output this year. Average utilization of its coal-fired plants would be -12% yoy to 5,100 hours in 2012E.

■ **Earnings hikes** – We raise our net profit estimates by 1%/7%/10% for FY12E/FY13E/FY14E on lower unit fuel cost, offsetting the dilution effect from 443m (8%) new shares issued in August. Our DCF TP remains at HK\$2.50, close to the exercise price of HK\$2.52/share for its Rmb1.14bn five-year CB issued in Aug 2012. If the CBs were converted, it would add 10% to its number of issued shares.

Asian Utilities

China Power Sector

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Statistical Abstract

FYE	Net Profit	EPS	P/E	D. Yld
31 Dec	(RmbM)	(Rmb)	(x)	(%)
2009A	519	0.142	11.5	2.8
2010A	667	0.131	12.4	2.8
2011A	505	0.100	16.4	2.8
2012E	1,216	0.231	7.1	6.4
2013E	1,365	0.246	6.6	6.9
2014E	1,486	0.268	6.1	7.5

Source: Powered by dataCentral

Earnings Revision

FYE	Net Profit	EPS	%	DPS
31 Dec	(RmbM)	(Rmb)	chg	(Rmb)
2012E	1,216	0.231	-2.9	0.105
Prev	1,207	0.238		0.107
2013E	1,365	0.246	-2.0	0.112
Prev	1,274	0.251		0.114
2014E	1,486	0.268	0.5	0.122
Prev	1,353	0.266		0.120

Source: Powered by dataCentral

2380.HK: Fiscal year end 31-Dec						Price: HK\$2.03; TP: HK\$2.50; Market Cap: HK\$11,267m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	14,622	16,317	18,266	20,574	24,496	PE (x)	12.4	16.4	7.1	6.6	6.1
Cost of sales	-8,293	-9,940	-9,886	-11,108	-13,545	PB (x)	0.7	0.6	0.6	0.6	0.5
Gross profit	6,329	6,376	8,380	9,466	10,952	EV/EBITDA (x)	9.9	11.0	8.0	7.5	7.0
Gross Margin (%)	43.3	39.1	45.9	46.0	44.7	FCF yield (%)	-21.8	-32.2	-26.7	-30.0	-24.8
EBITDA (Adj)	4,141	4,018	5,819	6,554	7,518	Dividend yield (%)	2.8	2.8	6.4	6.9	7.5
EBITDA Margin (Adj) (%)	28.3	24.6	31.9	31.9	30.7	Payout ratio (%)	34	45	45	45	45
Depreciation	-1,712	-1,810	-2,029	-2,565	-3,043	ROE (%)	5.4	4.0	8.7	8.9	9.2
Amortisation	0	0	0	0	0	Cashflow (Rmbm)					
EBIT (Adj)	2,429	2,208	3,790	3,989	4,475	EBITDA	4,141	4,018	5,819	6,554	7,518
EBIT Margin (Adj) (%)	16.6	13.5	20.8	19.4	18.3	Working capital	-600	-316	786	-227	-592
Net interest	-1,410	-1,463	-1,777	-1,785	-2,236	Other	-1,898	-2,343	-2,199	-2,260	-2,752
Associates	94	-28	122	168	321	Operating cashflow					
Non-op/Except	134	65	102	140	174	Capex	-3,449	-4,028	-6,697	-6,788	-6,418
Pre-tax profit	1,246	782	2,237	2,512	2,734	Net acq/disposals	-1,506	-1,221	0	0	0
Tax	-380	-194	-503	-565	-615	Other	-1,264	234	0	0	0
Extraord./Min.Int./Pref.div.	-199	-83	-518	-581	-633	Investing cashflow					
Reported net profit	667	505	1,216	1,365	1,486	Dividends paid	0	0	-230	-553	-621
Net Margin (%)	4.6	3.1	6.7	6.6	6.1	Financing cashflow					
Core NPAT	667	505	1,216	1,365	1,486	Net change in cash	-932	212	1,029	-244	165
Per share data						Free cashflow to s/holders					
Reported EPS (Rmb)	0.131	0.100	0.231	0.246	0.268		-1,806	-2,669	-2,291	-2,720	-2,244
Core EPS (Rmb)	0.131	0.100	0.231	0.246	0.268						
DPS (Rmb)	0.045	0.045	0.105	0.112	0.122						
CFPS (Rmb)	0.324	0.268	0.836	0.733	0.752						
FCFPS (Rmb)	-0.356	-0.526	-0.435	-0.490	-0.404						
BVPS (Rmb)	2.396	2.570	2.678	2.824	2.980						
Wtd avg ord shares (m)	5,107	5,107	5,268	5,550	5,550						
Wtd avg diluted shares (m)	5,077	5,077	5,268	5,550	5,550						
Growth rates											
Sales revenue (%)	32.7	11.6	12.0	12.6	19.1						
EBIT (Adj) (%)	94.9	-9.1	71.7	5.2	12.2						
Core NPAT (%)	28.5	-24.2	140.7	12.3	8.8						
Core EPS (%)	-7.2	-24.2	131.9	6.6	8.8						
Balance Sheet (Rmbm)											
Cash & cash equiv.	977	1,180	2,209	1,965	2,130						
Accounts receivables	1,717	3,411	3,819	4,302	5,122						
Inventory	336	731	727	817	996						
Net fixed & other tangibles	47,721	50,621	55,289	59,512	62,887						
Goodwill & intangibles	0	0	0	0	0						
Financial & other assets	6,039	7,449	7,591	7,809	8,204						
Total assets	56,790	63,392	69,635	74,404	79,339						
Accounts payable	461	875	871	978	1,193						
Short-term debt	8,914	7,600	9,368	9,368	11,486						
Long-term debt	29,169	32,739	33,768	36,798	37,709						
Provisions & other liab	3,352	5,689	6,883	7,121	7,314						
Total liabilities	41,896	46,903	50,890	54,265	57,702						
Shareholders' equity	12,238	13,125	14,863	15,675	16,540						
Minority interests	2,656	3,365	3,882	4,464	5,097						
Total equity	14,894	16,490	18,745	20,139	21,636						
Net debt	37,106	39,159	40,927	44,201	47,065						
Net debt to equity (%)	249.1	237.5	218.3	219.5	217.5						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Key Takeaways of Recent Company Update

Hydropower Output Surge in 2012E

CPI expects its hydropower net generation to be +40% yoy to 12M MWh in 2012E, including 10.6M MWh in 10M12, driven by more rainfall volume. Net profit of its hydropower arm, Wuling Hydropower Power Ltd, therefore was +207% yoy to Rmb491m at 9M12. CPI's average on-grid hydropower tariff would also be +0.5% yoy to Rmb284/MWh in 2012E on the generation mix change and cancellation of discounted tariffs at some plants for certain periods.

Figure 1. Financial data of Wu Ling Power - CPI's hydropower arm, being 63% held by CPI (PRC GAAP)

(Rmbm)	3M11	9M11	3M12	YoY	9M12	YoY
Revenue and other income	844	2,295	426	-50%	2,858	25%
Net profit	148	160	-258	-274%	491	207%
Total assets	30,690	32,440	33,347	9%	35,230	9%
Total liabilities	25,397	27,215	28,705	13%	29,838	10%

Source: Company Reports and Citi Research Estimates

Declining Unit Fuel Cost of Coal-Fired Plants

CPI would consume 15m tonnes of raw coal in 2012E. Its coal supply is ample with inventory average of 28 days at power plants in coastal provinces and 10 days at mine-mouth projects. The average unit fuel cost of CPI's coal-fired plants dropped from Rmb266/MWh in 1H12 to Rmb259/MWh in 10M12, including Rmb239/MWh in October. We keep our assumption for the unit fuel cost to be -1% yoy to Rmb254/MWh in FY12E. Its coal supply comprises:

- (i) 50% from key contracts with prices set by NDRC;
- (ii) 20% from normal contracts with prices set by negotiation between coal miners and CPI; and
- (iii) 30% from spot market.

CPI's average dark spread (= average tariff less unit fuel cost) of its coal-fired plants widened to Rmb145/MWh in Sep, from Rmb111/MWh in 1H12. The commissioning of its Shentou Power Plant Ph.1 (2x600MW) in 2013E would raise its average dark spread in view of the low coal unit cost and above company average dark spread there.

Reduced Utilization of Coal-Fired Plants

CPI estimated consolidated net generation of its coal-fired plants to be +3% yoy to 40M MWh in 2012E, including 30.5M MWh in 10M12. The new target is 7-8% below its target set at year beginning amid a weakened economy and substitution by high hydropower output this year. Average utilization of its coal-fired plants would be -12% yoy to 5,100 hours this year, with the sharpest decline in Henan and Sichuan provinces. From 2013E onwards, CPI expects PRC power demand growth at 1.0x GDP growth in future, down from 1.3x in 2006-10.

CPI's 50%-owned Xintang Thermal Plant in Guangdong province had its two 300MW units commence operation in Aug and Oct 2012. The company expects this plant to be around the break-even level only in 2H12E owing to the low tariff in its first year of operation. Its profitability should be better in 2013E at the higher standard benchmark tariff.

Relatively High Capex for Expansion

CPI's capex will be +25% yoy to Rmb6.8bn in 2012E, including Rmb3.2bn for coal-fired plants and Rmb3.6bn for hydropower. Its capex will remain at a similar level in the next year and net debt to equity ratio at around 223%.

Figure 2. CPI's attributable capacity expansion

	Location	Stake	Capacity	Fuel	2011 11,585	2012E 11,510	2013E 12,487	2014E 13,825	Completion date
Capacity at year start:									
Add - greenfield development									
Pingwei Ph.II	Anhui	100%	2X640	Coal					Unit 1: 19 Mar 07; unit 2: 24 Nov 07
Yaomeng Ph.II	Henan	100%	2X630	Coal					Unit 1: 26 Oct 07; unit 2: 29 Dec 07
Dabieshan	Hubei	93%	2X640	Coal					Unit 1: May 08; unit 2: Sep 08
Fuxi	Sichuan	51%	2X600	Coal	306	306			2011-2012
Xintang Thermal (associate)	Guangdong	50%	2X300	Coal		300			Aug 12, Oct 12
Shentou I	Shanxi	80%	2X600	Coal			960		2Q-3Q13E
Pingwei III	Anhui	51%	2X1,000	Coal				1,020	2H14E
Changshu (associate) **	Jiangsu	50%	2X1,000	Coal				1,000	2H14E
Wuling Hydro Plant *	Hunan/Guizhou	63%	n/a	Hydro	378	371	377	37	
Capacity at year end:					11,510	12,487	13,825	15,881	
- yoy					-1%	8%	11%	15%	

Source: Company Reports and Citi Research Estimates

* Hydropower addition of 371MW in 2012E and 398MW in 2013E via Wuling, which has 95% stake Baishi plant (3X140MW) to operate in 2012E, 95% stake Tuokou plant (4X200MW+2X15MW) in 2013E, except 1X200MW in 4Q12E, 100% owned Suoluogou I plant (10MW) and 100% held Jiesigou plant (24MW)

Figure 3. PRC IPPs: FY13E Net Profit Sensitivity

(Rmbm)	Huaneng	Datang	Huadian	CRP	CPI	Average
Base case net profit *	7,932	3,647	1,927	8,078	1,365	n/a
Case 1. net profit change with extra 1% cut in average coal price	454	157	280	291	71	n/a
% change	5.7%	4.3%	14.5%	3.6%	5.2%	6.7%
Case 2. net profit change with extra 1% hike in power tariff	746	299	626	460	125	n/a
% change	9.4%	8.2%	32.5%	5.7%	9.2%	13.0%
Case 3. net profit change with extra 1% cut in utilization hours	-272	-110	-121	-174	-73	n/a
% change	-3.4%	-3.0%	-6.3%	-2.2%	-5.3%	-4.0%
Case 4. net profit change with extra 25bps interest rate cut	242	213	287	205	43	n/a
% change	3.0%	5.8%	14.9%	2.5%	3.2%	5.9%

Source: Company Reports and Citi Research Estimates

* In Rmbm except CRP which is in HK\$m

** Key assumptions - (i) unit fuel cost ranges from -2% yoy (Huaneng, Datang, CRP & CPI) to -5% yoy (for Huadian) in 2013E; (ii) average 6.5% tariff hike on 1 Dec 2011, but nil in 2012-13E

Figure 4. CPI's earnings forecast – Citi vs. consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	1,216	1,267	-4%
FY13E	1,365	1,550	-12%
FY14E	1,486	1,657	-10%

Source: Bloomberg and Citi Research

DCF Calculation & Historical Valuations

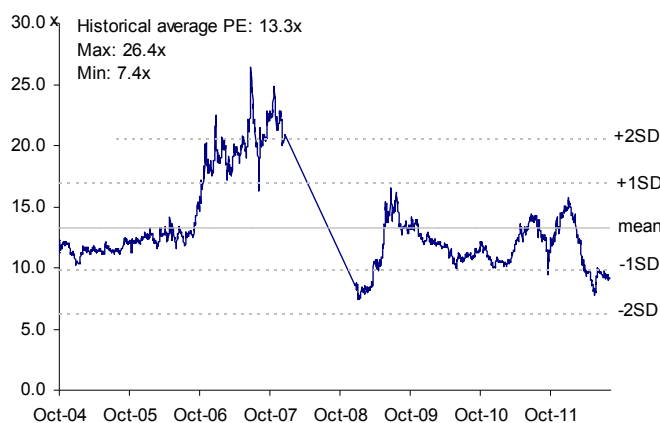
Figure 5. CPI's DCF calculation (Rmbm)

DCF – in the run up to 2010	2012	2013	2014	2015	2016	2017	2018	2019	2020
Net operating cashflow		4,068	4,174	4,306	4,766	5,152	5,370	5,576	5,795
Less: capex		-6,788	-6,418	-6,000	-945	-945	-945	-945	-945
FCF		-2,720	-2,244	-1,694	3,820	4,207	4,425	4,631	4,850
Discount factor		1.00	1.07	1.15	1.24	1.33	1.42	1.53	1.64
Net debt by end 2012E/NAV from 2013E	-40,927	-2,720	-2,091	-1,470	3,091	3,171	3,108	3,031	2,957
DCF - Terminal Value									
NAV in 2021E		5,044							
Expected terminal growth rate		4.0%							
Capital Employed on new capex		8.0%							
WACC		7.3%							
Terminal Value		75,933							
Discount factor		1.76							
NPV of terminal value		43,147							
Total DCF									
DCF - in the run up to 2020E	-31,851								
DCF- from 2021E		43,147							
Total DCF		11,296							
Exchange Rate		1.23							
Total DCF (in HK\$m)		13,876							
No. of share (m)		5,550							
DCF / share (HK\$)		2.50							

Source: Company Reports and Citi Research Estimates

Our DCF TP is HK\$2.50. At our DCF TP of HK\$2.50, CPI would be at 8.2x FY13E PER and 5.5% yield, which looks non excessive.

Figure 6. CPI's Historical Rolling PE Chart



Source: Citi Research

Figure 7. CPI's Historical Rolling PB Chart



Source: Citi Research

Financial Statement & Key Assumptions

Figure 8. Consolidated Income Statement

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12F	FY13F	FY14F
Power sale revenue	14,437	16,082	17,973	20,252	24,142
Other revenue	185	234	293	322	355
Total revenue	14,622	16,317	18,266	20,574	24,496
Coal consumption	-8,293	-9,940	-9,886	-11,108	-13,545
Repairs and maintenance	-480	-490	-501	-589	-738
Depreciation	-1,712	-1,810	-2,029	-2,565	-3,043
Labor cost	-747	-825	-935	-1,057	-1,219
S&A expenses	-180	-187	-214	-276	-361
Other operating expenses	-780	-856	-911	-990	-1,115
Operating profit	2,429	2,208	3,790	3,989	4,475
Interest income	104	109	113	132	130
Interest expense	-1,514	-1,572	-1,890	-1,917	-2,366
Profit from associates	94	-28	122	168	321
Dividend from Shanghai Electric Power	0	0	20	50	75
Other gain	134	65	82	90	99
Profit before taxation	1,246	782	2,237	2,512	2,734
Taxation	-380	-194	-503	-565	-615
Minority interest	-199	-83	-518	-581	-633
Net profit	667	505	1,216	1,365	1,486

Source: Company Reports and Citi Research Estimates

Figure 9. Consolidated Cash Flow

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12F	FY13F	FY14F
Cashflow from operations					
EBITDA	4,141	4,018	5,819	6,554	7,518
Working capital changes	-314	-121	1,193	256	228
Other operating cash items	-2,125	-2,380	-2,216	-2,310	-3,087
Gross operating cash flow	1,703	1,518	4,797	4,500	4,659
Net interest paid	104	109	113	132	130
Tax paid	-165	-277	-503	-565	-615
Net operating cash flow	1,642	1,350	4,406	4,068	4,174
Capex	-3,449	-4,028	-6,697	-6,788	-6,418
Free cash flow	-1,807	-2,678	-2,291	-2,720	-2,244
Cash from acquisitions	-1,506	-1,221	0	0	0
Disposal of fixed assets	25	94	0	0	0
Change in temp cash inv.	-1,289	140	0	0	0
Cashflow from investing	-6,220	-5,015	-6,697	-6,788	-6,418
Increase in Equity	106	222	752	0	0
Ordinary Dividend Paid	0	0	-230	-553	-621
Change in borrowings	3,539	3,646	2,798	3,030	3,030
Cash Flow from Financing	3,644	3,868	3,320	2,476	2,409
Beginning cash balance	1,911	977	1,180	2,209	1,965
Ending cash balance	977	1,180	2,209	1,965	2,130

Source: Company Reports and Citi Research Estimates

Figure 10. Consolidated Balance Sheet

As at 31 Dec (Rmbm)	FY10	FY11	FY12F	FY13F	FY14F
Property, plant and equipment	39,047	44,646	42,617	40,052	46,777
Construction in progress	5,903	4,704	11,401	18,189	14,840
Associated companies	1,723	1,876	2,018	2,235	2,631
Other tangible fixed assets	4,505	3,333	3,333	3,333	3,333
Goodwill	1,837	2,386	2,386	2,386	2,386
Total fixed assets	53,015	56,945	61,755	66,196	69,966
Cash and cash equivalents	977	1,180	2,209	1,965	2,130
Marketable securities	717	792	792	792	792
Inventories	336	731	727	817	996
Debtors	1,743	3,744	4,151	4,634	5,454
Others	1	1	1	1	1
Total current assets	3,775	6,447	7,880	8,208	9,372
Total assets	56,790	63,392	69,635	74,404	79,339
S/T loans	6,555	4,880	4,880	4,880	4,880
Current portion of L/T loans	2,359	2,720	4,488	4,488	6,606
Account payable	461	875	871	978	1,193
Other payable	2,432	4,124	4,124	4,124	4,124
Taxes payable	256	748	1,942	2,180	2,373
Total current liabilities	12,063	13,347	16,304	16,650	19,176
Long term bank loans	24,141	26,881	27,911	30,940	31,852
Other loans	5,692	6,675	6,675	6,675	6,675
Minority interests	2,656	3,365	3,882	4,464	5,097
Total L/T liabilities	32,489	36,920	38,468	42,079	43,623
Share capital	5,121	5,121	5,565	5,565	5,565
Other capital	4,303	4,303	4,612	4,612	4,612
Retained earnings	2,814	3,700	4,686	5,498	6,363
Total equity	12,238	13,125	14,863	15,675	16,540
Total liabilities & equity	56,790	63,392	69,635	74,404	79,339

Source: Company Reports and Citi Research Estimates

Figure 11. Key Ratios

Year-end 31 Dec (%)	FY10	FY11	FY12F	FY13F	FY14F
Operating margin	16.6	13.5	20.8	19.4	18.3
Net margin (%)	4.5	3.1	6.5	6.5	5.9
Net debt to total capital (x)	0.6	0.64	0.61	0.62	0.62
Net debt to equity (x)	2.54	2.42	2.23	2.24	2.21
Interest coverage ratio (x)	1.7	1.5	2.1	2.2	2.1
Cash interest coverage (x)	2.7	2.0	2.4	2.6	3.0
Debt to EBITDA ratio	9.4	10.2	0.0	0.0	0.0
ROA (%)	1.2	0.8	1.8	1.9	1.9
Three-year EPS CAGR (%)	17.7	23.2	39.1	4.0	2.1

Source: Company Reports and Citi Research Estimates

Figure 12. Key Assumptions

	FY10	FY11	FY12F	FY13F	FY14F
Capacity - year end (MW)					
Installed capacity	12,603	13,833	15,622	17,421	19,421
yoy change (%)	2.3	9.8	12.9	11.5	11.5
Consolidated capacity	11,146	12,346	14,166	15,996	18,654
yoy change (%)	6.6	10.8	14.7	12.9	16.6
Consolidated capacity - coal-fired	7,060	7,660	8,860	10,060	12,060
yoy change (%)	1.3	8.5	15.7	13.5	19.9
Consolidated capacity - hydro	4,086	4,686	5,306	5,936	6,594
yoy change (%)	17.2	14.7	13.2	11.9	11.1
Attributable capacity - year end *	11,585	11,510	12,487	13,824	15,881
yoy change (%)	3.6	-0.6	8.5	10.7	14.9
* Including Shanghai Electric Power					
Generation volume (M MWh)					
Gross generation	55.2	57.6	61.6	71.5	83.0
Net generation	52.3	54.4	58.2	67.5	78.3
yoy change (%)	27.5	3.9	7.1	16.0	16.0
Consolidated gross generation	48.6	50.1	53.4	60.3	65.9
Consolidated net generation	46.0	47.4	50.4	57.0	67.7
yoy change (%)	32.5	3.0	6.5	13.0	18.8
Consolidated net generation - coal-fired	35.2	38.8	39.0	44.8	54.2
Consolidated net generation - hydro	10.8	8.6	11.5	12.2	13.5
Utilization					
Utilization hours - consolidated coal-fired plants	5,329	5,783	5,116	5,052	5,175
yoy change (%)	5.1	8.5	-11.5	-1.3	2.4
Utilization rate (%)	60.8	66.0	58.4	57.7	59.1
Utilization hours - consolidated hydropower plants	3,190	2,502	3,300	3,000	3,000
yoy change (%)	-4.8	-21.6	31.9	-9.1	0.0
Utilization rate (%)	36.4	28.6	37.7	34.2	34.2
Tariff (Rmb/MWh)					
Average consolidated tariff (with 17% VAT)	367	397	417	416	417
yoy change (%)	-0.4	8.1	5.0	-0.2	0.4
Average consolidated tariff (ex-VAT)	314	339	356	355	357
yoy change (%)	-0.4	8.1	5.0	-0.2	0.4
Average consolidated tariff -coal-fired (ex-VAT)	328	348	377	375	375
yoy change (%)	4.2	6.1	8.3	-0.7	0.0
Average consolidated tariff - hydro (ex-VAT)	258	282	282	282	282
yoy change (%)	n/a	9.0	0.0	0.0	0.0
Coal price					
Unit fuel cost (Rmb/MWh) - consolidated plants	236	256	254	248	250
yoy change (%)	10.5	8.7	-1.0	-2.2	0.8
Net coal consumption rate (g/kWh)	325	319	319	319	319
yoy change (%)	-1.6	-1.6	0.0	0.0	0.0

Source: Company Reports and Citi Research Estimates

China Power International

Company description

Incorporated in Hong Kong in 1994, China Power International (CPI) is the flagship of the CPI Group, one of five power generating groups formed after the break-up of the former State Power Corporation (SPC) in December 2002. By end-FY11, CPI had attributable capacity of 11,510MW including 3,057 from Wuling Hydro. The company plans to expand by developing greenfield projects due to be completed in 2012E.

Investment strategy

We rate CPI shares Buy (1) with a HK\$2.50 DCF-based target price. We like the stock for its cheap valuation: FY13E P/B is at a large discount to the company's historical average. Its profit mix is diversified including about 70% from coal-fired plants and around 30% from hydropower ones in 2012E. Major concerns are higher-than-expected coal costs and the lower than expected hydropower generation volume, given the potential impact on CPI's net profit.

Valuation

Our target price of HK\$2.50 is based on a DCF model, which we think is suitable for power companies such as CPI that operate in a relatively stable industry with predictable cash flows. Our model factors in our earning forecasts up to 2020E and a terminal growth rate of 4% thereafter. Our WACC for CPI is 7.9%, which assumes (i) a risk-free rate of 5.2%, (ii) a market risk premium of 6.8%, (iii) an equity beta of 0.85x, according to Bloomberg, (iv) a cost of debt of 6.5%, (v) a target debt to capital ratio of 60%, and (vi) a corporate tax rate of 25%. At our DCF-based target price, CPI would trade at a 2012E PER of 8.2x.

Risks

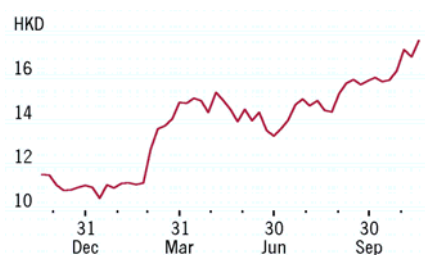
The key factors that could impede the stock from reaching our target price include: 1) Lower-than-expected generation volume; 2) Higher-than-anticipated coal prices; 3) Sharp tariff cuts from power pooling; and 4) Higher-than-expected interest-rate hikes.

China Resources Gas (1193.HK)

Slower EPS Growth after Recent Share Placement

Neutral	2
Price (28 Nov 12)	HK\$16.40
Target price	HK\$15.80
Expected share price return	-3.7%
Expected dividend yield	1.0%
Expected total return	-2.7%
Market Cap	HK\$36,474M
	US\$4,706M

Price Performance (RIC: 1193.HK, BB: 1193 HK)



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- **Maintain Neutral** — We maintain our Neutral rating on China Resources Gas (CR Gas), and keep our target price at HK\$15.80 to factor in higher gas volume growth and earnings dilution impact from recent share placement. Although CR Gas's earnings doubled from 2010 to 2012E, expensive valuation and potential policy impact to its Sichuan and Chongqing projects may cap near-term upside.
- **Rising gearing** — While recent top-up share placement should cover the shareholder's loan made to acquire the final batch of asset-injection from parent, we expect net gearing to rise to 76% by 2013E, from 50% in 2012E, driven by high investment costs (HK\$9bn in 2012) and increase in infrastructure borrowing.
- **How we differ** — 1) Despite significant growth in net earnings, recent asset injection and share placement may affect its ability to maintain 20%+ EPS growth; 2) Sichuan and Chongqing account for 48% of gas sales in 1H12 and 13% of net profit. A change in the gas pricing mechanism may affect profitability from its gas sales; 3) Tianjin project, after receiving final regulatory approval, may require 2-3 years to build out the pipelines before the joint venture turns profitable.
- **Latest developments** — After the company obtained MOFCOM approval for JV in Tianjin on Nov 6, CR Gas raised its shareholding in Zhongshan from 40% to 55% on Nov 23, with capital increase of Rmb88.86 million. The contribution is based on 14X Zhongshan's 2012 earnings, and should help Zhongshan develop gas projects.
- **Revising estimates** — CR Gas will likely remain a key player in the natural gas distributor sector, but in the near term, uncertainty to its growth may narrow its valuation premium to its peers. We revise up our 2012/13/14E earnings by 0.4%/1.4%/1.9% to factor in more aggressive gas volume growth.

Statistical Abstract

Year to 31 Dec	Net Profit (HK\$M)	Diluted EPS (HK\$)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	787	0.54	58.8	30.1	5.1	18.9	0.6
2011A	1,200	0.66	20.6	25.0	4.0	17.2	0.7
2012E	1,600	0.78	18.1	21.2	4.1	18.8	1.0
2013E	1,989	0.89	15.3	18.3	3.6	20.9	1.2
2014E	2,480	1.12	24.7	14.7	3.3	23.4	1.5

Source: Powered by dataCentral

1193.HK: Fiscal year end 31-Dec						Price: HK\$16.40; TP: HK\$15.80; Market Cap: HK\$36,474m; Recomm: Neutral					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	9,331	13,507	17,146	20,979	26,296	PE (x)	30.1	25.0	21.2	18.3	14.7
Cost of sales	-6,594	-9,534	-11,966	-14,767	-18,720	PB (x)	5.1	4.0	4.1	3.6	3.3
Gross profit	2,738	3,972	5,180	6,212	7,576	EV/EBITDA (x)	22.4	16.9	11.2	9.8	8.6
Gross Margin (%)	29.3	29.4	30.2	29.6	28.8	FCF yield (%)	-1.2	-2.4	-8.0	-3.4	-0.2
EBITDA (Adj)	1,582	2,109	3,280	4,018	4,857	Dividend yield (%)	0.6	0.7	1.0	1.2	1.5
EBITDA Margin (Adj) (%)	16.9	15.6	19.1	19.2	18.5	Payout ratio (%)	18	18	21	22	22
Depreciation	-375	-450	-967	-1,168	-1,311	ROE (%)	18.9	17.2	18.8	20.9	23.4
Amortisation	0	0	0	0	0	Cashflow (HK\$m)					
EBIT (Adj)	1,207	1,659	2,312	2,850	3,546	EBITDA	1,582	2,109	3,280	4,018	4,857
EBIT Margin (Adj) (%)	12.9	12.3	13.5	13.6	13.5	Working capital	137	159	88	-535	-57
Net interest	-82	-77	-263	-295	-323	Other	-94	-192	-372	-441	-542
Associates	16	117	119	136	157	Operating cashflow					
Non-op/Except	226	448	433	519	623	Capex	-1,903	-2,807	-5,705	-4,275	-4,318
Pre-tax profit	1,367	2,148	2,600	3,210	4,003	Net acq/disposals	0	0	0	0	0
Tax	-329	-551	-660	-802	-1,000	Other	7	238	-3,562	-1,162	-697
Extraord./Min.Int./Pref.div.	-252	-396	-340	-420	-524	Investing cashflow					
Reported net profit	787	1,200	1,600	1,989	2,480	Dividends paid	-136	-179	-330	-445	-556
Net Margin (%)	8.4	8.9	9.3	9.5	9.4	Financing cashflow					
Core NPAT	787	1,200	1,600	1,989	2,480	Net change in cash	3,540	148	1,956	-1,797	-1,784
Per share data						Free cashflow to s/holders					
Reported EPS (\$)	0.54	0.66	0.78	0.89	1.12		-279	-732	-2,709	-1,234	-60
Core EPS (\$)	0.54	0.66	0.78	0.89	1.12						
DPS (\$)	0.10	0.12	0.16	0.20	0.25						
CFPS (\$)	1.12	1.14	1.45	1.37	1.91						
FCFPS (\$)	-0.19	-0.40	-1.31	-0.55	-0.03						
BVPS (\$)	3.18	4.07	4.03	4.53	4.99						
Wtd avg ord shares (m)	1,445	1,828	2,064	2,224	2,224						
Wtd avg diluted shares (m)	1,445	1,828	2,064	2,224	2,224						
Growth rates											
Sales revenue (%)	127.0	44.7	26.9	22.4	25.3						
EBIT (Adj) (%)	135.6	37.5	39.3	23.3	24.4						
Core NPAT (%)	71.2	52.6	33.3	24.3	24.7						
Core EPS (%)	58.8	20.6	18.1	15.3	24.7						
Balance Sheet (HK\$m)											
Cash & cash equiv.	6,722	6,908	8,864	7,068	5,284						
Accounts receivables	1,771	2,714	3,161	3,867	4,919						
Inventory	269	412	777	960	1,229						
Net fixed & other tangibles	9,101	10,555	15,293	18,400	21,407						
Goodwill & intangibles	716	729	729	751	788						
Financial & other assets	2,200	4,088	5,872	6,918	7,264						
Total assets	20,779	25,406	34,695	37,964	40,892						
Accounts payable	3,667	4,068	5,279	5,410	6,687						
Short-term debt	534	1,536	1,336	1,603	1,635						
Long-term debt	5,156	4,964	11,364	12,500	12,375						
Provisions & other liab	3,438	4,219	4,893	5,095	5,307						
Total liabilities	12,795	14,787	22,872	24,608	26,005						
Shareholders' equity	5,832	8,102	8,967	10,079	11,087						
Minority interests	2,152	2,516	2,856	3,277	3,800						
Total equity	7,984	10,619	11,823	13,356	14,887						
Net debt	-1,032	-408	3,835	7,036	8,727						
Net debt to equity (%)	-12.9	-3.8	32.4	52.7	58.6						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Higher Earnings Risk

We maintain our Neutral rating on China Resources Gas (CR Gas), and keep our target price at HK\$15.80 to factor in higher gas volume growth and earnings dilution impact from recent share placement. While CR Gas has shown positive earnings growth in previous years, expensive valuation, potential policy impact to its Sichuan and Chongqing projects may cap near-term upside.

Latest Developments

CR Gas in November announced several new developments and also completed a share placement, in which the share price reacted poorly afterwards.

- CR Gas's joint venture with Tianjin obtained approval from The Ministry of Commerce (MOFCOM), which will allow CR Gas to gain access to the Tianjin market. The proposed joint venture will have Rmb5 billion in registered capital. CR Gas will hold 49% equity interest of the JV.
- Placement of 160 million shares at HK\$16.95, which represents 7.19% of the expanded issued share capital. CR Gas will receive net proceeds of HK\$2.7billion, which will be part of its US\$2 billion target of capital required to obtain city-gas projects in China, and organic expansions.
- Raised its stake in Zhongshan Public Gas from 40% to 55%, which was approved by the board on Nov 23. CR Gas will contribute RMB88.86 million, which equates to Zhongshan's 14X 2012 earnings. By increasing its holdings, CR Gas expects to gain better control of the company and to increase the return from the city gas portfolio.

Parent Asset Injection Complete

At the interim result, CR Gas announced injection of the 5th batch of projects from its parent, for HK\$2.42billion. Net asset value was HK\$1.35bn, and profit after tax in 2011 and Jan-July 12 was HK\$27.9mn and HK\$29.9mn respectively. The 16 projects include Fuzhou and Nanchang, which should benefit from rising gas availability from both expansion of LNG terminal and West-to-East II pipeline, and could grow to 1billion m3/year markets by 2015.

The projects have so far generated limited earnings, which will put the acquisition value at well over 40X PE. We are concerned that after this final round of acquisition, CR Gas may not maintain a valuation premium to its gas utilities peers, as its growth rate may match closer to the industry levels. Although CR Gas is still in position to obtain projects from the market, the company will need to compete with other gas distributors, which may push down its return metric.

Lowered Gas and Connection Growth

Based on our discussion with management, we believe the gas volume growth may still rise in 2H12. After accounting for the external growth, the company should see near 30% in gas sales growth in 2012, while organic gas volume growth may remain closer to 15-16% Y/Y. The growth rate is slower than earlier years, when it

routinely reached over 20%, but has seen a pickup in recent months. CR Gas also faces slower connection fee growth, given its significant scale in various markets.

With slower gas volume and gas connection growth, we believe the company will continue to focus on project acquisitions and development of its commercial and industrial customers, as it looks to improve the return metric of the company. In addition, the company will increasingly rely on shareholding increase in order to maintain its earnings growth.

Growth Drivers

After completing asset injection from its parent, the company will need to focus on developing and completing its joint-venture including Tianjin and Ningbo, both of which has been previously announced.

As for its Tianjin JV project, the company has obtained approval from NDRC but is still awaiting final confirmation from Tianjin City and Ministry of Commerce. Given the delay in the Tianjin project, we believe significant earnings contribution will take time, as the company will need to build out its development pipelines before the project can turn profitable.

As for the Ningbo project, the company is still in discussions and the JV may also take time before gas revenue can be recognized.

DCF Calculation

Figure 1. CR Gas – FCF Calculation

Actual value in the run up to FY20E:	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
FCF		-2,709	-1,234	-60	935	2,311	3,110	4,000	5,155	6,522
Discount factor		1.00	1.09	1.18	1.28	1.39	1.51	1.64	1.78	1.93
Net debt	-264	-2,709	-1,137	-51	731	1,665	2,064	2,446	2,904	3,385
Terminal Value:										
NAV in 2021E		3,486.4								
Expected terminal growth rate		3.0%								
Expected Capital Employed on new investments		15.0%								
WACC		8.5%								
Terminal Value		50,311								
Discount factor		1.93								
NPV of terminal value		26,111								
NPV (HK Millions)		35,145								
DCF/share (HK\$)		15.80								

Source: Citi Research

Figure 2. CR Gas – Key Statistics

	2008	2009	2010	2011	2012E	2013E	2014E
Coverage:							
Connected household (k)	2,199	3,422	8,765	10,498	11,546	12,609	13,688
y-y change (%)	19.6	55.6	156.1	19.8	10.0	9.2	8.6
Penetration rate (%)	21.7	25.5	35.5	38.7	38.7	40.2	41.5
Revenue mix*:							
- connection fee (HK\$m)	606	924	2,157	2,741	3,442	3,645	3,799
y-y change (%)	181.9	52.5	133.3	27.1	25.6	5.9	4.3
- NG sales (HK\$m)	1,919	2,822	7,175	10,766	13,704	17,335	22,497
y-y change (%)	45.2	47.1	154.2	50.0	27.3	26.5	29.8
Gas sale:							
Piped gas sales (mil m ³)	1,371	2,214	5,577	7,215	9,308	11,635	14,660
y-y change (%)	9.2	61.5	151.8	29.4	29.0	25.0	26.0
Average residential connection fee (RMB)		2,150	2,640	2,640	2,719	2,801	2,885
Average Tariff (RMB per m3)	2.03	2.15	2.24	2.46	2.58	2.66	2.74
y-y change (%)		6.0	4.2	9.8	5.0	3.0	3.0

Source: Citi Research estimates

Figure 3. CR Gas — Citi and Consensus Forecasts

	Citigroup	Consensus	% Difference
2012E Sales (HK\$m)	17,146	17,747	-3%
2012E Net Income (HK\$m)	1,600	1,562	2%
2012E EPS (HK\$)	\$0.78	0.78	0%
2013E Sales (HK\$m)	20,979	23,640	-11%
2013E Net Income (HK\$m)	1,989	1,985	0%
2013E EPS (HK\$)	\$0.89	0.95	-6%
2014E Sales (HK\$m)	26,296	30,179	-13%
2014E Net Income (HK\$m)	2,480	2,453	1%
2014E EPS (HK\$)	\$1.12	1.17	-5%

Source: Citi Research, Bloomberg Consensus

We revise our 2012/13/14E earnings estimates by 0.4%/1.4%/1.9% respectively to factor in better gas sales volume, and we also updated outstanding shares to factor in the impact of the recent share placement.

Figure 4. CR Gas – Earnings Revision

	Net Profit (HK\$ mils)			Diluted EPS (HK\$)		
	Old	New	% change	Old	New	% change
2012E	1,594	1,600	0.4%	0.80	0.78	-3.1%
2013E	1,961	1,989	1.4%	0.98	0.89	-9.1%
2014E	2,435	2,480	1.9%	1.22	1.12	-8.8%

Source: Citi Research estimates

Figure 5. CR Gas – Income Statement (2009-2014E, HK\$ Millions)

Year to 31 Dec	FY09	FY10	FY11	FY12E	FY13E	FY14E
Turnover	4,110	9,331	13,507	17,146	20,979	26,296
Gas connection	924	2,157	2,741	3,442	3,645	3,799
Piped gas sales and related products	2,822	7,175	10,766	13,704	17,335	22,497
Cost of sales	(2,858)	(6,594)	(9,534)	(11,966)	(14,767)	(18,720)
Gross profit	1,252	2,738	3,972	5,180	6,212	7,576
Other income	165	226	448	433	519	623
Distribution costs	(279)	(640)	(1,084)	(1,341)	(1,599)	(1,952)
Administration expenses	(461)	(890)	(1,229)	(1,526)	(1,763)	(2,078)
EBIT	677	1,433	2,107	2,745	3,369	4,169
Share of results of associates	7	16	117	119	136	157
Finance costs	(40)	(82)	(77)	(263)	(295)	(323)
Profit before taxation	644	1,367	2,148	2,600	3,210	4,003
Taxation	(85)	(329)	(551)	(660)	(802)	(1,000)
Minority interest	(99)	(252)	(396)	(340)	(420)	(524)
Net profits	438	787	1,200	1,600	1,989	2,480
% of chg	36.5%	79.8%	52.6%	33.3%	24.3%	24.7%

Source: Company, Citi Research estimates

Figure 6. CR Gas – Key Financial Ratios (2009-2014E)

	2009	2010	2011	2012E	2013E	2014E
Gross margin	30.5	29.3	29.4	30.2	29.6	28.8
Operating margin	16.5	15.4	15.6	16.0	16.1	15.9
Pre-tax margin	15.7	14.6	15.9	15.2	15.3	15.2
Net profit margin	11.2	8.4	8.9	9.3	9.5	9.4
Effective tax rate	13.3	24.1	25.7	25.4	25.0	25.0
Interest cover (x)	(33.6)	(50.0)	112.7	10.9	11.9	13.4
ROE (%)	18.4	13.5	14.8	17.8	19.7	22.4
Net debt-to-equity (%)	65.3	(6.3)	3.3	50.3	76.5	84.8

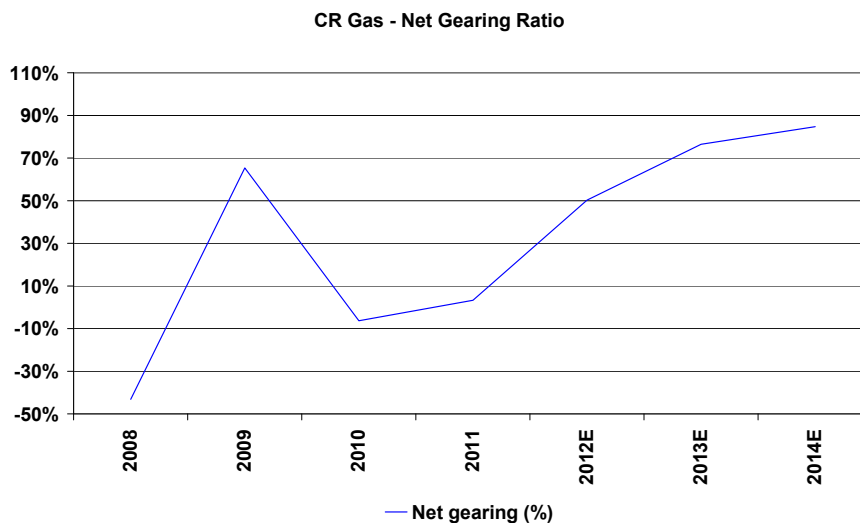
Source: Citi Research

Figure 7. CR Gas - Balance Sheet (2009-2014E, HK\$ Millions)

Year to 31 Dec	FY09	FY10	FY11	FY12E	FY13E	FY14E
Bank balances and cash	3,112	6,707	6,890	8,846	7,049	5,265
Pledged bank deposits	19	15	18	18	18	18
Account receivables	826	1,771	2,714	3,161	3,867	4,919
Inventories	200	269	412	777	960	1,229
Others	561	536	744	855	898	943
Current assets	4,719	9,298	10,777	13,657	12,793	12,375
PP&E	4,653	9,101	10,555	15,293	18,400	21,407
Others	2,015	1,664	3,345	5,017	6,020	6,321
Intangible asset	669	716	729	729	751	788
Non-current assets	7,337	11,481	14,628	21,038	25,171	28,517
Total assets	12,056	20,779	25,406	34,695	37,964	40,892
Account payables	2,221	3,667	4,068	5,279	5,410	6,687
Borrowings - amt due within one year	1,357	534	1,536	1,336	1,603	1,635
Derivative financial instruments	70	160	196	196	196	196
Others	1,566	2,629	3,369	4,042	4,244	4,457
Current liabilities	5,214	6,990	9,169	10,854	11,454	12,975
Borrowings - amt due after one year	3,383	5,805	5,618	12,018	13,155	13,030
Non-current liabilities	3,383	5,805	5,618	12,018	13,155	13,030
Total liabilities	8,597	12,795	14,787	22,872	24,608	26,005
Shareholders equity	2,492	5,832	8,102	8,967	10,079	11,087
Minority interest	967	2,152	2,516	2,856	3,277	3,800
Total Equity	3,459	7,984	10,619	11,823	13,356	14,887

Source: Company, Citi Research estimates

Figure 8. CR Gas – Net Gearing to Rise in 2012E



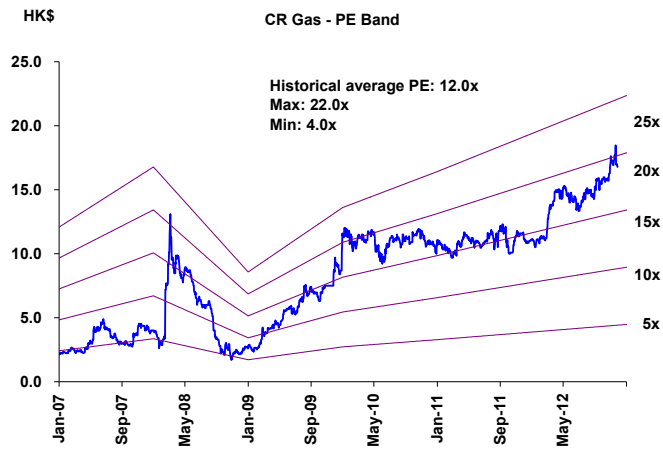
Source: Citi Research

Figure 9. CR Gas - Cash flow statement (2009-2014E, HK\$ Millions)

Year to 31 Dec	FY09	FY10	FY11	FY12E	FY13E	FY14E
Operating activities						
Profit before tax	644	1,367	2,148	2,600	3,210	4,003
Adjustments	(84)	(55)	(164)	263	295	323
EBIT	560	1,311	1,984	2,864	3,506	4,326
Exceptional items	(68)	28	14	0	0	0
Depreciation & amortisation	192	430	484	967	1,168	1,311
Change in working capital	322	137	159	88	(535)	(57)
Operating cash flow	1,090	1,906	2,640	3,919	4,138	5,580
Interest paid	17	26	(18)	(263)	(295)	(323)
Net operating cash flow	1,007	1,680	2,110	2,996	3,041	4,258
Capex	(961)	(1,903)	(2,807)	(5,705)	(4,275)	(4,318)
Free cash flow	46	(223)	(697)	(2,709)	(1,234)	(60)
Other investing activities	(959)	7	238	(3,562)	(1,162)	(697)
Investing cash flow	(1,919)	(1,896)	(2,569)	(9,267)	(5,437)	(5,015)
Change in equity	(254)	2,494	91	2,700	0	0
Change in debt	2,883	2,709	736	6,200	1,404	(93)
Dividends paid	(139)	(136)	(179)	(330)	(445)	(556)
Financing cash flow	2,603	3,732	245	8,227	599	(1,027)
Beginning cash	1,420	3,112	6,707	6,890	8,846	7,049
Change in cash	1,690	3,516	2,104	1,956	(1,797)	(1,784)
Ending cash	3,112	6,707	6,890	8,846	7,049	5,265

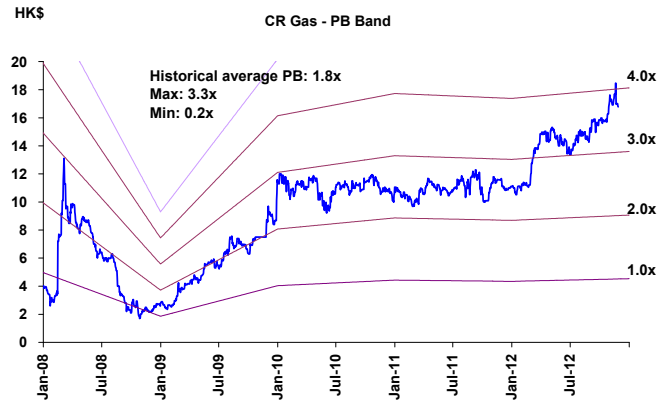
Source: Company, Citi Research estimates

Figure 10. CR Gas – Forward PE Band



Source: Citi Research

Figure 11. CR Gas – Forward PB Band



Source: Citi Research

China Resources Gas

Company description

CR Gas was first listed in 1994, and in 2008 the group acquired city-gas assets from China Resources (Holdings) for a consideration of about HK\$3.8billion. Since then, the company has focused its business on city gas distributors and disposed of its other non-core assets. As of September 2011, it operates 64 city gas projects in China. We expect it will expand its city-gas project footprint, and grow its existing projects organically by connecting new customers in commercial, industrial and consumer segments, and to pursue further project acquisitions.

Investment strategy

We rate CR Gas Neutral (2) with a DCF-based HK\$15.80 target price. After final asset injections from its parent, we believe CR Gas will face higher M&A costs to acquire projects from the market, as it scales up its operation. We anticipate CR Gas becoming a key city-gas operator in China, with organic and strategic growth drivers to result in significant earnings growth in the coming two years. However, CR Gas's high P/E multiple to its peers may lead to relative weaker performance. After getting approval for its JV with Tianjin Gas, we anticipate CR Gas will accelerate growth along the coastal region, taking advantage of increasing capacity from pipelines and LNG terminals. As for geographic coverage, we see company increasing targeting coastal projects that are currently held by local governments.

Valuation

Our target price for CR Gas of HK\$15.80 is based on a DCF valuation of its expected cash flows and outlook, and also equates to 18X 2013E earnings. Our model forecasts 25% earnings CAGR in the next two years, and we expect free cash flow to turn positive by 2015, after accounting for higher capex for the next two years. We use a WACC of 8.54% (13.49% cost of equity, factoring in a risk-free rate of 2.50%, a market risk premium of 8.8% and asset beta of 1.00) and assume a target gearing of 55%.

Risks

Upside risks to our target include if CR Gas can achieve faster than industry gas volume ramp-up, and if the company is able to secure projects at an inexpensive valuation. Downside risks to our target price include slower-than-expected gas volume sales, which are possible due to operational delays or supply constraints, changes in fees related to gas connection charges, increase in gas supplies, and less favorable government policy toward natural gas usage in China.

China Resources Power (0836.HK)

Coal Cost Cut Exceeds Market Expectations

Buy	1
Price (28 Nov 12)	HK\$17.16
Target price	HK\$21.00
<i>from HK\$20.00</i>	
Expected share price return	22.4%
Expected dividend yield	2.7%
Expected total return	25.1%
Market Cap	HK\$81,666M
	US\$10,538M

Price Performance (RIC: 0836.HK, BB: 836 HK)



Asian Utilities

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■ **Summary** – We raise our FY12-14E net profits by 2-5% and DCF TP +5% to HK\$21/share. The earning uplift is due to lower unit fuel cost (assumed -6% yoy in FY12E, from -4% before), more than enough to offset the negative effect of lower-than-targeted electricity and coal-output volumes. It should have positive free cashflow in FY12-14E, the first time since listing in 2003. Its yield should become more stable by switching from a rising payout ratio to increasing DPS. FY13E 10.1x PER and 1.4x PB look cheap, below historical means by 34% / 22% respectively.

■ **Higher generation volume ahead** – CRP's net power generation rose +5.2% yoy to 12.7M MWh in Oct, the first rise since March. It forecasts higher generation mom in Nov-Dec. Average utilization of its coal-fired plants on a same-plant basis would only slightly miss its target of 5,700 hours (-7% yoy) in 2012E, including 4,687 hours in 10M12, with the miss to be more than offset by lower unit fuel cost.

■ **Unit fuel cost cut likely to exceed company's minimum target** – CRP's 2012 coal supply includes 50% from contracts, 22% from mine-mouth mines, 22% from spot market and 6% from import. NDRC's plan not to set key contracted prices from Jan 2013 would not have an earnings impact as half of its contracted prices are above spot ones and vice versa. We lower CRP's unit fuel cost to -6% yoy, from -4%, in 2012E seeing as its peer Huaneng is lowering its unit fuel cost guidance after 9M12 results, and both companies have many of their plants in coastal regions.

■ **2012E coal output below target by 9%, but profit impacted to lesser extent** – We cut CRP's 2012E coal output by 9% to 16.86m tonnes. We have cut the output from CR Liansheng and CR Shanxi for safety reasons. where coal output levels in 10M12 were only 70% and 26% of CRP's annual targets. But the cut in CRP's profit from coal mines would be less than 9%, as the thermal coal output of these mines is of lower quality (4,00kcal/kg), and hence makes limited profit.

■ **Versus consensus** – Our FY12E net profit is 4% above consensus (Bloomberg).

Statistical Abstract

FYE	Net Profit	EPS	P/E	D. Yld
31 Dec	(HK\$M)	(HK\$)	(x)	(%)
2009A	5,317	1.17	14.6	2.2
2010A	4,904	1.04	16.6	2.0
2011A	4,451	0.94	18.3	1.7
2012E	6,988	1.47	11.6	2.7
2013E	8,078	1.71	10.1	3.2
2014E	8,613	1.82	9.4	3.4

Source: Powered by dataCentral

Earnings Revision

FYE	Net Profit	EPS	%	DPS
31 Dec	(HK\$M)	(HK\$)	chg	(HK\$)
2012E	6,988	1.47	2.0	0.47
Prev	6,848	1.45		0.46
2013E	8,078	1.71	3.0	0.54
Prev	7,843	1.66		0.53
2014E	8,613	1.82	5.3	0.58
Prev	8,179	1.73		0.55

Source: Powered by dataCentral

0836.HK: Fiscal year end 31-Dec						Price: HK\$17.16; TP: HK\$21.00; Market Cap: HK\$81,666m; Recomm: Buy					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	48,578	60,709	61,703	67,721	75,636	PE (x)	16.6	18.3	11.6	10.1	9.4
Cost of sales	-29,671	-38,383	-35,403	-37,641	-41,153	PB (x)	1.9	1.7	1.5	1.4	1.2
Gross profit	18,908	22,326	26,300	30,080	34,483	EV/EBITDA (x)	10.6	10.3	7.8	6.8	6.2
Gross Margin (%)	38.9	36.8	42.6	44.4	45.6	FCF yield (%)	-8.5	-7.3	3.2	0.3	2.0
EBITDA (Adj)	11,398	12,564	16,809	19,455	21,286	Dividend yield (%)	2.0	1.7	2.7	3.2	3.4
EBITDA Margin (Adj) (%)	23.5	20.7	27.2	28.7	28.1	Payout ratio (%)	33	32	32	32	32
Depreciation	-4,275	-5,502	-6,277	-6,561	-7,637	ROE (%)	12.3	9.9	13.9	14.4	13.9
Amortisation	0	0	0	0	0	Cashflow (HK\$m)					
EBIT (Adj)	7,123	7,062	10,532	12,894	13,649	EBITDA	11,398	12,564	16,809	19,455	21,286
EBIT Margin (Adj) (%)	14.7	11.6	17.1	19.0	18.0	Working capital	-555	-1,434	2,429	648	398
Net interest	-1,397	-2,220	-2,612	-2,618	-2,796	Other	-858	-2,060	-4,007	-4,826	-5,048
Associates	791	837	1,346	1,410	1,553	Operating cashflow					
Non-op/Except	0	1,184	240	0	0	Capex	-16,887	-15,029	-12,667	-15,008	-15,000
Pre-tax profit	6,517	6,862	9,506	11,686	12,406	Net acq/disposals	-439	0	0	0	0
Tax	-755	-1,243	-1,404	-2,386	-2,518	Other	-1,540	-4,326	1,059	1,059	1,059
Extraord./Min.Int./Pref.div.	-858	-1,169	-1,114	-1,222	-1,275	Investing cashflow					
Reported net profit	4,904	4,451	6,988	8,078	8,613	Dividends paid	-1,771	-1,549	-1,409	-2,212	-2,556
Net Margin (%)	10.1	7.3	11.3	11.9	11.4	Financing cashflow					
Core NPAT	4,904	4,451	6,988	8,078	8,613	Net change in cash	3,768	-2,614	911	65	-422
Per share data						Free cashflow to s/holders					
Reported EPS (\$)	1.04	0.94	1.47	1.71	1.82		-6,902	-5,959	2,565	269	1,636
Core EPS (\$)	1.04	0.94	1.47	1.71	1.82						
DPS (\$)	0.35	0.30	0.47	0.54	0.58						
CFPS (\$)	2.11	1.91	3.21	3.22	3.51						
FCFPS (\$)	-1.46	-1.26	0.54	0.06	0.35						
BVPS (\$)	9.05	10.11	11.30	12.55	13.84						
Wtd avg ord shares (m)	4,660	4,695	4,695	4,695	4,695						
Wtd avg diluted shares (m)	4,730	4,738	4,738	4,738	4,738						
Growth rates											
Sales revenue (%)	46.3	25.0	1.6	9.8	11.7						
EBIT (Adj) (%)	1.2	-0.9	49.1	22.4	5.9						
Core NPAT (%)	-7.8	-9.2	57.0	15.6	6.6						
Core EPS (%)	-11.6	-9.4	57.0	15.6	6.6						
Balance Sheet (HK\$m)											
Cash & cash equiv.	6,802	4,497	5,408	5,473	5,051						
Accounts receivables	10,763	16,123	16,387	17,985	20,087						
Inventory	2,006	3,593	3,314	3,523	3,852						
Net fixed & other tangibles	84,274	96,419	102,757	111,154	118,466						
Goodwill & intangibles	3,797	4,033	4,273	4,273	4,273						
Financial & other assets	35,369	43,702	49,457	49,208	49,013						
Total assets	143,011	168,366	181,597	191,617	200,742						
Accounts payable	14,682	19,306	21,584	23,689	26,458						
Short-term debt	20,668	26,418	8,051	8,561	8,202						
Long-term debt	54,243	56,569	78,518	78,957	78,755						
Provisions & other liab	3,157	4,501	4,637	4,988	5,048						
Total liabilities	92,750	106,794	112,790	116,195	118,463						
Shareholders' equity	42,164	47,473	53,053	58,920	64,976						
Minority interests	8,096	14,099	15,754	16,502	17,303						
Total equity	50,260	61,572	68,807	75,422	82,279						
Net debt	68,109	78,491	81,161	82,045	81,907						
Net debt to equity (%)	135.5	127.5	118.0	108.8	99.5						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Higher Power Generation Volume Ahead

CRP's net power generation rose +5.2% yoy to 12.7M MWh in Oct, the first rise since March, signaling economic improvement. The company forecasts higher generation mom in Nov-Dec. Regarding average utilization of its coal-fired power plants calculated on a same-plant basis, it expects to be slightly below its target of 5,700 hours in 2012E, which would be -7% yoy, including 4,687 hours in 10M12. But the miss would be relatively mild without much earning impact in our view. More important, the lower-than-expected net generation was driven by weaker-than-estimated power consumption, which also led to more coal cost cuts. Since CRP, like the other IPPs, has net profit being more sensitive to coal costs than to utilization, the benefit of more coal cost cuts should be more than enough to offset the negative effect of lower-than-expected generation volume.

Lower Unit Fuel Cost

Figure 1. CRP's raw coal supply mix (m tonne)

Source	2011	%	2012E	%	YoY
Contracts	n/a	n/a	39	50%	n/a
Mine mouth	n/a	n/a	17	22%	n/a
Spot market	n/a	n/a	17	22%	n/a
Import	n/a	n/a	5	6%	n/a
Total	65	100%	78	100%	20%

Source: Company Reports and Citi Research Estimates

CRP should consume 78m tonnes of raw coal in 2012E, including 50% from contracts, 22% from mine-mouth mines which are often without contracts, 22% from spot market and 6% from imports. CRP added that 25% of its coal supply, or about half of its supply from contracts, have contracted prices above local spot prices. We think the PRC government's plan not to set key contracted prices from Jan 2013 and to let them decide by market mechanisms won't have much impact on CRP's earnings next year, since the company has half of its contracted prices above spot ones and vice versa.

We lower CRP's unit fuel cost to -6% yoy, from -4% yoy, in 2012E. Huaneng Power reported strong 9M12 profit recently, driven by unit coal cost cuts of 4.3% yoy, exceeding its original target -3% yoy in 2012. After the results announcement, Huaneng revised down its 2012E unit fuel cost guidance from -3% yoy to -5% yoy. We think CR Power to have the same benefit as both companies have similar geographical presence with many plants in coastal regions – for CRP, it has 60% of its power plants in two coastal provinces; namely, Jiangsu and Guangdong. In mid Nov, coal inventory at CRP's power plants is ample at 20 days.

Mild Impact from Coal Output below Target

Figure 2. CRP's coal outputs from its mines in 2012E

(m tonne)	Province	Coal type	Stake	Old 2012E	yoy	Oct	10M12	% of 2012 old target	New 2012E	chg
Tianneng	Jiangsu	Thermal	100%	2.1	-19%	0.2	2.05	98%	2.5	19%
CR Liansheng	Lvliang, Shanxi	Thermal (2/3); coking (1/3)	51%	8.49	-16%	0.418	5.96	70%	7	-18%
CR Shanxi	Shanxi	Mainly coking	26%	1.5	n/a	0.05	0.39	26%	0.6	-60%
Henan Yonghua	Henan	Thermal	49%	1.91	nil	0.14	1.57	82%	1.91	0%
Hunan CR	Hunan	Thermal	100%	0.45	-1%	0.04	0.37	83%	0.45	0%
Shanxi Daning	Jincheng, Shanxi	Anthracite	51%	4	195%	0.44	3.52	88%	4.4	10%
Total				18.45	13%	1.3	13.86	75%	16.86	-9%

Source: Company Reports and Citi Research Estimates

We cut CRP's 2012E coal output by 9% to 16.86m tonnes, from previous company guidance of 18.45m tonnes, after our recent company update. We have cut the outputs from CR Liansheng and CR Shanxi where coal outputs in 10M12 were only 70% and 26% of CRP's annual targets. The output of these two mines missed expectations so far, since safety requirements have become more stringent due to 18th National Congress of the Communist Party of China being held on 8-14 Nov; both of them are re-opened old mines, hence safety levels have to be improved. We have lowered the 2012E output of these two mines by 18% and 60% respectively taking their 10M12 output into consideration. Meanwhile, we have raised the output of Tianneng and Shanxi Daning by 19% and 10%.

CRP's 2012E coal mine profit fall due to volume cuts would be less than the 9% volume cut. This is because the volume cuts are from two mines in Shanxi producing thermal coal of lower quality (4,000kal/kg for thermal coal there), hence of limited profit contribution, whereas the upward revision of volume includes that from the Daning coal mine producing high quality anthracite at high sale prices. The all-in cost of CRP's coal sales is an average Rmb300/tonne, including Rmb230/tonne of coal output cost at mines and Rmb70/tonne sales and marketing costs. The company has invested HK\$40bn in its coal mining business, including one-third from equity. The ROE of its coal mining business therefore is about 8% in FY12E, calculated from HK\$1bn attributable net profit.

CRP targets coal output of 22-23m tonnes in 2013E, with increments from CR Liansheng and CR Shanxi output is expected to grow when more mines there are qualified from a safety perspective.

Positive Free Cashflow & More Stable DPS

Figure 3. CRP's consolidated capex (HK\$bn)

Usage	2011	1H12	2012E	YoY	2013E
Coal-fired power plants	8.8	2.9	5	-43%	Up yoy
Renewable energy plants	5.5	0.8	3	-45%	Down yoy
Coal mines	7.1	3	4.6	-35%	Down yoy
Total	21.4	6.7	12.6	-41%	15.0

Source: Company Reports and Citi Research Estimates

CRP's consolidated capex target of -41% yoy to HK\$12.6bn in 2012E remains intact. The company would have positive free cashflow in 2012E, the first time since its listing in 2003. The company plans to increase its consolidated capex in 2013E,

primarily to invest more in coal-fired power plants in coastal provinces, while it also aims at keeping positive free cashflow in next year. We estimate its consolidated capex to +19% yoy to HK\$15bn in 2013E.

CRP utilizes its positive free cashflow for debt repayment. Its interest expense should drop hoh in 2H12E, owing to a Rmb interest rate total 50bps in 2Q12 and reduction of its outstanding loan amounts. Upon investor proposals, the company might consider changing its dividend policy to increase from a certain payout ratio (32% in 2012E) to a certain DPS (HK\$0.46 in 2012E); we think the possible change would be positive for investors, who would have a more stable dividend income.

Earning Drivers in 2013E

CRP should have earning increments in 2013E from the following:

- (i) Full-year contribution of new capacity added in 2012E – Though CRP will only have one coal-fired power generating unit (a 1,000MW unit at Puqi coal-fired plant Ph.II) to start operation in 2013E, its 2X1,000MW Hezhou coal-fired plant and another 1,000MW coal-fired unit at Puqi coal-fired plant Ph.II completed in 2H12E would also have full-year operation in 2013E. CRP added that NDRC of the PRC Government in Beijing remains selective in approving new coal-fired plant construction as there is no power shortage in China;

Figure 4. CRP's attributable capacity addition plan (MW)

	Fuel	Location	Stake	Capacity	FY11 19,358	FY12E 22,230	FY13E 25,062	FY14E 26,862	Status/completion date
Capacity at year start									
Add - new projects:									
Yueliangshan	Wind	Hebei	100%	33x1.5	99				2010
Yangxi Longgaoshan	Wind	Guangdong	100%	49	11				4Q10
Weihai ETD Zone	Wind	Shandong	100%	50	50				4Q10
Penglai Daluohang I	Wind	Shandong	100%	50	28				4Q10
Nanre (Nanjing Thermal)	Coal	Jiangsu	100%	2X600	600				21 Jan 10, 1H11
Heze	Coal	Shandong	100%	2X600	1,200				Mar 11, Jun 11
Dengfeng II	Coal	Henan	85%	2X600	510				Nov 11
Nanjing Chemical Industry Park *	Coal	Jiangsu	90%	2X300	270				2010-11
New wind power projects	Wind	n/a	100%		221				2011
Liangshan Yazuihe	Hydro	Sichuan	51%	260		133			6 Nov 2012
Hezhou	Coal	Guangxi	50%	2X1,000		1,000			Aug 12, Nov 12
Puqi II	Coal	Hubei	100%	2X1,000		1,000	1,000		Dec 12, 2Q13E
Qiaowan	Wind	Gansu	100%	800		700	800	800	2012-14E
Cangnan	Coal	Zhejiang	90%	2X1,000				1,800	2014E
Yichang cogeneration	Coal	Hubei	100%	2X300				600	2014E
Less - closure:									
Zhenjiang	Coal	Jiangsu	42.5%	4X138	-117				2011
Capacity at year end					22,230	25,062	26,862	30,062	
% increment					15%	13%	7%	12%	

Source: Company Reports and Citi Research Estimates

* Pending for NDRC approval

- (ii) Increase in coal output – CRP's coal output from its mines should rise to 22-23m tonnes in 2013E, from 16.86m tonnes in 2012E; and
- (iii) Increase in wind capacity – CRP has 1,500MW operational wind capacity whereas its parent has 600-700MW. The listed company might add over 800MW wind capacity in 2013E, including parental asset injection.

PER and PB below Historical Mean

Figure 5. CR Power – Forward PE Chart



Source: Citi Investment Research and Analysis

Figure 6. CR Power – Forward PB Chart



Source: Citi Investment Research and Analysis

PRC IPPs FY13E Net Profit Sensitivity

Figure 7. PRC IPPs: FY13E Net Profit Sensitivity

(Rmbm)	Huaneng	Datang	Huadian	CRP	CPI	Average
Base case net profit *	7,932	3,647	1,927	8,078	1,365	n/a
Case 1. net profit change with extra 1% cut in average coal price	454	157	280	291	71	n/a
% change	5.7%	4.3%	14.5%	3.6%	5.2%	6.7%
Case 2. net profit change with extra 1% hike in power tariff	746	299	626	460	125	n/a
% change	9.4%	8.2%	32.5%	5.7%	9.2%	13.0%
Case 3. net profit change with extra 1% cut in utilization hours	-272	-110	-121	-174	-73	n/a
% change	-3.4%	-3.0%	-6.3%	-2.2%	-5.3%	-4.0%
Case 4. net profit change with extra 25bps interest rate cut	242	213	287	205	43	n/a
% change	3.0%	5.8%	14.9%	2.5%	3.2%	5.9%

Source: Company Reports and Citi Research Estimates

* In Rmbm except CRP which is in HK\$m

** Key assumptions - (i) unit fuel cost ranges from -2% yoy (Huaneng, Datang, CRP & CPI) to -5% yoy (for Huadian) in 2013E; (ii) average 6.5% tariff hike on 1 Dec 2011, but nil in 2012-13E

CRP's DCF Calculation

Figure 8. CRP's DCF calculation

DCF of power plants

Value in the run up to FY20E

(Rmbm)	2012	2013	2014	2015	2016	2017	2018	2019	2020
Net operating cashflow		15,277	16,636	19,286	18,275	18,180	18,100	18,231	18,111
Capex		-15,008	-15,000	-13,000	-13,000	-4,470	-4,470	-4,470	-4,470
FCF		269	1,636	6,286	5,275	13,710	13,630	13,761	13,641
Discount factor		1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.8
Net debt in 2012/NAV from 2013E	-84,412	269	1,510	5,353	4,146	9,945	9,124	8,502	7,778

Terminal value

NAV in 2021E	14,187
Expected terminal growth rate	4%
Capital Employed on new capex	10.0%
WACC	8.4%
Terminal Value	195,372
Discount factor	1.8
NPV of terminal value	111,395

Total value

DCF - in the run up to 2020E	-37,785
DCF - from 2021E	111,395
Total DCF	73,610
Exchange Rate	1.22
Total DCF (in HK\$m)	89,878
No. of share (m)	4,662
DCF / share (HK\$)	19.28

DCF of coal mines

Value in the run up to FY20E

(Rmbm)	2012	2013	2014	2015	2016	2017	2018	2019	2020
FCF (profit)		638	838	999	1,087	1,098	1,109	1,120	1,132
Discount factor		1.00	1.08	1.17	1.27	1.38	1.49	1.62	1.75
Net debt in 2012/NAV from 2013E	-8,667	638	773	851	855	797	743	692	645

Terminal value

NAV in 2021E	1,177
Expected terminal growth rate	4.0%
Capital Employed on new capex	10.0%
WACC	8.4%
Terminal Value	16,206
Discount factor	1.8
NPV of terminal value	9,240

Total value

DCF - In the run up to 2020E	-2,673
DCF - from 2021E	9,240
Total DCF	6,568
Exchange Rate	1.22
Total DCF (in HK\$m)	8,019
No. of share (m)	4,662
DCF (Shanxi coal mine) / share	1.72
EV/ton of reserve (HK\$)	n/a

CRP's DCF

DCF - power plants (HK\$)	19.28
DCF - coal mines (HK\$)	1.72
Total DCF/share (HK\$)	21.00

Source: Company Reports and Citi Research Estimates

Financial Statements & Key Assumptions

Figure 9. CRP's income statement

Year-end 31 Dec (HK\$m)	FY10	FY11	FY12F	FY13F	FY14F
Sales of electricity	41,719	50,705	53,412	58,734	65,008
Heat supply	1,429	2,018	2,624	2,886	2,944
Sale of coal	5,430	7,985	5,667	6,101	7,684
Net revenues	48,578	60,709	61,703	67,721	75,636
Coal consumption	-29,671	-38,383	-35,403	-37,641	-41,153
Repairs and maintenance	-767	-1,015	-1,100	-1,224	-1,399
Depreciation	-4,275	-5,502	-6,277	-6,561	-7,637
Labor cost	-925	-3,247	-3,291	-3,564	-3,920
S&A expenses	-971	-425	-444	-493	-738
Other operating expenses	-4,847	-5,075	-4,656	-5,344	-7,141
Operating profit	7,123	7,062	10,532	12,894	13,649
Other income and gain/loss	1,130	1,295	1,342	1,366	1,358
Interest expense	-2,527	-3,516	-3,955	-3,985	-4,154
Profit from associates	790	740	1,249	1,314	1,456
Profit from jointly controlled entities	0	97	97	97	97
Fair value gain from interest rate swap	0	1,184	240	0	0
Profit before taxation	6,517	6,862	9,506	11,686	12,406
Taxation	-755	-1,243	-1,404	-2,386	-2,518
Minority interest	-858	-1,169	-1,114	-1,222	-1,275
Net profit	4,904	4,451	6,988	8,078	8,613

Source: Company Reports and Citi Research Estimates

Figure 10. Consolidated cashflow statement

Year-end 31 Dec (HK\$m)	FY10	FY11	FY12F	FY13F	FY14F
Cashflow from operations					
EBITDA	11,398	12,564	16,809	19,455	21,286
Working capital changes	-555	-1,434	2,429	648	398
Other operating cash items	594	2,421	1,533	1,701	1,790
Gross operating cash flow	11,437	13,551	20,771	21,804	23,474
Net interest paid	-819	-3,500	-4,135	-4,141	-4,319
Tax paid	-616	-976	-1,404	-2,386	-2,518
Net operating cash flow	10,002	9,075	15,232	15,277	16,636
Capex	-20,328	-15,029	-12,667	-15,008	-15,000
Free cash flow	-10,326	-5,953	2,565	269	1,636
Cash from acquisitions	-439	0	0	0	0
Disposal of fixed assets	517	240	0	0	0
Change in temp cash inv.	-2,058	-4,566	1,059	1,059	1,059
Cashflow from investing	-22,307	-19,355	-11,608	-13,949	-13,941
Ordinary Dividend Paid	-1,771	-1,549	-1,409	-2,212	-2,556
Change in borrowings	17,756	9,264	-1,304	949	-561
Other Financing Cash Flow	-3,140	259	0	0	0
Cash Flow from Financing	12,845	7,974	-2,712	-1,262	-3,117
Beginning cash balance	6,262	6,802	4,497	5,408	5,473
Ending cash balance	6,802	4,497	5,408	5,473	5,051

Source: Company Reports and Citi Research Estimates

Figure 11. Consolidated balance sheet

As at 31 Dec (HK\$m)	FY10	FY11	FY12F	FY13F	FY14F
Property, plant and equipment	59,881	66,262	69,838	70,379	72,642
Construction in progress	24,393	30,157	32,919	40,774	45,824
Associated companies	13,215	19,989	19,844	19,594	19,399
Other tangible fixed assets	17,391	22,119	28,019	28,019	28,019
Goodwill	3,797	4,033	4,273	4,273	4,273
Total fixed assets	118,676	142,560	154,894	163,041	170,158
Cash and cash equivalents	6,802	4,497	5,408	5,473	5,051
S/T bank deposits over 3 m/s	330	304	304	304	304
Inventories	2,006	3,593	3,314	3,523	3,852
Debtors	15,196	17,413	17,677	19,275	21,377
Total current assets	24,334	25,806	26,703	28,576	30,584
Total assets	143,011	168,366	181,597	191,617	200,742
Current portion of L/T loans	20,668	26,418	8,051	8,561	8,202
Trade creditors	14,682	19,306	21,584	23,689	26,458
Due to holding co's	1,209	2,706	2,706	2,706	2,706
Taxes payable	149	398	450	765	808
Dividends payable / Others	0	148	232	268	286
Total current liabilities	36,709	48,976	33,023	35,989	38,460
Long term bank loans	54,243	56,569	78,518	78,957	78,755
Other L-T liabilities	1,798	1,248	1,248	1,248	1,248
Minority interests	8,096	8,199	9,854	10,602	11,403
Perpetual capital securities	0	5,900	5,900	5,900	5,900
Total L/T liabilities	64,137	71,916	95,520	96,708	97,306
Share capital	4,720	4,745	4,745	4,745	4,745
Other capital	9,196	9,196	9,196	9,196	9,196
Retained earnings	28,249	33,532	39,112	44,978	51,034
Total equity	42,164	47,473	53,053	58,920	64,976
Total liabilities & equity	143,011	168,366	181,597	191,617	200,742

Source: Company Reports and Citi Research Estimates

Figure 12. Key Ratios

Year-end 31 Dec (%)	FY10	FY11	FY12F	FY13F	FY14F
Operating margin (%)	14.7	11.6	17.1	19.0	18.0
Net margin (%)	9.7	7.4	11.5	12.1	11.5
Net debt to total capital	48%	47%	45%	43%	41%
Net debt to equity	162%	165%	153%	139%	126%
Debt to EBITDA ratio (x)	6.7	6.7	5.2	4.6	4.1
Interest coverage (x)	5.3	4.5	5.0	5.6	5.8
Cash interest cover (x)	4.4	3.8	4.6	5.3	5.8
ROA (%)	3.7	3.0	4.2	4.6	4.6

Source: Company Reports and Citi Research Estimates

Figure 13. Key assumptions

	FY10	FY11	FY12F	FY13F	FY14F
Capacity (MW)					
Installed capacity - year end	29,421	32,034	36,044	37,844	41,244
yoy change (%)	7.6	8.9	12.5	5.0	9.0
Consolidated capacity - year end	18,921	21,809	23,819	25,619	29,019
yoy change (%)	13.3	15.3	9.2	7.6	13.3
Consolidated capacity - year average	17,835	20,559	22,522	24,818	27,418
yoy change (%)	23.2	15.3	9.5	10.2	10.5
Attributable capacity - year end	19,895	22,767	25,650	27,450	30,650
yoy change (%)	8.8	14.4	12.7	7.0	11.7
- in subsidiaries	15,401	18,390	21,272	23,072	26,272
- in associated companies	3,957	3,840	3,840	3,840	3,840
- in investments	538	538	538	538	538
Attributable capacity (ex-investment) - year end	19,358	22,230	25,112	26,912	30,112
Generation volume (M MWh)					
Gross generation	149.8	170.5	169.6	189.5	204.2
Net generation	140.5	158.8	158.2	177.6	191.5
yoy change (%)	23.0	13.0	-0.4	12.3	7.8
Consolidated gross generation	103.6	119.7	119.4	130.3	144.3
Consolidated net generation	96.8	112.1	112.1	122.8	136.0
yoy change (%)	28.6	15.8	0.0	9.6	10.7
Attri. gross generation	106.9	123.6	123.3	138.5	152.0
Attri. net generation	100.3	115.2	115.1	130.0	142.6
yoy change (%)	29.0	14.9	-0.1	12.9	9.8
Utilization hours - consolidated coal-fired plants	5,961	6,021	5,463	5,527	5,628
yoy change (%)	3.7	1.0	-9.3	1.2	1.8
Utilization rate (%)	68.0	68.7	62.4	63.1	64.3
Tariff (Rmb/MWh)					
Consolidated tariff before VAT	419	434	449	450	450
yoy change (%)	1.1	3.7	3.3	0.3	0.0
Consolidated coal-fired tariff before VAT	421	432	461	462	461
yoy change (%)	-0.8	2.5	6.7	0.2	-0.3
Coal price					
Raw coal price (Rmb/ton)	741	860	815	799	799
y-y change (%)	23.8	16.0	-5.3	-1.9	0.0
Unit fuel cost - consolidated coal-fired plants (Rmb/MWh)	253	280	263	258	257
yoy change (%)	21.4	10.9	-6.0	-2.0	-0.4
Coal consumption rate (g/kWh)	331	326	323	323	322
yoy change (%)	-1.7	-1.5	-0.8	-0.1	-0.4
Coal mine investment:					
Total coal output (m tonnes)	11	16	17	20	24
Attributable coal output (m tonnes)	7	10	10	11	13
Operating profit (Rmb/tonne)	188	220	100	100	100
Self sufficient ratio - on total basis	19%	24%	22%	23%	25%
Self sufficient ratio - on attributable basis	17%	20%	17%	17%	18%
Profit from coal business (HK\$m)	1,380	1,798	994	838	1,038
% of operating & associate profits from coal business	17%	40%	14%	10%	12%

Source: Company Reports and Citi Research Estimates

China Resources Power

Company description

Incorporated in Hong Kong on 27 August 2001, CR Power is the flagship power company and subsidiary of China Resource Holdings (CRH), which is the holding company of the China Resource Group, a major PRC state-owned conglomerate based in Hong Kong. With the exception of one power plant in Guangdong, all of the company's projects primarily use domestically manufactured equipment. Most of its generators are large, at 300MW or 600MW each. They are situated mostly in Eastern China, Central China and Southern China where power demand growth has been above the national average.

Investment strategy

We rate CR Power shares as Buy (1) even though the earnings outlook is challenging amid rising coal cost. FY13E PB is at the historical average. The company has raised its debt at fixed rates to 25% and in foreign currencies to mitigate interest rate risk. We believe CRP is better positioned than its PRC IPP peers given it has the highest net profit margin and hence earnings quality. It also has the highest coal self-sufficiency ratio among its peers.

Valuation

Our target price of HK\$21/share consists of: 1) HK\$19.28/share for power-generation business based on a DCF model, which we think is suitable for a power company such as CR Power that operates in a relatively stable industry with predictable cash flows; and 2) HK\$1.72/share for its coal mining business. Our model factors in our earnings forecasts up to 2015 and a terminal growth rate of 4.0% thereafter. Our WACC for CR Power is 8.4%, which assumes (i) a risk-free rate of 5.2%, (ii) a market risk premium of 6.8%, (iii) an equity beta of 1.31x, according to Bloomberg, (iv) a cost of debt of 6%, (v) a target debt-to-capital ratio of 60%, and (vi) a corporate tax rate of 25%. At our DCF-based target price, CR Power would trade at a FY13E PER of 11.2x.

Risks

Downside risks that could prevent shares from reaching our target price include: (1) Lower-than-expected generation volume; (2) Higher-than-anticipated coal prices; and (3) Lower-than-expected profit from coal mines.

China Suntien Green Energy (0956.HK)

Gas Pipelines Have Room to Grow; Wind Growth to Resume

Buy	1
Price (28 Nov 12)	HK\$1.51
Target price	HK\$1.95
<i>from HK\$2.10</i>	
Expected share price return	29.1%
Expected dividend yield	4.0%
Expected total return	33.2%
Market Cap	HK\$4,890M
	US\$631M

Price Performance

(RIC: 0956.HK, BB: 956 HK)



■ **Positive outlook** — We expect Suntien to see incremental growth driven by gas transmission and city-gas business, on rising retail and CNG usage, as its wind business sees more gradual growth driven by a centralized approval process. Maintain Buy but lower our target price to HK\$1.95 to reflect a more conservative earnings forecast.

■ **What's new** — Suntien targets to grow from 1.2-1.3 billion m³ in 2012E to 1.7 billion m³ by 2015, driven by the increase in gas transmissions. The slower 6% Y/Y growth in 2012E is partly driven by a reduction of gas sales to ENN Energy, which accounts for 5% of its gas sales volume. The company will focus on higher gas volumes in the retail and CNG segments, which can offer dollar margins of RMB0.66-0.73/sm³. In the first three quarters, retail and CNG accounts for ~50% of gas sales.

■ **Wind growth resumes** — Suntien's wind power generation in 3Q12 was up sharply Y/Y, and FY utilization hours may rise on the low-base effect. It also faced better grid curtailment to its peers. Grid curtailment was 3.8% in 2011 and only 5% in 2012E. Management expects the curtailment rate to drop in 2013E as new projects are in the southern part of Hebei with no grid curtailment issues. On the other hand, CER credit will likely be sharply lower, after the sharp fall in CDM prices.

■ **High capex** — The company targets Rmb1.8 billion capex in 2012E, well ahead of its operating cash flow, and it does expect potential debt funding to support its expansion in both gas projects and wind capacity installs.

■ **Revising estimates** — We lower our 2012/13/14E EPS estimates by 11%/7%/9% respectively to Rmb0.132/Rmb0.157/Rmb0.182 to factor in our more conservative gas volume growth forecasts.

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Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(RmbM)	(Rmb)	(%)	(x)	(x)	(%)	(%)
2010A	280	0.124	48.9	9.8	0.8	9.1	2.1
2011A	449	0.139	12.0	8.8	0.8	9.0	4.8
2012E	428	0.132	-4.6	9.2	0.8	8.3	5.0
2013E	510	0.157	19.0	7.7	0.7	9.7	5.3
2014E	589	0.182	15.6	6.7	0.7	10.9	5.5

Source: Powered by dataCentral

0956.HK: Fiscal year end 31-Dec						Price: HK\$1.51; TP: HK\$1.95; Market Cap: HK\$4,890m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	2,243	3,170	3,484	4,143	4,806	PE (x)	9.8	8.8	9.2	7.7	6.7
Cost of sales	-1,581	-2,257	-2,518	-2,983	-3,473	PB (x)	0.8	0.8	0.8	0.7	0.7
Gross profit	662	913	965	1,161	1,332	EV/EBITDA (x)	7.1	6.1	7.8	7.7	7.0
Gross Margin (%)	29.5	28.8	27.7	28.0	27.7	FCF yield (%)	-100.4	-61.0	-39.9	-7.5	-0.5
EBITDA (Adj)	844	1,193	1,257	1,468	1,709	Dividend yield (%)	2.1	4.8	5.0	5.3	5.5
EBITDA Margin (Adj) (%)	37.7	37.6	36.1	35.4	35.6	Payout ratio (%)	21	42	46	41	37
Depreciation	-237	-320	-384	-460	-553	ROE (%)	9.1	9.0	8.3	9.7	10.9
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	608	873	873	1,008	1,157	EBITDA	844	1,193	1,257	1,468	1,709
EBIT Margin (Adj) (%)	27.1	27.5	25.1	24.3	24.1	Working capital	-176	-283	83	37	27
Net interest	-168	-245	-336	-335	-355	Other	-15	-94	-630	-548	-628
Associates	50	73	85	97	102	Operating cashflow	654	816	709	957	1,108
Non-op/Except	0	0	0	0	0	Capex	-3,406	-3,213	-2,278	-1,253	-1,128
Pre-tax profit	490	701	621	770	903	Net acq/disposals	2	1	0	0	0
Tax	-58	-82	-42	-75	-99	Other	-57	-664	-133	-106	-96
Extraord./Min.Int./Pref.div.	-152	-170	-151	-185	-215	Investing cashflow	-3,461	-3,877	-2,411	-1,359	-1,224
Reported net profit	280	449	428	510	589	Dividends paid	0	-16	-188	-197	-207
Net Margin (%)	12.5	14.2	12.3	12.3	12.3	Financing cashflow	4,982	1,267	1,336	577	-106
Core NPAT	280	449	428	510	589	Net change in cash	2,145	-1,805	-366	175	-222
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-2,752	-2,397	-1,569	-296	-20
Reported EPS (Rmb)	0.124	0.139	0.132	0.157	0.182						
Core EPS (Rmb)	0.124	0.139	0.132	0.157	0.182						
DPS (Rmb)	0.026	0.058	0.061	0.064	0.067						
CFPS (Rmb)	0.290	0.252	0.219	0.295	0.342						
FCFPS (Rmb)	-1.218	-0.740	-0.485	-0.091	-0.006						
BVPS (Rmb)	1.486	1.608	1.588	1.641	1.686						
Wtd avg ord shares (m)	2,259	3,238	3,238	3,238	3,238						
Wtd avg diluted shares (m)	2,259	3,238	3,238	3,238	3,238						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	47.8	41.3	9.9	18.9	16.0						
EBIT (Adj) (%)	48.9	43.7	0.0	15.5	14.8						
Core NPAT (%)	68.2	60.5	-4.6	19.0	15.6						
Core EPS (%)	48.9	12.0	-4.6	19.0	15.6						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	2,475	920	553	728	507						
Accounts receivables	189	396	388	450	548						
Inventory	25	25	34	49	57						
Net fixed & other tangibles	6,079	7,713	9,608	10,401	10,976						
Goodwill & intangibles	1,423	2,449	2,816	2,957	3,105						
Financial & other assets	1,520	2,556	2,877	3,466	4,056						
Total assets	11,712	14,059	16,278	18,052	19,250						
Accounts payable	326	125	209	288	345						
Short-term debt	1,443	636	668	735	845						
Long-term debt	3,576	6,114	8,114	9,332	9,832						
Provisions & other liab	928	1,088	1,103	1,159	1,329						
Total liabilities	6,274	7,964	10,094	11,513	12,350						
Shareholders' equity	4,811	5,206	5,144	5,313	5,460						
Minority interests	628	889	1,040	1,225	1,440						
Total equity	5,439	6,095	6,184	6,538	6,900						
Net debt	2,544	5,831	8,229	9,338	10,170						
Net debt to equity (%)	46.8	95.7	133.1	142.8	147.4						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Stronger Growth in 2013

We expect Suntien to see incremental growth driven by gas transmission and city-gas business, on rising retail and CNG usage, as its wind business see more gradual growth, driven by a centralized approval process. Maintain Buy but lower our target price to HK\$1.95 (from HK\$2.10), to reflect a more conservative earnings forecast.

Growth Catalysts

After a difficult 12 months, China Suntien is starting to see better progress on its wind utilization rate in 2H12, and is making better progress on new wind projects. Management expects new project approvals will allow its geographical footprint to expand to other provinces in China. We see the following as key growth drivers going into 2013:

- Suntien obtained preliminary approval for a 200MW project and approval for another 200MW onshore project; management remains confident of adding 250MW in 2012 and in 2013. It added 1,650MW in reserve capacity in 1H12, and total reserved capacity reached 17,949MW;
- Gas volume will likely see limited growth in 2012E, after the reduction of gas usage by one of its key customers, but the volume growth is likely to resume in 2013 on increasing gas supplies from the Shaanxi-Beijing pipelines. We see the slowdown in growth rate by industrial customers to be temporary, especially on construction materials, where an improvement in the property sector may help resume gas consumption growth;
- Construction of Tangshan LNG project is 50% complete, and it continues to expand its city-gas business over its 370km city pipelines.
- For the wind capacity, we model flattish utilization hours in 2012E, but a stronger 4Q12 may lead to upside surprise to its full-year utilization hours.

Figure 1. China Suntien – Wind Capacity Increasing from Low Base

	2008	2009	2010	2011	2012E	2013E	2014E
New Capacity (MW)	173	173	448	346	250	250	300
Installed capacity (MW)	234	407	855	1,201	1,451	1,701	2,001
Attributable capacity (MW)	160	381	805	1,049	1,299	1,549	1,849
Gross power generation (MWh)	221,464	634,698	1,202,000	1,751,000	2,052,132	2,648,184	3,110,184
Net power delivered to grid (MWh)	166,639	554,841	1,099,473	1,672,331	1,870,313	2,413,555	2,834,622
Average utilization hours	2,130	2,276	2,360	2,048	2,050	2,100	2,100

Source: Company and Citi Research

Gas Growth in 2013 to Drive Re-rating

Suntien has two distinct business segments, natural gas distribution and wind power, making it more challenging to value the company. The stock currently trades at 9x 2012E earnings, much lower than downstream gas distributors' 17-20x 2012E earnings and wind operators' earnings.

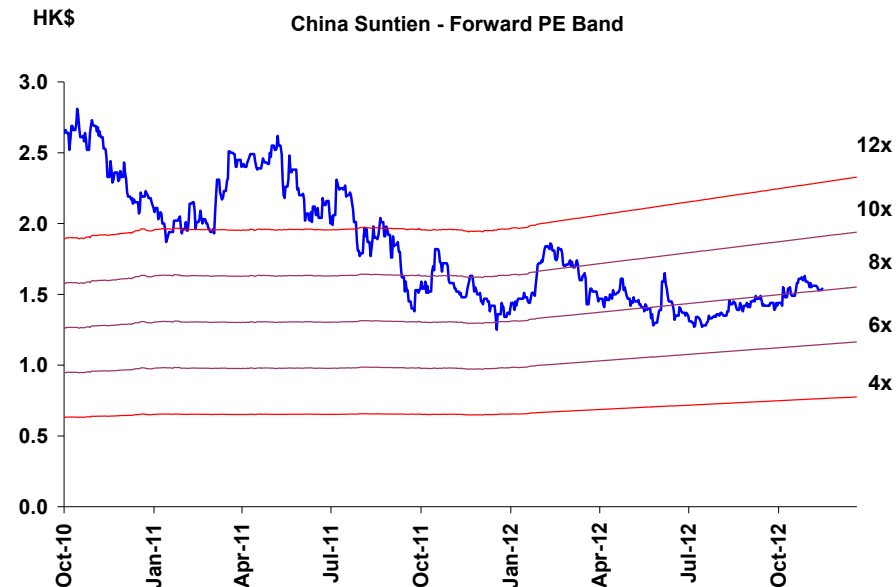
Our target price of HK\$1.95 is based on DCF, where we take into account a WACC of 8.93%, based on 58% target gearing, and a terminal growth rate of 3%. Our target price implies 10x 2013E earnings.

Figure 2. China Suntien – DCF Analysis

(Rmbm)	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
FCF		-1,569	-296	-20	197	563	972	1,398	1,890	2,336
Discount factor		1.00	1.09	1.19	1.29	1.41	1.53	1.67	1.82	1.98
Net debt	-6,031	-1,569	-272	-17	153	400	634	837	1,039	1,179
Terminal Value:										
NAV in 2021E		1,213.9								
Expected terminal growth rate		3.0%								
Expected Capital Employed on new investments		10.0%								
WACC		8.93%								
Terminal Value		14,337								
Discount factor		1.98								
NPV of terminal value		7,234								
DCF		3,586								
DCF/Share (HK\$)		1.95								

Source: Citi Research estimates

Figure 3. China Suntien – Forward PE Band (2010-2012E)



Source: Citi Research

While our 2012/13E earnings forecasts are 24% and 20% below consensus, we believe the gap could be due to limited coverage from analysts. We are factoring in lower gas volume growth on our latest forecast.

Figure 4. China Suntien – Earnings Revision

	Net Profit (Rmb mils)		Diluted EPS (Rmb)		% change
	Old	New	Old	New	
2012E	482	428	0.149	0.132	-11.2%
2013E	548	510	0.169	0.157	-6.9%
2014E	649	589	0.200	0.182	-9.2%

Source: Citi Research estimates

Figure 5. China Suntien – Citi Forecast vs. Bloomberg Consensus

Year to Dec	Citi	Consensus	% Difference
2012E Sales (Rmb mn)	3,484	3,824	-9%
2012E EBIT (Rmb mn)	873	1,055	-17%
2012E Net income (Rmb mn)	428	561	-24%
2012E EPS (Rmb)	0.13	0.17	-22%
2013E Sales (Rmb mn)	4,143	4,488	-8%
2013E EBIT (Rmb mn)	1,008	1,230	-18%
2013E Net income (Rmb mn)	510	635	-20%
2013E EPS (Rmb)	0.16	0.19	-18%
2014E Sales (Rmb mn)	4,806	5,190	-7%
2014E EBIT (Rmb mn)	1,157	1,551	-25%
2014E Net income (Rmb mn)	589	691	-15%
2014E EPS (Rmb)	0.18	0.21	-13%

Source: Bloomberg and Citi Research estimates

Figure 6. China Suntien – Key Financial Statistics

Year to Dec	2008	2009	2010	2011	2012E	2013E	2014E
Gross margin	24.6	28.1	29.5	28.8	27.7	28.0	27.7
Operating margin	21.0	26.9	27.1	27.5	25.1	24.3	24.1
Pre-tax margin	15.8	20.2	21.8	22.1	17.8	18.6	18.8
Net profit margin	8.5	11.0	12.5	14.2	12.3	12.3	12.3
Effective tax rate	6.2	6.1	11.9	11.7	6.8	9.8	11.0
Interest coverage (x)	4.1	3.9	3.6	3.6	2.6	3.0	3.3
ROE (%)	8.7	12.4	5.8	8.6	8.3	11.2	13.1
Net debt-to-equity (%)	198.3	200.5	52.9	115.9	163.9	209.3	230.8
EV/EBITDA (x)	18.9	10.9	7.6	9.7	10.6	9.8	8.9

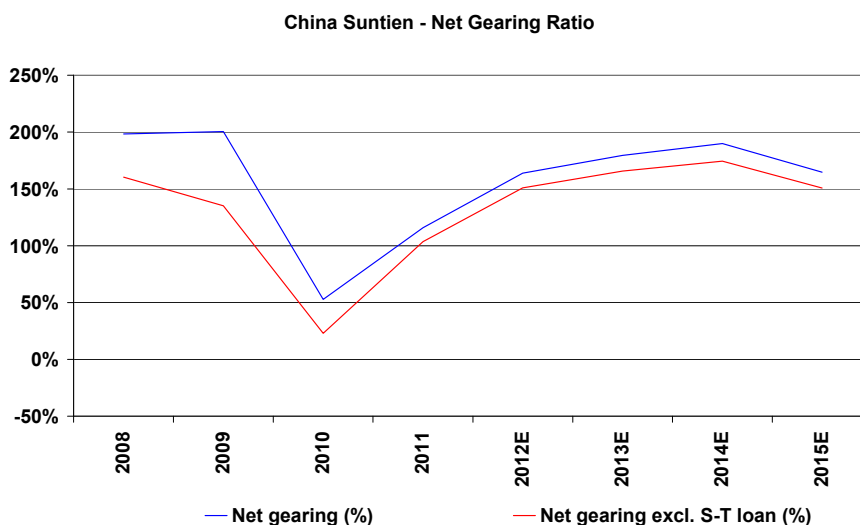
Source: Company and Citi Research estimates

Figure 7. China Suntien – Income Statement (Rmb m)

Year to Dec	FY09	FY10	FY11	FY12E	FY13E	FY14E
Turnover	1,517	2,243	3,170	3,484	4,143	4,806
Revenue from natural gas segment	1,253	1,727	2,310	2,617	3,015	3,467
Revenue from wind segment	265	516	860	863	1,125	1,334
Cost of sales	(1,091)	(1,581)	2,254	(2,518)	(2,983)	(3,473)
Gross profit	426	662	913	965	1,161	1,332
Other income	51	84	135	90	59	70
Distribution costs	(0)	(0)	(1)	(0)	(0)	(1)
Administration expenses	(69)	(101)	(152)	(165)	(190)	(216)
EBIT	408	608	873	873	1,008	1,157
Finance costs	(104)	(168)	(245)	(336)	(335)	(355)
Profit before taxation	306	490	701	621	770	903
Taxation	(19)	(58)	(82)	(42)	(75)	(99)
Minority interest	(121)	(152)	(170)	(151)	(185)	(215)
Net profits	166	280	449	428	510	589
Turnover YoY (%)	48.9%	47.8%	41.3%	9.9%	18.9%	16.0%
Net Profit YoY (%)	91.5%	68.2%	60.5%	-4.6%	19.0%	15.6%

Source: Company and Citi Research estimates

Figure 8. China Suntien – Net Gearing Ratio



Source: Company and Citi Research estimates

Figure 9. China Suntien – Balance Sheet (Rmb m)

As at 31 Dec	FY09	FY10	FY11	FY12E	FY13E	FY14E
Bank balances and cash	330	2,475	720	353	528	307
Pledged bank deposits	15	0	200	200	200	200
Account receivables	85	189	396	388	450	548
Inventories	22	25	25	34	49	57
Others	91	222	622	653	686	720
Current assets	542	2,911	1,962	1,629	1,913	1,832
PP&E	4,358	6,079	7,713	9,608	10,401	10,976
Investment & Others	881	1,298	1,934	2,224	2,781	3,337
Intangible asset	3	1,423	2,449	2,816	2,957	3,105
Non-current assets	5,241	8,801	12,097	14,649	16,139	17,418
Total assets	5,783	11,712	14,059	16,278	18,052	19,250
Account payables	440	326	125	209	288	345
Borrowings - amt due within one year	879	1,443	636	668	735	845
Taxation payable	14	26	14	16	17	19
Others	396	901	1,048	1,059	1,112	1,278
Current liabilities	1,728	2,696	1,824	1,951	2,152	2,487
Borrowings - amt due after one year	2,177	3,577	6,140	8,143	9,362	9,863
Non-current liabilities	2,177	3,577	6,140	8,143	9,362	9,863
Total liabilities	3,906	6,274	7,964	10,094	11,513	12,350
Shareholders equity	1,344	4,811	5,206	5,144	5,313	5,460
Minority interest	534	628	889	1,040	1,225	1,440
Total Equity	1,878	5,439	6,095	6,184	6,538	6,900

Source: Company and Citi Research estimates

Figure 10. China Suntien – Cash flow statement (Rmb m)

Year to Dec	FY09	FY10	FY11	FY12E	FY13E	FY14E
Operating activities						
Profit before tax	306	490	701	621	770	903
Adjustments	102	147	172	(336)	(335)	(355)
EBIT	408	637	873	285	435	548
Exceptional items	(2)	2	(0)	(0)	100	80
Depreciation & amortisation	151	237	370	384	460	553
Change in working capital	(43)	(176)	(283)	83	37	27
Operating cash flow	514	700	959	751	1,032	1,207
Net operating cash flow	502	654	866	709	957	1,108
Capex	(1,383)	(3,406)	(3,213)	(2,278)	(1,253)	(1,128)
Free cash flow	(882)	(2,752)	(2,347)	(1,569)	(296)	(20)
Change in equity	436	3,434	125	0	0	0
Change in debt	819	2,002	1,645	2,034	1,285	612
Other investing activities	10	(57)	(664)	(133)	(106)	(96)
Investing cash flow	(1,373)	(3,461)	(3,877)	(2,411)	(1,359)	(1,224)
Dividends paid	0	0	(16)	(188)	(197)	(207)
Financing cash flow	968	4,982	1,267	1,336	577	(106)
Beginning cash	233	330	2,475	720	353	528
Change in cash	97	2,176	(1,743)	(366)	175	(222)
Ending cash	330	2,475	720	353	528	307

Source: Company and Citi Research estimates

China Suntien Green Energy

Company description

China Suntien was established as a joint stock limited company in the PRC in Feb 2010, as part of the HECIC Group's clean energy business. It operates natural gas transmission and distribution facilities in Hebei province, and sells natural gas to wholesale, retail and CNG customers through its distribution channels. China Suntien also plans, develops, and operates wind farms for electricity generation. HECIC has been operating the business as a state-owned enterprise in Hebei province.

Investment strategy

We rate Suntien shares Buy with a target price of HK\$1.95. Suntien has stable city-gas operations, with gas volumes near 1.3bn m3 in 2012 and set to grow 20% yoy upon completion of trunk pipelines as well as securing of long-term gas supplies. The company's new wind farms focus on Hebei, where it faces less grid congestion with 1.2GW installed capacity. It could see 200-250MW/year new installments in 2012E. The stock trades at deep discounts to both China wind operators and gas distributors, given the uncertainty in wind project expansion. Near-term catalysts include higher gas volumes and approvals for new wind capacity.

Valuation

We value Suntien at HK\$1.95 a share based on a DCF approach. We assume a WACC of 8.93%, using a gearing ratio of 58% and a terminal growth rate of 3%. Our target price implies 10x 2013E earnings. In our view, a DCF approach can capture the company's growth outlook in both gas and wind segments, and take into account the long-term operating cash flow expected from its projects while balancing the capex needed to construct new city-gas pipelines and wind farms.

Risks

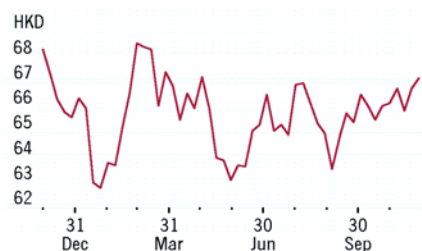
Downside risks that could prevent the stock from reaching our target price include: 1) a slower wind installation growth rate due to a protracted approval process or lack of grid connection; 2) lower utilization hours for its existing wind farm projects; 3) change in pricing mechanism for NG transmission and city-gas distribution; and 4) and higher-than-expected costs to expand its gas distribution and wind farm capacities.

CLP Holdings (0002.HK)

Sustainable Yield But With Moving Parts in 2013E

Neutral	2
Price (28 Nov 12)	HK\$67.10
Target price	HK\$70.00
from HK\$68.00	
Expected share price return	4.3%
Expected dividend yield	3.8%
Expected total return	8.2%
Market Cap	HK\$161,452M
	US\$20,832M

Price Performance
(RIC: 0002.HK, BB: 2 HK)



■ **Summary** – We cut FY13-14E net profit by 3-5% on more losses from the Indian Jhajjar coal-fired plant amid local coal shortage and lower margin of Australian retail power business in NSW from 1 July 2013 under new retail price calculation formula. But we mildly lift our DCF-based TP by 3% to HK\$70 after rollover. We do not expect a fundamental change in the HK business from SOC interim result in 2013E. At 4.0%, CLP's FY13E yield is decent amid prevailing low interest rate environment.

■ **Hong Kong business preparing interim review in 2013** – The key issue is to decide SOC capex in 2014-18E, after an expected HK\$39.9bn in 2009-13E in view of rising equipment costs, offsetting the lack of new plant construction and FGD installation completed in 2011. We expect CLP to put forward to build its 200MW off-shore wind farms in 2014-18E, but the scale might be cut to 100-150MW.

■ **Australian profit up hoh in 2H12E** – CLP renamed its Australian business into EnergyAustralia ('EA') in 3Q12 to facilitate possible local listing. EA's profit was up hoh in 2H12E for more gas demand in 3Q12 and an A\$257.5m cash compensation to Yallourn brown coal plant for carbon tax in 2H12-1H13E; EA's other plants have gained from carbon tax with tariff hike while below market average carbon intensity. EA's net profit would cut 3-5% from new NDW retail price calculation from Jul 2013.

■ **Indian coal shortage persists, but some improvement in 4Q12-1Q13E** – Utilization of the Jhajjar coal fired plant was below 40% in Aug-Oct. Execution of fuel supply agreement by Coal India was poor at below 50%. This plant has been approved to import 1m tonnes, of which 30-40% is to be used in 4Q12E and 60-70% in 1Q13E. We expect more losses from this plant in 2H12E than in 1H12 as coal shortage was exacerbated after commissioning of its unit 2 in July.

■ **Mild earning cuts** – FY13-14E NP is cut 3-5% on lower contributions from India and Australia. We do not see any catalysts for the share price until any EA listing, which media reports suggest could boost its financial resources by A\$2.5-3.0bn.

Asian Utilities

Hong Kong Utilities

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Statistical Abstract

FYE	Net Profit	EPS	P/E	D. Yld
31 Dec	(HK\$M)	(HK\$)	(x)	(%)
2009A	8,196	3.41	19.7	3.7
2010A	10,332	4.29	15.6	3.7
2011A	9,288	3.86	17.4	3.8
2012E	9,928	4.13	16.3	3.8
2013E	12,337	5.13	13.1	4.0
2014E	12,437	5.17	13.0	4.3

Source: Powered by dataCentral

Earnings Revision

FYE	Net Profit	EPS	%	DPS
31 Dec	(HK\$M)	(HK\$)	chg	(HK\$)
2012E	9,928	4.13	0.0	2.58
Prev	9,933	4.13		2.58
2013E	12,337	5.13	-3.1	2.66
Prev	12,728	5.29		2.76
2014E	12,437	5.17	-4.9	2.91
Prev	13,071	5.43		3.06

Source: Powered by dataCentral

0002.HK: Fiscal year end 31-Dec						Price: HK\$67.10; TP: HK\$70.00; Market Cap: HK\$161,452m; Recomm: Neutral					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	58,410	91,634	88,057	103,154	107,988	PE (x)	15.6	17.4	16.3	13.1	13.0
Cost of sales	-31,838	-35,663	-30,268	-33,982	-37,615	PB (x)	2.0	2.0	1.9	1.8	1.7
Gross profit	26,572	55,971	57,789	69,172	70,372	EV/EBITDA (x)	10.1	9.8	9.8	8.8	8.6
Gross Margin (%)	45.5	61.1	65.6	67.1	65.2	FCF yield (%)	1.1	0.3	3.7	7.4	7.8
EBITDA (Adj)	17,462	19,541	20,991	23,624	24,172	Dividend yield (%)	3.7	3.8	3.8	4.0	4.3
EBITDA Margin (Adj) (%)	29.9	21.3	23.8	22.9	22.4	Payout ratio (%)	58	65	62	52	56
Depreciation	-5,065	-6,353	-5,342	-5,429	-5,518	ROE (%)	13.7	11.5	11.9	13.9	13.2
Amortisation	0	0	0	0	0	Cashflow (HK\$m)					
EBIT (Adj)	12,397	13,188	15,649	18,196	18,654	EBITDA	17,462	19,541	20,991	23,624	24,172
EBIT Margin (Adj) (%)	21.2	14.4	17.8	17.6	17.3	Working capital	289	-1,782	-1,726	-1,718	-1,280
Net interest	-4,111	-5,859	-6,264	-6,351	-6,408	Other	-5,865	-5,135	-6,655	-3,254	-3,415
Associates	2,893	3,610	2,138	2,482	2,659	Operating cashflow	11,886	12,624	12,610	18,652	19,476
Non-op/Except	0	0	0	0	0	Capex	-10,137	-12,109	-6,644	-6,783	-6,926
Pre-tax profit	11,179	10,939	11,524	14,327	14,905	Net acq/disposals	-981	2,743	-4,044	-8,121	-8,119
Tax	-844	-1,650	-1,436	-1,841	-2,320	Other	855	-15,893	1,393	1,590	1,724
Extraord./Min.Int./Pref.div.	-3	-1	-159	-149	-148	Investing cashflow	-10,263	-25,259	-9,294	-13,314	-13,321
Reported net profit	10,332	9,288	9,928	12,337	12,437	Dividends paid	-5,984	-5,977	-6,148	-6,331	-6,777
Net Margin (%)	17.7	10.1	11.3	12.0	11.5	Financing cashflow	-4,915	11,809	-4,996	-5,242	-6,736
Core NPAT	10,332	9,288	9,928	12,337	12,437	Net change in cash	-3,292	-826	-1,681	96	-581
Per share data						Free cashflow to s/holders					
Reported EPS (\$)	4.29	3.86	4.13	5.13	5.17		1,749	515	5,966	11,870	12,551
Core EPS (\$)	4.29	3.86	4.13	5.13	5.17						
DPS (\$)	2.48	2.52	2.58	2.66	2.91						
CFPS (\$)	4.94	5.25	5.24	7.75	8.09						
FCFPS (\$)	0.73	0.21	2.48	4.93	5.22						
BVPS (\$)	33.11	33.77	35.49	38.06	40.23						
Wtd avg ord shares (m)	2,406	2,406	2,406	2,406	2,406						
Wtd avg diluted shares (m)	2,406	2,406	2,406	2,406	2,406						
Growth rates											
Sales revenue (%)	15.3	56.9	-3.9	17.1	4.7						
EBIT (Adj) (%)	14.3	6.4	18.7	16.3	2.5						
Core NPAT (%)	26.1	-10.1	6.9	24.3	0.8						
Core EPS (%)	26.1	-10.1	6.9	24.3	0.8						
Balance Sheet (HK\$m)											
Cash & cash equiv.	4,756	3,866	2,185	2,281	1,700						
Accounts receivables	11,118	19,419	20,006	23,698	25,973						
Inventory	751	1,470	1,475	1,435	1,416						
Net fixed & other tangibles	124,406	136,685	139,273	146,130	153,041						
Goodwill & intangibles	9,150	27,369	27,369	27,369	27,369						
Financial & other assets	29,174	25,479	26,922	27,502	28,124						
Total assets	179,355	214,288	217,231	228,416	237,623						
Accounts payable	11,509	17,133	15,826	17,306	17,840						
Short-term debt	11,795	16,893	16,129	17,809	19,299						
Long-term debt	36,807	52,925	54,994	54,892	53,925						
Provisions & other liab	39,486	45,985	44,635	46,435	49,216						
Total liabilities	99,597	132,936	131,585	136,442	140,281						
Shareholders' equity	79,661	81,259	85,393	91,572	96,793						
Minority interests	97	93	252	401	549						
Total equity	79,758	81,352	85,646	91,974	97,342						
Net debt	43,846	65,952	68,938	70,420	71,524						
Net debt to equity (%)	55.0	81.1	80.5	76.6	73.5						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

Rising Fuel Cost in HK Not a Concern

CLP does not need to build new plants in HK in the run-up to 2017E with over 30% reserve margin and being able to comply with the government's emission cut requirement from running existing power plants. In 2012E, CLP would consume 1.7bcm NG in HK, similar to that of a year ago, with the supply solely from Yacheng gas field in South China Sea at US\$6/mmbtu. In 2013E, its NG consumption volume would be the same as this year; but the supply source would be changed to have two-thirds from Yacheng gas field at US\$6/mmbtu and one-third from West-to-East pipeline No.2 at US\$18/mmbtu, in view of depleting reserve at Yacheng gas field. In 2015E, its NG consumption would double to 3.5bcm to raise its generation mix from NG and lower that from coal to comply with emission cap requirements set by the Hong Kong Government to lower CLP's emission in 2015E to not more than that in 2010; NG supply in 2015E would be solely from West-to-East pipeline No.2 assuming that the supply from Yacheng has been depleted by then. In 2017E, CLP would raise its generation mix from NG further and use more low sulphur coal to trim its emission by further 17% from that in 2015E, as per regulatory requirements.

The above fuel cost increment in HK would not affect CLP's earnings. Any fuel compensation shortfall would not affect its earnings as it would be reimbursed by end users. CLP's fuel clause account liability was less than previous expectation of HK\$800m by end 2012, due to (i) increase in generation mix from coal after emission cut works completion at Castle Peak power plant; and (ii) decline in coal price in 2012E. The company has submitted its plan to Hong Kong Government to raise its fuel clause surcharge effective 1 Jan 2013. The increment would be announced in mid Dec.

We expect CLP to put forward its plan to build off-shore wind farms in HK and have it included in SOC capex in 2014-18E to be decided by negotiation with Hong Kong Government in 2013E. Given the high unit cost (HK\$25m/MW), we expect this 200MW project to be built by phases to lessen tariff hike pressure, including 100-150MW to be built in 2014-18E.

Australian Recurring Profit up HoH in 2H12E

CLP renamed its Australian business into EnergyAustralia in 3Q12 to facilitate a potential local listing. The net debt to equity ratio of EnergyAustralia is not high at about 43%. But S&P's credit rating is only BBB from taking into consideration the competitive industry risk and counterparty risk of its forward contracts. CLP stipulated in its interim report that EnergyAustralia's listing in Australia is one of its options. We estimate it targets to float a 25-49% stake, which media reports suggest could boost its financial resources by A\$2.5-3.0bn and which would likely be recycled back to CLP's headquarters in Hong Kong for expansion elsewhere. But the possible listing looks difficult amid current stock market conditions and EA's competitive business landscape.

EnergyAustralia had higher profitability in 3Q12 than in 1H12, benefitting from stronger gas demand in winter. Its gas business there has higher profit margin than power business as pool prices of the latter are generally low. Power demand there is a relatively weak this year amid softened economy. Still, we expect EnergyAustralia's recurring profit in 2H12E to be higher than in 1H12 with the resumption of operation of the four units at Yallourn Power Plant after flooding.

EnergyAustralia has gained from carbon tax implementation in Australia since 1 July. The reasons are that (i) its brown coal fired Yallourn Power Plant will receive A\$257.5m cash compensation for carbon tax implementation for the first 12 months from 1 July 2012. The amount is to be amortized over the 12 months period (i.e.

from 1 July 2012 to 30 June 2013). To recap, CLP would have positive accounting benefit in the first five years of carbon tax implementation in Australia due to cash compensation in the first year and free carbon credit in the 2nd-5th years; and (ii) the average carbon intensity of its other power plants in Australia is slightly below market average hence could pass through the carbon tax from incremental target hike.

EnergyAustralia will have a few percentage net profit cut from 1 July 2013

when a new retail price path period will start in NSW. The retail price there is calculated taking into account various items, including a wholesale cost assumption. The wholesale cost assumption is 100% based on LRMC (low run marginal cost), which is primarily NG cost. From 1 July 2012, the wholesale cost assumption would be decided equal to the sum of 75% LRMC and 25% market cost. The market cost is similar to coal cost hence lower than LRMC. The change would lead to lower wholesale cost assumption, lower retail price, and therefore, lower retail sales margin of EnergyAustralia in NSW. With retail tariff up 20.6% after carbon tax implementation on 1 July 2012 and 40% in last four years, more regulatory policies to cut retail tariffs there could happen.

More Losses from India in 2H12E

Coal supply to CLP's Jhajjar coal-fired plant in India remains a key concern.

A fuel supply agreement was signed with Central Coalfield Limited under Coal India Limited for 5.21m tonnes pa of coal for 20 years, while CIL will not have a liability to pay liquefied damager for any shortage in next three years. The 5.21m tonnes coal supply would be sufficient for Jhajjar coal-fired plant to operate at 80% utilization. So far, the execution of this contract has been poor below 50%; hence utilization of the Jhajjar power plant was less than 40% in Aug-Oct.

Utilization of Jhajjar plant would be higher in 4Q12-1Q13E with imported coal.

This plant has been approved to import 1m tonnes on or before 31 March 2013, with the incremental cost to be passed through to grid companies. The import would start from Nov, including 30-40% of the 1m tonnes in 4Q12E and 60-70% in 1Q13E. We assume the utilization to rise to 50-60% in these periods with more ample coal supply.

China to Acquire Nuclear Plant Stake in 2013E

The Chinese Government announced in Oct to resume new nuclear project approval. We expect new project approval to resume in 1H13E. This would facilitate CLP to finalize its acquisition of 17% stake of Yanjiang nuclear power plant in Guangdong, China. To recap, the deal was suspended as potential ownership transfer was prohibited during PRC Government's suspension of new project approval. The acquisition cost of the 17% stake would be revisited and raised, from originally Rmb4.8bn, taking into account the increased capitalized interest expense. China Guangdong Nuclear Power Company Ltd., the developer of this project, is positive about CLP's investment in this project as both companies have two-decade cooperation at Daya Bay Nuclear Power Plant, also in Guangdong Province.

If the HK Government wants to raise the percentage of electricity from nuclear, CLP needs to invest in new nuclear units, such as Taishan Ph.2 Nuclear Plant. Output from Yanjiang Nuclear Plant is not planned to be sold to HK for long distance. Additionally, CLP's profit from Daya Bay nuclear plant would have upside from 2014E when the plant would be fully depreciated after operating for 20 years.

SOTP DCF

Figure 1. CLP - SOTP DCF

	Method	Value (HK\$m)	Per share (HK\$)	By project	By country
SOC	DCF	135,294	56.23	55%	55%
Non-SOC					
Australia:					
TXUA	DCF	53,582	22.27	22%	
Yallourn	DCF	9,700	4.03	4%	26%
China:					
Guangdong Nuclear Power	DCF	9,267	3.85	4%	
Guangxi Fangchenggang	DCF	5,540	2.30	2%	
CLP Guohua	DCF	1,901	0.79	1%	
Shandong JV	DCF	2,277	0.95	1%	
Guangzhou Pumped Storage	15x P/E	990	0.41	0%	
Shenmu II	DCF	554	0.23	0%	
Huaiji	15x P/E	494	0.21	0%	
Sichuan Jiangbian Hydro	DCF	1,767	0.73	1%	
Yunnan Dali Yang Er Hydro	DCF	294	0.12	0%	9%
India:					
GPEC	DCF	5,676	2.36	2%	
Jhajjar	DCF	5,362	2.23	2%	4%
Other countries:					
Taiwan (Ho-Ping)	DCF	5,145	2.14	2%	2%
Others	BV	10,263	4.27	4%	4%
Total		248,108	103.11	100%	100%
Net debt		-79,674	-33.11		
DCF		168,434	70.00		

Source: Company Reports and Citi Research Estimates

Figure 2. CLP's DCF sensitivity (HK\$/share)

Discount rate	6.47%	6.72%	6.95%
Terminal growth			
2.5%	\$68.1	\$64.7	\$61.7
3.0%	\$74.3	\$70.0	\$66.4
3.5%	\$82.6	\$76.9	\$72.3

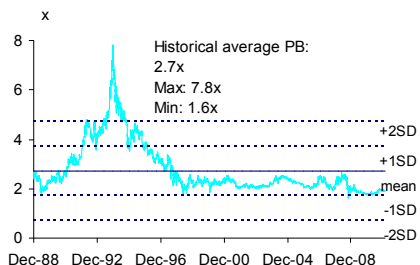
Source: Company Reports and Citi Research Estimates

Figure 3. CLP - Historical Rolling PE



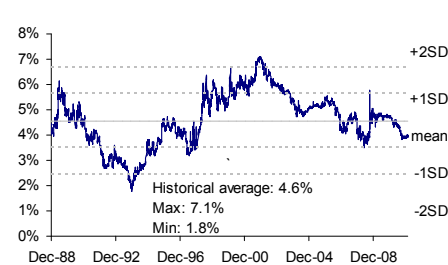
Source: Citi Research

Figure 4. CLP - Historical Rolling PB



Source: Citi Research

Figure 5. CLP - Dividend Yield



Source: Citi Research

Financial Statement & Key Assumptions

Figure 6. CLP - Income statements by business nature (HK\$m)

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
SOC earnings from Hong Kong	6,129	6,339	6,635	6,957	7,267
Non-SOC earnings	3,476	4,492	3,684	5,183	5,986
GNPJVC & PSDC	836	657	614	618	579
Other power projects in China	642	560	818	805	812
Power projects in AP	1,834	3,151	2,257	3,692	4,521
Sales to the Chinese mainland	47	53	59	64	71
Electricity-related businesses	117	70	-64	3	3
Total earnings	9,605	10,831	10,319	12,140	13,253
Less: Unallocated net finance costs	-18	-48	-69	-70	-71
Less: Unallocated group expenses	-439	-471	-581	-683	-746
Pre-exceptional earnings	9,148	10,312	9,669	11,386	12,437
Exceptional items	1,184	-1,024	259	950	0
Net profit	10,332	9,288	9,928	12,337	12,437
% chg	26.1%	-10.1%	6.9%	24.3%	0.8%
EPS	4.29	3.86	4.13	5.13	5.17
% chg	26.1%	-10.1%	6.9%	24.3%	0.8%

Source: Company Reports and Citi Research Estimates

Figure 7. CLP - Income statement in ordinary format (HK\$m)

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
SOC business	30,016	31,293	33,541	37,215	40,841
Other businesses	28,394	60,341	54,516	65,938	67,147
Total revenue	58,410	91,634	88,057	103,154	107,988
Purchases of electricity from CAPCO	-21,740	-23,494	-25,281	-28,995	-32,628
Purchases of nuclear electricity	-9,746	-11,817	-4,635	-4,635	-4,635
Pumped storage service fee	-352	-352	-352	-352	-352
Staff costs	-2,189	-2,623	-2,277	-2,323	-2,369
Other operating costs	-6,921	-33,807	-34,520	-43,224	-43,831
EBITDA	17,462	19,541	20,991	23,624	24,172
Depreciation	-5,065	-6,353	-5,342	-5,429	-5,518
EBIT	12,397	13,188	15,649	18,196	18,654
Net finance cost	-4,111	-5,859	-6,264	-6,351	-6,408
Share of Profits of JVs and associates	2,893	3,610	2,138	2,482	2,659
PBT	11,179	10,939	11,524	14,327	14,905
Tax	-844	-1,650	-1,436	-1,841	-2,320
Minority interest	-3	-1	-159	-149	-148
PAT	10,332	9,288	9,928	12,337	12,437
SOC transfer	0	0	0	0	0
Net profit	10,332	9,288	9,928	12,337	12,437
% chg	21.2%	14.4%	17.8%	17.6%	17.3%
EPS	4.29	3.86	4.13	5.13	5.17
% chg	26.1%	-10.1%	6.9%	24.3%	0.8%

Source: Company Reports and Citi Research Estimates

Figure 8. CLP's cashflow statement (HK\$m)

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Operating profit	12,397	13,188	15,649	18,196	18,654
Depreciation & goodwill	5,065	6,353	5,342	5,429	5,518
Other non-cash items	-941	958	168	0	0
Rebates	0	0	-1,039	0	0
Changes in working capital	289	-1,782	-1,726	-1,718	-1,280
Net finance cost	-4,100	-5,293	-3,297	-3,405	-3,452
Tax	-824	-800	-2,488	152	37
Operating CF	11,886	12,624	12,610	18,652	19,476
Capex	-10,137	-12,109	-6,644	-6,783	-6,926
Disposal/lease	267	490	-4,217	-8,433	-8,433
Investments	-3,324	-923	-984	-984	-984
Dividends received	2,594	3,230	2,377	2,574	2,708
Investing CF	-10,263	-25,259	-9,294	-13,314	-13,321
FCF	1,623	-12,635	3,316	5,338	6,155
Change in bank loans	1,069	17,786	1,151	1,089	41
Dividends paid	-5,984	-5,977	-6,148	-6,331	-6,777
Financing CF	-4,915	11,809	-4,996	-5,242	-6,736
Chg in cash & equiv	-3,292	-826	-1,681	96	-581

Source: Company Reports and Citi Research Estimates

Figure 9. Key assumptions

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
HK's tariff & sale volume:					
Basic tariff (HK\$/kWh)	0.800	0.801	0.842	0.884	0.928
Fuel clause rebate	0.115	0.141	0.178	0.208	0.278
Special Rebate	0.000	0.000	-0.033	0.000	0.000
Net tariff (HK\$/kWh)	0.915	0.942	0.987	1.092	1.206
yoy	2.7%	3.0%	4.8%	10.6%	10.5%
Local unit sales (m kWh)	30,929	31,168	31,480	31,794	32,112
% chg	1.2%	0.8%	1.0%	1.0%	1.0%
Export unit sales (m kWh)	2,609	2,957	3,253	3,578	3,936
% chg	-30.1%	13.3%	10.0%	10.0%	10.0%
SOC:					
Rate of return (%)	10.1%	10.1%	10.1%	10.1%	10.1%
Capex (HK\$m)	7,750	8,044	8,285	8,534	8,790
Permitted return (HK\$m)	6,129	6,339	6,635	6,957	7,267
- yoy	2.8%	3.4%	4.7%	4.9%	4.5%
Development fund balance (HK\$m)	1,505	637	283	110	548
Profit mix:					
% SOC earnings	59.3%	68.3%	66.8%	56.4%	58.4%
% non-SOC earnings	40.7%	31.7%	33.2%	43.6%	41.6%

Source: Company Reports and Citi Research Estimates

Figure 10. Key Ratios

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Interest coverage (x)	7.0	5.9	4.7	5.2	5.3
Dividend coverage (x)	2.9	3.1	3.4	3.7	3.5
ROA	6.2%	4.7%	4.6%	5.5%	5.3%
Net Debt/Equity	55.0%	81.1%	80.5%	76.6%	73.5%
Net Debt/Asset	24.4%	30.8%	31.7%	30.8%	30.1%
Total debt/EBITDA	2.8	3.7	3.4	3.1	3.0

Source: Company Reports and Citi Research Estimates

CLP Holdings

Company description

CLP is the larger of Hong Kong's two electric utilities, with a 74% share of the power market. It serves 2.3m customers in Kowloon, the New Territories and outlying islands, a franchise area with 80% of Hong Kong's population. CLP's generation assets in Hong Kong are owned by CAPCO, 40%-held by CLP Power HK and 60% by Exxon Mobil, while CLP wholly owns the T&D network in its service areas. CLP Power Hong Kong and CAPCO are governed by a Scheme of Control (SOC) agreement. The old agreement was signed in 1993 and the new one, which was signed in Jan 2008, is effective from Oct 2008 to Sep 2018. CLP also owns power projects in Australia, China, India, Taiwan and Thailand; it engages in non-electricity businesses such as property development and engineering services.

Investment strategy

We rate CLP shares as Neutral (2) with a target price of HK\$70. We expect more investment opportunities in Hong Kong given the government's fuel mix target by 2020. CLP would also look for overseas investment opportunities which fit its requirements. However, the company has limited financial flexibility for acquisitions given it is constrained by its high gearing, which could persist until the potential listing of EnergyAustralia in Australia. It is also possible that CLP may have to issue new shares if it raises its stake in CAPCO prior to the planned EnergyAustralia listing. Additionally, CLP looks vulnerable to political risk in Hong Kong as it targets to raise retail power tariffs in its HK operating region by 40% in the run-up to 2015E to offset higher total fuel cost.

Valuation

Our target price of HK\$70 is based on a DCF approach, factoring in our earnings forecasts up to 2015 and a terminal value thereafter. We think this method is suitable for power companies such as CLP Holdings that work in a relatively static industry with predictable cash flows. Our WACC for CLP is 6.7% (except 8% for Australian power plants, 9% for Australian power retail business, 9% for GPEC and Jhajjar, 8.5% for Chinese power plants and 8% for Ho Ping), which assumes (i) a risk-free rate of 5.2%, (ii) a market risk premium of 6.8%, (iii) an equity beta of 0.47x, according to Bloomberg, (iv) a cost of debt of 5.2%, (v) a target gearing ratio of 40%, and (vi) a 16.5% corporate tax rate. We assume a terminal growth rate of 3.0% beyond 2015.

Risks

Key factors that could cause the shares to trade above our target price include: (1) Unexpected sizable gain or loss from overseas assets. (2) Unexpected interest rate change; our DCF-based target price is highly sensitive to the risk-free rate, which depends on interest rates. Key downside risks to our target price include: 1) Regulatory risk in Australia regarding carbon pricing; and 2) Higher than expected coal prices in China.

Datang International Power Generation (0991.HK)

Possible Losses from Chemical and Coal-to-Gas Projects

Buy	1
Price (28 Nov 12)	HK\$2.73
Target price	HK\$3.30
Expected share price return	20.9%
Expected dividend yield	4.9%
Expected total return	25.8%
Market Cap	HK\$36,336M
	US\$4,688M

Price Performance (RIC: 0991.HK, BB: 991 HK)



■ **Summary** – Datang is our least favorable stock among PRC IPPs given (i) expected high capex in 2012-15E without positive free cashflow; and (ii) potential losses from Duolun coal-to-chemical project and Keqi coal-to-gas project. Our Buy rating is premised on our view that its incremental profits from coal fired plants amid coal price declines would be more than enough to offset the reduced profits from its coal mines. Valuations look inexpensive at 8.0x FY13E PER and 0.7x PB, below its historical means by 48% and 46% respectively.

■ **Declining unit coal cost** – We expect Datang would use 95m tonnes of raw coal in 2012E, with 35% from key contracts, 35% from spot market, 20% from its mines and 10% from import. Its unit fuel cost has continued to fall, from Rmb227/MWh in 9M12 to Rmb225/MWh in 10M12. Its key contracted prices are below the spot prices in those regions.

■ **Less profit from coal mines, but insignificant** – Datang's profit from coal mines fell 37% from Rmb727m in 2011 to Rmb457m in 9M12; its net profit from this business was also -12ppts to 25% in the same period. Its two associated coal mines are more profitable than the two consolidated ones since the former are larger in terms of output and with less intervention from regulators capping output for safety.

■ **Capex still high in 2012-15E** – Datang's capex should remain high at Rmb24bn (-8% yoy) in 2012E, with half of it for coal-to-gas projects, 20% for gas fired plants and 20% for others. We think Datang will remain in negative free cashflow in 2012-15E dragged by heavy capex (Rmb25bn each) from its two coal-to-gas projects.

■ **Loss making chemical & gas projects** – If Datang declares commercial operation of its coal-to-chemical project, it may incur a loss from it due to low utilization (unable to keep it above 70% consistently). Its Keqi coal-to-gas project that supplies NG to Beijing makes a loss at the prevailing Beijing city-gate NG sales price of Rmb1.8/m³; Beijing Enterprises (0392.HK; HK\$50.75; 1) has ceased investing in it.

Asian Utilities

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Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(RmbM)	(Rmb)	(%)	(x)	(x)	(%)	(%)
2010A	2,570	0.209	60.0	10.5	0.9	9.0	3.2
2011A	1,971	0.148	-29.1	14.8	0.7	5.6	5.0
2012E	3,184	0.239	61.5	9.2	0.7	8.0	4.9
2013E	3,647	0.274	14.5	8.0	0.7	8.7	5.6
2014E	4,023	0.302	10.3	7.3	0.6	9.1	6.2

Source: Powered by dataCentral

0991.HK: Fiscal year end 31-Dec						Price: HK\$2.73; TP: HK\$3.30; Market Cap: HK\$36,336m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	60,672	72,382	75,554	84,267	86,864	PE (x)	10.5	14.8	9.2	8.0	7.3
Cost of sales	-34,837	-44,484	-42,128	-45,849	-47,085	PB (x)	0.9	0.7	0.7	0.7	0.6
Gross profit	25,835	27,898	33,426	38,418	39,779	EV/EBITDA (x)	10.2	10.1	8.7	8.0	8.0
Gross Margin (%)	42.6	38.5	44.2	45.6	45.8	FCF yield (%)	-55.9	-60.5	-35.6	-23.8	-10.0
EBITDA (Adj)	16,586	18,158	22,872	26,729	27,617	Dividend yield (%)	3.2	5.0	4.9	5.6	6.2
EBITDA Margin (Adj) (%)	27.3	25.1	30.3	31.7	31.8	Payout ratio (%)	34	74	45	45	45
Depreciation	-7,382	-8,605	-9,176	-10,450	-10,472	ROE (%)	9.0	5.6	8.0	8.7	9.1
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	9,204	9,553	13,697	16,279	17,145	EBITDA	16,586	18,158	22,872	26,729	27,617
EBIT Margin (Adj) (%)	15.2	13.2	18.1	19.3	19.7	Working capital	2,707	-3,603	-1,467	-439	19
Net interest	-5,335	-6,992	-8,479	-9,162	-9,588	Other	-5,750	-6,202	-8,300	-9,635	-9,817
Associates	719	1,040	1,003	1,035	1,440	Operating cashflow	13,542	8,354	13,106	16,656	17,819
Non-op/Except	112	109	442	50	50	Capex	-28,618	-26,010	-23,495	-23,607	-20,724
Pre-tax profit	4,700	3,710	6,663	8,203	9,047	Net acq/disposals	0	0	0	0	0
Tax	-871	-668	-1,266	-1,558	-1,719	Other	0	-2,744	139	130	147
Extraord./Min.Int./Pref.div.	-1,259	-1,071	-2,213	-2,997	-3,305	Investing cashflow	-28,618	-28,754	-23,357	-23,478	-20,578
Reported net profit	2,570	1,971	3,184	3,647	4,023	Dividends paid	-825	-862	-1,464	-1,433	-1,641
Net Margin (%)	4.2	2.7	4.2	4.3	4.6	Financing cashflow	17,569	16,732	9,911	7,542	3,429
Core NPAT	2,570	1,971	3,184	3,647	4,023	Net change in cash	2,494	-3,643	-339	720	670
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-15,075	-17,657	-10,389	-6,952	-2,905
Reported EPS (Rmb)	0.209	0.148	0.239	0.274	0.302						
Core EPS (Rmb)	0.209	0.148	0.239	0.274	0.302						
DPS (Rmb)	0.070	0.110	0.108	0.123	0.136						
CFPS (Rmb)	1.100	0.628	0.985	1.251	1.339						
FCFPS (Rmb)	-1.225	-1.327	-0.781	-0.522	-0.218						
BVPS (Rmb)	2.506	2.926	3.055	3.221	3.400						
Wtd avg ord shares (m)	12,045	12,893	13,310	13,310	13,310						
Wtd avg diluted shares (m)	12,310	13,310	13,310	13,310	13,310						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	26.6	19.3	4.4	11.5	3.1						
EBIT (Adj) (%)	38.5	3.8	43.4	18.9	5.3						
Core NPAT (%)	67.2	-23.3	61.5	14.5	10.3						
Core EPS (%)	60.0	-29.1	61.5	14.5	10.3						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	3,443	4,467	3,990	4,580	5,103						
Accounts receivables	12,437	19,512	20,349	22,646	23,331						
Inventory	4,012	6,094	5,644	6,156	6,326						
Net fixed & other tangibles	179,234	200,923	215,243	228,400	238,652						
Goodwill & intangibles	6,548	7,826	7,826	7,826	7,826						
Financial & other assets	7,242	8,875	8,875	8,875	8,875						
Total assets	212,915	247,697	261,925	278,482	290,113						
Accounts payable	20,098	24,866	23,786	26,157	27,031						
Short-term debt	34,185	38,549	32,949	31,854	33,728						
Long-term debt	115,534	126,592	143,567	153,637	156,833						
Provisions & other liab	4,665	6,959	6,959	6,959	6,959						
Total liabilities	174,483	196,965	207,260	218,606	224,550						
Shareholders' equity	30,850	38,941	40,661	42,875	45,257						
Minority interests	7,583	11,791	14,004	17,001	20,306						
Total equity	38,433	50,732	54,665	59,876	65,563						
Net debt	146,276	160,673	172,526	180,910	185,457						
Net debt to equity (%)	380.6	316.7	315.6	302.1	282.9						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

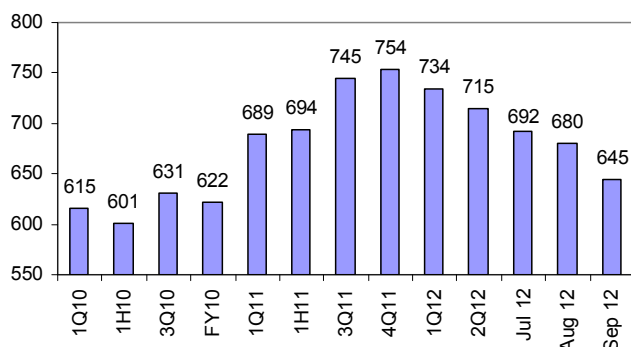
Other Key Takeaways from Company Visit

Generation Cut Narrowed in Oct

Datang targets gross generation of at least 200M MWh (-2% yoy) in 2012E.

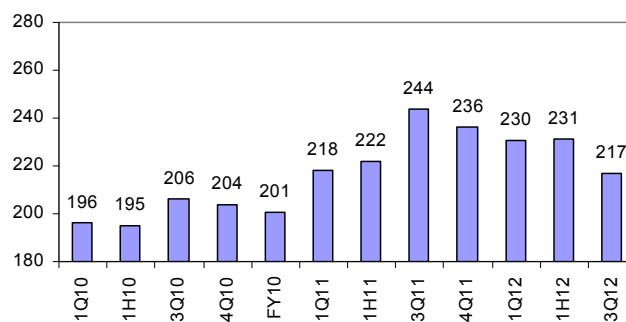
Utilization of its coal fired power plants should be about 5,300 hours this year, -4.3% yoy from 5,538 hours last year. Its gross generation decline was narrowed from -4.5% yoy in 3Q12 to below 1% yoy in Oct.

Figure 1. Datang's standard coal cost (Rmb/tonne)



Source: Citi Research

Figure 2. Datang's unit fuel cost (Rmb/MWh)



Source: Citi Research

Less Profits from Coal Mines

Figure 3. Outputs of Datang's coal mines (m tonnes)

Coal mine	stake	kcal/kg	1Q12	2Q12	3Q12	9M12	YoY	2012E	YoY
Yuzhou	50%	4,000	1.8	1.9	1.6	5.2	3%	7.0	-3%
Tashan	28%	5,000	6.2	6.3	7.0	19.4	9%	24.0	4%
Shengli	100%	3,000	2.2	2.7	1.0	5.8	-22%	7.0	-35%
Baoli	70%	n/a	0.5	0.3	0.2	1.0	-25%	1.3	-1%
Total			10.6	11.1	9.7	31.5	0%	39.3	-7%

Source: Company Reports and Citi Research Estimates

Datang's profit from coal mines fell 37% from Rmb727m in 2011 to Rmb457m in 9M12; its net profit from this business was also -12pts to 25% in the same period. Its two associated coal mines are more profitable than the two consolidated ones since the former are larger in terms of output and with less intervention from regulators capping output for safety reasons.

Figure 4. Datang's net profit from coal mines

Coal mines	Province	Stake	Heat content (kcal/kg)	2011	9M12
Tashan	Hebei	28%	5,500	574	400
Yuzhou	Shanxi	50%	4,000	80	39
Baoli	Jiangsu	70%	Various	13	13
Shengli East	Inner Mongolia	60%	3,000	60	6
Total				727	457
Datang's net profit				37%	25%

Source: Datang's Report and Estimations

The coal outputs of these mines are for power generation, except that of Shengli East for its coal-to-chemical project and coal-to-gas project, which Datang expects would use 6m and 7m tonnes respectively in 2013E. Datang is proceeding to double the production capacity of Shengli East to 20m tonnes in 2013-14E.

Possible Loss from Duolun Coal-to-Chemical Business

The project cost has overrun from Rmb11.8bn to Rmb24bn after completion was delayed by three years, and so far with an increment of Rmb3m capitalized interest per day. The breakeven cost per tonne of its product PP has gone up from Rmb4,000 to Rmb9,000. The completion delay was due to the plant being unable to run consistently at over 70% utilization.

Commercial operations of the project would be declared subject to Datang's parent company China Datang Group's decision. With the chairman of China Datang Group just promoted to SASAC, we think that the its parent might prefer to declare commercial operations of this project by year end so that the next chairman need not be responsible and borne by this burden (though the listed company prefers to declare commercial operations only after the project is able to achieve consistent utilization of over 70% to be profitable). We believe if it declares commercial operations in 4Q12, it would soon be loss making. The plant has had utilization rates of over 70%, but it has not been consistent. The problems are related to coal quality and its gasification facility. Initially, the plant was supposed to use coal from Datang's Shengli East coal mine (only 3,000kcal/kg in Ph.1), but the low coal quality has made the utilization low. Datang has tried blending the coal with high quality coal (4,000kcal/kg) procured from third parties. While this raised utilization, it also created new problems at its gasification facilities. The company has been trying to raise the utilization to facilitate commissioning, but with much uncertainty.

Figure 5. Comparison of Completed CTO Projects in China

	Shenhua Baotou	Shenhua Ningxia Coal	Datang International Power
Capex (Rmb bn)	~17	~18	>24 (original budget: 11.8)
Coal gasification technology, scale	Texaco (7 x 1500 t/d coal feed), 5 operating, 2 spare	Siemens (5 x 2000t/d coal feed) 4 operating, 1 spare	Shell (3 x 4000t/d coal feed) 3 operating, no spare
MTO technology	DMTO (Dalian Institute)	MTP (Lurgi)	MTP (Lurgi)
PE/PP technology	Dow / Unipol	Lummus / CB&I	Dow / Unipol
Air separation unit capacity	4 x 60,000Nm3/hr (local)	2 x 90,000Nm3/hr (Air Liquide)	3 x 58,000Nm3/hr (local)
Methanol capacity ('000 tonnes)	1,800	1,670	1,680
PE capacity	300	-	-
PP capacity	300	500	460
EPC contractors	LPEC, SSEC, SNEC (Sinopec affiliates)	Huanqiu (CNPC)	CNCEC
Startup status	On-spec methanol / MTO in 3Q10, commercial operation from Jan-11	On-spec PP in May-11	On-spec methanol / PP in Jun / Sep-11 shutdown in 2Q12 for repairs
Utilization	80-85% in 2011, 95% in 1H12	~170kt PP output in 2011; ~400kt target for 2012 (80% utilization)	Test-run (33kt output up to 15 Aug-12) Target 160kt in 2012 (70% run in Sep-Dec)

Source: Companies

In China, there are three MTO/MTP plants; namely, Shenhua Baotou, Shenhua Ningxia Coal (both owned by the parent of China Shenhua Energy), and Datang. The two projects under Shenhua Group are in operation, but commercial operation has not been declared for the project under Datang. We think the difference is due to the issues below:

- (i) Manpower - Shenhua uses similar technology with expertise acquired from Sinopec hence it has been able to run the technology successfully; but similar expertise seems much less in Datang's projects;
- (ii) Facilities - spare gasification funnel exists for Shenhua project, but not for Datang's one; and
- (iii) Raw material - coal of high quality is being used by Shenhua, but low quality by Datang in terms of energy, water and ash content.

Overall, the execution of the Shenhua Group seems much better than that of Datang. Shenhua Group is able to run its two plants at 80-95% utilization, whereas Datang's Duolun Coal-to-chemical project is only at around 70% utilization.

One-off Subsidies in 4Q12E

Datang's 90% owned Fuxin Coal-based NG Co. Ltd. was recently notified that it will be granted with two operational fiscal subsidies of Rmb379.5m and Rmb56m respectively, which will add up to a total sum of Rmb435.5m, to encourage Datang's investment which would be conducive to boost local economic growth. These government subsidies will be booked as non-operating income in 4Q12E, according to Datang. The amount will be used for the construction of a self-provided power plant and an air separation branch plant of the Fuxin project.

The Fuxin project has three phases with total annual NG output capacity of 4bn m³. Upon completion of its construction, it will mainly supply NG to Shenyang City of Liaoning Province and the surrounding cities such as Tieling, Fushun, Benxi and Fuxin. The construction of this project is in progress and scheduled to commence production of Ph.I in 2013. Though capex of this project is slightly higher than that of Keqi coal-to-gas project due to the longer transmission distance, Datang expects the incremental cost to be offset by higher NG sales price – currently, NG retail sales prices in Shenyang are Rmb3.3/m³ for residential and Rmb3.9/m³ for industrial users, much higher than in Beijing where the prices are Rmb2.05/m³ for residential and Rmb2.84/m³ for industrial users, which is served by the Keqi project.

Datang's DCF Calculation

Figure 6. DCF Calculation (Rmbm)

DCF of power plants

Value in the run up to FY20E	2011	2012E	2013E	2014E	2015E	2016E	2017E	2018E	2019E	2020E
Net operating cashflow - power plants		12,492	16,003	16,794	16,919	17,088	17,811	18,388	18,983	19,219
Capex of power plants		-21,592	-21,695	-19,046	-5,256	-3,881	-2,114	-2,114	-2,114	-2,114
FCF		-9,100	-5,692	-2,252	11,663	13,208	15,697	16,274	16,870	17,105
Discount factor		1.00	1.08	1.16	1.25	1.35	1.45	1.57	1.69	1.82
Net debt in 2011/NAV from 2012E	-153,052	-9,100	-5,281	-1,939	9,318	9,791	10,798	10,387	9,991	9,400

Terminal value	
NAV in 2021E	17,790
Expected terminal growth rate	4.0%
Capital Employed on new capex	8.5%
WACC	7.8%
Terminal Value	249,813
Discount factor	1.82
NPV of terminal value	137,288

Total value

DCF - in the run up to 2020E	-109,686
DCF - from 2021E	137,288
Total DCF	27,602
Exchange Rate	1.23
Total DCF (HK\$m)	33,950
No. of share (m)	13,310
DCF (power plants) / share (HK\$)	2.55

DCF of coal mines

Value in the run up to FY20E	2011	2012E	2013E	2014E	2015E	2016E	2017E	2018E	2019E	2020E
Net operating cashflow - coal mines		614	652	1,025	1,509	1,609	1,975	2,129	2,129	2,129
Capex of coal mines		-1,903	-1,912	-1,679	-463	-342	-186	-186	-186	-186
FCF		-1,289	-1,260	-654	1,046	1,267	1,788	1,943	1,943	1,943
Discount factor		1.00	1.08	1.16	1.25	1.35	1.45	1.57	1.69	1.82
Net debt in 2011/NAV from 2012E	-13,490	-1,289	-1,169	-563	836	940	1,230	1,240	1,151	1,068

Terminal value	
NAV in 2021E	2,010
Expected terminal growth rate	4.0%
Capital Employed on new capex	8.5%
WACC	7.8%
Terminal Value	28,230
Discount factor	1.82
NPV of terminal value	15,514

Total value

DCF - In the run up to 2020E	-10,047
DCF - from 2021E	15,514
Total DCF	5,467
Exchange Rate	1.23
Total DCF (HK\$m)	6,725
No. of share (m)	13,310
DCF (coal mines) / share	0.51

Datang's DCF

DCF - power plants (HK\$)	2.55
DCF - coal mines (HK\$)	0.51
DCF - chemical project (HK\$)	0.24
Total DCF/share (HK\$)	3.30

Source: Company Reports and Citi Research Estimates

Figure 7. Datang — Forward PER below historical mean



Source: Citi Research

Figure 8. Datang — Forward PB also below historical mean



Source: Company Reports and Citi Research Estimates

Figure 9. PRC IPPs: FY13E Net Profit Sensitivity

(Rmbm)	Huaneng	Datang	Huadian	CRP	CPI	Average
RIC	0902.HK	0991.HK	1071.HK	0836.HK	2380.HK	
Price (as of Nov 28)	HK\$6.30	HK\$2.73	HK\$2.39	HK\$17.16	HK\$2.03	
Rating	1	1	1	1	1	
Base case net profit *	7,932	3,647	1,927	8,078	1,365	n/a
Case 1. net profit change with extra 1% cut in average coal price	454	157	280	291	71	n/a
% change	5.7%	4.3%	14.5%	3.6%	5.2%	6.7%
Case 2. net profit change with extra 1% hike in power tariff	746	299	626	460	125	n/a
% change	9.4%	8.2%	32.5%	5.7%	9.2%	13.0%
Case 3. net profit change with extra 1% cut in utilization hours	-272	-110	-121	-174	-73	n/a
% change	-3.4%	-3.0%	-6.3%	-2.2%	-5.3%	-4.0%
Case 4. net profit change with extra 25bps interest rate cut	242	213	287	205	43	n/a
% change	3.0%	5.8%	14.9%	2.5%	3.2%	5.9%

Source: Company Reports and Citi Research Estimates

* In Rmbm except CRP which is in HK\$m

** Key assumptions - (i) unit fuel cost ranges from -2% yoy (Huaneng, Datang, CRP & CPI) to -5% yoy (for Huadian) in 2013E; (ii) average 6.5% tariff hike on 1 Dec 2011, but nil in 2012-13E

Figure 10. Datang: Attributable Capacity Expansion Plan (MW)

Datang's attributable generating capacity	Province	Capacity	Stake	FY11	FY12E	FY13E	FY14E	Completion
Capacity at year start:				25,504	27,272	29,020	30,347	
Add - coal-fired projects								
Linfen Hexi Cogen	Shanxi	2X600	100%	600				2011
Shaoxing Jiangbing	Zhejiang	2X452	70%		316	316		
Jiangshan Xingcheng	Zhejiang	2X100	100%		100	100		
Baocheng Gas	Shenzhen	2X180+2X400	51%		388	204		
Shizhu	Chongqing	2X350	70%			490		
Add - Hydropower projects								
Li Xian Jiang – Ma Lu Tang	Yunnan	100	70%	70				2,001
Li Xian Jiang – Shimenkan	Yunnan	2X65	70%	91				2011
Wulong Yinpan	Chongqing	4X150	51%	306				2011
Dandahe Hydro Power	Yunnan	100	100%		100			2012
Jintang River	Sichuan	380	51%		194			2012
Maliuwan	Yunnan	24	100%		24			1H12E
Dashuigou	n/a	24	100%		24			1H12E
Mengye River	Yunnan	68	100%		68			1H12E
Yubihe	Yunnan	100	100%		100			1H12E
Bohai Bay	Hebei	90	100%		90			1H12E
Tibet Botui Hydro power	Tibet	9.6	100%				10	2013E
Haokuoshui Hydro power	Chongqing	125	100%				125	2013E
Jiaomutang Hydro power	Guizhou	70	100%				70	2013E
Add - wind projects								
Datong Zuoyun I & II	Shanxi	99	100%					2009
Zhuozhi I, II & III	Inner Mongolia	282	100%	150				2010-2011
Liaoning	Liaoning	99	100%	49				
Datong Zuoyun III	Shanxi	21	100%	21				
Changle	Fujian	28	100%	28				
Qian-chatai	n/a	49.5	100%	50				2011
Hou-chatai	n/a	49.5	100%	50				2011
Chengdu Ligu Energy	Liaoning	150	54%	81				
Wanshengyong	Hebei	50	100%	50		100		2011-2013E
Siyanping	Chongqing	49.3	100%	49	50			2011-2012E
Hebei Wind power	Hebei	n/a	100%	345	150			2011-2012E
Faku	Liaoning	147	100%	50	48	49		2011-2013E
Jiangxi wind	Jiangxi	n/a	100%		96			2012E
Qingtongxia Wind	Ningxia	48	100%			48		2013E
Add - Solar								
Qingtongxia	Ningxia	50	100%	30		20		2011-2013E
Small capacity closed				-250				
Capacity at year end				27,272	29,020	30,347	30,552	
<i>yoy change</i>				<i>6.9%</i>	<i>6.4%</i>	<i>4.6%</i>	<i>0.7%</i>	

Source: Company Reports and Citi Estimates

Financial Statement & Key Assumptions

Figure 11. Datang Power — Consolidated Income Statements

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Sale of electricity	53,594	64,367	68,220	76,580	78,939
Heat supply	540	719	863	949	1,044
Coal sales	2,823	2,938	3,243	3,433	3,490
Chemical product sales	2,693	3,071	2,457	2,457	2,457
Others	1,023	1,287	772	850	935
Net revenues	60,672	72,382	75,554	84,267	86,864
Local government surcharges	-395	-569	-578	-596	-609
Fuel - power generation	-32,143	-42,153	-39,040	-42,580	-43,761
Fuel - coal sales	-2,694	-2,332	-3,088	-3,269	-3,324
Repairs and maintenance	-1,898	-1,899	-2,151	-2,444	-2,565
Depreciation	-7,382	-8,605	-9,176	-10,450	-10,472
Labor cost	-2,048	-2,367	-2,547	-2,851	-3,028
S&A expenses	-1,608	-1,693	-1,919	-2,162	-2,261
Other operating expenses	-3,300	-3,212	-3,358	-3,635	-3,699
Operating profit	9,204	9,553	13,697	16,279	17,145
Interest income	38	110	139	130	147
Interest expense	-5,373	-7,102	-8,618	-9,292	-9,735
Profit of JCEs	1	94	94	94	94
Profit from associates - coal mines	101	665	614	652	1,025
Profit from associates - power plants	618	281	295	289	321
Gain from investment securities	10	50	50	50	50
Exceptional gains	102	59	392	0	0
Profit before taxation	4,700	3,710	6,663	8,203	9,047
Taxation	-871	-668	-1,266	-1,558	-1,719
Minority interest	-1,259	-1,071	-2,213	-2,997	-3,305
Net profit	2,570	1,971	3,184	3,647	4,023

Source: Company Reports and Citi Research Estimates

Figure 12. Datang's net profit forecast – Citi vs. consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	3,184	3,141	+1%
FY13E	3,647	4,031	-10%
FY14E	4,023	4,820	-17%

Source: Company Reports and Citi Research Estimates

Figure 13. Key Ratios

Year-end 31 Dec (%)	FY10	FY11	FY12E	FY13E	FY14E
Gross margin (%)	31.7	27.3	33.3	34.2	34.6
Operating margin (%)	15.2	13.2	18.1	19.3	19.7
Net margin (%)	4.2	2.7	4.1	4.3	4.5
Net debt to total capital (x)	0.78	0.75	0.75	0.74	0.73
Net debt to equity (x)	3.81	3.17	3.16	3.02	2.83
Interest coverage ratio (x)	1.72	1.36	1.61	1.77	1.78
Cash interest coverage (x)	2.33	1.95	2.06	2.36	2.38
ROA (%)	1.3	0.9	1.2	1.3	1.4
ROE (%)	9.0	5.6	8.0	8.7	9.1
Three-year EPS CAGR (%)	22.4	9.5	26.8	10.3	5.0

Source: Company Reports and Citi Research Estimates

Figure 14. Datang Power — Consolidated Balance Sheet

As at 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Property plant and equipment	117,446	122,765	124,874	127,250	120,503
Construction in progress	61,788	78,158	90,369	101,149	118,149
Other fixed assets	13,790	16,701	16,701	16,701	16,701
Total fixed assets	193,024	217,624	231,943	245,100	255,353
Cash and cash equivalents	3,443	4,467	3,990	4,580	5,103
Inventories	4,012	6,094	5,644	6,156	6,326
Debtors	12,437	19,512	20,349	22,646	23,331
Total current assets	19,892	30,074	29,982	33,382	34,761
Total assets	212,915	247,697	261,925	278,482	290,113
S/T loans	19,375	22,924	22,924	22,924	22,924
Current portion of L/T loans	14,810	15,625	10,025	8,930	10,804
Trade creditors	18,930	23,940	22,172	24,183	24,853
Taxes payable	1,166	771	1,463	1,800	1,986
Dividends payable	2	155	152	174	192
Total current liabilities	54,283	63,415	56,735	58,011	60,759
Long term bank loans	115,534	126,592	143,567	153,637	156,833
Other loans	4,665	6,959	6,959	6,959	6,959
Minority interests	7,583	11,791	14,004	17,001	20,306
Total L/T liabilities	127,782	145,342	164,530	177,596	184,098
Share capital	12,310	13,310	13,310	13,310	13,310
Other capital	15,344	23,038	23,675	24,404	25,209
Retained earnings	3,196	2,593	3,676	5,161	6,738
Total equity	30,850	38,941	40,661	42,875	45,257
Total liabilities & equity	212,915	247,697	261,925	278,482	290,113

Source: Company Reports and Citi Research Estimates

Figure 15. Datang Power — Consolidated Cash Flow Statement

Year-end 31 Dec (Rmbm)	FY10	FY11E	FY12E	FY13E	FY14E
Cashflow from operations					
EBITDA	16,586	18,158	22,872	26,729	27,617
Working capital changes	2,707	-3,603	-1,467	-439	19
Other operating cash items	-161	-91	1,150	797	1,169
Gross operating cash flow	19,131	14,464	22,556	27,088	28,805
Net interest paid	-4,717	-6,240	-8,184	-8,873	-9,267
Tax paid	-871	-668	-1,266	-1,558	-1,719
Net operating cash flow	13,542	7,557	13,106	16,656	17,819
Capex	-28,618	-26,010	-23,495	-23,607	-20,724
Free cash flow	-15,075	-18,453	-10,389	-6,952	-2,905
Other investing cash flow	0	-3,325	0	0	0
Cashflow from investing	-28,618	-29,336	-23,495	-23,607	-20,724
Increase in Equity & MI	3,285	6,671	0	0	0
Ordinary Dividend Paid	-825	-862	-1,464	-1,433	-1,641
Change in borrowings	15,109	10,923	11,375	8,975	5,070
Other Financing Cash Flow	0	6,045	0	0	0
Cash Flow from Financing	17,569	22,777	9,911	7,542	3,429
Beginning cash balance	1,506	3,443	4,467	3,990	4,580
Ending cash balance	3,443	4,467	3,990	4,580	5,103

Source: Company Reports and Citi Research Estimates

Figure 16. Datang — Key Assumptions

	FY10	FY11	FY12F	FY13F	FY14F
Capacity (MW)					
Installed capacity - year end	36,201	38,485	41,326	42,894	43,099
y-y change (%)	13.3	6.3	7.4	3.8	0.5
Installed capacity - year average	33,170	37,918	39,529	43,062	43,297
y-y change (%)	18.7	14.3	4.2	8.9	0.5
Installed capacity (coal-fired) - year average	31,103	33,960	34,662	37,302	37,512
y-y change (%)	14.5	9.2	2.1	7.6	0.6
Consolidated capacity - year end	36,201	38,485	41,326	42,894	43,099
y-y change (%)	17.8	6.3	7.4	3.8	0.5
Attributable capacity (MW) - year end	25,504	27,272	29,070	30,397	30,602
y-y change (%)	20.5	6.9	6.6	4.6	0.7
Generation volume (M MWh)					
Gross generation	178	204	205	229	237
y-y change (%)	25.8	14.1	0.7	11.7	3.3
Gross generation (coal-fired)	169	188	182	201	207
y-y change (%)	24.2	11.1	-3.0	10.1	3.2
Net generation	168	192	194	219	226
y-y change (%)	26.0	14.2	1.1	12.6	3.4
Net generation (coal-fired)	145	163	173	192	198
y-y change (%)	19.0	13.0	5.6	11.2	3.2
Attri. gross generation	128	148	148	166	172
Attri. net generation	121	140	140	157	162
y-y change (%)	25.5	16.4	-0.3	12.2	3.5
Average utilization hours (coal-fired)	5,332	5,531	5,254	5,375	5,515
y-y change (%)	1.5	3.7	-5.0	2.3	2.6
Average utilization rates (coal-fired)	60.9%	63.1%	60.0%	61.4%	63.0%
Tariff (Rmb/MWh)					
Average tariff before VAT	373	392	411	410	409
y-y change (%)	1.3	5.1	4.9	-0.3	-0.3
Average tariff before VAT (coal-fired)	384	399	423	424	423
y-y change (%)	3.4	3.8	5.9	0.3	-0.3
Coal price					
Unit fuel cost (Rmb/MWh)	201	232	222	217	216
y-y change (%)	15.0	15.3	-4.0	-2.5	-0.5
Coal consumption (g/kWh)	324	320	318	317	315
y-y change (%)	-1.0	-1.2	-0.5	-0.5	-0.5

Source: Company Reports and Citi Research Estimates

Datang International Power Generation

Company description

Datang was established by North China Power Grid Company in 1994 to develop, construct, own and operate coal-fired power plants in North China, particularly in Beijing, Tianjin and Tangshan (BTT). Its parent became China Datang Group in December 2002 when the State Power Corporation was broken into five power-generating groups. By end-2011, the company had power plants with installed capacity of 38,484MW, with 84% coal-fired, 13% hydropower and 3% wind power. Datang has around a 38% share in the BTT areas, 10% in the North China Power Network and 2.5% in China. Its shares were listed on the Stock Exchange of Hong Kong and the London Stock Exchange in March 1997. Since September 2001, its American Depositary Receipts (ADRs) have been trading in the US over-the-counter market.

Investment strategy

We rate Datang shares Buy (1). We expect the company's coal mines and chemical project to be new earnings growth drivers in FY12-14E. Also, the company has adopted a more selective approach to building new power plants and has a greater focus on profitability rather than market share. This means the company is allocating resources on a more return-oriented basis.

Valuation

Our HK\$3.30 target price includes HK\$2.55/share assigned to Datang's power plants based on DCF, assuming a WACC of 7.8%, a risk-free rate of 5.2%, a market risk premium of 6.8%, an equity beta of 1.17x (according to Bloomberg), a cost of debt of 6.73%, a target debt to capital ratio of 67%, a corporate tax rate of 25%, and a terminal growth rate of 4%. We also include HK\$0.51/share for its coal mines and HK\$0.24/share for its chemical project. At our DCF-based target price, Datang would trade at a 2013E P/E of 9.8x. We use a sum-of-the-parts approach given the company's profile.

Risks

Key downside risks that could prevent the shares from reaching our target price include: 1) generation volumes coming in lower than our assumptions; 2) higher-than-expected coal prices; and 3) lower-than-expected profit from coal mines. We have not factored in any profit contribution from coal-to-chemical, coal-to-gas and aluminum production plants in our net profit estimates owing to inadequate data and track record, since these projects remain in the development phase.

Datang Renewable Power (1798.HK)

Lowest PB Among PRC Nationwide Wind-Farm Operators

Buy	1
Price (28 Nov 12)	HK\$0.84
Target price	HK\$1.08
Expected share price return	28.6%
Expected dividend yield	0.8%
Expected total return	29.4%
Market Cap	HK\$6,110M
	US\$788M

Price Performance (RIC: 1798.HK, BB: 1798 HK)



- **Summary** – We keep our Buy rating on Datang Renewables ('DR') for profit growth from new capacity, (ii) gradual improvements in grid curtailment ahead; (iii) inexpensive 0.5x FY13E PB. Unlike its peer Longyuan, DR does not have share-issuance risk and its CDM cut risk has also been mitigated after -64% yoy in 1H12.
- **More output despite of utilization cut** – DR's gross generation should add >25% yoy in 9M12 for more capacity, offsetting utilization -7.7% yoy to 1,200 hours amid grid curtailment (high at 26% in 1H12, losing 1.5M MWh output). It targets average utilization -7.7% yoy to 1,800 hours in FY12E, subject to wind resources in 4Q12E and assuming grid curtailment -34% hoh to 1.0M MWh in 2H12E.
- **More capacity in non-traditional regions** – DR had 5,259MW wind capacity by end 2011, including 500MW without output at that point as transmission lines were being built, operational by end 10M12. DR's capacity is 40% in Inner Mongolia, 20% in Northeast China, 20% in Central & Western China and 20% in Southeastern China. Its 700-800MW new capacity coming in 2012 has a higher proportion from Central & Western China and Southeastern China, minimizing curtailment.
- **CDM revenue dying down** – DR's CDM gross revenue was Rmb87m and net revenue, after netting direct cost, was Rmb71m (-64% yoy) in 1H12, based on floating price of average €3/tonne. Its CDM gross revenue was Rmb120m and estimated net revenue was Rmb98m in 9M12. If CDM spot price keeps at current low price of €1/tonne, its CDM revenue booked in 1H12 might have to be reversed in 2H12E for credit risk. DR expects CDM revenue to be low in 2013E.
- **No share issuance risk** – DR is an SOE, hence it can only issue new shares at not below 1.0x PB. It could not issue new shares in HK where its H-share price is at 0.5x FY12E PB. An A-share listing is unlikely in 2013 as (i) it was set up in Sep 2010 and A-share issuance could only be applied three years after establishment; and (ii) an application for A-share listing is time consuming, with a long queue.

Asian Utilities

Hong Kong Utilities

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Statistical Abstract

Year to 31 Dec	Net Profit (RmbM)	Diluted EPS (Rmb)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	456	0.064	28.5	10.6	0.6	7.5	0.0
2011A	730	0.101	58.4	6.7	0.5	8.4	5.8
2012E	202	0.028	-72.5	24.3	0.5	2.2	0.8
2013E	448	0.062	121.3	11.0	0.5	4.9	1.8
2014E	535	0.074	19.6	9.2	0.5	5.6	2.2

Source: Powered by dataCentral

1798.HK: Fiscal year end 31-Dec						Price: HK\$0.84; TP: HK\$1.08; Market Cap: HK\$6,110m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	2,748	4,347	4,670	5,591	6,340	PE (x)	10.6	6.7	24.3	11.0	9.2
Cost of sales	na	na	na	na	na	PB (x)	0.6	0.5	0.5	0.5	0.5
Gross profit	na	na	na	na	na	EV/EBITDA (x)	10.4	8.7	10.3	9.5	9.1
Gross Margin (%)	na	na	na	na	na	FCF yield (%)	-166.2	-163.8	-57.6	-40.4	-19.5
EBITDA (Adj)	2,389	3,790	3,953	4,768	5,401	Dividend yield (%)	0	5.8	0.8	1.8	2.2
EBITDA Margin (Adj) (%)	86.9	87.2	84.7	85.3	85.2	Payout ratio (%)	0	39	20	20	20
Depreciation	-886	-1,361	-1,788	-2,025	-2,262	ROE (%)	7.5	8.4	2.2	4.9	5.6
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	1,503	2,429	2,165	2,743	3,139	EBITDA	2,389	3,790	3,953	4,768	5,401
EBIT Margin (Adj) (%)	54.7	55.9	46.4	49.1	49.5	Working capital	-512	-1,246	-946	-888	-520
Net interest	-766	-1,431	-1,912	-2,101	-2,351	Other	-33	-59	32	-64	-94
Associates	-2	8	8	8	8	Operating cashflow	1,844	2,485	3,040	3,816	4,787
Non-op/Except	0	0	0	0	0	Capex	-9,855	-10,469	-5,869	-5,802	-5,746
Pre-tax profit	735	1,005	260	650	795	Net acq/disposals	0	0	0	0	0
Tax	-57	-35	32	-64	-94	Other	39	-1,198	34	31	33
Extraord./Min.Int./Pref.div.	-222	-240	-90	-138	-165	Investing cashflow	-9,815	-11,667	-5,835	-5,771	-5,713
Reported net profit	456	730	202	448	535	Dividends paid	0	0	-284	-40	-90
Net Margin (%)	16.6	16.8	4.3	8.0	8.4	Financing cashflow	12,498	8,449	1,270	2,828	526
Core NPAT	456	730	202	448	535	Net change in cash	4,500	-869	-1,525	873	-401
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-8,011	-7,984	-2,829	-1,986	-960
Reported EPS (Rmb)	0.064	0.101	0.028	0.062	0.074						
Core EPS (Rmb)	0.064	0.101	0.028	0.062	0.074						
DPS (Rmb)	0	0.039	0.006	0.012	0.015						
CFPS (Rmb)	0.258	0.344	0.418	0.525	0.658						
FCFPS (Rmb)	-1.122	-1.106	-0.389	-0.273	-0.132						
BVPS (Rmb)	1.169	1.250	1.238	1.294	1.356						
Wtd avg ord shares (m)	5,082	7,220	7,274	7,274	7,274						
Wtd avg diluted shares (m)	7,143	7,220	7,274	7,274	7,274						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	68.1	58.2	7.4	19.7	13.4						
EBIT (Adj) (%)	74.8	61.6	-10.9	26.7	14.4						
Core NPAT (%)	83.5	60.1	-72.3	121.3	19.6						
Core EPS (%)	28.5	58.4	-72.5	121.3	19.6						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	5,031	4,162	2,637	3,510	3,109						
Accounts receivables	1,495	2,694	3,478	4,213	4,811						
Inventory	10	12	12	12	12						
Net fixed & other tangibles	31,456	41,327	45,407	49,184	52,668						
Goodwill & intangibles	654	731	731	731	731						
Financial & other assets	2,714	5,356	5,607	6,686	7,565						
Total assets	41,360	54,283	57,874	64,337	68,897						
Accounts payable	85	504	504	504	504						
Short-term debt	3,619	5,667	5,667	5,667	5,667						
Long-term debt	21,957	29,514	33,014	38,014	41,014						
Provisions & other liab	5,143	6,719	6,801	7,720	8,669						
Total liabilities	30,804	42,404	45,986	51,905	55,854						
Shareholders' equity	8,353	9,089	9,007	9,414	9,860						
Minority interests	2,198	2,647	2,737	2,875	3,040						
Total equity	10,550	11,736	11,744	12,289	12,900						
Net debt	20,545	31,019	36,044	40,172	43,572						
Net debt to equity (%)	194.7	264.3	306.9	326.9	337.8						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Utilization Trough in 2012E

Figure 1. Average utilization of DR's wind farms by regions

	1H11	2H11	2011	YoY	1H12	YoY
Inner Mongolia	1,105	950	2,055	-5.0%	856	-23%
Northeastern	1,050	812	1,863	-12.7%	860	-18%
Central & Western	959	896	1,854	-5.6%	924	-4%
Southeastern Coastline	1,029	780	1,809	-12.9%	942	-8%
Total	1,064	887	1,951	-8.6%	880	-17%

Source: Company Reports and Citi Research Estimates

Figure 2. DR's wind capacity in China by regions (MW)

Region	2011	Mix	Add in 1H12	1H12	YoY	Mix
Inner Mongolia	2,309	44%	0.0	2,309	15%	42%
North-eastern	1,177	22%	48.0	1,225	12%	23%
Central and Western	1,008	19%	70.5	1,079	81%	20%
South-eastern Coastline	764	15%	4.5	769	76%	15%
Total	5,259	100%	123.0	5,382	30%	100%

Source: Company Reports and Citi Research Estimates

Figure 3. Grid curtailment ratio in China in 2011

	Wind Power Generation (bn KWh)	Curtailment Ratio	Curtailment Volume (bn KWh)
Northeast China Power Grid:			
Heilongjiang	4.4	14.5%	0.7
Jilin	4.0	14.9%	0.7
Liaoning	6.6	9.0%	0.7
East Inner Mongolia	8.8	25.3%	3.0
Sub-total	23.7	17.6%	5.1
North China Power Grid:			
Hebei	8.8	4%	0.4
Beijing	0.3	n/a	n/a
Tianjin	0.1	1%	0.0
Shanxi	1.3	n/a	n/a
Shandong	4.1	n/a	n/a
West Inner Mongolia	13.2	23%	0.4
Sub-total	27.9	14%	4.4
Northwest China Power Grid:			
Shaanxi	0.1	n/a	n/a
Gansu	7.1	27%	2.7
Qinghai	0.0	n/a	n/a
Ningxia	1.1	n/a	n/a
Xinjiang	2.9	3%	0.1
Sub-total	11.2	20%	2.8
Total of Three North Regions	63.5	16%	12.3

Source: China Wind Energy Association

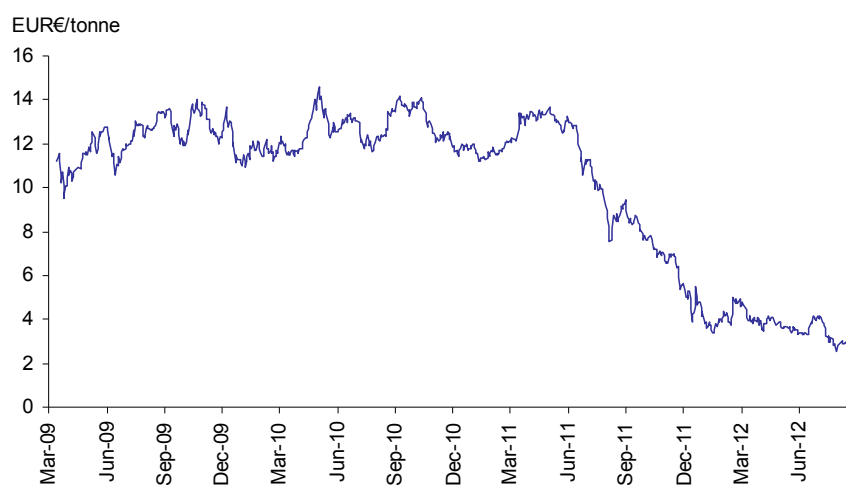
Less CDM Revenue for Lower Spot Prices

Figure 4. % of 1H12 profit from CDM revenue of PRC wind farm operators

Company	Stock code	CDM revenue (Rmbm)	Profit before tax (Rmbm)	%
Datang Renewables	1798 HK (Buy)	71	106	67%
Longyuan	916 HK (Buy)	447	1,988	22%
Huaneng Renewables	958 HK (Buy)	65	303	21%
Jingneng	579 HK (not rated)	83	538	15%
Suntien	956 HK (Buy)	68	526	13%
Average				18%

Source: Various companies

Figure 5. ICE Daily CER Index –CDM spot price



Source: Bloomberg (ticker - ICEDCER Index)

DCF at HK\$1.08/share

Figure 6. DR's DCF Calculation

(Rmbm)	2012E	2013E	2014E	2015E	2016E	2017E	2018E	2019E	2020E
Net operating cashflow		3,847	4,820	5,278	6,043	6,743	7,399	8,090	8,719
Less: capex		-5,802	-5,746	-5,691	-5,636	-5,581	-5,581	-5,528	-5,474
FCF		-1,955	-927	-412	407	1,161	1,817	2,562	3,245
Discount factor		1.00	1.07	1.15	1.23	1.32	1.42	1.52	1.63
Net debt by end 2009	-36,247	-1,955	-864	-359	331	880	1,284	1,689	1,996
DCF - Terminal Value									
NAV in 2021E	3,407								
Expected terminal growth rate	5.0%								
Capital Employed on new capex	9.0%								
WACC	7.2%								
Terminal Value	69,094								
Discount factor	1.74								
NPV of terminal value	39,642								
Total DCF									
DCF - in the run up to 2020E	-33,246								
DCF- from 2021E	39,642								
Total DCF	6,395								
Exchange Rate	1.23								
Total DCF (in HK\$m)	7,856								
No. of share (m)	7,274								
DCF / share (HK\$)	1.08								

Source: Company Reports and Citi Research Estimates

Figure 7. Net Profit Sensitivity Comparison in 2012E

	Longyuan	Huaneng Renewables	Datang Renewables
Base case (Rmbm)	2,869	546	202
Wind power capacity addition			
- wind power capacity addition in base case (MW)	1,600	700	700
- NP chg if capacity addition 5% above base case	+2.0%	+4.3%	+8.7%
- NP chg if capacity addition 5% below base case	-2.0%	-4.3%	-8.7%
On-grid tariffs of wind farms			
- average on-grid tariff of wind farms in base case (Rmb/MWh)	578	596	606
- NP chg if average on-grid tariff 5% above base case	+9.4%	+18.8%	+77.4%
- NP chg if average on-grid tariff 5% below base case	-9.4%	-18.8%	-77.4%
Average utilization hours of wind farms			
- average utilization hour in base case	1,924	1,667	1,700
- NP chg if average utilization hour 5% above base case	+9.9%	+30.1%	+77.4%
- NP chg if average utilization hour 5% below base case	-9.9%	-30.1%	-77.4%
Rmb borrowing rate			
- Rmb borrowing rate assumption in base case	5.2%	6.5%	6.5%
- NP chg if interest cost 1ppt above base case	-7.0%	-9.1%	+36.9%
- NP chg if interest cost 1ppt below base case	+7.0%	+9.1%	-36.9%
Corporate tax rate			
- Corporate tax rate assumption in base case	9.4%	7.0%	10.0%
- NP chg if tax rate 1ppt above base case	-1.1%	-1.2%	-1.2%
- NP chg if tax rate 1ppt below base case	+1.1%	+1.2%	+1.2%

Source: Citi Research

Peer Group Valuation and Historical Comp

Figure 8. Renewable Energy Companies — Valuation Comparison (Pricing date 28 Aug)

Company	Bloomberg code	Citi Rating	Share Price	Key business	Mkt cap (US\$m)	PER			PEG	PB			ROE		
						11	12E	13E		11	12E	13E	11	12E	13E
EDP Renovaveis	EDPR PL	Buy	EUR3.70	Wind generator	4,180	36.4	23.2	17.0	0.8	0.6	0.6	0.6	1.7%	2.6%	3.4%
Acciona	ANA SM	Neutral	EUR48.00	Renewable energy generator	3,560	17.6	20.3	13.6	n/a	0.6	0.5	0.5	3.7%	2.7%	3.8%
Theolia	TEO FP	NR	EUR1.38	Wind generator	116	n/a	n/a	n/a	n/a	n/a	0.4	0.4	n/a	4.1%	5.0%
Infigen Energy	IFN AU	NR	A\$0.26	Wind generator	208	n/a	n/a	n/a	n/a	n/a	n/a	0.3	n/a	n/a	-6.2%
China Power New Energy	735 HK	NR	HK\$0.40	Renewable energy generator	585	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Datang Renewables	1798 HK	Buy	HK\$0.84	Wind generator	788	6.7	24.3	11.0	-0.3	0.5	0.5	0.5	8.4%	2.2%	4.9%
Huaneng Renewables	958 HK	Buy	HK\$1.17	Wind generator	1,275	6.7	14.5	9.6	-0.7	0.7	0.7	0.6	12.3%	4.7%	6.8%
China Longyuan	0916.HK	Buy	HK\$5.19	Wind generator	4,999	11.8	10.9	11.5	10.8	1.2	1.1	1.0	10.8%	10.8%	9.3%
China Suntien	0956.HK	Buy	HK\$1.51	Wind generator	631	8.8	9.2	7.7	1.3	0.8	0.8	0.7	9.0%	8.3%	9.7%
Simple average						14.6	17.1	11.7	2.4	0.7	0.7	0.6	7.6%	5.0%	4.6%
Weighted average						17.7	16.2	12.3	3.5	0.8	0.7	0.7	6.2%	5.3%	5.6%

Source: Citi Research Estimates and I/B/E/S for non-rate (NR) companies

For FY13E, DR looks inexpensive with its 0.5x PB, slightly below sector average while its 4.9% ROE is lower than the sector average of 5.6%. And its 11.0x FY13x PER is slightly below the sector average of 12.3x.

Figure 9. DR's Forward PER Chart



Source: Citi Research

Figure 10. DR's Forward P/B Chart



Source: Company Reports and Citi Research Estimates

Financial Statements and Key Assumptions

Figure 11. Consolidated Income Statement

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Sales of electricity	2,378	3,656	4,171	5,082	5,820
Service concession construction revenue	0	24	25	27	28
CERs and VERs income	211	324	150	150	150
Other revenue	159	343	323	332	341
Total revenue	2,748	4,347	4,670	5,591	6,340
Service concession construction costs	0	-24	-25	-27	-28
Repairs and maintenance	-66	-69	-100	-112	-128
Depreciation	-886	-1,361	-1,788	-2,025	-2,262
Labor cost	-95	-188	-214	-254	-291
Other operating expenses	-198	-276	-377	-430	-492
Operating profit	1,503	2,429	2,165	2,743	3,139
Interest income	20	11	34	31	33
Interest expense	-786	-1,442	-1,946	-2,132	-2,384
Profit from associates and JVs	-2	8	8	8	8
Profit before taxation	735	1,005	260	650	795
Taxation	-57	-35	32	-64	-94
Minority interest	-222	-240	-90	-138	-165
Net profit	456	730	202	448	535

Source: Company Reports and Citi Research Estimates

Figure 12. DR's net profit forecasts – Citi vs. consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	202	474	-57%
FY13E	448	643	-30%
FY14E	535	924	-42%

Source: Bloomberg and Citi Research Estimates

Figure 13. Key Ratios

Year-end 31 Dec (%)	FY10	FY11	FY12E	FY13E	FY14E
Gross margin (%)	65.3	66.5	59.0	61.3	61.9
Operating margin (%)	54.7	55.9	46.4	49.1	49.5
Net margin (%)	16.5	16.7	4.3	8.0	8.4
Net debt to total capital (x)	0.5	0.6	0.7	0.7	0.7
Net debt to equity (%)	195	266	309	329	339
Interest coverage ratio (x)	1.9	1.7	1.1	1.3	1.3
Cash interest coverage (x)	2.0	2.0	1.7	1.9	1.9
Debt to EBITDA ratio	10.7	9.3	9.8	9.2	8.7
ROA (%)	1.4	1.5	0.4	0.7	0.8
ROE (%)	7.5	8.4	2.2	4.9	5.6
Three-year EPS CAGR (%)	-17.6	-1.2	-9.8	56.1	29.2

Source: Company Reports and Citi Research Estimates

Figure 14. Consolidated Balance Sheet

As at 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Property, plant and equipment	31,406	41,111	45,192	48,969	52,453
Land use rights	243	301	301	301	301
Associated companies	21	83	91	99	106
Other tangible fixed assets	110	657	657	657	657
Goodwill	403	422	422	422	422
Total fixed assets	32,181	42,575	46,663	50,448	53,939
Cash and cash equivalents	5,031	4,162	2,637	3,510	3,109
Inventories	10	12	12	12	12
Trade and bills receivable	1,495	2,694	3,478	4,213	4,811
Others	2,642	4,839	5,083	6,154	7,025
Total current assets	9,179	11,708	11,211	13,889	14,958
Total assets	41,360	54,283	57,874	64,337	68,897
S/T loans	3,619	5,667	5,667	5,667	5,667
Account payable	85	504	504	504	504
Other payable	5,031	6,637	6,719	7,637	8,587
Taxes payable	51	22	22	22	22
Total current liabilities	8,786	12,830	12,912	13,831	14,780
Long term bank loans	21,957	29,514	33,014	38,014	41,014
Other loans	66	203	203	203	203
Minority interests	2,198	2,647	2,737	2,875	3,040
Total L/T liabilities	24,221	32,364	35,954	41,092	44,257
Share capital	7,143	7,274	7,274	7,274	7,274
Other capital	530	473	473	473	473
Retained earnings	680	1,342	1,260	1,667	2,113
Total equity	8,353	9,089	9,007	9,414	9,860
Total liabilities & equity	41,360	54,283	57,874	64,337	68,897

Source: Company Reports and Citi Research Estimates

Figure 15. Consolidated Cash Flow Statement

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Cashflow from operations					
EBITDA	2,389	3,790	3,953	4,768	5,401
Working capital changes	121	-35	-161	-153	78
Other operating cash items	-634	-1,213	-784	-735	-598
Gross operating cash flow	1,876	2,542	3,007	3,881	4,881
Net interest paid	20	10	0	0	0
Tax paid	-52	-67	32	-64	-94
Net operating cash flow	1,844	2,485	3,040	3,816	4,787
Capex	-9,855	-10,469	-5,869	-5,802	-5,746
Free cash flow	-8,011	-7,984	-2,829	-1,986	-960
Disposal of fixed assets	23	0	0	0	0
Change in temp cash inv.	17	-1,199	34	31	33
Cashflow from investing	-9,815	-11,667	-5,835	-5,771	-5,713
Increase in Equity	4,510	437	0	0	0
Ordinary Dividend Paid	0	0	-284	-40	-90
Change in borrowings	9,175	3,613	3,500	5,000	3,000
Other Financing Cash Flow	-1,186	4,400	-1,663	-2,091	-2,295
Cash Flow from Financing	12,498	8,449	1,554	2,868	616
Beginning cash balance	531	5,031	4,162	2,637	3,510
Ending cash balance	5,031	4,162	2,637	3,510	3,109

Source: Company Reports and Citi Research Estimates

Figure 16. Key Assumptions

	FY10	FY11	FY12E	FY13E	FY14E
Capacity (MW)					
Total connected capacity - year end	4,028	5,259	5,959	6,659	7,359
yoy change (%)	53.8	30.6	13.3	11.7	10.5
Total generating capacity - year end	3,628	4,937	5,670	6,415	7,171
yoy change (%)	90.7	36.1	14.9	13.1	11.8
Consolidated capacity - year end	4,028	5,259	5,959	6,659	7,359
yoy change (%)	53.8	30.6	13.3	11.7	10.5
Attributable capacity - year end	3,322	4,495	5,153	5,825	6,511
yoy change (%)	58.9	35.3	14.6	13.0	11.8
Generation volume (M MWh)					
Consolidated gross generation	5.2	7.5	8.4	10.2	11.6
Wind power	5.2	7.5	8.4	10.2	11.6
Consolidated net generation	4.8	7.2	8.1	9.7	11.2
yoy change (%)	67.8	49.7	11.4	21.0	14.5
Wind power	4.8	7.2	8.1	9.7	11.2
Utilization					
Utilization hours - consolidated wind plants	2,134	1,951	1,700	1,790	1,812
yoy change (%)	-1.2	-8.6	-12.9	5.3	1.2
Utilization rate (%)	24.4	22.3	19.4	20.4	20.7
Tariff (Rmb/MWh)					
Average consolidated wind tariff (before 17% VAT)	576	591	606	610	610
yoy change (%)	2.4	2.7	2.4	0.7	0.0

Source: Company Reports and Citi Research Estimates

Datang Renewable Power

Company description

Datang Renewable Power is the second-largest listed wind power generator in China with a market share of 9% in terms of installed wind power capacity at end 2010. Its wind farms are mainly located in Northeastern China, Eastern China, Inner Mongolia, Central and Western China, and the Southeastern coastline. In FY11, the total installed capacity of Datang Renewable Power was 5,259 MW.

Investment strategy

We have a Buy (1) rating on Datang Renewable Power. We expect the company to achieve capacity growth of 700MW per annum, and believe its earnings quality will continue to improve as its new capacity becomes more geographically diversified (44% of its installed capacity was in Inner Mongolia at end 2011). While DR is less well placed than peers, Longyuan and Huaneng Renewables, in terms of grid connection risk and net debt/equity (266% at end 2011), this relative weakness looks to be already discounted at current valuations.

Valuation

Our target price for Datang Renewable Power of HK\$1.08 is based on a DCF model, which we believe is suitable for power companies operating in a relatively stable industry with predictable cash flows. Our model factors in earnings forecasts up to 2020 and a terminal growth rate of 5% thereafter. Our WACC for DR is 7.3%, which assumes a risk-free rate of 5.2%, a market risk premium of 6.8%, an equity beta of 1.0x, a cost of debt of 6.5%, a target debt-to-capital ratio of 67%, and a 25% corporate-tax rate. At our DCF-based target price, DR would trade at a 2012E P/B of 0.7x.

Risks

Key downside risks that could prevent the shares from reaching our target price include: 1) Utilization hours at wind farms being impacted by unpredictable weather conditions; 2) Less than expected capacity addition due to grid connection problems; 3) Poor transparency in the wind power tariff setting mechanism; and 4) RMB rate hikes.

Dongfang Electric Corporation (1072.HK)

Earnings Decline in 2013E

Sell	3
Price (28 Nov 12)	HK\$13.02
Target price	HK\$9.00
Expected share price return	-30.9%
Expected dividend yield	1.2%
Expected total return	-29.7%
Market Cap	HK\$30,517M
	US\$3,938M

Price Performance (RIC: 1072.HK, BB: 1072 HK)



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■ **Summary** – We maintain our Sell rating on DF given (i) its 2012E sales volume is likely to miss its target; (ii) GP margin would be under pressure from higher percentage of revenue from lower-margin EPC contracts; and (iii) fewer new orders in 2012E than a year ago. With 9M12 net profit at only 68% of consensus (Bloomberg) FY12E net profit, consensus numbers could be cut. Revenue, margin and net profit would drop further in FY13E, on our analysis.

■ **2012E sales target likely to be missed** – DF sold power equipment of 24,604MW in 9M12, or 70% of its targeted 35,000MW this year. It admitted the annual target would not be easy to achieve with reference to its production pipeline. Some planned nuclear projects would also be suspended under PRC's new nuclear policy.

■ **Lower GP margin** – DF's GP margin was -0.5ppts yoy to 19.0% in 3Q12 and -1.2 ppts yoy to 18.9% in 9M12 on margin cuts in wind equipment and engineering services. The company guides its net margin to be "stable" in 4Q12E, assuming it to fluctuate by less than 2ppts yoy. We assume GP margin to -1.8ppts to 18.5% in 2012E and -1.0ppt to 17.5% in 2013E on more margin cut in wind equipment sales and a higher proportion of revenue from lower GP margin (5-6%) of EPC orders.

■ **New orders cut** – DF's 3Q12 new orders were Rmb12.6bn, -2% yoy and -7% qoq. Its 9M12 new orders were Rmb34.6bn, -5% yoy, with the largest portions from coal-fired (36%) and gas equipment (28%); these orders were 85% from China and 15% from abroad. It expects its 2012E new orders to be below that (Rbm48bn) in 2011.

■ **Impact of new nuclear policy** – DF has Rmb3-4bn nuclear orders belonging to three planned inland projects which would not be built in 12th 5-year plan, according to PRC new nuclear policy. The company expects them to be transferred to other projects of the same developers to mitigate any financial impact. The orders are equal to 3% of DF's total orders on hand and <16% of its nuclear orders on hand.

Statistical Abstract

Year to 31 Dec	Net Profit (RmbM)	Diluted EPS (Rmb)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	2,577	1.286	37.3	8.1	1.9	26.2	1.2
2011A	3,056	1.525	18.6	6.9	1.5	24.6	1.5
2012E	2,446	1.221	-20.0	8.6	1.3	16.4	1.2
2013E	1,931	0.963	-21.1	10.9	1.2	11.4	0.9
2014E	1,832	0.914	-5.1	11.4	1.1	9.9	0.9

Source: Powered by dataCentral

1072.HK: Fiscal year end 31-Dec						Price: HK\$13.02; TP: HK\$9.00; Market Cap: HK\$30,517m; Recomm: Sell					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	38,080	42,917	40,289	39,241	40,323	PE (x)	8.1	6.9	8.6	10.9	11.4
Cost of sales	-30,380	-33,954	-32,609	-32,120	-33,215	PB (x)	1.9	1.5	1.3	1.2	1.1
Gross profit	7,700	8,962	7,680	7,121	7,108	EV/EBITDA (x)	10.5	8.7	10.3	11.2	11.3
Gross Margin (%)	20.2	20.9	19.1	18.1	17.6	FCF yield (%)	20.8	-8.2	-17.9	-9.5	7.3
EBITDA (Adj)	4,514	5,667	4,776	4,419	4,395	Dividend yield (%)	1.2	1.5	1.2	0.9	0.9
EBITDA Margin (Adj) (%)	11.9	13.2	11.9	11.3	10.9	Payout ratio (%)	10	10	10	10	10
Depreciation	-930	-1,229	-1,357	-1,377	-1,394	ROE (%)	26.2	24.6	16.4	11.4	9.9
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	3,584	4,438	3,418	3,041	3,001	EBITDA	4,514	5,667	4,776	4,419	4,395
EBIT Margin (Adj) (%)	9.4	10.3	8.5	7.8	7.4	Working capital	2,445	-6,511	-7,709	-5,496	-1,983
Net interest	40	88	67	-162	-191	Other	70	996	468	588	609
Associates	92	142	152	141	141	Operating cashflow	7,028	152	-2,465	-489	3,021
Non-op/Except	-871	-1,123	-739	-682	-682	Capex	-2,675	-1,862	-1,286	-1,500	-1,500
Pre-tax profit	2,845	3,544	2,898	2,338	2,268	Net acq/disposals	-560	-316	0	0	0
Tax	-169	-426	-377	-327	-340	Other	175	376	0	0	0
Extraord./Min.Int./Pref.div.	-99	-62	-75	-80	-96	Investing cashflow	-3,060	-1,802	-1,286	-1,500	-1,500
Reported net profit	2,577	3,056	2,446	1,931	1,832	Dividends paid	0	0	0	0	0
Net Margin (%)	6.8	7.1	6.1	4.9	4.5	Financing cashflow	-4,706	-296	-489	1,531	-125
Core NPAT	2,577	3,056	2,446	1,931	1,832	Net change in cash	-738	-1,945	-4,240	-458	1,397
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	4,353	-1,710	-3,750	-1,989	1,521
Reported EPS (Rmb)	1.286	1.525	1.221	0.963	0.914						
Core EPS (Rmb)	1.286	1.525	1.221	0.963	0.914						
DPS (Rmb)	0.130	0.160	0.123	0.097	0.092						
CFPS (Rmb)	3.507	0.076	-1.230	-0.244	1.507						
FCFPS (Rmb)	2.172	-0.853	-1.872	-0.992	0.759						
BVPS (Rmb)	5.509	6.888	7.985	8.851	9.673						
Wtd avg ord shares (m)	2,004	2,004	2,004	2,004	2,004						
Wtd avg diluted shares (m)	2,004	2,004	2,004	2,004	2,004						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	16.2	12.7	-6.1	-2.6	2.8						
EBIT (Adj) (%)	100.1	23.8	-23.0	-11.0	-1.3						
Core NPAT (%)	53.5	18.6	-20.0	-21.1	-5.1						
Core EPS (%)	37.3	18.6	-20.0	-21.1	-5.1						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	13,754	10,345	4,673	2,758	2,663						
Accounts receivables	13,608	16,931	15,356	14,957	15,369						
Inventory	34,580	35,860	36,505	35,078	36,273						
Net fixed & other tangibles	10,372	10,840	10,769	10,891	10,997						
Goodwill & intangibles	1,554	1,813	1,813	1,813	1,813						
Financial & other assets	8,384	6,653	6,120	5,985	6,125						
Total assets	82,253	82,443	75,236	71,481	73,240						
Accounts payable	16,619	20,180	14,945	12,515	11,647						
Short-term debt	39,397	34,156	30,219	25,192	25,824						
Long-term debt	276	166	166	2,166	2,166						
Provisions & other liab	14,177	13,334	13,025	12,913	13,164						
Total liabilities	70,469	67,836	58,355	52,785	52,801						
Shareholders' equity	11,039	13,803	16,001	17,736	19,383						
Minority interests	746	804	880	960	1,056						
Total equity	11,784	14,607	16,881	18,696	20,439						
Net debt	25,918	23,977	25,712	24,600	25,327						
Net debt to equity (%)	219.9	164.1	152.3	131.6	123.9						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

3Q12 Results Miss Consensus

Figure 1. DF's 3Q12 and 9M12 results highlight (PRC GAAP)

(in PRC GAAP)	3Q12	yoy	qoq	9M12	yoy	Comment
Revenue from operations	9,687	-6%	-7%	29,699	-1%	* Sales volume -23% yoy to 8,053MW in 3Q12
Less: Cost of sales	-7,769	-6%	-7%	-23,865	0%	
Less: Business tax & surcharge	-75	-6%	-25%	-234	34%	
Gross profit	1,844	-9%	-2%	5,601	-7%	
<i>GP margin</i>	<i>19.0%</i>	<i>-0.5ppts</i>	<i>+0.8ppts</i>	<i>18.9%</i>	<i>-1.2ppts</i>	* See GP Margin details in Fig.4, Fig.6, Fig.8 & Fig.10
Less: Distribution expenses	-191	-6%	-8%	-565	2%	
Less: Administrative expenses	-796	2%	-5%	-2,368	3%	
Less: Impairment loss	-233	61%	-5%	-645	-8%	* Increase in general bad debt provision due to aging of A/C receivables
Add: Investment gain	37	2%	-34%	113	23%	* Higher profit from JVs
Operating profit	662	-29%	1%	2,135	-18%	
<i>OP margin</i>	<i>6.8%</i>	<i>-2.2ppts</i>	<i>+0.5ppts</i>	<i>7.2%</i>	<i>-1.4ppts</i>	
Less: Finance expenses	0	-105%	-93%	11	-70%	* Less interest income for cash on hand -21% yoy by end 3Q12
						* Net debt to equity ratio +4ppts qoq to 213% in 3Q12 (including customer deposit as debts in our calculation)
Fair value gain	-11	15%	-78%	-67	-392%	
Add: Non-operating income	37	35%	63%	97	17%	* Government subsidies +25% yoy to Rmb91.8m in 9M12
Less: Non-operating expenses	-6	179%	-49%	-19	250%	
Profit before tax	682	-28%	12%	2,157	-21%	
Less: Income tax	-85	-59%	-7%	-297	-28%	
<i>Tax rate</i>	<i>12.5%</i>	<i>-9.0ppts</i>	<i>-2.4ppts</i>	<i>13.8%</i>	<i>-1.3ppts</i>	
Profit after tax	597	-20%	15%	1,860	-20%	
MI	-16	130%	-28%	-35	-7%	
Net profit	581	-22%	17%	1,825	-20%	
<i>NP margin</i>	<i>6.0%</i>	<i>-1.2ppts</i>	<i>+1.2ppts</i>	<i>6.1%</i>	<i>-1.5ppts</i>	

Source: Company Reports and Citi Research Estimates

Figure 2. DF's sales volume (MW)

	1Q12	2Q12	3Q12	yoy	Average unit size	9M12	yoy	Average unit size
Hydropower turbine generators	1,406	1,851	1,918	15%	240	5,175	24%	246
Steam turbine generators	6,910	6,260	5,930	-29%	456	19,100	-27%	516
Wind power generator	38	87	205	-50%	2	329	-75%	2
Sub-total	8,354	8,198	8,053	-23%	52	24,604	-22%	90
Power station boilers	5,300	6,180	5,256	-26%	404	16,736	-10%	356
Power station steam turbines	3,819	6,235	6,890	-16%	363	16,944	-43%	282

Source: Company Reports and Citi Research Estimates

3Q12 Segmental Performance Analysis

Figure 3. Revenue mix of High efficient & clean energy segment

(Rmbbn)	1H12	3Q12	YoY	9M12	YoY
Thermal power equipment	9.39	n/a	Down	n/a	Down
Nuclear power conventional island	1.31	n/a	n/a	n/a	n/a
Gas turbine equipment	0.72	n/a	Up	n/a	Up
Total	11.42	4.73	-21.2%	16.15	-8.1%

Source: Citi Research

Figure 4. GP Margin of High efficient & clean energy segment

	1H12	3Q12	YoY chg	9M12	YoY chg
Thermal power equipment	22.4%	n/a	Up	n/a	Up
Nuclear power conventional island	9.6%	n/a	n/a	n/a	n/a
Gas turbine equipment	16.1%	n/a	Up	n/a	Up
Total	20.5%	21.9%	+4.0ppts	20.9%	+1.8ppts

Source: Citi Research

Figure 5. Revenue mix of new energy segment

(Rmbbn)	1H12	3Q12	YoY	9M12	YoY
Wind Power	3.38	n/a	Flat	n/a	Flat
Nuclear power nuclear island	0.73	n/a	Up	n/a	Up
Total	4.11	2.63	27.1%	6.74	6.8%

Source: Citi Research

Figure 6. GP Margin of new energy segment

	1H12	3Q12	YoY chg	9M12	YoY chg
Wind Power	15.9%	n/a	Down	n/a	Down
Nuclear power nuclear island	35.0%	n/a	Up	n/a	Up
Total	18.2%	16.5%	-1.1ppts	17.5%	-1.3ppts

Source: Citi Research

Figure 7. Revenue mix of hydropower & environmental protection

(Rmbbn)	1H12	3Q12	YoY	9M12	YoY
Water Power	1.58	n/a	Up	n/a	Up
Environmental protection equipment	0.25	n/a	Up	n/a	Up
Total	1.83	1.22	49.3%	3.05	45.4%

Source: Citi Research

Figure 8. GP Margin of hydropower & environmental protection equipment segment

	1H12	3Q12	YoY chg	9M12	YoY chg
Water Power	20.6%	n/a	Up	n/a	Up
Environmental protection equipment	20.3%	n/a	Up	n/a	Up
Total	20.6%	23.9%	+11.9ppts	21.9%	+6.1ppts

Source: Citi Research

Figure 9. Revenue mix of projects & services segment

(Rmbbn)	1H12	3Q12	YoY	9M12	YoY
Engineering	2.06	n/a	Down	n/a	Down
Power plant services	0.45	n/a	n/a	n/a	n/a
Total	2.51	1.01	-22.5%	3.52	-5.0%

Source: Citi Research

Figure 10. GP Margin of Projects & services segment

	1H12	3Q12	YoY chg	9M12	YoY chg
Engineering	n/a	n/a	Down	n/a	Down
Power plant services	n/a	n/a	Down	n/a	Down
Total	15.0%	10.6%	-32.3ppts	13.7%	-18.7ppts

Source: Citi Research

Figure 11. DF's segmental operating profit and OP margin

Segment	Operating profit (Rmbm)			OP margin		
	9M12	Mix	yoy	2011	9M12	yoy
High efficiency & clean energy	3,380	59%	0.7%	19.5%	20.9%	1.8ppts
New energy	1,181	21%	-0.6%	21.2%	17.5%	-1.3ppts
Hydropower & environmental protection equipment	669	12%	100.2%	17.7%	21.9%	6.1ppts
Construction & services	483	8%	-60.4%	27.0%	13.7%	-18.7ppts
Total	5,713	100%	-6.3%	20.5%	19.4%	-1.1%

Source: Company Reports and Citi Research Estimates

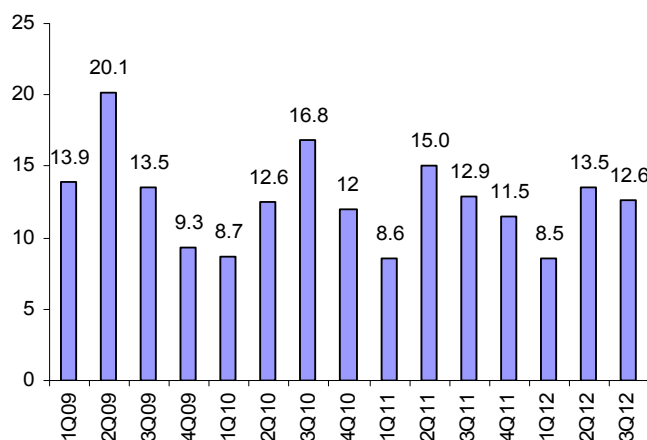
3Q12 New Orders Declined Slightly

Figure 12. DF's new orders (Rmbbn)

Segment	1Q11	2Q11	3Q11	4Q11	1Q12	2Q12	3Q12	YoY	Mix	9M12	Mix
High efficiency & clean energy	3.8	6.3	10.3	8.4	3.7	9.4	9.1	-12%	72%	22.1	64%
New energy	0.7	1.0	1.2	2.4	2.5	0.8	0.4	-66%	3%	3.7	11%
Hydropower & environmental protection equipment	0.1	1.2	0.2	0.4	1.3	1.1	1.1	442%	9%	3.5	10%
Construction & services	4.0	6.5	1.2	0.3	1.1	2.2	2.0	69%	16%	5.3	15%
Total	8.6	15.0	12.9	11.5	8.5	13.5	12.6	-2%	100%	34.6	100%
- Domestic	3.6	9.6	11.0	10.2	7.5	12.3	9.7	-12%	77%	29.5	85%
- Export	5.0	5.4	1.9	1.3	1.0	1.2	2.9	55%	23%	5.1	15%

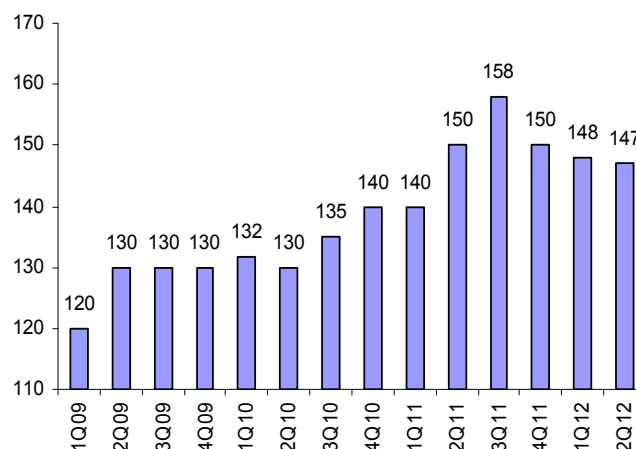
Source: Company Reports and Citi Research Estimates

Figure 13. DF's new orders (Rmbbn)



Source: Citi Research

Figure 14. DF's orders on hand (Rmbbn)

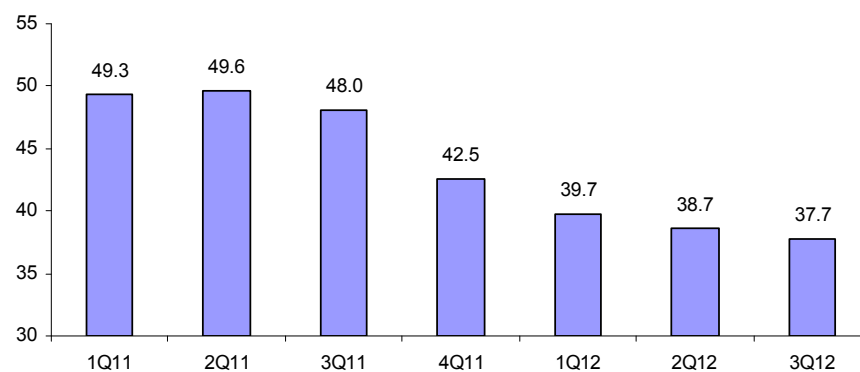


Source: Citi Research

DF's orders on hand were Rmb146.6bn by end 3Q12, including:

- (i) Rmb88.2bn or 60% from high efficiency & clean energy segment (of which Rmb67.7bn from coal fired Rmb10bn nuclear conventional island and Rmb10.7bn from gas equipment);
- (ii) Rmb19.8bn or 14% from new energy segment (of which Rmb15.4bn from nuclear island and Rmb4.4bn from wind equipment),
- (iii) Rmb14.6bn or 10% from hydropower & environmental protection equipment (including Rmb9.5bn from hydropower equipment & Rmb4.9bn from environmental protection equipment); and
- (iv) Rmb24bn or 16% from construction & services segment.

Figure 15. DF's declining customer receipt in advance (period end – Rmbbn)



Source: Company Reports and Citi Research Estimates

New Nuclear Policy Positive but Limited

On the summons of Premier Wen Jiabao, the State Council approved “Energy development 12th 5-year plan”, “Nuclear safety plan (2011-20)” and “Nuclear mid-long term development plan (2011-20)” on 24 Oct. The State Council agreed nuclear power development was essential in China, but safety is the first priority. The three key points about nuclear are:

- (4) **To progressively resume normal nuclear plant construction in China**, but no PRC nuclear capacity target for 2020 has been mentioned in the plan – DF expects new nuclear order flow to resume, after new nuclear project approvals were suspended since the Fukushima nuclear accident in March 2011. But the amount of new nuclear orders might be lower than those in 2008-10, in our view, as the future capacity addition would be progressive but not aggressive for safety reasons;
- (5) **To select site scientifically**, with new nuclear plants to be built in a few selected sites in coastal provinces, not in inland ones, in the 12th 5-year period (2011-15) – Of orders on hand, DF has Rmb3-4bn (or 3%) belonging to three planned inland nuclear projects; namely the nuclear island of Xianning nuclear project in Hubei under CGNPC, the nuclear island of Taohuajiang nuclear project in Hunan under CNNC and the conventional island of Pengze nuclear project in Jiangxi. DF has started production of the equipment of these orders and expects them to be transferred to other nuclear projects of these state-owned nuclear power plant developers, rather than cancelled. DF has got 10% customer deposit of these orders, which would help to mitigate financial impact, if any; and
- (6) **To approve the construction of new nuclear plants with 3G standards** in order to raise the entrance barrier and demand the highest global safety standards – This direction is more negative to DF than its peers as the majority of the DF’s orders are 2G while those of Shanghai Electric are mostly 3G.

Regarding nuclear exposure of PRC power equipment makers, in our calculation, gross profit from nuclear equipment is 9% for DF, 4% for Shanghai Electric and nil for Harbin Electric in 2012E.

Figure 16. PRC nuclear power plants in operation

Project	Province	Capacity (MW)	Type	Operator	Commercial operation
Daya Bay 1&2	Guangdong	1,968	PWR	CGNPC	1994
Qinshan Phase I	Zhejiang	310	PWR(CNP-300)	CNNC	Apr.1994
Qinshan Phase II, 1-2	Zhejiang	1,300	PWR(CNP-600)	CNNC	2002; 2004; 2010
Qinshan Phase II, 3-4	Zhejiang	1,320	CNP-600	CNNC	2010; 2012
Qinshan Phase III, 1&2	Zhejiang	1,400	PWR(Candu 6)	CNNC	2002; 2003
Ling Ao Phase I, 1&2	Guangdong	1,981	PWR	CGNPC	2002; 2003
Tian Wan, 1&2	Jiangsu	2,120	PWR(VVER-1000)	CNNC	2007
Ling Ao Phase II, 1	Guangdong	1,086	PWR(CPR-1000)	CGNPC	Sep.2010
Ling Ao Phase II, 2	Guangdong	1,086	CRP-1000	CGNPC	Aug.2011
Total		12,570			

Source: Company Reports and Citi Research Estimates

Figure 17. PRC nuclear power plants under construction

	Province	Capacity (MW)	Technology	Operator	Construction since	Planned operation
3rd Gen or above technology						
Sanmen, 1&2	Zhejiang	2,500	AP-1000	CNNC	Mar 2009, Dec 2009	Nov.2013; Sep.2014
Haiyang, 1&2	Shandong	2,500	AP-1000	CPI	Sep 2009, Jun 2010	May.2014; Mar.2015
Sub-total		5,000				
2nd Gen technology						
Hongyanhe, 1-4	Liaoning	4,480	CPR-1000	CGNPC	Aug.2007 - Aug.2009	Oct 2012; 2013; 2014
Ningde, 1-4	Fujian	4,360	CPR-1000	CGNPC and Datang	Feb.2008 - Sep.2010	Dec 2012; 2013; 2014; 2015
Fuqing, 1&2	Fujian	2,160	CPR-1000	CNNC	Nov 2008 - Jun 2010	Oct 2013; Aug 2014
Yangjiang	Guangdong	4,320	CPR-1000	CGNPC	Dec 2008 - Mar 2011	Aug 2013; 2014; 2015; 2016
Fangjiashan 1&2	Zhejiang	2,160	CPR-1000	CNNC	Dec 2008; Jul 2009	Dec 2013; Oct 2014
Hongyanhe, 5&6	Liaoning	2,160	CPR-1000	CGNPC	2011	2015
Fangchenggang, 1&2	Guangxi	2,160	CPR-1000	CGNPC	Jul.2010; 2011	2015; 2016
Fuqing, 3&4	Fujian	2,160	CPR-1000	CNNC	Jul.2010; 2012	Jul 2015; May 2016
Hongshiding (Rushan), 1&2	Shandong	2,160	CPR-1000	CNEC/CNNC	N/A	2015
Sub-total		26,120				
Total		31,120				

Source: Company Reports and Citi Research Estimates

Figure 18. PRC nuclear power plants in planning

	Province	Capacity (MW)	Technology	Operator	Construction since	Planned operation	Impact of new policy
3rd Gen in coastal areas							
Xudabao, 1&2	Liaoning	2,500	AP-1000	CNNC and Datang	N/A	N/A	Possible to be built in future
Sanmen, 3&4	Zhejiang	2,500	AP-1000	CNNC	N/A	N/A	Possible to be built in future
Haiyang, 3&4	Shandong	2,500	AP-1000	CPI	N/A	N/A	Possible to be built in future
Longyou(Zhexi), 1&2	Zhejiang	2,500	AP-1000	CNNC	N/A	N/A	Possible to be built in future
Zhangzhou, 1&2	Fujian	2,500	AP-1000	CNNC and Guodian	N/A	N/A	Possible to be built in future
Shaoquan, 1-4	Guangdong	5,000	AP-1000	CGNPC	N/A	N/A	Possible to be built in future
Taishan, 1&3	Guangdong	3,540	ERP	CGNPC	N/A	N/A	Possible to be built in future
Sanming, 1&2	Fujian	1,760	BN-800	CNNC	N/A	N/A	Possible to be built in future
Shandong Shidaowan	Shandong	210	HTR-PM	Huaneng	N/A	N/A	Possible to be built in future
Sub-total		23,010					
3rd Gen in inland areas							
Xianning(Dafan), 1&2	Hubei	2,500	AP-1000	CGNPC	N/A	N/A	Not build for in coastal province
Taohuaijiang, 1-4	Hunan	5,000	AP-1000	CNNC	N/A	N/A	Not build for in coastal province
Pengze, 1&2	Jiangxi	2,500	AP-1000	CPI	N/A	N/A	Not build for in coastal province
Xiaomoshan, 1&2	Hunan	2,500	AP-1000	CPI	N/A	N/A	Not build for in coastal province
Yanjiashan	Jiangxi	2,500	AP-1000	CNNC	N/A	N/A	Not build for in coastal province
Wuhu, 1&2	Anhui	2,500	AP-1000	CGNPC	N/A	N/A	Not build for in coastal province
Sub-total		17,500					
2nd Gen technology							
Ningde, 5&6	Fujian	2,160	CPR-1000	CGNPC	N/A	N/A	Not build for no 2G approval
Fuqing, 5&6	Fujian	2,160	CPR-1000/CNP1000	CNNC	N/A	N/A	Not build for no 2G approval
Lianyungang, 1&2	Jiangsu	2,160	CRP-1000	CGNPC	N/A	N/A	Not build for no 2G approval
Shanwei(Lufeng), 1&2	Guangdong	2,160	CRP-1001	CGNPC	N/A	N/A	Not build for no 2G approval
Tianwan, 3&4	Jiangsu	2,120	VVER-1000 (AES-91)	CNNC	N/A	N/A	Not build for no 2G approval
Tianwan, 5&6	Jiangsu	2,400	VVER-1200	CNNC	N/A	N/A	Not build for no 2G approval
Sub-total		13,160					
Total		53,670					

Source: Company Reports and Citi Research Estimates

Financial Statements & Key Assumptions

Figure 19. Consolidated Income Statement (Rmbm)

	FY10	FY11	FY12E	FY13E	FY14E
Revenue:					
High efficiency and clean energy	20,394	24,483	22,655	20,944	21,312
New energy	9,518	8,884	7,469	7,469	7,469
Hydro power and environmental protection equipments	2,958	3,242	3,566	3,922	4,315
Construction and services	4,734	5,834	6,126	6,432	6,754
Other income	476	474	474	474	474
Total revenue	38,080	42,917	40,289	39,241	40,323
COGS	-30,380	-33,954	-32,609	-32,120	-33,215
Business tax and surcharge	-227	-238	-238	-238	-238
Gross profit:					
High efficiency and clean energy	3,839	4,774	4,518	4,050	3,978
New energy	2,078	1,882	1,371	1,164	1,098
Hydro power and environmental protection equipments	517	573	677	745	819
Construction and services	1,061	1,576	957	1,005	1,055
Adjustment	-23	-81	-81	-81	-81
Total gross profit	7,473	8,724	7,442	6,883	6,871
<i>GP margin</i>	<i>19.6%</i>	<i>20.3%</i>	<i>18.5%</i>	<i>17.5%</i>	<i>17.0%</i>
Distribution expenses	-807	-889	-834	-813	-835
Administrative expenses	-3,082	-3,398	-3,190	-3,029	-3,035
Operating profits	3,584	4,438	3,419	3,042	3,001
OP margin	9.4%	10.3%	8.5%	7.8%	7.4%
Finance costs	-139	-148	-155	-271	-271
Assets written off	-987	-1,270	-870	-870	-870
Interest income	179	236	222	110	80
Associate	41	41	41	41	41
Jointly controlled entities	51	101	111	100	100
Fair value gain/loss	27	-41	-57	0	0
Non-operational gain/loss	89	188	188	188	188
Pre-tax profits	2,845	3,544	2,898	2,338	2,268
Income taxes	-169	-426	-377	-327	-340
Profit after tax	2,676	3,118	2,521	2,011	1,928
Minority interests	-99	-62	-75	-80	-96
Net Profit	2,577	3,056	2,446	1,931	1,832
<i>NP margin</i>	<i>6.8%</i>	<i>7.1%</i>	<i>6.1%</i>	<i>4.9%</i>	<i>4.5%</i>

Source: Company Reports and Citi Research Estimates

Figure 20. Key Ratios

	FY10	FY11	FY12E	FY13E	FY14E
Gross margin	19.6%	20.3%	18.5%	17.5%	17.0%
EBITDA margin	11.9%	13.2%	11.9%	11.3%	10.9%
Operating margin	9.4%	10.3%	8.5%	7.8%	7.4%
Net margin	6.8%	7.1%	6.1%	4.9%	4.5%
Net debt/equity (with customer advance)	220%	164%	152%	132%	124%
Net debt/equity (without customer advance)	-95%	-54%	-13%	9%	9%
ROE	26.2%	24.6%	16.4%	11.4%	9.9%
ROA	3.3%	3.7%	3.1%	2.6%	2.5%
Average working capital days					
Inventory	371	379	409	399	399
Receivables	207	186	186	186	186
Payables	94	84	84	84	84
Customer Deposits days	338	293	253	213	213

Source: Company Reports and Citi Research Estimates

Figure 21. Consolidated Balance Sheet

	FY10	FY11	FY12E	FY13E	FY14E
Intangible assets	131	131	131	131	131
Property, plant, and equipment	8,878	10,004	9,932	10,055	10,161
Construction in progress	1,494	836	836	836	836
Interests in associates/JCE	325	456	456	456	456
Investment securities/available for sale investments	412	338	338	338	338
Prepaid lease payments on land use rights	846	861	861	861	861
Deferred tax assets	708	952	952	952	952
Total non-current assets	12,794	13,578	13,507	13,629	13,735
Inventories	34,580	35,860	36,505	35,078	36,273
- Trade & bills receivables	13,091	16,480	14,948	14,559	14,960
- Other receivables	518	451	409	398	409
- Prepayments	7,516	5,728	5,196	5,060	5,200
Bank balances and cash	13,754	10,345	4,673	2,758	2,663
Total current assets	69,459	68,864	61,729	57,852	59,505
Total assets	82,253	82,443	75,236	71,481	73,240
Amount due to customers for contract work	7,799	7,799	7,490	7,377	7,629
Amount due to related parties	2,829	2,829	2,829	2,829	2,829
Trade and other payables	16,619	20,180	14,945	12,515	11,647
Customer receipt in advance	37,062	31,883	27,947	22,919	23,552
Other tax liabilities	1,350	448	448	448	448
Unsecured short term loans	2,335	2,272	2,272	2,272	2,272
Others	626	641	641	641	641
Total current liabilities	68,619	66,053	56,572	49,002	49,018
Deferred income	834	937	937	937	937
Unsecured long-term loans	276	166	166	2,166	2,166
Others	740	680	680	680	680
Total non-current liabilities	1,849	1,783	1,783	3,783	3,783
Net current assets	66,770	64,269	54,789	45,218	45,235
Net assets	11,784	14,607	16,881	18,696	20,439
Minority interests	746	804	880	960	1,056
Share capital	2,004	2,004	2,004	2,004	2,004
Reserves	9,035	11,799	13,997	15,733	17,379
Shareholders' equity	11,039	13,803	16,001	17,736	19,383

Source: Company Reports and Citi Research Estimates

Figure 22. Consolidated Cashflow Statement

	FY10	FY11	FY12E	FY13E	FY14E
PBT	2,845	3,544	2,898	2,338	2,268
Net finance cost	-40	-88	-67	162	191
Depreciations & amortization	930	2,145	1,357	1,377	1,394
Change in WC	2,445	-6,511	-7,709	-5,496	-1,983
Tax paid	-169	-426	-377	-327	-340
Others	-11	196	0	0	0
Operating CF	5,999	-1,139	-3,897	-1,946	1,530
Capex	-2,675	-1,862	-1,286	-1,500	-1,500
Purchase of associates & other LT assets	-560	-316	0	0	0
Others	175	376	0	0	0
Investing CF	-3,060	-1,802	-1,286	-1,500	-1,500
FCF	2,940	-2,940	-5,183	-3,446	30
Drawdown of loans	-4,828	2,692	-309	1,888	251
Repayment of loans	0	-2,780	0	0	0
Change in shareholder equity	-187	-292	-247	-195	-185
Change in MI	269	-3	0	0	0
Interest paid	40	88	67	-162	-191
Financing CF	-4,706	-296	-489	1,531	-125
Net cashflow	-954	-3,409	-5,672	-1,915	-95
Total cash at year end	13,754	10,345	4,673	2,758	2,663

Source: Company Reports and Citi Research Estimates

Figure 23. Key Assumptions

	FY10	FY11	FY12E	FY13E	FY14E
Sale volume (MW)					
- Coal-fired generators	26,673	32,763	n/a	n/a	n/a
% change	1.0%	22.8%	n/a	n/a	n/a
- Hydro generators	5,441	5,465	n/a	n/a	n/a
% change	9.9%	0.5%	n/a	n/a	n/a
- Wind generators	2,398	1,454	n/a	n/a	n/a
% change	112.3%	-39.4%	n/a	n/a	n/a
- Nuclear generators	n/a	n/a	n/a	n/a	n/a
- All generators	34,511	39,682	35,500	n/a	n/a
% change	6.3%	15.0%	-10.5%	n/a	n/a
Revenue mix					
High efficiency and clean energy	54%	57%	56%	53%	53%
New energy	25%	21%	19%	19%	19%
Hydro power and environmental protection equipments	8%	8%	9%	10%	11%
Construction and services	12%	14%	15%	16%	17%
GP margin					
High efficiency and clean energy	18.8%	19.5%	19.9%	19.3%	18.7%
New energy	21.8%	21.2%	18.4%	15.6%	14.7%
Hydro power and environmental protection equipments	17.5%	17.7%	19.0%	19.0%	19.0%
Construction and services	22.4%	27.0%	15.6%	15.6%	15.6%
Company total	19.6%	20.3%	18.5%	17.5%	17.0%

Source: Company Reports and Citi Research Estimates

Dongfang Electric Corporation

Company description

Dongfang is one of China's top three manufacturers of power-generation equipment. It acquired Dongfang Boiler and Dongfang Turbine in 2008 from its parent and now has the capability of making engines, turbines and boilers for coal-fired, hydro, nuclear and wind power plants.

Investment strategy

We have a Sell rating on Dongfang. We expect net profit to drop 20% pa in FY12-13E, with no sign of an earnings recovery in FY14E amid prevailing weak order flow, especially with the PRC Government unlikely to resume new nuclear projects soon. On our forecasts, DF's GP margin will narrow from 18.8% in 1H12 to 18.5% in 2H12E and to 17.5% in FY13E due to (i) cuts in sales prices of thermal equipment; (ii) low margin of conventional island without real economies of scale; (iii) more competition for nuclear orders; and (iv) overcapacity in wind equipment. Amid an industry downcycle, DF's current valuations look unattractive.

Valuation

Our target price of HK\$9.0 for Dongfang is based on a 2013E P/B of 0.8x, in line with the sector mean and reflecting a sharply declining ROE amid an industry downcycle. We believe P/B is a good primary valuation method for setting our TPs for Chinese power equipment companies as their product demand and profitability are cyclical.

Risks

The following upside risks could mean the stock trading above our target price: 1) higher-than-expected amount of domestic orders; 2) lower-than-expected raw material (mostly steel) cost, which accounts for 70-80% of total production costs for power-equipment makers; and 3) PRC Government resuming approvals of new nuclear projects.

Energy Development (EDC.PS)

Realizing Catalysts: Strong 3Q12 Profit; Progress on BacMan

Buy	1
Price (15 Nov 12)	P6.77
Target price	P7.65
from P7.00	
Expected share price return	13.0%
Expected dividend yield	2.4%
Expected total return	15.4%
Market Cap	P126,937M
	US\$3,084M

Price Performance
(RIC: EDC.PS, BB: EDC PM)



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■ **3Q/9M12 profits beat expectation; Reiterate Buy** – EDC posted 3Q12 recurring profit of P2.4bn (+32% YoY), bringing 9M12 figure to P6.3bn (+67% YoY). Positive surprise primarily came from stronger-than-expected hydro revenues (on robust sale of ancillary services) and lower-than-expected opex. We think key catalysts are materializing (ie delivering on earnings, commissioning of BacMan by yearend) and that the stock is still attractively valued -- 2013E PE of 13.7x on 2012-14E EPS CAGR of 20% (PEG of 0.7x). EDC remains our top pick in the Phils utilities space.

■ **BacMan on track for commissioning in Dec-12** – Generator rotors of Unit 1 and 2 (55 MW each) are now on site (back from rehabilitation in the UK) and re-blading of their turbines is ongoing. These units are set to be commissioned in Dec-12 for reliability runs, with commercial operations targeted for Jan-13. Unit 3 (20 MW) is still undergoing repair (will take 4-8 weeks more) and its generator rotor will need to be replaced to operate at full capacity. EDC has ordered the replacement rotor from Toshiba and will take 13 months to be delivered. Unit 3 will still be commissioned in 2013 using the existing rotor, but will be operating at a fraction of its rated capacity (ie 6-7 MW). EDC also is in the process of securing bilateral contracts for BacMan.

■ **Upward revisions to earnings and TP (to P7.65, from P7.00)** – We raise our 2012/13/14 earnings estimates by 15%/8%/6% mainly due to higher sale of ancillary services and lower opex (on easing rehab-related expenses) as evidenced in the 9M12 results. Partly offsetting these is the lower assumed load factor (to 75% from 80% previously) for BacMan in 2013E due to the issue with Unit 3.

■ **Attractive growth prospects** – We estimate a 2012-14E EPS CAGR of 20% driven by BacMan in 2013E, rehabilitation of the 305 MW Palinpinon-Tongonan geothermal plant (that will raise utilization rate from 71% to 82.5% in 2014E) and the 25MW Nasulo greenfield project in 2014. Upside in the medium to long-term could come from EDC's venture into wind power (starting with the 86 MW Burgos project – targeted for 2015) and overseas greenfield projects (Chile, Peru and Indonesia).

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(PM)	(P)	(%)	(x)	(x)	(%)	(%)
2010A	6,638	0.35	-7.2	19.1	4.2	22.4	1.8
2011A	4,451	0.24	-32.9	28.5	4.6	15.4	2.4
2012E	7,411	0.40	66.5	17.1	3.9	24.7	2.1
2013E	9,251	0.49	24.8	13.7	3.2	25.7	2.3
2014E	10,695	0.57	15.6	11.9	2.7	25.0	2.9

Source: Powered by dataCentral

EDC.PS: Fiscal year end 31-Dec						Price: P6.77; TP: P7.65; Market Cap: P126,938m; Recomm: Buy					
Profit & Loss (Pm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	24,902	25,263	28,050	31,453	34,326	PE (x)	19.1	28.5	17.1	13.7	11.9
Cost of sales	-7,084	-8,173	-7,462	-8,142	-8,719	PB (x)	4.2	4.6	3.9	3.2	2.7
Gross profit	17,818	17,090	20,588	23,311	25,607	EV/EBITDA (x)	12.1	12.7	9.9	8.7	7.9
Gross Margin (%)	71.6	67.6	73.4	74.1	74.6	FCF yield (%)	3.7	-0.6	0.5	1.5	1.7
EBITDA (Adj)	13,289	12,660	16,585	19,068	21,148	Dividend yield (%)	1.8	2.4	2.1	2.3	2.9
EBITDA Margin (Adj) (%)	53.4	50.1	59.1	60.6	61.6	Payout ratio (%)	34	67	35	32	35
Depreciation	-3,445	-3,442	-3,732	-4,234	-4,756	ROE (%)	13.9	-0.6	26.8	26.3	25.1
Amortisation	0	0	0	0	0	Cashflow (Pm)					
EBIT (Adj)	9,844	9,218	12,853	14,833	16,392	EBITDA	13,289	12,660	16,585	19,068	21,148
EBIT Margin (Adj) (%)	39.5	36.5	45.8	47.2	47.8	Working capital	694	1,275	-83	174	-1,063
Net interest	-3,361	-3,717	-3,535	-3,596	-3,698	Other	-3,373	-5,479	-4,745	-5,349	-5,545
Associates	0	0	0	0	0	Operating cashflow					
Non-op/Except	-1,311	-5,010	373	-310	-538	Capex	-5,891	-9,192	-11,114	-11,956	-12,417
Pre-tax profit	5,172	491	9,691	10,927	12,157	Net acq/disposals	-1,204	0	0	0	0
Tax	-777	124	-775	-1,157	-1,262	Other	0	0	0	0	0
Extraord./Min.Int./Pref.div.	-279	-782	-880	-319	-167	Investing cashflow					
Reported net profit	4,116	-167	8,036	9,450	10,728	Dividends paid	-2,496	-3,341	-2,633	-2,972	-3,708
Net Margin (%)	16.5	-0.7	28.6	30.0	31.3	Financing cashflow					
Core NPAT	6,638	4,451	7,411	9,251	10,695	Net change in cash	-5,063	6,335	2,961	-1,862	3,108
Per share data						Free cashflow to s/holders					
Reported EPS (P)	0.22	-0.01	0.43	0.50	0.57		4,718	-736	643	1,937	2,123
Core EPS (P)	0.35	0.24	0.40	0.49	0.57						
DPS (P)	0.12	0.16	0.14	0.16	0.20						
CFPS (P)	0.57	0.45	0.63	0.74	0.78						
FCFPS (P)	0.25	-0.04	0.03	0.10	0.11						
BVPS (P)	1.63	1.46	1.75	2.09	2.47						
Wtd avg ord shares (m)	18,750	18,750	18,750	18,750	18,750						
Wtd avg diluted shares (m)	18,750	18,750	18,750	18,750	18,750						
Growth rates											
Sales revenue (%)	12.8	1.5	11.0	12.1	9.1						
EBIT (Adj) (%)	12.5	-6.4	39.4	15.4	10.5						
Core NPAT (%)	-7.2	-32.9	66.5	24.8	15.6						
Core EPS (%)	-7.2	-32.9	66.5	24.8	15.6						
Balance Sheet (Pm)											
Cash & cash equiv.	6,158	12,493	15,454	13,592	16,700						
Accounts receivables	4,602	3,411	4,415	4,848	5,213						
Inventory	2,653	3,356	1,920	2,148	2,341						
Net fixed & other tangibles	57,774	58,764	66,242	74,060	81,818						
Goodwill & intangibles	4,543	4,705	4,609	4,513	4,417						
Financial & other assets	5,574	7,288	6,742	6,760	6,776						
Total assets	81,304	90,018	99,382	105,922	117,265						
Accounts payable	5,123	6,704	5,637	5,987	6,289						
Short-term debt	1,702	2,250	827	2,307	9,210						
Long-term debt	39,679	49,240	54,805	52,213	49,956						
Provisions & other liab	2,560	2,178	2,184	2,687	1,896						
Total liabilities	49,065	60,371	63,452	63,194	67,350						
Shareholders' equity	30,670	27,429	32,832	39,311	46,331						
Minority interests	1,569	2,218	3,097	3,417	3,584						
Total equity	32,239	29,647	35,930	42,727	49,914						
Net debt	35,223	38,996	40,178	40,928	42,466						
Net debt to equity (%)	109.3	131.5	111.8	95.8	85.1						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Catalysts are Materializing

We believe that the key catalysts we have identified for EDC – strong earnings performance and progress on the BacMan plant – are now being realized (see [Energy Development \(EDC.PS\) - Potential to Re-Rate](#)).

■ **Strong earnings performance.** EDC posted 9M12 recurring profit of P6.3bn (up 67% YoY) driven by a 20.5% jump in revenues and contained operating expenses (flat YoY). 9M12 profit came in close to our original FY12 profit estimate and management's initial earnings guidance of P6.8-7.0bn.

– Growth was largely driven by the following:

- 305 MW Palinpinon-Tongonan geothermal plant (revenues up 30.5% YoY) on the back of its contract re-pricing that became effective in July 2011 (which lifted average tariff by 14.8% YoY to P4.89/kWh) and higher sales volume (up 13.6% YoY) as new bilateral contracts were signed
- 132 MW Pantabangan-Masiway hydro plant (revenues up 188.8% YoY) largely driven by the sale of ancillary services (which comprised 49% of the plant's revenues)
- Better handle on operating expenses (cost of sales and services, GAEX), which came in flat YoY on the back of easing rehabilitation-related expenses. We could see an uptick in operating expenses in 4Q12 arising from higher personnel expenses and other back-ended expenses that are usually booked in the 4Q.

Figure 1. EDC's Profit and Loss (3Q12 and 9M12)

Pm	3Q12	3Q11	YoY change	9M12	9M11	YoY change
Revenues						
Sale of electricity	6,419	6,291	2.0%	21,443	17,699	21.1%
Revenue from drilling services	162	196	-17.3%	507	522	-2.9%
Revenues	6,581	6,486	1.5%	21,950	18,222	20.5%
Cost of sales and services*						
Cost of sales of electricity and steam	-1,776	-2,430	-26.9%	-7,503	-7,441	0.8%
Cost of drilling services	-88	-186	-52.7%	-299	-464	-35.6%
Cost of sales and services	-1,863	-2,616	-28.8%	-7,802	-7,905	-1.3%
GAEX*	-932	-872	6.9%	-3,129	-3,067	2.0%
Operating income	3,785	2,998	26.2%	11,019	7,250	52.0%
Operating margin	57.5%	46.2%		50.2%	39.8%	
Financial income/expense						
Interest income	94	63	48.0%	275	281	-2.4%
Interest expense	-949	-945	0.4%	-2,870	-3,260	-12.0%
Financial income/expense	-856	-882	-3.0%	-2,595	-2,979	-12.9%
Other income/expenses						
FX gain/losses	119	-286	-141.8%	814	-49	-1770.6%
Derivatives gains/losses	2	117	-98.4%	2	108	-98.2%
Miscellaneous	-18	6	-390.5%	-86	19	-555.9%
Loss on impairment of PP&E	0	0		0	-4,999	n.m.
Other income/expenses	103	-163	-163.2%	730	-4,921	-114.8%
Income before tax	3,033	1,954	55.2%	9,154	-650	-1508.5%
Tax expense	-197	-145	36.4%	-588	162	-462.5%
Net income before minority interest	2,835	1,809	56.7%	8,566	-488	-1856.4%
Minority interest	-348	-148	134.6%	-1,460	-183	700.2%
Reported net income	2,488	1,661	49.8%	7,106	-670	-1160.3%
Extraordinary items	-106	150	-170.7%	-795	4,454	-117.8%
Recurring net income	2,381	1,811	31.5%	6,311	3,784	66.8%

Source: Company Information; Note: *new presentation based on SRC Rule 68 issued by the Philippine SEC in Oct 2011; previously classified as operating expenses (O&M, Purchased services and utilities, GAEX and Deprecation/Amortization)

- We raise our earnings estimate by 15.1%, 8.1% and 5.7% in 2012E, 2013E and 2014E largely on the back of higher hydro revenues (driven by sale of ancillary services) and lower operating expenses.

Figure 2. Revisions to our Earnings Forecasts

Pm	New			Old			% change			
	2012E	2013E	2014E	2012E	2013E	2014E	2012E	2013E	2014E	
Revenues	28,050	31,453	34,326	27,484	31,257	34,591	2.1%	0.6%	-0.8%	1) Higher ancillary revenues for FG Hydro - we assume hydro plant can sell 500m kWh of ancillary services in 2012E (vs 450m kWh prev), 250m kWh in 2013E (vs 200m kWh prev) and 150m kWh in 2014E (vs 200m kWh prev) resulting in 2012E, 2013E and 2014E ancillary revenues of P2.3bn, P1.1bn and P690m (vs P2bn, P880m and P880m prev); we assume slowdown in 2013E and 2014E due to increased competition
Operating expenses	-15,197	-16,619	-17,934	-15,909	-17,557	-19,085	-4.5%	-5.3%	-6.0%	2) Impact of higher ancillary revenues in 2013 is partly offset by lower assumed load factor for BacMan from 80% to 75% to factor in lower utilization of Unit 3 (prior to rotor replacement) Easing rehabilitation-related expenses, lower depreciation expense
Operating income	12,853	14,833	16,392	11,576	13,701	15,506	11.0%	8.3%	5.7%	
Operating margin	45.8%	47.2%	47.8%	42.1%	43.8%	44.8%				
Financial income/expenses	-3,535	-3,596	-3,698	-3,391	-3,456	-3,567	4.2%	4.1%	3.7%	Lower interest income
Other income/expenses	373	-310	-538	373	-310	-318	0.0%	0.0%	69.0%	
Income before tax	9,691	10,927	12,157	8,558	9,934	11,621	13.2%	10.0%	4.6%	
Tax expense	-775	-1,157	-1,262	-856	-1,025	-1,230	-9.4%	13.0%	2.6%	Lower tax rate from 10% to 8% in 2012E following effective rate in 9M12; maintained at 10% in 2013E and 2014E
Income before minority interest	8,916	9,770	10,895	7,702	8,910	10,391	15.8%	9.6%	4.8%	
Minority interest	-880	-319	-167	-638	-154	-244	37.8%	107.0%	-31.7%	
Net profit	8,036	9,450	10,728	7,064	8,755	10,146	13.8%	7.9%	5.7%	
Extraordinary items	-625	-200	-33	-625	-200	-33	0.0%	0.0%	0.0%	
Recurring net profit	7,411	9,251	10,695	6,439	8,556	10,113	15.1%	8.1%	5.7%	

Source: Citi Research

■ **Progress on BacMan rehabilitation.** Management reported that the 130 MW BacMan plant is set to be commissioned in December 2012, following positive developments on its rehabilitation.

- Rehabilitation of the generator rotors of Units 1 and 2 (55 MW each) has been completed. These units were flown back to the Philippines in Oct/Nov (from the Alstom site in the UK). Blade replacement of their turbines is ongoing, with EDC engaging the services of US-based welders because of their technical expertise. Management targets to commission both units in December 2012 for reliability runs, with full commercial operations targeted for January 2013. Company has indicated FY revenues from these two units to reach P3.8bn.
- Rehabilitation of Unit 3 (20 MW) is ongoing and will take 4-8 more weeks. The generator stator (stationary part of a rotor), however, turned out to be defective. Thus, EDC will need to replace the generator rotor for the unit to operate at full capacity. EDC has already ordered the rotor replacement from Toshiba, which will take 13 months to be delivered. Until the time that the rotor is replaced, management indicated that the unit can only likely run at around 6-7 MW (which is what we have factored into our forecasts).
- We expect the operations of BacMan in 2013 to largely drive our estimated 25% recurring profit growth in 2013E.

We think stock is attractively valued considering its growth profile. It is trading at a PEG of 0.7x (based on 2013E PE of 13.7x and 2012-14E EPS CAGR of 20%). Potential upside to our valuation and medium to long-term earnings forecasts could come from 1) 86 MW Burgos wind project (which could lift our 2015E earnings estimate by around 5% and our fair value by around 1.3%) and 2) successful

expansion into overseas markets, particularly in Chile, Peru and Indonesia (company is undertaking exploration field works in Chile and Peru and has applied for preliminary survey rights covering pre-identified sites in Indonesia).

Figure 3. Comparable Valuations

		Price	Target		CIR	EPS growth	P/E (x)	P/B (x)	EV/EBITDA (x)	Yield (%)	ROE (%)
	RIC	15 Nov 2012	Price	Rating		FY13E FY14E	FY13E FY14E	FY13E FY14E	FY13E FY14E	FY13E FY14E	FY13E FY14E
Aboitiz Power	AP.PS	Php34.50	Php36.00	Buy	-5.2%	-4.4%	11.6 12.1	2.6 2.3	8.5 9.0	3.3% 3.0%	23.9% 19.9%
EDC	EDC.PS	Php6.77	Php7.65	Buy	24.8%	15.6%	13.7 11.9	3.2 2.7	8.7 7.9	2.3% 2.9%	26.3% 25.1%
Manila Electric	MER.PS	Php250.60	Php235.00	Sell	0.2%	-1.1%	17.6 17.8	3.9 3.7	7.7 7.6	4.2% 4.2%	22.2% 20.8%

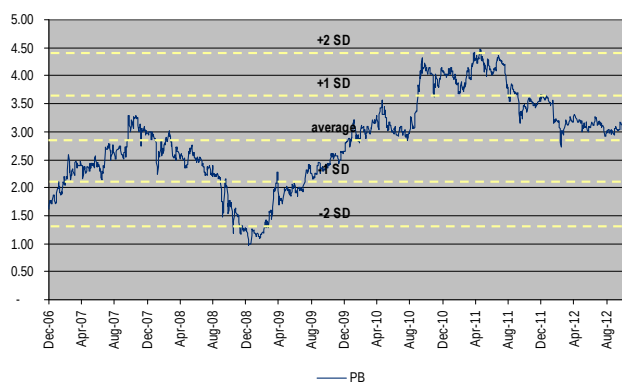
Source: Citi Research

Figure 4. EDC – 12m Forward PE Chart



Source: Citi Research

Figure 5. EDC – 12m Forward PBV Chart



Source: Citi Research

Energy Development

Company description

EDC is the largest geothermal producer in the Philippines (accounting for 60% of the country's installed geothermal capacity) and the second largest in the world. EDC is vertically integrated, operating both geothermal steamfields (1,199 MW) and power plants (1,199 MW). It also has a 60% stake in a 132 MW hydropower plant. Apart from steam production and power generation, the company is involved in drilling and consultancy services. EDC is 60%-owned by Red Vulcan Holdings Corp, which is under First Gen Corporation. EDC was listed in the Philippine Stock Exchange in December 2006.

Investment strategy

We have a Buy rating on EDC, the only listed pure renewable energy player in the Philippines. As the cost of fossil fuels rises, demand for clean energy will continue to grow, and we believe would-be competitors in the Philippines will struggle to replicate EDC's portfolio of geothermal assets. EDC has attractive growth prospects, both in the near- and long-term. We expect a 2012-15 CAGR of 21% in recurring earnings, driven by incremental contribution from acquisition. EDC is also undertaking several greenfield projects, which could start to contribute in 2014. Potential upside could come from EDC's planned venture into wind power development and overseas markets (Chile and Peru). Moreover, we estimate that EDC is currently trading at a 23% discount to replacement cost.

Valuation

Our price target of P7.65 is based on DCF models for the company's geothermal assets (100% owned) and its hydro plant operations (Pantabangan-Masiway, 60% owned). Our models deliver fair values of P136bn for the geothermal assets (using a WACC of 8.8% until 2031) and P5.9bn for the hydro assets (WACC of 8.8% and long-term growth of 3%). Our WACC assumes a risk-free rate of 5.8%, which is based on the three-month average of the government's 10-year bond yield, and an equity risk premium of 5%.

Risks

Key risks that could prevent the shares reaching our target price include: 1) Execution risk with regards to expansion and rehabilitation projects; and 2) Seasonality and hydrological conditions that could affect the performance of the hydro plant.

ENN Energy Holdings (2688.HK)

High Growth and Good Quality

Buy	1
Price (28 Nov 12)	HK\$33.75
Target price	HK\$39.00
from HK\$31.50	
Expected share price return	15.6%
Expected dividend yield	1.3%
Expected total return	16.8%
Market Cap	HK\$36,031M
	US\$4,649M

Price Performance (RIC: 2688.HK, BB: 2688 HK)



- **Summary** – We trim FY12-13E net profits by 6-8% factoring in latest management guidance but raise our DCF TP by 24% to HK\$39 on rollover. We are positive on ENN for high NG sales volume growth at 25% pa, expanded NG sales margin from new cheaper gas sources and resilient new household connections with less reliance on new properties. It would have no share issuance risk with positive free cashflow. FY13E PER of 16.0x is undemanding versus 20% FY12-14E EPS CAGR.
- **High NG sales volume growth** – ENN targets NG sale volume growth of 25% yoy to 6.2-6.3bcm in 2012E, including +23% yoy in 1H12. While some of its industrial customers had NG demand down 3-5% yoy year-to-date amid a weaker PRC economy, ENN has been selling its NG to new customers given NG is price competitive against such oil-related fuels as gasoline and diesel. ENN expects its NG sales volume to rise over 20% yoy in 2013-14E and to reach 15bcm in 2015E.
- **Expanded NG sales margin** – ENN's GP margin for NG sales was +3.9ppts yoy to 20% in 1H12 and gross profit per m³ was +Rmb0.07 or 8.1% yoy. This was due to the receipt of piped gas through West-to-East II and Tai-Qing-Wei pipelines in Henan, Anhui and Shandong, reducing the use of more expensive trucked gas and imported LNG. ENN expects the expanded margin to sustain in 2H12-FY13E.
- **Resilient new household connections** – ENN targets 1.05m new households in FY12E, including 560k in 1H12. The connections would include 50% from new properties (of which 3-5% from subsidized housing) and 50% from existing ones. The high % from existing buildings mitigates ENN's risk from slowdown of new property development in China. ENN's % revenue from connection fee is lowered to <25% in FY12E amid more NG sales.
- **Earning and TP revisions** – We trim our FY12-13E net profits by 6-8% factoring in the above new management guidance. Despite the earning cuts, we raise our DCF TP by 24% to HK\$39 on rollover.

Asian Utilities

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Statistical Abstract

FYE	Net Profit	EPS	P/E	D. Yld
31 Dec	(RmbM)	(Rmb)	(x)	(%)
2009A	803	0.774	35.0	0.7
2010A	1,013	0.954	28.4	0.9
2011A	1,253	1.180	23.0	1.1
2012E	1,519	1.410	19.2	1.3
2013E	1,831	1.700	16.0	1.6
2014E	2,176	2.020	13.4	1.9

Source: Powered by dataCentral

Earnings Revision

FYE	Net Profit	EPS	%	DPS
31 Dec	(RmbM)	(Rmb)	chg	(Rmb)
2012E	1,519	1.410	-6.8	0.351
Prev	1,607	1.514		0.375
2013E	1,831	1.700	-8.0	0.423
Prev	1,962	1.848		0.551
2014E	2,176	2.020	na	0.503
Prev	na	na		na

Source: Powered by dataCentral

2688.HK: Fiscal year end 31-Dec						Price: HK\$33.75; TP: HK\$39.00; Market Cap: HK\$36,031m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	11,215	15,068	17,644	21,896	26,982	PE (x)	28.4	23.0	19.2	16.0	13.4
Cost of sales	-8,203	-11,166	-12,887	-16,289	-20,310	PB (x)	4.7	4.0	3.4	2.9	2.5
Gross profit	3,012	3,902	4,756	5,607	6,672	EV/EBITDA (x)	15.3	12.1	10.2	9.0	7.9
Gross Margin (%)	26.9	25.9	27.0	25.6	24.7	FCF yield (%)	2.1	-1.8	0.5	1.4	2.4
EBITDA (Adj)	1,996	2,641	3,301	3,795	4,405	Dividend yield (%)	0.9	1.1	1.3	1.6	1.9
EBITDA Margin (Adj) (%)	17.8	17.5	18.7	17.3	16.3	Payout ratio (%)	25	25	25	25	25
Depreciation	-345	-388	-464	-526	-587	ROE (%)	18.2	19.2	19.6	19.9	20.2
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	1,651	2,253	2,837	3,269	3,818	EBITDA	1,996	2,641	3,301	3,795	4,405
EBIT Margin (Adj) (%)	14.7	15.0	16.1	14.9	14.1	Working capital	1,123	365	250	49	-177
Net interest	-311	-460	-618	-589	-559	Other	-362	-877	-816	-833	-922
Associates	282	367	352	419	453	Operating cashflow	2,757	2,129	2,735	3,011	3,305
Non-op/Except	189	167	251	301	331	Capex	-2,151	-2,650	-2,600	-2,600	-2,599
Pre-tax profit	1,811	2,327	2,821	3,400	4,042	Net acq/disposals	10	74	0	0	0
Tax	-410	-660	-800	-964	-1,146	Other	-24	-2,953	0	0	0
Extraord./Min.Int./Pref.div.	-388	-414	-502	-605	-719	Investing cashflow	-2,165	-5,529	-2,600	-2,600	-2,599
Reported net profit	1,013	1,253	1,519	1,831	2,176	Dividends paid	-418	-527	-309	-374	-451
Net Margin (%)	9.0	8.3	8.6	8.4	8.1	Financing cashflow	72	4,635	-603	-874	-951
Core NPAT	1,013	1,253	1,519	1,831	2,176	Net change in cash	663	1,235	-468	-463	-245
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	605	-521	134	411	706
Reported EPS (Rmb)	0.954	1.180	1.410	1.700	2.020						
Core EPS (Rmb)	0.954	1.180	1.410	1.700	2.020						
DPS (Rmb)	0.241	0.294	0.351	0.423	0.503						
CFPS (Rmb)	2.596	2.005	2.539	2.795	3.068						
FCFPS (Rmb)	0.570	-0.491	0.125	0.382	0.655						
BVPS (Rmb)	5.743	6.708	7.939	9.306	10.925						
Wtd avg ord shares (m)	1,050	1,050	1,066	1,066	1,066						
Wtd avg diluted shares (m)	1,062	1,062	1,077	1,077	1,077						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	33.3	34.4	17.1	24.1	23.2						
EBIT (Adj) (%)	18.6	36.5	25.9	15.2	16.8						
Core NPAT (%)	26.2	23.7	21.2	20.5	18.9						
Core EPS (%)	23.2	23.7	19.5	20.5	18.9						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	2,851	3,349	2,881	2,418	2,173						
Accounts receivables	1,356	1,837	1,940	2,408	2,967						
Inventory	249	272	391	494	617						
Net fixed & other tangibles	10,901	13,163	15,300	17,373	19,385						
Goodwill & intangibles	192	196	196	196	196						
Financial & other assets	4,090	8,071	8,245	8,900	9,902						
Total assets	19,640	26,888	28,953	31,790	35,240						
Accounts payable	4,937	6,131	6,774	8,045	9,548						
Short-term debt	2,379	3,213	3,213	3,213	3,213						
Long-term debt	3,884	7,459	6,959	6,459	5,959						
Provisions & other liab	900	1,245	1,248	1,252	1,255						
Total liabilities	12,100	18,048	18,195	18,969	19,975						
Shareholders' equity	6,031	7,046	8,463	9,919	11,645						
Minority interests	1,508	1,794	2,296	2,901	3,620						
Total equity	7,540	8,840	10,758	12,820	15,265						
Net debt	3,412	7,323	7,291	7,254	6,999						
Net debt to equity (%)	45.2	82.8	67.8	56.6	45.8						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

High NG Sales Volume Growth

ENN targets NG sale volume growth of 25% yoy to 6.2-6.3bcm in 2012E, including +23% yoy in 1H12. While some of its industrial customers had NG demand down 3-5% yoy year-to-date amid a weaker Chinese economy, ENN has been selling its NG to new customers with NG being price competitive against such oil related fuels as gasoline and diesel. The company expects NG sales volume to rise over 20% yoy in 2013-14E and to reach 15bcm in 2015E.

Expanded NG Sales Margin

Figure 1. ENN's NG unit sales prices and costs (Rmb/m3)

	1H11	FY11	1H12	yoy (amt)	yoy
Residential	2.45	2.49	2.55	0.10	4.1%
Commercial & Industrial	3.39	3.40	3.47	0.08	2.4%
Vehicles	3.76	3.85	3.94	0.18	4.8%
Company average	3.27	3.32	3.38	0.11	3.3%
NG cost	2.45	2.52	2.49	0.04	1.6%
Gross profit	0.82	0.80	0.89	0.07	8.1%

Source: Company Reports and Citi Research Estimates

ENN's GP margin of gas sales was 20% in 1H12, +3.9ppts yoy. The margin expansion was due to the receipt of piped gas through WE II and Tai-Qing-Wei pipeline for projects in Henan, Anhui and Shandong provinces, reducing the use of trucked gas/imported LNG which was more expensive.

ENN's expects the expanded 1H12 GP margin of its NG sales to be sustainable in 2H12-FY13E. Potential swing factors of the GP margin include (i) the impact of pricing reform, (ii) more usage of gas particularly in year-end which requires the use of more expensive gas from spot market, and (iii) more projects start to receive piped gas (Guangdong province), replacing expensive supply from truck gas/imported LNG.

Sustainable New Residential Connection

ENN targets to connect 1.05m households in FY12E, including 560k in 1H12. The connection would include 75-80% from commodity properties (of which 25-30% from new buildings and 50% from old buildings) and 20-25% from subsidized housing. They are from 90% from subsidiaries and 10% from associates & JCEs. ENN's % of revenue from connection fee keeps declining to below 25% in FY12E amid rising proportion of its revenue from recurrent NG sales.

Increasing Vehicle Refueling Stations

ENN plans to add 30-40 stations pa in coming years to cater for rising demand. By end 1H12, it had 236 CNG stations in operation and licenses on hand to add up to 440 stations.

CNG and LNG stations target different customer segments. CNG serves vehicles that run within a city, e.g. taxi/bus/private vehicles, while LNG serves heavy-duty trucks/inter-city buses that run long haul. ENN has been operating CNG stations since 2002, it's a relatively mature business compared with LNG stations, and it keeps growing and contributes to over 12% of its total revenue.

ENN started LNG station business in 2011 and it is in the beginning stage, so as the market itself, which has limited number of LNG vehicles in operation. Its strategy of developing LNG stations is to secure customers before constructing the station. i.e. signing contract with logistics company/public transport company which have LNG-fueled trucks/buses, then construct stations along their routes (e.g. Container terminals/mining zone/logistics hub). Risks will be the conversion rate of LNG vehicles remains sluggish, logistics/transport companies impacted by economy downturn.

ENN expects the returns from its investments in CNG and LNG refueling stations to be comparable. But in terms of gas sales volume, LNG stations will be a better growth driver – a typical LNG station supplies around 30,000 m³ of gas daily, double that of an ordinary CNG station.

Figure 2. ENN's vehicle gas refueling stations

CNG refueling stations since 2002		LNG refueling stations since 2011	
Number of approval obtained	440	Number of approval obtained	87
Aggregated no. of CNG refueling stations	234	Aggregated no. of LNG refueling stations	42
Potential no. of CNG refueling stations to be built	206	Potential no. of LNG refueling stations to be built	45
Example: Payback calculation of a CNG taxi		Example: Payback calculation of a New LNG truck	
Average gasoline price (Rmb/liter)	7.5	Average diesel price (Rmb/liter)	7.1
Average CNG price (Rmb/m3)	3.9	Average LNG price (Rmb/m3)	4.6
Average CNG price (Rmb/m3)	0.05	Diesel consumption per km (liter)	0.4
CNG consumption per km (m3)	0.06	LNG consumption per km (m3)	0.5
Cost saved per km (Rmb)	0.141	Cost saved per km (Rmb)	0.54
Average driving distance (km/day)	400	Average driving distance (km/day)	400
Daily average savings (Rmb)	56.4	Daily average savings (Rmb)	216
Monthly average savings (Rmb)	1,692	Monthly average savings (Rmb)	6,480
Conversion fee (Rmb)	3,500	Price difference of LNG truck & diesel truck (Rmb)	80,000
Monthly maintenance cost (Rmb)	175	Monthly maintenance cost (Rmb)	150
Payback period (months)	2	Payback period (months)	12

Source: ENN & Citi Research

Capex & Free Cashflow

ENN has budgeted capex of Rmb2.6bn pa in 2012-14E, including:

- (i) Rmb1.2-1.3bn recurrent capex for existing projects;
- (ii) Rmb500-600m for construction of LNG storage tanks to empower its gas storage capability to facilitate more gas sales longer term;
- (iii) Rmb400-500m for construction of new CNG/LNG stations;
- (iv) Rmb200-300m for M&As; and
- (v) Rmb100-200m for NG management and distributed energy projects.

ENN plans to have 8-10 new projects pa. ENN's expansion strategy is focusing on neighboring areas of its existing projects to create synergy as well as industrial parks. Industrial parks contain cluster of C/I customers with larger gas usage, ENN expect the industrial park projects acquired in 1H12 can reach 1bcm gas usage when they become mature in 3-4 years' time. Because of their concentrated location, it does not need to invest as much capex as a city gas project while able to enjoy decent volume growth afterwards. When acquiring industrial parks, ENN does not need to pay premium like acquiring large-scale city gas projects, thus, returns

can be higher than ordinary city gas projects. Risk will be the economy slows down impacting certain industries, but the company has a diversified portfolio of C/I customers to mitigate such risk. NG usage at industrial parks, around 0.15-0.15bcm each, has become more popular for emission cut reasons. ENN is also interested in acquisitions of deal size around HK\$1.5-2.0bn each, while able to meet its investment hurdle rate of 15% IRR. Furthermore, it might secure a relatively big project in Hebei province in coming months.

ENN expects to have slight positive free cashflow in 2012E. The company would raise its dividend payout ratio from prevailing 25% to 30% when its free cashflow exceeds Rmb1bn pa.

Regarding cost control, ENN's administrative cost against revenue was 9.2% in 2011 and 8.9% in 1H12. It would be around 9% in 2H12E. Its distribution cost against revenue was 1.9% in 2011-1H12 and would be around 2% in 2H12E.

ENN Energy Group Businesses

By end 2011, ENN Group had over 28,000 employees and Rmb40bn assets.

With 3bn tonnes coal reserve, ENN's group coal-to-chemical projects include:

1. Inner Mongolia Ordos Phase I with an annual production capacity of 0.6m tons of methanol and 0.4m tons of dimethyl ether was commissioned in July 2009;
2. Inner Mongolia Ordos Phase II 0.6m tons pa coal-based new material project – The total investment is Rmb11.55bn. This project could convert 0.6m tons of new material from methanol per annum and the construction work was started in May 2012;
3. Shanxi ENN and Shanxi Jiaotan Group jointly invested in (a) 0.3m tons coke oven gas to methanol and (b) 0.2m tons of dimethyl ether. Phase I which could convert 0.2m tons coke oven gas to methanol was commissioned in 2011
4. Jiangsu ENN and Canadian based Methanex jointly invested in the first phase of 0.2m tons of dimethyl ether project was commissioned in 2007
5. ENN's 20,000 tons pa dimethyl ether production facility in Bangbu, Anhui, was commissioned
6. ENN's underground coal gasification facility located at Ulanqab, Inner Mongolia
7. 0.72m methanol production plants at Tengzhou, Shandong – The project was jointly invested by ENN, Lenovo, China Oceanwide and other parties while ENN maintains the controlling interests. Phase I with an production capacity of 0.36m tons of methanol was commissioned in Dec 2009 while Phase II started the trail operation successfully in end 2011

Figure 3. ENN's earnings estimates – Citi vs. consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	1,519	1,544	-2%
FY13E	1,831	1,858	-1%
FY14E	2,176	2,161	+1%

Source: Bloomberg and CIRA Estimates

DCF Calculation & Historical Valuations

Figure 4. ENN's DCF calculation (Rmbm)

Actual value in the run up to FY20E:	2012	2013	2014	2015	2016	2017	2018	2019	2020
FCF		411	706	1,278	1,907	2,672	3,594	4,745	6,178
Discount factor		1.0	1.1	1.2	1.3	1.4	1.5	1.7	1.8
Net debt	-7,291	411	647	1,073	1,469	1,885	2,324	2,813	3,357
WACC	9.1%								
Terminal Value:									
NAV in 2021E	3,491.1								
Terminal growth rate	4%								
ROCE	15.0%								
WACC	9.1%								
Terminal Value	50,143								
Discount factor	1.84								
NPV of terminal value	27,244								
Total DCF	33,933								
Exchange rate (Rmb/HK\$)	1.24								
Total DCF (HK\$m)	42,013								
No. of issued shares (m)	1,077								
DCF/share (HK\$)	39.0								

Source: Company Reports and Citi Research Estimates

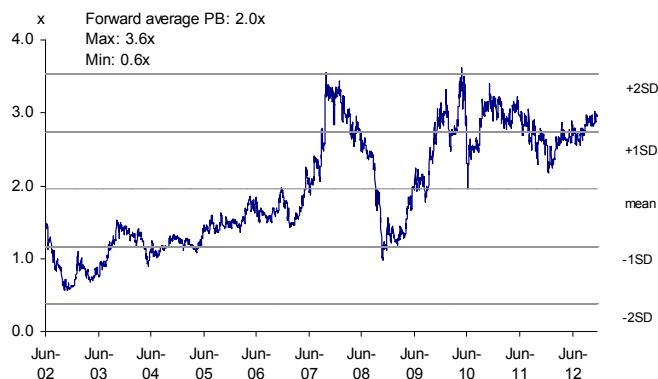
Valuations above Historical Mean

Figure 5. ENN's forward PER chart – above historical mean



Source: Citi Research

Figure 6. ENN's forward PB chart – above historical mean



Source: Citi Research

Compared to historical mean, ENN is trading around 1sd above mean. We think the market is rewarding the company for rising ROE driven by organic growth hence operating leverage from existing projects.

Financial Statements & Key Assumptions

Figure 7. Consolidated income statement (Rmbm)

Year to 31 Dec	2010	2011	2012F	2013F	2014F
Gas connection	2,635	3,415	3,754	4,100	4,458
Sales of piped gas	5,728	7,980	9,911	13,172	17,158
Bottled LPG sale	240	192	144	144	144
Sales of gas appliances	84	97	102	107	112
Vehicle gas refueling stations	1,209	1,620	2,117	2,645	3,259
Sales of materials	414	592	503	503	503
Wholesale of gas	905	1,172	1,113	1,225	1,347
Net Revenues	11,215	15,068	17,644	21,896	26,982
yoy	33.3%	34.4%	17.1%	24.1%	23.2%
COGS	-8,203	-11,166	-12,887	-16,289	-20,310
Connection expense	-968	-1,161	-1,306	-1,502	-1,628
Piped gas cost	-4,452	-6,476	-7,750	-10,365	-13,575
LPG cost	-228	-175	-130	-129	-128
Appliance sales	-62	-72	-75	-79	-83
Vehicle gas cost	-976	-1,281	-1,688	-2,114	-2,625
Sales of materials	-380	-557	-473	-473	-473
Wholesale of gas	-792	-1,056	-1,001	-1,101	-1,211
Depreciation	-345	-388	-464	-526	-587
Gross profits	3,012	3,902	4,756	5,607	6,672
GP margin	26.9%	25.9%	27.0%	25.6%	24.7%
Yoy	18.6%	29.6%	21.9%	17.9%	19.0%
Other revenue	189	167	251	301	331
Selling expenses	-213	-283	-331	-389	-480
Administration expenses	-1,169	-1,380	-1,588	-1,949	-2,374
Operating expense	-1,382	-1,663	-1,919	-2,338	-2,854
EBITDA	2,117	2,741	3,466	4,024	4,673
% change	12%	29%	26%	16%	16%
Operating profit	1,819	2,406	3,087	3,570	4,148
Yoy	12%	32%	28%	16%	16%
OP margin	16.2%	16.0%	17.5%	16.3%	15.4%
Other gains and losses	21	14	0	0	0
Share of results of associates	5	51	48	48	48
Share of results of JCEs	277	316	304	371	404
Finance costs	-311	-460	-618	-589	-559
Profit before tax	1,811	2,327	2,821	3,400	4,042
Taxation	-410	-660	-800	-964	-1,146
Profit after tax	1,401	1,667	2,021	2,436	2,895
Minority interest	-388	-414	-502	-605	-719
Net profits	1,013	1,253	1,519	1,831	2,176
yoy	26.2%	23.7%	21.2%	20.5%	18.9%

Source: Company Reports and Citi Research Estimates

Figure 8. Key Ratios

Y/e Dec, HK\$ m	2010	2011	2012F	2013F	2014F
Gross margin	26.9	25.9	27.0	25.6	24.7
Operating margin	16.2	16.0	17.5	16.3	15.4
Pre-tax margin	16.1	15.4	16.0	15.5	15.0
Net profit margin	9.0	8.3	8.6	8.4	8.1
Interest cover (x)	9.6	6.4	5.6	6.8	8.2
ROA (%)	5.2	4.7	5.2	5.8	6.2
Net debt-to-equity (%)	45.2	82.8	67.8	56.6	45.8
Cash flow-to-debt (x)	0.5	0.3	0.4	0.5	0.5
EBITDA margin (%)	19%	18%	20%	18%	17%

Source: Company Reports and Citi Research Estimates

Figure 9. Consolidated balance sheet (Rmbm)

As at 31 Dec	2010	2011	2012F	2013F	2014F
Cash	2,851	3,349	2,881	2,418	2,173
Accounts receivable	1,356	1,837	1,940	2,408	2,967
Inventories	249	272	391	494	617
Other current assets	622	3,486	3,660	4,315	5,318
Total current assets	5,079	8,944	8,872	9,636	11,075
Net fixed assets	10,901	13,163	15,300	17,373	19,385
Long-term investments	3,468	4,585	4,585	4,585	4,585
Goodwill	192	196	196	196	196
Total assets	19,640	26,888	28,953	31,790	35,240
Accounts payable	3,573	4,172	4,815	6,086	7,589
Other payables	1,364	1,959	1,959	1,959	1,959
Short term loans	2,379	3,213	3,213	3,213	3,213
Tax payable	172	176	179	183	186
Total current liabilities	7,488	9,520	10,167	11,441	12,947
Long-term debt	3,884	7,459	6,959	6,459	5,959
Deferred taxation and deferred Income	728	1,069	1,069	1,069	1,069
Total liabilities	12,100	18,048	18,195	18,969	19,975
Share capital	110	110	111	111	111
Reserves	5,922	6,936	8,351	9,808	11,534
Shareholders equity	6,031	7,046	8,463	9,919	11,645
Minority interest	1,508	1,794	2,296	2,901	3,620
Total equity	7,540	8,840	10,758	12,820	15,265
Total liabilities & equity	19,640	26,888	28,953	31,790	35,240

Source: Company Reports and Citi Research Estimates

Figure 10. Consolidated cashflow statement (Rmbm)

	2010	2011	2012F	2013F	2014F
Cashflow from operations					
EBITDA	2,117	2,741	3,466	4,024	4,673
Profit from associates and JCE	282	367	304	371	404
Working capital changes	1,123	365	250	49	-177
Other operating cash items	-138	-256	133	121	111
Gross operating cash flow	3,384	3,217	4,153	4,564	5,011
Net interest paid	-218	-428	-618	-589	-559
Tax paid	-410	-660	-800	-964	-1,146
Net operating cash flow	2,757	2,129	2,735	3,011	3,305
Capex	-2,151	-2,650	-2,600	-2,600	-2,599
Free cash flow	605	-521	134	411	706
Cash from acquisitions	-525	-737	0	0	0
Disposal of fixed assets	10	74	0	0	0
Change in temp cash inv.	-24	-2,953	0	0	0
Cashflow from investing	-2,690	-6,266	-2,600	-2,600	-2,599
Increase in Equity	8	71	206	0	0
Ordinary Dividend Paid	-418	-527	-309	-374	-451
Change in borrowings	402	4,370	-500	-500	-500
Other Financing Cash Flow	79	721	0	0	0
Cash Flow from Financing	72	4,635	-603	-874	-951
Beginning cash balance	2,713	2,851	3,349	2,881	2,418
Ending cash balance	2,851	3,349	2,881	2,418	2,173

Source: Company Reports and Citi Research Estimates

Figure 11. Key assumptions

	2010	2011	2012E	2013E	2014E
Total coverage:					
Connectable households (k)	15,623	17,714	18,641	19,147	19,668
New connection (k)	876	1,030	1,051	1,078	1,092
Acquired connection (k)	36	44	4	4	4
Total newly added connection	912	1,074	1,055	1,083	1,096
Connected household (k)	5,619	6,692	7,748	8,830	9,927
Total penetration rate (%)	36%	38%	42%	46%	50%
No. of vehicle refueling station	192.0	238	273	298	322
Revenue mix:					
Revenue from connection fee (Rmbm)	2,635	3,415	3,754	4,100	4,458
y-y change (%)	3.2	29.6	9.9	9.2	8.7
- from residential users (Rmbm)	1,886	2,442	2,457	2,417	2,423
y-y change (%)	-5.7	29.5	0.6	-1.6	0.2
- from C&I users (Rmbm)	1,163	973	1,297	1,683	2,035
y-y change (%)	10.3	-16.4	33.3	29.7	21.0
Revenue from piped gas sale (Rmbm)	5,728	7,980	9,911	13,172	17,158
y-y change (%)	40.5	39.3	24.2	32.9	30.3
Revenue from bottled LPG sale (Rmbm)	240	192	144	144	144
y-y change (%)	-73.2	-20.1	-25.0	0.0	0.0
Revenue from gas appliance sale (Rmbm)	84	97	102	107	112
y-y change (%)	-3.4	15.6	5.0	5.0	5.0
Revenue from vehicle gas refueling stations	1,209	1,620	2,117	2,645	3,259
y-y change (%)	51.6	34.0	30.7	25.0	23.2
Revenue from sales of materials	414.0	592.0	503.2	503.2	503.2
y-y change (%)	n/a	43.0	-15.0	0.0	0.0
Wholesale of gas	905.0	1,172.0	1,113.4	1,224.7	1,347.2
y-y change (%)	n/a	29.5	-5.0	10.0	10.0
Operating profit mix:**					
Connection fee	48.9%	44.9%	40.0%	33.8%	28.5%
Piped gas sales	43.7%	47.7%	52.7%	58.8%	64.1%
Others	7.4%	7.4%	7.4%	7.4%	7.4%
Gas sale:					
Total gas sale volume (mm ³)	4,149	5,374	6,572	8,060	9,854
y-y change (%)	41.1	29.5	22.3	22.6	22.3
- from residential users (mm ³)	641	824	1,004	1,142	1,280
y-y change (%)	23.3	28.6	21.8	13.7	12.1
- from C&I users (mm ³)	2,988	3,853	4,689	5,860	7,319
y-y change (%)	47.1	28.9	21.7	25.0	24.9
- from gas refueling stations (mm ³)	520	696	879	1,059	1,255
y-y change (%)	33.9	33.9	26.2	20.5	18.5
Unit prices:					
Average household connection fee (Rmb) #	2,602	2,602	2,602	2,602	2,602
y-y change (%)	-3.0	0.0	0.0	0.0	0.0
Average C&I connection fee (Rmb)	163	142	135	128	122
y-y change (%)	-15.0	-13.0	-5.0	-5.0	-5.0
Average gas cost (Rmb/m ³)	2.35	2.52	2.52	2.72	2.94
y-y change (%)	4.4	7.2	0.0	8.0	8.0
Average sale price (Rmb/m ³) (include 13% VAT)	3.05	3.18	3.23	3.49	3.70
y-y change (%)	5.0	4.6	1.5	8.0	6.0
Gas sale margin (Rmb/m ³)	0.70	0.66	0.71	0.77	0.76
Gas sale margin	23%	21%	22%	22%	21%

Source: Company Reports and Citi Research Estimates

* C&I: commercial and industrial

** Excludes unallocated corporate expenses

ENN Energy Holdings

Company description

ENN Energy (formerly known as Xinao Gas) specializes in distributing piped natural gas to residential households, commercial buildings and light industrial companies in China. The company was founded in 1993 with a single project in Langfang of Hebei province. It began expanding actively into other areas in 2002, after restrictions on foreign participation in piped gas distribution in urban cities were lifted. It currently holds over 100 exclusive city gas supply contracts in China, covering an urban population of 54m.

Investment strategy

We rate ENN Energy's shares as Buy (1), expecting strong natural gas sales growth of 20-25% pa in FY12-14E. ENN Energy is also financially strong with mild positive free cashflow in FY12E, which is expected to increase further in FY13E, and a reasonable net debt to equity ratio. There is upside potential to its dividend payout ratio (25% in FY12E) if its positive free cashflow exceeds Rmb1bn in a year.

Valuation

Our HK\$39 target price for ENN Energy uses an NAV approach for its gas distribution business based on DCF. We have not added any value for its DME production plant in Inner Mongolia due to the cyclical nature of its product prices, which are at low levels amid weakened demand. We think a DCF method is suitable for gas distribution companies such as ENN Energy that work in a relatively static industry with predictable cash flows. Our WACC for ENN Energy is 9.1%, which assumes: 1) a risk-free rate of 5.2%, 2) a market risk premium of 6.8%, 3) an equity beta of 0.96x, according to Bloomberg, 4) a cost of debt of 5.2%, 5) a target debt to capital ratio of 33.3%, 6) a 25% corporate tax rate, and 7) a terminal growth rate of 4% beyond 2015. At our DCF-based target price, ENN Energy would trade at a 2013E P/E of 18.7x.

Risks

Upside or downside risks that could prevent the shares from reaching our target price include: 1) higher or lower-than-expected revenue from connection fee with more provincial governments requesting amortization of connection fee income, and 2) higher or lower-than-expected gas sales volume.

GAIL (GAIL.BO)

Lacks Growth & Catalysts

Neutral	2
Price (27 Nov 12)	Rs339.40
Target price	Rs389.00
Expected share price return	14.6%
Expected dividend yield	2.7%
Expected total return	17.3%
Market Cap	Rs430,521M
	US\$7,728M

Price Performance

(RIC: GAIL.BO, BB: GAIL IN)



■ **Dearth of catalysts** — GAIL's near-term growth is likely to be constrained with vols for FY13E down ~6% yoy. Vols could only pick up by 4Q with the commissioning of the Dabhol (Dec/Jan) and Kochi (Jan/Mar) LNG terminals. While mgmt expects to exit FY13 with vols of c115-116 mmscmd, down from earlier estimates of c121-125, this would also depend on the extent of further KG declines. While GAIL stands to benefit from the recent gov't move to cap subsidised LPG cylinders per household at six every year (could lead to ~7-8% earnings upside for GAIL on a full-year basis), there have been recent talks of rolling back the decision.

■ **Dabhol and Kochi updates** — GAIL has guided to initial capacity of c1.2 MMTA (~4.5 mmscmd) at Dabhol (cannot operate during the four monsoon mths till the breakwater is commissioned in FY16E). Initial vols will flow through GAIL's Dabhol-Bangalore and Dabhol-Panvel pipelines. The first phase of the Kochi-Bangalore/ Mangalore pipeline to connect customers in the vicinity of PLNG's soon-to-be-commissioned Kochi terminal is complete. Initial demand potential is ~c4-5 mmscmd. Phase-II is likely to be completed by Dec-13. GAIL has recently signed short/long-term LNG deals (0.8 MMT from GDF Suez over 2 years starting Jan-13, 0.7 MMTA from GNF over 3 yrs starting Jan-13, 2.5 MMTA from Gazprom over 20 yrs starting FY19, 3.5 MMTA from Cheniere over 20 yrs starting 2017).

■ **Gas trading remains volatile** — GAIL's gas trading business remains highly volatile, with contribution to EBITDA swinging from 15-25% over the last few qtrs and impacting earnings quality. Gas trading vols in 2Q fell to 81 mmscmd, with lower spot LNG prices not benefiting the company due to lack of leverage to Dahej/Hazira utilisation.

■ **Maintain Neutral** — Valns for the gas business at ~11x P/E (vs. ~8x for its smaller peer) are not compelling enough given constrained near-term growth on the back of slight delay in commissioning of Dabhol/Kochi LNG terminals, falling KG vols, limited upside from PLNG's Dahej terminal, lack of leverage to Hazira vols, and delay in phase II of the Kochi-Bangalore/Mangalore pipeline to end-CY13.

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Statistical Abstract

Year to 31 Mar	Net Profit (RsM)	Diluted EPS (Rs)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2011A	35,611	28.07	13.4	12.1	2.2	19.8	2.2
2012A	36,609	28.86	2.8	11.8	2.0	17.9	2.6
2013E	37,945	29.91	3.7	11.3	1.8	16.6	2.7
2014E	39,291	30.97	3.5	11.0	1.6	15.5	2.9
2015E	43,050	33.94	9.6	10.0	1.5	15.4	3.2

Source: Powered by dataCentral

GCL-Poly Energy Holdings (3800.HK)

Sharp Drop in Poly and Wafer Shipments; Sold US Solar Farms

Sell/High Risk	3H
Price (15 Nov 12)	HK\$1.39
Target price	HK\$1.00
<i>from HK\$1.10</i>	
Expected share price return	-28.1%
Expected dividend yield	0.0%
Expected total return	-28.1%
Market Cap	HK\$21,510M
	US\$2,775M

Price Performance (RIC: 3800.HK, BB: 3800 HK)



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■ **Sharp drop in poly & wafer sales** — GCL-Poly reported 3Q12 operating data on 15 Nov. After a sharp production ramp in 1H12, polysilicon and wafer sales fell 89% and 42% QoQ, respectively, on reduced end-demand and further drop in pricing, as the sector underwent further de-stocking. We believe GCL losses in 2H12 may widen amid a weak pricing environment. Maintain Sell with revised TP of HK\$1.00.

■ **Rising inventories in 3Q12** — While GCL-Poly reduced poly and wafer production by 41% and 10% QoQ, respectively, poly and wafer sales dropped 89% and 42% QoQ, which will likely result in higher inventory and put stress on its cash flows. In 4Q12, poly prices dropped further with quotes near US\$15/kg, compared to 3Q12 ASP of US\$20.1/kg, and may lead to further inventory write-downs.

■ **US solar farm sale** — After GCL-Poly completed a 92MW solar farm sale in July, the 48MW solar farm sale in California announced today was sold for US\$50.5mn (HK\$391mn). Project acquisition/construction costs were not disclosed, and the solar farm disposal should provide capital for new project developments.

■ **China's solar developments** — Recent announcements by China to approve solar projects, with specific subsidy for rural usage, is positive to domestic demand, and GCL-Poly targets to obtain 2GW in new installs. However, construction projects will likely get competitive, as solar manufacturers are utilizing project developments as an exit for rising inventories, which may push down construction margins.

■ **Revising estimates** — In view of lower prices along the supply chain, we cut our 2012/13/14E EPS forecasts to HK\$(0.09)/HK\$0.01/HK\$0.09, respectively. Upside risks: 1) faster implementation of poly import-tariff in China, which may lead to a rebound in China's poly/wafer prices; 2) more aggressive production reduction by key poly makers; 3) significant improvements in order visibility from new markets for solar products, including Japan, US, and China, in view of weak European demand.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(HK\$M)	(HK\$)	(%)	(x)	(x)	(%)	(%)
2010A	4,024	0.26	na	5.4	1.3	29.0	3.7
2011A	4,275	0.28	6.2	5.0	1.0	23.3	4.0
2012E	-1,437	-0.09	-133.7	-15.0	1.2	-7.4	0.0
2013E	91	0.01	106.3	nm	1.4	0.5	0.1
2014E	1,452	0.09	na	14.8	1.4	9.4	1.4

Source: Powered by dataCentral

3800.HK: Fiscal year end 31-Dec						Price: HK\$1.39; TP: HK\$1.00; Market Cap: HK\$21,510m; Recomm: Sell/High Risk					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	18,472	25,506	18,101	21,252	24,953	PE (x)	5.4	5.0	-15.0	nm	14.8
Cost of sales	-11,661	-17,039	-16,794	-18,548	-20,778	PB (x)	1.3	1.0	1.2	1.4	1.4
Gross profit	6,811	8,466	1,307	2,704	4,175	EV/EBITDA (x)	3.4	3.7	39.5	16.0	11.2
Gross Margin (%)	36.9	33.2	7.2	12.7	16.7	FCF yield (%)	2.9	-71.9	3.5	1.2	8.7
EBITDA (Adj)	6,825	8,597	1,078	2,905	4,211	Dividend yield (%)	3.7	4.0	0	0.1	1.4
EBITDA Margin (Adj) (%)	36.9	33.7	6.0	13.7	16.9	Payout ratio (%)	20	20	0	20	20
Depreciation	-1,244	-2,125	-1,652	-1,985	-2,030	ROE (%)	29.0	23.3	-7.4	0.5	9.4
Amortisation	0	0	0	0	0	Cashflow (HK\$m)					
EBIT (Adj)	5,581	6,471	-574	920	2,180	EBITDA	6,825	8,597	1,078	2,905	4,211
EBIT Margin (Adj) (%)	30.2	25.4	-3.2	4.3	8.7	Working capital	1,595	-5,098	3,631	-1,939	-2,713
Net interest	-606	-1,166	-2,080	-1,863	-1,746	Other	-586	-735	1,268	1,335	1,318
Associates	11	3	120	222	372	Operating cashflow					
Non-op/Except	563	531	1,192	1,191	1,298	Capex	-7,208	-18,245	-5,225	-2,045	-945
Pre-tax profit	5,547	5,839	-1,343	469	2,105	Net acq/disposals	0	0	0	0	0
Tax	-1,159	-1,269	-44	-79	-352	Other	0	0	0	0	0
Extraord./Min.Int./Pref.div.	-364	-295	-50	-300	-300	Investing cashflow					
Reported net profit	4,024	4,275	-1,437	91	1,452	Dividends paid	0	0	0	0	0
Net Margin (%)	21.8	16.8	-7.9	0.4	5.8	Financing cashflow					
Core NPAT	4,024	4,275	-1,437	91	1,452	Net change in cash	4,764	378	1,341	-3,003	-3,151
Per share data						Free cashflow to s/holders					
Reported EPS (\$)	0.26	0.28	-0.09	0.01	0.09		626	-15,481	751	256	1,871
Core EPS (\$)	0.26	0.28	-0.09	0.01	0.09						
DPS (\$)	0.05	0.06	0	0.00	0.02						
CFPS (\$)	0.51	0.18	0.39	0.15	0.18						
FCFPS (\$)	0.04	-1.00	0.05	0.02	0.12						
BVPS (\$)	1.04	1.33	1.19	1.02	0.96						
Wtd avg ord shares (m)	15,472	15,472	15,474	15,474	15,474						
Wtd avg diluted shares (m)	15,501	15,501	15,474	15,474	15,474						
Growth rates											
Sales revenue (%)	273.7	38.1	-29.0	17.4	17.4						
EBIT (Adj) (%)	509.6	16.0	-108.9	260.1	137.1						
Core NPAT (%)	nm	6.2	-133.6	106.3	nm						
Core EPS (%)	nm	6.2	-133.7	106.3	nm						
Balance Sheet (HK\$m)											
Cash & cash equiv.	8,466	10,932	12,152	9,149	5,998						
Accounts receivables	2,599	7,040	7,972	9,360	10,990						
Inventory	1,647	3,627	4,495	5,066	5,675						
Net fixed & other tangibles	23,662	41,181	44,754	44,813	43,728						
Goodwill & intangibles	110	66	70	72	75						
Financial & other assets	4,096	4,641	2,901	3,300	4,036						
Total assets	40,581	67,488	72,343	71,760	70,502						
Accounts payable	4,384	7,628	8,208	9,066	10,156						
Short-term debt	6,537	12,212	17,012	16,193	14,637						
Long-term debt	7,379	19,535	19,935	20,134	18,121						
Provisions & other liab	4,901	5,979	7,163	8,627	10,441						
Total liabilities	23,202	45,354	52,319	54,021	53,354						
Shareholders' equity	16,152	20,567	18,408	15,822	14,931						
Minority interests	1,227	1,567	1,617	1,917	2,217						
Total equity	17,379	22,134	20,025	17,739	17,148						
Net debt	5,450	20,815	24,795	27,178	26,760						
Net debt to equity (%)	31.4	94.0	123.8	153.2	156.1						

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For definitions of the items in this table, please click [here](#).

Lowered Production Output

GCL-Poly reported 3Q12 operating data on 15 November. Polysilicon production was down 41% QoQ, as poly and wafer sales fell 89% and 42% QoQ respectively, on reduced end-demand and weak pricing. GCL sold two solar farms for US\$50.5mn, but we expect widened losses in 2H12 amid a weak pricing environment in the near term. Maintain Sell and lower our target price to HK\$1.00.

Figure 1. GCL-Poly – Quarterly Production Output and ASP

	1Q10	2Q10	3Q10	4Q10	1Q11	2Q11	3Q11	4Q11	1Q12	2Q12	3Q12
Polysilicon production (MT)	3,112	3,919	5,068	5,754	5,927	6,099	6,399	10,988	12,235	12,998	7,631
Polysilicon sales (MT)	2,584	2,810	3,405	1,708	914	594	359	945	3,041	5,971	657
Polysilicon ASP (US\$/kg)	50.00	50.00	51.70	60.10	62.17	61.70	44.00	26.20	25.98	21.80	20.10
Polysilicon production cost (US\$/kg)	35.00	31.50	25.40	22.90	22.30	21.80	20.90	19.30	19.70	18.20	
Wafer production (MW)					953	1,123	1,201	1,212	1,164	1,878	1,689
Wafer sales (MW)	31	198	435	787	959	1,160	1,089	1,242	1,234	1,952	1,130
Wafer ASP (US\$/W)	0.74	0.80	0.80	0.82	0.760	0.640	0.490	0.320	0.280	0.260	0.250
Wafer production cost (US\$/W)	0.63	0.57	0.58	0.55	0.364	0.341	0.295	0.262	0.247	0.229	
Wafer processing cost (US\$/W)					0.230	0.210	0.170	0.150	0.135	0.125	

Source: Citi Research and Company

Unlike previous quarters, the production cost figures were not disclosed. During the conference call, management did admit that the cost is likely higher than in 1H12, given production was at lowered utilization rates, which may remain in 4Q12 amid challenging market conditions.

While GCL-Poly reduced poly and wafer production by 41% and 10% QoQ, respectively, poly and wafer sales fell 89% and 42% QoQ, which is likely to result in higher inventory and put further stress on the company's cash flows. In 4Q12, poly prices dropped further with quotes near US\$15/kg, compared to 3Q12 ASP of US\$20.1/kg, and this may also result in pressure in inventory write-downs.

Solar Farm Disposals

After GCL-Poly completed a 92MW solar farm sale in July, the 48MW solar farm disposal in California announced on 15 November was sold for US\$50.5mn (HK\$391mn). Construction cost was not disclosed, and the solar farm disposal should provide capital for new project developments.

We believe the company will increasingly focus on new solar farm developments, especially in China, which may be a more profitable business segment at this time. Recent announcements by China to approve solar projects, with specific subsidies for rural usage, are likely positive to near-term demand. During the call, management also highlighted that GCL-Poly is confident in getting 2GW of projects from the government as a leader in the solar industry.

However, China's solar installation market may get more competitive, as supply-chain manufacturers are looking at project developments as an exit for rising inventories, and may push down construction margins.

Figure 2. China 12th Five Year Plan (12-5) on Solar Power Capacity

Category	Capacity 2010	Target Capacity 2015	Target Capacity 2020	Key focus in 2011-2015
Solar Power Station	45	1100	2300	
Photovoltaic (PV) Power Station	45	1000	2000	In Qinghai, Gansu, Xinjiang, Inner Mongolia, Tibet, Ningxia, Shaanxi, Yunnan, as well as regions suitable for building photovoltaic plants in North and Northeast part of China. The construction will be tied to large-scale hydro power and wind power developments
Solar Thermal Power Station	0	100	300	Build pilot solar-thermal power generation plants with plenty of solar, land and water resources
Distributed Power Production System	41	1000	2700	Build distributed on-grid photovoltaic system in industry park, economic development zone, and roof of large public buildings in the East China and Central China. Build independent PV systems in remote regions facing the electricity shortage in Tibet, Qinghai, Gansu, Shaanxi, Xinjiang, Yunnan and Sichuan as well as on outlying islands. Expanding electricity supply from PV system to city lighting and traffic signals systems
Total Power Generation	86	2100	5000	

Source: China National Energy Administration, Citi Research

Silane Development Offers Potential Growth

GCL-Poly recently announced trial success in its silane gas production, which is one of the components in FBR (fluidized bed) polysilicon production, an alternative to TCS Siemens technology.

Compared to GCL-Poly's existing manufacturing process, Silane FBR has a different operating metric from TCS-based polysilicon production, with higher equipment costs but lower energy usage, and can result in lower cash costs; for commercial production; only MEMC and REC have had success thus far. The investment into the Silane FBR process will allow GCL-Poly to become one of the few polysilicon producers with expertise in production using the low-energy consumption method. However, with the majority of GCL-Poly's capacity using the TCS Siemens-based process, we see limited near-term cost improvements.

During the call, management also highlighted the potential to sell Silane gas directly to the market, which may offer an additional revenue stream to support its struggling solar business. Based on industry data, Silane remain the largest specialty gas in China, and is mainly (80%) dominated by foreign players. If GCL-Poly is successful to ramp its Silane gas plant, utilizing a lower Silane cost would help reduce both the energy usage and the depreciation expenses related to polysilicon production.

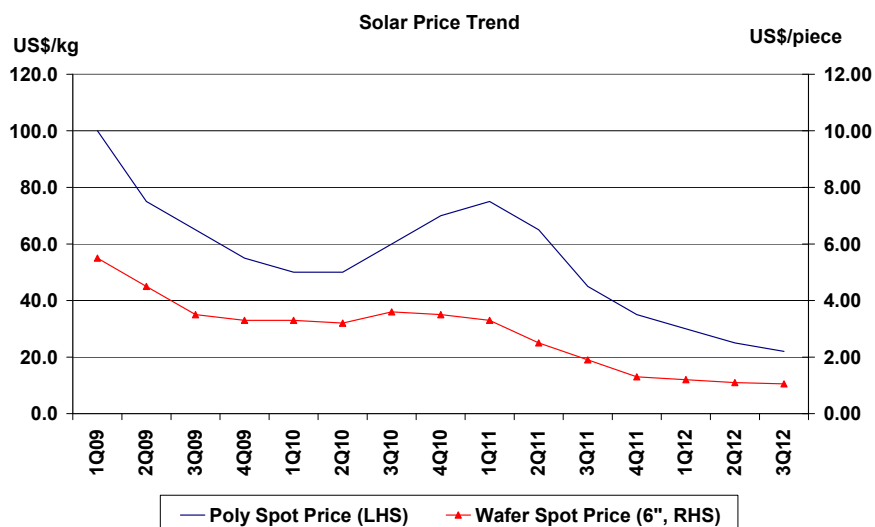
Potential Catalysts

While we believe the weak poly and wafer prices will result in further losses in 2H12, we do see a few potential upside risks that may affect GCL-Poly's share price performance:

- Faster implementation of poly import-tariff in China, where GCL-Poly's management expects it before Chinese New Year, may lead to a reduction in poly imports and could drive a rebound in China's poly/wafer prices.
- More aggressive production reduction by key poly makers, which may result in more stable demand/supply picture on the upstream market, as a number of tier-2 players have already shut down their production capacities.

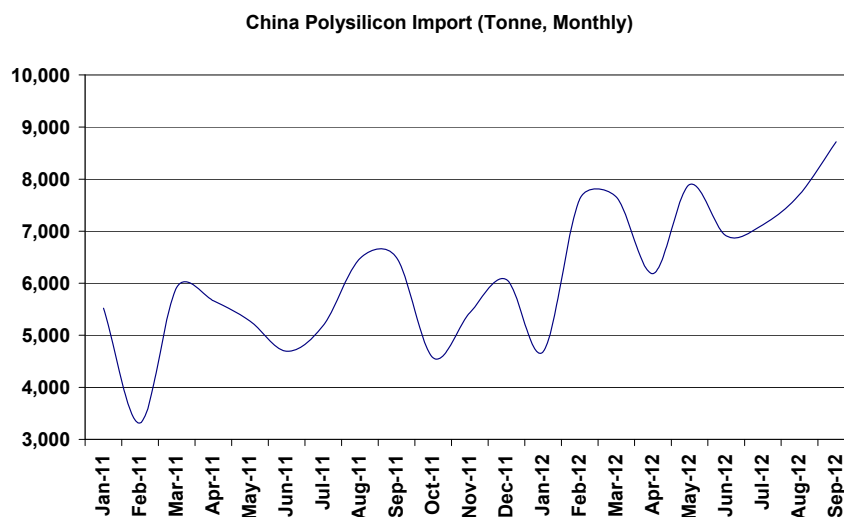
- Significant improvements in order visibility from new markets for solar products, including Japan, US, and China, in view of weak European demand. In recent weeks, we have seen a significant order pickup in Taiwan, mainly to fulfill year-end demand. If sustained, this may significantly alter the demand outlook in 2013.

Figure 3. Polysilicon Price Trend (1Q09-3Q12)



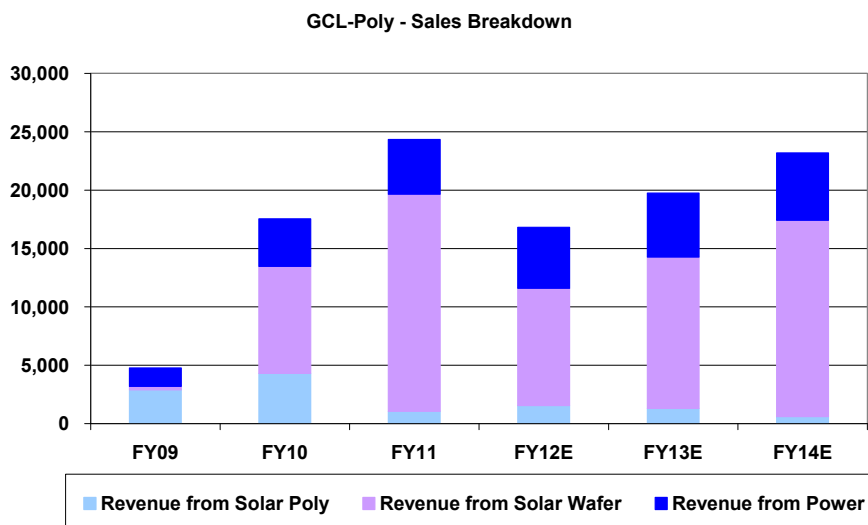
Source: Citi Research

Figure 4. China's Polysilicon Import Rose Sharply in 3Q12



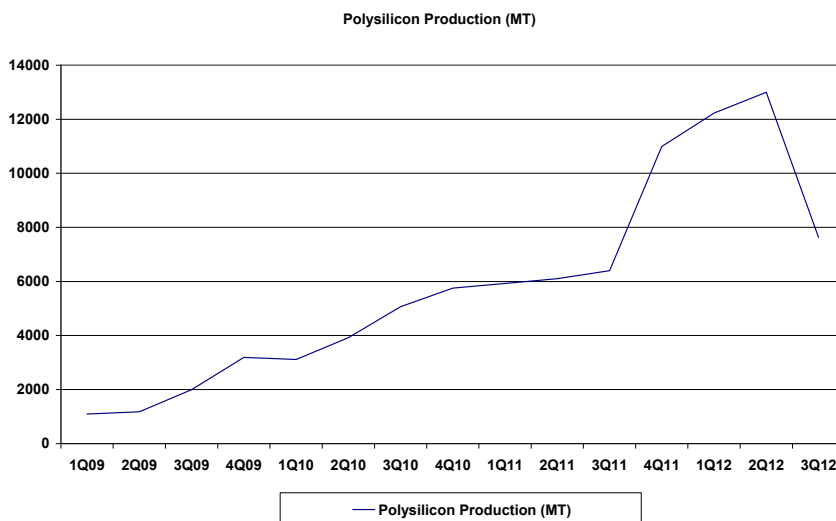
Source: Citi Research, Wind

Figure 5. GCL-Poly – Sales Breakdown



Source: Company, Citi Research Estimates

Figure 6. GCL-Poly – Sharp Drop in Poly Production in 3Q12



Source: Citi Research

We cut our production targets for 2012-14E sharply to reflect lowered utilization rates and as the company undergoes further production changes in the coming two years to meet the challenging pricing environment. For 2013-14E, we do factor in more robust polysilicon prices on the assumption of actions taken in China to protect domestic producers.

Figure 7. GCL-Poly – Key Statistics (Solar)

Y/e Dec. HKD m	2008	2009	2010	2011	2012E	2013E	2014E
Polysilicon capacity	5,000	13,000	21,000	46,000	65,000	65,000	75,000
Polysilicon utilization rate	35%	44%	85%	64%	60%	75%	87%
Polysilicon production	1,750	5,733	17,853	29,414	39,000	49,000	65,000
Polysilicon price (US\$/kg)	202	65	55	47.7	19.0	17.0	15.0
Polysilicon for internal use	0%	1%	44%	90%	73%	80%	92%
Wafer capacity	100	500	3,500	6,500	8,000	11,500	13,200
Wafer utilization rate	0%	10%	40%	69%	72%	67%	82%
Wafer production	-	49	1,400	4,488	5,761	7,704	10,846
Wafer blended ASP (US\$/Watt)	1.15	0.78	0.85	0.53	0.26	0.22	0.20

Source: Company, Citi Research Estimates

Figure 8. GCL-Poly – Earnings Revision

	Net Profit (HKD Mils)		Diluted EPS (HKD)		% change
	Old	New	Old	New	
2012E	(720)	(1,437)	(0.05)	(0.09)	-99.4%
2013E	1,344	91	0.09	0.01	-93.2%
2014E	1,456	1,452	0.09	0.09	-0.3%

Source: Citi Research estimates

Figure 9. GCL-Poly – Income Statement (HKD Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Turnover	4,944	18,472	25,506	18,101	21,252	24,953
Revenue from Solar Poly	2,880	4,293	1,044	1,553	1,293	605.28
Revenue from Solar Wafer	298	9,182	18,606	10,034	12,960	16,811
Revenue from Power	1,590	4,070	4,694	5,236	5,497	5,772
Cost of sales	(3,453)	(11,661)	(17,039)	(16,794)	(18,548)	(20,778)
Gross profit	1,491	6,811	8,466	1,307	2,704	4,175
Other income	219	575	613	619	248	310
Distribution costs	(7)	(46)	(57)	(42)	(50)	(59)
Administration expenses	(408)	(996)	(1,617)	(1,629)	(1,488)	(1,622)
Other operating expenses	(159)	(187)	(321)	(210)	(246)	(314)
EBIT	1,135	6,156	7,085	45	1,167	2,490
Gain / (Loss) on disposal	0	0	0	572	943	989
Share of results of associates	10	11	15	70	172	312
Finance costs	(349)	(606)	(1,166)	(2,080)	(1,863)	(1,746)
Change in fair value of derivatives	(853)	(13)	(82)	0	0	0
Profit before taxation	(57)	5,547	5,839	(1,343)	469	2,105
Taxation	(93)	(1,159)	(1,269)	(44)	(79)	(352)
Minority interest	(50)	(364)	(295)	(50)	(300)	(300)
Net profit	(200)	4,024	4,275	(1,437)	91	1,452
Turnover YoY (%)	40.4%	273.7%	38.1%	NA	17.4%	17.4%
Net Profit YoY (%)			6.2%	NA	NA	1500.1%

Source: Company, Citi Research Estimates

Figure 10. GCL-Poly – Balance Sheet (HKD Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Bank balances and cash	5,311	6,505	6,883	8,224	5,221	2,070
Pledged bank deposits	804	1,961	4,050	3,928	3,928	3,928
Account receivables	1,569	2,599	7,040	7,972	9,360	10,990
Inventories	727	1,647	3,627	4,495	5,066	5,675
Others	115	161	416	366	384	392
Current assets	8,526	12,873	22,015	24,985	23,959	23,055
PP&E	15,574	23,662	41,181	44,754	44,813	43,728
Others	2,038	3,936	4,225	2,535	2,915	3,644
Intangible asset	41	110	66	70	72	75
Non-current assets	17,652	27,708	45,473	47,359	47,800	47,447
Total assets	26,179	40,581	67,488	72,343	71,760	70,502
Account payables	2,396	4,384	7,628	8,208	9,066	10,156
Borrowings - amt due within one year	5,033	6,411	11,582	16,382	15,563	14,007
Amount due to related companies	139	126	630	630	630	630
Derivative financial instruments	27	568	80	96	115	139
Others	519	1,141	1,531	1,608	1,688	1,773
Current liabilities	8,114	12,630	21,452	26,925	27,063	26,704
Borrowings - amt due after one year	5,846	10,572	23,902	25,394	26,958	26,650
Non-current liabilities	5,846	10,572	23,902	25,394	26,958	26,650
Total liabilities	13,961	23,202	45,354	52,319	54,021	53,354
Shareholders equity	11,615	16,152	20,567	18,408	15,822	14,931
Minority interest	603	1,227	1,567	1,617	1,917	2,217
Total Equity	12,218	17,379	22,134	20,025	17,739	17,148

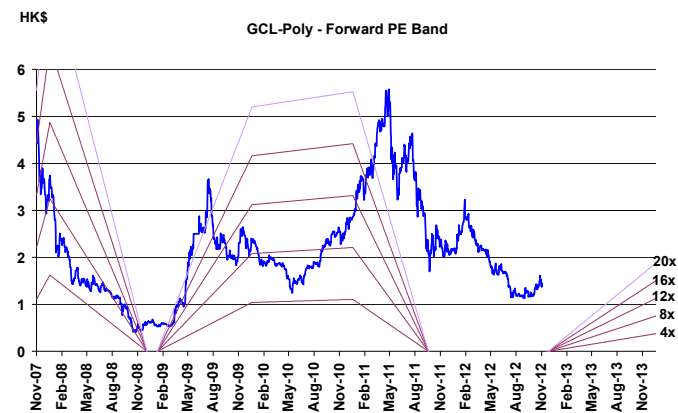
Source: Company, Citi Research Estimates

Figure 11. GCL-Poly – Cash Flow Statement (HKD Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Profit before tax	(57)	5,547	5,839	(1,343)	469	2,105
Adjustments	274	606	1,166	2,080	1,863	1,746
EBIT	217	6,154	7,005	737	2,333	3,851
Exceptional items	873	0	0	0	0	0
Depreciation & amortisation	253	1,244	2,125	1,652	1,985	2,030
Change in working capital	(862)	1,595	(5,098)	3,631	(1,939)	(2,713)
Operating cash flow	481	8,993	4,033	6,020	2,379	3,168
Interest paid	(206)	(606)	(1,166)	(2,080)	(1,863)	(1,746)
Net operating cash flow	352	7,834	2,764	5,976	2,300	2,815
Capex	(1,558)	(7,208)	(18,245)	(5,225)	(2,045)	(945)
Free cash flow	(1,206)	626	(15,481)	751	256	1,871
Change in equity	8,145	0	(0)	0	0	0
Change in debt	(2,346)	6,104	18,502	6,292	745	(1,864)
Other investing activities	0	0	0	0	0	0
Investing cash flow	(1,558)	(10,778)	(18,245)	(5,225)	(2,045)	(945)
Dividends paid	(26)	(797)	(855)	0	(18)	(290)
Financing cash flow	4,771	4,137	15,859	590	(3,259)	(5,022)
Beginning cash	1,746	5,311	6,505	6,883	8,224	5,221
Change in cash	3,565	1,194	378	1,341	(3,003)	(3,151)
Ending cash	5,311	6,505	6,883	8,224	5,221	2,070

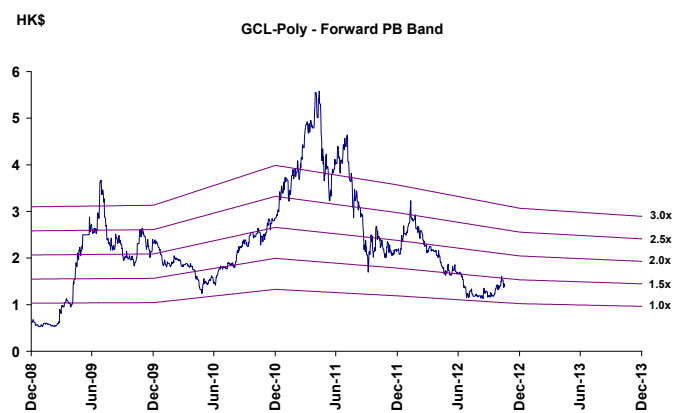
Source: Company, Citi Research Estimates

Figure 12. GCL-Poly – Forward PE Band



Source: Citi Research

Figure 13. GCL-Poly – Forward PB Band



Source: Citi Research

GCL-Poly Energy Holdings

Company description

GCL-Poly Energy Holdings is China's largest polysilicon producer and one of the world's leading wafer suppliers. The group has also started constructing wafer production facilities and targets to achieve 6.5GW of capacity. In addition, the group owns and invests in a total of 18 cogeneration power plants, one incineration power plant, one wind power plant and a 20MW solar farm in Xuzhou, Jiangsu province, which is now the largest solar farm in China.

Investment strategy

We rate GCL-Poly shares Sell / High Risk (3H) with a target price of HK\$1.00. We see the company facing a transition in 2012, as it shifts focus to develop solar farm projects. We are concerned that weak pricing for polysilicon and wafer may drag the company into losses in 2012E, despite the company's cost leadership in the sector. We believe the company will depend more on solar farm investment for earnings growth in 2012-13E, and we see more uncertainties given dependence on government programs, especially in newer markets such as the US and China. We believe its vertical alliance business model with solar cell/module makers in China will be supportive for its longer-term capacity expansion, and its leading cost structure will allow it to remain competitive amid sector oversupply and pricing concerns.

Valuation

Our HK\$1.00 target price equates 1.0x 2013E book and 11X 2014E earnings. As the company expands capacity and signs additional long-term contracts with customers, we believe a PB to ROE metric takes into account its growth metrics and equity return. Our target price is based on 1x PB to ROE and DCF. Compared with its longer-term ROE of 7%-10%, 1.0x book is reasonable given its cost leadership. On a DCF analysis, our valuation metric also equates to HK\$1.00, assuming further poly-silicon price reduction beyond 2012 and capex from solar farm investments. Our model takes into account further investment in polysilicon and wafer production capacity by 2013, given expected solar industry growth and its ambition to be one of the largest upstream suppliers. We use a WACC of 8.68% and assume target gearing of 65%. We have a relatively high equity beta of 1.80 given uncertainty to pricing in the solar sector.

Risks

We rate GCL-Poly High Risk given that the stock is deemed to be relatively volatile by our quantitative risk-rating model (based on stock price movements in the past year). We believe High Risk is also applicable given uncertainty over the company's new wafer capacity, which may take longer to improve the yield rate, while its low-cost polysilicon production should help it secure orders into 2012E. Upside risks that could help the stock trade above our target price are better pricing on polysilicon and wafer products, faster ramp-up of its production capacity, and faster revenue generation from its solar farm business.

Guangdong Investment (0270.HK)

Business on Track

Buy	1
Price (28 Nov 12)	HK\$6.20
Target price	HK\$7.00
from HK\$6.50	
Expected share price return	12.9%
Expected dividend yield	3.1%
Expected total return	16.0%
Market Cap	HK\$38,652M
	US\$4,987M

Price Performance (RIC: 0270.HK, BB: 270 HK)



China Water Sector

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■ **Summary** – We reiterate our Buy rating on Guangdong Investment (GDI), with a sum-of-the-parts target price of HK\$7.0 (+8% for rollover) as (i) recent business situation is on track; (ii) its earnings quality is high with over 70% from steady utility and infrastructure businesses; and (iii) positive catalysts from earnings uplifts from water distribution & property investments. Valuations are inexpensive at 11.7x FY13E PER and 3.2% yield. Net debt/equity ratio was 8% by end 1H12 with room for expansion.

■ **Business situation on track** – GDI guided its 3Q12 performance in line with that in 1H12. Its net profit was +15% yoy to HK\$1,871m in 1H12, contributing 67.9% (-0.4ppts yoy) from water distribution, 17.4% (+0.6ppts yoy) from property investment & development, 7.1% (-2.6ppts yoy) from department store operation, 3.0% (+0.7ppts yoy) from power generation, 2.8% (+1.1ppts yoy) from hotel business, and 2.2% (-0.2ppts yoy) from toll roads & bridges. Earnings quality was high with over 70% profit from steady utility and infrastructure businesses.

■ **Positive catalysts from water distribution and property investments** – GDI plans to focus on these two businesses ahead. Profits of water supply to HK is guaranteed by a 3-year contract with the HK government to have revenue growth of an average 5.8% pa in 2012-14E. Profit of property investments would gain from organic growth in FY12-13E and four new projects to be completed in FY14-17E.

■ **Attractive valuations** — GDI trades at 11.7x FY13E PER, which is 6% below its historical average since 2001 when the water business was injected into the company. Its FY12-13E PERs are similar to those of the HK-listed HK/China water names, without rewarding its high daily turnover of US\$5.0m on average in the past three months. Its 3.2% yield in FY13E, 1.0ppt above its historical average of 2.2%, seems attractive in view of the prevailing low interest rates environment as well as lower than most other HK utilities.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(HK\$M)	(HK\$)	(%)	(x)	(x)	(%)	(%)
2010A	2,420	0.39	18.2	16.0	2.0	13.4	2.4
2011A	2,994	0.48	23.6	12.9	1.8	14.7	2.9
2012E	3,526	0.56	17.8	11.0	1.7	15.7	3.1
2013E	3,323	0.53	-5.7	11.7	1.5	13.7	3.2
2014E	3,576	0.57	7.6	10.8	1.4	13.5	3.4

Source: Powered by dataCentral

0270.HK: Fiscal year end 31-Dec						Price: HK\$6.20; TP: HK\$7.00; Market Cap: HK\$38,652m; Recomm: Buy					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	6,352	7,161	7,832	8,264	8,743	PE (x)	16.0	12.9	11.0	11.7	10.8
Cost of sales	-2,159	-2,534	-2,873	-2,967	-3,088	PB (x)	2.0	1.8	1.7	1.5	1.4
Gross profit	4,192	4,628	4,958	5,297	5,655	EV/EBITDA (x)	9.2	8.6	7.5	6.6	6.0
Gross Margin (%)	66.0	64.6	63.3	64.1	64.7	FCF yield (%)	7.3	9.2	8.0	7.2	7.1
EBITDA (Adj)	4,407	4,568	4,975	5,270	5,585	Dividend yield (%)	2.4	2.9	3.1	3.2	3.4
EBITDA Margin (Adj) (%)	69.4	63.8	63.5	63.8	63.9	Payout ratio (%)	39	38	34	38	37
Depreciation	-165	-197	-197	-189	-182	ROE (%)	13.4	14.7	15.7	13.7	13.5
Amortisation	-815	-818	-818	-818	-818	Cashflow (HK\$m)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	3,808	4,330	4,906	4,736	4,975	EBITDA	4,795	5,469	5,921	5,743	5,975
EBIT Margin (Adj) (%)	60.0	60.5	62.6	57.3	56.9	Working capital	207	299	2,684	-505	-500
Net interest	-127	-84	138	-34	7	Other	-540	-973	-1,505	-1,109	-977
Associates	223	191	163	205	299	Operating cashflow	4,462	4,794	7,100	4,130	4,498
Non-op/Except	-151	-139	0	0	0	Capex	-1,638	-1,224	-4,012	-1,350	-1,758
Pre-tax profit	3,760	4,422	5,207	4,908	5,281	Net acq/disposals	61	-2,638	32	65	114
Tax	-942	-949	-1,118	-1,054	-1,134	Other	-349	1,255	0	0	0
Extraord./Min.Int./Pref.div.	-398	-478	-563	-531	-571	Investing cashflow	-1,926	-2,608	-3,980	-1,285	-1,645
Reported net profit	2,420	2,994	3,526	3,323	3,576	Dividends paid	-771	-1,151	-1,870	-1,247	-1,309
Net Margin (%)	38.1	41.8	45.0	40.2	40.9	Financing cashflow	-3,017	-1,341	-2,722	-1,942	-1,876
Core NPAT	2,420	2,994	3,526	3,323	3,576	Net change in cash	-380	957	398	903	978
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	2,823	3,570	3,088	2,780	2,740
Reported EPS (\$)	0.39	0.48	0.56	0.53	0.57	Segmental Sales (HK\$m)	2010	2011	2012E	2013E	2014E
Core EPS (\$)	0.39	0.48	0.56	0.53	0.57	Water distribution	4,067	4,493	4,855	5,112	5,383
DPS (\$)	0.15	0.18	0.19	0.20	0.21	Property investment and	859.5	939.5	1,000.3	1,054.6	1,111.7
CFPS (\$)	0.71	0.77	1.14	0.66	0.72	Department stores	543.8	649.3	760.7	887.9	1,014.5
FCFPS (\$)	0.45	0.57	0.49	0.44	0.44	Hotel	377.4	518.8	642.9	665.4	690.2
BVPS (\$)	3.07	3.47	3.73	4.06	4.43	Power generation	467.2	524.5	544.0	544.0	544.0
Wtd avg ord shares (m)	6,217	6,232	6,233	6,233	6,233	Toll roads and bridges	36.6	35.8	29.2	0.0	0.0
Wtd avg diluted shares (m)	6,246	6,253	6,253	6,253	6,253	Sales - total segments	6,352	7,161	7,832	8,264	8,743
Growth rates	2010	2011	2012E	2013E	2014E	Segmental EBIT (HK\$m)	2010	2011	2012E	2013E	2014E
Sales revenue (%)	7.4	12.7	9.4	5.5	5.8	Water distribution	2,381	2,586	2,773	3,000	3,240
EBIT (Adj) (%)	20.3	13.7	13.3	-3.5	5.0	Property investment and	1,053	1,481	1,688	1,256	1,204
Core NPAT (%)	18.4	23.7	17.8	-5.7	7.6	Department stores	234.9	257.8	245.7	280.6	315.4
Core EPS (%)	18.2	23.6	17.8	-5.7	7.6	Hotel	73.2	89.8	125.9	148.0	157.9
Balance Sheet (HK\$m)	2010	2011	2012E	2013E	2014E	Power generation	82.2	49.4	77.0	77.0	83.0
Cash & cash equiv.	3,841	3,543	3,941	4,844	5,822	Toll roads and bridges	21.1	-108.8	21.7	0.0	0.0
Accounts receivables	596	2,942	456	433	412	Others	-36.9	-25.3	-25.3	-25.3	-25.3
Inventory	60	61	72	84	96	EBIT - total segments	3,808	4,330	4,906	4,736	4,975
Net fixed & other tangibles	9,223	10,526	11,682	12,546	13,795						
Goodwill & intangibles	16,129	15,200	14,381	13,563	12,745						
Financial & other assets	2,072	2,560	5,710	6,351	6,924						
Total assets	31,920	34,832	36,241	37,821	39,794						
Accounts payable	1,957	2,545	2,479	2,186	1,929						
Short-term debt	720	2,484	577	470	383						
Long-term debt	3,222	1,346	2,546	2,075	1,692						
Provisions & other liab	4,113	4,008	3,972	3,814	3,676						
Total liabilities	10,012	10,383	9,574	8,546	7,681						
Shareholders' equity	19,117	21,599	23,255	25,331	27,599						
Minority interests	2,792	2,849	3,413	3,943	4,515						
Total equity	21,908	24,448	26,667	29,275	32,113						
Net debt	102	288	-818	-2,299	-3,747						
Net debt to equity (%)	0.5	1.2	-3.1	-7.9	-11.7						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Valuations vs. History & Peers

Figure 1. GDI — Sum-of-the-Parts Target Price

SOTP	Value (HK\$m)	Per share (HK\$)	By Business	Method
Water - Dongshen	28,583	4.59	66%	DCF @ 7.1% WACC
Water - Nansha	356	0.06	1%	DCF @ 8.1% WACC
Property Investment	9,534	1.53	22%	40% discount NAV
Department stores	1,679	0.27	4%	10x PER
Hotel	1,577	0.25	4%	40% discount NAV
Power generation	1,498	0.24	3%	DCF @ 8.1% WACC
Toll roads	1,222	0.20	3%	DCF @ 8.1% WACC
Net cash/ (debt)	-818	-0.13	-2%	
Total	43,631	7.00	100%	

Source: Company Reports and Citi Research Estimates

Figure 2. Valuation Comparison – GDI vs. HK Utilities (Price as at 28 Nov 2012)

	RIC	Rating	Price		PE		P/B		Net D/E		Yield	
			Current	Target	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E
HKCG	0003.HK	Sell	HK\$20.90	HK\$17.00	23.7	25.6	4.0	3.7	30.7%	26.8%	1.7%	2.0%
CLP	0002.HK	Neutral	HK\$67.10	HK\$70.00	16.3	13.1	1.9	1.8	81.2%	76.9%	3.8%	4.0%
Power Assets	0006.HK	Buy	HK\$67.85	HK\$77.00	14.7	13.7	2.3	2.2	30.2%	26.4%	3.7%	4.0%
CKI	1038.HK	Buy	HK\$46.30	HK\$54.50	11.8	10.7	1.8	1.7	14.3%	-1.2%	3.7%	3.9%
GDI	0270.HK	Buy	HK\$6.20	HK\$7.00	11.0	11.7	1.7	1.5	1.3%	-9.1%	3.1%	3.2%
Average					15.5	14.9	2.3	2.2	31.5%	24.0%	3.2%	3.4%

Source: Citi Research Estimates

Figure 3. Water Companies – Valuation Comparison (Price as at 28 Nov 2012)

Company Name	RIC	Rating	Price	Price	Mkt-Cap US\$M	ROE		PE		P/B		Yield	
			Currency	Current		FY12E	FY13E	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E
HK listed													
China Everbright	0257.HK	Buy	HKD	3.70	1,928	15.8%	14.6%	13.2	13.2	2.0	1.9	1.7%	1.7%
GDI	0270.HK	Buy	HKD	6.20	4,987	15.7%	13.7%	11.0	11.7	1.7	1.5	3.1%	3.2%
BJ Water Group	0371.HK	NR	HKD	1.81	1,614	9.4%	10.9%	15.2	12.5	1.4	1.3	2.0%	2.4%
China Water	0855.HK	NR	HKD	1.99	373	10.3%	6.9%	9.5	8.9	0.9	0.7	2.5%	2.7%
Average						12.8%	11.5%	12.2	11.5	1.5	1.3	2.3%	2.5%
PRC listed													
Xingrong Invest	000598.SZ	NR	CNY	7.02	1,300	16.4%	17.3%	12.3	10.7	1.9	1.9	2.7%	3.3%
Sound Environment	000826.SZ	NR	CNY	20.44	1,635	22.0%	23.5%	23.8	17.8	5.0	4.0	0.7%	0.9%
BJ Water Doctor	300055.SZ	NR	CNY	15.87	583	6.1%	7.9%	30.9	23.3	2.1	1.9	0.8%	0.9%
Originwater	300070.SZ	NR	CNY	32.71	2,888	15.1%	17.9%	32.2	22.6	4.7	4.0	0.2%	0.3%
CEEP	300172.SZ	NR	CNY	11.33	237	6.6%	7.1%	24.2	19.4	1.8	1.7	n/a	n/a
Safbon Water	300262.SZ	NR	CNY	14.74	316	7.0%	9.0%	56.7	42.1	4.3	3.9	0.2%	0.2%
Beijing Capital	600008.SS	NR	CNY	3.95	1,395	10.1%	11.7%	15.5	13.6	1.5	1.4	1.8%	1.0%
Wuhan Sanzhen	600168.SS	NR	CNY	4.97	352	8.9%	8.1%	15.1	15.1	1.3	1.2	n/a	n/a
H.IWTC	600187.SS	NR	CNY	6.54	449	11.1%	11.5%	28.9	25.2	2.3	2.9	n/a	n/a
Nanhai Dev	600323.SS	NR	CNY	5.94	553	9.7%	10.6%	17.2	14.0	1.4	1.5	1.9%	2.1%
Hongcheng Ww	600461.SS	NR	CNY	6.37	338	7.9%	8.1%	15.7	14.2	1.2	1.1	n/a	n/a
CQ Water	601158.SS	NR	CNY	4.93	3,800	13.8%	14.2%	13.5	12.0	1.9	1.7	2.2%	n/a
Jiangnan Water	601199.SS	NR	CNY	11.13	418	7.7%	7.9%	19.5	17.4	1.5	1.4	n/a	n/a
Average						10.9%	11.9%	23.5	19.0	2.4	2.2	1.3%	1.2%
Overall Average						11.4%	11.8%	20.8	17.3	2.2	2.0	1.6%	1.7%

Source: Citi Research Estimates & IBES for non rated (NR) companies

Financial Statements & Key Assumptions

Figure 4. GDI — Income Statement in Segment Format (HK\$m)

Year to 31 Dec	2010	2011	2012E	2013E	2014E
Revenue					
Water distribution	4,067	4,493	4,855	5,112	5,383
Property investment and development	860	940	1,000	1,055	1,112
Department stores	544	649	761	888	1,015
Hotel	377	519	643	665	690
Power generation	467	525	544	544	544
Toll roads and bridges	37	36	29	0	0
Total	6,352	7,161	7,832	8,264	8,743
Depreciation	165	197	197	189	182
Amortization	815	818	818	818	818
Fair value gain in investment properties	381	777	946	473	390
EBITDA	4,407	4,568	4,975	5,270	5,585
EBITDA margin	69.4%	63.8%	63.5%	63.8%	63.9%
EBIT					
Water distribution	2,381	2,586	2,773	3,000	3,240
Property investment and development	1,053	1,481	1,688	1,256	1,204
Department stores	235	258	246	281	315
Hotel	73	90	126	148	158
Power generation	82	49	77	77	83
Toll roads and bridges	21	-109	22	0	0
Others	-37	-25	-25	-25	-25
Total	3,808	4,330	4,906	4,736	4,975
EBIT margin	60.0%	60.5%	62.6%	57.3%	56.9%
yoy growth	20.3%	13.7%	13.3%	-3.5%	5.0%
Interest income	46	78	283	84	104
Other unallocated items	-144	-15	0	0	0
Finance cost	-173	-162	-145	-118	-96
Share of profit of a jointly-controlled entity	90	83	91	101	105
Share of profit of associates	134	108	72	104	194
Profit before tax	3,760	4,422	5,207	4,908	5,281
Income tax expense	-942	-949	-1,118	-1,054	-1,134
Profit after tax	2,818	3,472	4,089	3,854	4,147
Minority interests	-398	-478	-563	-531	-571
Net Profit	2,420	2,994	3,526	3,323	3,576
yoy growth	18.4%	23.7%	17.8%	-5.7%	7.6%
Non core profit	381	777	946	473	390
Non core profit - MI	-91	-186	-226	-113	-93
Non core profit - GDI	290	591	720	360	297
Core profit attributable to GDI's shareholders	2,130	2,403	2,806	2,963	3,279
yoy growth	12.7%	12.8%	16.8%	5.6%	10.7%

Source: Company Reports and Citi Research Estimates

Figure 5. GDI — Citi Net Profit vs. Consensus

Net profit (HK\$m)	Citi	Consensus	% Diff
FY12E	3,526	2,932	+20%
FY13E	3,323	3,142	+6%
FY14E	3,576	3,327	+7%

Source: Bloomberg and Citi Research Estimates

Figure 6. GDI — Income Statement in Ordinary Format (HK\$m)

Year to 31 Dec	2010	2011	2012E	2013E	2014E
Total Revenue	6,352	7,161	7,832	8,264	8,743
Sale of water and electricity	4,534	5,018	5,399	5,656	5,927
Sale of goods	56	62	73	85	97
Commissions from concessionaire sales	488	587	688	803	918
Hotel and rental income	1,237	1,458	1,643	1,720	1,802
Toll revenue	37	36	29	0	0
Cost of Sales	-2,159	-2,534	-2,873	-2,967	-3,088
Cost of inventories sold	-414	-481	-531	-552	-574
Amortization of intangible assets	-815	-818	-818	-818	-818
Others	-931	-1,235	-1,524	-1,597	-1,695
Gross Profit	4,192	4,628	4,958	5,297	5,655
GP margin	66.0%	64.6%	63.3%	64.1%	64.7%
Other income and gains/ (losses), net	-35	132	352	153	173
Interest income	46	78	283	84	104
Others	62	69	69	69	69
Changes in fair value of derivative	-144	-15	0	0	0
Changes in fair value of investment properties	381	777	946	473	390
Selling and distribution costs	-107	-150	-164	-173	-183
Administrative expenses	-695	-841	-875	-901	-928
Other operating expenses, net	-27	-153	-29	-29	-29
Impairment of PPE	-7	0	0	0	0
Impairment of intangible assets	0	-124	0	0	0
Others	-20	-29	-29	-29	-29
Operating profit	3,710	4,393	5,189	4,820	5,078
Finance cost	-173	-162	-145	-118	-96
Share of profit of a jointly-controlled entity	90	83	91	101	105
Share of profit of associates	134	108	72	104	194
Profit before tax	3,760	4,422	5,207	4,908	5,281
Income tax expense	-942	-949	-1,118	-1,054	-1,134
Profit after tax	2,818	3,472	4,089	3,854	4,147
Minority interests	-398	-478	-563	-531	-571
Net Profit	2,420	2,994	3,526	3,323	3,576

Source: Company Reports and Citi Research Estimates

Figure 7. GDI — Key Assumptions

As at 31 Dec	2010	2011	2012E	2013E	2014E
Profit from Dongshen Water Supply Project (HK\$m)	1,368	1,700	1,895	2,082	2,252
yoy	12.3%	24.3%	11.5%	9.8%	8.2%
<u>Yield</u>					
- Grade A office (GZ)	n/a	8.5%	8.5%	8.5%	8.5%
- Office (HK)	n/a	7.0%	8.0%	8.0%	8.0%
- Shopping mall (GZ)	n/a	7.5%	7.5%	7.5%	7.5%
<u>Same store sale growth (SSSG)</u>					
- Teemall	10.5%	14.8%	-5.0%	5.0%	5.0%
- Others	31.7%	24.3%	-5.0%	5.0%	5.0%
CPI	4%	4%	4%	3%	3%
RMB/HK\$	1.15	1.20	1.20	1.20	1.20

Source: Company Reports and Citi Research Estimates

Figure 8. GDI — Balance Sheet (HK\$m)

As at 31 Dec	2010	2011	2012E	2013E	2014E
Property, plant and equipment	3,165	3,294	3,164	3,043	2,929
Investment properties	5,934	7,107	8,397	9,388	10,754
Prepaid land lease payments	102	102	97	92	88
Goodwill	266	266	266	266	266
Interests in JCEs	859	807	708	598	485
Interests in associates	1,087	1,346	4,905	5,591	6,165
Intangible assets	15,862	14,933	14,115	13,297	12,478
Prepayment and deposit	0	343	0	0	0
Deferred tax assets	22	24	24	24	24
Total non-current assets	27,298	28,221	31,676	32,298	33,189
Properties for sale	0	0	32	97	211
Tax recoverable	3	0	0	0	0
Inventories	60	61	72	84	96
Receivables, prepayments and deposit	596	2,942	456	433	412
Derivative financial instruments	123	64	64	64	64
Cash and cash equivalents	3,841	3,543	3,941	4,844	5,822
Total current assets	4,622	6,610	4,565	5,523	6,605
Payables and accruals	1,957	2,545	2,479	2,186	1,929
Tax payable	323	400	400	400	400
Derivative financial instruments	425	265	265	265	265
Due to non-controlling shareholders of subsidiaries	402	318	318	318	318
Bank loans	720	2,484	577	470	383
Total current liabilities	3,827	6,012	4,039	3,639	3,296
Derivative financial instruments	95	0	0	0	0
Due to non-controlling shareholders of subsidiaries	5	0	0	0	0
Bank loans	3,222	1,346	2,546	2,075	1,692
Other non-current liabilities	1,566	1,370	1,334	1,177	1,038
Deferred tax liabilities	1,297	1,655	1,655	1,655	1,655
Total non-current liabilities	6,185	4,371	5,535	4,907	4,385
Net Assets	21,908	24,448	26,667	29,275	32,113
Share capital	3,115	3,116	3,116	3,116	3,116
Reserves	15,378	17,797	20,138	22,215	24,482
Proposed final dividends	623	686	0	0	0
Shareholders equity	19,117	21,599	23,255	25,331	27,599
Minority interests	2,792	2,849	3,413	3,943	4,515
Total equity	21,908	24,448	26,667	29,275	32,113

Source: Company Reports and Citi Research Estimates

Figure 9. GDI — Cash Flow Statement (HK\$m)

As at 31 Dec	2010	2011	2012E	2013E	2014E
Operating activities					
Profit before tax	3,760	4,422	5,207	4,908	5,281
Finance cost	173	162	145	118	96
Profit of JCEs	-90	-83	-91	-101	-105
Profit of associates	-134	-108	-72	-104	-194
Interest income	-46	-78	-283	-84	-104
Fair value change of derivatives	144	15	0	0	0
Depreciation	165	197	197	189	182
Recognition of prepaid land lease payments	4	5	5	5	5
Amortization of intangible assets	815	818	818	818	818
Fair value change of investment properties	-381	-777	-946	-473	-390
Impairment of PPE	7	0	0	0	0
Impairment intangible assets	0	124	0	0	0
Disposal loss/(gain) from intangible assets	0	1	0	0	0
Equity-settled share option expense	7	3	0	0	0
Interest income from available-for-sales investment	-3	-7	0	0	0
Operating profit before working capital changes	4,423	4,694	4,980	5,275	5,589
Chg in working capital	207	299	2,684	-505	-500
Cash generated from operations	4,629	4,993	7,664	4,770	5,090
Interest received	46	78	283	84	104
Dividends from JCEs	156	172	189	211	219
Dividends from associates	63	122	82	118	220
Hong Kong profits tax paid	-16	-10	0	0	0
Mainland China tax paid	-416	-562	-1,118	-1,054	-1,134
Net operating cash flow	4,462	4,794	7,100	4,130	4,498
Investing activities					
Purchase of available-for sale investments	-357	-1,440	0	0	0
Proceeds from disposals of investments	417	1,449	0	0	0
Purchase of PPE	-352	-244	-67	-68	-68
Purchase of intangible assets	-8	-11	0	0	0
Additions to investment properties	-474	-115	-345	-517	-977
Acquisitions of non-controlling interests	-2	-634	0	0	0
Acquisitions of an associate	-600	0	-700	-700	-600
Capital injection in associates	-202	-220	-2,868	0	0
Deposits for investment and loan receivables	0	-2,649	0	0	0
Proceeds from disposals of PPE	1	2	0	0	0
Decrease/ (increase) in non-pledge time deposit	-349	1,255	0	0	0
Investing cash flow	-1,926	-2,608	-3,980	-1,285	-1,645
Financing activities					
Equity issuance	35	4	0	0	0
Bank loans	400	609	0	0	0
Repayment of bank loans	-2,408	-729	-708	-577	-470
Finance costs paid	-273	-267	-145	-118	-96
Capital contribution from non-controlling interest	0	193	0	0	0
Dividend paid to minority interests	-87	-92	0	0	0
Dividend paid to shareholders	-684	-1,059	-1,870	-1,247	-1,309
Financing cash flow	-3,017	-1,341	-2,722	-1,942	-1,876
Chg in cash & equivalents	-482	845	398	903	978
Cash b/f	2,488	2,108	3,065	3,463	4,366
Exchange	102	112	0	0	0
Non-pledged time deposit with maturity > 3mths	1,733	478	478	478	478
Cash Balance	3,841	3,543	3,941	4,844	5,822
Capex	-1,638	-1,224	-4,012	-1,350	-1,758
Free Cash Flow	2,823	3,570	3,088	2,780	2,740

Source: Company Reports and Citi Research Estimates

Figure 10. GDI — Key Ratios

	2010	2011	2012E	2013E	2014E
Valuations					
Payout ratio	39%	37%	34%	38%	37%
Profitability					
EBITDA margin	69%	64%	64%	64%	64%
EBIT margin	60%	60%	63%	57%	57%
Net profit margin	38%	42%	45%	40%	41%
Interest coverage (x)	23.2	27.9	35.0	41.9	54.8
Dividend coverage (x)	2.6	2.7	3.0	2.7	2.7
Return					
ROA	7.7%	9.0%	9.9%	9.0%	9.2%
Gearing					
Net Debt/Equity (ex.MI)	0.5%	1.3%	-3.5%	-9.1%	-13.6%
Net Debt/Asset (ex.MI)	0.3%	0.8%	-2.3%	-6.1%	-9.4%

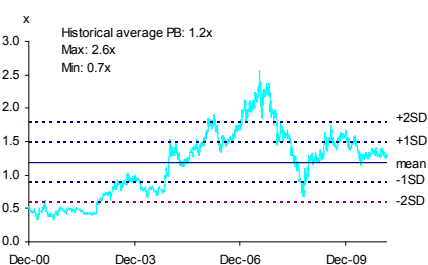
Source: Company Reports and Citi Research Estimates

Figure 11. GDI – Historical PER chart



Source: Citi Research

Figure 12. GDI – Historical PB chart



Source: Citi Research

Figure 13. GDI – Historical dividend yield chart



Source: Citi Research

Guangdong Investment

Company description

Guangdong Investment Limited (GDI) was established in Hong Kong and obtained its listing status on the Hong Kong Stock Exchange in 1973. Its immediate holding company, GDH Limited, which is ultimately controlled by Guangdong Provincial Government, currently owns 60.5% of GDI. The Group is principally engaged in (i) water supply to Hong Kong, Shenzhen and Dongguan, (ii) property investment and development, (iii) department store operations, (iv) hotel operation and management, (v) power generation business and also (vi) toll roads and bridges operations.

Investment strategy

We rate GDI as Buy (1) with a target price of HK\$7.0. Our positive investment view is based on GDI benefitting from resilient recurring earnings from its water distribution business which accounts for over 50% of group profit and generates approximately two-thirds of the group's operating cash flow. Its strong balance sheet and operating cash flow provide room for acquisitions. Potential positive catalysts are mainly acquisitions, which are expected to have low double-digit un-leveraged IRRs. We are also positive on GDI's key investment properties, Teemall and Teem Tower, amid their prime location, which provide potential upside to the group's NAV. GDI has four investment properties under construction to be completed in 2014-17E which underpin profit growth in the medium term.

Valuation

Our target price of HK\$7.0 for GDI is set equal to our sum of the parts-based NAV estimate of HK\$7.0 for FY13E. For the water distribution business, our fair value is derived by using a DCF with a WACC of 7.1% to discount our earning forecasts up to 2030 when the operating right of water supply expires. This method is suitable for the water business with much of its operations regulated and generating predictable cash flows. To value the power generation business, we use a DCF model with a 8.1% WACC, in line with China's IPPs. We also use a DCF method to value the toll road business, employing a WACC of 10.1%. For the property investment business, we use 7-8.5% cap rate with a 40% discount on NAV. For the hotel assets, a cap rate of 12-16% is employed with a 40% discount on NAV. For the department stores business, we use a 2012E P/E of 10.0x, lower than the sector average in view of keen competition in the Guangzhou market.

Risks

Downside risks to our investment thesis and target price on GDI include: (1) With property investment assets contributing ~25% of average operating profit over the next two years, any sudden slowdown in the Guangzhou and Hong Kong property markets would lead to negative rental reversions and decrease in its bottom line; (2) Exposure to the cyclical nature of the department stores/retail sector; (3) High coal price could act as a major negative catalyst to power generation business; (4) Pressure on toll road tariffs in the mainland; and (5) Failure to make value added acquisitions due to competition.

Harbin Electric (1133.HK)

Downcycle in 2013E

Buy	1
Price (28 Nov 12)	HK\$6.35
Target price	HK\$7.50
from HK\$8.30	
Expected share price return	18.1%
Expected dividend yield	2.7%
Expected total return	20.8%
Market Cap	HK\$4,290M
	US\$554M

Price Performance (RIC: 1133.HK, BB: 1133 HK)



Asian Utilities

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■ **Summary** – We lower our Harbin Electric (HBE) TP by 10% to HK\$7.50 (at 0.7x FY13E PB, from 0.8x FY12E before) in view of the postponed downcycle with revenue cut in 2012-13E (Rmb4.3bn in 3Q12), margin cut in the same period and weak new order flow (Rmb7bn in 3Q12). Our Buy rating is purely a valuation call as its 6.5x FY13E PER/0.6x PB are below historical mean by 30%/ 33% respectively.

■ **Declining revenue in 2012-13E** – We estimate HBE's revenue of Rmb4.3bn in 3Q12, equal to only 39% that in 1H12 due to low sales volume in the quarter since our company update. But we expect more sales in 4Q12E based on its production pipeline. HBE's revenue will drop yoy in 2013E on (i) weak coal-fired equipment demand and (ii) relatively small scale of its new energy power equipment business.

■ **Margin cuts in 2H12-FY13E** – HBE expects its GP margin to decline slightly in 2H12-FY13E owing to ASP cuts. PRC steel prices stabilized in Aug, with a mild rebound afterwards. Regarding nuclear equipment sales, owing to small volume and delayed new approvals after the Fukushima accident in March 2011, HBE's forecast is for breakeven only in 2014E, a year later than the previous forecast.

■ **Weak new order flow** – HBE got Rmb7bn new orders in 3Q12 based on its winning bids, equal to only 29% of its 1H12 new orders. The weak order flow was due to (i) capex cut on coal-fired plants of the PRC big five IPPs amid high gearing and more ROE-oriented investments; (ii) low success rate in winning gas turbine orders – year to date, HBE won only 2 out of its 20 bids; and (ii) lack of new nuclear projects.

■ **TP cut** – HBE's earnings downcycle in FY12-13E looks likely based on its production pipeline. Its profitability in FY14E will depend on 1H13E new orders, which are uncertain. Any recovery would be weaker for HBE than its peers owing to HBE's limited exposure to the new high-growth energy business. We lower our TP by 10% to HK\$7.50 (at 0.7x FY13E PB, vs. 0.8x FY12E before).

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(RmbM)	(Rmb)	(%)	(x)	(x)	(%)	(%)
2010A	1,001	0.727	65.1	7.0	0.7	10.9	2.7
2011A	1,229	0.892	22.7	5.7	0.7	12.1	2.7
2012E	1,210	0.879	-1.6	5.8	0.6	10.9	2.7
2013E	1,076	0.781	-11.1	6.5	0.6	8.9	2.7
2014E	1,142	0.829	6.1	6.2	0.5	8.8	2.7

Source: Powered by dataCentral

1133.HK: Fiscal year end 31-Dec						Price: HK\$6.35; TP: HK\$7.50; Market Cap: HK\$4,290m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	29,172	28,488	25,345	25,571	26,763	PE (x)	7.0	5.7	5.8	6.5	6.2
Cost of sales	-24,982	-22,777	-20,299	-20,562	-21,483	PB (x)	0.7	0.7	0.6	0.6	0.5
Gross profit	4,190	5,711	5,046	5,009	5,280	EV/EBITDA (x)	9.3	6.5	5.8	6.1	5.6
Gross Margin (%)	14.4	20.0	19.9	19.6	19.7	FCF yield (%)	-4.2	-17.5	-34.0	6.9	1.5
EBITDA (Adj)	1,690	1,940	2,092	2,019	2,179	Dividend yield (%)	2.7	2.7	2.7	2.7	2.7
EBITDA Margin (Adj) (%)	5.8	6.8	8.3	7.9	8.1	Payout ratio (%)	19	16	16	18	17
Depreciation	-492	-559	-635	-695	-755	ROE (%)	10.9	12.1	10.9	8.9	8.8
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	1,198	1,382	1,456	1,324	1,423	EBITDA	1,690	1,940	2,092	2,019	2,179
EBIT Margin (Adj) (%)	4.1	4.9	5.7	5.2	5.3	Working capital	-1,077	-1,662	-3,036	-99	-610
Net interest	138	180	80	48	56	Other	107	-509	-445	-432	-466
Associates	26	79	79	79	79	Operating cashflow	720	-231	-1,390	1,488	1,103
Non-op/Except	0	0	0	0	0	Capex	-1,012	-1,000	-1,000	-1,000	-1,000
Pre-tax profit	1,362	1,640	1,614	1,450	1,558	Net acq/disposals	0	0	0	0	0
Tax	-269	-257	-252	-239	-273	Other	2,918	261	165	133	141
Extraord./Min.Int./Pref.div.	-92	-155	-152	-135	-144	Investing cashflow	1,906	-739	-835	-867	-859
Reported net profit	1,001	1,229	1,210	1,076	1,142	Dividends paid	0	0	0	0	0
Net Margin (%)	3.4	4.3	4.8	4.2	4.3	Financing cashflow	-1,779	-2,217	-1,027	-76	-76
Core NPAT	1,001	1,229	1,210	1,076	1,142	Net change in cash	847	-3,187	-3,252	545	168
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-292	-1,231	-2,390	488	103
Reported EPS (Rmb)	0.727	0.892	0.879	0.781	0.829						
Core EPS (Rmb)	0.727	0.892	0.879	0.781	0.829						
DPS (Rmb)	0.140	0.140	0.140	0.140	0.140						
CFPS (Rmb)	0.523	-0.168	-1.009	1.081	0.801						
FCFPS (Rmb)	-0.212	-0.894	-1.736	0.355	0.075						
BVPS (Rmb)	7.046	7.700	8.439	9.080	9.769						
Wtd avg ord shares (m)	1,377	1,377	1,377	1,377	1,377						
Wtd avg diluted shares (m)	1,377	1,377	1,377	1,377	1,377						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	1.9	-2.3	-11.0	0.9	4.7						
EBIT (Adj) (%)	38.7	15.3	5.4	-9.1	7.5						
Core NPAT (%)	65.1	22.7	-1.6	-11.1	6.1						
Core EPS (%)	65.1	22.7	-1.6	-11.1	6.1						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	12,550	8,519	5,267	5,812	5,980						
Accounts receivables	16,807	18,467	20,556	20,739	21,706						
Inventory	12,825	11,616	12,021	12,177	12,722						
Net fixed & other tangibles	4,890	5,800	6,165	6,469	6,714						
Goodwill & intangibles	715	972	972	972	972						
Financial & other assets	2,969	5,022	5,301	5,390	5,556						
Total assets	50,756	50,396	50,282	51,558	53,650						
Accounts payable	13,142	16,059	17,247	17,452	18,170						
Short-term debt	12,519	8,065	6,545	6,598	6,879						
Long-term debt	9,176	7,651	6,701	6,701	6,701						
Provisions & other liab	4,757	6,097	6,097	6,097	6,097						
Total liabilities	39,592	37,873	36,589	36,848	37,847						
Shareholders' equity	9,701	10,602	11,618	12,501	13,450						
Minority interests	1,462	1,922	2,074	2,209	2,353						
Total equity	11,164	12,523	13,692	14,710	15,803						
Net debt	9,144	7,197	7,979	7,488	7,600						
Net debt to equity (%)	81.9	57.5	58.3	50.9	48.1						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

Declining Revenue in 2012-13E

Figure 1. HBE's Revenue Breakdown

Revenue	2011	1H12	yoy	Mix	3Q12	9M12	2012E	YoY
Thermal Power Equipment	17,295	6,539	-34%	59%	n/a	n/a	n/a	n/a
Hydro Power Equipment	3,163	1,735	21%	16%	n/a	n/a	n/a	n/a
Engineering services	5,105	1,499	-14%	13%	n/a	n/a	n/a	n/a
Ancillary equipment	719	392	-5%	4%	n/a	n/a	n/a	n/a
AD/DC motors and others	2,206	998	-7%	9%	n/a	n/a	n/a	n/a
Total	28,488	11,163	-24%	100%	4,337	15,500	25,345	-11%

Source: Company Reports and Citi Research Estimates

We estimate HBE's revenue of Rmb4.3bn in 3Q12, equal to only 39% that in 1H12, owing to low sales volume in the quarter since our recent company update. But we expect more sales volume in 4Q12E with reference to its production pipeline.

HBE's revenue will drop yoy in 2013E on (i) weak coal-fired power equipment demand and (ii) relatively small scale of its new energy power equipment business.

Margin Cuts in 2H12-FY13E

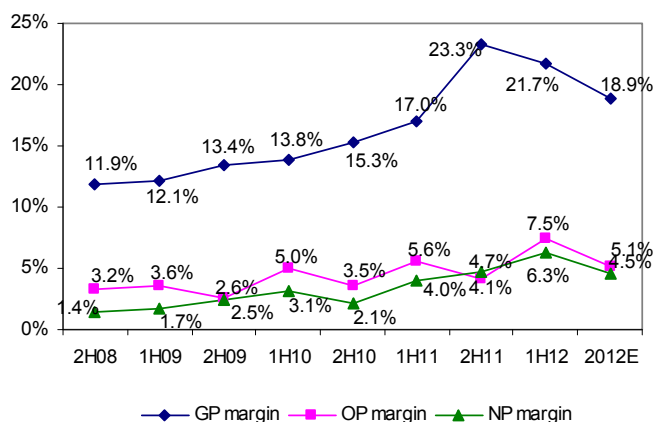
Figure 2. HBE's GP margins

GP margin	1H12	yoy	2012E	yoy	2013E	yoy
Thermal Power Equipment	21.6%	+5.1ppts	21.2%	-1.0ppt	20.7%	-0.5ppts
Hydro Power Equipment	35.6%	+6.2ppts	31.3%	+1.0ppt	30.8%	-0.5ppts
Engineering services	7.5%	-2.0ppts	6.5%	+1.0ppt	6.5%	flat
Ancillary equipment	42.7%	+18.1ppts	34.8%	-5.0ppts	34.8%	flat
AD/DC motors and others	11.3%	-2.6ppts	15.6%	flat	15.6%	flat
Company average	21.7%	+4.7ppts	19.9%	-0.1ppts	19.6%	-0.3ppts

Source: Company Reports and Citi Research Estimates

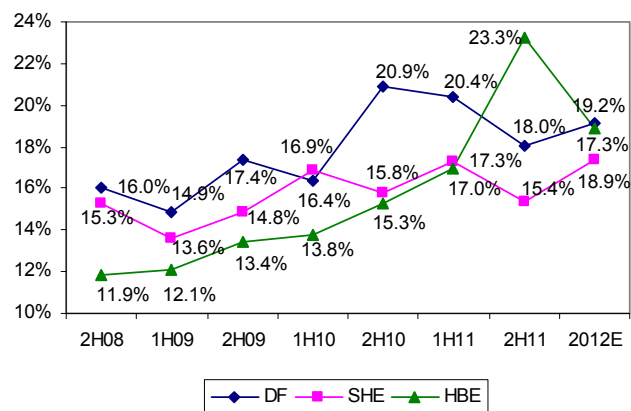
HBE expects its GP margin to decline slightly in 2H12-FY13E owing to ASP cuts. PRC steel prices stabilized in August, with a mild rebound afterwards. Regarding nuclear equipment sales, owing to the small volume and delayed new approvals following the Fukushima accident in March 2011, HBE forecasts breakeven only in 2014E, a year later than the previous forecast.

Figure 3. HBE's Margin Trends



Source: Citi Research

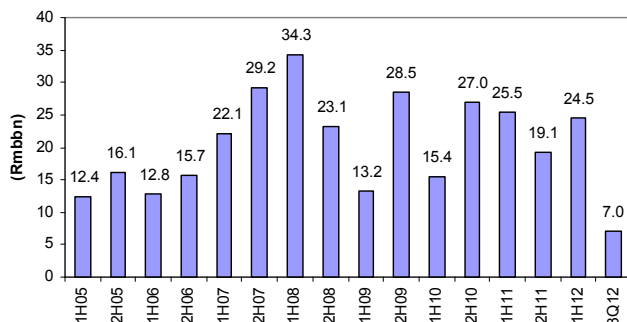
Figure 4. GP Margins of Three PRC Large Power Equipment Makers



Source: Citi Research

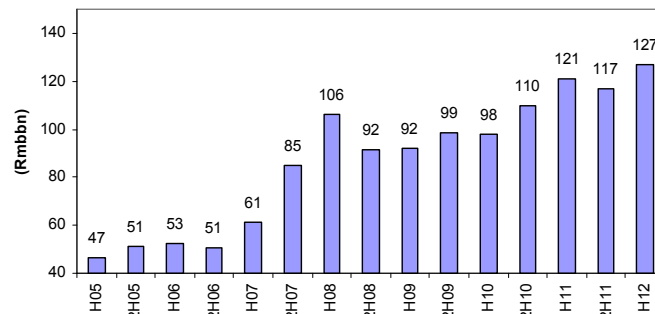
Weak New Order Flow

Figure 5. HBE's New Orders Flow (Rmbbn)



Source: Citi Research

Figure 6. HBE's Orders on Hand (Rmbbn)



Source: Citi Research

HBE received Rmb7bn new orders in 3Q12, based on our calculation and estimate after our company update, equal to only 29% of its new orders in 1H12. The weak new order flow was due to:

- (i) **Reduced capex of the PRC big five power generating groups**, which have nearly 50% of the PRC market share in terms of capacity, for coal-fired plant construction. While their profitability has improved in 2012 due to curtailed unit coal cost, some of their plants have huge accumulated loss. Meanwhile, their financial resources for new projects are relatively tight amid high gearing and the PRC government's more ROE-focused investment criteria. Furthermore, average utilization of PRC coal-fired plants has been cut in 2012E hence there is no urgency to build much new capacity;
- (ii) **Low success rate in winning gas turbine orders** – year to date, HBE won only 2 out of its 20 bids for gas turbine new orders; and

- (iii) **Lack of new nuclear equipment demand** amid PRC's more cautious attitude in new project approval after the Fukushima accident.

Figure 7. HBE's Trading Security Investments

Power company	Purchasing Date	Lock-up period	Volume (m)	Share price by End 2011 (Rmb/share)	Share price by end 1H12 (Rmb/share)	Mark-to-market Gain in 1H12 (Rmbm)	Value by end 1H12 (Rmbm)	Today share price (Rmb/share)
Huaneng	Dec-10	12 month	98	5.35	n/a	0	0	n/a
Guodian	Dec-10	Nil	160	2.79	2.7	-14	432	2.65
Datang	May-11	12 month	200	5.16	5.63	94	1,126	5.20
Inner Mongolia MengDian Huaneng	Mar-12	12 month	50	7.76	7.90	7	395	8.07
Total						87	1,953	

Source: Company Reports and Citi Research Estimates

HBE had no disposal of any trading security investments in July-Oct; we assume no disposal gain from this source in 2H12E. Its liquidity was tightened in 1H12 due to reduced customer deposits. In the period, its net cash (excluding customer deposits received but including advance from holding company in borrowing calculation) on hand was -48% to Rmb2.7bn (or HK\$2.14/share). If customer deposits received on the liability side of its balance sheet are included in the borrowing calculation, its net debt was +4% to Rmb13bn (or HK\$9.15/share) and net debt to equity ratio was +10bps to 79% in the same period. We think that the company might issue corporate debt in 1Q13E.

Figure 8. HBE's Cash Flow Statement (extract)

	1H11	1H12	YoY	Comment
Cashflow from operating activities	-3,132	-1,675	-47%	* Better collection in 1H12
Cashflow from investing activities	172	-979	-671%	* Much capex of new products such as nuclear power equipment has been made
Cashflow from financing activities	-252	-186	-26%	
Total	-3,212	-2,840	-12%	

Source: Company Reports and Citi Research Estimates

Financial Statements & Key Assumptions

Figure 9. Income Statement

Year ended Dec (Rmb m)	2010	2011	2012E	2013E	2014E
Main Thermal Power Equipment	17,984	17,295	14,152	14,340	15,311
Main Hydro Power Equipment	2,600	3,163	3,637	3,637	3,819
Engineering services for power stations	5,060	5,105	4,594	4,594	4,594
Ancillary equipment for power stations	919	719	755	793	832
AD/DC motors and others	2,609	2,206	2,206	2,206	2,206
Net sales	29,172	28,488	25,345	25,571	26,763
Cost of sales	-24,982	-22,777	-20,299	-20,562	-21,483
Gross Profit	4,190	5,711	5,046	5,009	5,280
<i>GP margin</i>	<i>14.4%</i>	<i>20.0%</i>	<i>19.9%</i>	<i>19.6%</i>	<i>19.7%</i>
Other income	690	633	317	317	332
Fair value gain of trading securities	-7	-395	87	0	0
Selling and distribution expenses	-509	-558	-623	-629	-658
Administrative expenses	-2,755	-3,040	-3,212	-3,215	-3,365
Other operating expenses	-411	-969	-158	-158	-166
Operating Profit	1,198	1,382	1,456	1,324	1,423
<i>OP margin</i>	<i>4.1%</i>	<i>4.9%</i>	<i>5.7%</i>	<i>5.2%</i>	<i>5.3%</i>
Depreciation and amortization	-492	-559	-635	-695	-755
EBITDA	1,690	1,940	2,092	2,019	2,179
Interest income	278	252	156	124	133
Interest expense	-140	-73	-76	-76	-76
Associates	26	79	79	79	79
Profit before tax	1,362	1,640	1,614	1,450	1,558
Tax	-269	-257	-252	-239	-273
Minority interests	-92	-155	-152	-135	-144
Net Profit	1,001	1,229	1,210	1,076	1,142
NP margin	3.4%	4.3%	4.8%	4.2%	4.3%
EPS	0.73	0.89	0.88	0.78	0.83
yoy	65.1%	22.7%	-1.6%	-11.1%	6.1%

Source: Company Reports and Citi Research Estimates

Figure 10. HBE's Net Profit Estimates – Citi vs. Consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	1,210	1,296	-7%
FY13E	1,076	1,274	-16%
FY14E	1,142	1,352	-16%

Source: Bloomberg and Citi Research Estimates

Figure 11. Balance Sheet

Year to 31 Dec (Rmb m)	2010	2011	2012E	2013E	2014E
Property, plants and equipments	4,890	5,800	6,165	6,469	6,714
Investment properties	5	5	5	5	5
Prepaid lease payments	419	433	433	433	433
Patent	90	262	262	262	262
Deferred tax assets	205	277	277	277	277
Interests in associates	329	547	617	687	757
Available-for-sale investments/Investment in securities	54	333	333	333	333
Total LT assets	5,992	7,656	8,091	8,466	8,780
Inventory	12,825	11,616	12,021	12,177	12,722
Receivables	17,702	20,316	22,615	22,816	23,880
Trade debtors	11,270	12,918	14,381	14,509	15,185
Other debtors, deposits and prepayments	5,524	5,537	6,163	6,218	6,508
Prepaid lease payments	13	12	12	12	12
Amounts due from customers for contract work	895	1,850	2,059	2,077	2,174
Amounts due from fellow subsidiaries	113	60	60	60	60
Available-for-sale investments	1,188	2,002	2,002	2,002	2,002
Derivative financial instruments	104	93	93	93	93
Restricted bank deposits	5	3	3	3	3
Pledged bank deposits	276	131	131	131	131
Bank deposits	1,091	247	247	247	247
Cash	11,459	8,272	5,020	5,564	5,732
Current assets	44,764	42,740	42,191	43,092	44,870
Total assets	50,756	50,396	50,282	51,558	53,650
Amount due to customers from contract work	1,274	1,413	1,413	1,413	1,413
Trade creditors	11,867	14,646	15,833	16,038	16,757
Other creditors and accrued charges	2,853	3,935	3,935	3,935	3,935
Deposits received	11,892	7,484	5,964	6,017	6,297
Amount due to ultimate holding company	24	36	36	36	36
Advance due from ultimate holding company	20	3	3	3	3
Tax liabilities	816	921	921	921	921
Derivative financial instrument	0	4	4	4	4
Current liabilities	28,747	28,441	28,108	28,366	29,366
ST borrowings	627	582	582	582	582
LT borrowings	1,583	1,314	1,314	1,314	1,314
Total debt	2,210	1,896	1,896	1,896	1,896
Other LT payables - Deposits received	7,593	6,337	5,387	5,387	5,387
Advance from ultimate holding company	1,043	1,199	1,199	1,199	1,199
Minority interests	1,462	1,922	2,074	2,209	2,353
Shareholders equity	9,701	10,602	11,618	12,501	13,450
Total equity and liability	50,756	50,396	50,282	51,558	53,650

Source: Company Reports and Citi Research Estimates

Figure 12. Cash Flow Statement

Year ended Dec (Rmb m)	2010	2011	2012E	2013E	2014E
Profit before tax	1,362	1,640	1,614	1,450	1,558
Associates	-26	-79	-79	-79	-79
Interest income	-278	-252	-156	-124	-133
Finance costs	140	73	76	76	76
EBIT	1,198	1,382	1,456	1,324	1,423
Depreciation	456	499	635	695	755
Amortization	25	0	0	0	0
Others	434	0	0	0	0
Allowance for bad and doubtful debts	558	0	0	0	0
Allowance for inventories	121	0	0	0	0
Gain on disposal of other assets	-151	0	0	0	0
Gain on fair value changes of derivative financial instruments	-97	0	0	0	0
Reversal of allowance for bad and doubtful debts & others	3	0	0	0	0
Change in WC	-1,077	-1,662	-3,036	-99	-610
Increase in inventories	1,191	1,209	-405	-156	-546
Increase in trade debtors	474	-1,648	-1,462	-128	-677
Increase in other debtors, deposits and prepayments	-57	-13	-627	-55	-290
Decrease in amounts due from customers for contract work	52	-955	-209	-18	-97
(Increase) decrease in amounts due from fellow subsidiaries	-34	53	0	0	0
(Dec) Inc in amounts due to customers for contract work	764	139	0	0	0
Increase in trade creditors	1,808	2,779	1,187	205	719
Inc (Dec) in other creditors and accrued charges	-189	1,082	0	0	0
Inc in deposits received	-3,996	-4,408	-1,520	53	281
Increase in due to related entities	-1,222	12	0	0	0
(Dec) Inc in VAT and other taxes payable	132	87	0	0	0
Cash from operations	1,036	218	-945	1,920	1,569
Dividend paid	-94	-193	-193	-193	-193
Income tax paid	-222	-257	-252	-239	-273
Net operating cashflow	720	-231	-1,390	1,488	1,103
Capex	-1,012	-1,000	-1,000	-1,000	-1,000
Dividend from associates	9	9	9	9	9
Others	2,910	252	156	124	133
Investing cashflow	1,906	-739	-835	-867	-859
Net cash after investing activities	2,627	-970	-2,225	621	245
Other loans raised	293	-1,100	-951	0	0
Advance from ultimate holding Co	229	0	0	0	0
Interest paid	-146	-73	-76	-76	-76
Distribution to minority shareholders	-225	0	0	0	0
Net cash from financing activities	151	-1,172	-1,027	-76	-76
Change in net debt	2,778	-2,142	-3,252	545	168
Repayments of borrowings	-2,584	-314	0	0	0
New borrowings raised	653	-731	0	0	0
Change of debt	847	-3,187	-3,252	545	168
Cash b/f	10,612	11,459	8,272	5,020	5,564
Cash, c/f	11,459	8,272	5,020	5,564	5,732

Source: Company Reports and Citi Research Estimates

Figure 13. Key Assumptions

Year ended 31-Dec	2010	2011	2012E	2013E	2014E
Price (Rmb000/MW)					
Main Thermal Power Equipment	829	882	882	882	882
yoy	-7.9%	6.4%	0.0%	0.0%	0.0%
Main Hydro Power Equipment	719	450	450	450	450
yoy	101.7%	-37.5%	0.0%	0.0%	0.0%
Sales volume (MW)					
Boilers	26,070	23,010	18,408	18,408	19,328
yoy	17.5%	-11.7%	-20.0%	0.0%	5.0%
Coal-fired turbines	20,827	20,257	16,206	16,206	17,016
yoy	0.0%	-2.7%	-20.0%	0.0%	5.0%
Coal-fired generators	17,940	15,297	12,238	12,238	12,849
yoy	-13.7%	-14.7%	-20.0%	0.0%	5.0%
Hydro generators	3,617	7,035	8,090	8,090	8,495
yoy	-31.9%	94.5%	15.0%	0.0%	5.0%
Gas turbines	256	256	1,280	1,920	2,880
yoy	0.2%	0.0%	400.0%	50.0%	50.0%
Revenue mix					
Main Thermal Power Equipment	61.6%	60.7%	55.8%	56.1%	57.2%
Main Hydro Power Equipment	8.9%	11.1%	14.4%	14.2%	14.3%
Engineering services for power stations	17.3%	17.9%	18.1%	18.0%	17.2%
Ancillary equipment for power stations	3.2%	2.5%	3.0%	3.1%	3.1%
AD/DC motors and others	8.9%	7.7%	8.7%	8.6%	8.2%
Gross profit margin					
Main Thermal Power Equipment	14.1%	22.2%	21.2%	20.7%	20.7%
Main Hydro Power Equipment	22.9%	30.3%	31.3%	30.8%	30.8%
Engineering services for power stations	3.4%	5.5%	6.5%	6.5%	6.5%
Ancillary equipment for power stations	33.5%	39.8%	34.8%	34.8%	34.8%
AD/DC motors and others	20.7%	15.6%	15.6%	15.6%	15.6%

Source: Company Reports and Citi Research Estimates

Figure 14. Key Ratios

	2010	2011	2012E	2013E	2014E
Profitability					
Gross margin	14.4%	20.0%	19.9%	19.6%	19.7%
EBITDA margin	5.8%	6.8%	8.3%	7.9%	8.1%
Operating margin	4.1%	4.9%	5.7%	5.2%	5.3%
PBT margin	4.7%	5.8%	6.4%	5.7%	5.8%
Net margin	3.4%	4.3%	4.8%	4.2%	4.3%
Net debt to equity (without customer advance)					
Net debt to equity (with customer advance)	82%	57%	58%	51%	48%
ROA	1.9%	2.4%	2.4%	2.1%	2.2%
Average working capital days					
Inventory	198	196	213	215	212
Receivables	227	244	309	324	318
Payables	218	288	370	378	369
Customer Deposits days	170	124	97	86	84

Source: Company Reports and Citi Research Estimates

Harbin Electric

Company description

Harbin Electric (HBE) is one of the top three Chinese power-generation equipment manufacturers. It produces boilers, turbines, and generators, as well as auxiliary equipment for both thermal and hydro power plants. It also provides engineering services for power plant construction. The company is investing in new production facilities to expand into nuclear and wind power equipment manufacturing.

Investment strategy

We have a Buy (1) rating on Harbin Electric. The company is closing its margin discount versus peers, hence deserves to be re-rated. Harbin Electric is a latecomer in wind and nuclear power equipment production and hence will have smaller exposure to these alternative energy products than its major peers. The company therefore makes over 80% of its money from coal-fired power equipment. Current valuations of Harbin Electric looks attractive.

Valuation

Our target price of HK\$7.50 is based on 0.7x 2013E P/B, a discount to the peer-group average of 2.1x. We use P/B because net profits will be lower in a downcycle. We believe a discount of c.60% is justified given HBE's high net profit sensitivity to margins, lower ROE and postponed nuclear development in China. The discount also factors in the weak demand for coal-fired power equipment, and the longer-than-expected recovery time.

Risks

Harbin Electric is one of the three largest PRC power equipment makers with large operating scale and over 25% PRC market share. Downside risk factors that could prevent the stock from reaching our target price include: (i) lower than expected order flow; (ii) lower than expected revenue; and (iii) lower than anticipated margins.

Hong Kong & China Gas (0003.HK)

Overvalued After Recent Rally

Sell	3
Price (28 Nov 12)	HK\$20.90
Target price	HK\$17.00
Expected share price return	-18.7%
Expected dividend yield	1.7%
Expected total return	-16.9%
Market Cap	HK\$181,634M
	US\$23,436M

Price Performance (RIC: 0003.HK, BB: 3 HK)



Asian Utilities

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■ **Summary** – While HKCG's fundamentals are solid, with resilient PRC NG sales growth at 15% yoy in 9M12 and profits from upstream new energy business to surge over 50% yoy in 2013E, these positive factors are probably reflected in its >20% share price rally since end-June. Valuations look rich at 25.6x 2013E PER, 35% above its historical mean of 19x since 2000 as well as well above average 15.8x of HK utilities and 17.6x of PRC gas names. We recommend switching to ENN (2688.HK; HK\$39.00; 1) for higher profit growth and/or CKI (1038.HK; HK\$54.50; 1) for lower (10.7x) FY13E PER but also for its defensive qualities.

■ **PRC NG sales growth sustainable in 9M12** – HKCG's PRC NG sales volume was +15% yoy to 8.6bcm in 9M12, including +14% yoy to 2.6bcm in 3Q12, gained from new customers. Its target is to sell at least 12bcm NG sales in 2012E, up 16-18% yoy. The company connected 908k new PRC residential households in 9M12, +1% yoy; it keeps its plan to connect 1.0-1.2m p.a. in 2012-14E, with no connection growth yoy in 2013E. Connections would be split c.80%/20% for new/old properties.

■ **New energy profit up >50% yoy in 2013E and >30% in 2014E** – We expect HKCG profit from its new energy business to increase +50% yoy in 2013E & +30% in 2014E, at 20-25% IRR. Profits could rise from 12% of the total in 2012E to 24% in 2014E. HKCG may also spend <HK\$1bn on a greenfield PRC shale gas project via a JV with a foreign partner that has the technology. The JV would sign production sharing contracts with a PRC firm with resources. A deal might be closed in 4Q12.

■ **Key earning assumptions** – We have assumed (i) a HK\$100m property revaluation gain in 2H12E on rising rental yields; (ii) a HK\$200m gain from investment securities (with 60% equities and 40% bonds) in 2H12E; and (iii) a 4% gas tariff hike in HK on 1 April 2013. Our DCF-based target price of HK\$17 is equal to 20.8x FY13E PER and 3.0x PB, which looks generous enough considering HKCG's decent track record but below-10% pa net profit CAGR in 2011-14E.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(HK\$M)	(HK\$)	(%)	(x)	(x)	(%)	(%)
2010A	5,585	0.71	6.6	29.6	4.4	15.7	1.7
2011A	6,150	0.74	4.9	28.2	4.4	15.6	2.3
2012E	7,322	0.88	19.1	23.7	4.0	16.8	1.7
2013E	7,107	0.82	-7.3	25.6	3.7	15.0	2.0
2014E	7,836	0.90	10.3	23.2	3.4	15.3	2.2

Source: Powered by dataCentral

0003.HK: Fiscal year end 31-Dec						Price: HK\$20.90; TP: HK\$17.00; Market Cap: HK\$181,634m; Recomm: Sell					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	19,375	22,331	24,979	27,924	31,078	PE (x)	29.6	28.2	23.7	25.6	23.2
Cost of sales	-9,421	-12,463	-13,433	-14,540	-16,220	PB (x)	4.4	4.4	4.0	3.7	3.4
Gross profit	9,955	9,868	11,545	13,384	14,858	EV/EBITDA (x)	29.8	25.4	21.4	19.8	18.2
Gross Margin (%)	51.4	44.2	46.2	47.9	47.8	FCF yield (%)	0.6	1.3	1.7	2.1	2.7
EBITDA (Adj)	5,830	6,890	8,208	8,778	9,395	Dividend yield (%)	1.7	2.3	1.7	2.0	2.2
EBITDA Margin (Adj) (%)	30.1	30.9	32.9	31.4	30.2	Payout ratio (%)	50	64	41	50	50
Depreciation	-1,152	-1,311	-1,715	-1,889	-2,036	ROE (%)	15.7	15.6	16.8	15.0	15.3
Amortisation	0	0	0	0	0	Cashflow (HK\$m)					
EBIT (Adj)	4,678	5,579	6,493	6,889	7,359	EBITDA	5,830	6,890	8,208	8,778	9,395
EBIT Margin (Adj) (%)	24.1	25.0	26.0	24.7	23.7	Working capital	95	-223	-552	-267	-291
Net interest	-711	-752	-952	-962	-954	Other	-691	-197	-439	-1,638	-1,206
Associates	2,418	2,652	2,793	2,741	3,088	Operating cashflow	5,234	6,470	7,217	6,873	7,899
Non-op/Except	702	590	1,101	400	400	Capex	-4,200	-4,200	-4,200	-3,000	-3,000
Pre-tax profit	7,087	8,069	9,435	9,067	9,893	Net acq/disposals	-1,822	-715	-810	-935	-1,090
Tax	-1,039	-1,344	-1,428	-1,296	-1,325	Other	-459	-3,465	1,298	1,435	1,635
Extraord./Min.Int./Pref.div.	-463	-575	-685	-665	-733	Investing cashflow	-6,481	-8,381	-3,713	-2,500	-2,455
Reported net profit	5,585	6,150	7,322	7,107	7,836	Dividends paid	-2,500	-4,148	-3,160	-3,553	-3,918
Net Margin (%)	28.8	27.5	29.3	25.5	25.2	Financing cashflow	-1,971	195	-4,157	-5,827	-5,628
Core NPAT	5,585	6,150	7,322	7,107	7,836	Net change in cash	-3,129	-1,716	-653	-1,454	-184
Per share data						Free cashflow to s/holders					
Reported EPS (\$)	0.71	0.74	0.88	0.82	0.90		1,034	2,270	3,017	3,873	4,899
Core EPS (\$)	0.71	0.74	0.88	0.82	0.90						
DPS (\$)	0.35	0.48	0.36	0.41	0.45						
CFPS (\$)	0.66	0.78	0.87	0.79	0.91						
FCFPS (\$)	0.13	0.27	0.36	0.45	0.56						
BVPS (\$)	4.74	4.78	5.26	5.67	6.12						
Wtd avg ord shares (m)	7,901	8,691	8,691	8,691	8,691						
Wtd avg diluted shares (m)	7,901	8,296	8,296	8,691	8,691						
Growth rates											
Sales revenue (%)	56.9	15.3	11.9	11.8	11.3						
EBIT (Adj) (%)	21.2	19.3	16.4	6.1	6.8						
Core NPAT (%)	5.9	10.1	19.1	-2.9	10.3						
Core EPS (%)	6.6	4.9	19.1	-7.3	10.3						
Balance Sheet (HK\$m)											
Cash & cash equiv.	11,338	11,736	11,083	9,629	9,445						
Accounts receivables	3,313	5,607	3,925	4,212	4,518						
Inventory	1,303	1,622	1,790	1,938	2,154						
Net fixed & other tangibles	33,694	40,352	42,837	43,948	44,912						
Goodwill & intangibles	0	0	0	0	0						
Financial & other assets	23,016	25,772	27,446	29,139	31,081						
Total assets	72,664	85,089	87,081	88,866	92,110						
Accounts payable	5,807	8,022	5,854	5,943	6,074						
Short-term debt	10,009	4,503	5,027	4,573	4,231						
Long-term debt	11,781	21,628	20,108	18,290	16,923						
Provisions & other liab	3,860	4,603	4,914	4,664	4,837						
Total liabilities	31,456	38,757	35,903	33,470	32,065						
Shareholders' equity	37,464	41,584	45,746	49,299	53,217						
Minority interests	3,745	4,748	5,432	6,096	6,829						
Total equity	41,209	46,332	51,178	55,395	60,046						
Net debt	10,451	14,396	14,053	13,234	11,708						
Net debt to equity (%)	25.4	31.1	27.5	23.9	19.5						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

PRC NG Sales Growth Sustainable in 9M12

HKCG PRC NG sales volume was +15% yoy to 8.6bcm in 9M12 and +14% to 1.6bcm in 3Q12; these growth rates were higher than Towngas China (TCCL)'s +10% yoy to 3.8bcm and +12% yoy to 1.2bcm in the same period. The growth rates of TCCL were lower due to below national average NG supply volume growth in Northeast China and the Sichuan province, where TCCL has many projects.

Sector-wide, the growth rates were faster for the commercial sector but lower for the residential and industrial sectors. Gas sales to vehicles are not significant, representing less than 5% of total NG sales. HKCG has 60-70 vehicle refueling stations in operation and has another 30-40 stations either under construction or pending development.

HKCG connected 0.91m households in 9M12, +1% yoy, including 0.48m under TCCL, +11% yoy. New household connection has higher growth for TCCL than HKCG due to the lower average household penetration rate of the former than the latter. HKCG targets to connect 1.0-1.2m residential households pa in 2012-14, including no growth yoy in 2013. The 9M12 connections included over 80% for new properties and below 20% for old ones. Among the 80% of connections coming from new buildings, this included 60% from commodity buildings and 20% from subsidized housing. HKCG added that the residential connection fee of units of both categories are similar, at around Rmb3,000/household.

HKCG and TCCL are targeting to get 8-10 downstream NG projects in 2012E, including six acquired so far.

Management Not Worried About NG Price Reform

The new method of setting citygate NG prices under PRC NG price reform has been tested in Guangdong and Guangxi provinces since 1H12. These two provinces are the first batch to test the new price-setting mechanism. Under this mechanism, the citygate prices – which are the prices NG is being sold at from such mid-stream long-distance pipeline and LNG terminal operators as PetroChina, Sinopec and CNOOC – are being set with reference to market prices of other energies, namely LPG and fuel oil. According to PetroChina Group, Sichuan and Chongqing will probably be the second batch of provinces testing this new regime, since these regions have many NG users with high usage volumes. We think the NG price reform would benefit upstream operators with higher profit margins, and expect the reform to be relatively neutral for downstream city gas distributors, as we assume they will be able to pass through the incremental cost to end users by raising retail prices in the long run, although increments of the latter might have a time delay that may result in temporary margin cuts.

HKCG believes PRC NG price could be raised by end-2012 or early-2013. Sichuan and Chongqing are regions with significant NG consumption. These regions account for NG sales of 35% for Towngas China and 15% for HKCG Enterprises, in terms of 100% sales volume of their NG sales of all their projects although they do not own a 100% stake in them. Much of HKCG's NG sales in these regions comes from Chengdu, which is owned 13% by HKCG and 30% by CR Gas.

Profit Surge of New Energy Businesses

Figure 1. HKCG — New Energy Businesses in China and Thailand

	Capex (Rmbm)	HKCG's Stake	Operation Since
China Upstream Gas-based Projects:			
1. Jilin Tianyuan NG exploration	140	50%	End 2007
2. Shanxi CBM Ph.1	154	70%	Oct-08
3. Shanxi CBM Ph.2	446	70%	Mar-11
4. Chongqing coal-mine-methane	420	50%	4Q13E
5. Shanxi Yangquan coal-mine-methane	450	60%	4Q13E
China Upstream Coal-based Projects:			
1. Inner Mongolia Xiaoyugou thermal coal mine	447	70%	1Q13E
2. Inner Mongolia coal to methanol	1,170	70%	1Q13E
3. Jiangxi Fengcheng coking coal mine	1,100	25%	4Q13E
4. Jiangxi Fengcheng coking plant	1,300	40%	4Q13E
5. Yunnan Long Chi coking coal mine	380	90%	2014E
6. Inner Mongolia Kejian thermal coal mine	450	100%	4Q11
Overseas Upstream Oil-based Projects:			
1. Thailand brownfield oil field	US\$170m	60%	3Q12

Source: Company Reports and Citi Research

HKCG expects its profits from upstream new energy businesses to rise 50% yoy in 2013E and +30% yoy in 2014E under the following assumptions:

- The company forecasts 20-25% IRR from its coal-mine investments. In its due diligence, coal prices in the next five years are estimated and are then assumed to rise with inflation;
- HKCG has two coal logistic projects. The project in Liaoning, which commenced operation in April 2011, made HK\$40m net profit in 1H12 and is expected to contribute similar profit in 2H12E. The project in Shandong started operation in July 2012 and is also forecast to contribute HK\$40m profit in 2H12E;
- HKCG is in the process of investing in a greenfield shale gas project in China. The Chinese government has identified two shale gas blocks for bidding. Being a foreign company, HKCG is not qualified to bid. Instead, it has formed a JV with a mid-sized foreign company with shale gas development technology. The JV is trying to sign production-sharing contracts with domestic companies with shale gas resources. The investment period is expected to be 2-3 years. The budgeted investment amount from HKCG is limited to below HK\$1bn. We believe a deal will be announced in coming months.

HKCG does not expect to spin off its new energy business by separately listing in 2012-14, given its limited scale. Any potential listing would only make sense if the new listing could be traded at higher valuations than HKCG, which would be difficult in view of HKCG's over-20x PER. HKCG perceives there is low risk associated with some upstream projects, such as CBM production, but high risk in others, such as shale gas extraction.

Financial Statement & Key Assumptions

Figure 2. HKCG — Consolidated Income Statement (HK\$m)

Year to 31 Dec	2010	2011	2012E	2013E	2014E
Total Town Gas Revenues	13,665	16,914	18,273	19,298	20,639
Equipment Sales	1,105	1,178	1,237	1,299	1,364
Maintenance & Services	323	332	335	338	342
Water sales	381	444	471	501	517
Property sales	167	0	0	0	0
Rental income	31	33	33	33	33
Upstream projects	0	0	1,064	1,819	2,452
Coal sale	0	734	749	1,233	1,782
Aviation fuel facility construction income	1,840	0	0	0	0
Other sales (incl. Towngas China)	1,864	2,696	2,817	3,402	3,950
Turnover	19,375	22,331	24,979	27,924	31,078
Stores and materials used	-8,230	-10,736	-11,302	-12,235	-13,601
Cost of Property Sale	-39	0	0	0	0
Cost of coal purchased	0	-416	-416	-416	-582
Salaries and wages	-1,467	-1,700	-1,955	-2,151	-2,323
Depreciation and amortization	-1,152	-1,311	-1,715	-1,889	-2,036
Other operating expenses	-3,810	-2,589	-3,097	-4,344	-5,176
EBIT	4,678	5,579	6,493	6,889	7,359
Investment income	702	590	500	400	400
Net finance cost	-711	-752	-952	-962	-954
Profit from associates/JVs	2,418	2,652	2,793	2,741	3,088
Disposal gain	0	0	601	0	0
PBT	7,087	8,069	9,435	9,067	9,893
Tax	-1,039	-1,344	-1,428	-1,296	-1,325
Minority interest	-463	-575	-685	-665	-733
Net Profit	5,585	6,150	7,322	7,107	7,836
% chg	5.9%	10.1%	19.1%	-2.9%	10.3%
EPS (HK\$)	0.71	0.74	0.88	0.82	0.90
% chg	6.6%	4.9%	8.2%	-7.3%	10.3%

Source: Company Reports and Citi Research Estimates

Figure 3. HKCG Net Profit Forecasts – Citi vs. Consensus

Net Profit (HK\$m)	Citi	Consensus	% Diff
FY12E	7,322	6,834	7%
FY13E	7,107	7,468	-5%
FY14E	7,836	8,086	-3%

Source: Bloomberg and Citi Research Estimates

Figure 4. HKCG — DCF Breakdown

	Value (HK\$m)	Per Share (HK\$)	Valuation Method	FY13E PER
HK core business	77,274	8.89	DCF	22.9
HK investment property	7,774	0.89	NAV	15.6
HK aviation fuel/landfill projects	4,617	0.53	DCF	20.1
China up/midstream gas projects	12,491	1.44	NAV	11.0
China downstream gas projects	43,125	4.96	DCF	19.3
China downstream water projects	2,461	0.28	NAV	19.3
Target Price	147,741	17.00		20.8

Source: Company Reports and Citi Research Estimates

Figure 5. Consolidated Balance Sheet (HK\$m)

As at 31 Dec	2010	2011	2012E	2013E	2014E
Fixed assets	26,890	32,255	34,740	35,852	36,816
Associates/JVs	18,571	21,672	23,345	25,038	26,980
Investment securities	3,441	3,111	3,111	3,111	3,111
Other LT assets	6,804	8,097	8,097	8,097	8,097
Total LT assets	55,707	65,134	69,293	72,097	75,003
Debtors	3,313	5,607	3,925	4,212	4,518
Inventories	1,303	1,622	1,790	1,938	2,154
Housing loans to staff	28	0	0	0	0
Cash and deposits	11,338	11,736	11,083	9,629	9,445
Current assets	16,958	19,955	17,788	16,769	17,107
Total assets	72,664	85,089	87,081	88,866	92,110
Provision for taxation	708	878	933	846	865
Creditors	5,807	8,022	5,854	5,943	6,074
ST loans & bank overdrafts	10,009	4,503	5,027	4,573	4,231
Current liabilities	16,523	13,403	11,814	11,362	11,170
Customers' deposits	1,134	1,281	1,384	1,461	1,563
Deferred tax	2,018	2,444	2,597	2,356	2,409
LT loans	11,781	21,628	20,108	18,290	16,923
LT liabilities	14,932	25,353	24,089	22,108	20,894
Minority interest	3,745	4,748	5,432	6,096	6,829
Share capital	1,796	1,975	2,173	2,173	2,173
Reserves	35,669	39,609	43,573	47,126	51,044
Shareholder equity	37,464	41,584	45,746	49,299	53,217
Total liability & shareholder equity	72,664	85,089	87,081	88,866	92,110

Source: Company Reports and Citi Research Estimates

Figure 6. Consolidated Cashflow Statement (HK\$m)

Year to 31 Dec	2010	2011	2012E	2013E	2014E
PBT	7,087	8,069	9,435	9,067	9,893
Profit from associates/JVs	-2,418	-2,652	-2,793	-2,741	-3,088
Depreciation	1,152	1,311	1,715	1,889	2,036
Chg in working capital	95	-223	-552	-267	-291
Other non-cash items	-286	0	0	0	0
Tax paid	-720	-34	-589	-1,075	-652
Operating CF	5,234	6,470	7,217	6,873	7,899
Purchase of PP&E	-4,166	-4,200	-4,200	-3,000	-3,000
Investments	-3,427	-5,354	-810	-935	-1,090
Dividend income	1,112	1,174	1,298	1,435	1,635
Investing CF	-6,481	-8,381	-3,713	-2,500	-2,455
FCF	376	-1,910	3,504	4,373	5,444
Share issuance/repurchase	8	0	0	0	0
Debt financing	1,277	4,342	-996	-2,273	-1,710
Dividend payout	-2,500	-4,148	-3,160	-3,553	-3,918
Others	-747	0	-1	-1	0
Financing CF	-1,963	195	-4,157	-5,827	-5,628
Chg in cash & equiv	-1,587	-1,716	-653	-1,454	-184

Source: Company Reports and Citi Research Estimates

Figure 7. Key Assumptions

Year to 31 Dec	2010	2011	2012E	2013E	2014E
HKCG's Net Profit Breakdown:					
- from HK gas business	3,042	3,327	3,255	3,373	3,270
- from PRC gas and water business	1,651	1,819	2,102	2,360	2,699
- from PRC new energy projects	0	272	593	1,028	1,495
- from PRC refueling stations	0	63	75	75	75
- from Thailand oil field	0	0	54	106	124
- from property sale	123	0	0	0	0
- from property revaluation	734	725	400	0	0
- from property investment	350	446	464	497	497
- from disposal gain	0	0	601	0	0
- from unallocated asset write-off	0	0	0	0	0
- from investment gain	300	20	500	400	400
- from other gains	140	0	0	0	0
- from HK aviation fuel business	96	230	230	230	230
- from net interest cost	-711	-752	-952	-962	-954
% of recurrent profit from China & new energy	37%	46%	48%	53%	59%
HK Towngas Business:					
Tariff (yoy)	14.3%	15.5%	5.0%	3.0%	1.0%
Sales volume (yoy)	1.1%	2.1%	1.0%	1.0%	1.0%
China Gas Business:					
Connectable households (m)	23.90	24.95	26.07	27.24	28.49
Connected households (m)	11.05	12.05	13.15	14.25	15.45
- yoy	9%	9%	9%	8%	8%
Piped gas penetration rate (%)	46.2%	48.3%	50.4%	52.3%	54.2%
Revenue from connection fee (Rmbm):	2,667	3,167	3,500	3,500	3,833
- yoy	48%	19%	11%	0%	10%
- from household	2,400	2,850	3,150	3,150	3,450
- from C&I customers	267	317	350	350	383
Revenue from gas sale (Rmbm)	19,688	26,226	30,824	38,458	46,421
- yoy	34%	33%	18%	25%	21%
- from household	4,173	5,214	5,993	7,310	8,756
- from C&I customers	15,516	21,012	24,831	31,148	37,666
Revenue mix:					
- connection fee	12%	11%	10%	8%	8%
- gas sale	88%	89%	90%	92%	92%
Connection fee per household (Rmb)	3,000	3,000	3,000	3,000	3,000
- yoy	20%	0%	0%	0%	0%
Gas sold (NG equiv, m m3)	8,535	10,300	12,092	13,954	15,967
- yoy	30%	21%	17%	15%	14%
- from household	2,176	2,472	2,842	3,209	3,592
- from C&I customers	6,358	7,828	9,251	10,745	12,374

Source: Company Reports and Citi Research Estimates

Figure 8. HKCG — Key Ratios

Year to 31 Dec	2010	2011	2012E	2013E	2014E
EBITDA margin	30.1%	30.9%	32.9%	31.4%	30.2%
EBIT margin	24.1%	25.0%	26.0%	24.7%	23.7%
Net margin	28.8%	27.5%	29.3%	25.5%	25.2%
ROE	15.7%	15.6%	16.8%	15.0%	15.3%
ROA	8.0%	7.8%	8.5%	8.1%	8.7%
Net debt/equity	25.4%	31.1%	27.5%	23.9%	19.5%
FCFPS (HK\$)	0.05	0.25	0.42	0.50	0.63

Source: Company Reports and Citi Research Estimates

Hong Kong & China Gas

Company description

HKCG is Hong Kong's sole producer, distributor and seller of piped gas, which it brands Towngas. It expanded into China in 2000. Armed with a first-mover advantage in China and robust cash inflow of more than HK\$2bn pa from its cash cow Hong Kong operation, HKCG has secured 101 downstream gas distribution projects in China, including those from Towngas China. Most of these projects are in solid commercial and industrial development areas, which should account for about 76% of total NG demand in 2011. HKCG is expanding vertically and horizontally into other utilities businesses such as water supply, especially in cities where it has established strong ties with local governments through its gas distribution projects, as well as up/midstream projects including coal bed methane liquefaction facility, coal-to-menthol plant and coal mine.

Investment strategy

We rate HKCG as Sell (3). We think the PER premium HKCG trades at against its peer group average will narrow in view of its rising PRC business risk. Negative catalysts include: 1) Higher regulatory risk with the forthcoming natural gas price reform; and 2) Higher risk from expansion into upstream gas businesses.

Valuation

Our sum-of-the-parts target price is HK\$17, including HK\$8.89 from the HK gas business calculated using DCF, HK\$0.89 from other HK businesses based on NAV, and HK\$7.21 from the China gas, water and new energy businesses determined primarily by DCF. The DCF values are determined with reference to earnings forecasts up to 2020 and a terminal value thereafter. This method is appropriate for gas distribution companies such as HKCG in a relatively static industry with predictable cash flows. Our WACC for HKCG is 7.85%, which assumes: 1) a risk-free rate of 5.2%; 2) a market risk premium of 6.8%; 3) an equity beta of 0.60x, according to Bloomberg for the last two years; 4) cost of debt of 5.2%; 5) a target debt to capital ratio of 28.6%; and 6) a 16.5% corporate tax rate. We assume a terminal growth rate of 1.0% for HK and 3.0% for China beyond 2020.

Risks

Upside risks that could prevent the stock from achieving our target price include: 1) Higher- and earlier-than-expected profit contribution from upstream gas businesses; 2) Higher-than-expected gas sales growth; and 3) Higher-than-expected profit from connection fees.

Huadian Power International (1071.HK)

Historical Burden Paid; Higher Profit from 4Q12E

Buy	1
Price (28 Nov 12)	HK\$2.39
Target price	HK\$2.80
Expected share price return	17.2%
Expected dividend yield	2.1%
Expected total return	19.2%
Market Cap	HK\$17,617M US\$2,273M

Price Performance (RIC: 1071.HK, BB: 1071 HK)



■ **Summary** – We reiterate our Buy on Huadian with a DCF-based target price of HK\$2.80, expecting: (i) better profitability ahead with historical burden paid in 9M12; (ii) net profit to double yoy in FY13E on lower unit fuel cost and more non-operating income; and (iii) attractive 7.3x FY13E PER and 4.2% yield. The share price has underperformed its peers due to historical burden, which should not be a concern ahead.

■ **Historical burden paid** – Huadian's 9M12 net profit was below expectation at Rmb415m, dragged by historical burden including overdue coal and maintenance costs incurred in 2010-11 when the company tried to avoid net loss to facilitate new A-share issuance in the Shanghai Stock Exchange. Without the historical burden, Huadian's net profit would surge to Rmb503m in 4Q12, from Rmb415m in 9M12.

■ **Declining unit fuel cost** – We expect Huadian's standard coal cost to -5.3% yoy to Rmb783/tonne in 2012E and -5.0% yoy to Rmb744/tonne in 2013E. Its standard coal cost fell from Rmb808/tonne in 9M12 to Rmb715/tonne in November. We also assume its unit fuel cost to -5.2% yoy in 2012E and -4.7% yoy in 2013E. Its coal cost risk in this winter is mitigated by high coal inventory of 35 days at its plants.

■ **More non-operating income ahead** – Huadian's net non-operating gains were Rmb206m in 2Q12 and Rmb57m in 3Q12. Its net non-operating gain would add Rmb200m qoq in 4Q12E as: (i) Rmb80m non-operating expense paid by Lingwu Power Plant in Ningxia in 3Q12 would not be repeated in 4Q12E; and (ii) the government subsidy from heat supply would be more in the winter in 4Q12E.

■ **High net profit sensitivity** – Huadian has the highest net profit sensitivity among its peers to coal cost, power tariffs, utilization and interest cost, given its thin net margin (2.9% in FY13E). Our FY13E net profit is 6% above current consensus (Bloomberg) and we expect the share price to react positively to 4Q12E results due to be released in March-2013 on improving profitability without historical burden.

Asian Utilities

Hong Kong Utilities

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Statistical Abstract

Year to 31 Dec	Net Profit (RmbM)	Diluted EPS (Rmb)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	170	0.025	-86.8	76.5	0.8	1.1	0.0
2011A	74	0.011	-56.6	176.0	0.8	0.5	0.0
2012E	918	0.130	na	14.8	0.8	5.5	2.1
2013E	1,927	0.261	101.2	7.3	0.8	10.7	4.2
2014E	1,965	0.267	1.9	7.2	0.7	10.1	4.3

Source: Powered by dataCentral

1071.HK: Fiscal year end 31-Dec						Price: HK\$2.39; TP: HK\$2.80; Market Cap: HK\$17,617m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	45,197	54,178	58,002	65,582	67,769	PE (x)	76.5	nm	14.8	7.3	7.2
Cost of sales	-32,959	-38,871	-37,237	-40,269	-41,562	PB (x)	0.8	0.8	0.8	0.8	0.7
Gross profit	12,238	15,307	20,765	25,313	26,207	EV/EBITDA (x)	13.4	11.2	8.4	7.3	7.4
Gross Margin (%)	27.1	28.3	35.8	38.6	38.7	FCF yield (%)	-87.7	-101.8	-71.2	-44.8	-40.9
EBITDA (Adj)	6,403	8,729	12,929	15,967	16,667	Dividend yield (%)	0	0	2.1	4.2	4.3
EBITDA Margin (Adj) (%)	14.2	16.1	22.3	24.3	24.6	Payout ratio (%)	0	0	31	31	31
Depreciation	-4,677	-5,574	-6,064	-7,405	-7,553	ROE (%)	1.1	0.5	5.5	10.7	10.1
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	1,726	3,155	6,865	8,562	9,114	EBITDA	6,403	8,729	12,929	15,967	16,667
EBIT Margin (Adj) (%)	3.8	5.8	11.8	13.1	13.4	Working capital	2,178	536	111	362	225
Net interest	-3,289	-4,925	-6,135	-6,052	-6,516	Other	-6,501	-4,905	-7,131	-7,294	-7,675
Associates	501	1,007	768	786	842	Operating cashflow	2,080	4,360	5,909	9,035	9,218
Non-op/Except	1,265	928	300	312	315	Capex	-13,469	-17,592	-15,565	-15,374	-15,000
Pre-tax profit	202	165	1,798	3,608	3,756	Net acq/disposals	-6,566	0	0	0	0
Tax	-117	-30	-478	-835	-873	Other	998	1,007	768	786	842
Extraord./Min.Int./Pref.div.	84	-61	-402	-846	-918	Investing cashflow	-19,037	-16,585	-14,797	-14,588	-14,158
Reported net profit	170	74	918	1,927	1,965	Dividends paid	-360	0	0	0	-597
Net Margin (%)	0.4	0.1	1.6	2.9	2.9	Financing cashflow	16,666	13,101	9,233	5,861	5,052
Core NPAT	170	74	918	1,927	1,965	Net change in cash	-291	876	345	308	112
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-11,388	-13,232	-9,656	-6,339	-5,782
Reported EPS (Rmb)	0.025	0.011	0.130	0.261	0.267						
Core EPS (Rmb)	0.025	0.011	0.130	0.261	0.267						
DPS (Rmb)	0	0	0.040	0.081	0.083						
CFPS (Rmb)	0.307	0.644	0.836	1.226	1.250						
FCFPS (Rmb)	-1.682	-1.954	-1.366	-0.860	-0.784						
BVPS (Rmb)	2.389	2.405	2.334	2.557	2.742						
Wtd avg ord shares (m)	6,771	6,771	7,067	7,371	7,371						
Wtd avg diluted shares (m)	6,771	6,771	7,067	7,371	7,371						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	24.0	19.9	7.1	13.1	3.3						
EBIT (Adj) (%)	-59.6	82.8	117.6	24.7	6.5						
Core NPAT (%)	-85.3	-56.6	nm	109.8	1.9						
Core EPS (%)	-86.8	-56.6	nm	101.2	1.9						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	1,236	2,112	2,456	2,480	2,591						
Accounts receivables	6,512	7,707	7,656	7,754	7,794						
Inventory	1,760	2,778	2,661	2,877	2,970						
Net fixed & other tangibles	102,548	115,534	125,853	134,541	142,588						
Goodwill & intangibles	0	0	0	0	0						
Financial & other assets	16,506	20,928	20,928	20,928	20,928						
Total assets	128,561	149,059	159,554	168,580	176,872						
Accounts payable	7,740	6,684	6,543	6,810	6,919						
Short-term debt	34,574	38,860	40,231	39,443	40,264						
Long-term debt	55,506	65,598	73,459	80,109	84,938						
Provisions & other liab	8,878	14,117	14,200	14,610	14,859						
Total liabilities	106,699	125,259	134,434	140,972	146,979						
Shareholders' equity	16,176	16,285	17,203	18,846	20,213						
Minority interests	5,687	7,515	7,917	8,762	9,680						
Total equity	21,863	23,800	25,120	27,608	29,893						
Net debt	88,844	102,346	111,234	117,072	122,610						
Net debt to equity (%)	406.4	430.0	442.8	424.0	410.2						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsSch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Key Takeaways from Recent Company Visit

1. **Huadian's gross generation was flat yoy at 11.9M MWh in Oct**, reaching 116.5M MWh in 10M12. The company forecast its gross generation of 160M MWh in 2012E, assuming some increment in the last two months of the year from commissioning of cogeneration units. Average utilization of its coal-fired plants would -3.5% yoy to about 5,300 hours in 2012E, including 3,827 hours in the first nine months of the year.
2. **Huadian is not aware of any NDRC relaxation in approving new coal-fired power projects**, probably without shortage and for emission cut concern. The company's consolidated capex would -10% yoy to around Rmb15-16bn in 2012E, including Rmb14bn in 10M12.
3. **Huadian forecast PRC power demand growth of 5% yoy in 2012E** (including 4.9% in 10M12) and 6-7% yoy in 2013E, lower than the average 11.6% since 2000 and 10.1% in the last five years, since the PRC government has had more concerns about energy efficiency and emission control.
4. **Huadian thinks its net non-operating gain to add Rmb200m qoq in 4Q12E**, from Rmb206m in 2Q12 and Rmb57m in 3Q12, because: (i) a Rmb80m non-operating expense paid by its Lingwu Power Plant in Ningxia province in 3Q12 would not be repeated in 4Q12E, and (ii) government subsidy from heat supply would be more during winter time in 4Q12E.

Figure 1. Huadian — Attributable Capacity Addition Plan (MW)

Power Plant	Province	Capacity	Fuel	Stake	FY11	FY12E	FY13E	FY14E	Completion Date
Capacity at year start					23,989	25,785	30,973	32,104	
Add - new projects:									
Lingwu II	Ningxia	2x1,000	Coal	65%	728				2011
Luhua Cogeneration	Shijiazhuang	2X300	Coal	81%	540				2011
Za-gunao	Sichuan	363	Hydro	49%	77				2011
Kezuozhongqi (Dailiji)	Inner Mongolia	49.5	Wind	100%	50				2011
Luding	Sichuan	4X230	Hydro	100%	460	460			2011-2012
Ningdong Ph. III & IV	Ningxia	49.5*6	Wind	100%	50	149			2011-2012
Laizhou	Shandong	2X1,000	Coal	100%		2,000			Nov 12, Dec 12
Hanzhou Banshan Gas-fired	Zhejiang	3X350	Gas	64%		672			2012
Shuiluohe Ninglang Hydro	Sichuan	2x38	Hydro	100%		424			2012
Liupanshan Wind Power Co	n/a	396	Wind	100%		396			2H12
Guyuan Ph. II & III	Hebei	100 + 49.5	Wind	100%		150			2012
Wind power of Yu County	Hebei	49.5	Wind	100%		99			2012
Yueliangshan	Hebei	2x49.5	Wind	100%		99			2012
Lixian Xinghe Hydro	Sichuan	67	Hydro	100%		67			2012
Zhangjiakou Kangbao	Hebei	100	Wind	100%		50			2012
Ningxia Solar	Ningxia	10	Solar	60%		6			2H12
Zibo	Shandong	2X300	Coal	100%		300	300		4Q12, 1H13
Qudong	Henan	2X300	Coal	90%		270	270		Nov 12, 1H13
Laizhou Jincheng	Shandong	2x48	Wind	100%		48	48		2012-13
Tianjin Wuqing	Tianjin	2X200	Gas	100%			400		2013
Hangzhou Xiasha Gas	Zhejiang	2X100	Gas	56%			112		2013
Nanjiang	Tianjin	2X300+900	Coal	50%				750	2014
Shouzhou	Shanxi	2X300	Coal	100%				600	2014
Luan	Anhui	600	Coal	100%				600	2H14
Capacity at year end					25,785	30,973	32,104	34,054	
% yoy increment					7.5%	20.1%	3.6%	6.1%	

Source: Company Reports and Citi Research Estimates

DCF Calculation & Historical Valuations

Figure 2. Huadian — DCF Calculation

DCF - in the run up to FY15E (RmbM)

	2012E	2013E	2014E	2015E	2016E	2017E	2018E	2019E	2020E
Net operating cashflow		9,035	9,218	9,211	8,885	8,951	9,049	9,193	9,360
Profit from associates		582	638	682	672	665	659	654	652
Profit from coal business		204	204	204	204	204	204	204	204
Less: capex		-15,374	-15,000	-800	-800	-800	-800	-800	-800
FCF		-5,554	-4,940	9,297	8,961	9,020	9,112	9,251	9,416
Discount factor		1.0	1.1	1.2	1.2	1.3	1.4	1.5	1.6
Net debt in 2012/NAV from 2013E	-111,234	-5,554	-4,606	8,082	7,263	6,816	6,420	6,077	5,766

DCF - Terminal Value

NAV in 2021E	9,792
Expected terminal growth rate	4%
Capital Employed on new capex	8.5%
WACC	7.26%
Terminal Value	159,219
Discount factor	1.63
NPV of terminal value	97,509

Total DCF

DCF - In the run up to 2020E	-80,971
DCF from 2021E	97,509
Total DCF	16,538
Exchange Rate	1.25
Total DCF (in HK\$M)	20,639
No. of share (m)	7,371
DCF / share (HK\$)	2.80

Source: Company Reports and Citi Research Estimates

Figure 3. Huadian — Citi vs. Consensus Earnings Forecasts

Net profit (RmbM)	Citi	Consensus	% Diff
FY12E	918	1,194	-23.1%
FY13E	1,927	1,820	5.9%
FY14E	1,965	1,988	-1.1%

Source: Bloomberg and Citi Research Estimates

Figure 4. Huadian — Forward PB



Source: Citi Research

Figure 5. Huadian — Historical PE



Source: Citi Research

Financial Statement & Key Assumptions

Figure 6. Huadian — Consolidated Income Statement

Year-end 31 Dec (RmbM)	FY10	FY11	FY12E	FY13E	FY14E
Electricity sale	43,530	51,125	54,680	62,001	63,910
Sale of heat	1,668	2,698	2,968	3,190	3,430
Net Revenues	45,197	54,178	58,002	65,582	67,769
Sales related taxes	-179	-198	-229	-234	-236
Coal consumption	-32,959	-38,871	-37,237	-40,269	-41,562
Major overhaul expenses	-511	-619	-712	-806	-807
Repairs and maintenance	-640	-705	-862	-1,110	-1,145
Depreciation	-4,677	-5,574	-6,064	-7,405	-7,553
Labor cost	-1,949	-2,460	-2,764	-3,258	-3,318
S&A expenses	-1,179	-1,304	-1,526	-1,850	-1,895
Other operating expenses	-1,377	-1,291	-1,745	-2,088	-2,140
Operating profit	1,726	3,155	6,865	8,562	9,114
Interest income	27	66	44	47	49
Interest expense	-3,315	-4,991	-6,179	-6,099	-6,565
Other incomes/(expenses)	1,265	928	300	312	315
Profit from associates	323	557	564	582	638
Profit from JCEs	5	0	0	0	0
Profit from coal business	173	450	204	204	204
Profit before taxation	202	165	1,798	3,608	3,756
Taxation	-117	-30	-478	-835	-873
Minority interest	84	-61	-402	-846	-918
Net profit	170	74	918	1,927	1,965

Source: Company Reports and Citi Research Estimates

Figure 7. Huadian — Consolidated Cashflow Statement

Year-end 31 Dec (RmbM)	FY10	FY11	FY12E	FY13E	FY14E
Cashflow from operations					
EBITDA	6,403	8,729	12,929	15,967	16,667
Working capital changes	2,178	536	111	362	225
Other operating cash items	-4,334	-812	-773	-672	-551
Gross operating cash flow	4,248	8,453	12,266	15,657	16,341
Net interest paid	-2,029	-4,063	-5,879	-5,787	-6,250
Tax paid	-117	-30	-478	-835	-873
Net operating cash flow	2,103	4,360	5,909	9,035	9,218
Capex	-13,469	-17,592	-15,565	-15,374	-15,000
Free cash flow	-11,366	-13,232	-9,656	-6,339	-5,782
Cash from acquisitions	-6,566	557	564	582	638
Disposal of fixed assets	87	0	0	0	0
Others	889	450	204	204	204
Cashflow from investing	-19,059	-16,585	-14,797	-14,588	-14,158
Ordinary Dividend Paid	-360	0	0	-285	-597
Change in borrowings	16,616	13,101	9,233	5,861	5,649
Other Financing Cash Flow	694	0	0	0	0
Cash Flow from Financing	16,950	13,101	9,233	5,577	5,052
Beginning cash balance	1,242	1,236	2,112	2,456	2,480
Ending cash balance	1,236	2,112	2,456	2,480	2,591

Source: Company Reports and Citi Research Estimates

Figure 8. Huadian — Consolidated Balance Sheet

As at 31 Dec (RmbM)	FY10	FY11	FY12E	FY13E	FY14E
Property plant and equipment	79,401	96,349	118,506	111,101	103,549
Construction in progress	23,147	19,185	7,347	23,440	39,040
Investments	9,573	10,783	10,783	10,783	10,783
Intangible assets	6,212	7,768	7,768	7,768	7,768
Others long term assets	624	1,973	1,973	1,973	1,973
Total fixed assets	118,957	136,057	146,376	155,064	163,112
Cash and cash equivalent	1,236	2,112	2,456	2,480	2,591
Inventories	1,760	2,778	2,661	2,877	2,970
Debtors	6,609	8,113	8,061	8,159	8,199
Total current assets	9,605	13,002	13,178	13,516	13,760
Total assets	128,561	149,059	159,554	168,580	176,872
S/T loans	15,818	21,128	21,128	21,128	21,128
Current portion of L/T loans	15,747	14,181	15,552	14,764	15,584
Trade creditors	7,740	6,684	6,543	6,810	6,919
Other payables and accruals	7,292	11,885	10,078	8,986	9,073
Taxes payable	64	126	2,016	3,519	3,680
Total current liabilities	46,660	54,004	55,317	55,206	56,384
Long term bank loans	44,019	50,792	58,654	65,303	70,132
Other loans	11,487	14,806	14,806	14,806	14,806
Other payables	2,464	3,167	3,167	3,167	3,167
Deferred tax liabilities	2,068	2,490	2,490	2,490	2,490
Minority interests	5,687	7,515	7,917	8,762	9,680
Total L/T liabilities	65,725	78,770	87,033	94,528	100,275
Share capital	6,771	6,771	7,371	7,371	7,371
Other capital	2,210	2,221	1,759	2,048	2,342
Retained earnings	7,195	7,293	8,074	9,427	10,500
Total equity	16,176	16,285	17,203	18,846	20,213
Total liabilities & equity	128,561	149,059	159,554	168,580	176,872

Source: Company Reports and Citi Research Estimates

Figure 9. Huadian — Key Ratios

Year-end 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Gross margin (%)	14.2	15.5	22.6	24.4	24.6
Operating margin (%)	3.8	5.8	11.8	13.1	13.4
Net margin (%)	0.4	0.1	1.6	2.9	2.9
Current ratio (x)	0.21	0.24	0.24	0.24	0.24
Liabilities (ex-MI) to asset ratio (x)	0.83	0.84	0.84	0.84	0.83
Net debt to total capital ratio (x)	0.78	0.77	0.78	0.77	0.77
Net debt to equity ratio (x)	3.93	4.15	4.29	4.11	3.98
Interest cover (x)	0.63	0.76	1.21	1.51	1.49
Cash interest cover (x)	1.88	1.75	1.98	2.48	2.47
Debt to EBITDA ratio (x)	13.60	11.56	8.52	7.27	7.30
Debt service ratio (x)	0.11	0.33	0.46	0.53	0.56
ROA (%)	0.1	0.1	0.6	1.2	1.1

Source: Company Reports and Citi Research Estimates

Figure 10. Huadian — Key Assumptions

	FY10	FY11	FY12E	FY13E	FY14E
Capacity					
Installed capacity (MW)	32,319	34,611	40,222	41,470	44,170
yoy change (%)	11.7	7.1	16.2	3.1	6.5
Consolidated capacity (MW) - year end	27,418	29,818	35,352	36,600	37,800
yoy change (%)	19.2	8.8	18.6	3.5	3.3
Consolidated capacity (MW) - year average	24,809	27,797	30,502	35,342	36,116
yoy change (%)	9.1	12.0	9.7	15.9	2.2
Attributable capacity (MW) - year end	23,989	25,785	30,974	32,104	34,054
yoy change (%)	12.7	7.5	20.1	3.6	6.1
Generation					
Total gross generation (M MWh)	141.4	158.1	162.1	183.9	193.7
Total net generation (M MWh)	132.4	148.5	152.4	173.7	183.3
yoy change (%)	20.8	12.2	2.6	14.0	5.5
Consolidated gross generation (M MWh)	130.3	150.8	155.1	176.9	182.4
Consolidated net generation (M MWh)	121.3	140.8	144.9	165.3	170.4
yoy change (%)	21.4	16.1	2.9	14.1	3.1
Attributable gross generation (M MWh)	109.0	121.1	125.2	145.9	153.1
Attributable net generation (M MWh)	101.5	113.1	117.0	136.3	143.1
yoy change (%)	22.3	11.4	3.4	16.5	5.0
Average utilization hours of coal-fired plants	5,539	5,494	5,296	5,322	5,366
yoy change (%)	11.8	-0.8	-3.6	0.5	0.8
Attributable average utilization rate	63.2%	62.7%	60.5%	60.8%	61.3%
Tariff					
Consolidated tariff (Rmb/MWh)	420	425	442	439	439
yoy change (%)	1.6	1.1	4.0	-0.6	0.0
Coal cost					
Unit coal cost (Rmb/MWh)	264	270	256	244	243
yoy change (%)	19.4	2.3	-5.2	-4.7	-0.3
Coal consumption rate (g/KWh)	321	316	316	316	316
yoy change (%)	-0.6	-1.6	0.0	0.0	0.0
Standard coal cost (Rmb/ton)	799	827	783	744	751
yoy change (%)	22.9	3.5	-5.3	-5.0	1.0

Source: Company Reports and Citi Research Estimates

Figure 11. PRC IPPs: FY13E Net Profit Sensitivity

(RmbM)	Huaneng	Datang	Huadian	CRP	CPI	Average
Base case net profit *	7,932	3,647	1,927	8,078	1,365	n/a
Case 1. net profit change with extra 1% cut in average coal price	454	157	280	291	71	n/a
% change	5.7%	4.3%	14.5%	3.6%	5.2%	6.7%
Case 2. net profit change with extra 1% hike in power tariff	746	299	626	460	125	n/a
% change	9.4%	8.2%	32.5%	5.7%	9.2%	13.0%
Case 3. net profit change with extra 1% cut in utilization hours	-272	-110	-121	-174	-73	n/a
% change	-3.4%	-3.0%	-6.3%	-2.2%	-5.3%	-4.0%
Case 4. net profit change with extra 25bps interest rate cut	242	213	287	205	43	n/a
% change	3.0%	5.8%	14.9%	2.5%	3.2%	5.9%

Source: Citi Research Estimates

* In RmbM except CRP which is in HK\$M

** Key assumptions - (i) unit fuel cost ranges from -2% yoy (Huaneng, Datang, CRP & CPI) to -5% yoy (for Huadian) in 2013E; (ii) average 6.5% tariff hike on 1 Dec 2011, but nil in 2012-13E

Huadian Power International

Company description

Huadian Power International is the largest IPP in Shandong, China, with a market share in the province of about 18%. After an official announcement of a reform in China's power industry structure in Dec 2002, Shandong International Power, along with Heilongjiang Electric Company, was placed under China Huadian Group. The company was renamed Huadian International Power Development in May 2003. Most of the key projects of the China Huadian Group will be owned, built and operated by the listed company.

Investment strategy

We have a Buy rating on Huadian shares. The stock was the worst performer among PRC IPPs in the last three months owing to lower-than-expected profit dragged by historical cost burden. However, we think that its profitability will improve ahead as: (i) the burden has been paid; (ii) unit fuel cost is declining; and (iii) more non-operating income.

Valuation

Our target price of HK\$2.80 is based on a DCF model, which we think is suitable for power companies such as Huadian that operate in a largely stable industry with predictable cash flows. Our model factors in our earnings forecasts up to 2015E and a terminal growth rate of 4% thereafter. Our WACC for Huadian is 7.26%, assuming (1) a risk-free rate of 5.2%, (2) a market risk premium of 6.8%, (3) an equity beta of 1.0x, according to Bloomberg, (4) 6.5% cost of debt, (5) a target debt to capital ratio of 66.67%, and (6) a corporate tax rate of 25%. At our DCF-based target price, Huadian would trade at 0.9x 2013E PB.

Risks

Downside risks to our investment thesis and target price for Huadian include: 1) Higher-than-expected coal price; 2) Lower-than-forecast power generation; and 3) More-than-anticipated interest expense.

Huaneng Power International (0902.HK)

Positive Free Cashflow in 2012-13E, the First Time Since 2003

Buy	1
Price (28 Nov 12)	HK\$6.28
Target price	HK\$7.50
Expected share price return	19.4%
Expected dividend yield	4.4%
Expected total return	23.8%
Market Cap	HK\$91,533M
	US\$11,811M

Price Performance
(RIC: 0902.HK, BB: 902 HK)



■ **Summary** – We reiterate our Buy on Huaneng given its improving profitability from: (i) unit fuel cost to drop at least 5% yoy in 2012E and 2% yoy in 2013E; (ii) curtailed interest cost from capex cut (-46% yoy to Rmb15bn in FY12E) and lowered average interest cost (5.8% in 9M12). FY13E valuations of 8.9x PER and 1.2x PB look attractive at below historical mean. Positive free cashflow in 2012-13E, the first time since 2003, should mean no share issuance is necessary unless related to M&A.

■ **Generation started showing positive growth in October** – Huaneng targets at least 300M MWh consolidated gross generation in China in 2012E. Generation was +0.9% yoy in October, ending a six month decline. The company expects average utilization of its coal fired plants to be about 5,170 hours in 2012E, -8.7% yoy, while average utilization of all power plants is expected to drop by similar extent from 5,531 hours last year.

■ **Declining unit fuel cost** – Huaneng's unit fuel cost is targeted to drop at least 5% yoy in FY12E and was down mom in October. It is expected to consume 140m tonnes of raw coal in 2012E, including 27% from key contracts with prices set by NDRC, 46% from domestic spot market, 6% from coal mines of its groups, and 21% from import. It had 21 coal inventory days at its power plants in mid-November.

■ **Curtailed capex in FY12-13E** – Huaneng's consolidated capex could be below its budget of Rmb20.8bn as the actual amount was only Rmb10.3bn in 9M12. We estimate capex to fall 46% yoy to Rmb15bn in FY12E, with 61% for coal-fired plants, 16% for technical upgrades, 9% for wind farms, 7% for port facilities, 3% for hydropower, and 4% for others. Its capex may also be relatively low in FY13E when only 2.5GW of new capacity would start operation, -63% yoy from 6.7GW in FY12E.

■ **Positive free cashflow, the first time since 2003** – We expect Huaneng to have positive free cashflow in FY12-13E, the first time since FY03. Its interest cost could drop HoH in 2H12E due to lowered average interest cost from 50bps, RMB interest rate cut in 2Q12, and curtailed net debt to equity ratio amid positive free cashflow.

Asian Utilities

Hong Kong Utilities

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Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(RmbM)	(Rmb)	(%)	(x)	(x)	(%)	(%)
2010A	3,348	0.277	-32.4	18.2	1.3	7.0	4.0
2011A	1,181	0.084	-69.6	60.1	1.4	2.3	1.0
2012E	6,273	0.446	431.3	11.3	1.3	12.0	4.4
2013E	7,932	0.564	26.5	8.9	1.2	14.2	5.6
2014E	8,173	0.581	3.0	8.7	1.1	13.6	5.8

Source: Powered by dataCentral

0902.HK: Fiscal year end 31-Dec						Price: HK\$6.28; TP: HK\$7.50; Market Cap: HK\$91,533m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	104,170	132,937	135,685	144,803	150,689	PE (x)	18.2	60.1	11.3	8.9	8.7
Cost of sales	-67,892	-90,546	-84,228	-88,801	-92,690	PB (x)	1.3	1.4	1.3	1.2	1.1
Gross profit	36,279	42,391	51,458	56,002	58,000	EV/EBITDA (x)	9.7	9.6	7.1	6.5	6.4
Gross Margin (%)	34.8	31.9	37.9	38.7	38.5	FCF yield (%)	-4.3	-22.3	3.9	7.2	7.6
EBITDA (Adj)	19,076	20,614	29,703	32,744	33,349	Dividend yield (%)	4.0	1.0	4.4	5.6	5.8
EBITDA Margin (Adj) (%)	18.3	15.5	21.9	22.6	22.1	Payout ratio (%)	72	60	50	50	50
Depreciation	-10,447	-11,867	-11,413	-12,240	-12,399	ROE (%)	7.0	2.3	12.0	14.2	13.6
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	8,629	8,748	18,290	20,504	20,950	EBITDA	19,076	20,614	29,703	32,744	33,349
EBIT Margin (Adj) (%)	8.3	6.6	13.5	14.2	13.9	Working capital	-40	707	257	350	-37
Net interest	-5,194	-7,570	-8,842	-8,484	-8,709	Other	-970	-8,270	-11,487	-11,729	-11,892
Associates	569	704	725	745	751	Operating cashflow	18,067	13,052	18,473	21,364	21,420
Non-op/Except	160	169	0	0	0	Capex	-20,704	-28,893	-15,708	-16,263	-16,000
Pre-tax profit	4,164	2,051	10,173	12,765	12,991	Net acq/disposals	0	0	0	0	0
Tax	-843	-869	-2,645	-3,245	-3,183	Other	103	352	363	373	375
Extraord./Min.Int./Pref.div.	27	-1	-1,255	-1,587	-1,636	Investing cashflow	-20,601	-28,541	-15,346	-15,890	-15,625
Reported net profit	3,348	1,181	6,273	7,932	8,173	Dividends paid	-2,811	-703	-3,136	-3,966	-4,087
Net Margin (%)	3.2	0.9	4.6	5.5	5.4	Financing cashflow	13,063	14,455	430	-4,984	-5,468
Core NPAT	3,348	1,181	6,273	7,932	8,173	Net change in cash	10,579	-985	3,607	540	378
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-2,638	-15,841	2,765	5,102	5,420
Reported EPS (Rmb)	0.277	0.084	0.446	0.564	0.581						
Core EPS (Rmb)	0.277	0.084	0.446	0.564	0.581						
DPS (Rmb)	0.200	0.050	0.223	0.282	0.291						
CFPS (Rmb)	1.492	0.929	1.314	1.520	1.524						
FCFPS (Rmb)	-0.218	-1.127	0.197	0.363	0.386						
BVPS (Rmb)	3.827	3.620	3.843	4.125	4.416						
Wtd avg ord shares (m)	12,107	14,055	14,055	14,055	14,055						
Wtd avg diluted shares (m)	12,107	14,055	14,055	14,055	14,055						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	35.8	27.6	2.1	6.7	4.1						
EBIT (Adj) (%)	-5.9	1.4	109.1	12.1	2.2						
Core NPAT (%)	-32.1	-64.7	431.3	26.5	3.0						
Core EPS (%)	-32.4	-69.6	431.3	26.5	3.0						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	9,548	8,670	12,228	12,717	13,045						
Accounts receivables	10,909	15,378	15,696	16,751	17,431						
Inventory	5,190	7,526	6,912	7,197	7,490						
Net fixed & other tangibles	155,225	177,968	182,263	186,285	189,886						
Goodwill & intangibles	12,641	13,890	13,890	13,890	13,890						
Financial & other assets	34,425	33,984	34,347	34,719	35,095						
Total assets	227,938	257,416	265,335	271,560	276,837						
Accounts payable	19,555	25,768	23,666	24,644	25,645						
Short-term debt	62,900	69,378	54,998	58,360	60,171						
Long-term debt	79,016	97,700	115,681	111,300	108,109						
Provisions & other liab	4,041	5,013	7,041	7,753	7,688						
Total liabilities	165,513	197,858	201,385	202,057	201,612						
Shareholders' equity	53,789	50,883	54,019	57,985	62,072						
Minority interests	8,636	8,675	9,930	11,518	13,153						
Total equity	62,425	59,558	63,949	69,503	75,225						
Net debt	132,368	158,407	158,451	156,943	155,235						
Net debt to equity (%)	212.0	266.0	247.8	225.8	206.4						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

Capacity Expansion Plan

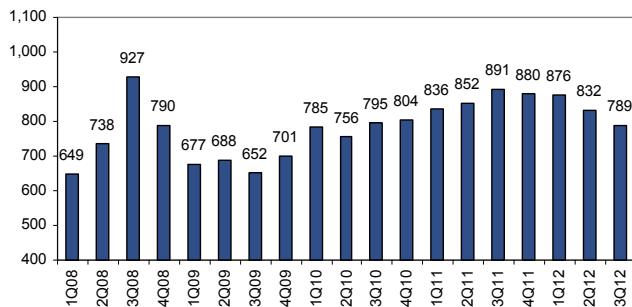
Figure 1. Huaneng — Attributable Installed Capacity

	Province	Type	Capacity	Stake	FY11 48,155	FY12E 53,828	FY13E 58,996	FY14E 60,699	Completion
Capacity at year beginning									
Add - greenfield development:									
Ganhekou Wind	Gansu	Wind	502	100%	502				2011
Huade Wind	Inner Mongolia	Wind	33X1.5	100%	49				2011
Dalian Wafangdian Wind	Liaoning	Wind	48	100%	48				2011
Zuoquan	Shanxi	Coal	2X673	60%	404	404			2011-12
Yueyang Ph.III	Hunan	Coal	2X600	55%	330	330			2011-12
Fuzhou Ph.III	Fujian	Coal	2X660	100%	660	660			2011-12
Xiangqi Hydro	Hunan	Hydro	1X80	100%	20	60			2012
Jinling Thermal	Jiangsu	Coal	2X1,030	60%		618			2011-12
Haimen Ph.II	Guangdong	Coal	2X1,000	100%		2,000			2012
Qinbei Ph.III	Henan	Coal	1X1,000	60%		600			2012
Jiuquan Wind	Gansu	Wind	501	100%		501			2012
Suzihe	Liaoning	Hydro	3X12.5	100%		38			2012
Qidong Wind	Jiangsu	Wind	44	65%		29			2012
Changtu Wind	Liaoning	Wind	94	100%		50	44		2012-13
Weihai	Shandong	Coal	2X600	60%			720		2013
Nantong	Jiangsu	Coal	1000	70%			700		2013
Jinling Gas	Jiangsu	Gas	382	51%			195		2013
Wudong	Jiangsu	Wind	48	90%			43		2013
Liangjiang Gas	Chongqing	Gas	5X300	100%				1,500	2014
Chuxiong Gas	Yunnan	Gas	2X300	100%				600	2014
Add - acquisition:									
Diandong Energy	Yunnan	Coal	4X600	100%	2,400				
Diandong Yuwang	Yunnan	Coal	2X600	100%	1,200				
Others					60				
Less - closure									
Yushe Power	Shaanxi	Coal	2X100	60%		-120			
Capacity at year end					53,828	58,996	60,699	62,799	
- yoy					14.9%	9.6%	2.9%	3.5%	

Source: Company Reports and Citi Research Estimates

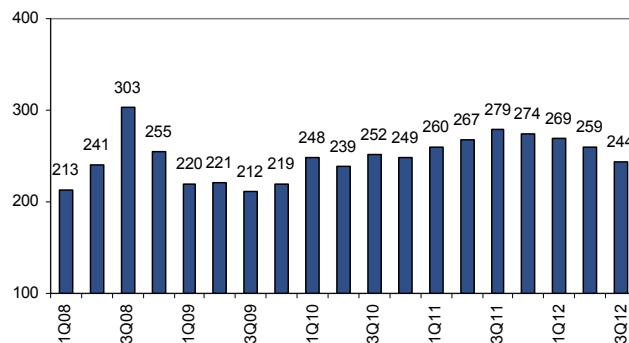
* Huaneng would have a total of 3,414.5MW wind capacity by end-2012E

Figure 2. Huaneng — Quarterly Standard Coal Cost, Declining in 2012



Source: Company Reports

Figure 3. Huaneng — Quarterly Unit Fuel Cost, Declining in 2012



Source: Company Reports

DCF Calculation & Historical Valuation

Figure 4. Huaneng — DCF Calculation (Rmbm)

	2011	2012E	2013E	2014E	2015E	2016E	2017E	2018E	2019E	2020E
Net operating cashflow		18,473	21,364	21,420	22,087	22,108	22,181	22,216	22,180	22,098
Less: capex		-15,346	-15,890	-15,625	-2,307	-2,311	-2,316	-2,322	-2,349	-2,324
FCF		3,128	5,474	5,795	19,780	19,797	19,865	19,895	19,832	19,774
Discount factor		1.00	1.08	1.16	1.26	1.36	1.46	1.58	1.70	1.84
Net debt in 2011 / NAV from 2012E onwards	-158,443	3,128	5,074	4,978	15,747	14,607	13,585	12,609	11,649	10,765

DCF - Terminal Value

NAV in 2021E	20,565
Expected terminal growth rate	4.0%
Capital Employed on new capex	8.5%
WACC	7.9%
Terminal Value	279,410
Discount factor	1.84
NPV of terminal value	152,119

Total DCF

DCF in the run up to 2020E	-66,300
DCF from 2021E	152,119
Total DCF	85,819
Exchange Rate	1.23
Total DCF (HK\$m)	105,415
No. of share (m)	14,055
Total DCF/Share (HK\$)	7.50

Source: Company Reports and Citi Research Estimates

Our DCF-based target price of HK\$7.50 is equal to 11x FY13E PER and 1.5x FY13E PB, versus historical means of 12x and 1.3x respectively. We think Huaneng is deserving of a re-rating for its prevailing positive free cashflow, which has not happened since 2003.

Figure 5. Huaneng — Forward Rolling PER



Source: Citi Research

Figure 6. Huaneng — Forward Rolling PB



Source: Citi Research

Financial Statement & Key Assumptions

Figure 7. Huaneng — Consolidated Income Statement

Year-ended 31 Dec (RmbM)	FY10	FY11	FY12E	FY13E	FY14E
Net Revenues	104,170	132,937	135,685	144,803	150,689
Operating costs:					
Coal cost	-65,060	-85,949	-78,937	-82,200	-85,538
Gas cost	-2,832	-4,597	-5,291	-6,602	-7,152
Repairs and maintenance	-2,302	-2,529	-2,638	-2,926	-3,287
Depreciation & amortization	-10,447	-11,867	-11,413	-12,240	-12,399
Labor cost	-4,067	-4,622	-4,867	-5,316	-5,633
Purchase of electricity	-5,557	-8,613	-8,264	-8,347	-8,431
Services fees to HIPDC	-141	-141	-146	-154	-161
Other operating expenses	-5,135	-5,872	-5,838	-6,515	-7,139
Operating profit	8,629	8,748	18,290	20,504	20,950
Interest income	89	166	191	196	203
Interest expense	-5,283	-7,736	-9,032	-8,681	-8,912
Exchange losses	88	76	0	0	0
Profit from associates	569	704	725	745	751
Exceptional gains	72	93	0	0	0
Profit before taxation	4,164	2,051	10,173	12,765	12,991
Taxation	-843	-869	-2,645	-3,245	-3,183
Minority interest	27	-1	-1,255	-1,587	-1,636
Net profit	3,348	1,181	6,273	7,932	8,173

Source: Company Reports and Citi Research Estimates

Figure 8. Huaneng — Citi vs. Consensus Net Profit Forecasts

Net Profit (RmbM)	Citi	Consensus	% Diff
FY12E	6,273	6,019	4.2%
FY13E	7,932	7,681	3.3%
FY14E	8,173	7,993	2.3%

Source: Bloomberg and Citi Research Estimates

Figure 9. Huaneng — Key Ratios

Year end 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Gross Margin (%)	22.6	21.1	27.6	28.2	28.1
Operating Margin (%)	8.3	6.6	13.5	14.2	13.9
Net Margin (%)	3.2	0.9	4.6	5.5	5.4
Interest coverage ratio (x)	1.8	1.3	2.1	2.5	2.5
Cash interest coverage (x)	3.3	2.5	3.2	3.6	3.7
Debt to EBITDA ratio	7.4	8.1	5.7	5.2	5.0
Net Debt to Equity (x)	2.12	2.66	2.48	2.26	2.06
ROA (%)	1.6	0.5	2.4	3.0	3.0

Source: Company Reports and Citi Research Estimates

Figure 10. Huaneng — Consolidated Balance Sheet

As at 31 Dec (RmbM)	FY10	FY11	FY12E	FY13E	FY14E
Property plant and equipment	146,178	176,653	175,569	172,798	160,399
Construction in progress	9,047	1,315	6,694	13,487	29,487
Associated companies	11,973	13,588	13,951	14,323	14,699
Goodwill	12,641	13,890	13,890	13,890	13,890
Other fixed assets	16,543	15,552	15,552	15,552	15,552
Total fixed assets	196,382	220,999	225,656	230,051	234,027
Cash and bank	9,548	8,670	12,228	12,717	13,045
Account receivable	16,818	20,222	20,540	21,594	22,275
Other current assets	5,190	7,526	6,912	7,197	7,490
Total current assets	31,556	36,417	39,679	41,509	42,810
Total assets	227,938	257,416	265,335	271,560	276,837
Bank loans	57,830	59,116	54,998	58,360	60,171
Account payable	19,826	25,998	23,877	24,864	25,874
Convertible notes	5,157	10,298	0	0	0
Taxes payable	744	1,019	3,101	3,804	3,731
Dividends payable	80	168	168	168	168
Total current liabilities	83,637	96,598	82,143	87,196	89,943
Long term bank loans	65,185	79,845	97,826	93,446	90,254
Convertible notes	13,831	17,855	17,855	17,855	17,855
Other L/T liabilities	2,860	3,561	3,561	3,561	3,561
Total L/T liabilities	81,876	101,261	119,242	114,861	111,670
Minority interest	8,636	8,675	9,930	11,518	13,153
Share capital	14,055	14,055	14,055	14,055	14,055
Other capital	25,389	24,830	25,928	27,316	28,222
Retained earnings	14,344	11,997	14,036	16,614	19,795
Total equity	53,789	50,883	54,019	57,985	62,072
Total liabilities & equity	227,938	257,416	265,335	271,560	276,837

Source: Company Reports and Citi Research Estimates

Figure 11. Huaneng — Cash Flow Statement

Year-end 31 Dec (RmbM)	FY10	FY11	FY12E	FY13E	FY14E
Cashflow from operations					
EBITDA	19,076	20,614	29,703	32,744	33,349
Working capital changes	-40	707	257	350	-37
Other operating cash items	87	169	0	0	0
Gross operating cash flow	19,123	21,491	29,960	33,094	33,312
Net interest paid	55	-7,570	-8,842	-8,484	-8,709
Tax paid	-1,111	-869	-2,645	-3,245	-3,183
Net operating cash flow	18,067	13,052	18,473	21,364	21,420
Capex	-26,981	-28,541	-15,346	-15,890	-15,625
Free cash flow	-8,914	-15,489	3,128	5,474	5,795
Cashflow from investing	-26,981	-28,541	-15,346	-15,890	-15,625
Increase in Equity	284	0	0	0	0
Ordinary Dividend Paid	-2,811	-703	-3,136	-3,966	-4,087
Change in borrowings	15,591	15,158	3,566	-1,018	-1,381
Cash Flow from Financing	13,063	14,455	430	-4,984	-5,468
Exchange rate changes	50	50	50	50	50
Beginning cash balance	5,227	9,426	8,549	12,106	12,596
Ending cash balance	9,426	8,549	12,106	12,596	12,924

Source: Company Reports and Citi Research Estimates

Figure 12. Huaneng – Key Assumptions

	FY10	FY11	FY12E	FY13E	FY14E
Capacity (MW)					
Operational capacity (MW) - year end	55,696	62,786	69,987	72,687	74,787
y-y change (%)	4.1	12.7	11.5	3.9	3.0
Consolidated capacity (MW) - year end	50,152	58,702	65,903	66,903	69,003
yoy change (%)	1.5	17.0	12.3	1.5	3.1
Consolidated capacity (MW) - year average	47,428	55,095	60,831	63,733	65,283
y-y change (%)	10.7	16.2	10.4	4.8	2.4
Attributable capacity (MW) - year end	48,155	53,808	59,227	59,927	62,027
y-y change (%)	4.9	11.7	10.1	1.2	3.5
Generation (M MWh)					
Consolidated gross generation	257.0	313.6	304.3	327.8	343.3
Consolidated net generation	241.8	298.8	289.9	312.4	327.3
y-y change (%)	25.4	23.6	-3.0	7.7	4.8
Attri. gross generation	229.3	276.2	265.5	284.5	298.3
Attri. net generation	216.0	263.2	252.9	271.0	284.3
y-y change (%)	24.9	21.8	-3.9	7.2	4.9
Utilization hours of coal-fired plants	5,565	5,816	5,056	5,202	5,301
y-y change (%)	6.6	4.5	-13.1	2.9	1.9
Utilization rate	63.5%	66.4%	57.7%	59.4%	60.5%
Tariff (Rmb/MWh)					
Consolidated tariff before VAT	422	430	458	461	460
y-y change (%)	4.8	2.0	6.4	0.7	-0.1
Coal cost					
Unit fuel cost (Rmb/MWh)	248	270	257	251	254
y-y change (%)	14.8	9.0	-5.0	-2.1	1.0
Coal consumption per unit generation (g/kWh)	298	294	291	291	291
y-y change (%)	-1.2	-1.1	-1.0	0.0	0.0
Coal consumption per unit sale (g/kWh)	316	312	309	309	309
y-y change (%)	-1.4	-1.1	-1.0	0.0	0.0
Average price of natural coal (Rmb/ton, excluding tax)	517.2	572.2	542.5	531.6	531.6
y-y change (%)	12.2	10.6	-5.2	-2.0	0.0

Source: Company Reports and Citi Research Estimates

Figure 13. PRC IPPs: FY13E Net Profit Sensitivity

(RmbM)	Huaneng	Datang	Huadian	CRP	CPI	Average
Base case net profit *	7,932	3,647	1,927	8,078	1,365	n/a
Case 1. net profit change with extra 1% cut in average coal price	454	157	280	291	71	n/a
% change	5.7%	4.3%	14.5%	3.6%	5.2%	6.7%
Case 2. net profit change with extra 1% hike in power tariff	746	299	626	460	125	n/a
% change	9.4%	8.2%	32.5%	5.7%	9.2%	13.0%
Case 3. net profit change with extra 1% cut in utilization hours	-272	-110	-121	-174	-73	n/a
% change	-3.4%	-3.0%	-6.3%	-2.2%	-5.3%	-4.0%
Case 4. net profit change with extra 25bps interest rate cut	242	213	287	205	43	n/a
% change	3.0%	5.8%	14.9%	2.5%	3.2%	5.9%

Source: Citi Research Estimates

* In RmbM except CRP which is in HK\$M

** Key assumptions - (i) unit fuel cost ranges from -2% yoy (Huaneng, Datang, CRP & CPI) to -5% yoy (for Huadian) in 2013E; (ii) average 6.5% tariff hike on 1 Dec 2011, but nil in 2012-13E

Huaneng Power International

Company description

Huaneng is the largest Chinese power producer with 55,350MW attributable installed capacity as at end 2011, equivalent to a national market share of 5%. About one-third of the company's capacity was manufactured overseas in the late 1990s, so it generally operates at higher efficiency levels than most of its domestic peers. Of Huaneng's attributable capacity, 95% is coal-fired, with the exceptions being the 70%-owned 3x390 MW gas-fired Shanghai Power Plant and the 49%-owned 2,200 MW hydropower units at Sichuan. The company's power plants are in all regional power networks, with the highest proportion in Eastern China, Northern China and Central China.

Investment strategy

We rate Huaneng as Buy given: 1) we see improving profitability from unit fuel cost drop in 2012-13E and also from curtailed interest cost from capex cut; 2) consensus appears to have under-estimated the benefits from declining coal prices; 3) generation, which started showing positive yoy growth in Oct 2012 after six months of decline, is expected to accelerate; and 4) positive free cashflow in 2012-13E, the first time since 2003, should mean that no share issuance is necessary unless related to M&A. Our Huaneng DCF-based target price is HK\$7.50.

Valuation

Our target price of HK\$7.50 is based on a DCF model, which we believe is suitable for power companies such as Huaneng that operate in a relatively stable industry with predictable cash flows. Our model factors in our earnings forecasts up to 2015 and a terminal growth rate of 4% thereafter. Our WACC for Huaneng is 7.9%, which assumes a risk-free rate of 5.2%, a market risk premium of 6.8%, an equity beta of 1.06x (according to Bloomberg), a cost of debt of 6.5%, a target debt-to-capital ratio of 60%, and a 25% corporate-tax rate. At our DCF-based target price, Huaneng would trade at a 2013E P/E of 11x and P/B of 1.5x.

Risks

Key downside risks that could cause the shares to deviate from our target price include: 1) Lower-than-expected power demand; 2) Higher-than-expected coal prices; and 3) Unexpected power tariff cut.

Huaneng Renewables (0958.HK)

Inexpensive; Probable Trough in Earnings in 2012

Buy	1
Price (28 Nov 12)	HK\$1.17
Target price	HK\$1.25
Expected share price return	6.8%
Expected dividend yield	1.4%
Expected total return	8.2%
Market Cap	HK\$9,883M
	US\$1,275M

Price Performance (RIC: 0958.HK, BB: 958 HK)



Asian Utilities

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- **Summary** – We keep our Buy rating on Huaneng Renewables ('HR') for profit growth from new capacity, (ii) gradual improvements in grid curtailment ahead; (iii) inexpensive 0.6x FY13E PB. Unlike its peer Longyuan, HR does not have share issuance risk and its CDM-cut risk has also been mitigated after -71% yoy in 1H12.
- **More output despite utilization cut** – HR's gross generation was +15% yoy to 5.48M MWh in 9M12 for more capacity (+39% yoy to 4,904MW in 2011 and +5% to 5,151MW in 9M12), offsetting utilization -15% yoy to 1,186 hours on grid curtailment (at 12% in 2011, 24% in 1H12 and 19% in 9M12). It plans gross output +23% yoy to 8.5M MWh in 2012E, with average utilization -13.4% yoy to 1,700 hours. 700MW wind farms are to be added each year over 2012-13, with 247MW in 9M12.
- **Parent asset injection on hold** – HR has 65% of the wind farms of China Huaneng Group which has 800MW wind farms which meet HR's investment hurdle rate of 12% IRR, based on 20-year operation at stable tariffs and utilization. But a parental asset injection would not occur soon, as the parental asset could only be disposed of at above 1.0x net book value, which would be too expensive versus HR's 0.6x.
- **CDM revenue dying down** – HR's CDM revenue in 2H12E would be less than that (Rmb65m, -71% yoy) in 1H12, which was based on its 10% CDM contracts at a fixed price average €10/tonne. Provision might have to be made ahead if these fixed-priced contracts are not honored. HR expects CDM revenue to be low in 2013.
- **No share-issuance dilution risk** – HR is a state-owned enterprise hence could only issue new shares at not less than 1.0x PB. It would not be able to issue new shares in Hong Kong where its H-share price is at only 0.7x FY12E PB. Any listing in the A-share market is unlikely in 2013 as (i) the company was set up on 10 Aug 2010 and A-share issuance could only be applied for three years after establishment; and (ii) an application for A-share listing is time consuming, with a long queue.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(RmbM)	(Rmb)	(%)	(x)	(x)	(%)	(%)
2010A	528	0.091	99.8	10.3	1.0	13.4	0.0
2011A	1,023	0.141	54.5	6.7	0.7	12.3	0.0
2012E	546	0.065	-54.1	14.5	0.7	4.7	1.4
2013E	827	0.098	51.6	9.6	0.6	6.8	2.1
2014E	939	0.111	13.5	8.5	0.6	7.2	2.4

Source: Powered by dataCentral

0958.HK: Fiscal year end 31-Dec						Price: HK\$1.17; TP: HK\$1.25; Market Cap: HK\$9,883m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	2,018	3,878	4,334	5,088	5,771	PE (x)	10.3	6.7	14.5	9.6	8.5
Cost of sales	na	na	na	na	na	PB (x)	1.0	0.7	0.7	0.6	0.6
Gross profit	na	na	na	na	na	EV/EBITDA (x)	13.3	8.1	8.3	8.0	7.7
Gross Margin (%)	na	na	na	na	na	FCF yield (%)	-180.9	-94.4	-42.5	-20.2	-13.9
EBITDA (Adj)	1,768	3,521	3,914	4,586	5,184	Dividend yield (%)	0	0	1.4	2.1	2.4
EBITDA Margin (Adj) (%)	87.6	90.8	90.3	90.1	89.8	Payout ratio (%)	0	0	20	20	20
Depreciation	-634	-1,169	-1,619	-1,837	-2,063	ROE (%)	13.4	12.3	4.7	6.8	7.2
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	1,134	2,351	2,295	2,749	3,121	EBITDA	1,768	3,521	3,914	4,586	5,184
EBIT Margin (Adj) (%)	56.2	60.6	52.9	54.0	54.1	Working capital	-485	-921	-961	136	45
Net interest	-493	-1,017	-1,568	-1,688	-1,876	Other	-727	-1,354	-97	-106	-162
Associates	0	0	0	0	0	Operating cashflow	555	1,246	2,856	4,616	5,067
Non-op/Except	0	-195	-50	0	0	Capex	-10,418	-7,699	-6,231	-6,222	-6,172
Pre-tax profit	641	1,139	677	1,061	1,245	Net acq/disposals	1	0	0	0	0
Tax	-32	-32	-47	-106	-162	Other	698	-88	123	85	79
Extraord./Min.Int./Pref.div.	-81	-84	-84	-127	-144	Investing cashflow	-9,720	-7,787	-6,108	-6,137	-6,093
Reported net profit	528	1,023	546	827	939	Dividends paid	0	0	0	-109	-165
Net Margin (%)	26.2	26.4	12.6	16.3	16.3	Financing cashflow	9,673	12,896	26	1,118	880
Core NPAT	528	1,023	546	827	939	Net change in cash	509	6,208	-3,226	-404	-147
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-9,863	-6,453	-3,375	-1,607	-1,105
Reported EPS (Rmb)	0.091	0.141	0.065	0.098	0.111						
Core EPS (Rmb)	0.091	0.141	0.065	0.098	0.111						
DPS (Rmb)	0	0	0.013	0.020	0.022						
CFPS (Rmb)	0.096	0.171	0.338	0.546	0.600						
FCFPS (Rmb)	-1.701	-0.887	-0.400	-0.190	-0.131						
BVPS (Rmb)	0.911	1.342	1.406	1.491	1.583						
Wtd avg ord shares (m)	5,800	7,272	8,447	8,447	8,447						
Wtd avg diluted shares (m)	5,800	7,272	8,447	8,447	8,447						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	101.1	92.2	11.8	17.4	13.4						
EBIT (Adj) (%)	110.2	107.3	-2.4	19.8	13.5						
Core NPAT (%)	99.8	93.7	-46.6	51.6	13.5						
Core EPS (%)	99.8	54.5	-54.1	51.6	13.5						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	1,310	7,577	4,352	3,948	3,801						
Accounts receivables	960	2,010	2,514	2,982	3,405						
Inventory	1	2	2	2	2						
Net fixed & other tangibles	30,571	40,890	45,501	49,887	53,996						
Goodwill & intangibles	395	381	381	381	381						
Financial & other assets	208	669	669	669	669						
Total assets	33,445	51,531	53,420	57,869	62,255						
Accounts payable	0	0	0	0	0						
Short-term debt	5,050	7,042	6,759	6,759	6,759						
Long-term debt	14,970	21,641	23,641	26,641	29,641						
Provisions & other liab	7,306	10,688	10,231	10,835	11,303						
Total liabilities	27,326	39,371	40,631	44,235	47,703						
Shareholders' equity	5,284	11,332	11,877	12,596	13,369						
Minority interests	835	828	912	1,039	1,183						
Total equity	6,119	12,159	12,789	13,634	14,552						
Net debt	18,709	21,106	26,048	29,452	32,599						
Net debt to equity (%)	305.8	173.6	203.7	216.0	224.0						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

DCF TP at HK\$1.25/share

Figure 1. HR's DCF Calculation (Rmbm)

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Net operating cashflow		2,979	4,701	5,146	5,516	6,796	7,361	7,895	8,425	8,965
Less: capex		-6,231	-6,222	-6,172	-6,166	-6,166	-6,166	-6,166	-6,166	-6,166
FCF		-3,252	-1,522	-1,026	-650	630	1,195	1,729	2,259	2,799
Discount factor		1.00	1.07	1.15	1.23	1.32	1.42	1.52	1.63	1.74
Net debt by end 2011	-22,654	-3,252	-1,420	-893	-528	477	844	1,140	1,389	1,606
DCF - Terminal Value										
NAV in 2021E		2,939								
Expected terminal growth rate		5.0%								
Capital Employed on new capex		9.0%								
WACC		7.2%								
Terminal Value		59,577								
Discount factor		1.87								
NPV of terminal value		31,886								
Total DCF										
DCF - in the run up to 2020E	-23,290									
DCF- from 2021E		31,886								
Total DCF		8,596								
Exchange Rate		1.23								
Total DCF (in HK\$m)		10,559								
No. of share (m)		8,447								
DCF / share (HK\$)		1.25								

Source: Company Reports and Citi Research Estimates

At our DCF TP, HR would be traded at 10.6x FY13E PER and 0.7x PB, which does not look excessive.

Figure 2. HR's net profit forecasts – Citi vs. consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	546	776	-30%
FY13E	827	1,108	-25%
FY14E	939	1,538	-39%

Source: Bloomberg and Citi Research Estimates

Valuation against Peers

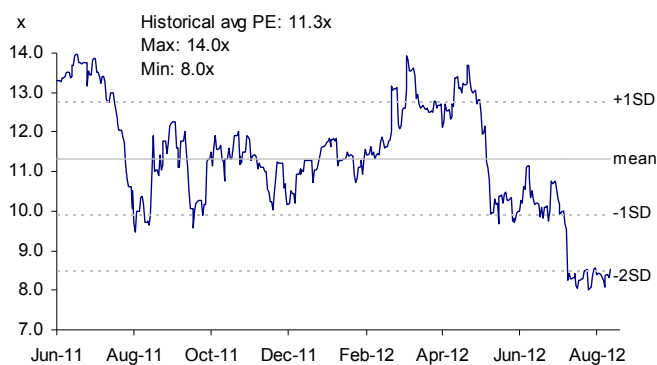
Figure 3. Renewable Energy Companies — Valuation Comparison (Pricing date 28 Aug)

Company	Bloomberg code	Citi Rating	Share Price	Key business	Mkt cap (US\$m)	PER			PEG	PB			ROE		
						11	12E	13E		11	12E	13E	11	12E	13E
EDP Renovaveis	EDPR PL	Buy	EUR3.70	Wind generator	4,180	36.4	23.2	17.0	0.8	0.6	0.6	0.6	1.7%	2.6%	3.4%
Acciona	ANA SM	Neutral	EUR48.00	Renewable energy generator	3,560	17.6	20.3	13.6	n/a	0.6	0.5	0.5	3.7%	2.7%	3.8%
Theolia	TEO FP	NR	EUR1.38	Wind generator	116	n/a	n/a	n/a	n/a	n/a	0.4	0.4	n/a	4.1%	5.0%
Infigen Energy	IFN AU	NR	A\$0.26	Wind generator	208	n/a	n/a	n/a	n/a	n/a	n/a	0.3	n/a	n/a	-6.2%
China Power New Energy	735 HK	NR	HK\$0.40	Renewable energy generator	585	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Datang Renewables	1798 HK	Buy	HK\$0.84	Wind generator	788	6.7	24.3	11.0	-0.3	0.5	0.5	0.5	8.4%	2.2%	4.9%
Huaneng Renewables	958 HK	Buy	HK\$1.17	Wind generator	1,275	6.7	14.5	9.6	-0.7	0.7	0.7	0.6	12.3%	4.7%	6.8%
China Longyuan	916 HK	Buy	HK\$5.19	Wind generator	4,999	11.8	10.9	11.5	10.8	1.2	1.1	1.0	10.8%	10.8%	9.3%
China Suntien	956 HK	Buy	HK\$1.51	Wind generator	631	8.8	9.2	7.7	1.3	0.8	0.8	0.7	9.0%	8.3%	9.7%
Simple average						14.6	17.1	11.7	2.4	0.7	0.7	0.6	7.6%	5.0%	4.6%
Weighted average						17.7	16.2	12.3	3.5	0.8	0.7	0.7	6.2%	5.3%	5.6%

Source: Citi Research Estimates and I/B/E/S for non-rated (NR) companies

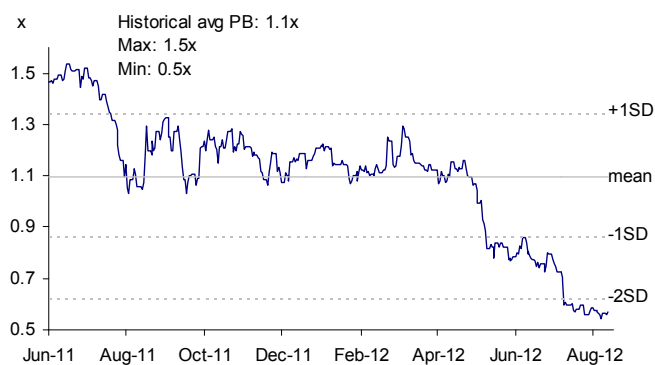
HR's 14.5x FY12E PER and 0.7x PB are similar to the global sector average of 16.2x PER and 0.7x PB. Such a PE is inexpensive in our view given that its 6.8% FY13E ROE is higher than the industry average of 5.6%.

Figure 4. HR's forward PER at historical trough level



Source: Citi Research

Figure 5. HR's forward P/B at historical trough level



Source: Citi Research

HR's 0.7x FY12E PB is low and much less than its historical mean of 1.1x since listing in mid 2011.

Financial Statements and Key Assumptions

Figure 6. Income Statement

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Sales of electricity	1,759	3,160	3,959	4,700	5,371
CERs and VERs income	165	484	130	130	130
Other revenue	95	235	246	257	270
Total revenue	2,018	3,878	4,334	5,088	5,771
Repairs and maintenance	-28	-30	-39	-51	-69
Depreciation	-634	-1,169	-1,619	-1,837	-2,063
Labor cost	-79	-118	-139	-165	-191
Other operating expenses	-143	-210	-242	-286	-328
Operating profit	1,134	2,351	2,295	2,749	3,121
Interest income	22	48	123	85	79
Interest expense	-515	-1,065	-1,690	-1,773	-1,955
Forex gain (loss)	0	-195	-50	0	0
Profit before taxation	641	1,139	677	1,061	1,245
Taxation	-32	-32	-47	-106	-162
Minority interest	-81	-84	-84	-127	-144
Net profit	528	1,023	546	827	939

Source: Company Reports and Citi Research Estimates

Figure 7. Balance Sheet

As at 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Property, plant and equipment	27,803	36,956	41,568	45,953	50,063
Lease prepayments	65	102	102	102	102
Associated companies	0	85	85	85	85
Other tangible fixed assets	2,703	3,831	3,831	3,831	3,831
Goodwill	395	381	381	381	381
Total fixed assets	30,966	41,356	45,968	50,353	54,463
Cash and cash equivalents	1,309	7,518	4,292	3,888	3,742
Restricted deposits	1	59	59	59	59
Inventories	1	2	2	2	2
Trade and bills receivable	960	2,010	2,514	2,982	3,405
Others	208	584	584	584	584
Total current assets	2,479	10,174	7,452	7,516	7,793
Total assets	33,445	51,531	53,420	57,869	62,255
S/T loans	4,818	6,759	6,759	6,759	6,759
Other payable	6,487	9,122	8,381	8,971	9,439
Taxes payable	6	18	19	32	32
Total current liabilities	11,311	15,899	15,158	15,762	16,230
Long term bank loans	13,201	19,257	21,257	24,257	27,257
Other loans	2,813	4,215	4,215	4,215	4,215
Minority interests	835	828	912	1,039	1,183
Total L/T liabilities	16,849	24,300	26,384	29,511	32,655
Share capital	5,800	8,447	8,447	8,447	8,447
Retained earnings	-516	2,885	3,431	4,149	4,923
Total equity	5,284	11,332	11,877	12,596	13,369
Total liabilities & equity	33,445	51,531	53,420	57,869	62,255

Source: Company Reports and Citi Research Estimates

Figure 8. Cash Flow Statement

Year-end 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Cashflow from operations					
EBITDA	1,768	3,521	3,914	4,586	5,184
Working capital changes	100	106	-457	604	468
Other operating cash items	-593	-1,043	-601	-574	-585
Gross operating cash flow	1,275	2,584	2,856	4,616	5,067
Net interest paid	-705	-1,316	0	0	0
Tax paid	-15	-22	0	0	0
Net operating cash flow	555	1,246	2,856	4,616	5,067
Capex	-10,418	-7,699	-6,231	-6,222	-6,172
Free cash flow	-9,863	-6,453	-3,375	-1,607	-1,105
Cash from acquisitions	40	42	0	0	0
Disposal of fixed assets	1	0	0	0	0
Change in temp cash inv.	658	-130	123	85	79
Cashflow from investing	-9,720	-7,787	-6,108	-6,137	-6,093
Increase in Equity	2,259	5,408	0	0	0
Ordinary Dividend Paid	0	0	0	-109	-165
Change in borrowings	7,083	7,916	1,717	3,000	3,000
Other Financing Cash Flow	331	-428	-1,690	-1,664	-1,789
Cash Flow from Financing	9,673	12,896	26	1,227	1,045
Beginning cash balance	819	1,309	7,518	4,292	3,888
Ending cash balance	1,309	7,518	4,292	3,888	3,742

Source: Company Reports and Citi Research Estimates

Figure 9. Key Ratios

Year-end 31 Dec (%)	FY10	FY11	FY12E	FY13E	FY14E
Gross margin (%)	67.2	69.1	61.7	62.9	63.1
Operating margin (%)	56.2	60.6	52.9	54.0	54.1
Net margin (%)	25.9	31.0	13.4	16.0	16.1
Net debt to total capital (x)	0.7	0.4	0.6	0.6	0.6
Net debt to equity (%)	319	186	218	229	237
Interest coverage ratio (x)	2.2	2.1	1.4	1.6	1.6
Cash interest coverage (x)	2.2	2.5	2.1	2.4	2.5
Debt to EBITDA ratio	11.8	8.6	8.2	7.7	7.4
ROA (%)	2.1	2.9	1.1	1.5	1.6
ROE (%)	13.4	14.7	5.1	6.8	7.2
Three-year EPS CAGR (%)	15.7	2.5	-12.8	23.7	17.4

Source: Company Reports and Citi Research Estimates

Figure 10. Key Assumptions

	FY10	FY11	FY12E	FY13E	FY14E
Capacity (MW)					
Total Installed capacity - year end	3,523	4,929	5,654	6,354	7,054
yoy change (%)	127.3	39.9	14.7	12.4	11.0
Total connected capacity - year end	3,523	4,929	5,654	6,354	7,054
% of total installed capacity	100.0	100.0	100.0	100.0	100.0
yoy change (%)	127.3	39.9	14.7	12.4	11.0
Total generating capacity - year end	2,240	4,904	5,654	6,354	7,054
yoy change (%)	95.4	118.9	15.3	12.4	11.0
Consolidated capacity - year end	3,523	4,929	5,654	6,354	7,054
yoy change (%)	127.3	39.9	14.7	12.4	11.0
Attributable capacity - year end	3,340	4,745	5,469	6,170	6,870
yoy change (%)	140.8	42.1	15.3	12.8	11.3
Wind power attributable capacity	3,340	4,721	5,421	6,121	6,821
yoy change (%)	140.8	41.4	14.8	12.9	11.4
Generation volume (M MWh)					
Consolidated gross generation	3.8	6.9	8.6	10.2	11.7
Wind power	3.8	6.8	8.5	10.1	11.6
Others	0.0	0.0	0.0	0.1	0.1
Consolidated net generation	3.4	6.2	7.7	9.2	10.5
yoy change (%)	111.9	81.7	25.0	18.8	14.4
Wind power	3.4	6.2	7.7	9.1	10.4
Others	0.0	0.0	0.0	0.1	0.1
Utilization					
Utilization hours - consolidated wind plants	2,265	1,962	1,667	1,736	1,774
yoy change (%)	-4.2	-13.4	-15.0	4.1	2.2
Utilization rate (%)	25.9	22.4	19.0	19.8	20.3
Tariff (Rmb/MWh)					
Average consolidated tariff	604	596	596	595	594
yoy change (%)	-2.0	-1.4	-0.1	-0.1	-0.1

Source: Company Reports and Citi Research Estimates

Huaneng Renewables

Company description

Huaneng Renewables is the third largest listed wind power generator in China with a market share of 8% in terms of installed wind power capacity at end-2011. Its wind farms are mainly located in Northeast China, East China, Inner Mongolia, South China, North China, and Xinjiang. In FY11, the total installed capacity of Huaneng Renewables was 4,904 MW.

Investment strategy

We rate Huaneng Renewables Buy (1). We expect the company to achieve capacity growth of 700MW per annum in 2012-14E, and believe its earnings quality will continue to improve as its new capacity in the next three years will primarily come from South China and North China, which would diversify grid connection risk. Of the three largest listed wind power generators in China, HR has the highest net profit CAGR and the highest ROE in FY11-13E, which is not reflected in its valuations (being at par with Datang Renewables and at a significant discount to Longyuan on FY12E PER).

Valuation

Our target price for Huaneng Renewables of HK\$1.25 is based on a DCF model, which we believe is suitable for power companies that operate in a relatively stable industry with predictable cash flows. Our model factors in earnings forecasts up to 2020 and a terminal growth rate of 5% thereafter. Our WACC for Huaneng Renewables is 7.2%, which assumes a risk-free rate of 5.2%, a market risk premium of 6.8%, an equity beta of 0.90x, a cost of debt of 6.5%, a target debt-to-capital ratio of 60%, and a 25% corporate-tax rate. At our DCF-based target price, Huaneng Renewables would trade at a 2012E P/E of 15.7x and P/B of 0.7x.

Risks

Key downside risks that could prevent the shares from reaching our target price include: 1) Utilization hours of the wind farms being negatively impacted by unpredictable weather conditions; 2) Less-than-expected capacity addition due to grid connection problems; 3) Poor transparency in the wind power tariff setting mechanism; and 4) RMB rate hikes.

KEPCO (015760.KS)

Some One-off Losses in 4Q12E, But Better Profits in 2013E

Buy	1
Price (28 Nov 12)	W26,700
Target price	W35,000
Expected share price return	31.1%
Expected dividend yield	0.0%
Expected total return	31.1%
Market Cap	W17,140,440M
	US\$15,790M

Price Performance (RIC: 015760.KS, BB: 015760 KS)



Regional Utilities

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■ **Summary** – We keep our Buy call on KEPCO on (i) strong 3Q12 net profit +2.5x yoy to W940bn; (ii) unit fuel cost to drop qoq in 4Q12E; and (iii) a 5% tariff hike in mid-2013E post presidential election in Dec 2012. 4Q12E might be loss-making on one-off losses of W400bn for nuclear unit suspension and up to W850bn decommissioning cost which would not occur in 2013E. Adj PB of 0.5x looks cheap.

■ **Strong 3Q12 results** – KEPCO's net profit was +2.5x yoy to W940bn in 3Q12 and improved from net loss in 2Q12, driven by higher tariff (+11.6% yoy to W107.5/kWh from two tariff hikes and progress rate of residential tariffs). Average unit fuel cost was -14.2% qoq to W50.9/kWh in 3Q12 (with unit coal, LNG and oil costs -11.4% qoq to W116k/tonne, -7.5% qoq to W1,048k/tonne & -1.4% qoq to W874/litre respectively), though +13.5% yoy, offsetting higher purchased power cost. Net forex gain was W306bn in 3Q12 on KRW appreciation, vs. W19bn net forex loss in 3Q11.

■ **Mixed outlook for 4Q12E** – KEPCO guides its unit coal, LNG and oil costs to drop qoq in 4Q12E. Fuel cost represented 50% of 9M12 operating expense. But such benefit would be offset by higher purchased power expense from buying more output from IPPs at high cost (W176.7/kWh in 3Q12) amid tight supply reserve margin (3.8% at end 3Q12). We see recurrent profit at -50% qoq in 4Q12E.

■ **Two potential one-off losses in 4Q12E** – We include a one-off W400bn for fuel cost and power purchase costs in 4Q12E on suspension of reactor units 5 and 6 at Yonggwang nuclear plant replacing unapproved items from 5 Nov to year-end. Subject to government decision, KEPCO might have to book up to W850bn decommissioning cost in 4Q12E, vs. W350bn in 4Q11, on rising standards after Fukushima accident. We assume W500bn decommissioning cost in 4Q12E.

■ **Better profitability in 2013E** – We assume a 5% tariff hike in mid 2013E, after presidential election in Dec 2012. The potential risk of higher decommissioning cost would not happen in 2013E as relevant government review is once every two years.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(WM)	(W)	(%)	(x)	(x)	(%)	(%)
2010A	-72,156	-112	25.4	nm	0.4	-0.2	0.0
2011A	-3,370,464	-5,253	na	-5.1	0.3	-7.1	0.0
2012E	-1,822,045	-2,840	45.9	-9.4	0.3	-3.5	0.0
2013E	1,803,381	2,811	199.0	9.5	0.3	3.4	3.3
2014E	3,540,446	5,518	96.3	4.8	0.3	6.5	6.4

Source: Powered by dataCentral

015760.KS: Fiscal year end 31-Dec						Price: W26,700; TP: W35,000; Market Cap: W17,140,440m; Recomm: Buy					
Profit & Loss (Wm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	39,425,960	43,532,302	49,582,310	55,538,737	60,851,023	PE (x)	nm	-5.1	-9.4	9.5	4.8
Cost of sales	-34,022,447	-42,557,979	-47,924,588	-48,406,269	-51,104,803	PB (x)	0.4	0.3	0.3	0.3	0.3
Gross profit	5,403,513	974,323	1,657,721	7,132,468	9,746,220	EV/EBITDA (x)	7.6	12.6	12.9	6.9	5.7
Gross Margin (%)	13.7	2.2	3.3	12.8	16.0	FCF yield (%)	-30.2	-40.9	-44.1	-22.0	-9.6
EBITDA (Adj)	7,022,999	4,776,383	5,349,642	10,894,914	13,795,868	Dividend yield (%)	0	0	0	3.3	6.4
EBITDA Margin (Adj) (%)	17.8	11.0	10.8	19.6	22.7	Payout ratio (%)	0	0	0	31	31
Depreciation	-4,976,107	-5,658,261	-5,826,377	-6,177,753	-6,704,399	ROE (%)	-0.2	-7.1	-3.5	3.4	6.5
Amortisation	-466,739	-466,739	-466,739	-466,739	-466,739	Cashflow (Wm)					
EBIT (Adj)	1,580,153	-1,348,617	-943,474	4,250,422	6,624,730	EBITDA	7,022,999	4,776,383	5,349,642	10,894,914	13,795,868
EBIT Margin (Adj) (%)	4.0	-3.1	-1.9	7.7	10.9	Working capital	-464,241	361,370	2,453,882	1,755,888	1,462,130
Net interest	-1,505,347	-1,831,954	-2,579,680	-2,873,875	-3,018,286	Other	-920,595	-993,024	-444,558	-1,461,946	-2,116,845
Associates	75,477	123,095	236,212	248,979	260,702	Operating cashflow					
Non-op/Except	286,883	584,350	1,006,881	884,329	951,218	Capex	-10,813,733	-11,144,294	-14,909,000	-14,955,000	-14,793,000
Pre-tax profit	437,166	-2,473,126	-2,280,061	2,509,855	4,818,364	Net acq/disposals	100,766	61,704	0	0	0
Tax	-457,021	-819,871	547,215	-602,365	-1,156,407	Other	-175,713	-113,103	-371,226	-41,435	-39,727
Extraord./Min.Int./Pref.div.	-52,301	-77,467	-89,199	-104,109	-121,511	Investing cashflow					
Reported net profit	-72,156	-3,370,464	-1,822,045	1,803,381	3,540,446	Dividends paid	0	-44,663	0	0	-541,014
Net Margin (%)	-0.2	-7.7	-3.7	3.2	5.8	Financing cashflow					
Core NPAT	-72,156	-3,370,464	-1,822,045	1,803,381	3,540,446	Net change in cash	640,393	-709,364	1,796,846	912,261	487,251
Per share data						Free cashflow to s/holders					
Reported EPS (W)	-112	-5,253	-2,840	2,811	5,518		-5,175,570	-6,999,565	-7,550,035	-3,766,144	-1,651,848
Core EPS (W)	-112	-5,253	-2,840	2,811	5,518						
DPS (W)	0	0	0	869	1,706						
CFPS (W)	8,788	6,460	11,470	17,440	20,483						
FCFPS (W)	-8,067	-10,910	-11,768	-5,870	-2,575						
BVPS (W)	63,934	83,031	80,191	83,002	87,678						
Wtd avg ord shares (m)	642	642	642	642	642						
Wtd avg diluted shares (m)	642	642	642	642	642						
Growth rates											
Sales revenue (%)	16.0	10.4	13.9	12.0	9.6						
EBIT (Adj) (%)	-7.9	-185.3	30.0	550.5	55.9						
Core NPAT (%)	25.4	nm	45.9	199.0	96.3						
Core EPS (%)	25.4	nm	45.9	199.0	96.3						
Balance Sheet (Wm)											
Cash & cash equiv.	2,097,285	1,387,921	3,184,767	4,097,028	4,584,279						
Accounts receivables	5,026,846	6,267,050	7,138,029	7,995,535	8,760,309						
Inventory	3,479,690	3,851,751	4,387,057	4,914,084	5,384,117						
Net fixed & other tangibles	78,899,513	112,384,881	120,902,505	129,108,827	136,617,622						
Goodwill & intangibles	2,412,127	5,099,233	5,242,012	5,373,062	5,507,388						
Financial & other assets	7,694,937	7,477,014	7,875,330	7,957,714	8,042,163						
Total assets	99,610,398	136,467,850	148,729,699	159,446,250	168,895,877						
Accounts payable	3,047,273	3,132,744	5,955,951	6,010,393	6,334,489						
Short-term debt	6,769,293	7,025,910	12,180,670	13,156,638	13,732,606						
Long-term debt	33,107,540	39,403,578	43,769,374	47,513,246	49,657,118						
Provisions & other liab	15,197,691	33,101,668	34,752,601	38,787,379	42,072,129						
Total liabilities	58,121,797	82,663,900	96,658,595	105,467,656	111,796,342						
Shareholders' equity	41,017,863	53,270,296	51,448,251	53,251,631	56,251,063						
Minority interests	470,738	533,654	622,853	726,962	848,473						
Total equity	41,488,601	53,803,950	52,071,104	53,978,594	57,099,536						
Net debt	37,779,548	45,041,567	52,765,277	56,572,856	58,805,444						
Net debt to equity (%)	91.1	83.7	101.3	104.8	103.0						

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For definitions of the items in this table, please click [here](#)

4Q12E and 2013E Business Outlook

Figure 1. KEPCO's unit fuel cost

	1Q12	2Q12	3Q12	2012E	% of KEPCO's fuel cost in 2012E
Coal (W'000/tonne)	134.4	131.6	116.6	117.0	38%
LNG (W'000/tonne)	995.0	1,133.1	1047.6	1,000.0	46%
Oil (W'000/litre)	884.6	886.2	837.6	859.0	12%
Nuclear	n/a	n/a	n/a	n/a	4%
Weighted average/total	n/a	n/a	n/a	n/a	100%

Source: Company Reports and Citi Research Estimates

Nuclear decommissioning cost should increase as long as KEPCO has more nuclear power plants. To determine the decommissioning cost, certain standards exist. The government is trying to raise the standards after the Fukushima nuclear accident. The decommissioning cost, included in other operating expense, was W350bn in 4Q11, W22.5bn in 1Q12, W24.4bn in 2Q12 and W23.9bn in 3Q12. KEPCO expects it to rise to as much as W850bn in 4Q12E. In case the government can not decide the decommissioning cost on or before 31 March 2013, the big decommissioning cost would be postponed to 2013E. The decommissioning cost is reviewed by Ministry of Knowledge and Economy once every two years.

Figure 2. KEPCO: 7 out of 23 operational nuclear units undergoing overhauls

Plant	Capacity	Today's Available Capacity	Completion of O/H (Expected)
Gori#1	587	607	
Gori#2	650	667	
Gori#3	950	0	TBD
Gori#4	950	1,048	
ShinGori#1	1,000	1,052	
ShinGori#2	1,000	1,051	
Younggwang#1	950	1,009	
Younggwang#2	950	999	
Younggwang#3	1,000	0	till Nov.23,2012
Younggwang#4	1,000	1,047	
Younggwang#5	1,000	0	TBD
Younggwang#6	1,000	0	TBD
Uljin#1	950	1,009	
Uljin#2	950	1,009	
Uljin#3	1,000	1,044	
Uljin#4	1,000	0	till Aug 31, 2013
Uljin#5	1,000	1,049	
Uljin#6	1,000	0	till Nov.30,2012
Wolseong#1	679	0	TBD
Wolseong#2	700	707	
Wolseong#3	700	717	
Wolseong#4	700	696	
ShinWolseong#1	1,000	1,039	
23 units	20,716	14,750	7 units under O/H

Source: Company Reports and Citi Research Estimates

KEPCO added 2X1,000MW (Shingor #2, Shinwolsung #1) nuclear power generating units in Aug 2012; it had over 90% utilization in 3Q12. It will have another 2,400MW nuclear unit (1,000MW Shinwolsung#2, 1,400MW Shingor #3) to be added in Apr-Sep 2013E as well as another 1,400MW (Shingor #4) in Sep 2014E, subject to approvals. Commissionings had a few months of delay due to approval and safety checks. The company also has plans to add more nuclear units from 2017E, when the schedule has been delayed from 2016 to 2017E; construction of some of them has started. Addition of nuclear capacity would help

KEPCO mitigate its loss making power purchase business as power purchase unit cost (W176.7/kWh in 3Q12) is more expensive than retail tariff (average W107.47/kWh in 3Q12) there. KEPCO has little choice and needs to buy much output from IPPs due to low supply reserve margin of only 3.8% by end 3Q12.

Kim Joong-kyum, the CEO of KEPCO, has resigned. According to an official at the Ministry of Knowledge Economy, it might be due to issues with the government over an electricity rate spike. An EGM will be held on 17 Dec to re-elect a new CEO. We doubt the change of CEO will have much of a financial impact on the company but it highlights how KEPCO's operating environment is vulnerable to the political climate.

Figure 3. KEPCO's previous tariff hikes

Time	Magnitude
Nov 2008	4.5%
Jun 2009	3.9%
Aug 2010	3.5%
Aug 2011	4.9%
Dec 2011	4.5%
Aug 2012	4.9%

Source: Company Reports and Citi Research Estimates

Figure 4. KEPCO's net profit forecast – Citi vs. consensus

Net profit (Wbn)	Citi	Consensus	% diff
4Q12E	-439	-48	n/a
FY12E	-1,822	-1,643	11%
FY13E	1,803	846	113%
FY14E	3,540	2,062	72%

Source: Bloomberg and Citi Research Estimates

Citi is more optimistic than consensus on KEPCO's earnings in 2013-14E expecting benefit from lower fuel costs driven by reduced commodity prices and commissioning of four new nuclear power generating units in 2H12-FY13E. But our 2H12E net profits based on our recent due diligence are below consensus.

Figure 5. KEPCO's FY13E Net Profit Sensitivity (base case: W1,803bn net profit)

Items	variable	Amount (Wtrn)	Change from base case
Tariff	+1%	+388	+22%
Demand growth *	+1ppt	-497	-28%
Unit coal cost	+1%	-53	-3%
Unit LNG cost	+1%	-98	-5%
Oil price	+1%	-24	-1%
KRW Interest rate	+1ppt	-249	-14%
W/US\$ appreciation (year average)	W10	+296	+16%

* Assuming the additional demand to be procured from IPPs

Source: Company Reports and Citi Research Estimates

DCF & Historical Valuations

Figure 6. KEPCO's DCF calculation

(Wbn)	2012	2013	2014	2015	2016	2017	2018	2019	2020
Operating free cashflow		11,189	13,141	12,756	13,448	14,366	15,271	16,217	17,223
Investing cashflow		-14,955	-14,793	-14,688	-10,435	-10,435	-10,435	-10,435	-10,435
Free cashflow		-3,766	-1,652	-1,932	3,012	3,930	4,836	5,782	6,787
Discount factor		1.00	1.07	1.13	1.21	1.29	1.37	1.46	1.56
PV		-3,766	-1,551	-1,702	2,491	3,051	3,524	3,955	4,358
PV (2013-20E)	10,361								
Net debt	-51,995								
Terminal Value	64,089								
Total	22,455								
No. of share (m)	642								
DCF (W)	35,000								
Assumptions:									
WACC	6.5%								
Cost of equity	8.0%								
Cost of debt (after tax)	4.5%								
Equity beta	0.70								
Risk free rate	4.5%								
Market risk premium	5.0%								
Target debt to capital ratio	33%								
Corporate tax rate	20.0%								
Capital employed on new investments	6.0%								
Terminal growth rate (from 2020E)	2.0%								

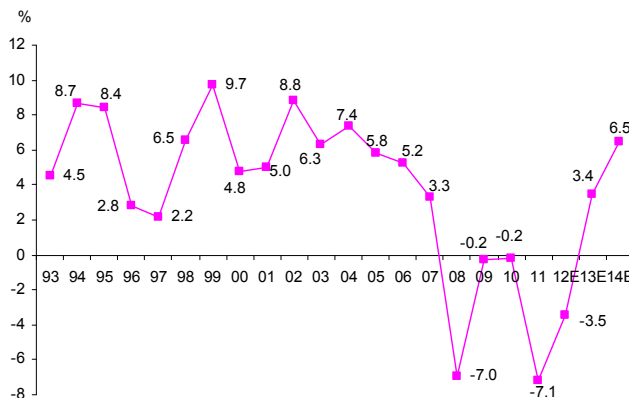
Source: Company Reports and Citi Research Estimates

Figure 7. KEPCO – PB discounted to historical mean



Source: Company Reports and Citi Research Estimates

Figure 8. KEPCO – Improving ROE in FY12-14E



Source: Company Reports and Citi Research Estimates

We think PB is a good reference point for KEPCO as its tariff adjustment is supposed to be determined with reference to its ROIC. Superficially, its FY13E PB is 0.3x; but we adjust it by stripping out a W20.6trn asset revaluation gain from the use of IFRS effective on 1 Jan 2011 since such a gain is excluded from ROIC in tariff setting. After making this adjustment, KEPCO is trading at 0.5x FY13E PB, same as historical average since FY00.

Financial Table & Key Assumptions

Figure 9. Consolidated Income Statement (Wm)

Year to Dec	2010	2011	2012E	2013E	2014E
Operating revenues					
Sale of electric power	37,518,463	40,860,000	45,507,980	50,056,093	53,872,870
Other operating revenues	1,907,497	2,672,302	4,074,329	5,482,644	6,978,153
Total revenue	39,425,960	43,532,302	49,582,310	55,538,737	60,851,023
Fuel	-18,114,518	-21,716,500	-25,047,620	-23,054,181	-24,206,890
Labor	-2,198,633	-2,590,851	-2,806,106	-3,033,493	-3,173,664
Power purchased for resale	-5,382,428	-7,500,100	-9,959,480	-11,525,108	-12,101,363
Depreciation	-4,976,107	-5,658,261	-5,826,377	-6,177,753	-6,704,399
Maintenance	-1,299,733	-1,428,000	-1,536,228	-1,640,500	-1,735,909
Decommissioning	-196,164	-485,000	-680,000	-714,000	-749,700
Research and development	-545,901	-524,000	-596,824	-668,522	-732,466
Amortization of nuclear fuel	-466,739	-466,739	-466,739	-466,739	-466,739
Other	-1,388,125	-2,712,528	-1,602,039	-1,794,495	-1,966,139
Cost of electric power per kWh	-34,568,348 79.0	-43,081,979 93.8	-48,521,413 102.6	-49,074,791 99.4	-51,837,269 100.0
Other operating costs	-1,618,402	-47,243	-53,809	-60,273	-66,038
Selling, general and admin expenses	-1,659,057	-1,751,697	-1,950,562	-2,153,251	-2,322,986
Operating expenses	-37,845,807	-44,880,919	-50,525,783	-51,288,315	-54,226,294
Other revenue		598,303	681,454	763,318	836,330
Operating income	1,580,153	-750,314	-262,020	5,013,740	7,461,059
<i>Operating margin</i>	<i>4.0%</i>	<i>-1.7%</i>	<i>-0.5%</i>	<i>9.0%</i>	<i>12.3%</i>
Other income/expenses					
Interest income	119,477	291,625	146,730	211,749	245,337
Interest expense	-1,624,824	-2,123,579	-2,726,410	-3,085,624	-3,263,623
Investment income from affiliates	75,477	123,095	236,212	248,979	260,702
Foreign exchange translation gains/losses	119,919	-173,627	197,551	0	0
Gain on disposal of investments	4,482	0	0	0	0
Gain on currency swaps	-206,084	75,000	0	0	0
Donations	-30,884	-100,352	-57,149	-64,015	-70,138
Other	399,450	185,026	185,026	185,026	185,026
Income before tax	437,166	-2,473,126	-2,280,061	2,509,855	4,818,364
Income tax					
Current	-321,373	-561,022	547,215	-602,365	-1,156,407
Effective tax rate	-73.5%	22.7%	-24.0%	-24.0%	-24.0%
Deferred	-135,648	-258,849	0	0	0
Effective tax rate	-31.0%	10.5%	0.0%	0.0%	0.0%
Income after tax	-19,855	-3,292,997	-1,732,846	1,907,490	3,661,956
Minority interests	-52,301	-77,467	-89,199	-104,109	-121,511
Net income	-72,156	-3,370,464	-1,822,045	1,803,381	3,540,446
<i>Net margin</i>	<i>-0.2%</i>	<i>-7.7%</i>	<i>-3.7%</i>	<i>3.2%</i>	<i>5.8%</i>

Source: Company Reports and Citi Research Estimates

Figure 10. Consolidated Balance Sheet (Wm)

Year to Dec	2010	2011	2012E	2013E	2014E
Share Capital	3,207,839	3,209,820	3,209,820	3,209,820	3,209,820
Capital surplus	14,764,631	843,758	843,758	843,758	843,758
Retained earnings	23,333,052	49,958,207	48,136,162	49,939,542	52,938,974
Capital adjustment accounts	-287,659	-741,489	-741,489	-741,489	-741,489
Minority interests	470,738	533,654	622,853	726,962	848,473
Shareholders Funds	41,488,601	53,803,950	52,071,104	53,978,594	57,099,536
Long-term debt	33,107,540	39,403,578	43,769,374	47,513,246	49,657,118
Other Liabilities					
Severance and retirement benefits	1,688,189	1,942,994	1,730,000	1,870,091	1,956,404
Reserve for decommissioning costs	5,976,326	7,000,235	7,680,235	8,394,235	9,143,935
Reserve for self-insurance	123,098	0	0	0	0
Currency and interest rate swaps	209,448	464,295	559,500	606,699	633,897
Deferred tax liabilities	990,116	6,786,779	6,786,779	6,786,779	6,786,779
Other long-term liabilities	4,439,694	9,324,852	10,620,796	11,896,695	13,034,615
Total other liabilities	13,426,871	25,519,155	27,377,310	29,554,498	31,555,630
Current liabilities					
Short-term borrowings	457,426	1,173,568	1,213,568	1,253,568	1,293,568
Current portion of long-term debt	6,311,867	5,852,342	10,967,102	11,903,070	12,439,038
Trade notes and accounts payable	3,047,273	3,132,744	5,955,951	6,010,393	6,334,489
Income tax payable	256,198	505,154	-492,722	542,380	1,041,249
Accrued expenses	605,595	1,142,440	1,108,273	1,118,700	1,179,206
Other current liabilities	909,027	5,934,919	6,759,739	7,571,800	8,296,044
Total current liabilities	11,587,386	17,741,167	25,511,911	28,399,912	30,583,594
Total shareholders Funds and liabilities	99,610,398	136,467,850	148,729,699	159,446,250	168,895,877
Fixed assets					
Land	6,402,061	13,161,848	14,505,033	15,890,247	17,260,456
Buildings	6,805,887	11,176,328	12,316,888	13,493,136	14,656,643
Structures	31,862,985	47,437,594	52,278,666	57,271,219	62,209,690
Nuclear fuel	1,090,091	1,482,022	1,873,953	2,265,884	2,657,815
Machinery	17,546,198	37,017,308	34,519,436	40,188,638	46,181,587
Vehicles	32,775	38,217	42,117	46,139	50,118
Construction in progress	19,744,150	19,911,800	27,978,678	28,065,003	27,760,989
Other	2,409,078	11,872,378	13,083,971	14,333,475	15,569,444
Sub-total	135,524,615	142,097,495	156,598,742	171,553,742	186,346,742
Less accumulated depreciation	-49,631,390	-29,304,861	-35,696,237	-42,444,915	-49,729,120
Less construction grants	-6,993,712	-407,753	0	0	0
Net fixed assets	78,899,513	112,384,881	120,902,505	129,108,827	136,617,622
Investments	4,764,710	4,485,356	4,831,910	4,866,784	4,902,535
Deferred income tax assets	1,145,548	372,478	372,478	372,478	372,478
Currency and interest rate swaps	375,757	0	0	0	0
Other non-current assets	2,036,370	5,099,233	5,242,012	5,373,062	5,507,388
Cash and cash equivalents	2,097,285	1,387,921	3,184,767	4,097,028	4,584,279
Short-term financial instruments	671,967	770,539	770,539	770,539	770,539
Accounts receivable less doubtful debts	5,026,846	6,267,050	7,138,029	7,995,535	8,760,309
Inventories	3,479,690	3,851,751	4,387,057	4,914,084	5,384,117
Other current assets	1,112,712	1,848,641	1,900,403	1,947,913	1,996,611
Total current assets	12,388,500	14,125,902	17,380,795	19,725,099	21,495,855
Total assets	99,610,398	136,467,850	148,729,699	159,446,250	168,895,877

Source: Company Reports and Citi Research Estimates

Figure 11. Consolidated Cashflow Statement (Wm)

Year to Dec	2010	2011F	2012E	2013E	2014E
Net profit	-72,156	-3,370,464	-1,822,045	1,803,381	3,540,446
Depreciation and Amortization	6,318,534	6,877,340	6,391,376	6,748,678	7,284,206
Gain on disposal of investments	-4,482	0	0	0	0
Unrealized foreign currency translation losses	-91,187	226,012	-197,551	0	0
Investment income from affiliates	-37,739	-120,398	-118,106	-124,489	-130,351
Minority interests in subsidiaries	52,301	77,467	89,199	104,109	121,511
Change in Working Capital					
Accounts receivable	-481,419	-1,006,645	-870,979	-857,506	-764,774
Inventories	-842,584	-1,411,785	-535,306	-527,026	-470,033
Other current assets	-76,684	-151,043	-51,762	-47,510	-48,698
Trade notes and accounts payable	210,622	710,831	2,823,207	54,442	324,096
Income tax payable	173,832	248,956	-997,876	1,035,102	498,869
Accrued expenses	85,695	30,386	-34,167	10,427	60,505
Other current liabilities	-4,437	907,855	824,820	812,062	724,244
Deferred taxes	562,342	506,819	0	0	0
Other long-term liabilities	-91,608	525,996	1,295,944	1,275,899	1,137,920
Severance and retirement payments	-579,624	-354,774	-212,994	140,091	86,313
Decommissioning costs	-275,238	1,023,909	680,000	714,000	749,700
Other	791,995	-575,733	95,206	47,198	27,198
Cash Flow From Operations	5,638,163	4,144,729	7,358,965	11,188,856	13,141,152
Capital Expenditure	-10,813,733	-11,144,294	-14,909,000	-14,955,000	-14,793,000
Asset Sales	100,766	61,704	0	0	0
Investments	-1,155,326	15,169	-228,448	89,615	94,600
Short-term financial instruments	-205,340	-82,616	0	0	0
Other non-current assets	1,184,953	-45,656	-142,779	-131,050	-134,327
Cash Flow From Investing	-10,888,680	-11,195,693	-15,280,226	-14,996,435	-14,832,727
Long-term debt	5,723,091	6,083,017	9,678,107	4,679,840	2,679,840
Cash dividends	-32,498	-44,663	0	0	-541,014
Short-term borrowings	-207,492	114,015	40,000	40,000	40,000
Minority interests in subsidiaries and others	375,311	189,231	0	0	0
Cash Flow From Financing	5,858,412	6,341,600	9,718,107	4,719,840	2,178,826

Source: Company Reports and Citi Research Estimates

Figure 12. Key assumptions

	FY10	FY11	FY12E	FY13E	FY14E
Korea GDP growth (%)	8.6%	3.6%	3.0%	4.4%	5.0%
Inflation (%)	3.1%	4.0%	2.8%	2.5%	2.5%
Won/US\$, year average	1,165	1,165	1,140	1,140	1,140
Won/US\$, year end	1,150	1,153	1,140	1,140	1,140
National installed capacity (MW)	74,078	76,540	79,342	81,642	85,478
% chg	2.8%	3.3%	3.7%	2.9%	4.7%
Peak demand (MWh)	71,308	72,550	74,509	77,459	80,945
% chg	6.8%	1.7%	2.7%	4.0%	4.5%
Facility reserve margin (%)	6.2%	5.5%	5.2%	5.4%	5.6%
KEPCO's installed capacity (MW)	65,560	66,993	68,866	71,608	75,748
IPP as % of power generation	8.3%	10.7%	11.0%	11.0%	11.0%
Electricity sales (GWh)	437,823	459,328	473,108	493,924	518,621
% chg	11.0%	4.9%	3.0%	4.4%	5.0%
Industrial as % of total sales	54.0%	55.7%	55.7%	55.7%	55.7%
Commercial as % of total sales	22.2%	21.7%	21.7%	21.7%	21.7%
Average tariff (W/KWh)	86.3	90.3	96.2	101.3	103.9
% chg	2.5%	4.6%	6.6%	5.4%	2.5%
Yoy change of unit fuel cost:					
- company average (in W)	13.9%	15.0%	12.4%	-11.8%	0.0%
- coal cost (W'000/ton)	102	119	117	110	110
- LNG (W'000/ton)	779	889	1,000	950	950
- oil (W'000/ton)	613	792	859	829	829
- uranium (Wbn)	960	1,100	1,184	1,243	1,305
Effective tax rate (%)	104.5%	-33.2%	24.0%	24.0%	24.0%
Dividend payout ratio (%)	0.0%	0.0%	0.0%	30.0%	30.0%
Capital expenditure (Wbn)	10,814	11,144	14,909	14,955	14,793
as % of sales	27.4%	25.6%	30.1%	26.9%	24.3%

Source: Company Reports and Citi Research Estimates

KEPCO

Company description

Established in 1961, KEPCO is Korea's monopoly electricity generation, transmission and distribution company. The government holds a 51% stake and the company is supervised by the Ministry of Knowledge Economy (formerly known as Ministry of Commerce, Industry and Energy). At end 2011, KEPCO held 67,006MW or 90% of national installed generating capacity, with the rest owned by a few small-scale IPPs. Industrial customers accounted for 56% of its sales, commercial 22% and residential 18% in 2011. KEPCO's generation mix is 45% from coal, 35% from nuclear, 17% from LNG, 2% from oil and 1% from others. It also purchased 10.7% of its power sale capacity from IPPs in 2011. Meanwhile, it owns 24.5% of Korea Gas, the world's largest LNG importer.

Investment strategy

We rate KEPCO a Buy. KEPCO's FY13E PB is at a sizable discount to its historical average. Also, we expect positive news flow from: 1) a tariff hike in 2013E; and 2) an easing of unit fuel costs from the peak in 2H12 on lower commodity prices and the commissioning of new nuclear units in 2H12-FY13E.

Valuation

Our W35,000 target price is based on DCF, factoring in our earnings forecasts up to 2020 and a terminal value thereafter. We think this method is appropriate for power companies such as KEPCO that work in a relatively static industry with predictable cash flows. Our WACC for KEPCO is 6.5%, which assumes: 1) a risk-free rate of 4.5%; 2) a market risk premium of 5.0%; 3) an equity beta of 0.7x, according to Bloomberg for the last two years; 4) a cost of debt of 4.5%; 5) a target debt to capital ratio of 33.3%; and 6) a 20% corporate tax rate. Beyond 2020, we assume a terminal growth rate of 2%. At our target price, KEPCO would trade at a P/B of 0.42x for 2013E.

Risks

Downside risk factors that could prevent the stock from reaching our target are: 1) higher-than-assumed fuel costs; 2) higher-than-expected power demand growth in Korea, increasing KEPCO's power purchases from IPPs; and 3) later-than-expected tariff hike.

NTPC (NTPC.BO)

Business Remains Steady

Neutral	2
Price (27 Nov 12)	Rs158.60
Target price	Rs173.00
Expected share price return	9.1%
Expected dividend yield	2.6%
Expected total return	11.7%
Market Cap	Rs1,307,731M
	US\$23,474M

Price Performance (RIC: NTPC.BO, BB: NTPC IN)



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- **2QFY13 good but not as good as the headline numbers** — NTPC's 2QFY13 reported PAT at Rs31.4bn was up 30% YoY. However, this included Rs10.2bn of prior period sales (vs Rs7.6bn of prior period sales in 2QFY12). Adjusting for the same recurring PAT at Rs23.6bn was up 24% YoY vs Citi at Rs22.4bn. 1HFY13 Recurring PAT at Rs47.3bn, up 8% YoY, is more or less on track with our FY13E growth expectations of 9%.
- **Regulatory equity block increase in line with expectations** — We had expected a strong increase in commercial capacity and a consequent increase in the regulatory equity block by Rs52.7bn in FY13E. Vis-à-vis our expectations, the regulatory equity block has increased Rs29.1bn already in 1HFY13.
- **But this would be neutralized by lower PLF and PAF** — Coal PLF declined 348bps YoY in 2QFY13 post a decline of 40bps YoY in 1QFY13 and coal PAFs declined 330bps YoY in 2QFY13 post a decline of 149bps YoY in 1QFY13. This would keep overall company RoE under check below 13%. Lower PAFs imply lower availability incentives and lower PLFs imply lower fuel cost savings.
- **Glimmer of hope – Improving coal supply** — Coal India's production up to 15 Oct in FY13 is up 9% YoY partly on account of Coal India's alacrity, and largely on account of the relatively delayed and lesser monsoons this year. This is reflective in NTPC's domestic coal supply +10% YoY in 1QFY13 and +24% YoY in 2QFY13. This provides a glimmer of hope that PLF and PAF could improve in 2HFY13.
- **Business remains steady** — It's around five years since the R-Power IPO and, in the interim period, private IPPs/ investors have questioned NTPC's ability to compete. NTPC saw the impending inflationary risks and chose to make reasonable bids. While NTPC continues to work on adding 14GW of capacity in the XIIth Plan, most private IPPs are making a bee-line to CERC pleading for tariff hikes in PPAs which do not allow such increases post the bid.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Mar	(RsM)	(Rs)	(%)	(x)	(x)	(%)	(%)
2011A	87,236	10.58	7.3	15.0	1.9	13.4	2.4
2012A	87,711	10.64	0.5	14.9	1.8	12.4	2.5
2013E	95,931	11.63	9.4	13.6	1.7	12.6	2.6
2014E	102,955	12.49	7.3	12.7	1.5	12.6	2.8
2015E	109,864	13.32	6.7	11.9	1.4	12.4	2.9

Source: Powered by dataCentral

Petronet LNG (PLNG.BO)

Buy: L(ONG) Story Remains Intact

Buy	1
Price (27 Nov 12)	Rs160.15
Target price	Rs180.00
Expected share price return	12.4%
Expected dividend yield	1.9%
Expected total return	14.3%
Market Cap	Rs120,113M
	US\$2,156M

Price Performance (RIC: PLNG.BO, BB: PLNG IN)



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- **At the forefront of LNG capacity build-out in India** — The demand for gas, esp. in LNG form, is expected to remain robust in India over the next decade. India's total LNG capacity is slated to increase from ~13 MMTPA now to ~50 MMTPA by FY17. PLNG, which received ~80% of India's LNG imports in FY12, will remain a big driver of this growth, controlling ~50% of the expanded capacity after five years.
- **Pricing power, operating leverage key advantages** — PLNG has a first-mover advantage in the Indian LNG space, which has afforded it a high degree of pricing power and the ability to operate at high levels of utilisation (making good use of operating leverage to earn higher margins). PLNG's margins are less a function of concessions/exclusivities and more due to its first mover advantage, which could eventually be driven down by free market forces.
- **Long-term regulatory risks** — While there have been some concerns on the regulation of marketing margins by the PNGRB, we would be less worried on this account for PLNG given that LNG is out of the PNGRB's purview. There has recently been talk of instituting eligibility criteria for new terminals as well as reservation of capacity on 'open access' basis in LNG terminals. While this could be more of a threat to new players, it could lead to fears of tariff regulations for incumbents, which could remain an overhang.
- **Near-term earnings growth subdued, but remain long-term buyers** — PLNG is currently in expansion mode and earnings could remain under pressure given the upcoming commissioning of the Kochi LNG terminal in 4QFY13 (we assume marginal vols of 0.3/1.0 MMT in FY13/14E). This will be followed by the second jetty at Dahej in 2HFY14E (driving meaningful earnings growth in FY15E) and the brownfield 5 MMTPA expansion at Dahej by end-CY15. The 5 MMTPA Gangavaram terminal is in the pre-project activity stage. We remain long-term buyers of the stock, given our long term +ve thesis on the Indian LNG sector and our view of limited regulatory risk for the company. Our TP of Rs180 is based on a Mar-13E DCF value (with Dahej vols at 10.7/10.7 MMT in FY13/14E).

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Mar	(RsM)	(Rs)	(%)	(x)	(x)	(%)	(%)
2011A	6,196	8.26	53.2	19.4	4.5	25.2	1.2
2012A	10,575	14.10	70.7	11.4	3.4	34.1	1.6
2013E	10,309	13.75	-2.5	11.7	2.8	26.4	1.9
2014E	10,245	13.66	-0.6	11.7	2.4	21.9	1.9
2015E	14,455	19.27	41.1	8.3	1.9	25.6	1.9

Source: Powered by dataCentral

PT Perusahaan Gas Negara (PGAS.JK)

Regulatory Uncertainty Persists

Sell	3
Price (26 Nov 12)	Rp4,650
Target price	Rp4,375
from Rp3,400	
Expected share price return	-5.9%
Expected dividend yield	4.1%
Expected total return	-1.8%
Market Cap	Rp112,723,012M
	US\$11,727M

Price Performance (RIC: PGAS.JK, BB: PGAS IJ)



Published on 27 November 2012

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- **Maintain Sell with TP revised +27%** — We maintain Sell based on 1) regulatory uncertainty impacting volumes; and 2) valuations at +1std. The disbandment of BP MIGAS (Upstream Regulator) by the Constitutional Court is likely to disrupt exploration and production activities in Indonesia. We have raised FY12-14E earnings and TP to Rp4,375 (from Rp3,400) on higher spread of USD4.2/btu (prev. USD3.5/btu) following successful increase in tariffs in West Java. However, the stock price in our opinion, has over adjusted, limiting upside from current levels. Based on our analysis, a change of USD0.25/btu in spread and 50mmcf in volume would impact earnings by 7%, each. Reporting currency has been changed to USD, from Rp, and we have updated our numbers accordingly.
- **BP MIGAS disbandment creates volume uncertainty** — The newspaper Jakarta Post has quoted concerns from both former BP MIGAS and Indonesia Petroleum Association (representing foreign E&P companies) officials. They expect this to disrupt both ongoing and new projects, including LNG imports. The employees and responsibilities of BP MIGAS have been transferred to the Ministry of Energy.
- **Next tariff change in April 2013** — The 35% increase in tariff for Western Java customers has been in effect since Sep 2012. The remaining 11.1% (cumulative 50% from Aug '12 level) is due in April 2013. We do not expect any resistance from customers at the time of the second revision, and assume distribution spread of USD4.2/btu.
- **Volume forecast for 2013E revised down to 850mmcf (from 900cf)** — This is based on regulatory uncertainty. However, the 2013E volume is 6% higher than 2012 YTD and in line with mgmt's guidance of 850mmcf for Q4 CY12. We forecast FY13 LNG volume at 100 btu, as per management they will not aggressively market it until full tariff increase is absorbed. Current LNG prices of USD17-20/btu are more expensive than piped gas tariff of USD9.74/btu. We see upside risks if existing E&P companies decide to accelerate production to lock-in profits now, due to uncertainty over their contracts.

Statistical Abstract

FYE	Net Profit	EPS	P/E	D. Yld
31 Dec	(US\$M)	(US¢)	(x)	(%)
2009A	606	2.7	18.1	3.3
2010A	687	2.8	17.1	3.5
2011A	676	2.8	17.4	3.5
2012E	800	3.3	14.7	4.1
2013E	876	3.6	13.4	4.5
2014E	973	4.0	12.1	5.0

Source: Powered by dataCentral

Earnings Revision

FYE	Net Profit	EPS	%	DPS
31 Dec	(US\$M)	(US¢)	chg	(US¢)
2012E	800	3.3	36.7	2.0
Prev	586	2.4		1.4
2013E	876	3.6	35.3	2.2
Prev	648	2.7		1.6
2014E	973	4.0	28.9	2.4
Prev	755	3.1		1.9

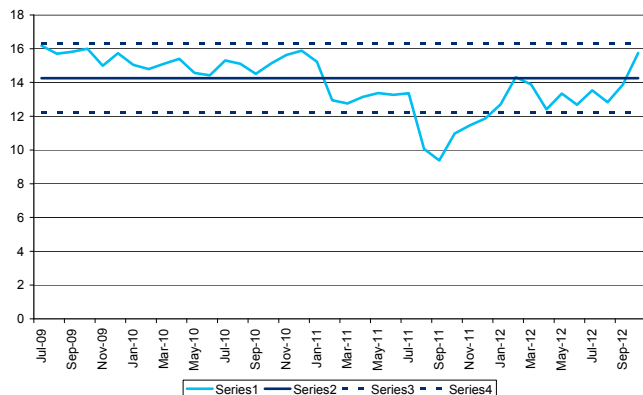
Source: Powered by dataCentral

PGAS.JK: Fiscal year end 31-Dec						Price: Rp4,650; TP: Rp4,375; Market Cap: Rp112,723,012m; Recomm: Sell					
Profit & Loss (US\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	2,176	2,231	2,565	3,295	3,645	PE (x)	17.1	17.3	14.7	13.4	12.0
Cost of sales	-795	-889	-1,120	-1,710	-1,925	PB (x)	7.7	6.8	5.6	4.6	3.9
Gross profit	1,381	1,342	1,445	1,586	1,720	EV/EBITDA (x)	11.0	12.0	10.1	8.8	7.6
Gross Margin (%)	63.5	60.2	56.3	48.1	47.2	FCF yield (%)	2.9	3.2	5.4	4.8	6.1
EBITDA (Adj)	1,106	991	1,151	1,265	1,400	Dividend yield (%)	3.5	3.5	4.1	4.5	5.0
EBITDA Margin (Adj) (%)	50.8	44.4	44.9	38.4	38.4	Payout ratio (%)	60	60	60	60	60
Depreciation	-111	-111	-179	-186	-194	ROE (%)	49.9	41.6	41.9	37.7	35.0
Amortisation	0	0	0	0	0	Cashflow (US\$m)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	995	880	972	1,079	1,206	EBITDA	1,106	991	1,151	1,265	1,400
EBIT Margin (Adj) (%)	45.7	39.5	37.9	32.7	33.1	Working capital	57	-46	61	-70	-30
Net interest	-14	7	-1	12	23	Other	-271	-181	-141	-191	-219
Associates	0	0	0	0	0	Operating cashflow	892	764	1,071	1,003	1,151
Non-op/Except	-94	-15	43	15	0	Capex	-547	-386	-436	-437	-437
Pre-tax profit	888	873	1,014	1,106	1,229	Net acq/disposals	427	377	301	300	300
Tax	-176	-175	-206	-221	-246	Other	-82	2	-56	-11	-12
Extraord./Min.Int./Pref.div.	-25	-21	-8	-9	-10	Investing cashflow	-202	-7	-191	-149	-149
Reported net profit	687	676	800	876	973	Dividends paid	-366	-412	-411	-483	-530
Net Margin (%)	31.6	30.3	31.2	26.6	26.7	Financing cashflow	-184	-828	-548	-509	-635
Core NPAT	687	676	800	876	973	Net change in cash	532	-59	332	349	370
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	345	378	635	566	713
Reported EPS (¢)	2.8	2.8	3.3	3.6	4.0						
Core EPS (¢)	2.8	2.8	3.3	3.6	4.0						
DPS (¢)	1.7	1.7	2.0	2.2	2.4						
CFPS (¢)	3.7	3.2	4.4	4.1	4.7						
FCFPS (¢)	1.4	1.6	2.6	2.3	2.9						
BVPS (¢)	6.3	7.2	8.6	10.6	12.4						
Wtd avg ord shares (m)	24,235	24,235	24,235	24,235	24,235						
Wtd avg diluted shares (m)	24,235	24,235	24,235	24,235	24,235						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	24.6	2.5	15.0	28.5	10.6						
EBIT (Adj) (%)	33.3	-11.5	10.4	10.9	11.8						
Core NPAT (%)	13.3	-1.5	18.3	9.5	11.1						
Core EPS (%)	6.1	-1.5	18.3	9.5	11.1						
Balance Sheet (US\$m)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	1,232	1,172	1,509	1,858	2,228						
Accounts receivables	210	219	274	344	373						
Inventory	3	2	2	2	2						
Net fixed & other tangibles	1,998	1,894	1,907	1,869	1,824						
Goodwill & intangibles	0	0	0	0	0						
Financial & other assets	97	114	66	79	87						
Total assets	3,540	3,402	3,758	4,152	4,514						
Accounts payable	72	61	152	160	166						
Short-term debt	227	98	96	96	77						
Long-term debt	1,195	994	889	793	716						
Provisions & other liab	396	368	380	388	393						
Total liabilities	1,890	1,521	1,517	1,436	1,351						
Shareholders' equity	1,515	1,734	2,089	2,561	3,005						
Minority interests	136	147	152	155	158						
Total equity	1,651	1,881	2,241	2,716	3,163						
Net debt	190	-80	-525	-970	-1,435						
Net debt to equity (%)	11.5	-4.2	-23.4	-35.7	-45.4						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

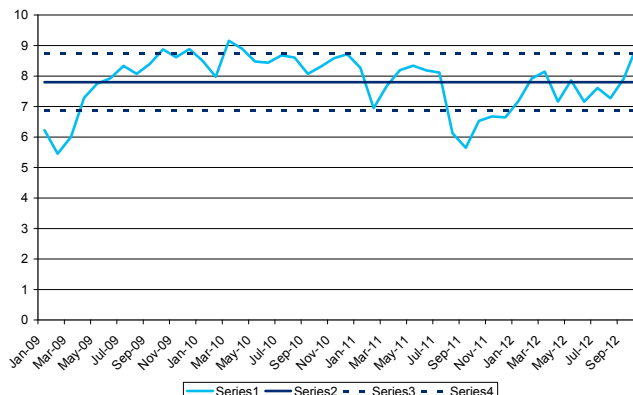
Valuation Trends

Figure 1. Consensus Forward PER (x)



Source: MSCI Factset

Figure 2. Consensus Forward EV/EBITDA (x)



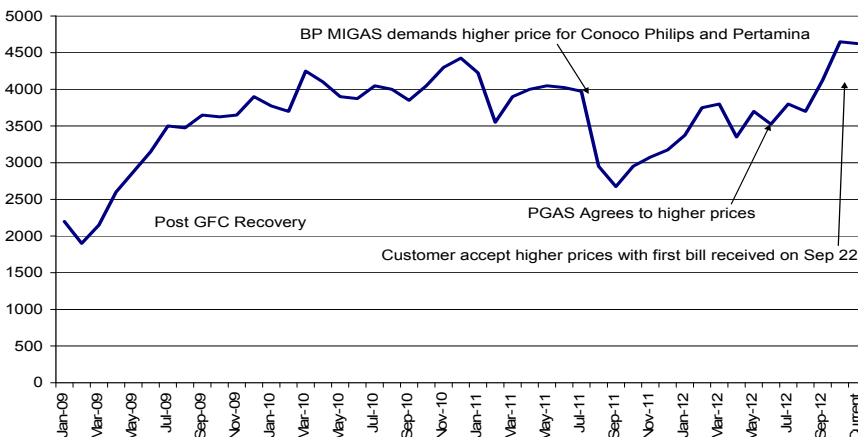
Source: MSCI Factset

Figure 3. Forward EV/EBITDA Premium /(Disc)- Relative to MSCI Asia Pac-Utility



Source: MSCI Factset

Figure 4. Share Price Trend (Rp) and Events



Source: Bloomberg and Citi Research

Target Price Calculation

We use forward EV/EBITDA for our Target Price calculation. We believe using the mean multiple from 2009 onwards is more appropriate (vs. the mean over 2005-11 previously) as this period 1) has relatively less volatility, 2) incorporates regulatory uncertainties; and 3) has adjusted for slower volume growth.

Figure 5. Target Price Calculation

Mean EV/EBITDA	7.8
1 Std	0.9
Target Multiple (Based on Mean)	7.8
EBITDA (2013F) – USD Mn	1,265
Enterprise Value – USD Mn	9,866
Net Cash – USD Mn	962
Value of Equity – USD Mn	10,828
Shares Issue (Bn)	24.2
Per Share (USD)	0.45
Exchange Rate (IDR/USD) – Citi forecast	9800
Rp/Share	4,375

Source: Company Reports and Citi Research Estimates

Earnings Model

We believe the key earnings driver is now volume, as we expect spread has settled at USD4.2/btu.

Figure 6. Key Earnings Assumptions

		2011A	2012A (9M)	2012E	2013E
Distribution Volume	mmcf/d	795	800	819	850
Selling Price	USD/btu	7.1	7.6	8.02	9.71
Gas Cost	USD/btu	3.1	3.4	3.80	5.51
Spread	USD/btu	4.0	4.2	4.2	4.2
Transmission Volume	mmcf/d	845	873	885	885
Margin	USD/btu	0.57	0.62	0.61	0.61
LNG Volume	mmcf/d				100
Spread	USD/btu				3.0

Source: Company Reports and Citi Research Estimates

Figure 7. Earnings Sensitivity to Spreads – 2013E

In USD Mn	Base Case	+ USD0.25/btu	Change
Spread Assumption	4.20	4.20	
EBITDA	1,265	1,342	6.1%
Net Profit	876	937	7.0%

Source: Citi Research Estimates

Figure 8. Earnings Sensitivity to Volume – 2013E

In USD Mn	Base Case	+ 50mmcf/d	Change
Volume Assumption	850.0	900.0	
EBITDA	1,265	1,341	6.0%
Net Profit	876	936	6.8%

Source: Citi Research Estimates

PGAS and the Ministry of Energy have reached an understanding that the selling prices will be raised in two phases, to compensate for higher purchase price to Pertamina and Conoco Phillips. Cumulative increase for customers will be 50%, of which 35% was successfully implemented in Sep. The remaining will be due in April 2013.

Figure 9. Tariff Revision as per agreement with BP Migas

Producer	Buyer	Volume btu	Current Tariff USD	Revised Tariff USD
Pertamina	PGN	250	2.23	5.5 (2012), 6 (2013), +4% pa till 2017
ConocoPhillips (Grissik)	PGN	412	1.85	5.61 (till March 2013), 6 (till March 2014), 6.5, +3% pa till 2023
ConocoPhillips Indonesia	Petronas	263	3.10	6, +2.5%pa
PHE WMO	PLN	123	3.00	6.05, +3% pa
Medco E&P Indonesia	PLN Indralaya	19-24	2.68	4.3-5.3
Pertamina EP	Pusri	166	3.64	5.89, +3% pa
JOB Pertamina Talisman	Pembangunan	10	5.40	5.75, +3% pa

Source: Bisnis Indonesia

BP MIGAS Abolishment

The Constitutional Court of Indonesia on November 13, 2012 abolished the Upstream Regulator, BP MIGAS. The decision was taken because the Court believes that BP MIGAS instead of running natural resources for the people's welfare, was handing the natural resources over to state-owned or private companies instead, through production-sharing contracts.

These developments in our opinion, are likely to raise concerns for foreign exploration and production companies. While there are both negatives and positives for PGAS, we believe the uncertainty is negative given the risk/return profile of the sector, particularly with valuations already at +1std.

Government ministers have reassured that the decision will not impact the current agreements. Both staff and functions of BP MIGAS have been transferred to the Ministry of Energy.

However, both ex-officials of BP MIGAS and Indonesia Petroleum Association have raised concerns over near and long term impacts. They even stated that this may also hinder liquefied natural gas (LNG) ship shipments from Indonesia. Quoting Jakarta Post, there are over 29 concessions slated to expire between 2013 and 2021 such as Mahakam Block in East Kalimantan.

From the perspective of PGAS, we believe:

- Negatives include: New E&P activity should slow down impacting additional future gas. International LNG exporters may be more cautious dealing with Indonesian importers.
- Potential positives include: Existing E&P companies may want to maximize their production in the near future, due to uncertainty of further changes. Some may opt to sell their Indonesian assets at lower valuations, with PGAS as potential acquirer.

Quants View – Glamour

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Data as of: 23-Nov-12

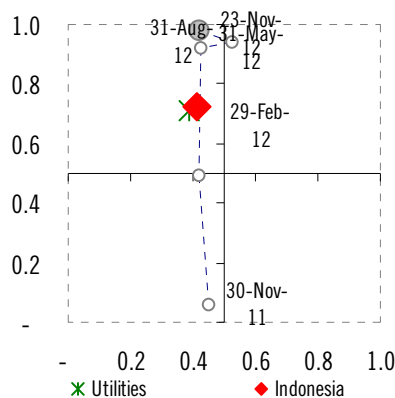
Radar Screen Quadrant Definitions

Glamour <i>Poor relative value but superior relative momentum</i>	Attractive <i>Superior relative value and superior relative momentum</i>
Unattractive <i>Poor relative value and poor relative momentum</i>	Contrarian <i>Superior relative value but poor relative momentum</i>

Perusahaan Gas Negara Pt lies in the Glamour quadrant of our Value-Momentum map with strong momentum but relatively weak value scores. The stock has moved from the Attractive quadrant to the Glamour quadrant in the past three months indicating strong momentum with weakening valuation scores. Compared with its peers in the Utilities sector, Perusahaan Gas Negara Pt fares better on the valuation metric and on the momentum metric. Similarly, compared with its peers in its home market of Indonesia, Perusahaan Gas Negara Pt fares better on the valuation metric and on the momentum metric.

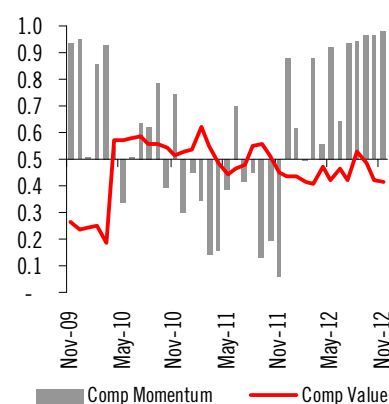
From a macro perspective, Perusahaan Gas Negara Pt has a low beta to the region, so can be expected to hold its own given a decline in the regional market. It is also likely to benefit from a weaker US dollar.

Figure 10. Radar Quadrant Chart History



Source: Citi Research

Figure 11. Radar Valuation and Momentum Scores



Source: Citi Research

Figure 12. Radar Model Inputs

IBES EPS (Actual and Estimates)

FY(-2)	262.00	Implied Trend Growth (%)	6.25
FY(-1)	257.00	Trailing PE (x)	15.67
FY0	245.00	Implied Cost of Debt (%)	3.96
FY1	291.26	Standardised MCap	0.59
FY2	331.50		

Note: Standardised MCap calculated as a Z score – (mkt cap - mean)/std dev – capped at 3

Source: Citi Research, I/B/E/S

Figure 13. Stock Performance Sensitivity to Key Macro Factors

Region	0.68	Commodity ex Oil	(0.02)
Widening APACxJ CDS	(0.11)	Rising Oil Prices	(0.13)
Growth	(0.42)	Rising Asian IR's	(0.03)
Value	(0.71)	Rising EM Yields	(0.23)
Small Caps Outperform Large Caps	0.02	Weaker US\$ (vs Asia)	1.60
Widening US Credit Spreads	0.07	Weaker ¥ (vs US\$)	0.40

Source: Citi Research

PT Perusahaan Gas Negara

Company description

Perusahaan Gas Negara listed on the Jakarta and Surabaya Stock Exchanges on 15 December 2003. It is the leading gas transmitter and distributor in Indonesia. The transmission pipelines primarily serve Grissik-Duri (north Sumatera) and Grissik-Singapore. The distribution business is in Java and Sumatera. Revenues for transmission are on a toll-like basis. P Gas is paid for volume of gas transported. It acts as a merchant on the distribution business, sourcing for supply and on-selling to commercial and retail customers. Pricing is primarily in US\$.

Investment strategy

We rate the shares Sell based on valuations and regulatory uncertainty. The disbandment of Upstream Regulator by the Constitutional Court is likely to disrupt exploration and production activities in Indonesia. Spread uncertainty has reduced as customers have accepted the 35% increase in tariffs from 20 Sep, with the spread maintained at USD4.2/btu. Another 15% increase is due from April and we assume the spread will remain at USD4.2/btu (prev. USD3.5/btu). The share price has over adjusted on the tariff hikes, in our opinion. It trades at 1std above mean on both PER and EV/EBITDA. We use valuations from 2009 onwards as prior data was extremely volatile, with large standard deviations.

Valuation

Our Rp4,375 target price is based on forward EV/EBTIDA as we believe this captures the near term earnings trend, while there is significant uncertainty over long term volume trends. Our target EV/EBITDA multiple of 7.8x is the mean since 2009 (to reflect reduced volatility since then, with standard deviation at 0.9x).

Risks

Upside risks to our recommendation are: 1) Higher volumes from existing suppliers 2) Higher volumes of imported LNG; and 3) Acquisition of cheap assets from large cash balance of USD1.4bn. These risks could prevent the stock from attaining our target price.

Power Assets (0006.HK)

Proxy for Bond Investments

Buy	1
Price (28 Nov 12)	HK\$67.85
Target price	HK\$77.00
<i>from HK\$68.50</i>	
Expected share price return	13.5%
Expected dividend yield	3.7%
Expected total return	17.2%
Market Cap	HK\$144,810M
	US\$18,685M

Price Performance (RIC: 0006.HK, BB: 6 HK)



Asian Utilities

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■ **Summary** – We prefer Power Assets Holdings ('PAH') to CLP as we expect PAH would gain from adding a new NG fired unit in HK in 2014-18E, and potential M&As should increase FY13E EPS by up to 11.3%. SOC mid-term review in 2013 should largely reflect the status quo, hence we see limited risks. Despite richer valuations after the recent rally, we view PAH, which has the lowest equity beta among all HSI Constituent stocks, as attractive with expanded yield premium against high quality bonds. Our DCF TP is +12% to HK\$77 on valuation base year rollover.

■ **To add a new NG power unit in 2014-18E** – PAH's HK generation mix is 70% from coal and 30% from NG; the latter comes from 710MW base load units, including L9 unit (335MW) and GT5-7 units (3X125MW) built in 2006 and 1991 respectively. PAH could meet local emissions cap in the run up to 2017 but would need a new L10 NG unit in the next capex period (2014-18) to fulfill its 2020 emissions target. We assume PAH's SOC capex of HK\$14.5bn in 2014-18, 18% above that in 2009-13E. But HK Govt might propose lower returns vs. current 9.99% ROA for the new unit.

■ **Mild profit cut due to timing difference in Australia** – We cut PAH's 2012E net profit by HK\$100m or 1% factoring a one-off smart meter installation expense in 2H12E under its power grid company CHEDHA in Australia. The installation cost will likely be paid by end users later, hence profitability should recover in 2013E.

■ **Catalysts from potential acquisitions** – PAH's FY13E EPS could +11.3% by spending HK\$30bn on M&As, while keeping its 'A+' rating, at 14% ROE and 4% funding cost. We forecast its consolidated net debt/equity and see-through gearing (incl. HK\$40bn non-resource debts) would be 26% & 86% respectively by end 2012.

■ **Attractive yield vs. high quality bonds** – PAH has low equity beta (0.4x based on Bloomberg) as it operates regulated utilities in developed countries. Its 4.0% FY13E yield is above the 10-year US\$ Treasury note yield and HK\$ exchange bond fund note by 204bps and 308bps respectively, versus an average of 137bps and 220bps in the last five years, and 119bps and 170bps in the last 10 years.

Statistical Abstract

FYE	Net Profit	EPS	P/E	D. Yld
31 Dec	(HK\$M)	(HK\$)	(x)	(%)
2009A	6,697	3.14	21.6	3.1
2010A	7,194	3.37	20.1	3.1
2011A	9,075	4.25	16.0	3.4
2012E	9,827	4.60	14.7	3.7
2013E	10,590	4.96	13.7	4.0
2014E	10,850	5.08	13.3	4.1

Source: Powered by dataCentral

Earnings Revision

FYE	Net Profit	EPS	%	DPS
31 Dec	(HK\$M)	(HK\$)	chg	(HK\$)
2012E	9,827	4.60	-0.9	2.53
Prev	9,921	4.65		2.56
2013E	10,590	4.96	0.3	2.73
Prev	10,563	4.95		2.72
2014E	10,850	5.08	0.1	2.80
Prev	10,842	5.08		2.79

Source: Powered by dataCentral

0006.HK: Fiscal year end 31-Dec						Price: HK\$67.85; TP: HK\$77.00; Market Cap: HK\$144,810m; Recomm: Buy					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	10,371	10,201	10,418	10,836	11,489	PE (x)	20.1	16.0	14.7	13.7	13.3
Cost of sales	-3,938	-4,032	-4,477	-4,323	-4,500	PB (x)	2.6	2.5	2.3	2.2	2.0
Gross profit	6,433	6,169	5,942	6,513	6,989	EV/EBITDA (x)	17.5	18.9	18.8	17.0	15.5
Gross Margin (%)	62.0	60.5	57.0	60.1	60.8	FCF yield (%)	3.6	3.7	2.2	4.2	4.3
EBITDA (Adj)	7,512	6,780	6,691	7,217	7,609	Dividend yield (%)	3.1	3.4	3.7	4.0	4.1
EBITDA Margin (Adj) (%)	72.4	66.5	64.2	66.6	66.2	Payout ratio (%)	63	55	55	55	55
Depreciation	-1,734	-1,777	-1,837	-1,850	-1,861	ROE (%)	13.3	15.9	16.4	16.4	15.6
Amortisation	0	0	0	0	0	Cashflow (HK\$m)					
EBIT (Adj)	5,778	5,003	4,854	5,367	5,748	EBITDA	7,512	6,780	6,691	7,217	7,609
EBIT Margin (Adj) (%)	55.7	49.0	46.6	49.5	50.0	Working capital	99	530	-1,746	224	153
Net interest	513	692	900	1,024	1,099	Other	-139	518	555	530	587
Associates	1,899	4,193	4,721	5,108	5,325	Operating cashflow	7,472	7,828	5,501	7,971	8,349
Non-op/Except	0	0	0	0	0	Capex	-2,238	-2,480	-2,310	-1,922	-2,191
Pre-tax profit	8,190	9,888	10,475	11,499	12,172	Net acq/disposals	-9,354	-55	-2,533	-83	-81
Tax	-937	-858	-867	-963	-1,032	Other	-3,625	2,653	2,642	2,739	2,854
Extraord./Min.Int./Pref.div.	-59	45	219	54	-291	Investing cashflow	-15,217	118	-2,200	735	582
Reported net profit	7,194	9,075	9,827	10,590	10,850	Dividends paid	-4,503	-4,503	-5,405	-5,824	-5,967
Net Margin (%)	69.4	89.0	94.3	97.7	94.4	Financing cashflow	8,889	-7,307	736	-6,628	-7,730
Core NPAT	7,194	9,075	9,827	10,590	10,850	Net change in cash	1,140	639	4,037	2,078	1,201
Per share data						Free cashflow to s/holders					
Reported EPS (\$)	3.37	4.25	4.60	4.96	5.08		5,234	5,348	3,191	6,049	6,158
Core EPS (\$)	3.37	4.25	4.60	4.96	5.08						
DPS (\$)	2.11	2.32	2.53	2.73	2.80						
CFPS (\$)	3.50	3.67	2.58	3.74	3.91						
FCFPS (\$)	2.45	2.51	1.50	2.83	2.89						
BVPS (\$)	26.31	27.12	29.19	31.42	33.71						
Wtd avg ord shares (m)	2,134	2,134	2,134	2,134	2,134						
Wtd avg diluted shares (m)	2,134	2,134	2,134	2,134	2,134						
Growth rates											
Sales revenue (%)	-0.2	-1.6	2.1	4.0	6.0						
EBIT (Adj) (%)	-1.0	-13.4	-3.0	10.6	7.1						
Core NPAT (%)	7.4	26.1	8.3	7.8	2.5						
Core EPS (%)	7.4	26.1	8.3	7.8	2.5						
Balance Sheet (HK\$m)											
Cash & cash equiv.	5,839	6,123	9,639	11,248	11,930						
Accounts receivables	1,182	1,101	1,126	1,171	1,241						
Inventory	747	1,115	1,147	980	981						
Net fixed & other tangibles	48,910	49,659	50,216	50,370	50,781						
Goodwill & intangibles	0	0	0	0	0						
Financial & other assets	36,025	36,732	41,515	43,813	46,101						
Total assets	92,703	94,730	103,643	107,583	111,034						
Accounts payable	1,865	3,669	2,236	2,268	2,309						
Short-term debt	8,461	617	4,727	5,955	5,795						
Long-term debt	17,312	23,009	25,041	23,010	21,408						
Provisions & other liab	8,928	9,562	9,343	9,290	9,580						
Total liabilities	36,566	36,857	41,348	40,523	39,091						
Shareholders' equity	56,137	57,873	62,295	67,060	71,943						
Minority interests	0	0	0	0	0						
Total equity	56,137	57,873	62,295	67,060	71,943						
Net debt	19,934	17,503	20,129	17,717	15,273						
Net debt to equity (%)	35.5	30.2	32.3	26.4	21.2						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Net Profit Breakdown by Geography

Figure 1. PAH's net profit breakdown

	2011	1H12	2H12E	2012E
Overseas profit:				
Australia	843	435	326	761
UK:				
- Northern Gas Network	500	325	216	541
- Western Gas Network			67	67
- Power Networks	2,731	1,751	1,519	3,270
- Seabank Power Station	327	66	106	172
- UK total	3,558	2,142	1,907	4,049
China	507	203	319	522
Other infrastructures	410	137	243	380
Other activities	-723	-320	-190	-510
Sub-total	4,595	2,597	2,605	5,202
Domestic profit from HK	4,480	1,736	2,888	4,624
Total	9,075	4,333	5,494	9,827

Source: Company Reports and Citi Research Estimates

Potential Benefit from Acquisitions

Figure 2. Scenario Analysis: PAH's potential 11.3% FY13E net profit enhancement effect from potential acquisitions

Capex of potential acquisitions in FY12E	FY13E net profit	Profit uplift from base case	Net debt/equity ratio by end FY13E (excluding off B/S debt)	Net debt/equity ratio by end FY13E (including off B/S debt)
Nil (base case)	10,590	N/A	26.4%	85.5%
HK\$10bn	10,990	3.8%	32.4%	100.4%
HK\$20bn	11,390	7.6%	38.3%	115.3%
HK\$30bn	11,790	11.3%	44.3%	130.2%

* Assuming capex to be financed 60% by debt and 40% by equity; ** Assuming 14% ROE for potential acquisitions and 4% PAH equity funding cost

*** PAH had HK\$35bn off balance sheet borrowing by end FY11

Source: Company Reports and Citi Research Estimates

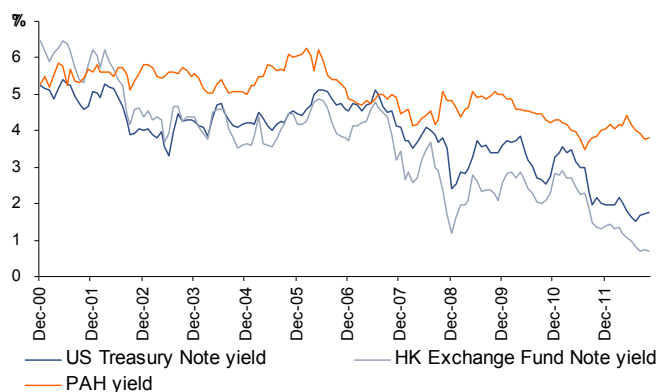
Figure 3. PAH's proportional off balance debts under its project companies

	Country	Loans in local currencies (m)	Exchange rate vs.HK\$	Stake	Loans in HK\$m	S&P rating
Northern Gas Network	The UK	1,100	12.19	41.3%	5,536	BBB+
UK Power Network	The UK	3,200	12.19	40%	17,060	BBB+
WWU	The UK	1,312	12.19	30%	4,798	A-/BBB
Australian power grids	Australia	5,500	7.94	27.9%	12,197	A-
Total					39,591	

Source: Company Reports and Citi Research Estimates

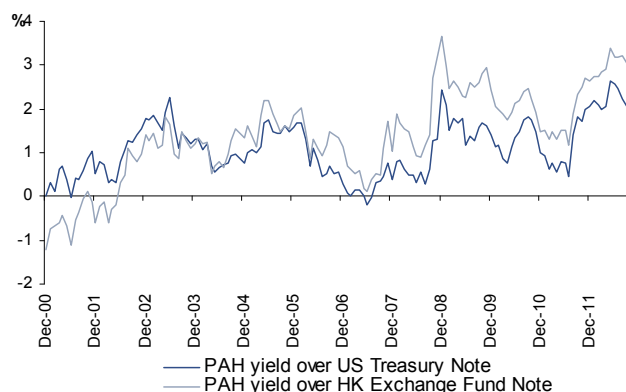
Expanded Yield vs. High Quality Bonds

Figure 4. Comparison of Power Assets' yield, US\$ treasury note yield & HK\$ exchange fund notes yield



Source: Citi Research, Federal Reserve, Hong Kong Monetary Authority

Figure 5. Widened spread of Power Asset's yield against US\$ treasury note yield and HK\$ exchange fund notes yield



Source: Citi Research

The premium of PAH's dividend yield against 10-year US\$ treasury note yield and HK\$ exchange bond fund note has been widened to 204bps and 308bps respectively, versus an average of 137bps and 220bps in the last five years as well as 119bps and 170bps in the last 10 years.

Figure 6. HK Utilities – Valuation Comparison (Price as at 28 Nov 2012)

	RIC	Rating	Price		PE		P/B		Net D/E		Yield	
			Current	Target	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E	FY12E	FY13E
HKCG	0003.HK	Sell	HK\$20.90	HK\$17.00	23.7	25.6	4.0	3.7	30.7%	26.8%	1.7%	2.0%
CLP	0002.HK	Neutral	HK\$67.10	HK\$70.00	16.3	13.1	1.9	1.8	81.2%	76.9%	3.8%	4.0%
Power Asset	0006.HK	Buy	HK\$67.85	HK\$77.00	14.7	13.7	2.3	2.2	30.2%	26.4%	3.7%	4.0%
CKI	1038.HK	Buy	HK\$46.30	HK\$54.50	11.8	10.7	1.8	1.7	14.3%	-1.2%	3.7%	3.9%
Average					16.6	15.8	2.5	2.3	39.1%	32.2%	3.2%	3.5%

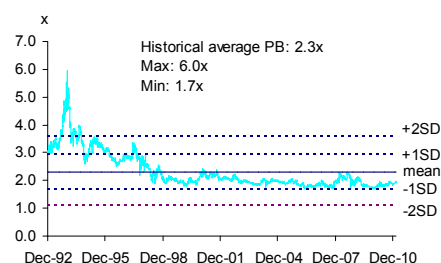
Source: Citi Research Estimates

Figure 7. PAH's historical rolling PER Chart



Source: Company Reports and Citi Research Estimates

Figure 8. PAH's historical rolling PB Chart



Source: Company Reports and Citi Research Estimates

Figure 9. PAH's historical rolling div. yield



Source: Company Reports and Citi Research Estimates

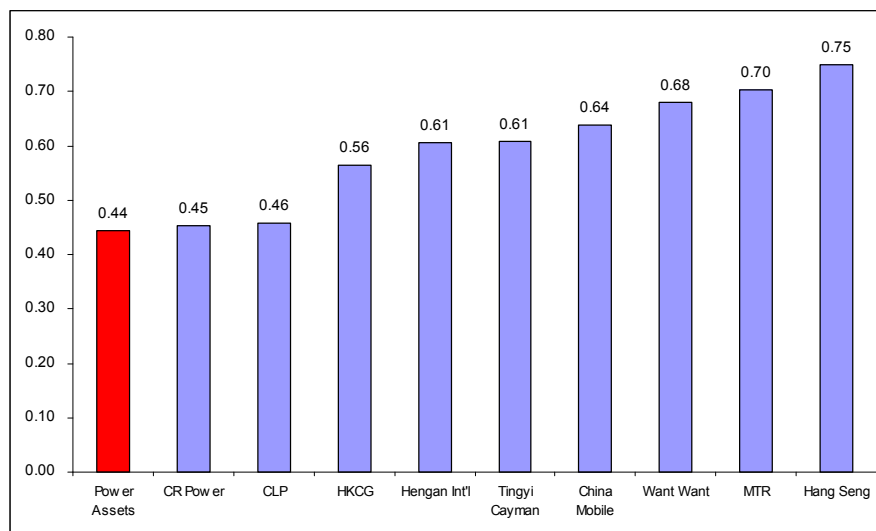
DCF Calculation

Figure 10. PAH's DCF TP breakdown

	Value (HK\$m)	Per share (HK\$)	By project	By country
Hong Kong:				
SOC	84,112	39.4	51%	51%
Others				
Australia:				
ETSA Utilities	8,660	4.1	5%	
Citipower	4,333	2.0	3%	
Powercor	10,119	4.7	6%	14%
Others				
The UK:				
UK Power Grid	46,523	21.8	28%	
Seabank	3,868	1.8	2%	
Northern Gas Network	14,411	6.8	9%	
Western Gas Network	3,166	1.5	2%	41%
Others				
China:				
Zhuhai Power Plant (Unit 1 & 2)	1,782	0.8	1%	
Jinwan Power Plant (Unit 3&4)	1,527	0.7	1%	
Zhuhai Power Plant (Unit 5 & 6)	0	0.0	0%	
Siping Cogeneration Plant	231	0.1	0%	
Dafengba Wind Farm (Dali)	0	0.0	0%	
Laoting Wind Farm	0	0.0	0%	2%
Others				
New Zealand - power grids	1,314	0.6	1%	1%
Canada - power generation	1,406	0.7	1%	1%
Thailand - power generation	3,016	1.4	2%	2%
Others				
Sub-total	184,468	86.4	112%	112%
Net debt	-20,129	-9.4	-12%	-12%
Total	164,338	77.0	100.0%	100%

Source: Company Reports and Citi Research Estimates

Figure 11. The 10 HSI Constituent Stocks with lowest beta



Source: Bloomberg

PAH has high earnings quality with 95% recurring profits from regulated utility businesses in OEDC countries. Its equity beta is only 0.44, according to Bloomberg.

Financial Statements & Key Assumptions

Figure 12. Consolidated income statement (HK\$m)

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Gross turnover	10,371	10,201	10,418	10,836	11,489
Direct costs	-3,938	-4,032	-4,477	-4,323	-4,500
Net turnover	6,433	6,169	5,942	6,513	6,989
Other revenue	1,063	1,625	1,894	2,063	2,121
Other operating costs	-814	-1,482	-1,514	-1,575	-1,670
Finance costs	-391	-617	-568	-611	-594
Operating profit	6,291	5,695	5,754	6,391	6,847
Profit from associates	1,391	3,717	4,199	4,639	4,806
Profit from JCEs (PRC power plants)	508	476	522	469	520
PBT	8,190	9,888	10,475	11,499	12,172
Tax	-937	-858	-867	-963	-1,032
PAT	7,253	9,030	9,608	10,536	11,140
SOC transfers					
- Development Fund	-58	46	230	63	-280
- Rate Reduction Reserve	-1	-1	-11	-9	-10
Net profit to shareholders	7,194	9,075	9,827	10,590	10,850
% chg	7.4%	26.1%	8.3%	7.8%	2.5%
EPS (HK\$)	3.37	4.25	4.60	4.96	5.08
% Change	7.4%	26.1%	8.3%	7.8%	2.5%

Source: Company Reports and Citi Research Estimates

Figure 13. Consolidated balance sheet (HK\$m)

As at 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Fixed assets	47,924	48,799	49,356	49,510	49,921
Associates	35,456	35,697	40,224	42,593	45,064
Other LT assets	986	860	860	860	860
Total non-current assets	84,366	85,356	90,440	92,963	95,845
Inventories	747	1,115	1,147	980	981
Trade and other receivables	1,182	1,101	1,126	1,171	1,241
Fuel clause account	569	1,035	1,291	1,221	1,037
Bank balances and other liquid funds	5,839	6,123	9,639	11,248	11,930
Total current assets	8,337	9,374	13,203	14,620	15,189
Total assets	92,703	94,730	103,643	107,583	111,034
Trade and other payables	1,865	3,669	2,236	2,268	2,309
Bank overdrafts	2	0	2	2	2
Bank loans (current portion)	8,459	617	4,725	5,953	5,793
Total current liabilities	10,326	4,286	6,963	8,223	8,103
Long-term debt	17,312	23,009	25,041	23,010	21,408
Other long-term liabilities	8,249	8,707	8,707	8,707	8,707
Rate Reduction Reserve	4	1	12	22	32
Development Fund	543	497	267	204	484
Total long-term liabilities	26,240	32,571	34,385	32,299	30,988
Share Capital	2,134	2,134	2,134	2,134	2,134
Reserves	54,003	55,739	60,161	64,926	69,809
Total shareholders' equity	56,137	57,873	62,295	67,060	71,943
Total shareholders' equity & liabilities	92,703	94,730	103,643	107,583	111,034

Source: Company Reports and Citi Research Estimates

Figure 14. Consolidated cashflow statement (HK\$m)

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
Operating profit	6,291	5,695	5,754	6,391	6,847
Depreciation	1,734	1,777	1,837	1,850	1,861
Chg. in working capital	99	530	-1,746	224	153
Rebates	-11	-4	0	0	0
Other non-cash items	-573	-915	-900	-1,024	-1,099
net interest paid	301	833	900	1,024	1,099
Tax	-877	-564	-867	-963	-1,032
Operating CF	7,043	7,431	4,979	7,502	7,829
Purchase of PP&E	-11,592	-2,535	-2,394	-2,005	-2,272
Investments	0	0	-2,448	0	0
Loans to associates	-4,522	349	-94	0	0
Dividends from associates	897	2,304	2,736	2,739	2,854
Investing CF	-15,184	-1,439	-599	735	582
FCF	-8,141	5,992	4,379	8,237	8,411
Debt financing	13,320	-2,857	6,141	-804	-1,763
Others	72	53	0	0	0
Dividend paid	-4,503	-4,503	-5,405	-5,824	-5,967
Financing CF	8,889	-7,307	736	-6,628	-7,730
Net chg in cash & equiv.	748	-1,315	5,115	1,609	681

Source: Company Reports and Citi Research Estimates

Figure 15. Key assumptions

Year to 31 Dec	FY10	FY11	FY12E	FY13E	FY14E
HKE's tariff & sales volume:					
Basic tariff (HK\$/kWh)	0.95	0.93	0.94	0.97	1.02
Fuel clause rebate (HK\$/kWh)	0.25	0.30	0.37	0.39	0.39
Special rebate (HK\$/kWh)	0.00	0.00	0.00	0.00	0.00
Net tariff (HK\$/kWh)	1.20	1.23	1.31	1.36	1.41
- yoy	-0.1%	2.8%	6.5%	3.7%	3.6%
Unit sales growth (%)	0.0%	-0.3%	1.0%	1.0%	1.0%
SOC:					
Rate of return	10.0%	10.0%	10.0%	10.0%	10.0%
Capex (HK\$m)	2,431	2,887	2,310	1,922	2,191
Permitted return (HK\$m)	4,633	4,584	4,624	4,631	4,673
- yoy	1.9%	-1.1%	0.9%	0.1%	0.9%
Development fund balance (HK\$m)	543	497	267	204	484
Profit mix:					
% SOC earnings	65%	50%	43%	44%	43%
% non-SOC earnings	35%	50%	57%	56%	57%

Source: Company Reports and Citi Research Estimates

Figure 16. Key ratios

	FY10	FY11	FY12E	FY13E	FY14E
EBIT margin	55.7%	49.0%	46.6%	49.5%	50.0%
Net profit margin	69.4%	89.0%	94.3%	97.7%	94.4%
Interest coverage (x)	16.7	9.9	10.3	10.4	11.3
Dividend coverage (x)	1.7	1.4	1.2	1.2	1.3
ROA	8.6%	9.7%	9.9%	10.0%	9.9%
Net Debt/Equity	35.5%	30.2%	32.3%	26.4%	21.2%
Net Debt/Asset	21.5%	18.5%	19.4%	16.5%	13.8%
Total debt/ EBITDA	3.4	3.5	4.4	4.0	3.6

Source: Company Reports and Citi Research Estimates

Power Assets

Company description

Power Assets (formerly called Hongkong Electric) is one of Hong Kong's two fully integrated power companies and sole supplier of electricity to Hong Kong Island, Ap Lei Chau and Lamma Island in Hong Kong. In FY11, it had 566,000 customers and supplied 26% of total electricity in the Hong Kong SAR. It generates electricity solely at its Lamma Power Station, with total installed capacity of 3,736MW. Power Assets owns overseas utility assets in Australia, Canada, New Zealand, Thailand and the UK.

Investment strategy

We rate Power Assets as Buy (1) with a HK\$77 DCF-based target price. We expect more investment opportunities to reach the HK government's fuel mix target by 2020. Also, Power Assets gearing is low and is financially well resourced for potential expansion in the UK, Australia, New Zealand and Canada, in our view.

Valuation

Our DCF-based target price of HK\$77 factors in our earnings forecasts up to 2015 and a terminal value thereafter. This method is suitable for power companies such as Power Assets, which operate in a relatively static industry with predictable cash flows. Our WACC is 6.4% (however, 8.0% for Australian and New Zealand power distribution, UK gas distribution, Canadian power generation and China power generation, and 10% for Thai power generation), which assumes: 1) a risk-free rate of 5.2%; 2) a market risk premium of 6.8%; 3) an equity beta of 0.4x, according to Bloomberg; 4) cost of debt of 5.2%; 5) a target debt to capital ratio of 37.5%; and 6) a 16.5% corporate tax rate. We assume a terminal growth rate of 2-4% beyond 2015. At our DCF-based target price, Power Assets would trade at a 15.5x PER and 3.5% yield in FY13E.

Risks

Negative risk factors include: 1) failure to meet emission targets, resulting in a lower rate of return; 2) an unfavorable regulatory regime for the Australian operations. If these risk factors have a greater impact than we anticipate, shares will likely have difficulty attaining our target price.

Power Grid Corporation of India (PGRD.BO)

Top India Electric Utility Pick

Buy	1
Price (27 Nov 12)	Rs118.40
Target price	Rs140.00
Expected share price return	18.2%
Expected dividend yield	1.9%
Expected total return	20.2%
Market Cap	Rs548,159M
	US\$9,840M

Price Performance

(RIC: PGRD.BO, BB: PWGR IN)



■ **Top India Electric Utility pick** — PGCIL is set to deliver 17% EPS growth over FY12-15E with average RoEs of 15.8%. The company remains the safest play in the Indian Electric Utilities sector, which has been riddled by shortages of coal, falling merchant prices and deteriorating SEB finances. The above risks could impede PGCIL's growth due to delayed capacity additions. However, we believe our estimates reflect the same (XIIth Plan capex of Rs825bn vs targets of Rs1,000bn).

■ **Achieved XIth Plan targets - XIIth Plan target 2x** — PGCIL achieved its XIth Plan targets with capex of Rs553bn. The target has almost doubled to Rs1,000bn for the XIIth Plan. Of this, Rs840bn already has investment approvals from the Board and PGCIL has already awarded contracts for Rs700bn of projects.

■ **Does PGCIL plan to raise equity?** — If PGCIL does Rs1,000bn in the XIIth plan there might be marginal equity shortfall in FY13E/14E. But in these years, the company may do capex at leverage > 70:30 and make up for the same over FY15E-17E. From FY15E onwards there is no mismatch. There is a regulatory cap of 75:25 and PGCIL does not want to fully utilize this leeway.

■ **Does Government of India (GoI) plan to sell down stake?** — GoI has discussed the same with company, but management has tried to impress on GoI that it should not sell down the stake. The stake is not that high and it will become tricky if it sells down the stake now and five years later company needs to raise equity for growth.

■ **Ministry of Power also does not want stake sell down** — According to *Hindustan Times*, in a letter to finance minister, the outgoing power minister said that disinvestment of PGCIL has already taken place twice. Any further disinvestment in PGCIL may be deferred by GoI, as this will be detrimental to company's growth.

■ **Significant recurring PAT beat in 2Q** — PGCIL's 2QFY13 recurring PAT at Rs11.1bn was up 57% YoY, which was 28% ahead of Citi at Rs8.7bn.

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Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Mar	(RsM)	(Rs)	(%)	(x)	(x)	(%)	(%)
2011A	25,437	5.49	11.9	21.5	2.6	13.6	1.5
2012A	32,710	7.07	28.6	16.8	2.3	14.6	1.8
2013E	39,092	8.44	19.5	14.0	2.1	15.7	1.9
2014E	45,143	9.75	15.5	12.1	1.9	16.3	2.0
2015E	51,624	11.15	14.4	10.6	1.7	16.5	2.1

Source: Powered by dataCentral

Shanghai Electric Group (2727.HK)

Management Guides Better Outlook in 4Q12E

Buy	1
Price (28 Nov 12)	HK\$3.10
Target price	HK\$3.50
Expected share price return	12.9%
Expected dividend yield	3.2%
Expected total return	16.1%
Market Cap	HK\$39,753M
	US\$5,129M

Price Performance (RIC: 2727.HK, BB: 2727 HK)



■ **Summary** — Shanghai Electric (SHE) is our preferred play in the PRC power equipment sector as (i) we expect revenue to rise qoq in 4Q12E on more deliveries; (ii) its new orders should increase qoq in 4Q12E and rise over 25% yoy in 2H12E; and (iii) its 9.6x FY13E PER and 0.9x PB look inexpensive, and are below historical means by 36% and 47% respectively. We expect its net profit decline in 2013E would be the least among its peers given that it has the most diversified profit mix. Furthermore, the PRC govt's recent decision not to build inland nuclear projects has not affected its orders on hand.

■ **3Q profit missed expectations mainly due to lower revenue** — SHE's net profit was Rmb1,028m in 3Q12, -15% yoy and -9% qoq due to (i) lower revenue (-10% yoy, -14% qoq to Rmb17.3bn) on smaller sales volume; (ii) more operating costs (selling & distribution costs +19% yoy, +16% qoq to Rmb638m while admin costs +27% yoy but -2% qoq, to Rmb1,681m); (iii) less non-operating income (-33% yoy but +29% qoq to Rmb175m); (iv) higher MI (+24% yoy and +1% qoq to Rmb447m).

■ **Revenue to rebound qoq in 4Q12E** — SHE's revenue was -10% qoq and -14% yoy to Rmb17.3bn in 3Q12. Its No.1 revenue source was the high efficiency and clean energy segment, contributing 40% to 3Q revenue. Revenue of this source, booked on a delivery basis, was -37% yoy to Rmb7.4bn in 3Q12, when less volume was delivered. It expects revenue of the segment to rise in 4Q12 with more deliveries.

■ **New orders to rise qoq in 4Q12E** — SHE's new orders flow has recovered from a trough in 1H12 with coal fired equipment demand from China. Its new orders flow rose from Rmb2.6bn in 1Q12 to Rmb8.4bn in 2Q12 and further to Rmb14.1bn in 3Q12. It expects new orders to rise further in 4Q12E with two sizeable PRC orders close to be signed. New orders are likely to rise >25% yoy to >Rmb30bn in 2H12E.

■ **Profit guidance** — In the 3Q12 results investor call, SHE said it targets net profit growth of 6-8% yoy in 2012 and flat yoy in 2013.

Asian Utilities

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Statistical Abstract

Year to 31 Dec	Net Profit (RmbM)	Diluted EPS (Rmb)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	2,784	0.219	11.7	11.4	1.2	11.3	2.6
2011A	3,267	0.255	16.3	9.8	1.1	11.6	3.1
2012E	3,461	0.270	5.9	9.2	1.0	11.3	3.2
2013E	3,337	0.260	-3.6	9.6	0.9	10.1	3.1
2014E	3,645	0.284	9.2	8.8	0.9	10.3	3.4

Source: Powered by dataCentral

2727.HK: Fiscal year end 31-Dec						Price: HK\$3.10; TP: HK\$3.50; Market Cap: HK\$39,753m; Recomm: Buy					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	62,957	67,918	69,652	71,256	77,650	PE (x)	11.4	9.8	9.2	9.6	8.8
Cost of sales	-52,679	-55,740	-56,381	-58,278	-63,518	PB (x)	1.2	1.1	1.0	0.9	0.9
Gross profit	10,279	12,178	13,271	12,977	14,132	EV/EBITDA (x)	4.7	3.3	3.3	3.4	2.8
Gross Margin (%)	16.3	17.9	19.1	18.2	18.2	FCF yield (%)	5.0	-3.8	-1.9	10.4	11.5
EBITDA (Adj)	4,528	5,726	5,842	5,719	6,236	Dividend yield (%)	2.6	3.1	3.2	3.1	3.4
EBITDA Margin (Adj) (%)	7.2	8.4	8.4	8.0	8.0	Payout ratio (%)	30	30	30	30	30
Depreciation	-1,263	-1,286	-1,163	-1,267	-1,370	ROE (%)	11.3	11.6	11.3	10.1	10.3
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	3,264	4,441	4,679	4,452	4,865	EBITDA	4,528	5,726	5,842	5,719	6,236
EBIT Margin (Adj) (%)	5.2	6.5	6.7	6.2	6.3	Working capital	-373	-5,944	-3,496	512	398
Net interest	-52	-83	-71	-93	-115	Other	666	1,065	-774	-715	-776
Associates	814	734	1,100	1,128	1,241	Operating cashflow	4,820	847	1,572	5,516	5,858
Non-op/Except	0	0	0	0	0	Capex	-3,245	-2,068	-2,184	-2,184	-2,184
Pre-tax profit	4,025	5,091	5,708	5,487	5,991	Net acq/disposals	-1,612	525	0	0	0
Tax	-228	-715	-877	-830	-905	Other	499	835	-210	-231	-255
Extraord./Min.Int./Pref.div.	-1,014	-1,109	-1,369	-1,320	-1,442	Investing cashflow	-4,358	-708	-2,394	-2,415	-2,438
Reported net profit	2,784	3,267	3,461	3,337	3,645	Dividends paid	-1,344	-1,526	-1,664	-1,698	-1,722
Net Margin (%)	4.4	4.8	5.0	4.7	4.7	Financing cashflow	2,366	-168	-1,235	-1,291	-1,336
Core NPAT	2,784	3,267	3,461	3,337	3,645	Net change in cash	2,767	-61	-2,057	1,811	2,084
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	1,576	-1,221	-612	3,333	3,675
Reported EPS (Rmb)	0.219	0.255	0.270	0.260	0.284						
Core EPS (Rmb)	0.219	0.255	0.270	0.260	0.284						
DPS (Rmb)	0.065	0.076	0.081	0.078	0.085						
CFPS (Rmb)	0.379	0.066	0.123	0.430	0.457						
FCFPS (Rmb)	0.124	-0.095	-0.048	0.260	0.287						
BVPS (Rmb)	2.106	2.282	2.475	2.654	2.861						
Wtd avg ord shares (m)	12,709	12,824	12,824	12,824	12,824						
Wtd avg diluted shares (m)	12,709	12,824	12,824	12,824	12,824						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	9.3	7.9	2.6	2.3	9.0						
EBIT (Adj) (%)	22.4	36.0	5.4	-4.8	9.3						
Core NPAT (%)	13.5	17.4	5.9	-3.6	9.2						
Core EPS (%)	11.7	16.3	5.9	-3.6	9.2						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	18,943	19,088	17,031	18,842	20,926						
Accounts receivables	27,239	33,264	35,730	36,553	39,834						
Inventory	19,872	20,885	22,670	23,433	25,540						
Net fixed & other tangibles	13,461	13,618	14,638	15,555	16,368						
Goodwill & intangibles	4,036	4,729	4,836	4,952	5,078						
Financial & other assets	14,662	15,130	16,252	17,387	18,656						
Total assets	98,212	106,715	111,158	116,722	126,401						
Accounts payable	56,067	60,664	61,427	63,496	69,206						
Short-term debt	396	685	685	685	685						
Long-term debt	2,021	701	1,201	1,701	2,201						
Provisions & other liab	5,226	7,010	7,023	7,060	7,163						
Total liabilities	63,709	69,059	70,336	72,941	79,255						
Shareholders' equity	27,002	29,257	31,739	34,038	36,682						
Minority interests	7,500	8,399	9,083	9,743	10,464						
Total equity	34,502	37,656	40,822	43,781	47,146						
Net debt	-16,526	-17,703	-15,146	-16,456	-18,040						
Net debt to equity (%)	-47.9	-47.0	-37.1	-37.6	-38.3						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Business Review in 3Q12

Figure 1. SHE's revenue breakdown by business segment (PRC GAAP)

Revenue	3Q12	YoY	QoQ	9M12	YoY
New Energy	1,880	-14%	8%	5,118	0%
High efficiency and clean energy	7,426	4%	-37%	26,333	15%
Industrial equipment	7,454	-3%	38%	17,140	-3%
Modern services	4,743	-2%	-2%	13,380	18%
Eliminations *	-3,115	109%	-15%	-7,363	157%
Net sales	18,388	-10%	-9%	54,608	1%

* Higher elimination amount due to more intra-group transactions mainly related to equipment procurement for EPC and BTG projects

Source: Company Reports and Citi Research Estimates

Figure 2. SHE's gross profit by segment (Rmbm)

Segment	3Q12	YoY	QoQ	9M12	YoY
New Energy	263	20%	-4%	679	5%
High efficiency and clean energy	1,542	1%	-37%	5,445	15%
Industrial equipment	1,446	-8%	42%	3,359	-6%
Modern services	484	28%	0%	1,366	54%
Eliminations	16	-108%	-66%	-310	-28%
Total	3,751	8%	-12%	10,539	12%

Source: Citi Research

Figure 3. SHE's gross profit margin

Segment	3Q12	YoY	QoQ	9M12	YoY
New Energy	14.0%	+4.5ppts	-1.1ppts	13.3%	+0.7ppts
High efficiency and clean energy	20.8%	-0.6ppts	+0.1ppts	20.7%	same
Industrial equipment	19.4%	-1.0ppts	+0.5ppts	19.6%	-0.5ppts
Modern services	10.2%	+2.4ppts	+0.2ppts	10.2%	+2.4ppts
Total	20.4%	+3.3ppts	-0.8ppts	19.3%	+2.0ppts

Source: Citi Research

For domestic coal fired equipment demand, SHE won new orders from projects in Guangdong and Xinjiang. Among its coal-fired equipment orders, large-scale 1,000MW coal-fired units account for 50% of its orders on hand and over 60% of new orders in 3Q12. SHE estimates China to start construction of 50-60GW coal fired capacity in 2013E, less than before, owing to more emphasis and clean energy for emission cuts.

For gas turbine, the production volume would rise from 1 in 2011 to 12 in 2012E and 14 in 2013E. Revenue rose to Rmb3bn in 9M12, from immaterial a year ago; SHE expects to make Rmb4bn revenue from this product in 2012E. The GP margin of this segment rose from 7.2% in 2011 to 14.2% in 1H12 boosted by an engineering work done in 1H12. SHE expects the GP margin to rise from 8% in 3Q12 to 20% in 5 years' time. New order flow of gas turbine was also strong with 7 new units ordered since June end.

For nuclear equipment, revenue of nuclear island sales was +22% yoy to Rmb2.2bn in 9M12, while its GP margin was +2.5ppts yoy to 21.4% in the same period. Revenue of conventional island sales has been insignificant with the first unit delivered in 2011 (note: revenue is booked on delivery basis for conventional island); GP margin was -8% in 9M12 for low sales volume hence without much economies of scale. Additionally, SHE claimed it has not included any orders from proposed inland nuclear projects; Chinese government's decision not to build inland nuclear projects in 12th 5-year period therefore not affecting its Rmb246bn orders on hand. The company is able to produce the whole range of equipment for nuclear power plants using AP1000 technology which has been chosen by the PRC Government for new projects. But the new volume would not be exponential since China would develop new nuclear projects progressively, not aggressively, putting safety as first priority.

For wind turbine sales, revenue was about 15% yoy to Rmb2.2bn in 9M12 for reduced sales volume. Its GP margin was -0.4ppts yoy to 0.3% in the same period.

For export, SHE made revenue of Rmb10.4bn in 9M12 for equipment sales to primarily India (3-4 projects), Vietnam (2X600MW) and Iran (4X300MW). It had no new orders from overseas in 9M12 due to coal shortage and import tax issues in India.

Figure 4. SHE's quarterly new order (Rmbbn)

Product type	1Q12	2Q12	3Q12	QoQ
Coal-fired	0.5	1.4	8.6	511%
Engineering (domestic)	1.3	1.1	3.3	196%
Gas turbine	0.0	3.0	1.2	-60%
Wind	0.4	1.7	0.8	-55%
Nuclear - nuclear island	0.0	0.0	0.2	7900%
Forging	0.3	1.2	0.0	-100%
Nuclear - conventional island	0.0	0.0	0.0	n/a
Engineering (overseas)	0.0	0.0	0.0	n/a
Total	2.6	8.4	14.1	68%

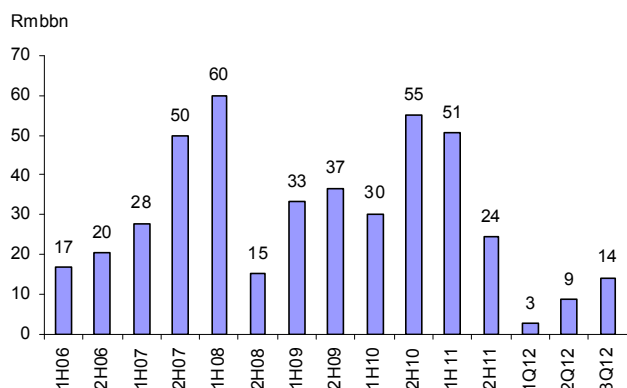
Source: Citi Research

Figure 5. SHE's orders on hand (Rmbbn)

Product type	1Q12	2Q12	3Q12	QoQ
Engineering (overseas)	100.7	97.0	95.0	-2%
Coal-fired	85.8	79.7	83.5	5%
Nuclear - nuclear island	18.9	17.8	17.0	-4%
Nuclear - conventional island	15.7	14.4	14.9	3%
Engineering (domestic)	7.0	9.6	13.9	45%
Gas turbine	7.8	8.6	9.7	13%
Others	10.0	5.5	7.4	35%
Wind	6.1	5.4	5.2	-4%
Total	252.0	238.0	246.6	4%

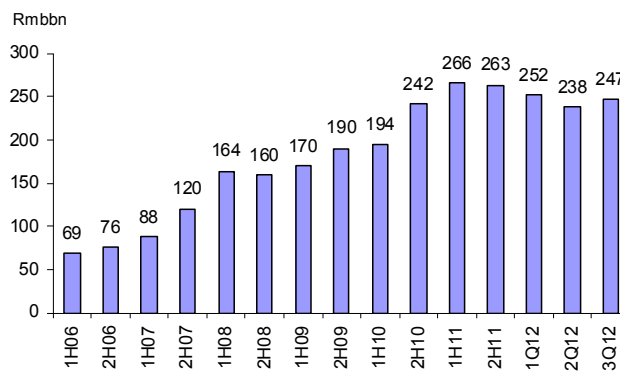
Source: Citi Research

Figure 6. SHE's quarterly new orders (Rmbbn)



Source: Citi Research

Figure 7. SHE's orders on hand (Rmbbn)



Source: Citi Research

Figure 8. SHE's forward PE chart



Source: Citi Research

Figure 9. SHE's forward PB chart



Source: Citi Research

Financial Statement & Key Assumptions

Figure 10. Consolidated Income Statement

Year ended Dec (Rmbm)	2010	2011	2012E	2013E	2014E
New Energy	6,200	7,170	6,899	7,276	7,598
High Efficiency and Clean Energy	27,161	30,172	32,948	30,777	33,421
Industrial Equipment	18,572	19,846	20,221	21,380	23,469
Modern Service	12,894	15,183	17,461	19,207	21,128
Other business	2,522	1,502	1,652	1,817	1,999
Eliminations	-4,392	-5,955	-9,530	-9,202	-9,964
Net sales	62,957	67,918	69,652	71,256	77,650
Cost of sales	-52,679	-55,740	-56,381	-58,278	-63,518
Gross Profit	10,279	12,178	13,271	12,977	14,132
Other revenue and gains	1,328	894	894	804	804
Selling and distribution expenses	-1,554	-1,732	-1,915	-1,888	-1,980
Administrative expenses	-3,833	-4,173	-4,976	-5,162	-5,548
Other operating expenses	-2,955	-2,726	-2,594	-2,279	-2,544
Operating Profit	3,264	4,441	4,679	4,452	4,865
Depreciation and amortization	1,263	1,286	1,163	1,267	1,370
EBITDA	4,527	5,726	5,842	5,719	6,236
Interest expense	-52	-83	-71	-93	-115
Associates and JVE	814	734	1,100	1,128	1,241
Profit before tax	4,025	5,091	5,708	5,487	5,991
Tax	-228	-715	-877	-830	-905
Minority interests	-1,014	-1,109	-1,369	-1,320	-1,442
Net Profit	2,784	3,267	3,461	3,337	3,645
Ex extraordinary	2,784	3,267	3,461	3,337	3,645
EPS (Rmb)	0.22	0.25	0.27	0.26	0.28
Change in Basic EPS	11.7%	16.3%	5.9%	-3.6%	9.2%

Source: Company Reports and Citi Research Estimates

Figure 11. SHE's earnings estimates – Citi vs. consensus

Net profit (Rmbm)	Citi	Consensus	% diff
FY12E	3,461	3,508	-1%
FY13E	3,337	3,664	-9%
FY14E	3,645	4,018	-9%

Source: Bloomberg and Citi Research Estimates

Figure 12. Key ratios

	2010	2011	2012E	2013E	2014E
Gross margin	16.3%	17.9%	19.1%	18.2%	18.2%
EBITDA margin	7.4%	8.4%	6.6%	3.8%	1.8%
Operating margin	5.2%	6.5%	6.7%	6.2%	6.3%
PBT margin	6.4%	7.5%	8.2%	7.7%	7.7%
Net margin	4.4%	4.8%	5.0%	4.7%	4.7%
Net debt/equity	net cash	net cash	net cash	net cash	net cash
ROE	11.3%	11.6%	11.3%	10.1%	10.3%
ROA	3.0%	3.2%	3.2%	2.9%	3.0%
Average working capital days					
Inventory	137	133	141	144	141
Receivables	158	163	143	87	39
Payables	379	382	395	391	381

Source: Company Reports and Citi Research Estimates

Figure 13. Consolidated Cashflow Statement

Year end Dec (Rmbm)	2010	2011	2012E	2013E	2014E
Profit before tax	4,025	5,091	5,708	5,487	5,991
Associates	-814	-734	-1,100	-1,128	-1,241
Finance costs	52	83	71	93	115
EBIT	3,264	4,441	4,679	4,452	4,865
Depreciation	1,123	1,178	1,163	1,267	1,370
Amortization	102	74	153	165	179
Investment valuation gain	-574	-757	0	0	0
Provision	1,589	1,891	0	0	0
Disposal gain	-253	-63	0	0	0
Write-down of inventories to net realizable value	361	443	0	0	0
Others	302	324	-50	-50	-50
Change in inventories	-1,251	-1,875	-1,785	-763	-2,107
Change in construction contracts	161	166	-22	-7	-29
Change in trade/other receivables	-3,363	-7,361	-2,466	-823	-3,280
Change in other long-term assets	43	34	0	0	0
Change in payables (trade, bills, other payable & accruals)	4,546	3,884	776	2,105	5,814
Utilization and reversal of warranty provision and other provisions	-509	-792	0	0	0
Cash from operations	5,542	1,587	2,449	6,347	6,763
Income tax paid	-722	-740	-877	-830	-905
Net operating cashflow	4,820	847	1,572	5,516	5,858
Capex	-3,245	-2,068	-2,184	-2,184	-2,184
Investments	-102	-33	-210	-231	-255
Dividend from associates	601	868	0	0	0
Others	-1,612	525	0	0	0
Investing cashflow	-4,358	-708	-2,394	-2,415	-2,438
Net cash after investing activities	462	139	-822	3,101	3,420
Increase in issued capital/paid-up capital	2,221	0	0	0	0
Capital injection from minority interests	1,018	580	0	0	0
Dividend paid to minorities/Subsidiaries	-589	-692	-685	-660	-721
Dividend paid by the Company	-755	-835	-980	-1,038	-1,001
Share issue expenses	0	0	0	0	0
Increase in customer deposit/repurchase agreement	777	847	0	0	0
Interest paid	-108	-159	-71	-93	-115
Net cash from financing activities	2,564	-259	-1,735	-1,791	-1,836
Change in net debt	3,027	-120	-2,557	1,311	1,584
Repayments of borrowings	-489	-352	500	500	500
New borrowings raised	291	443	0	0	0
Change of cash	2,829	-29	-2,057	1,811	2,084

Source: Company Reports and Citi Research Estimates

Figure 14. Consolidated Balance Sheet

Year to 31 Dec (Rmbm)	2010	2011	2012E	2013E	2014E
Operating	11,226	11,384	12,404	13,320	14,133
Construction in progress	2,234	2,234	2,234	2,234	2,234
PP&E	13,461	13,618	14,638	15,555	16,368
Prepaid land premium/land lease prepayments	1,345	1,462	1,462	1,462	1,462
Goodwill	16	12	12	12	12
Intangible assets	684	774	831	897	973
Investments in associates	2,969	2,992	4,092	5,220	6,461
Long-term investments	309	681	681	681	681
Other Long-term assets	593	753	753	753	753
Loans receivable	876	978	1,028	1,078	1,128
Deferred tax assets	1,115	1,503	1,503	1,503	1,503
Total LT assets	21,367	22,773	25,000	27,161	29,341
Inventory	19,872	20,885	22,670	23,433	25,540
Construction contracts	456	290	312	319	347
Trade receivables	15,977	18,365	19,726	20,181	21,992
Bill receivables	2,375	4,451	4,781	4,891	5,330
Prepayments	8,886	10,449	11,223	11,482	12,512
Loan receivables	1,704	953	953	953	953
Discounted bills related to finance co	82	1,633	1,633	1,633	1,633
Short-term investments & central bank statutory reserve	7,896	7,198	7,198	7,198	7,198
Pledged bank deposits	653	629	629	629	629
Cash	18,943	19,088	17,031	18,842	20,926
Current assets	76,844	83,942	86,158	89,561	97,061
Total assets	98,212	106,715	111,158	116,722	126,401
Trade and bills payables	17,507	23,650	23,988	24,837	27,116
Other payables and accruals	38,560	37,013	37,439	38,658	42,090
Tax payable	829	1,120	1,133	1,170	1,274
Customer deposits	1,931	2,784	2,784	2,784	2,784
Other Current liabilities	1,656	2,131	2,131	2,131	2,131
Current liabilities	60,483	66,700	67,476	69,581	75,395
ST borrowings	396	685	685	685	685
Debentures & repos	1,000	0	0	0	0
LT borrowings	1,021	701	1,201	1,701	2,201
Total debt	2,417	1,386	1,886	2,386	2,886
Other LT payables	476	637	637	637	637
Deferred tax liabilities	333	337	337	337	337
Minority interests	7,500	8,399	9,083	9,743	10,464
Shareholders equity	27,002	29,257	31,739	34,038	36,682
Total equity and liability	98,212	106,715	111,158	116,722	126,401

Source: Company Reports and Citi Research Estimates

Figure 15. Key Assumptions

	2010	2011	2012E	2013E	2014E
Average selling price (Rmb'000/MW)					
Turbines	299	299	299	284	299
% change	8%	0%	0%	-5%	5%
Boilers	409	421	421	400	420
% change	8%	3%	0%	-5%	5%
Turbine generators	108	112	112	106	111
% change	8%	3%	0%	-5%	5%
Wind power equipment	4,539	3,404	3,234	3,234	3,234
% change	-10%	-25%	-5%	0%	0%
Elevators (per unit)	256	260	260	266	273
% change	2%	1%	0%	2%	2%
Sales volume (MW)					
Turbines	21,585	22,125	24,338	23,121	24,277
% change	-6%	2%	10%	-5%	5%
Boilers	24,620	27,082	29,790	28,300	29,715
% change	-10%	10%	10%	-5%	5%
Generators	24,530	26,983	29,681	28,197	29,607
% change	-10%	10%	10%	-5%	5%
Wind power units (completed)	662	989	875	901	901
% change	168%	49%	-12%	3%	0%
Elevators (in unit)	39,809	43,790	45,979	50,577	55,635
% change	19%	10%	5%	10%	10%
Revenue mix					
Power Equipment	43%	43%	36%	20%	9%
Electromechanical Equipment	22%	22%	30%	47%	61%
New Energy	9%	10%	9%	9%	9%
High Efficiency and Clean Energy	39%	39%	37%	34%	34%
Industrial Equipment	28%	28%	28%	29%	29%
Modern Service	20%	22%	24%	26%	26%
Gross profit margin					
New Energy	14.1%	9.8%	10.8%	10.8%	10.8%
High Efficiency and Clean Energy	17.3%	21.8%	21.3%	20.8%	20.8%
Industrial Equipment	19.1%	19.3%	18.8%	18.3%	18.3%
Modern Service	7.1%	5.7%	8.7%	8.7%	8.7%
Other business	8.9%	16.6%	11.6%	11.6%	11.6%
Total	16.3%	17.9%	19.1%	18.2%	18.2%

Source: Company Reports and Citi Research Estimates

Shanghai Electric Group

Company description

Shanghai Electric, through its numerous subsidiaries and JVs, manufactures power-generation equipment, elevators, heavy machinery, machine tools, printing and packaging equipment, diesel engines, urban transit railcars and environmental protection equipment.

Investment strategy

We rate Shanghai Electric a Buy with a target price of HK\$3.50. We expect its 1) total revenue to rise 6-8% yoy in FY12E, driven by nuclear and gas equipment sales and 2) GP margin to drop 0.5-1.0ppts yoy in FY12E. The stock's valuations look inexpensive relative to historical levels.

Valuation

Our HK\$3.50 target price for SHE is based on 1.2x FY12E P/B, a 30-35% discount to the historical average given the more selective new power plant capacity approval requirements of the National Development and Reform Commission in China. We use P/B as net profits are thin in downcycles. The discount also factors in the weak demand for coal-fired power equipment, but expect the PRC government to resume approval of new nuclear projects by end-2012E.

Risks

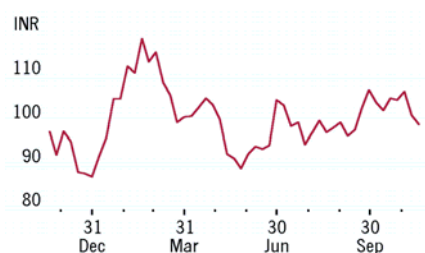
Factors that could prevent the stock from reaching our target price include: 1) lower-than-expected order flow; 2) lower-than-anticipated margins with unexpected raw material price hikes; and 3) lower-than-expected returns from a potential asset injection and the JV with Siemens.

Tata Power (TTPW.BO)

Facing Multiple Headwinds

Sell	3
Price (27 Nov 12)	Rs103.35
Target price	Rs85.00
Expected share price return	-17.8%
Expected dividend yield	1.3%
Expected total return	-16.5%
Market Cap	Rs245,257M
	US\$4,402M

Price Performance (RIC: TTPW.BO, BB: TPWR IN)



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- **Top India Electric Utility Sell** — Adani Power and Tata Power are our top sells in this space and the 2QFY13 result has only reinforced our views. Tata Power's 2QFY13 cons. recurring PAT at Rs1.4bn declined 49% YoY. 1HFY13 recurring PAT at Rs4.1bn declined 36% YoY.
- **Stock price perhaps already factors in Mundra UMPP tariff hike** — Trading at P/E multiples of 17x – 20x over FY13E-15E and P/BV of 1.7x FY14E with average RoEs of 9.5% over the next three years, Tata Power is expensive and is perhaps already factoring in a favorable outcome in its tariff hike plea for Mundra UMPP with CERC. It is very difficult to justify why Tata Power should trade at a on par or at premium to our top India Electric Utility pick, PGCIL, which has RoEs of 15%+. Further, 50-60% of FY13E-15E profits are from coal mining which should intrinsically trade at lower P/E multiples than those that Electric Utilities trade at.
- **Coal sheen wearing off** — Post declining 52% YoY in 1QFY13, coal EBIT has once again declined 49% YoY on account of 3% YoY fall in volumes, 18% YoY decline in ASPs despite a marginal decline in cost of production.
- **Mundra UMPP continues to bleed** — After writing off Rs18bn in FY12, Tata Power has once again written off Rs2.5bn in 2QFY13 in this project. Recurring losses in 1HFY13 stand at Rs3.8bn vs our FY13E estimate of Rs5.6bn. The write-offs do benefit future profits as they reduces future depreciation.
- **Maithon losses continue** — In 1HFY13 Maithon has reported losses of Rs592mn and is an example of how a largely regulated RoE project can end up making losses due to poor project management. We have optimistically modeled in 2HFY13 profits of Rs1.3bn in our estimates, with hopes that this project would turn around.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Mar	(RsM)	(Rs)	(%)	(x)	(x)	(%)	(%)
2011A	8,716	3.67	9.5	28.1	2.3	8.4	1.2
2012A	10,573	4.45	21.3	23.2	2.2	9.6	1.2
2013E	11,543	4.86	9.2	21.2	2.0	9.8	1.3
2014E	12,598	5.31	9.1	19.5	1.9	10.0	1.3
2015E	13,924	5.87	10.5	17.6	1.7	10.3	1.3

Source: Powered by dataCentral

Tenaga Nasional (TENA.KL)

Compensation to Continue, Tariff Adjustment only Post-GE

Buy	1
Price (28 Nov 12)	RM6.93
Target price	RM8.75
Expected share price return	26.3%
Expected dividend yield	2.9%
Expected total return	29.1%
Market Cap	RM38,280M
	US\$12,536M

Price Performance (RIC: TENA.KL, BB: TNB MK)



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- **Reiterating Buy** — We reiterate our Buy recommendation on Tenaga as we believe Tenaga's outlook will continue to improve with the imminent increase in gas supply and assurance of continued government support.
- **Positive on tariff adjustment, but only after GE** — Management has been engaging the government and Petronas to work towards a holistic fuel cost pass-through formula that would include prices of coal (now fixed at US\$85/mt) and natural gas. Based on management estimates, a 12% tariff hike would be needed if [1] regulated gas price is raised by RM3/mmbtu to RM16.70; [2] new gas supply is purchased at RM44; and coal cost is raised from US\$85/mt to market price. The last time Tenaga raised electricity tariffs was in Jun 2011 by an average of 7.12%. Although management is optimistic the government would proceed with the tariff revision, management believes the rate hike will not happen before the general elections (GE).
- **Melaka Plant Delayed to 1Q2013, Coal Price to be Below US\$100/tonne** — Tenaga has been receiving an average of 1,000mmscfd of gas supply from Petronas, vs. the minimum pre-agreed of 1,250mmscfd at subsidized rate. Management is optimistic that new supplies from Petronas' LNG re-gasification plant would have neutral impact on Tenaga. Management guided for coal cost to be <US\$100/tonne for FY13E (vs the US\$103.6/tonne in FY12A). Coal consumption is seen to have peaked at 20.8m MT in FY12.
- **Limited details on Prai and PPA Extensions** — Management said that it is still in negotiations with the EC regarding the Prai power plant and guided for the agreement to be finalized by mid-November. Similarly, details remain scant regarding any savings from lower capacity payments as a result of the PPA extensions but it will likely be neutral for Tenaga as the savings will be passed back.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Aug	(RMM)	(RM)	(%)	(x)	(x)	(%)	(%)
2011A	727	0.13	-70.8	52.1	1.2	2.3	0.5
2012A	3,168	0.58	334.9	12.0	1.1	9.3	2.9
2013E	3,759	0.68	17.8	10.2	1.0	9.9	2.9
2014E	4,130	0.75	9.9	9.3	0.9	10.0	3.2
2015E	4,337	0.79	5.0	8.8	0.8	9.9	3.4

Source: Powered by dataCentral

Towngas China (1083.HK)

Stronger Earnings Growth in 2013E on Rising Gas Supplies

Buy	1
Price (23 Nov 12)	HK\$6.28
Target price	HK\$7.20
<i>from HK\$6.05</i>	
Expected share price return	14.6%
Expected dividend yield	1.1%
Expected total return	15.8%
Market Cap	HK\$15,451M
	US\$1,993M

Price Performance (RIC: 1083.HK, BB: 1083 HK)



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■ **Better growth in 2013** — Towngas China has demonstrated significant earnings and ROE improvement, and we expect TCCL to resume 20+% earnings growth in 2013E, driven by higher gas supplies, especially in the Northeast. We raise our target to HK\$7.20 to reflect our more optimistic long-term earnings growth metric.

■ **What's new** — After TCCL reported its interim result, with net profit +18% Y/Y, EnerChina announced the sale of 8 million shares during Sep 3-7 (of its 195 million) as part of its long-term exit strategy. We believe its gradual exit will not impact TCCL's operations, and HKCG as the major shareholder will continue to use TCCL as its platform to invest in new China gas projects. After adding 4 projects in 1H12, we expect the company to acquire 5-6 more projects in 2H12. We remain optimistic on its gas volume growth reaching 16% Y/Y in 2012E and 21% Y/Y in 2013E, driven by higher supplies from the gas pipelines and ramp-up of LNG terminals in Dalian.

■ **New gas supplies** — We expect TCCL to see 16% gas sales volume growth in 2012E, driven by incremental gas supplies amid weaker demand from existing industrial gas users. It will likely benefit from new gas pipelines, including the West to East Shenyang line. Ramp-up of LNG terminal in Dalian and gas supplies from Datang coal-to-gas should also increase gas supplies in the Northeast region.

■ **Upside catalysts** — We see a ramp-up of new projects and scale to help TCCL achieve a sustainable re-rating. As the minority shareholder sells down its stakes, the company's liquidity may also improve. Other catalysts are acquisition of new industrial projects, completion of mid-stream pipelines and completion of residential price adjustments for the remaining 8-10 projects in 2H12.

■ **Revising estimates** — We revise up our 2012/13/14E EPS estimates by 3%/5%/5% respectively to reflect more optimistic gas volume assumptions.

Statistical Abstract

Year to 31 Dec	Net Profit (HK\$M)	Diluted EPS (HK\$)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	436	0.20	47.4	31.5	1.6	5.8	0.5
2011A	709	0.29	44.6	21.8	1.6	7.8	0.8
2012E	810	0.33	14.3	19.1	1.5	8.1	0.9
2013E	1,001	0.41	23.6	15.4	1.4	9.4	1.1
2014E	1,194	0.49	19.2	12.9	1.3	10.3	1.3

Source: Powered by dataCentral

1083.HK: Fiscal year end 31-Dec						Price: HK\$6.28; TP: HK\$7.20; Market Cap: HK\$15,451m; Recomm: Buy					
Profit & Loss (HK\$m)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	2,981	4,321	4,936	6,235	8,115	PE (x)	31.5	21.8	19.1	15.4	12.9
Cost of sales	-2,291	-3,341	-3,779	-4,810	-6,310	PB (x)	1.6	1.6	1.5	1.4	1.3
Gross profit	690	980	1,157	1,424	1,804	EV/EBITDA (x)	27.8	13.7	11.2	9.2	7.4
Gross Margin (%)	23.1	22.7	23.4	22.8	22.2	FCF yield (%)	-0.9	-0.3	1.7	3.2	4.3
EBITDA (Adj)	528	1,058	1,302	1,578	1,915	Dividend yield (%)	0.5	0.8	0.9	1.1	1.3
EBITDA Margin (Adj) (%)	17.7	24.5	26.4	25.3	23.6	Payout ratio (%)	15	17	17	17	17
Depreciation	-198	-471	-580	-692	-802	ROE (%)	5.8	7.8	8.1	9.4	10.3
Amortisation	0	0	0	0	0	Cashflow (HK\$m)					
EBIT (Adj)	330	586	722	887	1,113	EBITDA	528	1,058	1,302	1,578	1,915
EBIT Margin (Adj) (%)	11.1	13.6	14.6	14.2	13.7	Working capital	204	-80	-194	-254	-384
Net interest	-142	-142	-153	-139	-130	Other	-130	-125	166	209	193
Associates	300	394	547	633	671	Operating cashflow	601	853	1,274	1,534	1,725
Non-op/Except	138	195	65	75	83	Capex	-719	-894	-1,005	-1,046	-1,056
Pre-tax profit	626	1,034	1,181	1,457	1,737	Net acq/disposals	0	0	0	0	0
Tax	-136	-257	-294	-361	-430	Other	-156	-136	-300	-330	-363
Extraord./Min.Int./Pref.div.	-54	-68	-78	-95	-113	Investing cashflow	-875	-1,031	-1,305	-1,376	-1,419
Reported net profit	436	709	810	1,001	1,194	Dividends paid	-64	-123	-140	-174	-207
Net Margin (%)	14.6	16.4	16.4	16.1	14.7	Financing cashflow	693	1,025	160	-174	-507
Core NPAT	436	709	810	1,001	1,194	Net change in cash	465	847	128	-15	-201
Per share data						Free cashflow to s/holders	-117	-41	269	489	669
Reported EPS (\$)	0.20	0.29	0.33	0.41	0.49						
Core EPS (\$)	0.20	0.29	0.33	0.41	0.49						
DPS (\$)	0.03	0.05	0.06	0.07	0.08						
CFPS (\$)	0.28	0.35	0.52	0.62	0.70						
FCFPS (\$)	-0.05	-0.02	0.11	0.20	0.27						
BVPS (\$)	3.92	3.91	4.18	4.52	4.92						
Wtd avg ord shares (m)	2,186	2,458	2,458	2,458	2,458						
Wtd avg diluted shares (m)	2,187	2,460	2,460	2,460	2,460						
Growth rates											
Sales revenue (%)	47.2	44.9	14.2	26.3	30.2						
EBIT (Adj) (%)	34.7	77.6	23.1	22.9	25.5						
Core NPAT (%)	64.4	62.6	14.3	23.6	19.2						
Core EPS (%)	47.4	44.6	14.3	23.6	19.2						
Balance Sheet (HK\$m)											
Cash & cash equiv.	1,429	2,051	2,179	2,164	1,964						
Accounts receivables	531	852	1,082	1,366	1,778						
Inventory	148	388	432	557	744						
Net fixed & other tangibles	5,074	6,128	6,553	6,907	7,161						
Goodwill & intangibles	3,563	4,031	4,031	4,031	4,031						
Financial & other assets	4,138	4,733	5,033	5,363	5,726						
Total assets	14,883	18,183	19,311	20,389	21,404						
Accounts payable	531	852	932	1,088	1,304						
Short-term debt	2,818	1,708	1,708	1,708	1,708						
Long-term debt	904	3,374	3,674	3,674	3,374						
Provisions & other liab	1,494	1,951	1,951	1,951	1,951						
Total liabilities	5,748	7,884	8,264	8,420	8,336						
Shareholders' equity	8,563	9,615	10,285	11,113	12,099						
Minority interests	572	684	761	856	969						
Total equity	9,135	10,299	11,046	11,969	13,068						
Net debt	2,293	3,030	3,202	3,217	3,118						
Net debt to equity (%)	25.1	29.4	29.0	26.9	23.9						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citi.com
For definitions of the items in this table, please click [here](#).

Stronger Gas Growth in 2013; Raise Target

Towngas China has demonstrated significant earnings and ROE improvement, and we expect TCCL to resume 20+% earnings growth in 2013E, driven by higher gas supplies, especially in the Northeast. We raise our target to HK\$7.20 to reflect our more optimistic long-term earnings growth metric.

New Projects Announced in 1H12

The company has been making consistent progress in new project acquisitions. The four new projects in 1H12 include three city-gas projects in Wafangdian City in Dalian, Binhai Science and Technology Park in Zhaoyuan, Yifeng County in Yichun, and a mid-stream project in Wafangdian.

The three city-gas projects are expected to add 217mn m³ in gas sales in five years' time.

New Growth Catalysts

- Management remains confident it can maintain 15% gas volume growth in 2H12 as a group. Based on our latest discussion, gas volume growth already started to pick up in 3Q12 to 11-12% for TCCL. In the coming year, we believe the company will source from three new gas supplies: 1) West to East Shenyang line, which should commence operation by the end of 2012; 2) Dalian LNG terminal, which will commission in 2012; and 3) Datang coal to gas, which is expected to commission by 2013.
- Slight margin improvement was driven by residential price hikes, and the company expects to complete residential price adjustments for the remaining 8-10 projects in 2H12, which should help TCCL to maintain or slightly improve its dollar margin for gas sales.
- TCCL will focus on acquiring new industrial projects, in which TCCL can obtain a higher shareholding, given its location in new industrial zones. The company also lowered its guidance for capex in 2012, which may be near RMB1 billion as each of the ten projects will likely require capex of around RMB80-100million.
- Gas connections may see sustained demand, driven by government (affordable) housing, which is able to help offset the slowdown in private housing demand. Based on our recent discussion with management, the company should see sustained growth in 2H12, driven by projects that commenced since 2011 and the company does not see a sharp slowdown in 2H12. As for 2013, management is optimistic that connection fees can remain steady compared with 2012, which implies a stable earnings contribution to the company's total net profits.

Figure 1. Towngas China – Key Statistics

	2009	2010	2011	2012E	2013E	2014E
Revenue mix:						
- connection fee (HK\$m)	492	695	1,034	1,112	1,180	1,355
y-y change (%)	9.7	41.4	48.7	7.6	6.1	14.9
- NG sale (HK\$m)	1,534	2,286	3,288	3,824	5,055	6,760
y-y change (%)	30.0	49.1	43.8	16.3	32.2	33.7
Gas sale:						
Piped gas sale (mil m ³)	3,163	4,025	4,670	5,285	6,336	7,684
y-y change (%)	16.8	27.2	16.0	13.2	19.9	21.3
Unit prices:						
Average connection fee per household	3,041	3,041	3,132	3,226	3,323	3,422
y-y change (%)	-8%	0%	3%	3%	3%	3%
Average NG sale price (RMB/m ³)	2.0	2.1	2.2	2.3	2.5	2.6
y-y change (%)	0%	5%	5%	5%	5%	5%

Source: Company data, Citi Research estimates

Revising estimates

We revise our 2012/13/14E EPS estimates by 3%/5%/5% to HK\$0.33/HK\$0.41/HK\$0.49 to reflect stronger gas volume growth and slightly better gas sales margins.

Figure 2. Towngas China – Earnings Revision

	Net Profit (HK\$ mils)		Diluted EPS (HK\$)		
	Old	New	Old	New	% change
2012E	790	810	0.32	0.33	3%
2013E	953	1,001	0.39	0.41	5%
2014E	1,137	1,194	0.46	0.49	5%

Source: Citi Research estimates

Figure 3. Towngas China – Income Statements (HK\$ Millions, 2009-2014E)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Turnover	2,025	2,981	4,321	4,936	6,235	8,115
Gas pipeline construction	492	695	1,034	1,112	1,180	1,355
Piped gas retail sales	1,534	2,286	3,288	3,824	5,055	6,760
Cost of sales	(1,509)	(2,291)	(3,341)	(3,779)	(4,810)	(6,310)
Gross profit	516	690	980	1,157	1,424	1,804
Other income	65	138	195	65	75	83
Operating expenses	(271)	(360)	(394)	(435)	(537)	(691)
EBIT	245	330	586	722	887	1,113
Share of results of associates	137	183	195	273	322	348
Finance costs	(127)	(142)	(142)	(153)	(139)	(130)
Profit before taxation	394	626	1,034	1,181	1,457	1,737
Taxation	(92)	(136)	(257)	(294)	(361)	(430)
Minority interest	(47)	(54)	(68)	(78)	(95)	(113)
Net profits	255	436	709	810	1,001	1,194
% of chg	26.2%	70.7%	62.6%	14.3%	23.6%	19.2%

Source: Company data, Citi Research estimates

Figure 4. Towngas China – Balance Sheet (HK\$ Millions, 2009-2014E)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Bank balances and cash	964	1,429	2,051	2,179	2,164	1,964
Account receivables	98	531	852	1,082	1,366	1,778
Inventories	102	148	388	432	557	744
Others	412	105	207	207	207	207
Current assets	1,575	2,214	3,498	3,900	4,294	4,693
PP&E	4,077	5,074	6,128	6,553	6,907	7,161
Others	2,743	4,032	4,526	4,826	5,156	5,519
Intangible asset	2,935	3,563	4,031	4,031	4,031	4,031
Non-current assets	9,755	12,669	14,685	15,411	16,095	16,711
Total assets	11,330	14,883	18,183	19,311	20,389	21,404
Account payables	215	531	852	932	1,088	1,304
Borrowings - amt due within one year	562	2,792	1,513	1,513	1,513	1,513
Amts due to minority shareholders	83	26	195	195	195	195
Others	1,294	1,352	1,731	1,731	1,731	1,731
Current liabilities	2,153	4,701	4,291	4,371	4,526	4,742
Borrowings - amt due after one year	2,289	1,046	3,593	3,893	3,893	3,593
Non-current liabilities	2,289	1,046	3,593	3,893	3,893	3,593
Total liabilities	4,442	5,748	7,884	8,264	8,420	8,336
Shareholders equity	6,434	8,563	9,615	10,285	11,113	12,099
Minority interest	455	572	684	761	856	969
Total Equity	6,888	9,135	10,299	11,046	11,969	13,068

Source: Company data, Citi Research estimates

Figure 5. Towngas China – Cash flow statement (HK\$ Millions, 2009-2014E)

	FY09	FY10	FY11E	FY12E	FY13E	FY14E
Operating activities						
Profit before tax	394	626	1,034	1,181	1,457	1,737
Adjustments	(110)	(208)	(294)	153	139	130
EBIT	284	418	739	1,334	1,596	1,867
Exceptional items	(1)	0	0	0	0	0
Depreciation & amortisation	185	198	247	580	692	802
Change in working capital	179	204	(80)	(194)	(254)	(384)
Operating cash flow	647	819	906	1,721	2,034	2,286
Interest paid	(122)	(134)	(140)	(153)	(139)	(130)
Taxation	(64)	(84)	(138)	(294)	(361)	(430)
Dividends	(39)	(64)	(123)	(140)	(174)	(207)
Net operating cash flow	461	601	628	1,274	1,534	1,725
Capex	(516)	(719)	(894)	(1,005)	(1,046)	(1,056)
Investing cash flow	(836)	(875)	(1,031)	(1,305)	(1,376)	(1,419)
Free cash flow	(55)	(117)	(266)	269	489	669
Change in equity	14	36	46	0	0	0
Change in debt	461	840	1,044	300	0	(300)
Other investing activities	(320)	(156)	(136)	(300)	(330)	(363)
Financing cash flow	470	693	1,025	160	(174)	(507)
Beginning cash	864	964	1,429	2,051	2,179	2,164
Change in cash	95	420	622	128	(15)	(201)
Effect of foreign exchange rate changes	4	45	0	0	0	0
Ending cash	964	1,429	2,051	2,179	2,164	1,964

Source: Company data, Citi Research estimates

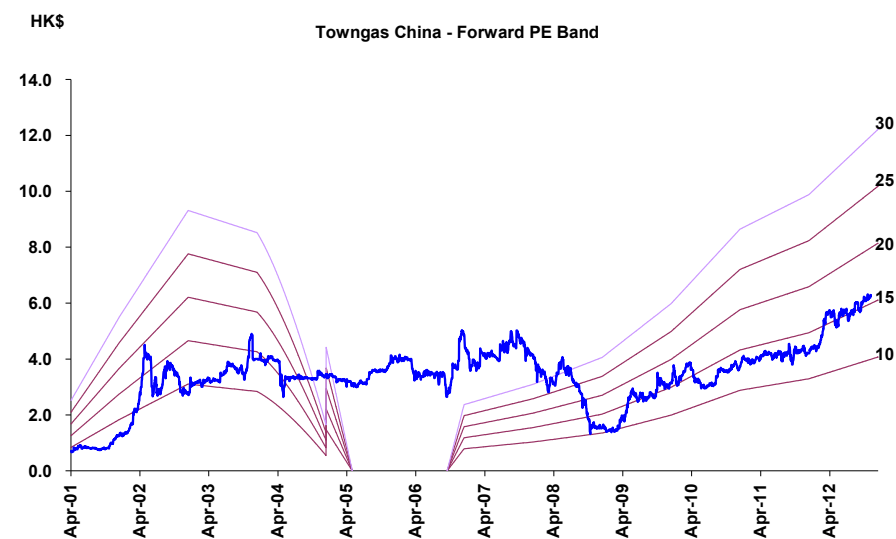
DCF Calculation

Figure 6. Towngas China – DCF Calculation (HKD Millions)

Actual value in the run up to FY15E:	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
FCF		269	489	669	1,107	1,345	1,524	1,909	2,309	2,798
Discount factor		1.00	1.08	1.17	1.27	1.37	1.48	1.60	1.73	1.87
Net debt	-3,250	269	452	572	875	983	1,030	1,192	1,334	1,494
Terminal Value:										
NAV in 2016E		1,539								
Expected terminal growth rate		3.0%								
Expected Capital Employed on new investments		15.0%								
WACC		8.2%								
Terminal Value		23,876								
Discount factor		1.87								
NPV of terminal value		12,751								
NPV in HK\$		17,702								
DCF/share (HK\$)		7.20								

Source: Citi Research estimates

Figure 7. Towngas China – Forward PE Band



Source: Citi Research

Towngas China

Company description

Towngas China focuses on providing city-gas to mainland Chinese cities. Together with Towngas and Hua Yan Water, it formed a significant stake in the Mainland Utilities for Hong Kong and China Gas. The company operates 64 projects in China as of mid-2011, mostly along the coastal region and in Sichuan/Chongqing area. While HKCG operates mainly in Jiangsu, Guangdong area, Towngas China has focused on developments in Sichuan, Shandong and Liaoning, and is expanding its presence in other cities as well.

Investment strategy

We rate Towngas China at Buy (1), expecting it to benefit from organic growth as it raises residential penetration, with potential asset injection from HKCG in the next 1-2 years. Towngas China may benefit from an increase in gas supplies as further long-distance pipelines are connected. With attractive valuation to peers, the stock should outperform, in our view, after completing its shareholder restructuring.

Valuation

Our target price of HK\$7.20 is based on DCF. Our model forecasts earnings growth of 14% and 24% in 2012 and 2013 respectively, and we expect TCCL's free cash flow to improve over the next two years. We have modeled in a similar capex level into 2013E, as the company looks to ramp up capacity, increase its penetration rate in cities, and invest in city-gas project assets from its parent group. We use a WACC of 8.16%, assuming gearing of 50%, cost of equity of 12.6%, equity beta of 1.26, risk-free rate of 2.50% and market risk premium of 8%.

Risks

The main downside risks for TCCL include difficulties in improving operations in its existing city-gas projects, and getting projects from its parent at unattractive pricing. In the past, it paid relatively high earnings multiples for such projects. Other risks include 1) any government measure that affects the company's ability to pass through upstream gas price increases, 2) slower than expected growth in gas volumes, and 3) difficulty in obtaining new projects due to more aggressive bidding by competitors. These risks could impede the stock from achieving our target price.

Xinjiang Goldwind Science & Technology (2208.HK)

Seasonal Recovery in 4Q12E Not Sustainable into 1H13E

Sell	3
Price (28 Nov 12)	HK\$3.01
Target price	HK\$2.30
from HK\$2.20	
Expected share price return	-23.6%
Expected dividend yield	0.8%
Expected total return	-22.8%
Market Cap	HK\$8,111M
	US\$1,047M

Price Performance

(RIC: 2208.HK, BB: 2208 HK)



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■ **Summary** – We assume an L-shaped recovery of the PRC WTG industry with limited sales volume growth in 2013 and low profit margins on WTG sales amid over-capacity in China. Goldwind's profitability in 4Q12E might have some recovery driven by increased sales volume for seasonal reasons and disposal gains from wind farm sales. These positive factors look unsustainable in 1H13E. Valuations are unattractive at 21.7x FY13E PER. Maintain Sell.

■ **More sales volume in 4Q12E for seasonal reasons** – We calculate Goldwind's sales volume -41% yoy to 1,355MW in 9M12, including -40% yoy to 634MW in 3Q12, on (i) no new approvals and slow construction in grid curtailment areas; & (ii) high gearing of PRC wind farm operators. It often sells more in 4Q for installation before winter. Goldwind plans to sell 2,500MW in FY12, with 1,145MW (+42% yoy) in 4Q12, which looks too high to us. And we expect limited volume growth yoy in FY13.

■ **100MW wind farm sales in 4Q12E** – Goldwind plans to sell 100MW wind farms every six months (100MW in 1H11, 140MW in 2H11 & 118MW in 1H12). With no such sales in 3Q12, we assume 100MW to be sold in 4Q12, making a Rmb100m disposal gain, vs. Rmb302m in 1H12. Profit from wind farm sales is hard to predict.

■ **Industry cycle trough, but no rise yet** – WTG prices have bottomed with most makers at losses. But overproduction persists with >40% PRC idle capacity. We have raised our FY12-14E net profit by 2-11% for a mild margin uplift assuming less competition after the exit of some inefficient and small PRC WTG makers.

■ **Unattractive valuation** – Goldwind's 21.7x FY13E PER looks high. We view 0.5x FY13E PB as unattractive against its low ROE of 2.3% in the same year.

Statistical Abstract

Year to	Net Profit	Diluted EPS	EPS growth	P/E	P/B	ROE	Yield
31 Dec	(RmbM)	(Rmb)	(%)	(x)	(x)	(%)	(%)
2010A	2,290	0.849	9.0	2.8	0.5	24.8	14.1
2011A	607	0.225	-73.5	10.7	0.5	4.6	2.1
2012E	249	0.092	-59.0	26.2	0.5	1.9	0.8
2013E	301	0.112	20.9	21.7	0.5	2.3	1.0
2014E	526	0.195	75.0	12.4	0.5	3.9	1.8

Source: Powered by dataCentral

2208.HK: Fiscal year end 31-Dec						Price: HK\$3.01; TP: HK\$2.30; Market Cap: HK\$8,111m; Recomm: Sell					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	17,475	12,756	10,674	11,558	13,258	PE (x)	2.8	10.7	26.2	21.7	12.4
Cost of sales	-13,454	-10,699	-9,054	-9,734	-10,933	PB (x)	0.5	0.5	0.5	0.5	0.5
Gross profit	4,021	2,057	1,620	1,825	2,325	EV/EBITDA (x)	0.7	2.8	8.4	9.5	8.2
Gross Margin (%)	23.0	16.1	15.2	15.8	17.5	FCF yield (%)	-36.3	-109.5	-42.5	-40.8	-18.2
EBITDA (Adj)	3,039	1,162	961	1,182	1,662	Dividend yield (%)	14.1	2.1	0.8	1.0	1.8
EBITDA Margin (Adj) (%)	17.4	9.1	9.0	10.2	12.5	Payout ratio (%)	40	22	22	22	22
Depreciation	-167	-197	-296	-386	-476	ROE (%)	24.8	4.6	1.9	2.3	3.9
Amortisation	0	0	0	0	0						
EBIT (Adj)	2,872	965	665	796	1,186	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT Margin (Adj) (%)	16.4	7.6	6.2	6.9	8.9	EBITDA	3,039	1,162	961	1,182	1,662
Net interest	-88	-204	-391	-448	-500	Working capital	-2,186	-4,674	-660	-749	330
Associates	16	103	96	106	116	Other	-667	-621	-71	-93	-179
Non-op/Except	0	0	0	0	1	Operating cashflow	186	-4,133	230	340	1,813
Pre-tax profit	2,800	864	370	453	803	Capex	-2,552	-3,006	-3,000	-3,000	-3,000
Tax	-416	-146	-81	-104	-192	Net acq/disposals	246	-585	0	0	0
Extraord./Min.Int./Pref.div.	-94	-111	-40	-48	-85	Other	-189	235	66	37	19
Reported net profit	2,290	607	249	301	526	Investing cashflow	-2,495	-3,356	-2,934	-2,963	-2,981
Net Margin (%)	13.1	4.8	2.3	2.6	4.0	Dividends paid	-950	-926	-135	-55	-67
Core NPAT	2,290	607	249	301	526	Financing cashflow	7,279	5,905	-592	459	414
						Net change in cash	4,863	-1,688	-3,296	-2,163	-754
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-2,365	-7,140	-2,770	-2,660	-1,187
Reported EPS (Rmb)	0.849	0.225	0.092	0.112	0.195						
Core EPS (Rmb)	0.849	0.225	0.092	0.112	0.195						
DPS (Rmb)	0.340	0.050	0.020	0.025	0.043						
CFPS (Rmb)	0.069	-1.534	0.085	0.126	0.673						
FCFPS (Rmb)	-0.878	-2.649	-1.028	-0.987	-0.441						
BVPS (Rmb)	4.931	4.777	4.819	4.910	5.080						
Wtd avg ord shares (m)	2,695	2,695	2,695	2,695	2,695						
Wtd avg diluted shares (m)	2,695	2,695	2,695	2,695	2,695						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	63.8	-27.0	-16.3	8.3	14.7						
EBIT (Adj) (%)	41.8	-66.4	-31.1	19.6	49.0						
Core NPAT (%)	31.2	-73.5	-59.0	20.9	75.0						
Core EPS (%)	9.0	-73.5	-59.0	20.9	75.0						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	9,658	7,707	4,411	2,248	1,492						
Accounts receivables	7,583	10,299	12,746	13,802	14,015						
Inventory	4,391	5,148	5,773	6,206	5,473						
Net fixed & other tangibles	4,141	4,893	7,656	10,328	12,910						
Goodwill & intangibles	610	716	658	599	541						
Financial & other assets	2,014	3,666	3,752	3,846	3,950						
Total assets	28,398	32,430	34,994	37,029	38,382						
Accounts payable	8,130	7,581	9,873	10,613	10,423						
Short-term debt	1,502	5,467	1,467	1,467	1,467						
Long-term debt	1,465	2,575	6,575	7,575	8,575						
Provisions & other liab	3,670	3,539	3,657	3,657	3,657						
Total liabilities	14,767	19,162	21,572	23,313	24,123						
Shareholders' equity	13,289	12,874	12,988	13,233	13,692						
Minority interests	342	394	434	483	567						
Total equity	13,631	13,269	13,422	13,716	14,259						
Net debt	-6,691	335	3,631	6,794	8,550						
Net debt to equity (%)	-49.1	2.5	27.1	49.5	60.0						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

3Q12 Results Highlight

Figure 1. Goldwind's 3Q12 results highlight

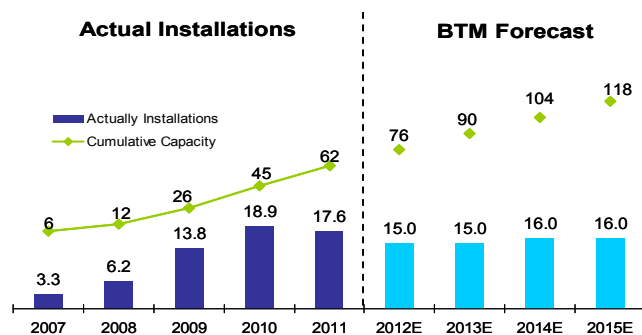
	3Q12	yoy	qoq	9M12	yoy	Comment
Revenue	2,463	-42%	38%	5,946	-37%	* Lowered sales volume of WTGs * Goldwind plans to raise its revenue from overseas from 9.7% in 2011 to 10-15% in 2012E and 30% in 2015E.
Cost of sales	-2,056	-42%	38%	-5,059	-34%	
Business tax and surcharge	-8	14%	57%	-23	51%	* Increase in VAT payment
Gross Profit	399	-38%	38%	865	-50%	
GP margin	16.2%	+1.0ppt	same	14.5%	-4.0ppts	* GP margin +1.0ppt yoy to 16.2% in 3Q12 for (i) most low priced orders were delivered in 4Q11-1Q12; and (ii) cost control
Selling expense	-197	-18%	82%	-424	-34%	* Less transportation, maintenance & spare costs for reduced WTG sales
Administrative expense	-163	-12%	11%	-476	8%	
Asset impairment loss	-10	118%	-76%	-60	19%	* Increased bad debt provisions and slower recovery of accounts receivable
Operating profit	29	-87%	-489%	-96	-116%	
OP margin	1.2%	-3.9ppts	+1.6ppts	-1.6%	-8.0ppts	
Investment income	-6	-107%	-103%	382	21%	* A disposal loss of Rmb6m in 3Q12 from a blade company
	17	-12%	-48%	82	-31%	* Reduction of government subsidies to only Rmb5.5m in 3Q12
Non-operating expense	-1	5%	1%	-4	-35%	
Finance cost	-80	97%	-33%	-295	118%	* Issuance of Rmb3bn corporate bond in March 2012 * Net debt to equity ratio -10ppts to 3% in 3Q12
Profit before tax	-41	-115%	-140%	70	-92%	
Tax	9	-116%	-129%	-22	-88%	
Effective tax rate	21.4%	+1.4ppts	-7.6ppts	31.1%	+10.6ppts	
Profit after tax	-32	-115%	-144%	48	-93%	
MI	-1	-96%	-86%	-10	-89%	
Net profit	-34	-118%	-151%	39	-94%	
NP margin	-1.4%	-5.9ppts	-5.1pts	0.6%	-5.9ppts	
EPS (Rmb)	-0.01	-118%	-151%	0.01	-94%	

Source: Company Reports and Citi Research Estimates

Goldwind reported net loss of Rmb34m in 3Q12, the first loss-making quarter since its listing in Oct 2010. Its net profit was also -94% yoy to Rmb39m in 9M12, close to the low end of its previous profit warning to drop 50-100%. The loss was due to (i) reduced product demand hence lowered sales volume of its WTG (revenue -42% yoy to Rmb2,463m in 3Q12); (ii) lack of disposal gains from wind farm sales in the quarter; and (iii) an increase in finance costs (+97% yoy to Rmb80m in 3Q12) after the issuance of an Rmb3bn corporate bond in Mar 2012.

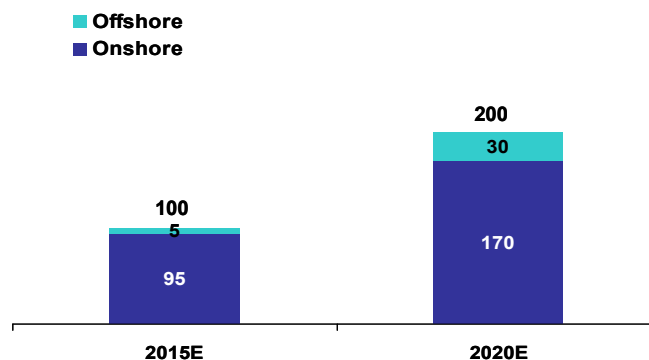
Sales Volume Outlook

Figure 2. China annual installed wind capacity (GW)



Source: Citi Research

Figure 3. Target Connected Wind Capacity (GW)



Source: Citi Research

China had over 55GW grid-connected wind capacity by end 3Q12, up from 44GW by end 2011; in other words, over 11GW were added in 9M12. BTM forecast PRC annual grid connected wind capacity to add 15GW pa in 2012-13E, less than 17.6GW in 2011.

Figure 4. Grid curtailment ratio of wind power by province in 2011

	Real Power Generation (bn KWh)	Abandonment Ratio	Abandonment Volume (bn KWh)
Northeast China Power Grid:			
Heilongjiang	4.4	14.5%	0.7
Jilin	4.0	14.9%	0.7
Liaoning	6.6	9.0%	0.7
East Inner Mongolia	8.8	25.3%	3.0
Sub-total	23.7	17.6%	5.1
North China Power Grid:			
Heibei	8.8	4%	0.4
Beijing	0.3	n/a	n/a
Tianjin	0.1	1%	0.0
Shanxi	1.3	n/a	n/a
Shandong	4.1	n/a	n/a
West Inner Mongolia	13.2	23%	0.4
Sub-total	27.9	14%	4.4
Northwest China Power Grid:			
Shaanxi	0.1	n/a	n/a
Gansu	7.1	27%	2.7
Qinhai	0.0	n/a	n/a
Ningxia	1.1	n/a	n/a
Xinjiang	2.9	3%	0.1
Sub-total	11.2	20%	2.8
Total of Three North Regions	63.5	16%	12.3

Source: Company Reports and Citi Research Estimates

We calculate Goldwind's sales volume -41% yoy to 1,355MW in 9M12, including -40% yoy to 634MW in 3Q12, assuming stable sales prices in 3Q12. Management attributed the reduced sales volume to:

- (i) No new approvals of wind farm construction in such serious grid curtailment provinces as Inner Mongolia, Gansu and those in Northeast China;
- (ii) Construction of wind farms approved in these regions is generally slow in view of grid curtailment and high gearing of PRC wind farm operators; and
- (iii) PRC wind farm operators have become more selective in capacity expansion being more profit-oriented, rather than capacity-focused.

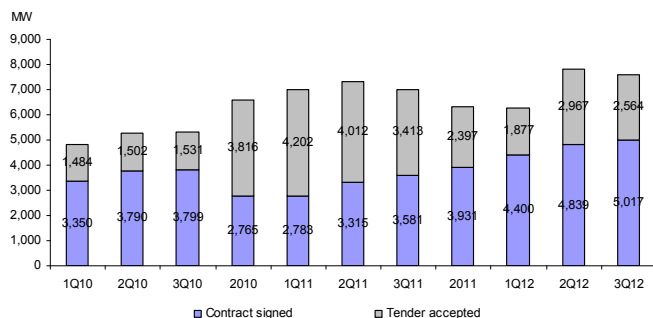
Additionally, Goldwind does not expect fast development of distributed wind farms in view of (i) their small scale; and (ii) the current system will block the development.

Goldwind targets to sell 2,500MW WTGs in 2012E, -20% yoy; we think the target aggressive as it needs to sell 1,145MW WTGs, +42% yoy, in 4Q12E to achieve this target. PRC WTG product demand is constrained by grid connection and curtailment that accounted for over 90% of its sales in 9M12. We assume Goldwind's 2012E sale volume target to be missed.

Additionally, Goldwind sold 101.8MW overseas in 9M12, representing 7.5% its sales volume in the period. It expects 10-15% WTG sales from overseas in 2012, higher than 9.7% in 2011, and targets 30% WTG sales from abroad in 2013-15.

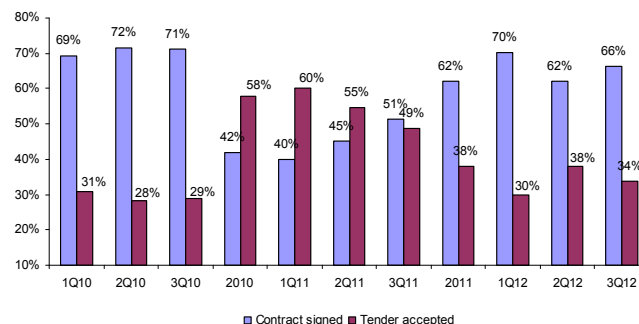
New Order Flow

Figure 5. Goldwind's orders on hand (MW) – period end



Source: Goldwind

Figure 6. Goldwind's orders on hand mix – period end



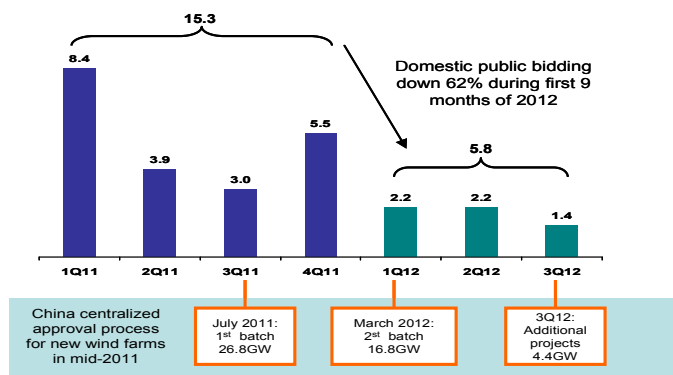
Source: Goldwind

Goldwind's total orders on hand were -2.9% yoy to 7,581MW in 3Q12. The orders on hand include two types:

- (i) Those with contracts signed were 5,017MW (+3.7% qoq) by end 3Q12, including 79% being 1.5MW units, 20% being 2.5MW units and 1% being 3.0MW units. Geographically, they were 93% from China and 7% (+4ppts yoy) from overseas; and
- (ii) Those only with tender accepted but without contract signed were 2,564MW (-13.6% qoq) by end 3Q12.

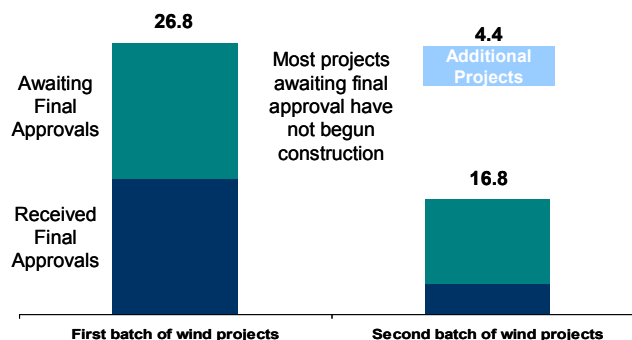
The orders on hand could take long time to be delivered. As per our understanding from some PRC wind-farm operators such as Huaneng Renewables, they have placed their orders in batches for their demand in coming years. The orders placed depend on the on-going development progress of the planned projects, hence could not be ascertained previously. And, generally speaking, no customer deposit has been paid for these orders on hand amid excessive WTG production capacity in China.

Figure 7. Quarterly Domestic Public Tenders (GW)



Source: Goldwind and Citi Research

Figure 8. Domestic Wind Farms Projects Proposed by NEA (GW)



Source: Goldwind and Citi Research

NDRC's approval of new wind farm projects has been slowing down. According to China Wind Energy Association, this was in view of:

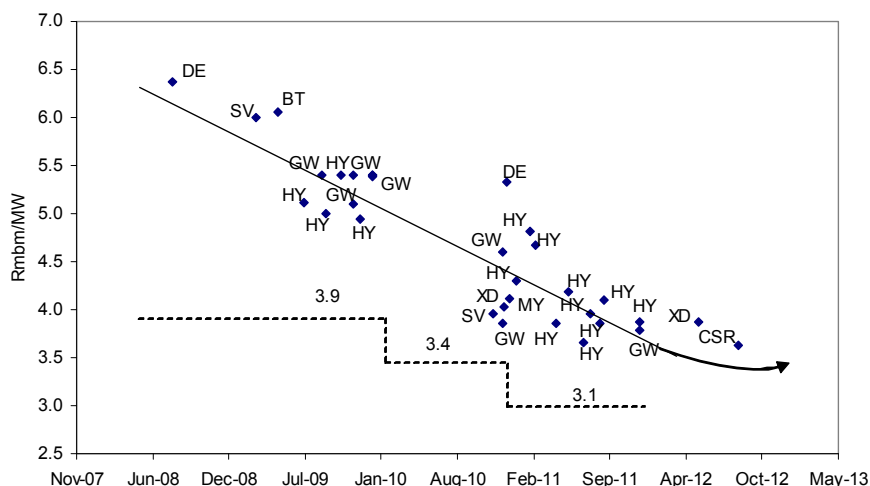
- (iv) Tightened credit for PRC wind-farm operators;

- (v) Elimination of unqualified and small PRC WTG makers; and
- (vi) Slowdown in new installation to ease grid curtailment – according to Goldwind, 80% wind turbine generator orders tendered in China in 9M12 are for installation at low wind speed regions.

The association expects more regulatory approval in 2014-15E when the small and inefficient WTG manufacturers have exited and grid curtailment might be mitigated after the completion of new ultra-high voltage transmission lines.

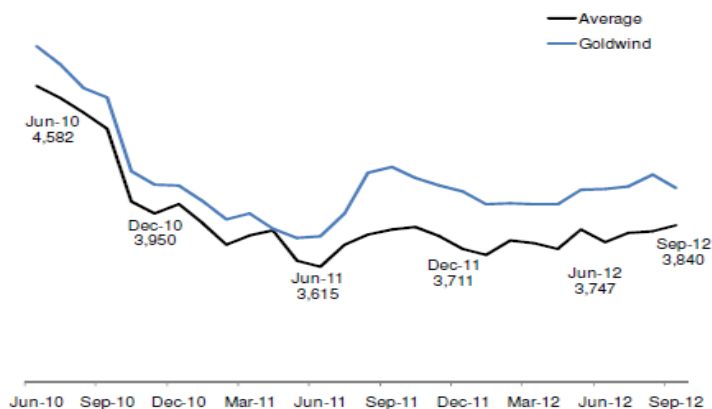
Sales Prices Outlook

Figure 9. PRC WTG bidding prices (Rmbm/MW)



Source: Dongfang, Sinovel, Baoding Tianwei, Huayi Electric, Goldwind, Mingyang, China Hydro, XEMC

Figure 10. Average Domestic Public Bidding Price (RMB/kW)



Source: Goldwind and Citi Research

Market WTG sales prices have seen some recovery in 2012, from the trough of Rmb3,200-3,400/kW, but production and labor costs have also gone up, leading to WTG sales in 3Q12 making a loss. Goldwind claimed its sales prices are 5-7% above those of its peers, but its production cost is also higher as it uses direct-drive technology.

Improved Operating Cashflow

Figure 11. Goldwind's cashflow statement (extract)

(Rmbm)	1Q12	2Q12	3Q12	YoY	9M12	YoY
Cashflow from operating activities	-1,587	971	1,912	-187%	1,295	-119%
Cashflow from investing activities	-332	-766	-502	-33%	-1,601	-13%
Cashflow from financing activities	1,356	-1,699	-1,283	-192%	-1,626	-137%
Total	-563	-1,495	127	-108%	-1,931	-54%

Source: Company Reports and Citi Research Estimates

Goldwind had positive operating cash inflow of Rmb1.9bn in 3Q12 and Rmb1.3bn in 9M12, improving on the operating cash outflow of the same periods last year. We attribute the improvement mostly to an increase in bills payable (+34% qoq to Rmb3.40bn in 3Q12) and trade payables (+11% yoy to Rmb6.16bn in 3Q12). We are uncertain if this improvement is sustainable. Its cash outflow of investing activities was -33% yoy to Rmb502m in 3Q11 with no capacity expansion this year in view of weak product demand. Some short-term debts were repaid, lowering its net debt to equity ratio by 10ppts to 3% in 3Q12.

Figure 12. Goldwind's net profit revision – New vs. Old

	New	Old	% change
Revenue (Rmbm)			
FY12E	10,674	10,674	0.0%
FY13E	11,558	11,503	0.5%
FY14E	13,258	13,194	0.5%
Gross profit (Rmbm)			
FY12E	1,620	1,457	11.2%
FY13E	1,825	1,644	11.0%
FY14E	2,325	2,122	9.6%
GP margin			
FY12E	15.2%	13.7%	1.5%
FY13E	15.8%	14.3%	1.5%
FY14E	17.5%	16.1%	1.5%
Operating profit (Rmbm)			
FY12E	665	613	8.5%
FY13E	796	734	8.4%
FY14E	1,186	1,103	7.5%
OP margin			
FY12E	6.2%	5.7%	0.5%
FY13E	6.9%	6.4%	0.5%
FY14E	8.9%	8.4%	0.6%
Net profit (Rmbm)			
FY12E	249	243	2.3%
FY13E	301	275	9.3%
FY14E	525	474	10.8%
NP margin			
FY12E	2.3%	2.3%	0.1%
FY13E	2.6%	2.4%	0.2%
FY14E	4.0%	3.6%	0.4%
EPS (Rmb)			
FY12E	0.09	0.09	2.5%
FY13E	0.11	0.10	9.4%
FY14E	0.19	0.18	10.7%
DCF (HK\$)	2.30	2.20	4.5%
Rating	SELL	SELL	

Source: Company Reports and Citi Research Estimates

Figure 13. Goldwind's net profits – Citi vs. consensus

Net profit (Rmbm)	CIR	Consensus	% diff
FY12E	249	202	23.5%
FY13E	301	301	0.1%
FY14E	525	398	32.0%

Source: Bloomberg and Citi Research Estimates

Net Profit Sensitivity Table

Figure 14. Goldwind's net profit sensitivity

	FY12E NP 249	FY13E NP 301
Base case (Rmbm)		
Sale volume		
- sale volume assumption in base case (MW)	2,300	2,456
- NP chg if sale volume 5% above base case	+14%	+13%
- NP chg if sale volume 5% below base case	-14%	-13%
Average sale price		
- average sale price assumption in base case (yoy)	4.3	4.3
- NP chg if average sale price 5% above base case	+131%	+115%
- NP chg if average sale price 5% below base case	-131%	-115%
Average component costs		
- average component cost assumption in base case (Rmb/MW)	3.2	3.2
- NP chg if component cost 5% above base case	-103%	-90%
- NP chg if component cost 5% below base case	+103%	+90%
Warranty provision		
- amount, included in selling & distribution expenses (Rmbm)	675	816
- NP chg if warrant provision 1ppt above base case	-27%	-23%
- NP chg if warrant provision 1ppt below base case	+27%	+23%
Rmb borrowing rate		
- Rmb borrowing rate assumption in base case	6.0%	6.0%
- NP chg if component cost 1ppt above base case (Rmbm)	-22%	-19%
- NP chg if component cost 1ppt below base case (Rmbm)	+22%	+19%
Source: Company Reports and Citi Research Estimates		

Goldwind's net profit is most sensitive to average sales prices and its production cost due to its thin net margin (2.3-2.4% in FY12-13E).

DCF & Historical Valuations

Figure 15. Goldwind's DCF Calculation

(Rmbm)	2012	2013	2014	2015	2016	2017	2018	2019	2020
Net operating cashflow		340	1,812	3,033	3,372	1,628	1,818	1,903	1,993
Less: capex		-3,000	-3,000	-700	-700	-700	-700	-700	-700
FCF		-2,660	-1,188	2,333	2,672	928	1,118	1,203	1,293
Discount factor		1.00	1.10	1.20	1.32	1.44	1.58	1.73	1.89
Net cash by end 2011E / NAV from 2012E	-3,742	-2,660	-1,085	1,944	2,032	644	708	696	682

DCF - Terminal Value

NAV in 2021E	1,345
Expected terminal growth rate	4.0%
Capital Employed on new capex	8.0%
WACC	9.6%
Terminal Value	12,094
Discount factor	2.08
NPV of terminal value	5,826

Total DCF

DCF - In the run up to 2020E	-780
DCF from 2020E	5,826
Total DCF	5,047
Exchange Rate	1.23
Total DCF (HK\$m)	6,199
No. of share (m)	2,695
DCF / share (HK\$)	2.30

Source: Company Reports and Citi Research Estimates

Figure 16. Goldwind's historical PE chart



Source: Citi Research

Figure 17. Goldwind's historical PB chart



Source: Company Reports and Citi Research Estimates

Goldwind's PER is high due to its depressed earnings, which we do not expect to improve in 2H12-FY13E. The company therefore deserves to trade at a low PB in our view.

Financial Statement & Key Assumptions

Figure 18. Consolidated Income Statement

Year ended Dec (Rmb m)	FY10	FY11	FY12E	FY13E	FY14E
Wind turbine & component sales	17,005	11,700	10,009	10,745	12,275
Wind power services	293	363	456	563	708
Wind power generation	178	139	208	250	275
Total revenue	17,475	12,756	10,674	11,558	13,258
- Raw material and components	-14,688	-10,502	-7,676	-8,193	-9,135
- Labor	-113	-121	-86	-92	-102
- Depreciation and amortization	-122	-121	-86	-92	-102
- Other production costs	-316	-544	-388	-415	-462
- Inventory changes and transferred fixed assets	1,784	846	603	644	718
Wind turbine sales - component	n/a	-257	-1,020	-1,095	-1,251
Wind power services	n/a	n/a	-297	-366	-460
Wind power generation	n/a	n/a	-104	-125	-137
Total cost of sales	-13,454	-10,699	-9,054	-9,734	-10,933
Gross profit	4,021	2,057	1,620	1,825	2,325
<i>GP margin</i>	<i>23.0%</i>	<i>16.1%</i>	<i>15.2%</i>	<i>15.8%</i>	<i>17.5%</i>
Other income	637	716	637	649	662
Selling and distribution expenses	-1,096	-990	-675	-711	-779
Administrative expenses	-418	-739	-798	-838	-880
Other operating expenses	-272	-79	-118	-130	-143
Operating Profit	2,872	965	665	796	1,186
<i>OP margin</i>	<i>16.4%</i>	<i>7.6%</i>	<i>6.2%</i>	<i>6.9%</i>	<i>8.9%</i>
Interest income	29	54	66	37	19
Interest expense	-117	-258	-457	-486	-519
Jointly controlled entities	5	2	10	11	12
Associates	11	101	86	94	104
Profit before tax	2,800	864	370	453	802
Tax	-416	-146	-81	-104	-192
Tax rate	14.9%	16.9%	22.0%	23.0%	24.0%
Minority interests	-94	-111	-40	-48	-85
Net profit	2,290	607	249	301	525
<i>NP margin</i>	<i>13.1%</i>	<i>4.8%</i>	<i>2.3%</i>	<i>2.6%</i>	<i>4.0%</i>
<i>NP yoy</i>	<i>31%</i>	<i>-74%</i>	<i>-59%</i>	<i>21%</i>	<i>75%</i>

Source: Company Reports and Citi Research Estimates

Figure 19. Key Ratios

	FY10	FY11	FY12E	FY13E	FY14E
Profitability					
Gross margin	23.0%	16.1%	15.2%	15.8%	17.5%
EBITDA margin	17.4%	9.1%	9.0%	10.2%	12.5%
Operating margin	16.4%	7.6%	6.2%	6.9%	8.9%
PBT margin	16.0%	6.8%	3.5%	3.9%	6.0%
Net margin	13.1%	4.8%	2.3%	2.6%	4.0%
Average working capital days					
Inventory	98	163	233	233	183
Receivables	110	256	436	436	386
Payables	161	268	398	398	348
Working capital cycle	47	151	271	271	221

Source: Company Reports and Citi Research Estimates

Figure 20. Consolidated Balance Sheet

Year to 31 Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Property, plants and equipments	3,783	4,580	7,342	10,015	12,597
Investment properties	88	85	85	85	85
Prepaid land lease payments	260	215	215	215	215
Goodwill	257	326	326	326	326
Other intangible assets	353	390	332	273	215
Interests in jointly-controlled entities	173	393	393	393	393
Interests in associates	90	253	339	433	537
Available-for-sale investments	60	258	258	258	258
Deferred tax assets	399	465	465	465	465
Others	10	33	33	33	33
Prepayment	88	65	65	65	65
Total LT assets	5,562	7,063	9,853	12,562	15,189
Inventory	4,391	5,148	5,773	6,206	5,473
Trade and bills receivables	7,583	10,299	12,746	13,802	14,015
Prepayment, deposits and other receivables	1,204	2,212	2,212	2,212	2,212
Pledged bank deposits	335	110	110	110	110
Cash and cash equivalents	9,324	7,597	4,301	2,137	1,382
Current assets	22,836	25,367	25,141	24,467	23,192
Total assets	28,398	32,430	34,994	37,029	38,382
Trade and bills payables	8,130	7,581	9,873	10,613	10,423
Other payables	1,952	1,952	1,952	1,952	1,952
Derivative financial instruments	8	1	1	1	1
Short term borrowings	1,502	5,467	1,467	1,467	1,467
Tax payable	423	119	119	119	119
Provision	442	593	711	711	711
Current liabilities	12,456	15,713	14,123	14,864	14,674
Long term borrowings	1,465	2,575	6,575	7,575	8,575
Deferred tax liabilities	69	44	44	44	44
Provision	574	677	677	677	677
Government grants	187	153	153	153	153
Other long-term liabilities	15	0	0	0	0
Total non current liabilities	2,311	3,449	7,449	8,449	9,449
Minority interests	342	394	434	483	567
Issued share capital	2,695	2,695	2,695	2,695	2,695
Reserves	9,678	10,045	10,293	10,539	10,997
Proposed final dividend	916	135	0	0	0
Shareholders equity	13,289	12,874	12,988	13,233	13,692
Total equity and liability	28,398	32,430	34,994	37,029	38,382

Source: Company Reports and Citi Research Estimates

Figure 21. Consolidated Cash Flow Statement

Year ended Dec (Rmbm)	FY10	FY11	FY12E	FY13E	FY14E
Profit before tax	2,800	864	370	453	802
Finance costs	117	258	457	486	519
Foreign exchange difference	40	11	0	0	0
Interest income	-8	-25	-66	-37	-19
Profit of associates	-16	-103	-86	-94	-104
Depreciation	113	138	238	328	418
Amortization	54	58	58	58	58
Gain on disposal of subsidiaries	-415	-294	0	0	0
Impairment of trade and other receivables	187	17	0	0	0
Others	-50	-38	0	0	0
Sub-total	2,794	778	972	1,193	1,674
Change in working capital					
Increase in inventories	-1,443	-780	-624	-433	733
Increase in trade debtors	-4,416	-2,816	-2,446	-1,056	-213
Increase in other debtors, deposits and prepayments	-340	-357	0	0	0
Increase in trade creditors	3,638	-182	2,292	741	-190
Inc (Dec) in other creditors and accrued charges	-210	-792	0	0	0
Increase in provision	584	254	119	0	0
Sub-total	-2,186	-4,674	-660	-749	330
Cash from operations	608	-3,896	311	445	2,004
Income tax paid	-422	-237	-81	-104	-192
Net operating cashflow	186	-4,133	230	340	1,812
Capex	-2,552	-3,006	-3,000	-3,000	-3,000
Free cash flow	-2,365	-7,140	-2,770	-2,660	-1,188
Additions of prepaid land lease payments	-31	-11	0	0	0
Additions of other intangible assets	-84	-206	0	0	0
Acquisitions of subsidiaries	64	-325	0	0	0
Purchase of jointly controlled entities	-25	-147	0	0	0
Purchase of associates	-23	-37	0	0	0
Purchase of available-for-sale investments	-3	-198	0	0	0
Disposals of subsidiaries	213	199	0	0	0
Disposals of associates	16	-276	0	0	0
Decrease / (increase) in pledged deposits	-116	318	0	0	0
Others	45	331	66	37	19
Net cash from investing activities	-2,495	-3,356	-2,934	-2,963	-2,981
Sub-total	-2,309	-7,490	-2,704	-2,622	-1,170
Borrowings	3,659	11,883	0	1,000	1,000
Repayment	-2,023	-4,906	0	0	0
Interest paid	-153	-252	-457	-486	-519
Proceeds from issue of shares	7,039	0	0	0	0
Share issuance expense	-269	0	0	0	0
Capital contribution from minority shareholders	-24	105	0	0	0
Dividend paid to equity holders	-924	-916	-135	-55	-67
Dividend paid to minority shareholders	-26	-10	0	0	0
Net cash from financing activities	7,279	5,905	-592	459	414
Foreign exchange changes	-106	-103	0	0	0
Change in cash and cash equivalent	4,863	-1,688	-3,296	-2,163	-755
Cash, year start	4,379	9,242	7,555	4,258	2,095
Cash, year end	9,242	7,555	4,258	2,095	1,340
Cash, year start	9,324	7,597	4,301	2,137	1,382
Non-pledged deposits	-81	-42	-42	-42	-42
Cash, year end	9,242	7,555	4,258	2,095	1,340

Source: Company Reports and Citi Research Estimates

Figure 22. Key assumptions

Year ended 31-Dec	FY10	FY11	FY12E	FY13E	FY14E
Sales Price (Rmbm/MW, incl. 17% VAT)					
- 0.75MW unit	4.1	3.9	3.9	3.9	4.0
- 1.5MW unit	4.9	4.4	4.3	4.3	4.4
yoy	-22%	-10%	-3%	0%	2%
- 2.5MW unit	5.8	4.5	4.5	4.5	4.6
yoy	n/a	-22%	0%	0%	2%
- 3.0MW unit	n/a	4.5	4.7	4.7	4.8
- 5.0MW unit	n/a	n/a	4.5	4.5	4.6
- all units	4.9	4.2	4.3	4.3	4.4
Sales volume (unit)					
- 0.75MW unit	205	1	0	0	0
- 1.5MW unit	2,567	1,921	1,479	1,571	1,759
- 2.5MW unit	1	87	30	32	36
- 3.0MW unit	0	2	2	3	3
- 5.0MW unit	0	0	0	3	3
Total	2,773	2,011	1,511	1,608	1,801
Sales volume (MW)					
- 0.75MW unit	154	1	0	0	0
- 1.5MW unit	3,851	2,882	2,218	2,356	2,639
- 2.5MW unit	3	218	75	80	89
- 3.0MW unit	0	6	7	8	8
- 5.0MW unit	0	0	0	13	14
Total	4,007	3,106	2,300	2,456	2,751
yoy	97%	-22%	-26%	7%	12%
Revenue mix					
Wind turbine sales					
- 0.75MW unit	637	3	0	0	0
- 1.5MW unit	18,829	12,679	9,465	10,107	11,546
- 2.5MW unit	14	979	338	360	412
- 3.0MW unit	0	23	33	35	40
- 5.0MW unit	0	0	0	56	64
- intersegment sales	-2,476	-1,428	-1,027	-1,102	-1,259
Sub-total	17,005	12,254	8,809	9,457	10,804
Wind power components	n/a	554	1,200	1,288	1,472
Wind power services	293	363	456	563	708
Wind power generation	178	139	208	250	275
Total	17,475	13,310	10,674	11,558	13,258
Cost of sales per MW (ex-VAT)	3.4	3.1	3.2	3.2	3.2
yoy	-14%	-8%	5%	0%	0%
Warranty provision against WTG sales	4.2%	3.0%	3.0%	3.0%	3.0%
GP margin					
- 0.75MW unit	27.7%	32.9%	n/a	n/a	n/a
- 1.5MW unit	22.9%	14.6%	12.2%	12.6%	14.3%
- 2.5MW unit	8.5%	3.0%	7.4%	9.7%	14.2%
- 3.0MW unit	n/a	-15.3%	-0.9%	4.1%	10.7%
- 5.0MW unit	n/a	n/a	n/a	-0.7%	6.2%
Wind turbine sales - component	0.0%	0.0%	15.0%	15.0%	15.0%
Wind power services	0.0%	0.0%	35.0%	35.0%	35.0%
Wind power generation	0.0%	0.0%	50.0%	50.0%	50.0%
Company average	23.0%	16.1%	15.2%	15.8%	17.5%

Source: Company Reports and Citi Research Estimates

Xinjiang Goldwind Science & Technology

Company description

Goldwind was founded in February 1998 and was ranked the second-largest wind turbine maker in China in 2009 in terms of newly installed wind capacity. The company was listed on Shenzhen Stock Exchange on 26 Dec 2007 and then on the Hong Kong Stock Exchange on 8 Oct 2010.

Investment strategy

We rate Goldwind shares Sell (3). The company faces lower sales volume and selling prices amid competition, industry overcapacity and slower growth in China's wind power capacity additions. The company is also vulnerable to higher steel costs, and will likely find it difficult to pass on incremental costs to customers in a competitive environment. Despite the significant share price fall since its listing in October 2010, we expect the de-rating to continue.

Valuation

Our target price of HK\$2.20 is based on DCF. We use DCF in deriving a fair value range for Goldwind as we believe the company, as a leading wind turbine maker, does not face any significant issues as a long-term going concern and will operate for a long period ahead. We think DCF is the appropriate method to capture the long-term potential return on the company. Assumptions in our DCF include: 1) WACC of 9.2%; 2) equity beta of 0.93; 3) risk-free rate of 5.2%; 4) market risk premium of 6.8%; 5) gearing ratio of 33.3%; 6) tax rate of 15% and terminal growth rate of 4%. At our target price, FY12E P/B is 0.4x.

Risks

Key upside risks that could sustain the shares above our target price include: 1) Higher-than-expected new orders with unexpected favorable government policies; and 2) Higher-than-expected margin with greater sales in high-margin products.

Yingde Gases (2168.HK)

Weak Merchant Market Likely to Drive Weaker 2H12 Earnings

Neutral	2
Price (28 Nov 12)	HK\$7.58
Target price	HK\$7.00
Expected share price return	-7.7%
Expected dividend yield	2.1%
Expected total return	-5.5%
Market Cap	HK\$13,696M
	US\$1,767M

Price Performance (RIC: 2168.HK, BB: 2168 HK)



■ **Maintain Neutral** — We maintain our cautious view on Yingde Gases and target price of HK\$7.00. While Yingde's key customers are able to maintain production, a tough merchant gas market and uncertainty with smaller steel makers remain an overhang, and we view its share price at 13.9x 2013 PE as fairly valued.

■ **Weak merchant sales** — Yingde faced weak merchant sales in 1H12, which caused its profitability to fall 20% Y/Y. We believe 2H12 merchant gas sales will remain under pressure, given customers, especially steel and solar equipment producers, face challenging market conditions due to weak pricing.

■ **Sun is shining... in some places** — We visited Yingde's plant in Rizhao, China in mid-Oct. Yingde's plant in Rizhao was running at full utilization, servicing mainly the on-site steel producer. Yingde's management remains optimistic on gas usage from most on-site customers and is also confident to add oxygen capacity of 200,000 Nm³/hour per year over the next 2-3 years.

■ **Potential catalysts** — 1) The company's chemical project in Inner Mongolia, which accounts for ~25% of its capacity. It faced a one-month shutdown in 2H12, but has since resumed operation; 2) Merchant gas sales volume is 90% driven from direct sales, which suggests weak distributor demand. Argon ASP has remained at trough since May but could see better demand as China sees a rise in domestic solar projects; 3) Several projects that fell below minimum take-or-pay (MTO) in May have recovered, and a sustained recovery could drive stronger gas sales.

■ **Developing profitable, long-term opportunities** — Yingde has been expanding its operation to develop merchant customers, and if successful, should drive more sustainable gas sales. The company has been able to secure a strong pipeline, that should support its capacity growth in 2013-14E. As for new supply contracts, given weaker visibility for its customers, we believe the company may face more competitive pricing, which will lead to a lower project return metric.

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Statistical Abstract

Year to 31 Dec	Net Profit (RmbM)	Diluted EPS (Rmb)	EPS growth (%)	P/E (x)	P/B (x)	ROE (%)	Yield (%)
2010A	577	0.319	-16.0	19.1	2.5	14.2	1.6
2011A	831	0.459	43.9	13.3	2.2	17.8	2.1
2012E	652	0.361	-21.5	16.9	2.1	12.8	2.6
2013E	791	0.437	21.2	13.9	2.0	15.0	3.2
2014E	936	0.517	18.3	11.8	1.9	16.8	4.0

Source: Powered by dataCentral

2168.HK: Fiscal year end 31-Dec						Price: HK\$7.58; TP: HK\$7.00; Market Cap: HK\$13,696m; Recomm: Neutral					
Profit & Loss (Rmbm)	2010	2011	2012E	2013E	2014E	Valuation ratios	2010	2011	2012E	2013E	2014E
Sales revenue	3,005	4,240	4,703	5,791	7,072	PE (x)	19.1	13.3	16.9	13.9	11.8
Cost of sales	-1,841	-2,789	-3,333	-4,102	-5,060	PB (x)	2.5	2.2	2.1	2.0	1.9
Gross profit	1,164	1,451	1,370	1,689	2,011	EV/EBITDA (x)	11.1	8.2	10.6	9.1	8.4
Gross Margin (%)	38.7	34.2	29.1	29.2	28.4	FCF yield (%)	-10.9	-9.7	-12.5	-11.5	-7.9
EBITDA (Adj)	945	1,371	1,177	1,572	1,900	Dividend yield (%)	1.6	2.1	2.6	3.2	4.0
EBITDA Margin (Adj) (%)	31.5	32.3	25.0	27.1	26.9	Payout ratio (%)	31	28	43	45	47
Depreciation	-159	-279	-235	-376	-460	ROE (%)	14.2	17.8	12.8	15.0	16.8
Amortisation	0	0	0	0	0	Cashflow (Rmbm)	2010	2011	2012E	2013E	2014E
EBIT (Adj)	786	1,091	942	1,195	1,440	EBITDA	945	1,371	1,177	1,572	1,900
EBIT Margin (Adj) (%)	26.2	25.7	20.0	20.6	20.4	Working capital	-220	-162	-83	-64	-125
Net interest	-118	-110	-143	-203	-249	Other	7	-250	-240	-366	-474
Associates	0	0	0	0	0	Operating cashflow	732	959	854	1,142	1,301
Non-op/Except	0	0	0	0	0	Capex	-1,930	-2,030	-2,230	-2,410	-2,169
Pre-tax profit	668	981	799	993	1,191	Net acq/disposals	0	0	0	0	0
Tax	-135	-146	-143	-197	-250	Other	-510	-159	-143	-146	-149
Extraord./Min.Int./Pref.div.	44	-5	-4	-5	-6	Investing cashflow	-2,440	-2,189	-2,373	-2,557	-2,318
Reported net profit	577	831	652	791	936	Dividends paid	0	-181	-235	-282	-353
Net Margin (%)	19.2	19.6	13.9	13.7	13.2	Financing cashflow	593	1,201	1,156	997	981
Core NPAT	577	831	652	791	936	Net change in cash	-1,132	-30	-363	-418	-37
Per share data	2010	2011	2012E	2013E	2014E	Free cashflow to s/holders	-1,197	-1,071	-1,375	-1,269	-868
Reported EPS (Rmb)	0.319	0.459	0.361	0.437	0.517						
Core EPS (Rmb)	0.319	0.459	0.361	0.437	0.517						
DPS (Rmb)	0.100	0.130	0.156	0.195	0.244						
CFPS (Rmb)	0.405	0.530	0.472	0.631	0.719						
FCFPS (Rmb)	-0.662	-0.592	-0.760	-0.701	-0.480						
BVPS (Rmb)	2.406	2.759	2.870	2.976	3.163						
Wtd avg ord shares (m)	1,809	1,809	1,809	1,809	1,809						
Wtd avg diluted shares (m)	1,809	1,809	1,809	1,809	1,809						
Growth rates	2010	2011	2012E	2013E	2014E						
Sales revenue (%)	45.5	41.1	10.9	23.1	22.1						
EBIT (Adj) (%)	26.7	38.8	-13.6	26.9	20.5						
Core NPAT (%)	8.9	43.9	-21.5	21.2	18.3						
Core EPS (%)	-16.0	43.9	-21.5	21.2	18.3						
Balance Sheet (Rmbm)	2010	2011	2012E	2013E	2014E						
Cash & cash equiv.	1,438	1,117	737	319	282						
Accounts receivables	749	1,029	1,147	1,428	1,744						
Inventory	30	32	48	81	100						
Net fixed & other tangibles	4,847	6,069	8,063	10,097	11,807						
Goodwill & intangibles	13	13	14	14	15						
Financial & other assets	558	1,564	1,408	1,268	1,331						
Total assets	7,635	9,825	11,415	13,207	15,278						
Accounts payable	1,244	1,363	1,355	1,667	2,057						
Short-term debt	784	1,116	1,339	1,406	1,476						
Long-term debt	1,132	2,085	3,235	4,435	5,685						
Provisions & other liab	108	258	279	294	311						
Total liabilities	3,268	4,821	6,207	7,802	9,529						
Shareholders' equity	4,353	4,991	5,193	5,384	5,723						
Minority interests	14	12	16	21	27						
Total equity	4,367	5,003	5,209	5,405	5,750						
Net debt	478	2,083	3,837	5,522	6,878						
Net debt to equity (%)	10.9	41.6	73.7	102.2	119.6						

For further data queries on Citi's full coverage universe please contact Citi Research Data Services at CitiRsch.DataServices.Global@citigroup.com
For definitions of the items in this table, please click [here](#).

Customers Remain Under Pressure

A number of Yingde's customers remain under margin pressure, given the challenging pricing environment in the steel sector, and as one of its major steel customers also faces pressure to consolidate. We expect its customers to honor its long-term contracts, but new projects may face weaker pricing and project ramp may slow in the next 12-18 months. As for merchant gas sales, both pricing and volumes remain challenged, and we are cautious on demand from distributors in 2H12.

Figure 1. Yingde Gases – Key Statistics

Y/e Dec, RMB m	2009	2010	2011	2012E	2013E	2014E
Total Installed Capacity (Nm3/hour)						
Oxygen (Nm3/hour)	424,300	692,800	937,400	1,167,400	1,334,900	1,554,900
Argon (Nm3/hour)	14,760	18,000	21,760	28,650	36,963	42,463
Nitrogen (Nm3/hour)	689,500	950,000	1,121,900	1,390,700	1,761,700	1,981,700
Sales Volume (Oxygen Total, FY forecast)	2,987,000	3,491,000	6,067,989	7,082,770	7,879,878	9,393,481
Sales Volume Growth (Y/Y, %)	60%	17%	74%	17%	11%	19%
Sales Volume (on-site, FY forecast)						
Oxygen (Nm3 '000s)	2,786,998	3,254,630	5,772,464	6,866,030	7,562,838	9,010,757
Argon (Nm3 '000s)	22,955	25,350	49,446	52,725	71,216	83,450
Nitrogen (Nm3 '000s)	2,273,300	2,786,882	3,485,025	4,259,936	5,548,580	6,327,092
On-site Sales Volume Growth (Y/Y, %)	64%	19%	53%	20%	18%	17%
Sales Volume (merchant, FY forecast)						
Oxygen (Nm3 '000s)	200,002	236,370	295,525	216,740	317,039	382,724
Argon (Nm3 '000s)	61,045	80,650	70,389	69,310	90,314	106,865
Nitrogen (Nm3 '000s)	38,700	112,118	186,246	209,404	290,175	323,844
Merchant Sales Volume Growth (Y/Y, %)	57%	43%	29%	-10%	41%	17%
Merchant Market ASP						
Oxygen (RMB/Nm3)	0.74	0.91	1.05	0.99	1.01	1.03
Oxygen ASP (Y/Y, %)	-6%	23%	16%	-6%	2%	2%
Argon (RMB/Nm3)	1.62	4.03	3.37	1.67	1.71	1.74
Argon ASP (Y/Y, %)	-50%	149%	-16%	-50%	2%	2%
Nitrogen (RMB/Nm3)	0.68	0.72	0.83	0.79	0.81	0.85
Nitrogen ASP (Y/Y, %)	-11%	6%	16%	-5%	2%	5%

Source: Company and Citi Research estimates

Figure 2. Yingde Gases – DCF Calculation (Rmb Millions)

Actual value in the run up to FY20E:	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
FCF		-1,375	-1,269	-868	281	724	1,034	1,311	1,595	1,858
Discount factor		1.00	1.07	1.15	1.24	1.33	1.43	1.54	1.66	1.78
Net debt	-2,242	-1,375	-1,181	-752	226	543	721	851	963	1,045
Terminal Value:										
NAV in 2021E		1,081.1								
Expected terminal growth rate		3.5%								
Expected capital employed on new investments		14.0%								
WACC		7.47%								
Terminal value		20,446								
Discount factor		1.78								
NPV of terminal value		11,494								
DCF/Share (HK\$)		7.00								

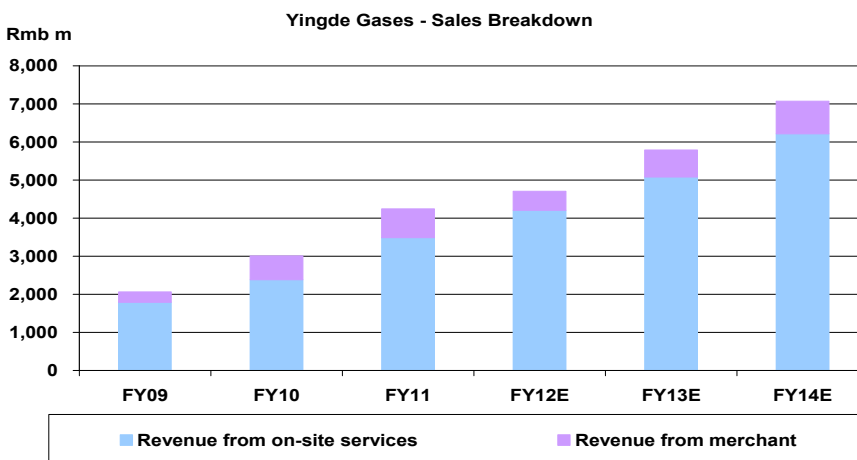
Source: Citi Research estimates

Figure 3. Yingde Gas – Income Statement (2009-2014E, Rmb Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Turnover	2,066	3,005	4,240	4,703	5,791	7,072
Revenue from on-site services	1,792	2,384	3,488	4,206	5,081	6,216
Revenue from merchant	273	621	752	497	709	855
Cost of sales	(1,291)	(1,841)	(2,789)	(3,333)	(4,102)	(5,060)
Gross profit	775	1,164	1,451	1,370	1,689	2,011
Other income	1	2	11	20	22	27
Distribution costs	(19)	(58)	(132)	(145)	(160)	(186)
Administration expenses	(137)	(322)	(239)	(302)	(355)	(412)
EBIT	620	786	1,091	942	1,195	1,440
Finance costs	(61)	(128)	(127)	(167)	(231)	(283)
Profit before taxation	566	668	981	799	993	1,191
Taxation	(34)	(135)	(146)	(143)	(197)	(250)
Minority interest	(2)	44	(5)	(4)	(5)	(6)
Net profits	530	577	831	652	791	936
Turnover YoY (%)	46.3%	45.5%	41.1%	10.9%	23.1%	22.1%
Net Profit YoY (%)	23.2%	8.9%	43.9%	-21.5%	21.2%	18.3%

Source: Company Reports and Citi Research estimates

Figure 4. Yingde Gases – Sales Breakdown (2009-2014E, RMB Millions)



Source: Citi Research

Figure 5. Yingde Gases – Key Financial Ratios

Y/e Dec, Rmb m	2009	2010	2011	2012E	2013E	2014E
Gross margin	37.5	38.7	34.2	29.1	29.2	28.4
Operating margin	30.0	26.2	25.7	20.0	20.6	20.4
Pre-tax margin	27.4	22.2	23.1	17.0	17.1	16.8
Net profit margin	25.7	19.2	19.6	13.9	13.7	13.2
Effective tax rate	6.0	20.2	14.9	17.9	19.9	21.0
Interest cover (x)	6.9	5.0	8.2	5.6	6.1	6.7
ROE (%)	14.0	13.3	16.6	12.6	14.7	16.3
Receivable days	65.7	91.0	88.6	89.0	90.0	90.0
Inventory days	1.9	6.0	4.2	5.2	7.2	7.2
Payable days	135.5	246.7	178.4	148.4	148.4	148.4
Net debt-to-equity (%)	(21.1)	21.7	44.9	77.0	105.5	123.0

Source: Company, Citi Research estimates

Figure 6. Yingde Gases – Balance Sheet (2009-2014E, Rmb Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Bank balances and cash	2,102	970	958	578	160	123
Pledged bank deposits	115	468	159	159	159	159
Account receivables	372	749	1,029	1,147	1,428	1,744
Inventories	7	30	32	48	81	100
Others	5	3	9	8	8	8
Current assets	2,601	2,220	2,187	1,939	1,836	2,134
PP&E	2,873	4,847	6,069	8,063	10,097	11,807
Others	191	555	1,556	1,400	1,260	1,323
Intangible asset	10	13	13	14	14	15
Non-current assets	3,073	5,415	7,637	9,477	11,371	13,144
Total assets	5,674	7,635	9,825	11,415	13,207	15,278
Account payables	479	1,244	1,363	1,355	1,667	2,057
Borrowings - amt due within one year	783	784	1,116	1,339	1,406	1,476
Taxation payable	10	56	74	81	89	98
Others	3	3	11	11	12	12
Current liabilities	1,275	2,087	2,563	2,786	3,174	3,643
Borrowings - amt due after one year	550	1,160	2,205	3,367	4,573	5,830
Non-current liabilities	550	1,160	2,205	3,367	4,573	5,830
Total liabilities	1,825	3,247	4,768	6,152	7,747	9,473
Shareholders equity	3,776	4,353	4,991	5,193	5,384	5,723
Minority interest	49	14	12	16	21	27
Total Equity	3,825	4,367	5,003	5,209	5,405	5,750

Source: Company, Citi Research estimates

Figure 7. Yingde Gases – Cash Flow Statement (2009-2014E, Rmb Millions)

	FY09	FY10	FY11	FY12E	FY13E	FY14E
Profit before tax	566	668	981	799	993	1,191
Adjustments	(86)	(186)	(110)	(143)	(203)	(249)
EBIT	480	483	872	655	790	942
Exceptional items	55	310	102	152	151	154
Depreciation & amortisation	115	159	297	235	376	460
Change in working capital	(171)	(220)	(162)	(83)	(64)	(125)
Operating cash flow	564	918	1,219	1,103	1,456	1,680
Net operating cash flow	478	732	976	854	1,142	1,301
Capex	(888)	(1,930)	(2,030)	(2,230)	(2,410)	(2,169)
Free cash flow	(410)	(1,197)	(1,053)	(1,375)	(1,269)	(868)
Change in equity	1,966	(13)	0	0	0	0
Change in debt	404	601	1,376	1,385	1,274	1,327
Other investing activities	115	(510)	(159)	(143)	(146)	(149)
Investing cash flow	(774)	(2,440)	(2,189)	(2,373)	(2,557)	(2,318)
Dividends paid	0	0	(181)	(235)	(282)	(353)
Financing cash flow	2,369	593	1,201	1,156	997	981
Beginning cash	28	2,102	970	941	578	160
Change in cash	2,074	(1,114)	(30)	(363)	(418)	(37)
Ending cash	2,102	970	941	578	160	123

Source: Company, Citi Research estimates

Yingde Gases

Company description

Yingde Gases is the largest domestic independent industrial gas supplier specializing in on-site gas supply in China. It has been providing mainly air separation units (ASU) to customers in a number of locations in China, which the company operates to separate the gas from air. On-site gas supply generated 89% of Yingde's total revenue in 2010, with its customers mainly being in the steel and chemical industries.

Investment strategy

We rate Yingde Gases Neutral (2) with a DCF-based target price of HK\$7.00. Yingde focuses on industrial gas production, with significant exposure to steel makers and is expanding its coverage to chemical and non-ferrous segments. While we believe its strong on-site business model will allow a significant expansion of its gas volume in the next two years, earnings growth may slow in 2012 given the weaker utilization rate and also rising ramp-up costs. We are also concerned about a slowdown in new project ramp-up, given rising competition for industrial gas outsourcing.

Valuation

Our HK\$7.00 target price for Yingde is based on a DCF valuation model and equates to 16X FY12E earnings. We use DCF as our primary valuation methodology to factor in its predictable cash flows driven by its take-or-pay contracts and to account for the company's increasing project scale. Our DCF valuation assumes a WACC of 7.47% and a terminal growth rate of 3.1%. Our WACC is calculated using a risk-free rate of 3.5%; a market-risk premium of 8.0%; an equity beta of 1.47; a cost of debt of 5.50%; a target gearing of 70%; and a 25% tax rate.

Risks

We see the main risks to our Neutral rating and target price as being unexpected changes to the company's off-take agreements and higher merchant-gas demand. While the company's existing projects have take-or-pay agreements, it can see better returns by tapping into the merchant market to sell excess capacity. As such, changes in merchant prices can have an impact on its overall profitability. Other risks include the ability to raise prices on existing project, demand for its products, new contracts for specialty gas projects, merchant gas prices.

Appendix A-1

Analyst Certification

The research analyst(s) primarily responsible for the preparation and content of this research report are named in bold text in the author block at the front of the product except for those sections where an analyst's name appears in bold alongside content which is attributable to that analyst. Each of these analyst(s) certify, with respect to the section(s) of the report for which they are responsible, that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc and its affiliates. No part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendation(s) or view(s) expressed by that research analyst in this report.

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Due to Citi's involvement in the acquisition of China Gas Holdings (the Company), Citi Research restricted publication of new research reports, and suspended its rating and target price on December 12, 2011 (the Suspension Date). While Citi Research may continue to publish research on the Company, it will not express a view about the proposed transaction, nor will its financial model(s) take into account the transaction. Additionally, the Company price chart available on Citi Research's disclosure website, is current only through the Suspension Date and, accordingly, does not reflect that Citi Research suspended its rating and target price on the Suspension Date.

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Citi is acting as financial advisor to China Petroleum & Chemical Corp and ENN Energy Holdings in the proposed takeover of China Gas Holdings.

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Gas Berhad, Power Grid Corporation of India, Petronet LNG, Suez Environnement, SP Ausnet, Tenaga Nasional, Tata Power, Veolia Environnement, Vestas Wind System (Den), YTL Power International, YTL Corp.

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<i>Data current as of 5 Oct 2012</i>	12 Month Rating			Relative Rating		
	Buy	Hold	Sell	Buy	Hold	Sell
Citi Research Global Fundamental Coverage	51%	38%	11%	7%	85%	7%
% of companies in each rating category that are investment banking clients	50%	47%	45%	59%	47%	50%

Guide to Citi Research Fundamental Research Investment Ratings:

Citi Research stock recommendations include an investment rating and an optional risk rating to highlight high risk stocks.

Risk rating takes into account both price volatility and fundamental criteria. Stocks will either have no risk rating or a High risk rating assigned.

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